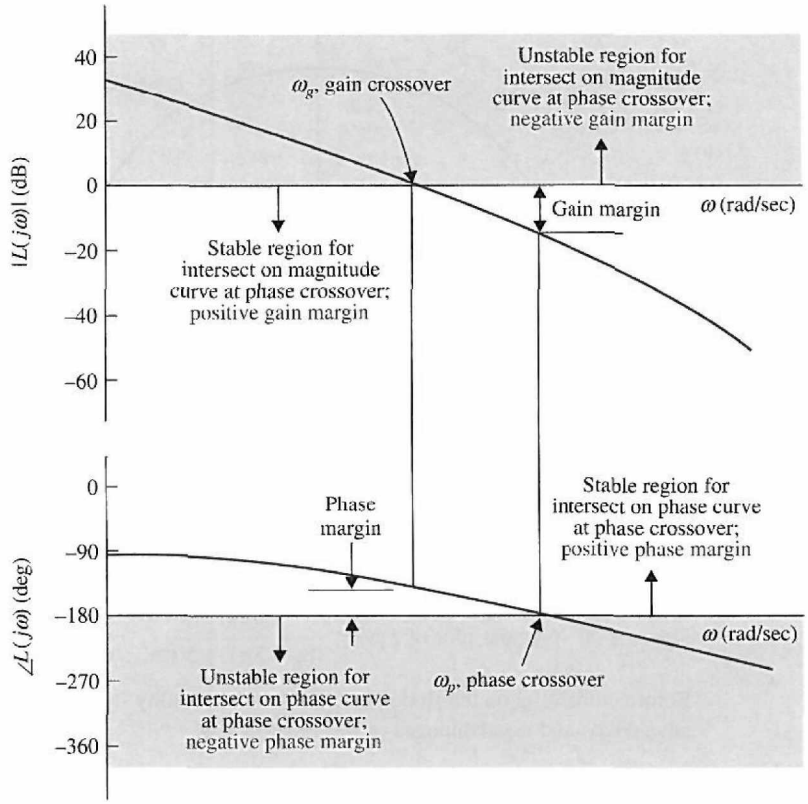


Figure 8-41 Determination of gain margin and phase margin on the Bode plot.

Disadvantage of the Bode Plot

1. Absolute and relative stability of only minimum-phase systems can be determined from the Bode plot. For instance, there is no way of telling what the stability criterion is on the Bode plot.

With reference to the definitions of gain margin and phase margin given in Figs. 8-37 and 8-39, respectively, the interpretation of these parameters from the Bode diagram is illustrated in Fig. 8-41 for a typical minimum-phase loop transfer function. The following observations can be made on system stability with respect to the properties of the Bode plot:

- Bode plots are useful only for stability studies of systems with minimum-phase loop transfer functions.

1. The gain margin is positive and the system is stable if the magnitude of $L(j\omega)$ at the phase crossover is negative in dB. That is, the gain margin is measured below the 0-dB-axis. If the gain margin is measured above the 0-dB-axis, the gain margin is negative, and the system is unstable.
2. The phase margin is positive and the system is stable if the phase of $L(j\omega)$ is greater than $-180°$ at the gain crossover. That is, the phase margin is measured above the $-180°$-axis. If the phase margin is measured below the $-180°$-axis, the phase margin is negative, and the system is unstable.

▶ **EXAMPLE 8-11-1** Consider the loop transfer function given in Eq. (8-98); the Bode plot of the function is drawn as shown in Fig. 8-42. The following results are observed easily from the magnitude and phase plots.

The gain crossover is the point where the magnitude curve intersects the 0-dB axis. The gain-crossover frequency ω_g is 6.22 rad/sec. The phase margin is measured at the gain crossover. The phase

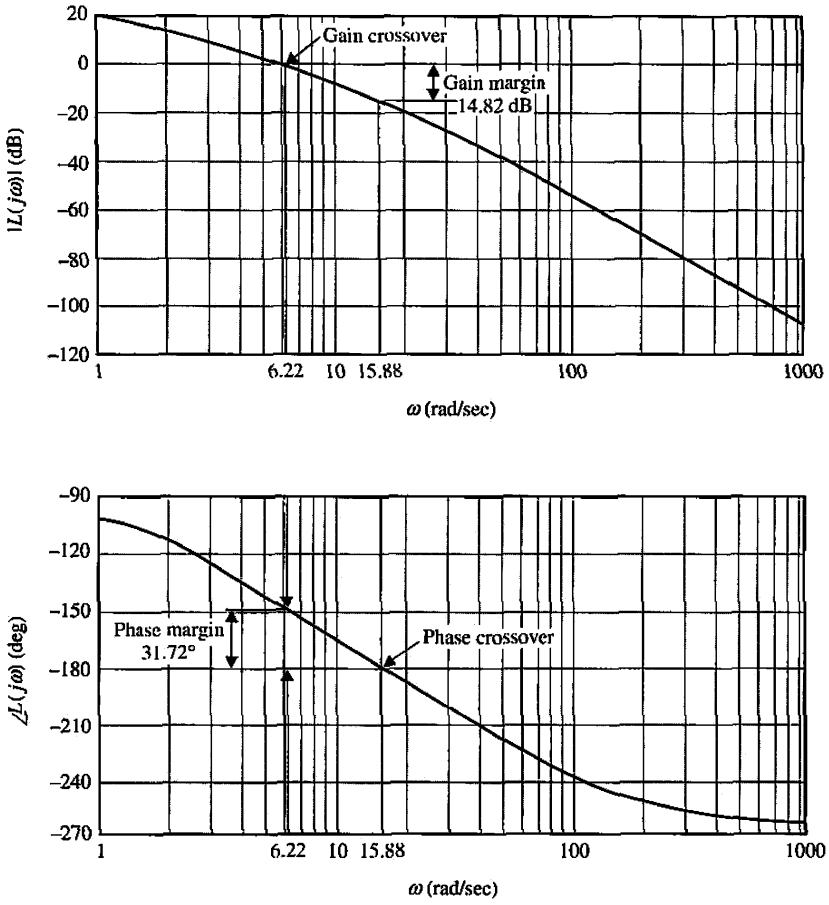


Figure 8-42 Bode plot of $L(s) = \dfrac{2500}{s(s+5)(s+50)}$.

margin is measured from the $-180°$-axis and is 31.72°. Because the phase margin is measured above the $-180°$-axis, the phase margin is positive, and the system is stable.

The phase crossover is the point where the phase curve intersects the $-180°$-axis. The phase-crossover frequency is $\omega_p = 15.88$ rad/sec. The gain margin is measured at the phase crossover and is 14.8 dB. Because the gain margin is measured below the 0-dB-axis, the gain margin is positive, and the system is stable.

Toolbox 8-11-1

MATLAB statements for Fig. 8-42

```
G = zpk([],[0 -1 -1],2500)
margin(G)

grid
```

The reader should compare the Nyquist plot of Fig. 8-40 with the Bode plot of Fig. 8-42, and the interpretation of ω_g, ω_p, GM, and PM on these plots. ◄

8-11-1 Bode Plots of Systems with Pure Time Delays

The stability analysis of a closed-loop system with a pure time delay in the loop can be conducted easily with the Bode plot. Example 8-11-2 illustrates the standard procedure.

► **EXAMPLE 8-11-2** *Consider that the loop transfer function of a closed-loop system is*

$$L(s) = \frac{Ke^{-T_d s}}{s(s+1)(s+2)} \tag{8-101}$$

Fig. 8-43 shows the Bode plot of $L(j\omega)$ with $K = 1$ and $T_d = 0$. The following results are obtained:

Gain-crossover frequency = 0.446 rad/sec

Phase margin = 53.4°

Phase-crossover frequency = 1.416 rad/sec

Gain margin = 15.57 dB

Thus, the system with the present parameters is stable.

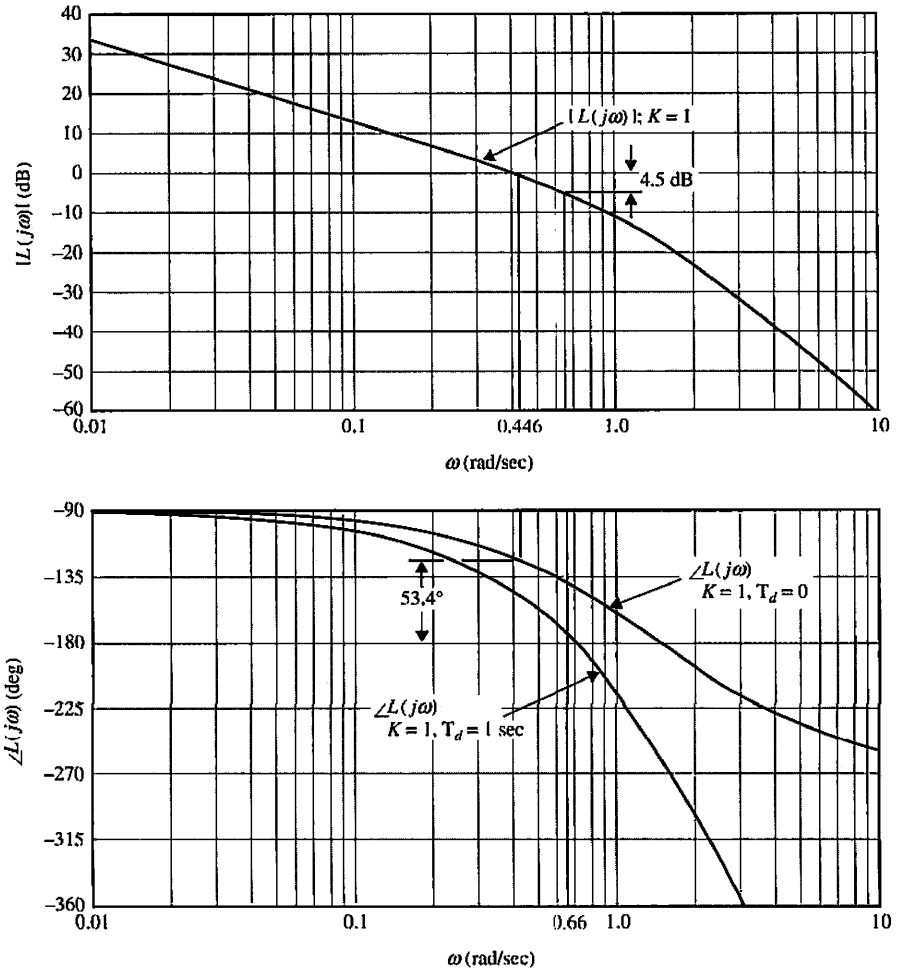


Figure 8-43 Bode plot of $L(s) = \dfrac{Ke^{-T_d s}}{s(s+1)(s+2)}$.

The effect of the pure time delay is to add a phase of $-T_d\omega$ radians to the phase curve while not affecting the magnitude curve. The adverse effect of the time delay on stability is apparent, because the negative phase shift caused by the time delay increases rapidly with the increase in ω. To find the critical value of the time delay for stability, we set

$$T_d\omega_g = 53.4^\circ \frac{\pi}{180^\circ} = 0.932 \quad \text{radians} \tag{8-102}$$

Solving for T_d from the last equation, we get the critical value of T_d to be 2.09 seconds.

Continuing with the example, we set T_d arbitrarily at 1 second and find the critical value of K for stability. Fig. 8-43 shows the Bode plot of $L(j\omega)$ with this new time delay. With K still equal to 1, the magnitude curve is unchanged. The phase curve droops with the increase in ω, and the following results are obtained:

Phase-crossover frequency = 0.66 rad/sec

Gain margin = 4.5 dB

Thus, using the definition of gain margin of Eq. (8-91), the critical value of K for stability is $10^{4.5/20} = 1.68$.

8-12 RELATIVE STABILITY RELATED TO THE SLOPE OF THE MAGNITUDE CURVE OF THE BODE PLOT

In addition to GM, PM, and M_p as relative stability measures, the slope of the magnitude curve of the Bode plot of the loop transfer function at the gain crossover also gives a qualitative indication on the relative stability of a closed-loop system. For example, in Fig. 8-42, if the loop gain of the system is decreased from the nominal value, the magnitude curve is shifted downward, while the phase curve is unchanged. This causes the gain-crossover frequency to be lower, and the slope of the magnitude curve at this frequency is less negative; the corresponding phase margin is increased. On the other hand, if the loop gain is increased, the gain-crossover frequency is increased, and the slope of the magnitude curve is more negative. This corresponds to a smaller phase margin, and the system is less stable. The reason behind these stability evaluations is quite simple. For a minimum-phase transfer function, the relation between its magnitude and phase is unique. Because the negative slope of the magnitude curve is a result of having more poles than zeros in the transfer function, the corresponding phase is also negative. In general, the steeper the slope of the magnitude curve, the more negative the phase. Thus, if the gain crossover is at a point where the slope of the magnitude curve is steep, it is likely that the phase margin will be small or negative.

8-12-1 Conditionally Stable System

The illustrative examples given thus far are uncomplicated in the sense that the slopes of the magnitude and phase curves are monotonically decreasing as ω increases. The following example illustrates a **conditionally stable system** that is capable of going through stable/unstable conditions as the loop gain varies.

EXAMPLE 8-12-1 Consider that the loop transfer function of a closed-loop system is

$$L(s) = \frac{100K(s+5)(s+40)}{s^3(s+100)(s+200)} \tag{8-103}$$

The Bode plot of $L(j\omega)$ is shown in Fig. 8-44 for $K = 1$. The following results on the system stability are obtained:

Gain-crossover frequency = 1 rad/sec

Phase margin = -78°

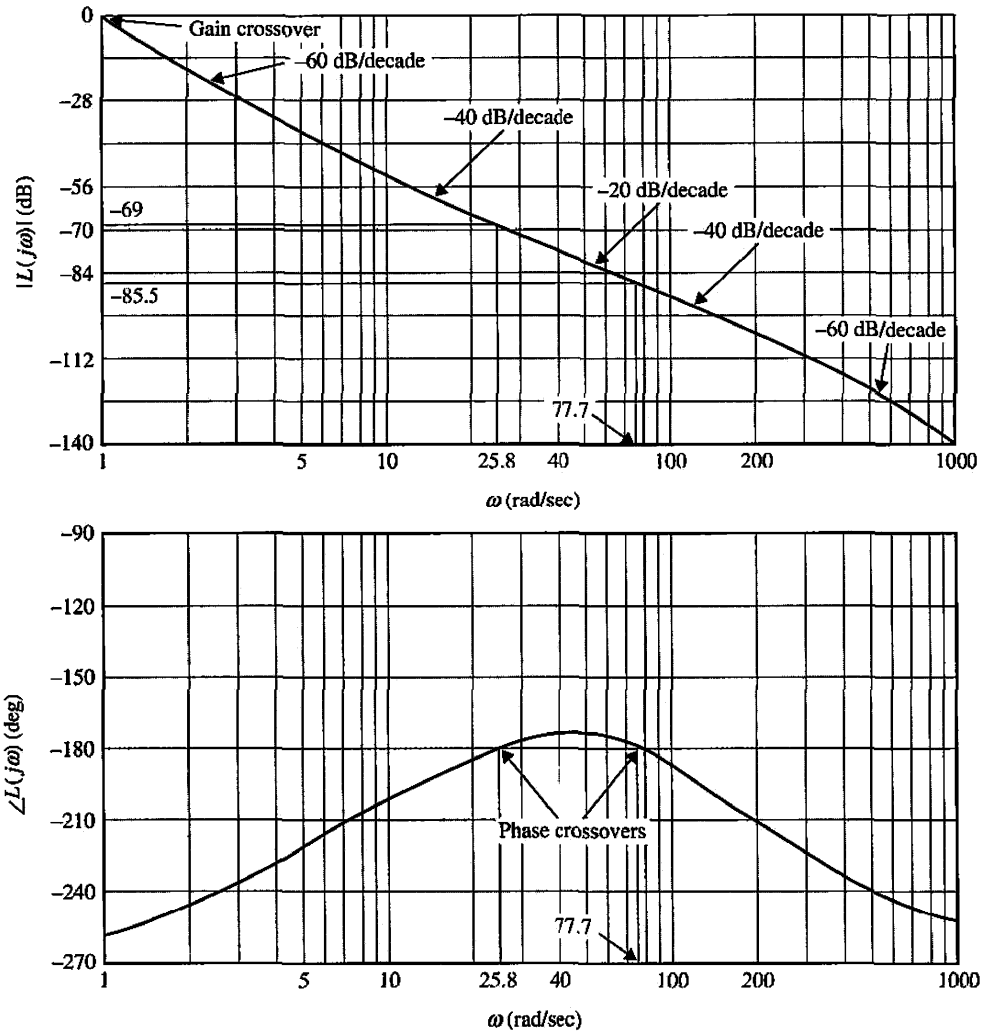


Figure 8-44 Bode plot of $L(s) = \dfrac{100K(s+5)(s+40)}{s^3(s+100)(s+200)}$, $K = 1$.

There are two phase crossovers: one at 25.8 rad/sec and the other at 77.7 rad/sec. The phase characteristics between these two frequencies indicate that, if the gain crossover lies in this range, the system would be stable. From the magnitude curve, the range of K for stable operation is found to be between 69 and 85.5 dB. For values of K above and below this range, the phase of $L(j\omega)$ is less than $-180°$, and the system is unstable. This example serves as a good example of the relation between relative stability and the slope of the magnitude curve at the gain crossover. As observed from Fig. 8-44, at both very low and very high frequencies, the slope of the magnitude curve is -60 dB/decade; if the gain crossover falls in either one of these two regions, the phase margin is negative, and the system is unstable. In the two sections of the magnitude curve that have a slope of -40 dB/decade, the system is stable only if the gain crossover falls in about half of these regions, but even then the phase margin is small. If the gain crossover falls in the region in which the magnitude curve has a slope of -20 dB/decade, the system is stable.

Fig. 8-45 shows the Nyquist plot of $L(j\omega)$. It is of interest to compare the results on stability derived from the Bode plot and the Nyquist plot. The root-locus diagram of the system is shown in Fig. 8-46. The root loci give a clear picture on the stability condition of the system with respect to K. The number of crossings of the root loci on the $j\omega$-axis of the s-plane equals the number of crossings

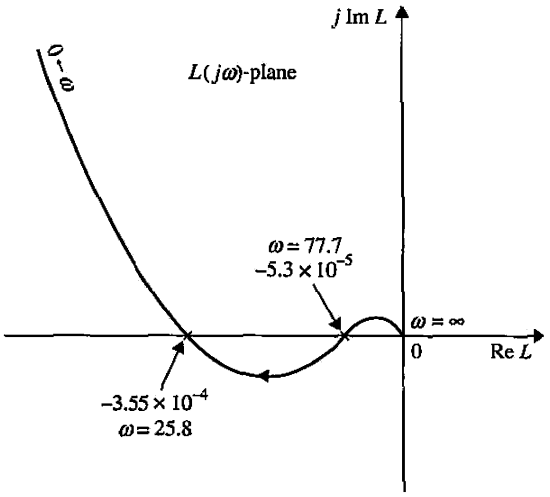


Figure 8-45 Nyquist plot of $L(s) = \dfrac{100K(s+5)(s+40)}{s^3(s+100)(s+200)}$, $K = 1$.

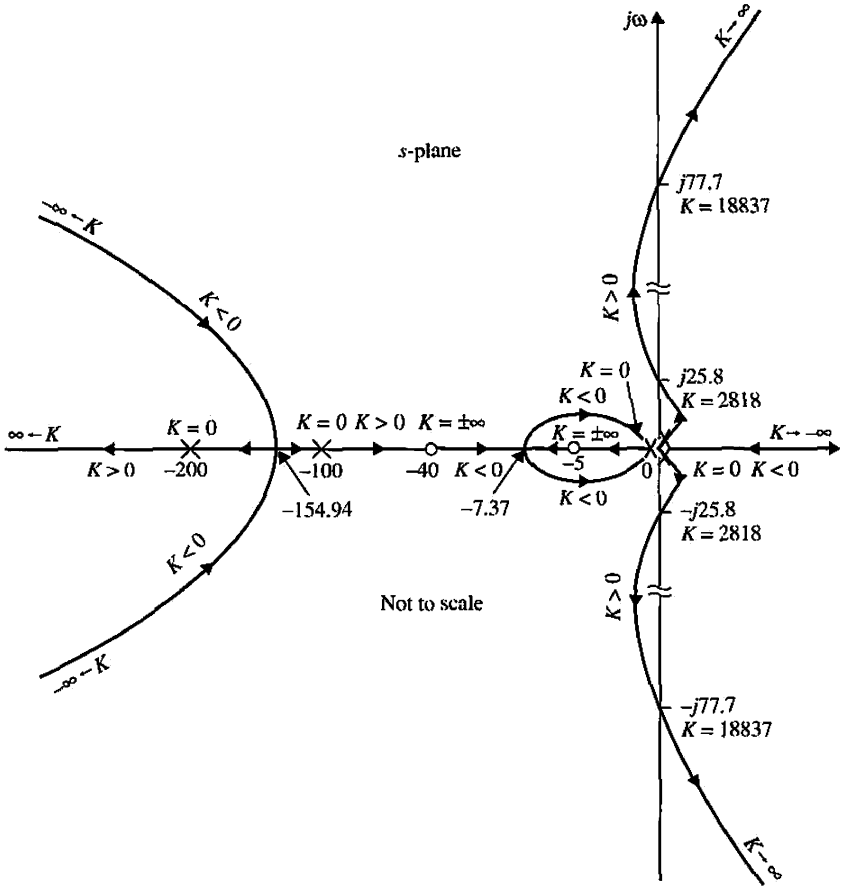


Figure 8-46 Root loci of $G(s) = \dfrac{100K(s+5)(s+40)}{s^3(s+100)(s+200)}$.

of the phase curve of $L(j\omega)$ of the $-180°$ axis of the Bode plot and the number of crossings of the Nyquist plot of $L(j\omega)$ with the negative real axis. The reader should check the gain margins obtained from the Bode plot and the coordinates of the crossover points on the negative real axis of the Nyquist plot with the values of K at the $j\omega$-axis crossings on the root loci. ◀

8-13 STABILITY ANALYSIS WITH THE MAGNITUDE-PHASE PLOT

The magnitude-phase plot described in Chapter 2 is another form of the frequency-domain plot that has certain advantages for analysis and design in the frequency domain. The magnitude-phase plot of a transfer function $L(j\omega)$ is done in $|L(j\omega)|$(dB) versus $\angle L(j\omega)$ (degrees). The magnitude-phase plot of the transfer function in Eq. (8-98) is constructed in Fig. 8-47 by use of the data from the Bode plot of Fig. 8-42. The gain and phase crossovers and the gain and phase margins are clearly indicated on the magnitude-phase plot of $L(j\omega)$.

- The critical point is the intersect of the 0-dB-axis and the $-180°$-axis.
- The phase crossover is where the locus intersects the $-180°$-axis.

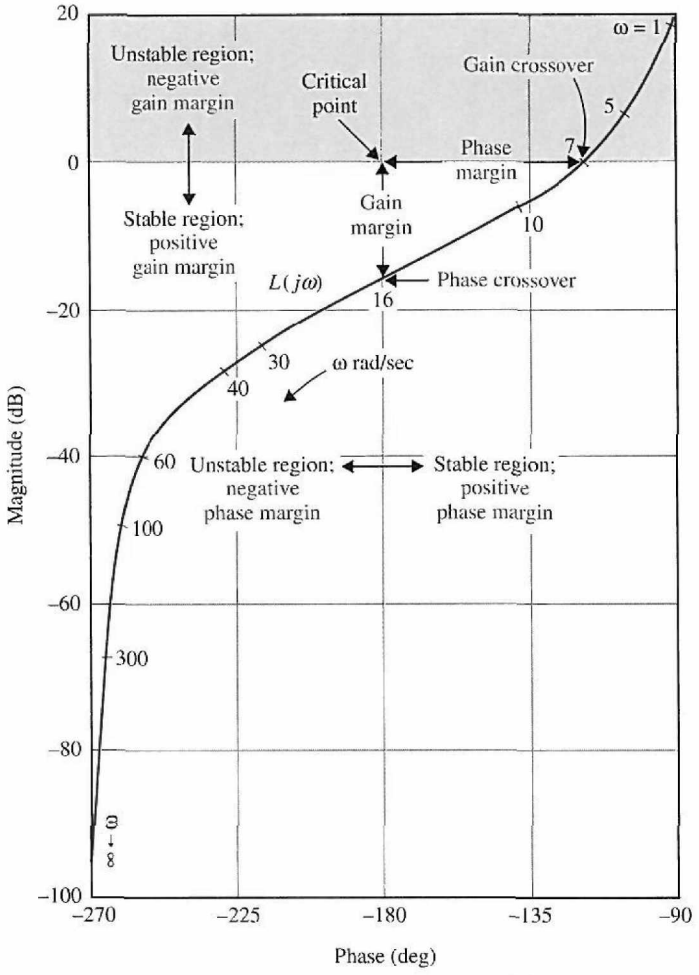


Figure 8-47 Gain-phase plot of $L(s) = \dfrac{10}{s(1 + 0.2s)(1 + 0.02s)}$.

- The gain crossover is where the locus intersects the 0-dB-axis.
- The gain margin is the vertical distance in dB measured from the phase crossover to the critical point.
- The phase margin is the horizontal distance measured in degrees from the gain crossover to the critical point.

The regions in which the gain and phase crossovers should be located for stability are also indicated. Because the vertical axis for $|L(j\omega)|$ is in dB, when the loop gain of $L(j\omega)$ changes, the locus is simply shifted up and down along the vertical axis. Similarly, when a constant phase is added to $L(j\omega)$, the locus is shifted horizontally without distortion to the curve. If $L(j\omega)$ contains a pure time delay T_d, the effect of the time delay is to add a phase equal to $-\omega T_d \times 180°/\pi$ along the curve.

Another advantage of using the magnitude-phase plot is that, *for unity-feedback systems*, closed-loop system parameters such as M_r, ω_r, and BW can all be determined from the plot with the help of the constant-M loci. These closed-loop performance parameters are not represented on the Bode plot of the forward-path transfer function of a unity-feedback system.

8-14 CONSTANT-M LOCI IN THE MAGNITUDE-PHASE PLANE: THE NICHOLS CHART

It was pointed out earlier that, analytically, the resonant peak M_r and bandwidth BW are difficult to obtain for high-order systems, and the Bode plot provides information on the closed-loop system only in the form of gain margin and phase margin. It is necessary to develop a graphical method for the determination of M_r, ω_r, and BW using the forward-path transfer function $G(j\omega)$. As we shall see in the following development, the method is directly applicable only to unity-feedback systems, although with some modification it can also be applied to nonunity-feedback systems.

Consider that $G(s)$ is the forward-path transfer function of a unity-feedback system. The closed-loop transfer function is

$$M(s) = \frac{G(s)}{1 + G(s)} \tag{8-104}$$

For sinusoidal steady state, we replace s with $j\omega$; $G(s)$ becomes

$$\begin{aligned} G(j\omega) &= ReG(j\omega) + jImG(j\omega) \\ &= x + jy \end{aligned} \tag{8-105}$$

where, for simplicity, x denotes $\mathrm{Re}G(j\omega)$ and y denotes $\mathrm{Im}G(j\omega)$. The magnitude of the closed-loop transfer function is written

$$|M(j\omega)| = \left|\frac{G(j\omega)}{1 + G(j\omega)}\right| = \frac{\sqrt{x^2 + y^2}}{\sqrt{(1+x)^2+y^2}} \tag{8-106}$$

For simplicity of notation, let M denote $|M(j\omega)|$; then Eq. (8-106) leads to

$$M\sqrt{(1+x)^2+y^2} = \sqrt{x^2 + y^2} \tag{8-107}$$

Squaring both sides of Eq. (8-107) gives

$$M^2\left[(1+x)^2+y^2\right] = x^2 + y^2 \tag{8-108}$$

Rearranging Eq. (8-108) yields

$$(1-M^2)x^2+(1-M^2)y^2-2M^2x=M^2 \tag{8-109}$$

This equation is conditioned by dividing through by $(1-M^2)$ and adding the term $[M^2/(1-M^2)]^2$ on both sides. We have

$$x^2+y^2-\frac{2M^2}{1-M^2}x+\left(\frac{M^2}{1-M^2}\right)^2=\frac{M^2}{1-M^2}+\left(\frac{M^2}{1-M^2}\right)^2 \tag{8-110}$$

which is finally simplified to

$$\left(x-\frac{M^2}{1-M^2}\right)^2+y^2=\left(\frac{M}{1-M^2}\right)^2 \qquad M\neq 1 \tag{8-111}$$

For a given value of M, Eq. (8-111) represents a circle with the center at

$$x=\text{Re}G(j\omega)=\frac{M^2}{1-M^2} \qquad y=0 \tag{8-112}$$

The radius of the circle is

$$r=\left|\frac{M}{1-M^2}\right| \tag{8-113}$$

When M takes on different values, Eq. (8-111) describes in the $G(j\omega)$-plane a family of circles that are called the **constant-M loci**, or the **constant-M circles**. Fig. 8-48 illustrates a typical set of constant-M circles in the $G(j\omega)$-plane. These circles are symmetrical with

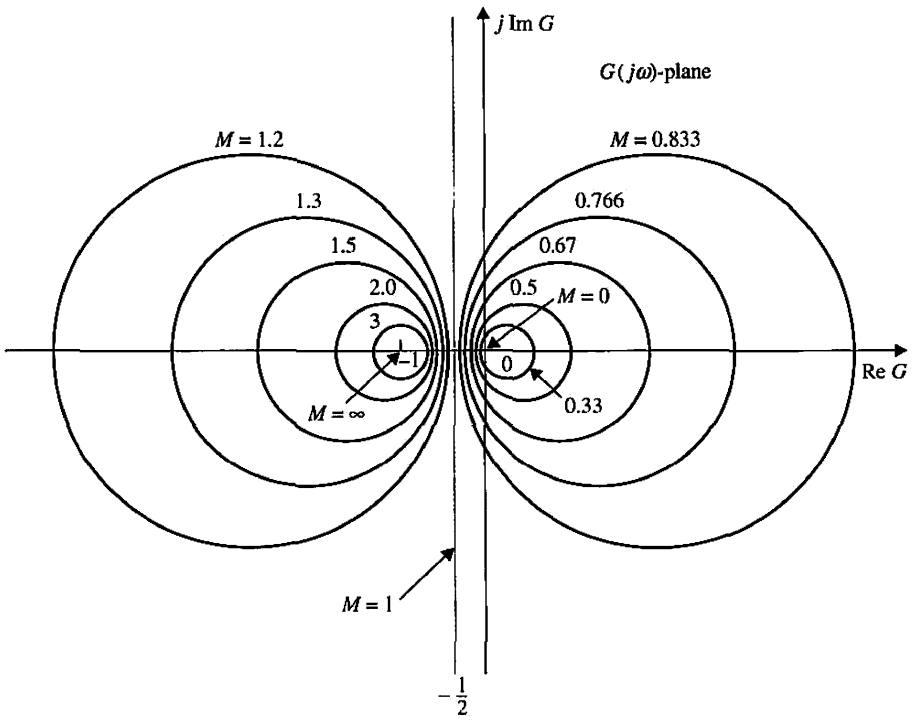
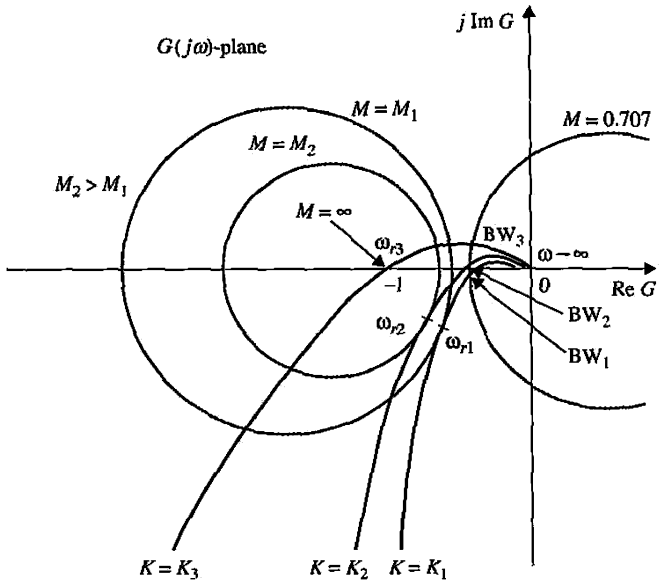


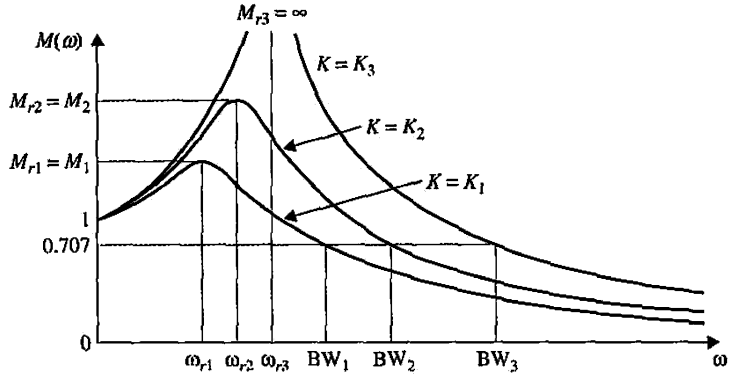
Figure 8-48 Constant-M circles in polar coordinates.

respect to the $M = 1$ line and the real axis. The circles to the left of the $M = 1$ locus correspond to values of M greater than 1, and those to the right of the $M = 1$ line are for M less than 1. Eqs. (8-111) and (8-112) show that, when M becomes infinite, the circle degenerates to a point at $(-1, j0)$. Graphically, the intersection of the $G(j\omega)$ curve and the constant-M circle gives the value of M at the corresponding frequency on the $G(j\omega)$ curve. If we want to keep the value of M_r less than a certain value, the $G(j\omega)$ curve must not intersect the corresponding M circle at any point and at the same time must not enclose the $(-1, j0)$ point. The constant-M circle with the smallest radius that is tangent to the $G(j\omega)$ curve gives the value of M_r, and the resonant frequency ω_r is read off at the tangent point on the $G(j\omega)$ curve.

Fig. 8-49(a) illustrates the Nyquist plot of $G(j\omega)$ for a unity-feedback control system together with several constant-M loci. For a given loop gain $K = K_1$, the intersects between the $G(j\omega)$ curve and the constant-M loci give the points on the $|M(j\omega)|$-versus-ω curve. The resonant peak M_r is found by locating the smallest circle that is tangent to the $G(j\omega)$



(a)



(b)

Figure 8-49 (a) Polar plots of $G(s)$ and constant-M loci. (b) Corresponding magnification curves.

curve. The resonant frequency is found at the point of tangency and is designated as ω_{r1}. If the loop gain is increased to K_2, and if the system is still stable, a constant-M circle with a smaller radius that corresponds to a larger M is found tangent to the $G(j\omega)$ curve, and thus the resonant peak will be larger. The resonant frequency is shown to be ω_{r2}, which is closer to the phase-crossover frequency ω_p than ω_{r1}. When K is increased to K_3, so that the $G(j\omega)$ curve now passes through the $(-1, j0)$ point, the system is marginally stable, and M_r is infinite; ω_{p3} is now the same as the resonant frequency ω_r.

When enough points of intersection between the $G(j\omega)$ curve and the constant-M loci are obtained, the magnification curves of $|M(j\omega)|$-versus-ω are plotted, as shown in Fig. 8-49(b).

The bandwidth of the closed-loop system is found at the intersect of the $G(j\omega)$ curve and the $M = 0.707$ locus. For values of K beyond K_3, the system is unstable, and the constant-M loci and M_r no longer have any meaning.

• When the system is unstable, the constant-M loci and M_r no longer have any meaning.

A major disadvantage in working in the polar coordinates of the Nyquist plot of $G(j\omega)$ is that the curve no longer retains its original shape when a simple modification such as the change of the loop gain is made to the system. Frequently, in design situations, not only must the loop gain be altered, but a series controller may have to be added to the system. This requires the complete reconstruction of the Nyquist plot of the modified $G(j\omega)$. For design work involving M_r and BW as specifications, it is more convenient to work with the magnitude-phase plot of $G(j\omega)$, because when the loop gain is altered, the entire $G(j\omega)$ curve is shifted up or down vertically without distortion. When the phase properties of $G(j\omega)$ are changed independently, without affecting the gain, the magnitude-phase plot is affected only in the horizontal direction.

• BW is the frequency where the $G(j\omega)$ curve intersects the $M = -3$ dB locus of the Nichols chart.

For that reason, the constant-M loci in the polar coordinates are plotted in magnitude-phase coordinates, and the resulting loci are called the **Nichols chart**. A typical Nichols chart of selected constant-M loci is shown in Fig. 8-50. Once the $G(j\omega)$ curve of the system is constructed in the Nichols chart, the intersects between the constant-M loci and the $G(j\omega)$ trajectory give the value of M at the corresponding frequencies of $G(j\omega)$. The resonant

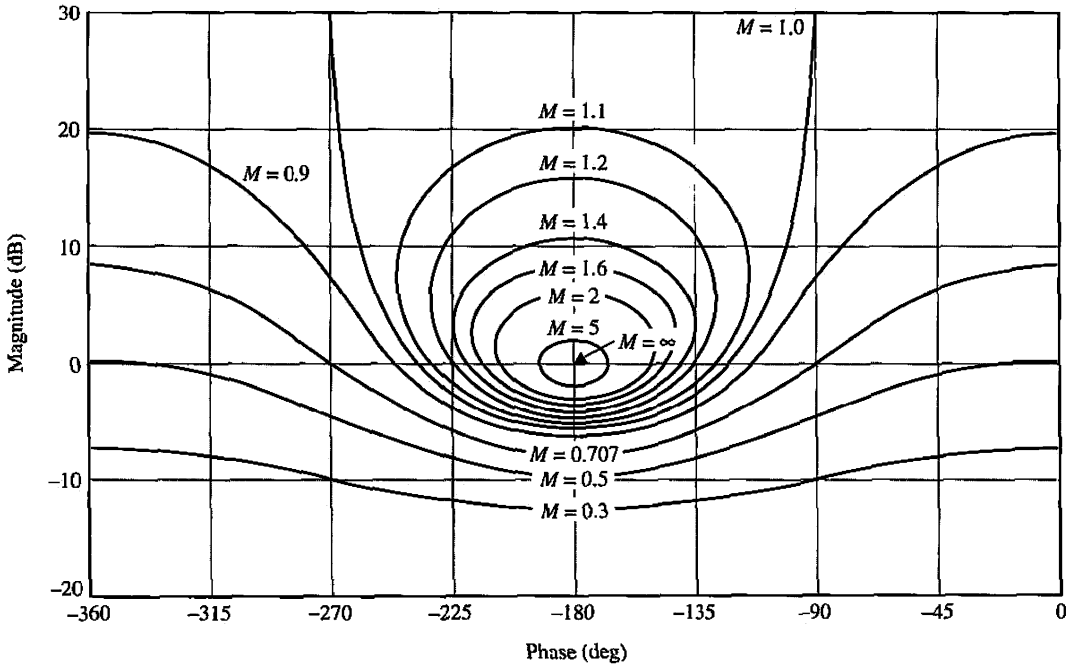


Figure 8-50 Nichols chart.

peak M_r is found by locating the smallest of the constant-M locus ($M \geq 1$) that is tangent to the $G(j\omega)$ curve from above. The resonant frequency is the frequency of $G(j\omega)$ at the point of tangency. *The bandwidth of the closed-loop system is the frequency at which the* G(jω) *curve intersects the* M = 0.707 *or* M = −3 *dB locus.*

The following example illustrates the relationship among the analysis methods using the Bode plot and the Nichols chart.

▶ EXAMPLE 8-14-1 Consider the position-control system of the control surfaces of the airship analyzed in Section 5-8. The forward-path transfer function of the unity-feedback system is given by Eq. (5-153), and is repeated here:

$$G(s) = \frac{1.5 \times 10^7 K}{s(s + 400.26)(s + 3008)} \tag{8-114}$$

The Bode plots for $G(j\omega)$ are shown in Fig. 8-51 for $K = 7.248, 14.5, 181.2$, and 273.57. The gain and phase margins of the closed-loop system for these values of K are determined and shown on

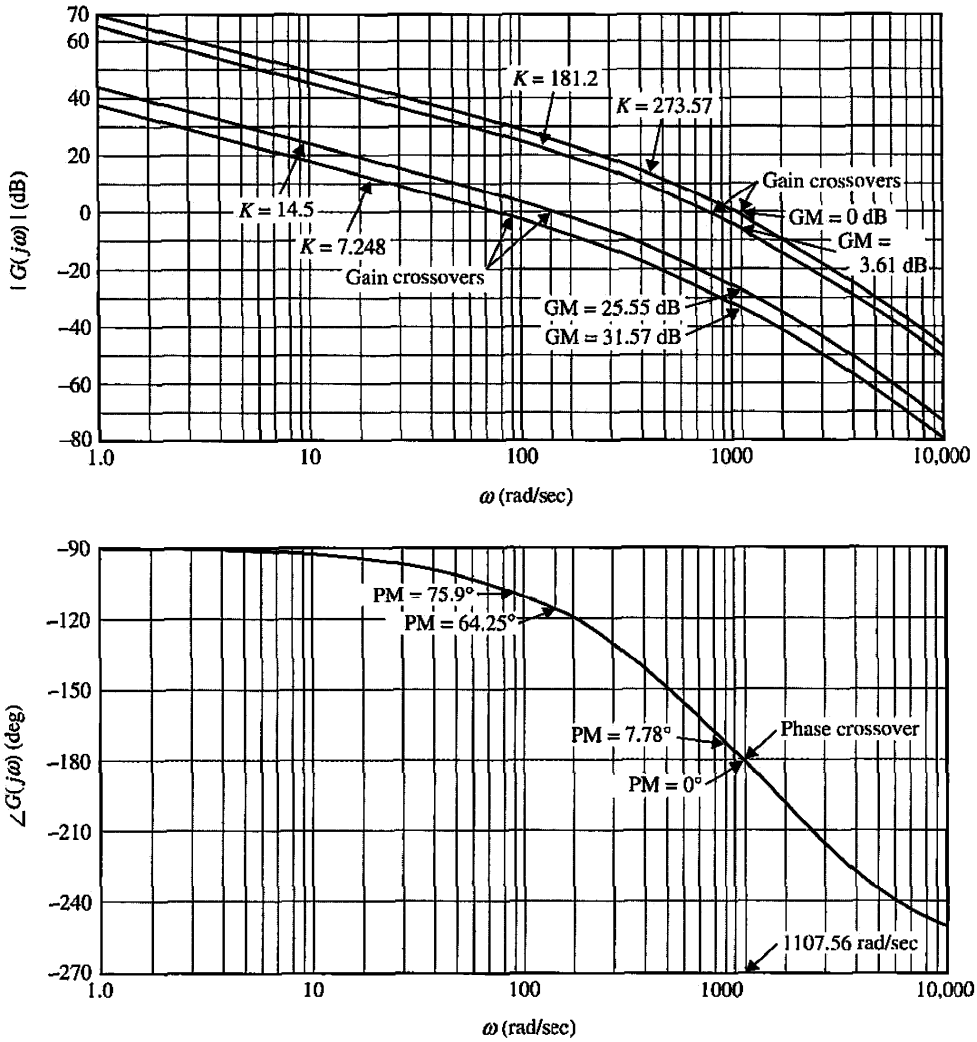


Figure 8-51 Bode diagrams of the system in Example 8-14-1.

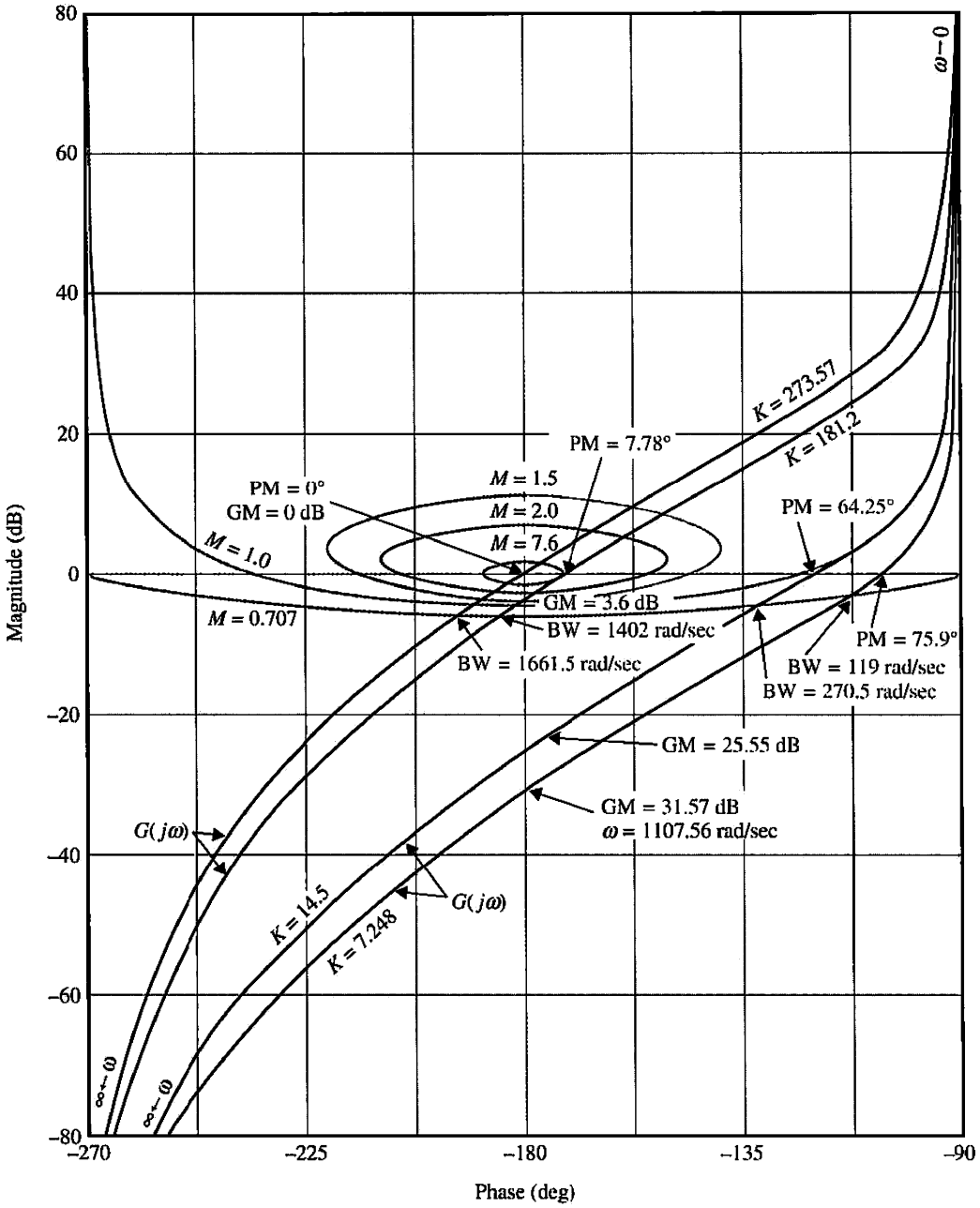


Figure 8-52 Gain-phase plots and Nichols chart of the system in Example 8-14-1.

the Bode plots. The magnitude-phase plots of $G(j\omega)$ corresponding to the Bode plots are shown in Fig. 8-52. These magnitude-phase plots, together with the Nichols chart, give information on the resonant peak M_r, resonant frequency ω_r, and the bandwidth BW. The gain and phase margins are also clearly marked on the magnitude-phase plots. Fig. 8-53 shows the closed-loop frequency responses. Table 8-2 summarizes the results of the frequency-domain analysis for the four different values of K together with the time-domain maximum overshoots determined in Section 5-8.

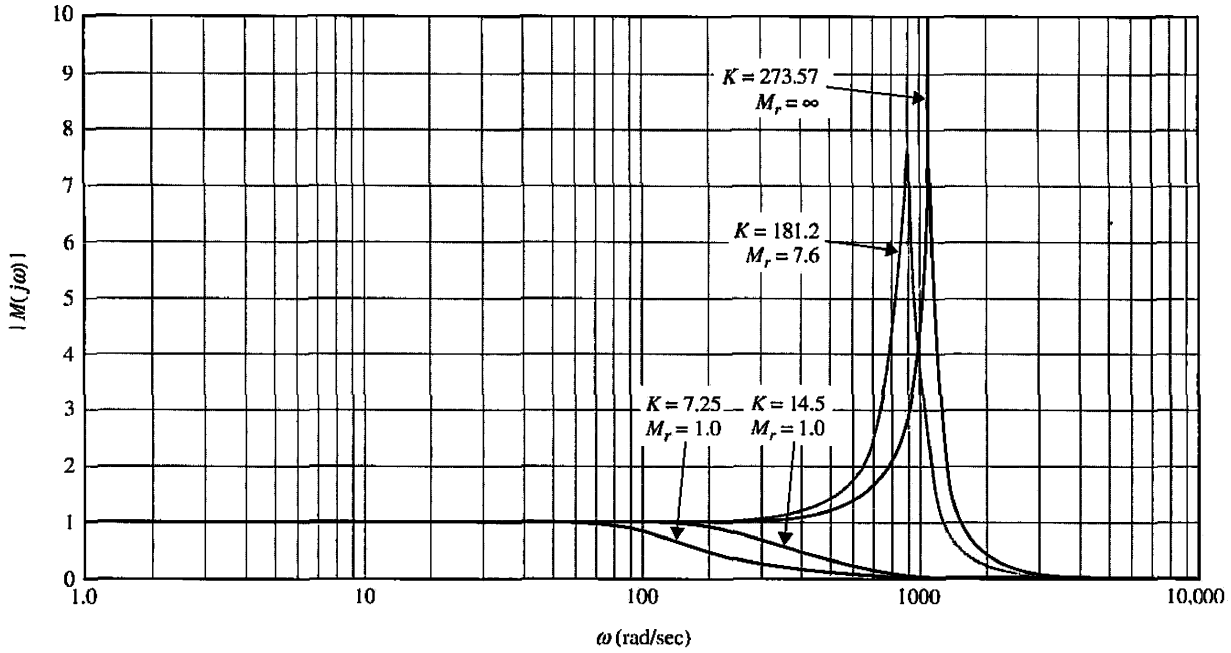


Figure 8-53 Closed-loop frequency response of the system in Example 8-14-1.

TABLE 8-2 Summary of Frequency-Domain Analysis

K	Maximum Overshoot (%)	M_r	ω_r (rad/sec)	Gain Margin (dB)	Phase Margin (deg)	BW (rad/sec)
7.25	0	1.0	1.0	31.57	75.9	119.0
14.5	4.3	1.0	43.33	5.55	64.25	270.5
181.2	15.2	7.6	900.00	3.61	7.78	1402.0
273.57	100.0	∞	1000.00	0	0	1661.5

8-15 NICHOLS CHART APPLIED TO NONUNITY-FEEDBACK SYSTEMS

The constant-M loci and the Nichols chart presented in the preceding sections are limited to closed-loop systems with unity feedback whose transfer function is given by Eq. (8-104). When a system has nonunity feedback, the closed-loop transfer function of the system is expressed as

$$M(s) = \frac{G(s)}{1 + G(s)H(s)} \tag{8-115}$$

where $H(s) \neq 1$. The constant-M loci and the Nichols chart cannot be applied directly to obtain the closed-loop frequency response by plotting $G(j\omega)H(j\omega)$, since the numerator of $M(s)$ does not contain $H(j\omega)$.

By proper modification, the constant-M loci and Nichols chart can still be applied to a nonunity-feedback system. Let us consider the function

$$P(s) = H(s)M(s) = \frac{G(s)H(s)}{1 + G(s)H(s)} \tag{8-116}$$

Apparently, Eq. (8-116) is of the same form as Eq. (8-104). The frequency response of $P(j\omega)$ can be determined by plotting the function $G(j\omega)H(j\omega)$ in the amplitude-phase coordinates along with the Nichols chart. Once this is done, the frequency-response information for $M(j\omega)$ is obtained as follows.

$$|M(j\omega)| = \frac{|P(j\omega)|}{|H(j\omega)|} \tag{8-117}$$

or, in terms of dB,

$$|M(j\omega)|(dB) = |P(j\omega)|(dB) - |H(j\omega)|(dB) \tag{8-118}$$

$$\phi_m(j\omega) = \angle M(j\omega) = \angle P(j\omega) - \angle H(j\omega) \tag{8-119}$$

▶ 8-16 SENSITIVITY STUDIES IN THE FREQUENCY DOMAIN

• Sensitivity study is easily carried out in the frequency domain.

The advantage of using the frequency domain in linear control systems is that higher-order systems can be handled more easily than in the time domain. Furthermore, the sensitivity of the system with respect to parameter variations can be easily interpreted using frequency-domain plots. We shall show how the Nyquist plot and the Nichols chart can be utilized for the analysis and design of control systems based on sensitivity considerations.

Consider a linear control system with unity feedback described by the transfer function

$$M(s) = \frac{G(s)}{1 + G(s)} \tag{8-120}$$

The sensitivity of $M(s)$ with respect to the loop gain K, which is a multiplying factor in $G(s)$, is defined as

$$S_G^M(s) = \frac{\dfrac{dM(s)}{M(s)}}{\dfrac{dG(s)}{G(s)}} = \frac{dM(s)}{dG(s)} \frac{G(s)}{M(s)} \tag{8-121}$$

Taking the derivative of $M(s)$ with respect to $G(s)$ and substituting the result into Eq. (8-121) and simplifying, we have

$$S_G^M(s) = \frac{1}{1 + G(s)} = \frac{1/G(s)}{1 + 1/G(s)} \tag{8-122}$$

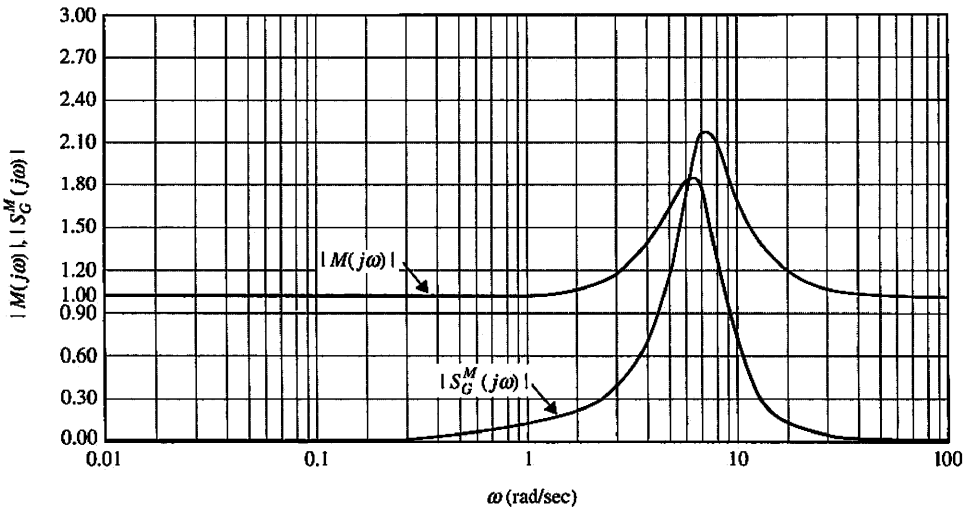


Figure 8-54 $|M(j\omega)|$ and $|S_G^M(j\omega)|$ versus ω for $G(s) = \dfrac{2500}{s(s+5)(s+2500)}$.

Clearly, the sensitivity function $S_G^M(s)$ is a function of the complex variable s. Fig. 8-54 shows the magnitude plot of $S_G^M(s)$ when $G(s)$ is the transfer function given in Eq. (8-98). It is interesting to note that the sensitivity of the closed-loop system is inferior at frequencies greater than 4.8 rad/sec to the open-loop system whose sensitivity to the variation of K is always unity. In general, it is desirable to formulate a design criterion on sensitivity in the following manner:

$$|S_G^M(j\omega)| = \frac{1}{|1 + G(j\omega)|} = \frac{|1/G(j\omega)|}{|1 + 1/G(j\omega)|} \leq k \tag{8-123}$$

where k is a positive real number. This sensitivity criterion is in addition to the regular performance criteria on the steady-state error and the relative stability.

Eq. (8-123) is analogous to the magnitude of the closed-loop transfer function, $|M(j\omega)|$, given in Eq. (8-106), with $G(j\omega)$ replaced by $1/G(j\omega)$. Thus, the sensitivity function of Eq. (8-123) can be determined by plotting $1/G(j\omega)$ in the magnitude-phase coordinates with the Nichols chart. Fig. 8-55 shows the magnitude-phase plots of $G(j\omega)$ and $1/G(j\omega)$ of Eq. (8-98). Notice that $G(j\omega)$ is tangent to the $M = 1.8$ locus from below,

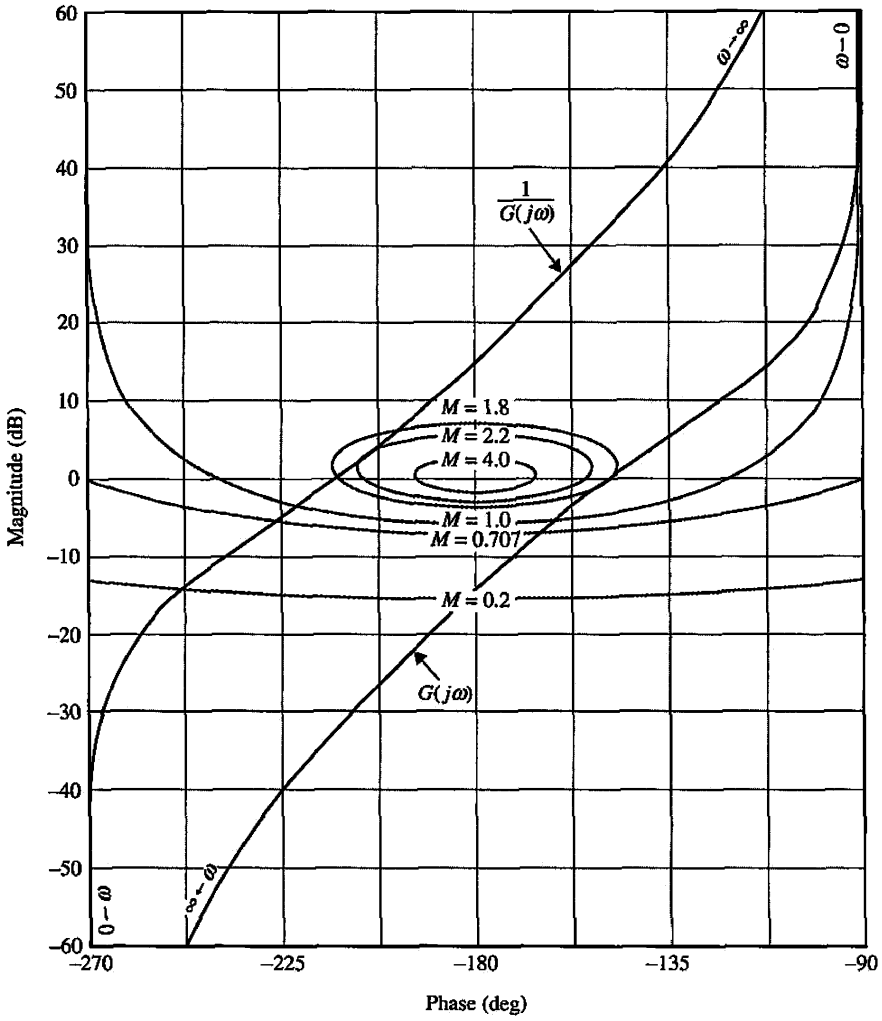


Figure 8-55 Magnitude-phase plots of $G(j\omega)$ and $1/G(j\omega)$ for $G(s) = \dfrac{2500}{s(s+5)(s+50)}$.

which means that M_r of the closed-loop system is 1.8. The $1/G(j\omega)$ curve is tangent to the $M = 2.2$ curve from above and, according to Fig. 8-54, is the maximum value of $|S_G^M(s)|$.

Eq. (8-123) shows that, for low sensitivity, the loop gain of $G(j\omega)$ must be high, but it is known that, in general, high gain could cause instability. Thus, the designer is again challenged by the task of designing a system with both a high degree of stability and low sensitivity.

The design of robust control systems (low sensitivity) with the frequency-domain methods is discussed in Chapter 9.

8-17 MATLAB TOOLS AND CASE STUDIES

Apart from the MATLAB toolboxes in this chapter, this chapter does not contain any software. In Chapter 9 we will introduce the Automatic Control Systems MATLAB tools. The Automatic Control Systems software (**ACSYS**) consists of a number of m-files and GUIs (graphical user interface) for the analysis of simple control engineering transfer functions. All the frequency response topics may also be solved utilizing **ACSYS**.

8-18 SUMMARY

The chapter began by describing typical relationships between the open-loop and closed-loop frequency responses of linear systems. Performance specifications such as the resonance peak M_r, resonant frequency ω_r, and bandwidth BW were defined in the frequency domain. The relationships among these parameters of a second-order prototype system were derived analytically. The effects of adding simple poles and zeros to the loop transfer function on M_r and BW were discussed.

The Nyquist criterion for stability analysis of linear control systems was developed. The stability of a single-loop control system can be investigated by studying the behavior of the Nyquist plot of the loop transfer function $G(s)H(s)$ for $\omega = 0$ to $\omega = \infty$ with respect to the critical point. If $G(s)H(s)$ is a minimum-phase transfer function, the condition of stability is simplified so that the Nyquist plot will not enclose the critical point.

The relationship between the root loci and the Nyquist plot was described in Section 8-7. The discussion should add more perspective to the understanding of both subjects.

Relative stability was defined in terms of gain margin and phase margin. These quantities were defined in the polar coordinates as well as on the Bode diagram. The gain-phase plot allows the Nichols chart to be constructed for closed-loop analysis. The values of M_r and BW can be easily found by plotting the $G(j\omega)$ locus on the Nichols chart.

The stability of systems with pure time delay is analyzed by use of the Bode plot.

Sensitivity function $S_G^M(j\omega)$ was defined as a measure of the variation of $M(j\omega)$ due to variations in $G(j\omega)$. It was shown that the frequency-response plots of $G(j\omega)$ and $1/G(j\omega)$ can be readily used for sensitivity studies.

Finally, using the MATLAB toolboxes developed in this chapter or the **ACSYS** software, described in detail in Chapter 9, the reader may practice all the concepts discussed here.

REVIEW QUESTIONS

1. Explain why it is important to conduct frequency-domain analyses of linear control systems.
2. Define resonance peak M_r of a closed-loop control system.
3. Define bandwidth BW of a closed-loop system.
4. List the advantages and disadvantages of studying stability with the Nyquist plot.
5. List the advantages and disadvantages of carrying out frequency-domain analysis with the Bode plot.

6. List the advantages and disadvantages of carrying out frequency-domain analysis with the magnitude-phase plot.

7. The following quantities are defined:
Z = number of zeros of $L(s)$ that are in the right-half s-plane
P = number of poles of $L(s)$ that are in the right-half s-plane
P_ω = number of poles of $L(s)$ that are on the $j\omega$-axis

Give the conditions on these parameters for the system to be (a) open-loop stable and (b) closed-loop stable.

8. What condition must be satisfied by the function $L(j\omega)$ so that the Nyquist criterion is simplified to investigating whether the $(-1, j0)$ point is enclosed by the Nyquist plot?

9. Give all the properties of a minimum-phase transfer function.

10. Give the definitions of gain margin and phase margin.

11. By applying a sinusoidal signal of frequency ω_0 to a linear system, the steady-state output of the system will also be of the same frequency. **(T) (F)**

12. For a prototype second-order system, the value of M_r depends solely on the damping ratio ζ. **(T) (F)**

13. Adding a zero to the loop transfer function will always increase the bandwidth of the closed-loop system. **(T) (F)**

14. The general effect of adding a pole to the loop transfer function is to make the closed-loop system less stable while decreasing the bandwidth. **(T) (F)**

15. For a minimum-phase loop transfer function $L(j\omega)$, if the phase margin is negative, then the closed-loop system is always unstable. **(T) (F)**

16. Phase-crossover frequency is the frequency at which the phase of $L(j\omega)$ is 0°. **(T) (F)**

17. Gain-crossover frequency is the frequency at which the gain of $L(j\omega)$ is 0 dB. **(T) (F)**

18. Gain margin is measured at the phase-crossover frequency. **(T) (F)**

19. Phase margin is measured at the gain-crossover frequency. **(T) (F)**

20. A closed-loop system with a pure time delay in the loop is usually less stable than one without a time delay. **(T) (F)**

21. The slope of the magnitude curve of the Bode plot of $L(j\omega)$ at the gain crossover usually gives indication on the relative stability of the closed-loop system. **(T) (F)**

22. Nichols chart can be used to find BW and M_r information of a closed-loop system. **(T) (F)**

23. Bode plot can be used for stability analysis for minimum- as well as nonminimum-phase transfer functions. **(T) (F)**

Answers to these review questions can be found on this book's companion Web site: www.wiley.com/college/golnaraghi.

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2. R. W. Brockett and J. L. Willems, "Frequency Domain Stability Criteria—Part I," *IEEE Trans. Automatic Control*, Vol. AC-10, pp. 255–261, July 1965.
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Sensitivity Function

6. A. Gelb."Graphical Evaluation of the Sensitivity Function Using the Nichols Chart," *IRE Trans. Automatic Control*, Vol. AC-7, pp. 57–58, July 1962.

PROBLEMS

8-1. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{K}{s(s + 6.54)}$$

Analytically, find the resonance peak M_r, resonant frequency ω_r, and bandwidth BW of the closed-loop system for the following values of K:

(a) $K = 5$

(b) $K = 21.39$

(c) $K = 100$

Use the formulas for the second-order prototype system given in the text.

8-2. Use MATLAB to verify your answer to Problem 8-1.

8-3. The transfer function of a system is

$$G(s) = \frac{s + \dfrac{1}{A_1}}{s + \dfrac{1}{A_2}}$$

Determine when the system is a lead-network and lag-network.

8-4. Use MATLAB to solve the following problems. Do not attempt to obtain the solutions analytically. The forward-path transfer functions of unity-feedback control systems are given in the following equations. Find the resonance peak M_r, resonant frequency ω_r, and bandwidth BW of the closed-loop systems. (Reminder: Make certain that the system is stable.)

(a) $G(s) = \dfrac{5}{s(1 + 0.5s)(1 + 0.1s)}$

(b) $G(s) = \dfrac{10}{s(1 + 0.5s)(1 + 0.1s)}$

(c) $G(s) = \dfrac{500}{(s + 1.2)(s + 4)(s + 10)}$

(d) $G(s) = \dfrac{10(s + 1)}{s(s + 2)(s + 10)}$

(e) $G(s) = \dfrac{0.5}{s(s^2 + s + 1)}$

(f) $G(s) = \dfrac{100e^{-s}}{s(s^2 + 10s + 50)}$

(g) $G(s) = \dfrac{100e^{-s}}{s(s^2 + 10s + 100)}$

(h) $G(s) = \dfrac{10(s + 5)}{s(s^2 + 5s + 5)}$

8-5. The specifications on a second-order unity-feedback control system with the closed-loop transfer function

$$M(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

are that the maximum overshoot must not exceed 10% and the rise time must be less than 0.1 sec. Find the corresponding limiting values of M_r and BW analytically.

8-6. Repeat Problem 8-5 for maximum overshoot $\leq 20\%$ and $t_r \leq 0.2$ sec.

8-7. Repeat Problem 8-5 for maximum overshoot $\leq 30\%$ and $t_r \leq 0.2$ sec.

8-8. Consider the forward-path transfer function of a unity-feedback control system given by

$$G(s) = \frac{0.5K}{s(0.25s^2 + 0.375s + 1)}$$

(a) Analytically find K such that the closed-loop bandwidth is about 1.5 rad/s (0.24 Hz).

(b) Use MATLAB to verify your answer to part (a).

8-9. Repeat Problem 8-8 with a resonance peak of 2.2.

8-10. The closed-loop frequency response $|M(j\omega)|$-versus-frequency of a second-order prototype system is shown in Fig 8P-10. Sketch the corresponding unit-step response of the system; indicate the values of the maximum overshoot, peak time, and the steady-state error due to a unit-step input.

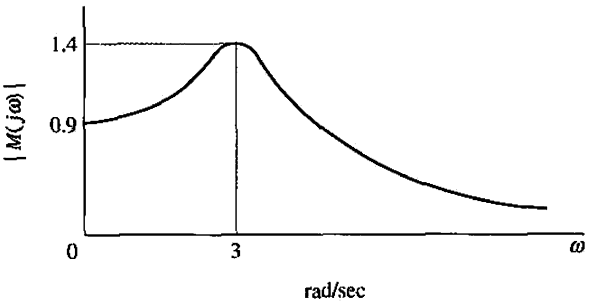


Figure 8P-10

8-11. The forward-path transfer function of a system with an integral control $H(s) = \dfrac{K}{s}$ is

$$G(s) = \frac{1}{10s + 1}$$

(a) Find K when the closed-loop resonance peak is 1.4.

(b) Determine the frequency at resonance, overshoot for step input, phase margin, and closed-loop BW according to the result of part (a).

8-12. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{1 + Ts}{2s(s^2 + s + 1)}$$

Use MATLAB to find the values of BW and M_r of the closed-loop system for $T = 0.05$, 1, 2, 3, 4, and 5.

8-13. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{1}{2s(s^2 + s + 1)(1 + Ts)}$$

Use MATLAB to find the values of BW and M_r of the closed-loop system for $T = 0$, 0.5, 1, 2, 3, 4, and 5. Use MATLAB to find the solutions.

8-14. If a loop transfer function of a system is given by

$$G(s)H(s) = \frac{0.5K}{0.25s^3 + 0.375s^2 + s + 0.5k}$$

(a) Use the second-order approximation to find the BW and the damping ratio.

(b) If BW = 1.5 rad/s, find K and the damping ratio.

(c) Use MATLAB to verify your answer to part (b).

8-15. The loop transfer functions $L(s)$ of single-feedback-loop systems are given below. Sketch the Nyquist plot of $L(j\omega)$ for $\omega = 0$ to $\omega = \infty$. Determine the stability of the closed-loop system. If the system is unstable, find the number of poles of the closed-loop transfer function that are in the right-half s-plane. Solve for the intersect of $L(j\omega)$ on the negative real axis of the $L(j\omega)$-plane analytically. You may construct the Nyquist plot of $L(j\omega)$ using MATLAB.

(a) $L(s) = \dfrac{20}{s(1 + 0.1s)(1 + 0.5s)}$

(b) $L(s) = \dfrac{10}{s(1 + 0.1s)(1 + 0.5s)}$

(c) $L(s) = \dfrac{100(1 + s)}{s(1 + 0.1s)(1 + 0.2s)(1 + 0.5s)}$

(d) $L(s) = \dfrac{10}{s^2(1 + 0.2s)(1 + 0.5s)}$

(e) $L(s) = \dfrac{3(s+2)}{s(s^3+3s+1)}$

(f) $L(s) = \dfrac{0.1}{s(s+1)(s^2+s+1)}$

(g) $L(s) = \dfrac{100}{s(s+1)(s^2+2)}$

(h) $L(s) = \dfrac{10(s+10)}{s(s+1)(s+100)}$

8-16. The loop transfer functions of single-feedback-loop control systems are given in the following equations. Apply the Nyquist criterion and determine the values of K for the system to be stable. Sketch the Nyquist plot of $L(j\omega)$ with $K = 1$ for $\omega = 0$ to $\omega = \infty$. You may use a computer program to plot the Nyquist plots.

(a) $L(s) = \dfrac{K}{s(s+2)(s+10)}$

(b) $L(s) = \dfrac{K(s+1)}{s(s+2)(s+5)(s+15)}$

(c) $L(s) = \dfrac{K}{s^2(s+2)(s+10)}$

(d) $L(s) = \dfrac{K}{(s+5)(s+2)^2}$

(e) $L(s) = \dfrac{K(s+5)(s+1)}{(s+50)(s+2)^3}$

8-17. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{K}{(s+5)^n}$$

Determine by means of the Nyquist criterion, the range of $K(-\infty < K < \infty)$ for the closed-loop system to be stable. Sketch the Nyquist plot of $G(j\omega)$ for $\omega = 0$ to $\omega = \infty$.

(a) $n = 2$

(b) $n = 3$

(c) $n = 4$

8-18. Sketch the Nyquist plot for the controlled system shown in Fig. 8P-18.

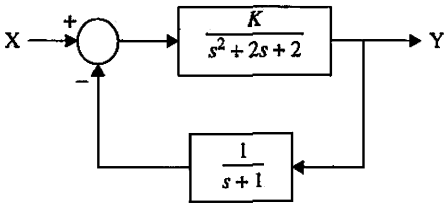


Figure 8P-18

Determine by means of the Nyquist criterion, the range of $K(-\infty < K < \infty)$ for the closed-loop system to be stable.

8-19. The characteristic equation of a linear control system is given in the following equation.

$$s(s^3 + 2s^2 + s + 1) + K(s^2 + s + 1) = 0$$

(a) Apply the Nyquist criterion to determine the values of K for system stability.

(b) Check the answers by means of the Routh-Hurwitz criterion.

8-20. Repeat Problem 8-19 for $s^3 + 3s^2 + 3s + 1 + K = 0$.

8-21. The forward-path transfer function of a unity-feedback control system with a PD (proportional-derivative) controller is

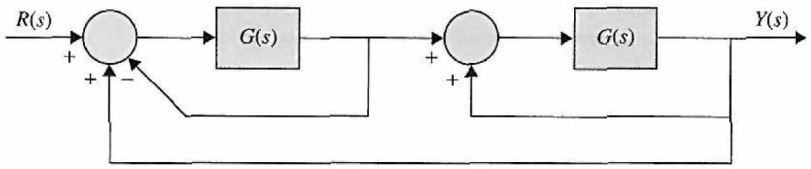
$$G(s) = \frac{10(K_P - K_D s)}{s^2}$$

Select the value of K_P so that the parabolic-error constant K_a is 100. Find the equivalent forward-path transfer function $G_{eq}(s)$ for $\omega = 0$ to $\omega = \infty$. Determine the range of K_D for stability by the Nyquist criterion.

8-22. The block diagram of a feedback control system is shown in Fig. 8P-22.

(a) Apply the Nyquist criterion to determine the range of K for stability.

(b) Check the answer obtained in part (a) with the Routh-Hurwitz criterion.



$$G(s) = \frac{K}{(s+4)(s+5)}$$

Figure 8P-22

8-23. The forward-path transfer function of the liquid-level control system in Problem 5-42 is

$$G(s) = \frac{K_a K_i n K_f N}{s(R_a J s + K_i K_b)(As + K_o)}$$

The following system parameters are given: $K_a = 50$, $K_i = 10$, $K_f = 50$, $J = 0.006$, $K_b = 0.0706$, $n = 0.01$, and $R_a = 10$. The values of A, N, and K_o are variable.

(a) For $A = 50$ and $K_o = 100$, sketch the Nyquist plot of $G(j\omega)$ for $\omega = 0$ to ∞ with N as a variable parameter. Find the maximum integer value of N so that the closed-loop system is stable.

(b) Let $N = 10$ and $K_o = 100$. Sketch the Nyquist plot of an equivalent transfer function $G_{eq}(j\omega)$ that has A as a multiplying factor. Find the critical value of K_o for stability.

(c) For $A = 50$ and $N = 10$, sketch the Nyquist plot of an equivalent transfer function $G_{eq}(j\omega)$ that has K_o as a multiplying factor. Find the critical value of K_o for stability.

8-24. The block diagram of a dc-motor control system is shown in Fig. 8P-24. Determine the range of K for stability using the Nyquist criterion when K_t has the following values:

(a) $K_t = 0$

(b) $K_t = 0.01$

(c) $K_t = 0.1$

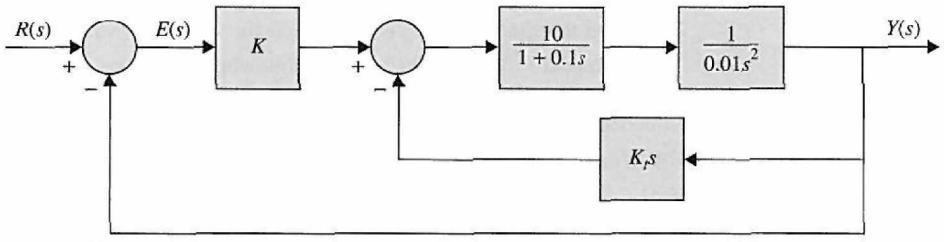


Figure 8P-24

8-25. For the system shown in Fig. 8P-24, let $K = 10$. Find the range of K_t for stability with the Nyquist criterion.

8-26. Fig. 8P-26 shows the block diagram of a servomotor.

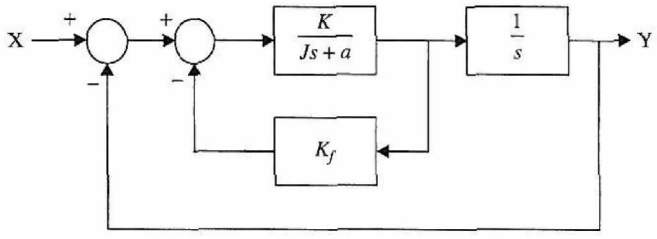


Figure 8P-26

Assume $J = 1\,\text{kg-m}^2$ and $B = 1$ N-m/rad/sec. Determine the range of K for stability using the Nyquist criterion when K_f has the following values:

(a) $K_f = 0$

(b) $K_f = 0.1$

(c) $K_f = 0.2$

8-27. For the system shown in Fig. 8P-26, let $K = 10$. Find the range of K_f for stability with the Nyquist criterion.

8-28. For the controlled system shown in Fig. 8P-28, draw the Nyquist plot and apply the Nyquist criterion to determine the range of K for stability and determine the number of roots in the right-half s-plane for the values of K where the system is unstable.

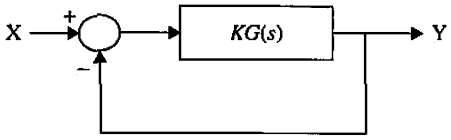


Figure 8P-28

(a) $G(s) = \dfrac{s+1}{(s-1)^2}$

(b) $G(s) = \dfrac{s-1}{(s+1)^2}$

8-29. The steel-rolling control system shown in Fig. 4P-18 has the forward-path transfer function

$$G(s) = \frac{100Ke^{-T_d s}}{s(s^2 + 10s + 100)}$$

(a) When $K = 1$, determine the maximum time delay T_d in seconds for the closed-loop system to be stable.

(b) When the time delay T_d is 1 sec, find the maximum value of K for system stability.

8-30. Repeat Problem 8-29 with the following conditions.

(a) When $K = 0.1$, determine the maximum time delay T_d in seconds for the closed-loop system to be stable.

(b) When the time delay T_d is 0.1 sec, find the maximum value of K for system stability.

8-31. The open-loop transfer function of a system is given by

$$G(s)H(s) = \frac{K}{s(\tau_1 s + 1)(\tau_2 s + 1)}$$

Study the stability of the system for the following:

(a) K is small.

(b) K is large.

8-32. The system schematic shown in Fig. 8P-32 is devised to control the concentration of a chemical solution by mixing water and concentrated solution in appropriate proportions. The transfer function of the system components between the amplifier output e_a (V) and the valve position x (in.) is

$$\frac{X(s)}{E_a(s)} = \frac{K}{s^2 + 10s + 100}$$

When the sensor is viewing pure water, the amplifier output voltage e_a is zero; when it is viewing concentrated solution, $e_a = 10$ V; and 0.1 in. of the valve motion changes the output concentration from zero to maximum. The valve ports can be assumed to be shaped so that the output

concentration varies linearly with the valve position. The output tube has a cross-sectional area of 0.1 in.^{2}, and the rate of flow is 10^3 in./sec regardless of the valve position. To make sure the sensor views a homogeneous solution, it is desirable to place it at some distance D in. from the valve.

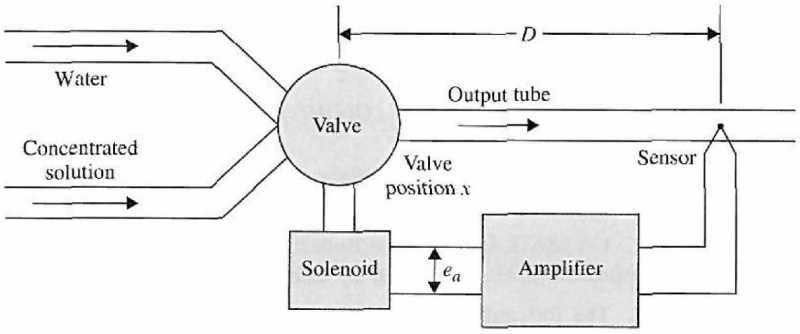


Figure 8P-32

(a) Derive the loop transfer function of the system.

(b) When $K = 10$, find the maximum distance D (in.) so that the system is stable. Use the Nyquist stability criterion.

(c) Let $D = 10$ in. Find the maximum value of K for system stability.

8-33. For the mixing system described in Problem 8-32, the following system parameters are given:

When the sensor is viewing pure water, the amplifier output voltage $e_s = 0$ V; when it is viewing concentrated solution, $e_a = 1$ V; and 0.1 in. of the valve motion changes the output concentration from zero to maximum. The rest of the system characteristics are the same as in Problem 8-32. Repeat the three parts of Problem 8-32.

8-34. Figure 8P-34 shows the block diagram of a control system.

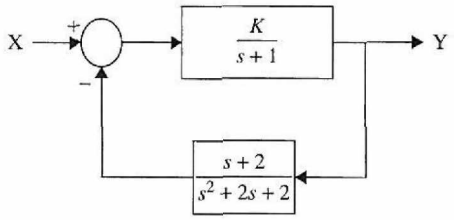


Figure 8P-34

(a) Draw the Nyquist plot and apply the Nyquist criterion to determine the range of K for stability.

(b) Determine the number of roots in the right-half s-plane for the values of K where the system is unstable.

(c) Use Routh's criterion to determine the range of K for stability.

8-35. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{1000}{s(s^2 + 105s + 600)}$$

(a) Find the values of M_r, ω_r, and BW of the closed-loop system.

(b) Find the parameters of the second-order system with the open-loop transfer function

$$G_L(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

that will give the same values for M_r and ω_r as the third-order system. Compare the values of BW of the two systems.

8-36. Sketch or plot the Bode diagrams of the forward-path transfer functions given in Problem 8-4. Find the gain margin, gain-crossover frequency, phase margin, and the phase-crossover frequency for each system.

8-37. The loop transfer function of a system is given by

$$G(s)H(s) = \frac{25(s+1)}{s(s+2)(s^2+2s+16)}$$

Use MATLAB to plot the Bode diagrams of the system and find the phase margin and gain margin of the system.

8-38. Use MATLAB to plot the Bode diagrams of the system shown in Fig. 8P-34 when $K = 1$, and determine the stable range of K by using phase margin and gain margin.

8-39. The forward-path transfer functions of unity-feedback control systems are given in the following equations. Plot the Bode diagram of $G(j\omega)/K$, and do the following: **(1)** Find the value of K so that the gain margin of the system is 20 dB. **(2)** Find the value of K so that the phase margin of the system is 45°.

(a) $G(s) = \dfrac{K}{s(1+0.1s)(1+0.5s)}$

(b) $G(s) = \dfrac{K(s+1)}{s(1+0.1s)(1+0.2s)(1+0.5s)}$

(c) $G(s) = \dfrac{K}{(s+3)^3}$

(d) $G(s) = \dfrac{K}{(s+3)^4}$

(e) $G(s) = \dfrac{Ke^{-s}}{s(1+0.1s+0.01s^2)}$

(f) $G(s) = \dfrac{K(1+0.5s)}{s(s^2+s+1)}$

8-40. The forward-path transfer functions of unity-feedback control systems are given in the following equations. Plot $G(j\omega)/K$ in the gain-phase coordinates of the Nichols chart, and do the following: **(1)** Find the value of K so that the gain margin of the system is 10 dB. **(2)** Find the value of K so that the phase margin of the system is 45°. **(3)** Find the value of K so that $M_r = 1.2$.

(a) $G(s) = \dfrac{10K}{s(1+0.1s)(1+0.5s)}$

(b) $G(s) = \dfrac{5K(s+1)}{s(1+0.1s)(1+0.2s)(1+0.5s)}$

(c) $G(s) = \dfrac{10K}{s(1+0.1s+0.01s^2)}$

(d) $G(s) = \dfrac{10Ke^{-s}}{s(1+0.1s+0.01s^2)}$

8-41. The forward-path of a unity-feedback system is given by

$$G(s)H(s) = \frac{K(s+1)(s+2)}{s^2(s+3)(s^2+2s+25)}$$

(a) Plot the Bode diagram.

(b) Plot the root locus.

(c) Find the gain and frequency where instability occurs.

(d) Find the gain at the phase margin of 20°.

(e) Find the gain margin when the phase margin is 20°.

8-42. The Bode diagram of the forward-path transfer function of a unity-feedback control system is obtained experimentally (as shown in Fig. 8P-42) when the forward gain K is set at its nominal value.

(a) Find the gain and phase margins of the system from the diagram as best you can read them. Find the gain- and phase-crossover frequencies.

(b) Repeat part (a) if the gain is doubled from its nominal value.

(c) Repeat part (a) if the gain is 10 times its nominal value.

(d) Find out how much the gain must be changed from its nominal value if the gain margin is 40 dB.

(e) Find out how much the loop gain must be changed from its nominal value if the phase margin is 45°.

(f) Find the steady-state error of the system if the reference input to the system is a unit-step function.

(g) The forward path now has a pure time delay of T_d sec, so that the forward-path transfer function is multiplied by $e^{-T_d s}$. Find the gain margin and the phase margin for $T_d = 0.1$ sec. The gain is set at nominal.

(h) With the gain set at nominal, find the maximum time delay T_d the system can tolerate without going into instability.

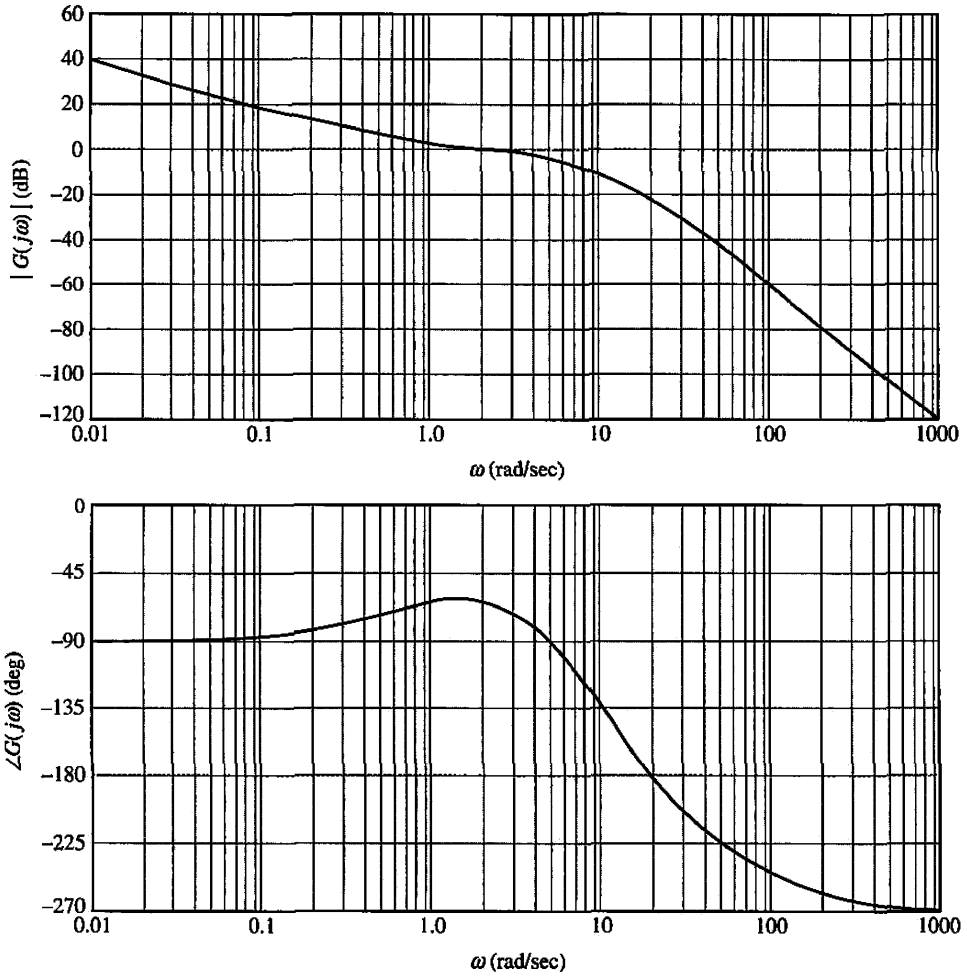


Figure 8P-42

8-43. Repeat Problem 8-42 using Fig. 8P-42 for the following parts.

(a) Find the gain and phase margins if the gain is four times its nominal value. Find the gain- and phase-crossover frequencies.

(b) Find out how much the gain must be changed from its nominal value if the gain margin is 20 dB.

(c) Find the marginal value of the forward-path gain for system stability.

(d) Find out how much the gain must be changed from its nominal value if the phase margin is 60°.

(e) Find the steady-state error of the system if the reference input is a unit-step function and the gain is twice its nominal value.

(f) Find the steady-state error of the system if the reference input is a unit-step function and the gain is 20 times its nominal value.

(g) The system now has a pure time delay so that the forward-path transfer function is multiplied by $e^{-T_d s}$. Find the gain and phase margins when $T_d = 0.1$ sec. The gain is set at its nominal value.

(h) With the gain set at 10 times its nominal, find the maximum time delay T_d the system can tolerate without going into instability.

8-44. The forward-path transfer function of a unity-feedback control system is

$$G(s)H(s) = \frac{80e^{-0.1s}}{s(s+4)(s+10)}$$

(a) Draw the Nyquist plot of the system.

(b) Plot the Bode diagram of the system.

(c) Find the phase margin and gain margin of the system.

8-45. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{K(1+0.2s)(1+0.1s)}{s^2(1+s)(1+0.01s)^2}$$

(a) Construct the Bode and Nyquist plots of $G(j\omega)/K$ and determine the range of K for system stability.

(b) Construct the root loci of the system for $K \geq 0$. Determine the values of K and ω at the points where the root loci cross the $j\omega$-axis, using the information found from the Bode plot.

8-46. Repeat Problem 8-45 for the following transfer function:

$$G(s) = \frac{K(s+1.5)(s+2)}{s^2(s^2+2s+2)}$$

8-47. Repeat Problem 8-45 for the following transfer function:

$$G(s)H(s) = \frac{16000(s+1)(s+5)}{s(s+0.1)(s+8)(s+20)(s+50)}$$

8-48. The forward-path transfer function of the dc-motor control system described in Fig. 3P-11 is

$$G(s) = \frac{6.087 \times 10^8 K}{s(s^3+423.42s^2+2.6667 \times 10^6 s+4.2342 \times 10^8)}$$

Plot the Bode diagram of $G(j\omega)$ with $K = 1$, and determine the gain margin and phase margin of the system. Find the critical value of K for stability.

8-49. The transfer function between the output position $\Theta_L(s)$ and the motor current $I_a(s)$ of the robot arm modeled in Fig. 4P-53 is

$$G_p(s) = \frac{\Theta_L(s)}{I_a(s)} = \frac{K_i(Bs+K)}{\Delta_o}$$

where

$$\Delta_o(s) = s\{J_L J_m s^3 + [J_L(B_m+B)+J_m(B_L+B)]s^2 + [B_L B_m + (B_L+B_m)B + (J_m+J_L)K]s + K(B_L+B_m)\}$$

The arm is controlled by a closed-loop system, as shown in Fig. 8P-49. The system parameters are $K_a = 65, K = 100, K_i = 0.4, B = 0.2, J_m = 0.2, B_L = 0.01, J_L = 0.6$, and $B_m = 0.25$.

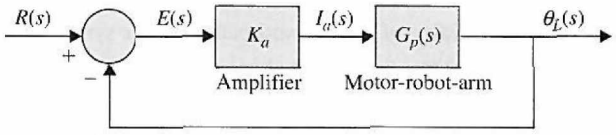


Figure 8P-49

(a) Derive the forward-path transfer function $G(s) = \Theta_L(s)/E(s)$.

(b) Draw the Bode diagram of $G(j\omega)$. Find the gain and phase margins of the system.

(c) Draw $|M(j\omega)|$ versus ω, where $M(s)$ is the closed-loop transfer function. Find M_r, ω_r, and BW.

8-50. For the ball-and-beam system described in Problem 4-11 and shown in Fig. 8P-50, assume the following:

$m = 0.11\,\text{kg}$	mass of the ball	$I = 9.99 \times 10^{-6}\,\text{kg-m}^2$	ball's moment of inertia
$r = 0.015$	radius of the ball	P	ball position coordinate
$d = 0.03\,\text{m}$	lever arm offset	α	beam angle coordinate
$g = 9.8\,\text{m/s}^2$	gravitational acceleration	θ	servo gear angle
$L = 1.0\,\text{m}$	length of the beam		

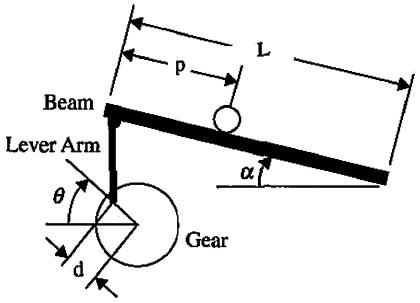


Figure 8P-50

If the system is controlled by a proportional controller in a unity-feedback control system,

(a) Find the transfer function from the gear angle (θ) to the ball position (P).

(b) Find the closed-loop transfer function.

(c) Find the range of K for stability.

(d) Plot the Bode diagram for the system for $K = 1$, and find the gain and phase margins of the system.

(e) Draw $|M(j\omega)|$ versus ω, where $M(s)$ is the closed-loop transfer function. Find M_r, ω_r, and BW.

8-51. The gain-phase plot of the forward-path transfer function of $G(j\omega)/K$ of a unity-feedback control system is shown in Fig. 8P-51. Find the following performance characteristics of the system.

(a) Gain-crossover frequency (rad/sec) when $K = 1$.

(b) Phase-crossover frequency (rad/sec) when $K = 1$.

(c) Gain margin (dB) when $K = 1$.

(d) Phase margin (deg) when $K = 1$.

(e) Resonance peak M_r when $K = 1$.

(f) Resonant frequency ω_r (rad/sec) when $K = 1$.

(g) BW of the closed-loop system when $K = 1$.

(h) The value of K so that the gain margin is 20 dB.

(i) The value of K so that the system is marginally stable. Find the frequency of sustained oscillation in rad/sec.

(j) Steady-state error when the reference input is a unit-step function.

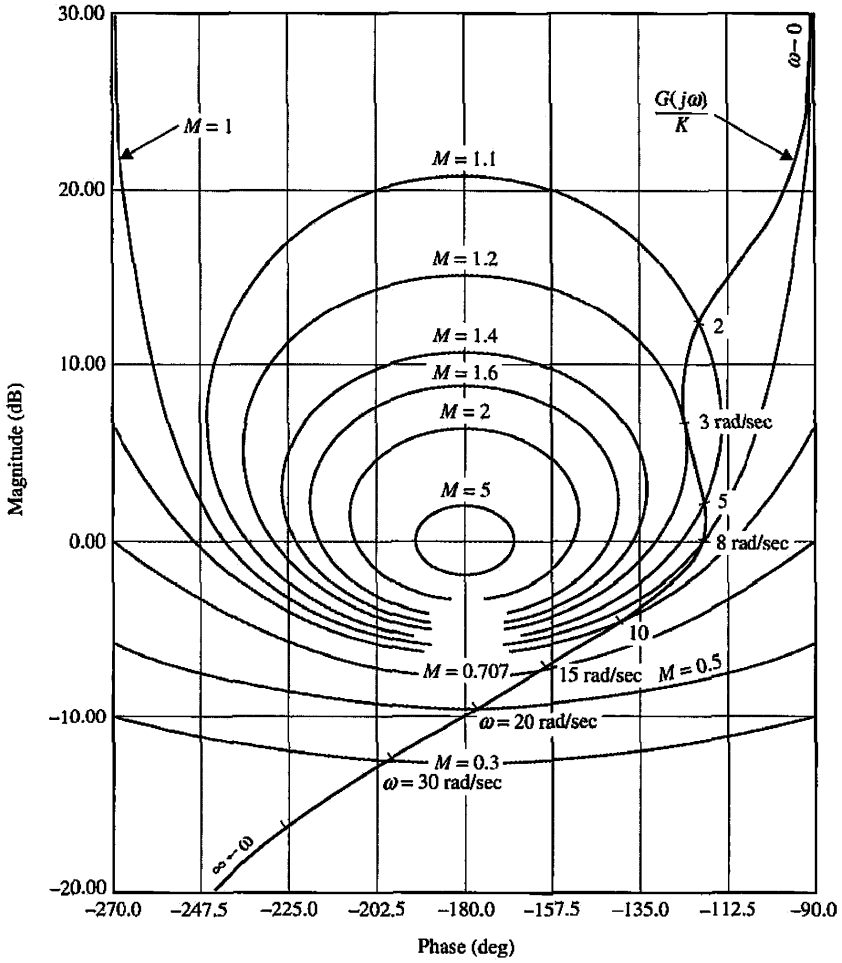


Figure 8P-51

8-52. Repeat parts (a) through (g) of Problem 8-51 when $K = 10$. Repeat part (h) for gain margin = 40 dB.

8-53. For the system in Problem 8-44, plot the Nichols chart and find magnitudes and phase angles of the closed-loop frequency response. Then plot the Bode diagram of the closed-loop system.

8-54. Use ACSYS or MATLAB to analyze the frequency response of the following unity-feedback control systems. Plot the Bode diagrams, polar plots, and gain-phase plots, and compute the phase margin, gain margin, M_r, and BW.

(a) $G(s) = \dfrac{1 + 0.1s}{s(s+1)(1+0.01s)}$

(b) $G(s) = \dfrac{0.5(s+1)}{s(1+0.2s)(1+s+0.5s^2)}$

(c) $G(s) = \dfrac{(s+1)}{s(1+0.2s)(1+0.5s)}$

(d) $G(s) = \dfrac{1}{s(1+s)(1+0.5s)}$

(e) $G(s) = \dfrac{50}{s(s+1)(1+0.5s^2)}$

(f) $G(s) = \dfrac{(1+0.1s)e^{-0.1s}}{s(s+1)(1+0.01s)}$

(g) $G(s) = \dfrac{10e^{-0.1s}}{s^2+2s+2}$

8-55. For the gain-phase plot of $G(j\omega)/K$ shown in Fig. 8P-51, the system now has a pure time delay of T_d in the forward path, so that the forward-path transfer function becomes $G(s)e^{-T_d s}$.

(a) With $K = 1$, find T_d so that the phase margin is 40°.

(b) With $K = 1$, find the maximum value of T_d so that the system will remain stable.

8-56. Repeat Problem 8-55 with $K = 10$.

8-57. Repeat Problem 8-55 so that the gain margin is 5 dB when $K = 1$.

8-58. The block diagram of a furnace-control system is shown in Fig. 8P-58. The transfer function of the process is

$$G_p(s) = \frac{1}{(1 + 10s)(1 + 25s)}$$

The time delay T_d is 2 sec.

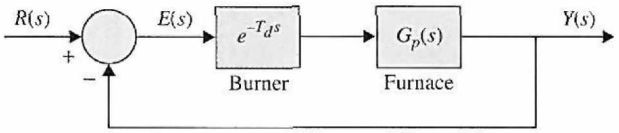


Figure 8P-58

(a) Plot the Bode diagram of $G(s) = Y(s)/E(s)$, and find the gain-crossover and phase-crossover frequencies. Find the gain margin and the phase margin.

(b) Approximate the time delay by [Eq. (4-223)]

$$e^{-T_d s} \quad \frac{1}{1 + T_d s + T_d s^2/2}$$

and repeat part (a). Comment on the accuracy of the approximation. What is the maximum frequency below which the polynomial approximation is accurate?

(c) Repeat part (b) for approximating the time delay term by [Eq. (4-224)]

$$e^{-T_d s} \cong \frac{1 - T_d s/2}{1 + T_d s/2}$$

8-59. Repeat Problem 8-58 with $T_d = 1$ sec.

8-60. Plot the $|S_G^M(j\omega)|$-versus-ω plot for the system described in Problem 8-49 for $K = 1$. Find the frequency at which the sensitivity is maximum and the value of the maximum sensitivity.

8-61. Fig. 8P-61 shows the pitch controller system for an aircraft, as described in Problem 4-12.

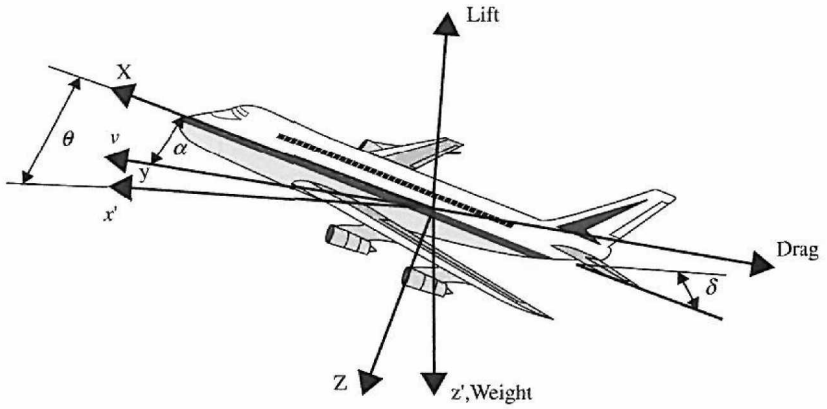


Figure 8P-61

If the system is controlled by a proportional controller in a unity-feedback control system,

(a) Find the transfer function between pitch angle and elevator deflection angle.

(b) Find the closed-loop transfer function.

(c) Find the range of K for stability.

(d) Plot the Bode diagram for the system for $K = 1$, and find the gain and phase margins of the system.

(e) Draw $|M(j\omega)|$ versus ω, where $M(s)$ is the closed-loop transfer function. Find M_r, ω_r, and BW.

CHAPTER 9

Design of Control Systems

9-1 INTRODUCTION

All the foundations of analysis that we have laid in the preceding chapters led to the ultimate goal of design of control systems. Starting with the controlled process such as that shown by the block diagram in Fig. 9-1, control system design involves the following three steps:

1. Determine what the system should do and how to do it (design specifications).
2. Determine the controller or compensator configuration, relative to how it is connected to the controlled process.
3. Determine the parameter values of the controller to achieve the design goals.

These design tasks are explored further in the following sections.

9-1-1 Design Specifications

We often use design specifications to describe what the system should do and how it is done. These specifications are unique to each individual application and often include specifications about **relative stability, steady-state accuracy (error), transient-response characteristics**, and **frequency-response characteristics**. In some applications there may be additional specifications on **sensitivity to parameter variations**, that is, **robustness**, or **disturbance rejection**.

The design of linear control systems can be carried out in either the time domain or the frequency domain. For instance, **steady-state accuracy** is often specified with respect to a step input, a ramp input, or a parabolic input, and the design to meet a certain requirement is more conveniently carried out in the time domain. Other specifications such as **maximum overshoot, rise time**, and **settling time** are all defined for a unit-step input and, therefore, are used specifically for time-domain design. We have learned that relative stability is also measured in terms of **gain margin, phase margin**, and M_r. These are typical frequency-domain specifications, which should be used in conjunction with such tools as the Bode plot, polar plot, gain-phase plot, and Nichols chart.

We have shown that, for a second-order prototype system, there are simple analytical relationships between some of these time-domain and frequency-domain specifications. However, for higher-order systems, correlations between time-domain and frequency-domain specifications are difficult to establish. As pointed out earlier, the analysis and

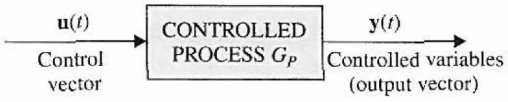


Figure 9-1 Controlled process.

design of control systems is pretty much an exercise of selecting from several alternative methods for solving the same problem.

Thus, the choice of whether the design should be conducted in the time domain or the frequency domain depends often on the preference of the designer. We should be quick to point out, however, that in most cases, time-domain specifications such as maximum overshoot, rise time, and settling time are usually used as the final measure of system performance. To an inexperienced designer, it is difficult to comprehend the physical connection between frequency-domain specifications such as gain and phase margins and resonance peak to actual system performance. For instance, does a gain margin of 20 dB guarantee a maximum overshoot of less than 10%? To a designer it makes more sense to specify, for example, that the maximum overshoot should be less than 5% and a settling time less than 0.01 sec. It is less obvious what, for example, a phase margin of 60° and an M_r of less than 1.1 may bring in system performance. The following outline will hopefully clarify and explain the choices and reasons for using time-domain versus frequency-domain specifications.

1. Historically, the design of linear control systems was developed with a wealth of graphical tools such as the Bode plot, Nyquist plot, gain-phase plot, and Nichols chart, which are all carried out in the frequency domain. The advantage of these tools is that they can all be sketched by following approximation methods without detailed plotting. Therefore, the designer can carry out designs using frequency-domain specifications such as **gain margin, phase margin, M_r,** and the like. High-order systems do not generally pose any particular problem. For certain types of controllers, design procedures in the frequency domain are available to reduce the trial-and-error effort to a minimum.
2. Design in the time domain using such performance specifications as **rise time, delay time, settling time, maximum overshoot**, and the like is possible *analytically* only for second-order systems or for systems that can be approximated by second-order systems. General design procedures using time-domain specifications are difficult to establish for systems with an order higher than the second.

The development and availability of high-powered and user-friendly computer software, such as MATLAB, is rapidly changing the practice of control system design, which until recently had been dictated by historical development. Now with MATLAB, the designer can go through a large number of design runs using the time-domain specifications within a matter of minutes. This diminishes considerably the historical edge of the frequency-domain design, which is based on the convenience of performing graphical design manually.

Throughout the chapter, we have incorporated small MATLAB toolboxes to help your understanding of the examples, and, at the end of the chapter, we present the Automatic Control Systems software package (**ACSYS**)—it is easy to use and fully graphics based to eliminate the user's need to write code.

Finally, it is generally difficult (except for an experienced designer) to select a meaningful set of frequency-domain specifications that will correspond to the desired time-domain performance requirements. For example, specifying a phase margin of 60° would be meaningless unless we know that it corresponds to a certain maximum overshoot. As it turns out, to control maximum overshoot, usually one has to specify at least phase margin and M_r. Eventually, establishing an intelligent set of frequency-domain specifications becomes a trial-and-error process that precedes the actual design, which often is also a

trial-and-error effort. However, frequency-domain methods are still valuable in interpreting noise rejection and sensitivity properties of the system, and, most important, they offer another perspective to the design process. Therefore, in this chapter the design techniques in the time domain and the frequency domain are treated side by side, so that the methods can be easily compared.

9-1-2 Controller Configurations

In general, the dynamics of a linear controlled process can be represented by the block diagram shown in Fig. 9-1. The design objective is to have the controlled variables, represented by the output vector $\mathbf{y}(t)$, behave in certain desirable ways. The problem essentially involves the determination of the control signal $\mathbf{u}(t)$ over the prescribed time interval so that the design objectives are all satisfied.

Most of the conventional design methods in control systems rely on the so-called **fixed-configuration design** in that the designer at the outset decides the basic configuration of the overall designed system and decides where the controller is to be positioned relative to the controlled process. The problem then involves the design of the elements of the controller. Because most control efforts involve the modification or compensation of the system-performance characteristics, the general design using fixed configuration is also called **compensation**.

Fig. 9-2 illustrates several commonly used system configurations with controller compensation. These are described briefly as follows.

- ***Series (cascade) compensation:*** Fig. 9-2(a) shows the most commonly used system configuration with the controller placed in series with the controlled process, and the configuration is referred to as **series** or **cascade compensation**.
- ***Feedback compensation:*** In Fig. 9-2(b), the controller is placed in the minor feedback path, and the scheme is called **feedback compensation**.
- ***State-feedback compensation:*** Fig. 9-2(c) shows a system that generates the control signal by feeding back the state variables through constant real gains, and the scheme is known as **state feedback**. The problem with state-feedback control is that, for high-order systems, the large number of state variables involved would require a large number of transducers to sense the state variables for feedback. Thus, the actual implementation of the state-feedback control scheme may be costly or impractical. Even for low-order systems, often not all the state variables are directly accessible, and an **observer** or **estimator** may be necessary to create the estimated state variables from measurements of the output variables.

The compensation schemes shown in Figs. 9-2(a), (b), and (c) all have one degree of freedom in that there is only one controller in each system, even though the controller may have more than one parameter that can be varied. The disadvantage with a one-degree-of-freedom controller is that the performance criteria that can be realized are limited. For example, if a system is to be designed to achieve a certain amount of relative stability, it may have poor sensitivity to parameter variations. Or if the roots of the characteristic equation are selected to provide a certain amount of relative damping, the maximum overshoot of the step response may still be excessive because of the zeros of the closed-loop transfer function. The compensation schemes shown in Figs. 9-2(d), (e), and (f) all have two degrees of freedom.

- ***Series-feedback compensation:*** Fig. 9-2(d) shows the series-feedback compensation for which a series controller and a feedback controller are used.

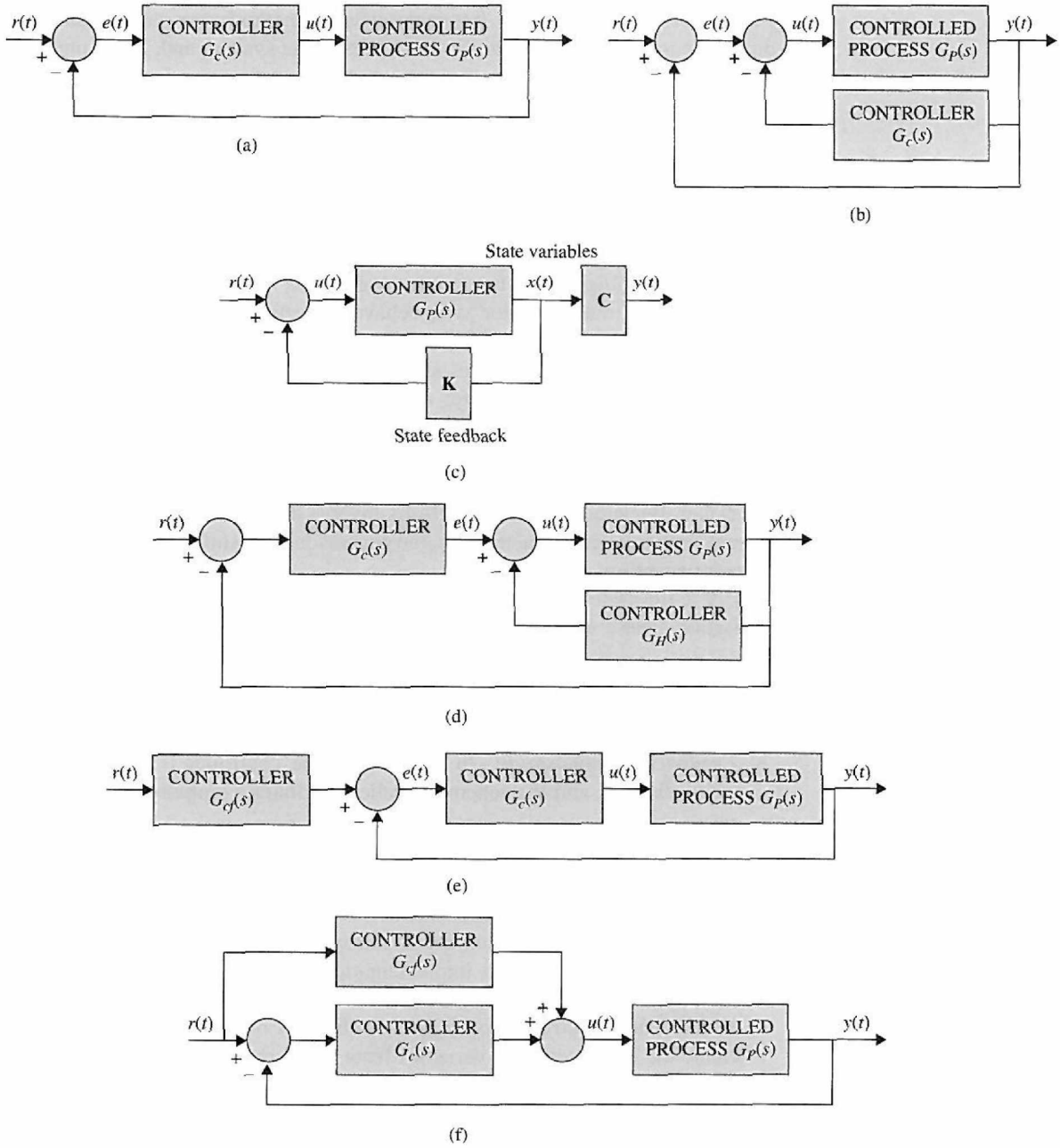


Figure 9-2 Various controller configurations in control-system compensation. (a) Series or cascade compensation. (b) Feedback compensation. (c) State-feedback control. (d) Series-feedback compensation (two degrees of freedom). (e) Forward compensation with series compensation (two degrees of freedom). (f) Feedforward compensation (two degrees of freedom).

- ***Feedforward compensation:*** Figs. 9-2(e) and (f) show the so-called **feedforward compensation**. In Fig. 9-2(e), the feedforward controller $G_{cf}(s)$ is placed in series with the closed-loop system, which has a controller $G_c(s)$ in the forward path. In Fig. 9-2(f), the feedforward controller $G_{cf}(s)$ is placed in parallel with the forward path. The key to the feedforward compensation is that the controller $G_{cf}(s)$ is not in

the loop of the system, so it does not affect the roots of the characteristic equation of the original system. The poles and zeros of $G_{cf}(s)$ may be selected to add or cancel the poles and zeros of the closed-loop system transfer function.

One of the commonly used controllers in the compensation schemes just described is a PID controller, which applies a signal to the process that is proportional to the actuating signal in addition to adding integral and derivative of the actuating signal. Because these signal components are easily realized and visualized in the time domain, PID controllers are commonly designed using time-domain methods. In addition to the PID-type controllers, lead, lag, lead-lag, and notch controllers are also frequently used. The names of these controllers come from properties of their respective frequency-domain characteristics. As a result, these controllers are often designed using frequency-domain concepts. Despite these design tendencies, however, all control system designs will benefit by viewing the resulting design from both time- and frequency-domain viewpoints. Thus, both methods will be used extensively in this chapter.

It should be pointed out that these compensation schemes are by no means exhaustive. The details of these compensation schemes will be discussed in later sections of this chapter. Although the systems illustrated in Fig. 9-2 are all for continuous-data control, the same configurations can be applied to discrete-data control, in which case the controllers are all digital, with the necessary interfacings and signal converters.

9-1-3 Fundamental Principles of Design

After a controller configuration is chosen, the designer must choose a controller type that, with proper selection of its element values, will satisfy all the design specifications. The types of controllers available for control-system design are bounded only by one's imagination. Engineering practice usually dictates that one choose the simplest controller that meets all the design specifications. In most cases, the more complex a controller is, the more it costs, the less reliable it is, and the more difficult it is to design. Choosing a specific controller for a specific application is often based on the designer's past experience and sometimes intuition, and it entails as much *art* as it does *science*. As a novice, you may initially find it difficult to make intelligent choices of controllers with confidence. By understanding that confidence comes only through experience, this chapter provides guided experiences that illustrate the basic elements of control system designs.

After a controller is chosen, the next task is to choose controller parameter values. These parameter values are typically the coefficients of one or more transfer functions making up the controller. The basic design approach is to use the analysis tools discussed in the previous chapters to determine how individual parameter values influence the design specifications and, finally, system performance. Based on this information, controller parameters are selected so that all design specifications are met. While this process is sometimes straightforward, more often than not it involves many design iterations since controller parameters usually interact with each other and influence design specifications in conflicting ways. For example, a particular parameter value may be chosen so that the maximum overshoot is satisfied, but in the process of varying another parameter value in an attempt to meet the rise-time requirement, the maximum overshoot specification may no longer be met! Clearly, the more design specifications there are and the more controller parameters there are, the more complicated the design process becomes.

In carrying out the design either in the time domain or the frequency domain, it is important to establish some basic guidelines or design rules. Keep in mind that time-domain design usually relies heavily on the s-plane and the root loci. Frequency-domain

design is based on manipulating the gain and phase of the loop transfer function so that the specifications are met.

In general, it is useful to summarize the time-domain and frequency-domain characteristics so that they can be used as guidelines for design purposes.

1. Complex-conjugate poles of the closed-loop transfer function lead to a step response that is underdamped. If all system poles are real, the step response is overdamped. However, zeros of the closed-loop transfer function may cause overshoot even if the system is overdamped.
2. The response of a system is dominated by those poles closest to the origin in the s-plane. Transients due to those poles farther to the left decay faster.
3. The farther to the left in the s-plane the system's dominant poles are, the faster the system will respond and the greater its bandwidth will be.
4. The farther to the left in the s-plane the system's dominant poles are, the more expensive it will be and the larger its internal signals will be. While this can be justified analytically, it is obvious that striking a nail harder with a hammer drives the nail in faster but requires more energy per strike. Similarly, a sports car can accelerate faster, but it uses more fuel than an average car.
5. When a pole and zero of a system transfer function nearly cancel each other, the portion of the system response associated with the pole will have a small magnitude.
6. Time-domain and frequency-domain specifications are loosely associated with each other. Rise time and bandwidth are inversely proportional. Larger phase margin, larger gain margin, and lower M_r will improve damping.

9-2 DESIGN WITH THE PD CONTROLLER

In all the examples of control systems we have discussed thus far, the controller has been typically a simple amplifier with a constant gain K. This type of control action is formally known as **proportional control**, because the control signal at the output of the controller is simply related to the input of the controller by a proportional constant.

Intuitively, one should also be able to use the derivative or integral of the input signal, in addition to the proportional operation. Therefore, we can consider a more general continuous-data controller to be one that contains such components as adders (addition or subtraction), amplifiers, attenuators, differentiators, and integrators. The designer's task is to determine which of these components should be used, in what proportion, and how they are connected. For example, one of the best-known controllers used in practice is the PID controller, where the letters stand for **proportional, integral,** and **derivative**. The integral and derivative components of the PID controller have individual performance implications, and their applications require an understanding of the basics of these elements. To gain an understanding of this controller, we consider just the PD portion of the controller first.

Fig. 9-3 shows the block diagram of a feedback control system that arbitrarily has a second-order prototype process with the transfer function

$$G_p(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \tag{9-1}$$

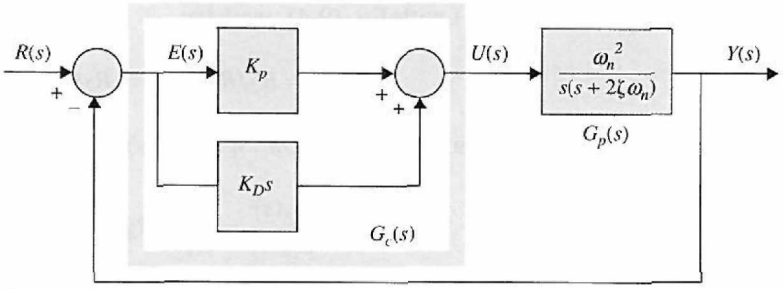


Figure 9-3 Control system with PD controller.

The series controller is a proportional-derivative (PD) type with the transfer function

$$G_c(s) = K_P + K_D s \tag{9-2}$$

Thus, the control signal applied to the process is

$$u(t) = K_P e(t) + K_D \frac{de(t)}{dt} \tag{9-3}$$

where K_P and K_D are the proportional and derivative constants, respectively. Using the components given in Table 4-4, two electronic-circuit realizations of the PD controller are shown in Fig. 9-4. The transfer function of the circuit in Fig. 9-4(a) is

$$\frac{E_o(s)}{E_{in}(s)} = \frac{R_2}{R_1} + R_2 C_1 s \tag{9-4}$$

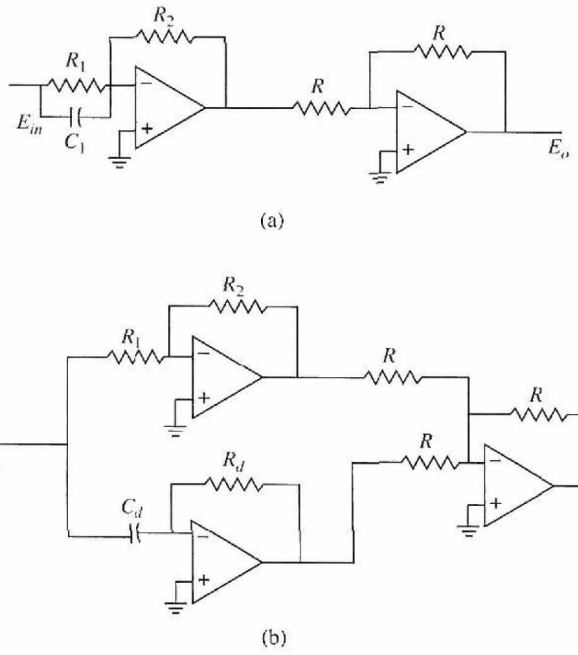


Figure 9-4 Op-amp circuit realization of the PD controller.

Comparing Eq. (9-2) with Eq. (9-4), we have

$$K_P = R_2/R_1 \quad K_D = R_2C_1 \tag{9-5}$$

The transfer function of the circuit in Fig. 9-4(b) is

$$\frac{E_o(s)}{E_{in}(s)} = \frac{R_2}{R_1} + R_dC_ds \tag{9-6}$$

Comparing Eq. (9-2) with Eq. (9-6), we have

$$K_P = R_2/R_1 \quad K_D = R_dC_d \tag{9-7}$$

The advantage with the circuit in Fig. 9-4(a) is that only two op-amps are used. However, the circuit does not allow the independent selection of K_P and K_D because they are commonly dependent on R_2. An important concern of the PD controller is that, if the value of K_D is large, a large capacitor C_1 would be required. The circuit in Fig. 9-4(b) allows K_P and K_D to be independently controlled. A large K_D can be compensated by choosing a large value for R_d, thus resulting in a realistic value for C_d. Although the scope of this text does not include all the practical issues involved in controller transfer function implementation, these issues are of the utmost importance in practice.

The forward-path transfer function of the compensated system is

- PD control adds a simple zero at $s = -K_P/K_D$ to the forward-path transfer function.

$$G(s) = \frac{Y(s)}{E(s)} = G_c(s)G_p(s) = \frac{\omega_n^2(K_P + K_Ds)}{s(s + 2\zeta\omega_n)} \tag{9-8}$$

which shows that the PD control is equivalent to adding a simple zero at $s = -K_P/K_D$ to the forward-path transfer function.

9-2-1 Time-Domain Interpretation of PD Control

The effect of the PD control on the transient response of a control system can be investigated by referring to the time responses shown in Fig. 9-5. Let us assume that the unit-step response of a stable system with only proportional control is as shown in Fig. 9-5(a), which has a relatively high maximum overshoot and is rather oscillatory. The corresponding error signal, which is the difference between the unit-step input and the output $y(t)$, and its time derivative $de(t)/dt$ are shown in Figs. 9.5(b) and (c), respectively. The overshoot and oscillation characteristics are also reflected in $e(t)$ and $de(t)/dt$. For the sake of illustration, we assume that the system contains a motor of some kind with its torque proportional to $e(t)$. The performance of the system with proportional control is analyzed as follows.

1. During the time interval $0 < t < t_1$: The error signal $e(t)$ is positive. The motor torque is positive and rising rapidly. The large overshoot and subsequent oscillations in the output $y(t)$ are due to the excessive amount of torque developed by the motor and the lack of damping during this time interval.
2. During the time interval $t_1 < t < t_3$: The error signal $e(t)$ is negative, and the corresponding motor torque is negative. This negative torque tends to slow down the output acceleration and eventually causes the direction of the output $y(t)$ to reverse and undershoot.

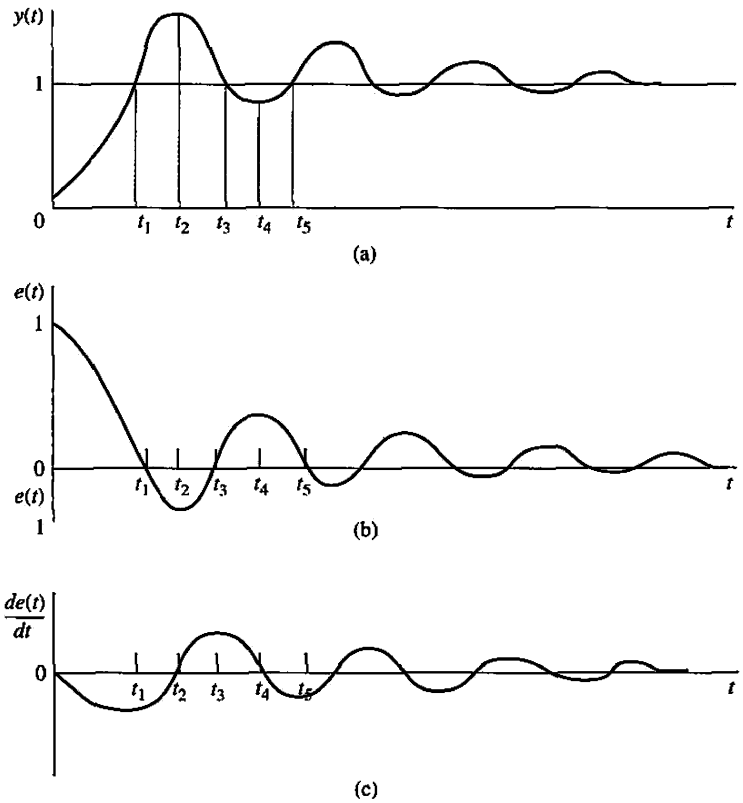


Figure 9-5 Waveforms of $y(t)$, $e(t)$, and $de(t)/dt$, showing the effect of derivative control. (a) Unit-step response. (b) Error signal. (c) Time rate of change of the error signal.

3. During the time interval $t_3 < t < t_5$: The motor torque is again positive, thus tending to reduce the undershoot in the response caused by the negative torque in the previous time interval. Because the system is assumed to be stable, the error amplitude is reduced with each oscillation, and the output eventually settles to its final value.

Considering the above analysis of the system time response, we can say that the contributing factors to the high overshoot are as follows:

1. The positive correcting torque in the interval $0 < t < t_1$ is too large.
2. The retarding torque in the time interval $t_1 < t < t_2$ is inadequate.

Therefore, to reduce the overshoot in the step response, without significantly increasing the rise time, a logical approach would be to

1. Decrease the amount of positive correcting torque during $0 < t < t_1$.
2. Increase the retarding torque during $t_1 < t < t_2$.

Similarly, during the time interval, $t_2 < t < t_4$, the negative corrective torque in $t_2 < t < t_3$ should be reduced, and the retarding torque during $t_3 < t < t_4$, which is now in the positive direction, should be increased to improve the undershoot of $y(t)$.

The PD control described by Eq. (9-2) gives precisely the compensation effect required. Because the control signal of the PD control is given by Eq. (9-3), Fig. 9-5(c) shows the following effects provided by the PD controller:

1. For $0 < t < t_1$, $de(t)/dt$ is negative; this will reduce the original torque developed due to $e(t)$ alone.
2. For $t_1 < t < t_2$, both $e(t)$ and $de(t)/dt$ are negative, which means that the negative retarding torque developed will be greater than that with only proportional control.
3. For $t_2 < t < t_3$, $e(t)$ and $de(t)/dt$ have opposite signs. Thus, the negative torque that originally contributes to the undershoot is reduced also.

Therefore, all these effects will result in smaller overshoots and undershoots in $y(t)$.

• PD is essentially an anticipatory control.

Another way of looking at the derivative control is that since $de(t)/dt$ represents the slope of $e(t)$, the PD control is essentially an *anticipatory* control. That is, by knowing the slope, the controller can anticipate direction of the error and use it to better control the process. Normally, in linear systems, if the slope of $e(t)$ or $y(t)$ due to a step input is large, a high overshoot will subsequently occur. The derivative control measures the instantaneous slope of $e(t)$, predicts the large overshoot ahead of time, and makes a proper corrective effort before the excessive overshoot actually occurs.

• Derivative or PD control will have an effect on a steady-state error only if the error varies with time.

Intuitively, derivative control affects the steady-state error of a system only if the steady-state error varies with time. If the steady-state error of a system is constant with respect to time, the time derivative of this error is zero, and the derivative portion of the controller provides no input to the process. But if the steady-state error increases with time, a torque is again developed in proportion to $de(t)/dt$, which reduces the magnitude of the error. Eq. (9-8) also clearly shows that the PD control does not alter the system type that governs the steady-state error of a unity-feedback system.

9-2-2 Frequency-Domain Interpretation of PD Control

For frequency-domain design, the transfer function of the PD controller is written

$$G_c(s) = K_P + K_D s = K_P\left(1 + \frac{K_D}{K_P}s\right) \tag{9-9}$$

• The PD controller is a high-pass filter.

so that it is more easily interpreted on the Bode plot. The Bode plot of Eq. (9-9) is shown in Fig. 9-6 with $K_P = 1$. In general, the proportional-control gain K_P can be combined with a series gain of the system, so that the zero-frequency gain of the PD controller can be regarded as unity. The high-pass filter characteristics of the PD controller are clearly shown by the Bode plot in Fig. 9-6. The phase-lead property may be utilized to improve the phase margin of a control system. Unfortunately, the magnitude characteristics of the PD controller push the gain-crossover frequency to a higher value. Thus, *the design principle of the PD controller involves the placing of the corner frequency of the controller, $\omega = K_P/K_D$, such that an effective improvement of the phase margin is realized at the new gain-crossover frequency.* For a given system, there is a range of values of K_P/K_D that is optimal for improving the damping of the system. Another practical consideration in selecting the values of K_P and K_D is in the physical implementation of the PD controller. Other apparent effects of the PD control in the frequency domain are that, due to its high-pass characteristics, in most cases it will increase the BW of the system and reduce the rise time of the step response. The practical disadvantage of the PD controller is that the differentiator portion is a high-pass filter, which usually accentuates any high-frequency noise that enters at the input.

• The PD controller has the disadvantage that it accentuates high-frequency noise.

• The PD controller will generally increase the BW and reduce the rise time of the step response.

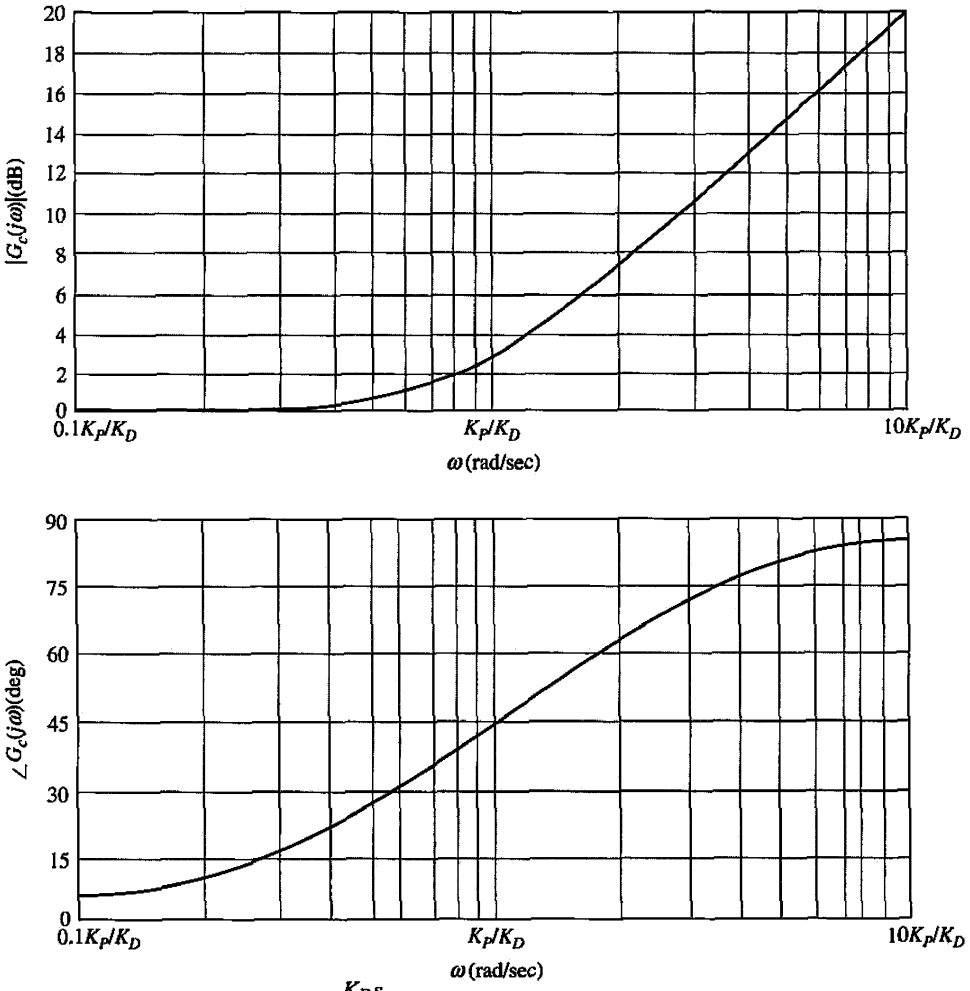


Figure 9-6 Bode diagram of $1+\frac{K_D s}{K_P}, K_P = 1$.

9-2-3 Summary of Effects of PD Control

Though it is not effective with lightly damped or initially unstable systems, a properly designed PD controller can affect the performance of a control system in the following ways:

1. Improving damping and reducing maximum overshoot.
2. Reducing rise time and settling time.
3. Increasing BW.
4. Improving GM, PM, and M_r.
5. Possibly accentuating noise at higher frequencies.
6. Possibly requiring a relatively large capacitor in circuit implementation.

The following example illustrates the effects of the PD controller on the time-domain and frequency-domain responses of a second-order system.

Toolbox 9-2-1

The Bode diagram for Fig. 9-6 is obtained by the following sequence of MATLAB functions

```
%%%
```

The steady-state error due to a unit-ramp input is $e_{ss} = 1/K_v = 0.000443/K_P$.

Eq. (9-12) shows that the effects of the PD controller are as follows:

1. Adding a zero at $s = -K_P/K_D$ to the closed-loop transfer function
2. Increasing the *damping term*, which is the coefficient of the s term in the denominator, from 361.2 to $361.2 + 815,265K_D$

The characteristic equation is written

$$s^2 + (361.2 + 815,265K_D)s + 815,265K_P = 0 \tag{9-14}$$

We can set $K_P = 1$, which is acceptable from the steady-state error requirement. The damping ratio of the system is

$$\zeta = \frac{361.2 + 815,265K_D}{1805.84} = 0.2 + 451.46K_D \tag{9-15}$$

which clearly shows the positive effect of K_D on damping. If we wish to have critical damping, $\zeta = 1$, Eq. (9-15) gives $K_D = 0.001772$. We should quickly point out that Eq. (9-12) no longer represents a prototype second-order system, since the transient response is also affected by the zero of the transfer function at $s = -K_P/K_D$. It turns out that, for this second-order system, as the value of K_D increases, the zero will move very close to the origin and effectively cancel the pole of $G(s)$ at $s = 0$. Thus, as K_D increases, the transfer function in Eq. (9-12) approaches that of a first-order system with the pole at $s = -361.2$, and the closed-loop system will not have any overshoot. In general, for higher-order systems, however, the zero at $s = -K_P/K_D$ may increase the overshoot when K_D becomes very large.

We can apply the root-contour method to the characteristic equation in Eq. (9-14) to examine the effect of varying K_P and K_D. First, by setting K_D to zero, Eq. (9-14) becomes

$$s^2 + 361.2s + 815,265K_P = 0 \tag{9-16}$$

The root loci of the last equation as K_P varies between 0 and ∞ are shown in Fig. 9-7. When $K_D \neq 0$, the characteristic equation in Eq. (9-14) is conditioned as

$$1 + G_{eq}(s) = 1 + \frac{815,265K_Ds}{s^2 + 361.2s + 815,265K_P} = 0 \tag{9-17}$$

The root contours of Eq. (9-14) with K_P = constant and K_D varying are constructed based on the pole–zero configuration of $G_{eq}(s)$ and are shown in Fig. 9-8 for $K_P = 0.25$ and $K_P = 1$. We see that, when $K_P = 1$ and $K_D = 0$, the characteristic equation roots are at $-180.6 + j884.67$ and $-180.6 - j884.67$, and the damping ratio of the closed-loop system is 0.2. When the value of K_D is increased, the two characteristic equation roots move toward the real axis along a circular arc. When K_D is increased to 0.00177, the roots are real and equal at -902.92, and the damping is critical. When K_D is increased beyond 0.00177, the two roots become real and unequal, and the system is overdamped.

When K_P is 0.25 and $K_D = 0$, the two characteristic equation roots are at $-180.6 + j413.76$ and $-180.6 - j413.76$. As K_D increases in value, the root contours again show the improved damping due to the PD controller. Fig. 9-9 shows the unit-step responses of the closed-loop system without PD control and with $K_P = 1$ and $K_D = 0.00177$. With the PD

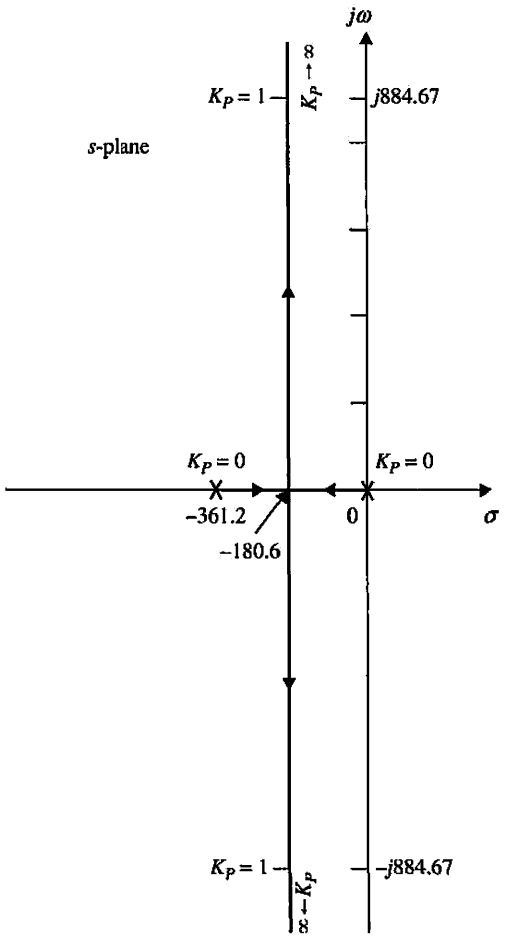


Figure 9-7 Root loci of Eq. (9-16).

control, the maximum overshoot is 4.2%. In the present case, although K_D is chosen for critical damping, the overshoot is due to the zero at $s = -K_P/K_D$ of the closed-loop transfer function. Table 9-1 gives the results on maximum overshoot, rise time, and settling time for $K_P = 1$ and $K_D = 0$, 0.0005, 0.00177, and 0.0025. The results in Table 9-1 show that the performance requirements are all satisfied with $K_D \geq 0.00177$. It should be kept in mind that K_D should only be large enough to satisfy the performance requirements. Large K_D corresponds to large BW, which may cause high-frequency noise problems, and there is also the concern of the capacitor value in the op-amp-circuit implementation.

TABLE 9-1 Attributes of the Unit-Step Responses of the System in Example 9-2-1 with PD Controller

K_D	t_r (sec)	t_s (sec)	Maximum Overshoot (%)
0	0.00125	0.0151	52.2
0.0005	0.0076	0.0076	25.7
0.00177	0.00119	0.0049	4.2
0.0025	0.00103	0.0013	0.7

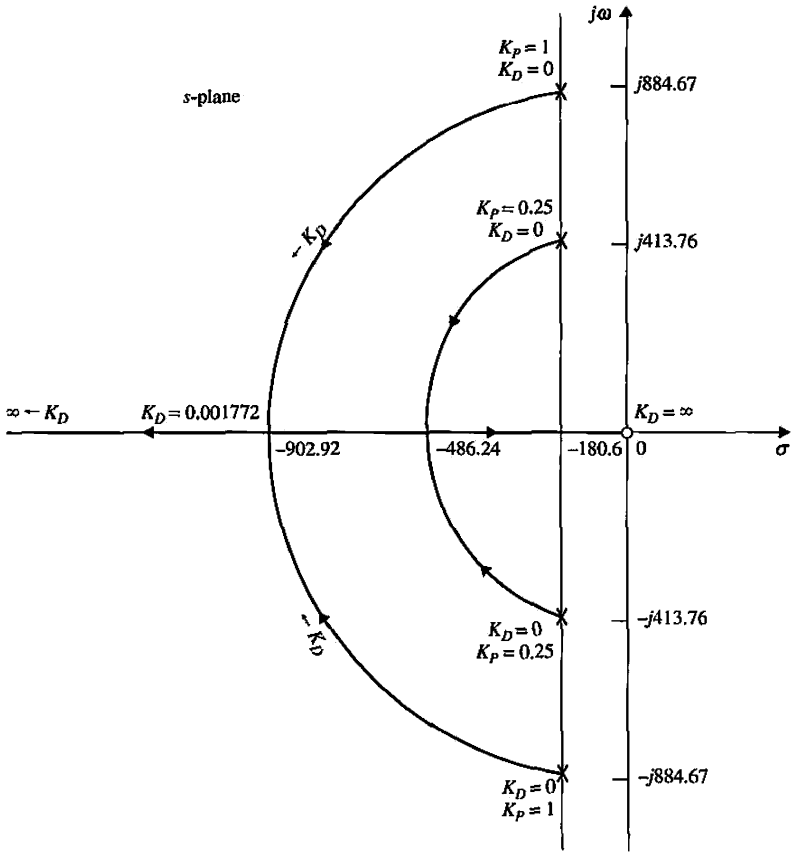


Figure 9-8 Root contours of Eq. (9-14) when $K_P = 0.25$ and 1.0; K_D varies.

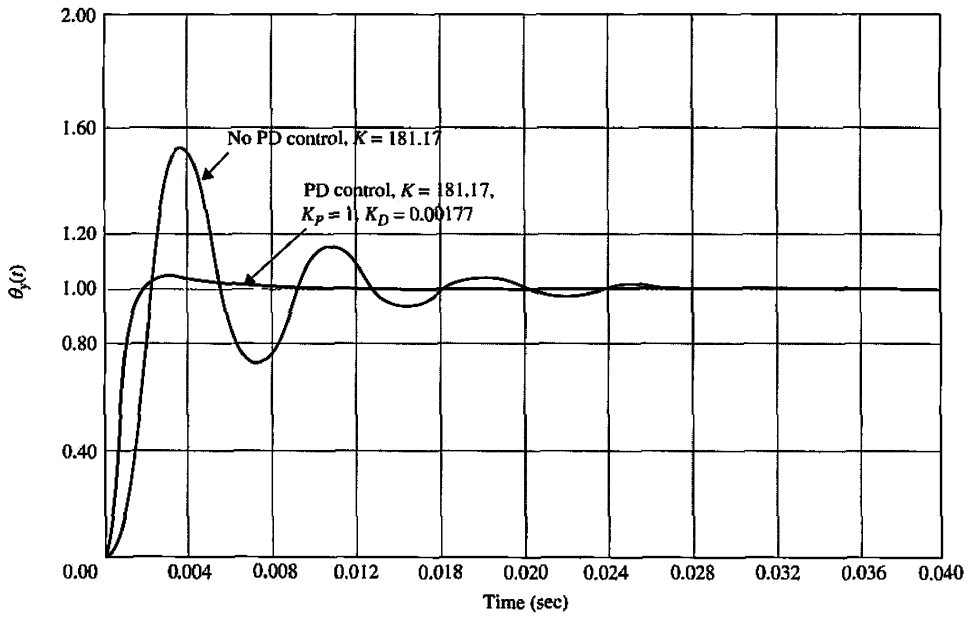


Figure 9-9 Unit-step response of the attitude control system in Fig. 5-29 with and without PD controller.

Toolbox 9-2-2

Root loci of Eq. (9-16) shown in Fig. 9-7 are obtained by the following sequence of MATLAB functions

```
KP = 1;
den = [1 361.2 815265*KP];
num = [1];
rlocus(num,den)
hold on
KP = 0;
den = [1 361.2 815265*KP];
num = [1];
rlocus(num,den)
```

The general conclusion is that the PD controller decreases the maximum overshoot, the rise time, and the settling time.

Another analytic way of studying the effects of the parameters K_P and K_D is to evaluate the performance characteristics in the parameter plane of K_P and K_D. From the characteristic equation of Eq. (9-14), we have

$$\zeta = \frac{0.2 + 451.46K_D}{\sqrt{K_P}} \tag{9-18}$$

Applying the stability requirement to Eq. (9-14), we find that, for system stability,

$$K_P > 0 \quad \text{and} \quad K_D > -0.000443$$

Toolbox 9-2-3

Root contours of Eq. (9-14) shown in Fig. 9-8 are obtained by the following sequence of MATLAB functions

```
KP = 1; KD = 1;
den = [1 361.2+815265*KD 815265*KP]; num = [1];
rlocus(num,den)
hold on
KD = 1e-6;
for i = 1:1:260
    den = [1 361.2+815265*KD 815265*KP];
    num = [1];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    T = tf(numCL,denCL);
    PoleData(:,i)=pole(T);
KD = KD +3e-5;
end
i=60; %%% for continuation of graph
PoleData(1,i+1) = - sqrt((real(PoleData(1,i))^2)+(imag(PoleData(1,i))^2));
PoleData(2,i+1) = - sqrt((real(PoleData(2,i))^2)+(imag(PoleData(2,i))^2));
plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\
KP = 0.25;
KD = 1;
```

```
den = [1 361.2+815265*KD 815265*KP];
num = [1];
rlocus(num,den)
hold on
KD = 1e-6;
for i = 1:1:260
    den = [1 361.2+815265*KD 815265*KP];
    num = [1];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    T = tf(numCL,denCL);
    PoleData(:,i)=pole(T);
    KD = KD +3e-5;
end
i=23; %%% for continuation of graph
PoleData(1,i+1) = - sqrt((real(PoleData(1,i))^2)+(imag(PoleData(1,i))^2));
PoleData(2,i+1) = - sqrt((real(PoleData(2,i))^2)+(imag(PoleData(2,i))^2));
plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));
%%%
```

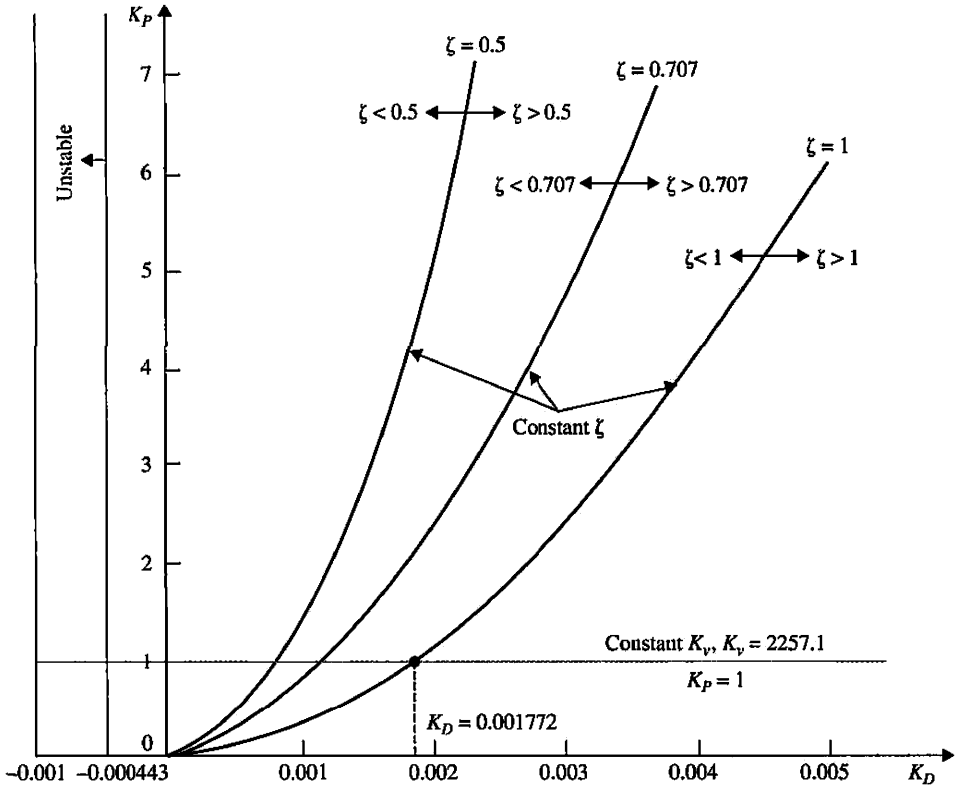


Figure 9-10 K_P-versus-K_D parameter plane for the attitude control system with a PD controller.

the uncompensated system is 22.68°, and the resonant peak M_r is 2.522. These values correspond to a lightly damped system. Let us give the following performance criteria:

Steady-state error due to a unit-ramp input ≤ 0.00443

Phase margin $\geq 80°$

Resonant peak $M_r \leq 1.05$

BW ≤ 2000 rad/sec

The Bode plots of $G(s)$ for $K_P = 1$ and $K_D = 0$, 0.005, 0.00177, and 0.0025 are shown in Fig. 9-11. The performance measures in the frequency domain for the compensated system with these controller parameters are tabulated in Table 9-2, along with the time-domain attributes for comparison. The Bode plots as well as the performance data were easily generated by using MATLAB tools. Use **ACSYS** component controls to reproduce the results in Table 9-2.

The results in Table 9-2 show that the gain margin is always infinite, and thus the relative stability is measured by the phase margin. This is one example where the gain margin is not an effective measure of the relative stability of the system. When $K_D = 0.00177$, which corresponds to critical damping, the phase margin is 82.92°, the resonant peak M_r is 1.025, and BW is 1669 rad/sec. The performance requirements in the frequency domain are all satisfied. Other effects of the PD control are that the BW and the gain-crossover frequency are increased. The phase-crossover frequency is always infinite in this case.

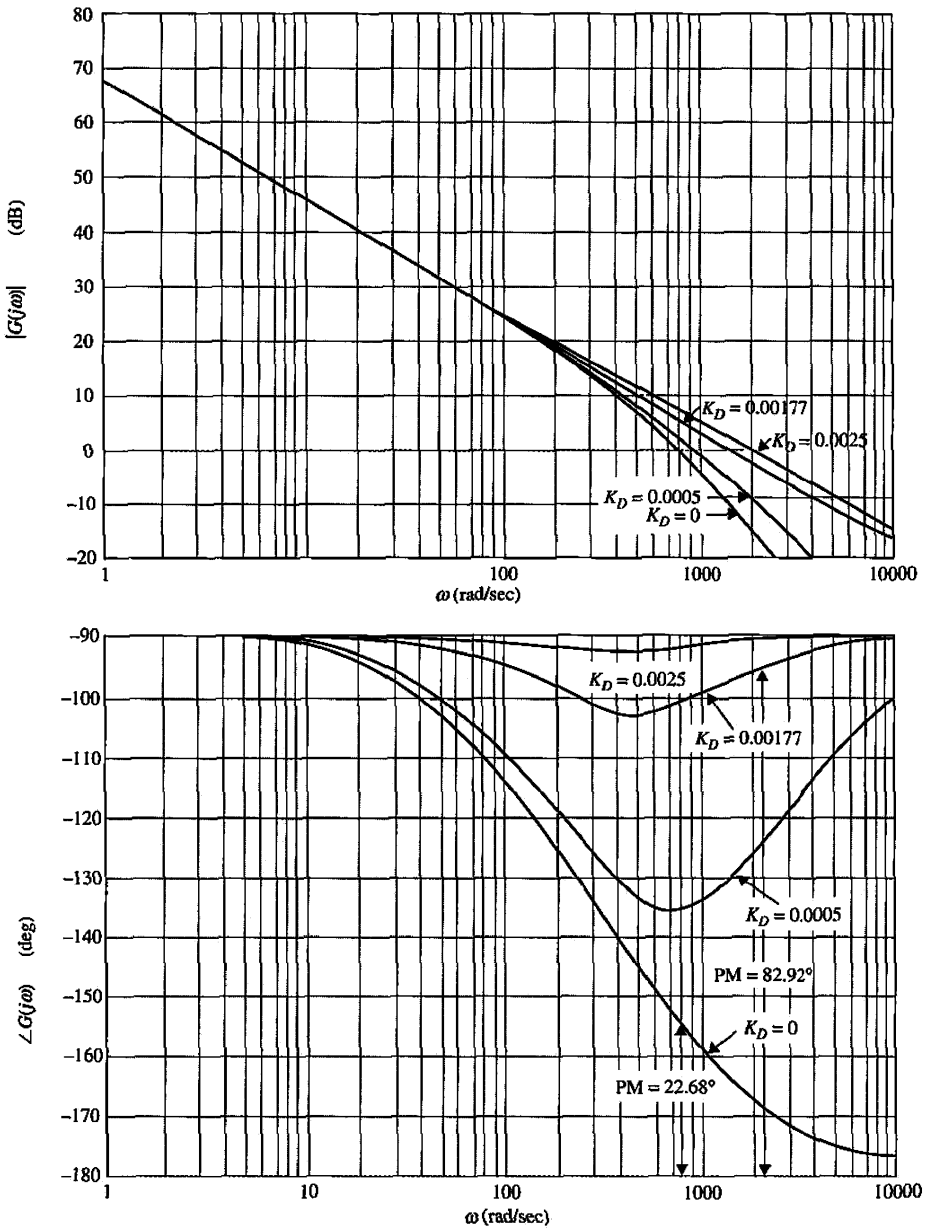


Figure 9-11 Bode plot of $G(s) = \dfrac{815,265(1 + K_D s)}{s(s + 361.2)}$.

TABLE 9-2 Frequency-Domain Characteristics of the System in Example 9-2-1 with PD Controller

K_D	GM (dB)	PM (deg)	Gain CO (rad/sec)	BW (rad/sec)	M_r	t_r (sec)	t_s (sec)	Maximum Overshoot (%)
0	∞	22.68	868	1370	2.522	0.00125	0.0151	52.2
0.0005	∞	46.2	913.5	1326	1.381	0.0076	0.0076	25.7
0.00177	∞	82.92	1502	1669	1.025	0.00119	0.0049	4.2
0.0025	∞	88.95	2046	2083	1.000	0.00103	0.0013	0.7

EXAMPLE 9-2-2 Consider the third-order aircraft attitude control system discussed in Chapter 5 with the forward-path transfer function given in Eq. (5-153),

$$G(s) = \frac{1.5 \times 10^7 K}{s(s^2 + 3408.3s + 1,204,000)} \tag{9-19}$$

The same set of time-domain specifications given in Example 9-2-1 is to be used. It was shown in Chapter 5 that, when $K = 181.17$, the maximum overshoot of the system is 78.88%.

Let us attempt to meet the transient performance requirements by use of a PD controller with the transfer function given in Eq. (9-2). The forward-path transfer function of the system with the PD controller and $K = 181.17$ is

$$G(s) = \frac{2.718 \times 10^9 (K_P + K_D s)}{s(s^2 + 3408.3s + 1,204,000)} \tag{9-20}$$

You may also use our MATLAB toolbox **ACSYS** to solve this problem. See Section 9-19.

Time-Domain Design

Setting $K_P = 1$ arbitrarily, the characteristic equation of the closed-loop system is written

$$s^3 + 3408.3s^2 + (1,204,000 + 2.718 \times 10^9 K_D)s + 2.718 \times 10^9 = 0 \tag{9-21}$$

To apply the root-contour method, we condition Eq. (9-21) as

$$1 + G_{eq}(s) = 1 + \frac{2.718 \times 10^9 K_D s}{s^3 + 3408.2s^2 + 1,204,000s + 2.718 \times 10^9} = 0 \tag{9-22}$$

where

$$G_{eq}(s) = \frac{2.718 \times 10^9 K_D s}{(s + 3293.3)(s + 57.49 + j906.6)(s + 57.49 - j906.6)} \tag{9-23}$$

Toolbox 9-2-5

Root contours of Fig. 9-12 are obtained by the following sequence of MATLAB functions

You may wish to use clc, close all, or clear all before running the following

```
% Root contours
kd=0.005;

num = [2.718*10^9*kd 0];
den = [1 3408.2 1204000 2.718*10^9];
rlocus(num,den)
```

• If a system has very low damping or is unstable, the PD control may not be effective in improving the stability of the system.

The root contours of Eq. (9-21) are plotted as shown in Fig. 9-12, based on the pole–zero configuration of $G_{eq}(s)$. The root contours of Fig. 9-12 reveal the effectiveness of the PD controller for the improvement on the relative stability of the system. Notice that, as the value of K_D increases, one root of the characteristic equation moves from -3293.3 toward the origin, while the two complex roots start out toward the left and eventually approach the vertical asymptotes that intersect at $s = -1704$. The immediate assessment of the situation is that, if the value of K_D is too large, *the two complex roots will actually have reduced*

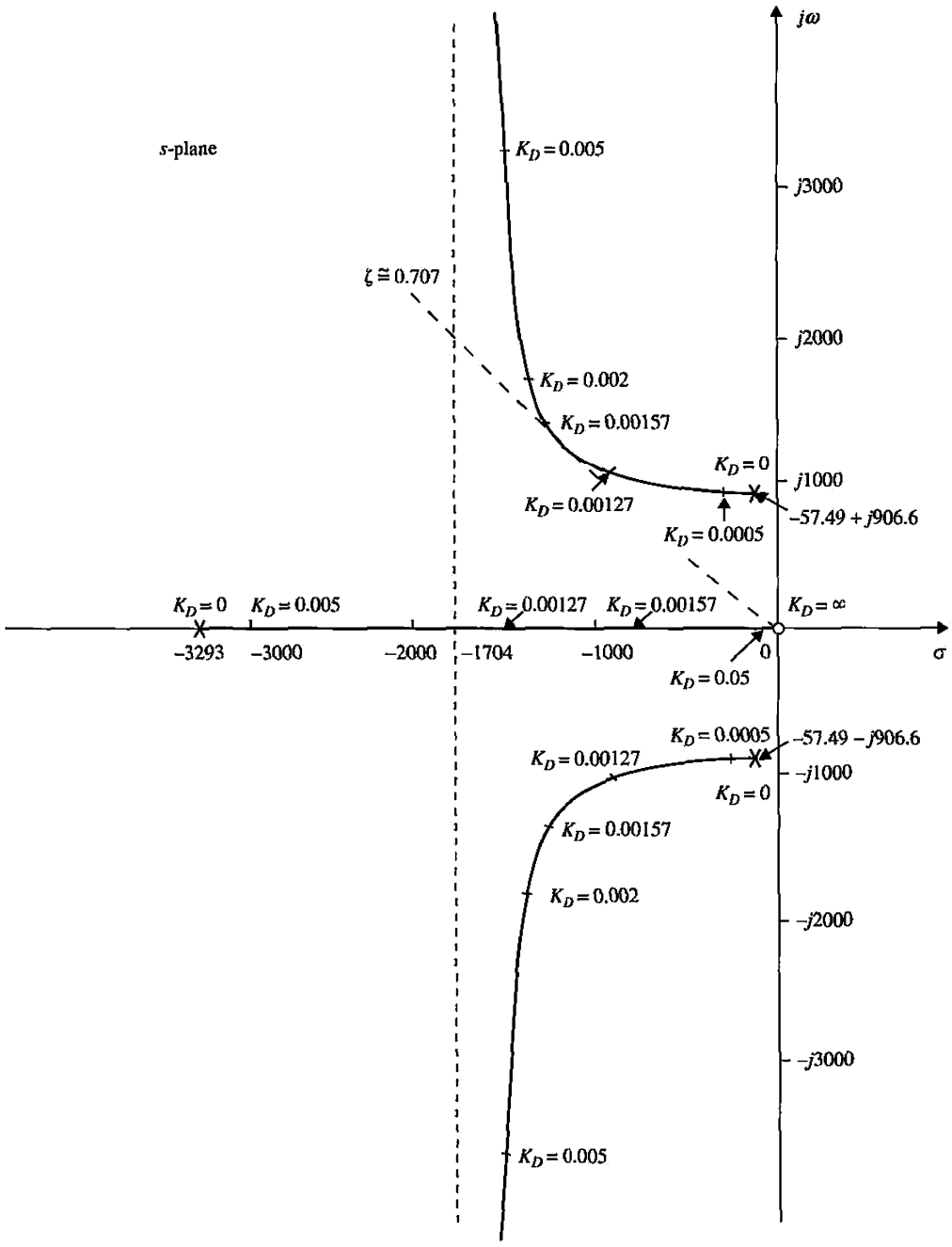


Figure 9-12 Root contours of
$s^3 + 3408.3s^2 + (1,204,000 + 2.718 \times 10^9 K_D)s + 2.718 \times 10^9 = 0.$

damping while increasing the natural frequency of the system. It appears that the ideal location for the two complex characteristic equation roots, from the standpoint of relative stability, is near the bend of the root contour, where the relative damping ratio is approximately 0.707. The root contours of Fig. 9-12 clearly show that, if the original system has low damping or is unstable, the zero introduced by the PD controller may not be able to add sufficient damping or even stabilize the system.

TABLE 9-3 Time-Domain Attributes of the Third-Order System in Example 9-2-2 with PD Controller

K_D	% Maximum Overshoot	t_r (sec)	t_s (sec)	Characteristic Equation Roots	
0	78.88	0.00125	0.0495	−3293.3,	$-57.49 \pm j906.6$
0.0005	41.31	0.00120	0.0106	−2843.07,	$-282.62 \pm j936.02$
0.00127	17.97	0.00100	0.00398	−1523.11,	$-942.60 \pm j946.58$
0.00157	14.05	0.00091	0.00337	−805.33,	$-1301.48 \pm j1296.59$
0.00200	11.37	0.00080	0.00255	−531.89,	$-1438.20 \pm j1744.00$
0.00500	17.97	0.00042	0.00130	−191.71,	$-1608.29 \pm j3404.52$
0.01000	31.14	0.00026	0.00093	−96.85,	$-1655.72 \pm j5032$
0.05000	61.80	0.00010	0.00144	−19.83,	$-1694.30 \pm j11583$

Table 9-3 gives the results of maximum overshoot, rise time, settling time, and the roots of the characteristic equation as functions of the parameter K_D. The following conclusions are drawn on the effects of the PD controller on the third-order system.

1. The minimum value of the maximum overshoot, 11.37%, occurs when K_D is approximately 0.002.
2. Rise time is improved (reduced) with the increase of K_D.
3. Too high a value of K_D will actually increase the maximum overshoot and the settling time substantially. The latter is because the damping is reduced as K_D is increased indefinitely.

Fig. 9-13 shows the unit-step responses of the system with the PD controller for several values of K_D. The conclusion is that, while the PD control does improve the damping of the system, it does not meet the maximum-overshoot requirement.

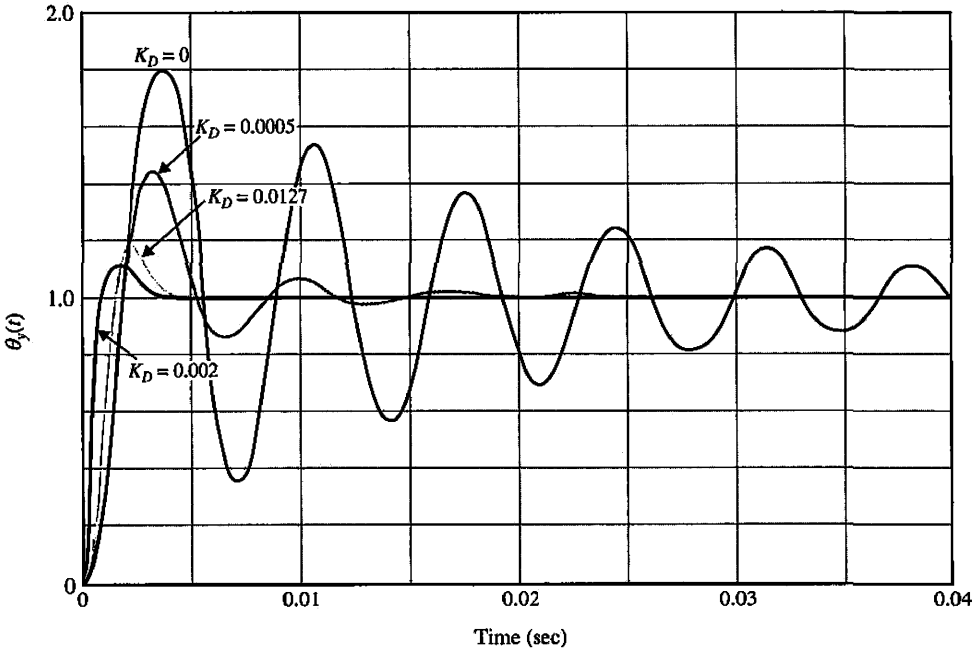


Figure 9-13 Unit-step responses of the system in Example 9-2-2 with PD controller.

Frequency-Domain Design

The Bode plot of Eq. (9-20) is used to conduct the frequency-domain design of the PD controller. Fig. 9-14 shows the Bode plot for $K_P = 1$ and $K_D = 0$. The following performance data are obtained for the uncompensated system:

Gain margin = 3.6 dB

Phase margin = 7.77°

Resonant Peak $M_r = 7.62$

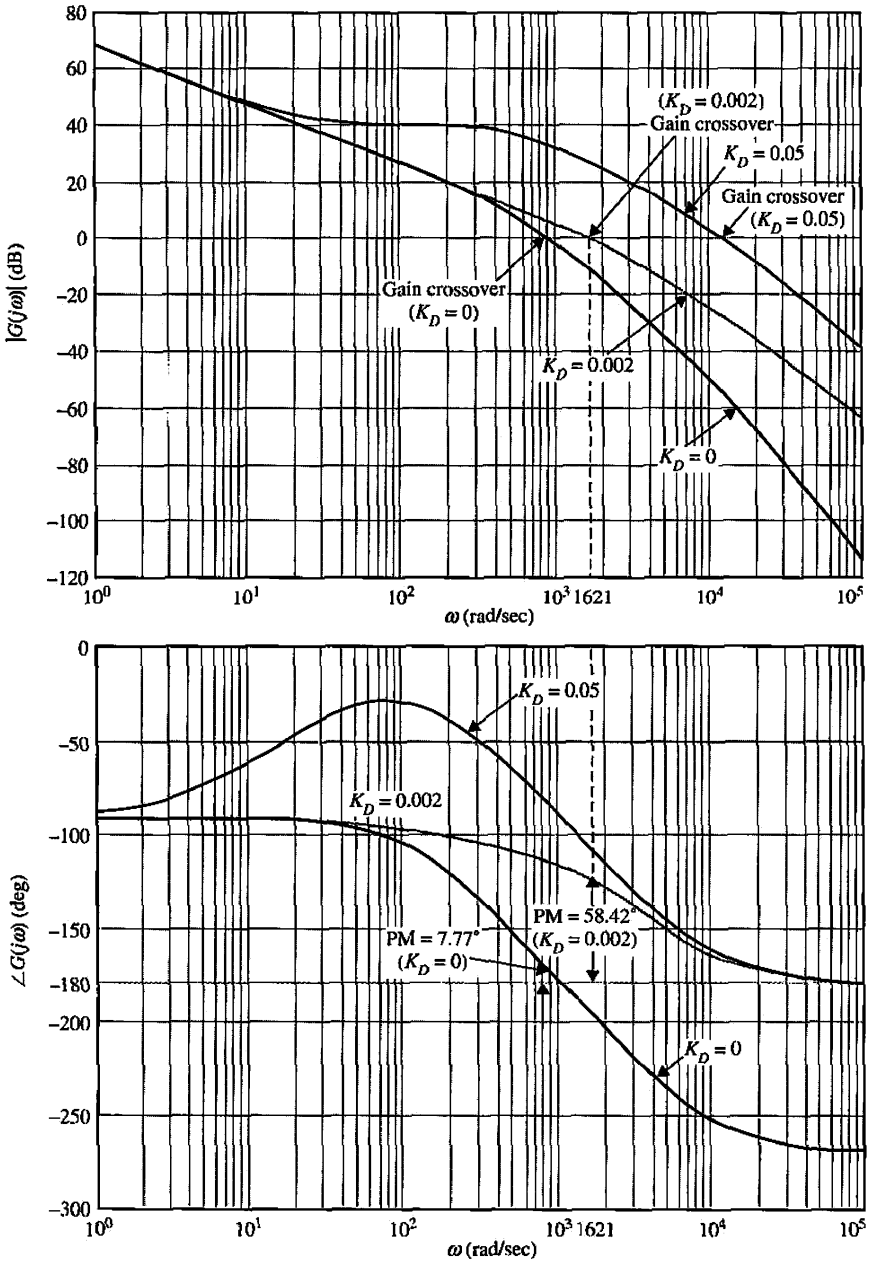


Figure 9-14 Bode diagram of $G(s)$ of the system in Example 9-2-2 with PD controller.

TABLE 9-4 Frequency-Domain Characteristics of the Third-Order System in Example 9-2-2 with PD Controller

K_D	GM (dB)	PM (deg)	M_r	BW (rad/sec)	Gain CO (rad/sec)	Phase CO (rad/sec)
0	3.6	7.77	7.62	1408.83	888.94	1103.69
0.0005	∞	30.94	1.89	1485.98	935.91	∞
0.00127	∞	53.32	1.19	1939.21	1210.74	∞
0.00157	∞	56.83	1.12	2198.83	1372.30	∞
0.00200	∞	58.42	1.07	2604.99	1620.75	∞
0.00500	∞	47.62	1.24	4980.34	3118.83	∞
0.01000	∞	35.71	1.63	7565.89	4789.42	∞
0.0500	∞	16.69	3.34	17989.03	11521.00	∞

Bandwidth BW = 1408.83 rad/sec

Gain crossover (GCO) = 888.94 rad/sec

Phase crossover (PCO) = 1103.69 rad/sec

Let us use the same set of frequency-domain performance requirements listed in Example 9-2-1. The logical way to approach this problem is to first examine how much additional phase is needed to realize a phase margin of 80°. Because the uncompensated system with the gain set to meet the steady-state requirement is only 7.77°, the PD controller must provide an additional phase of 72.23°. This additional phase must be placed at the gain crossover of the compensated system in order to realize a PM of 80°. Referring to the Bode plot of the PD controller in Fig. 9-6, we see that the additional phase is always accompanied by a gain in the magnitude curve. As a result, the gain crossover of the compensated system will be pushed to a higher frequency at which the phase of the uncompensated system would correspond to an even smaller PM. Thus, we may run into the problem of diminishing returns. This symptom is parallel to the situation illustrated by the root-contour plot in Fig. 9-12, in which case the larger K_D would simply push the roots to a higher frequency, and the damping would actually be decreased. The frequency-domain performance data of the compensated system with the values of K_D used in Table 9-3 are obtained from the Bode plots for each case, and the results are shown in Table 9-4. The Bode plots of some of these cases are shown in Fig. 9-14. Notice that the gain margin becomes infinite when the PD controller is added, and the phase margin becomes the dominant measure of relative stability. This is because the phase curve of the PD-compensated system stays above the −180°-axis, and the phase crossover is at infinity.

Toolbox 9-2-6

Fig. 9-13 is obtained by the following sequence of MATLAB functions

```
KP = 1;
KD = [ 0.0005 0.0127 0.002];

for i =1:length(KD)
   num =[2.718e9*KD(i) 2.718e9*KP];
   den = [1 3408.3 0 0];
   tf(num,den);
```

```
    [numCL,denCL]=cloop(num,den);
    step(numCL,denCL)
    hold on
end
axis([0 0.04 0 2])
```

When $K_D = 0.002$, the phase margin is at a maximum of 58.42°, and M_r is also minimum at 1.07, which happens to agree with the optimal value obtained in the time-domain design summarized in Table 9-3. When the value of K_D is increased beyond 0.002, the phase margin decreases, which agrees with the findings from the time-domain design that large values of K_D actually decrease damping. However, the BW and the gain crossover increase continuously with the increase in K_D. The frequency-domain design again shows that the PD control falls short in meeting the performance requirements imposed on the system. Just as in the time-domain design, we have demonstrated that if the original system has very low damping, or is unstable, PD control may not be effective in improving the stability of the system. Another situation under which PD control may be ineffective is if the slope of the phase curve near the gain-crossover frequency is steep, in which case the rapid decrease of the phase margin due to the increase of the gain crossover from the added gain of the PD controller may render the additional phase ineffective.

Toolbox 9-2-7

Bode diagram of G(s) in Example 9-2 in Fig. 9-14 is obtained by the following sequence of MATLAB functions

```
KD = [0 0.002 0.05];
KP=1;

for i = 1:length(KD)
    num =[2.718e9*KD(i) 2.718e9*KP];
    den = [1 3408.3 1204000 0];
    bode(num,den);
    hold on;
end
```

9-3 DESIGN WITH THE PI CONTROLLER

We see from Section 9-2 that the PD controller can improve the damping and rise time of a control system at the expense of higher bandwidth and resonant frequency, and the steady-state error is not affected unless it varies with time, which is typically not the case for step-function inputs. Thus, the PD controller may not fulfill the compensation objectives in many situations.

The integral part of the PID controller produces a signal that is proportional to the time integral of the input of the controller. Fig. 9-15 illustrates the block diagram of a prototype second-order system with a series PI controller. The transfer function of the PI controller is

$$G_c(s) = K_P + \frac{K_I}{s} \tag{9-24}$$

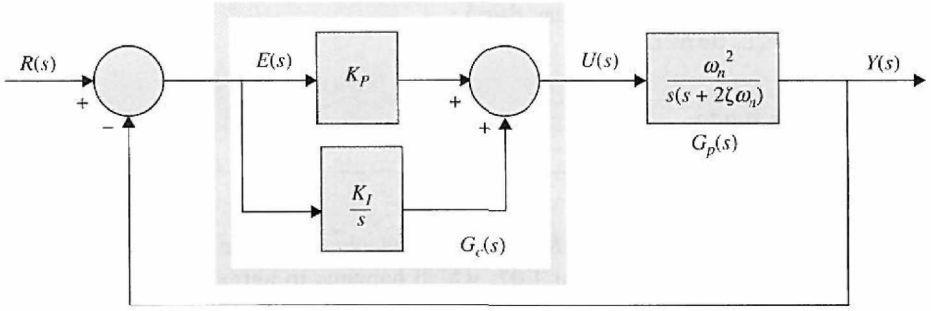


Figure 9-15 Control system with PI controller.

Using the circuit elements given in Table 4-4, two op-amp-circuit realizations of Eq. (9-24) are shown in Fig. 9-16. The transfer function of the two-op-amp circuit in Fig. 9-16(a) is

$$G_c(s) = \frac{E_o(s)}{E_{in}(s)} = \frac{R_2}{R_1} + \frac{R_2}{R_1 C_2 s} \tag{9-25}$$

Comparing Eq. (9-24) with Eq. (9-25), we have

$$K_P = \frac{R_2}{R_1} \quad K_I = \frac{R_2}{R_1 C_2} \tag{9-26}$$

The transfer function of the three-op-amp circuit in Fig. 9-16(b) is

$$G_c(s) = \frac{E_o(s)}{E_{in}(s)} = \frac{R_2}{R_1} + \frac{1}{R_i C_i s} \tag{9-27}$$

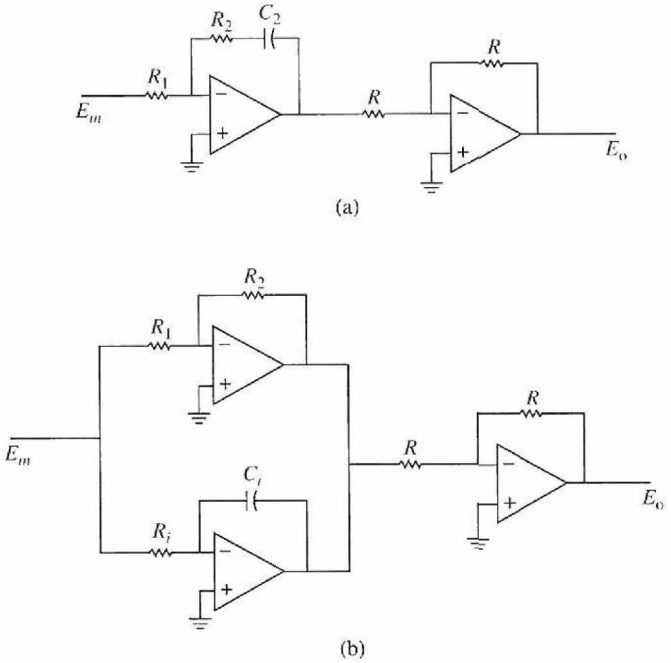


Figure 9-16 Op-amp-circuit realization of the PI controller, $G_c(s) = K_P + \frac{K_I}{s}$.
(a) Two-op-amp circuit.
(b) Three-op-amp circuit.

Thus, the parameters of the PI controller are related to the circuit parameters as

$$K_P = \frac{R_2}{R_1} \quad K_I = \frac{1}{R_i C_i} \tag{9-28}$$

The advantage with the circuit in Fig. 9-16(b) is that the values of K_P and K_I are independently related to the circuit parameters. However, in either circuit, K_I is inversely proportional to the value of the capacitor. Unfortunately, effective PI-control designs usually result in small values of K_I, and thus we must again watch out for unrealistically large capacitor values.

The forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{\omega_n^2(K_P s + K_I)}{s^2(s + 2\zeta\omega_n)} \tag{9-29}$$

Clearly, the immediate effects of the PI controller are as follows:

1. Adding a zero at $s = -K_I/K_P$ to the forward-path transfer function.
2. Adding a pole at $s = 0$ to the forward-path transfer function. This means that the system type is increased by 1 to a type 2 system. Thus, the steady-state error of the original system is improved by one order; that is, if the steady-state error to a given input is constant, the PI control reduces it to zero (provided that the compensated system remains stable).

The system in Fig. 9-15, with the forward-path transfer function in Eq. (9-29), will now have a zero steady-state error when the reference input is a ramp function. However, because the system is now of the third order, *it may be less stable* than the original second-order system or even become *unstable* if the parameters K_P and K_I are not properly chosen.

In the case of a type 1 system with a PD control, the value of K_P is important because the ramp-error constant K_v is directly proportional to K_P, and thus the magnitude of the steady-state error is inversely proportional to K_P when the input is a ramp. On the other hand, if K_P is too large, the system may become unstable. Similarly, for a type 0 system, the steady-state error due to a step input will be inversely proportional to K_P.

When a type 1 system is converted to type 2 by the PI controller, K_P no longer affects the steady-state error, and the latter is always zero for a stable system with a ramp input. The problem is then to choose the proper combination of K_P and K_I so that the transient response is satisfactory.

9-3-1 Time-Domain Interpretation and Design of PI Control

The pole–zero configuration of the PI controller in Eq. (9-24) is shown in Fig. 9-17. At first glance, it may seem that PI control will improve the steady-state error at the expense of stability. However, we shall show that, if the location of the zero of $G_c(s)$ is selected properly, both the damping and the steady-state error can be improved. Because the PI controller is essentially a low-pass filter, the compensated system usually will have a slower rise time and longer settling time. *A viable method of designing the PI control is to select the zero at* $s = -K_I/K_P$ *so that it is relatively close to the origin and away from the most significant poles of the process; the values of* K_P *and* K_I *should be relatively small.*

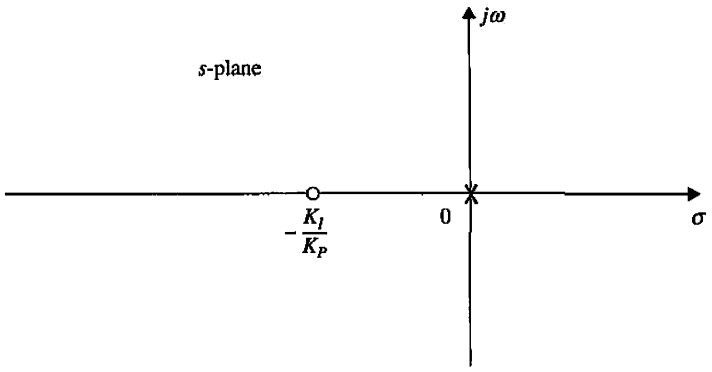


Figure 9-17 Pole–zero configuration of a PI controller.

9-3-2 Frequency-Domain Interpretation and Design of PI Control

For frequency-domain design, the transfer function of the PI controller is written

$$G_c(s) = K_P + \frac{K_I}{s} = \frac{K_I\left(1 + \frac{K_P}{K_I}s\right)}{s} \tag{9-30}$$

The Bode plot of $G_c(j\omega)$ is shown in Fig. 9-18. Notice that the magnitude of $G_c(j\omega)$ at $\omega = \infty$ is 20 $\log_{10} K_P$ dB, which represents an attenuation if the value of K_P is less than 1. This attenuation may be utilized to improve the stability of the system. The phase of $G_c(j\omega)$ is always negative, which is detrimental to stability. Thus, we should place the corner frequency of the controller, $\omega = K_I/K_P$, as far to the left as the bandwidth requirement allows, so the phase-lag properties of $G_c(j\omega)$ do not degrade the achieved phase margin of the system.

The frequency-domain design procedure for the PI control to realize a given phase margin is outlined as follows:

1. The Bode plot of the forward-path transfer function $G_p(s)$ of the uncompensated system is made with the loop gain set according to the steady-state performance requirement.
2. The phase margin and the gain margin of the uncompensated system are determined from the Bode plot. For a specified phase margin requirement, the new gain-crossover frequency ω_g' corresponding to this phase margin is found on the Bode plot. The magnitude plot of the compensated transfer function must pass through the 0-dB-axis at this new gain-crossover frequency in order to realize the desired phase margin.
3. To bring the magnitude curve of the uncompensated transfer function down to 0 dB at the new gain-crossover frequency ω_g', the PI controller must provide the amount of attenuation equal to the gain of the magnitude curve at the new gain-crossover frequency. In other words, set

$$\left|G_P\left(j\omega_g'\right)\right|_{\text{dB}} = -20\log_{10}K_P \text{ dB} \quad K_P < 1 \tag{9-31}$$

from which we have

$$K_P = 10^{-\left|G_P\left(j\omega_g'\right)\right|_{\text{dB}}/20} \quad K_P < 1 \tag{9-32}$$

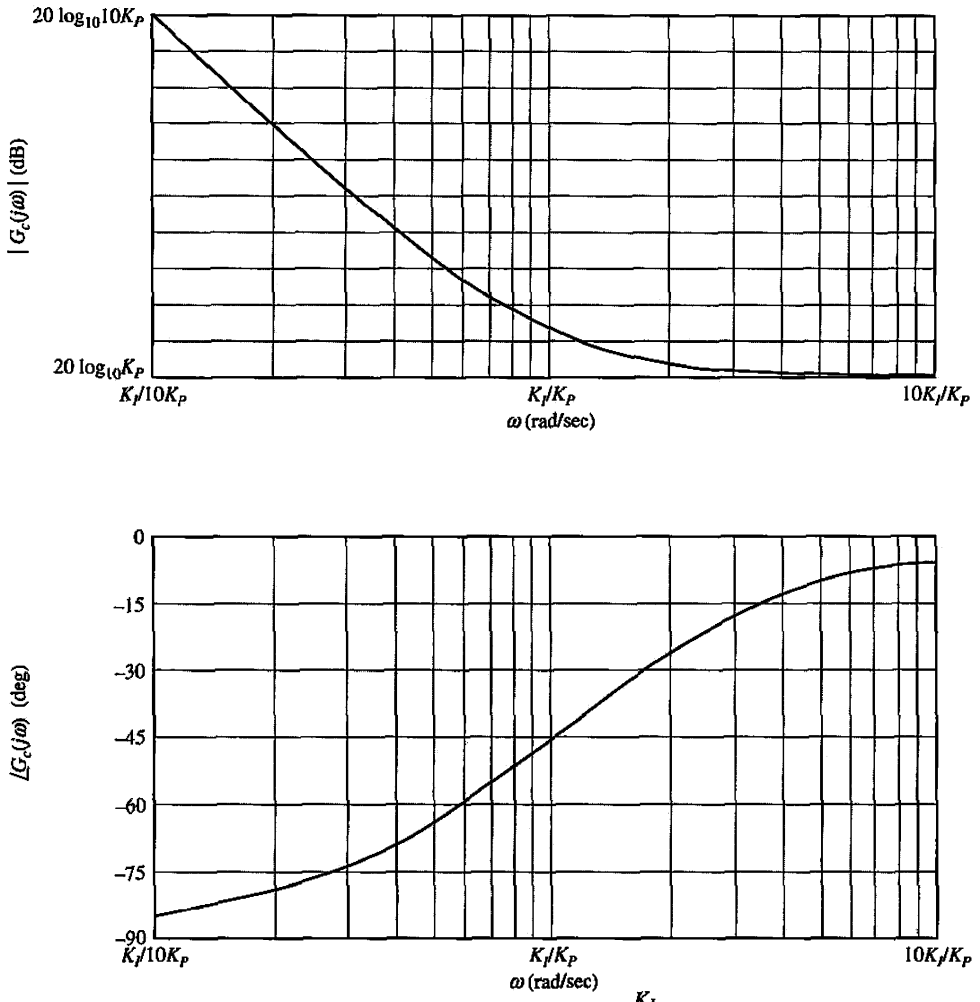


Figure 9-18 Bode diagram of the PI controller. $G_c(s) = K_P + \frac{K_I}{s}$.

Once the value of K_P is determined, it is necessary only to select the proper value of K_I to complete the design. Up to this point, we have assumed that, although the gain-crossover frequency is altered by attenuating the magnitude of $G_c(j\omega)$ at ω_g', the original phase is not affected by the PI controller. This is not possible, however, since, as shown in Fig. 9-18, the attenuation property of the PI controller is accompanied with a phase lag that is detrimental to the phase margin. It is apparent that, if the corner frequency $\omega = K_I/K_P$ is placed far below ω_g', the phase lag of the PI controller will have a negligible effect on the phase of the compensated system near ω_g'. On the other hand, the value of K_I/K_P should not be too small or the bandwidth of the system will be too low, causing the rise time and settling time to be too long. As a general guideline, K_I/K_P should correspond to a frequency that is at least one decade, sometimes as much as two decades, below ω_g'. That is, we set

$$\frac{K_I}{K_P} = \frac{\omega_g'}{10} \text{ rad/sec} \tag{9-33}$$

Within the general guideline, the selection of the value of K_I/K_P is pretty much at the discretion of the designer, who should be mindful of its effect on BW and its practical implementation by an op-amp circuit.

4. The Bode plot of the compensated system is investigated to see if the performance specifications are all met.
5. The values of K_I and K_P are substituted in Eq. (9-30) to give the desired transfer function of the PI controller.

If the controlled process $G_P(s)$ is type 0, the value of K_I may be selected based on the ramp-error-constant requirement, and then there would only be one parameter, K_P, to determine. By computing the phase margin, gain margin, M_r, and BW of the closed-loop system with a range of values of K_P, the best value for K_P can be easily selected.

Based on the preceding discussions, we can summarize the advantages and disadvantages of a properly designed PI controller as the following:

1. Improving damping and reducing maximum overshoot.
2. Increasing rise time.
3. Decreasing BW.
4. Improving gain margin, phase margin, and M_r.
5. Filtering out high-frequency noise.

It should be noted that in the PI controller design process, selection of a proper combination of K_I and K_P, so that the capacitor in the circuit implementation of the controller is not excessively large, is more difficult than in the case of the PD controller.

The following examples will illustrate how the PI control is designed and what its effects are.

EXAMPLE 9-3-1 Consider the second-order attitude-control system discussed in Example 9-2-1. Applying the PI controller of Eq. (9-24), the forward-path transfer function of the system becomes

$$G(s) = G_c(s)G_P(s) = \frac{4500KK_P(s + K_I/K_P)}{s^2(s + 361.2)} \tag{9-34}$$

You may use **ACSYS** to solve this problem.

Time-Domain Design

Let the time-domain performance requirements be

Steady-state error due to parabolic input $t^2u_s(t)/2 \leq 0.2$

Maximum overshoot $\leq 5\%$

Rise time $t_r \leq 0.01$ sec

Settling time $t_s \leq 0.02$ sec

We have to relax the rise time and settling time requirements from those in Example 9-2-1 so that we will have a meaningful design for this system. The significance of the requirement on the steady-state error due to a parabolic input is that it indirectly places a minimum requirement on the speed of the transient response.

The parabolic-error constant is

$$K_a = \lim_{s \to 0} s^2 G(s) = \lim_{s \to 0} s^2 \frac{4500KK_P(s + K_I/K_P)}{s^2(s + 361.2)} = \frac{4500KK_I}{361.2} = 12.46KK_I \tag{9-35}$$

The steady-state error due to the parabolic input $t^2 u_s(t)/2$ is

$$e_{ss} = \frac{1}{K_a} = \frac{0.08026}{KK_I} (\leq 0.2) \tag{9-36}$$

Let us set $K = 181.17$, simply because this was the value used in Example 9-2-1. Apparently, to satisfy a given steady-state error requirement for a parabolic input, the larger the K, the smaller K_I can be. Substituting $K = 181.17$ in Eq. (9-36) and solving K_I for the minimum steady-state error requirement of 0.2, we get the minimum value of K_I to be 0.002215. If necessary, the value of K can be adjusted later.

With $K = 181.17$, the characteristic equation of the closed-loop system is

$$s^3 + 361.2s^2 + 815,265K_P s + 815,265K_I = 0 \tag{9-37}$$

Applying Routh's test to Eq. (9-37) yields the result that the system is stable for $0 < K_I/K_P < 361.2$. This means that the zero of $G(s)$ at $s = -K_I/K_P$ cannot be placed too far to the left in the left-half s-plane, or the system will be unstable. Let us place the zero at $-K_I/K_P$ relatively close to the origin. For the present case, the most significant pole of $G_P(s)$, besides the pole at $s = 0$, is at -361.2. Thus, K_I/K_P should be chosen so that the following condition is satisfied:

$$\frac{K_I}{K_P} \ll 361.2 \tag{9-38}$$

The root loci of Eq. (9-37) with $K_I/K_P = 10$ are shown in Fig. 9-19. Notice that, other than the small loop around the zero at $s = -10$, these root loci for the most part are very similar to those shown in Fig. 9-7, which are for Eq. (9-16). With the condition in Eq. (9-38) satisfied, Eq. (9-34) can be approximated by

$$G(s) \cong \frac{815,265K_P}{s(s + 361.2)} \tag{9-39}$$

where the term K_I/K_P in the numerator is neglected when compared with the magnitude of s, which takes on values along the operating points on the complex portion of the root loci that correspond to, say, a relative damping ratio in the range of $0.7 < \zeta < 1.0$. Let us assume that we wish to have a relative damping ratio of 0.707. From Eq. (9-39), the required value of K_P for this damping ratio is 0.08. This should also be true for the third-order system with the PI controller if the value of K_I/K_P satisfies Eq. (9-38). Thus, with $K_P = 0.08$, $K_I = 0.8$; the root loci in Fig. 9-19 show that the relative damping ratio of the two complex roots is approximately 0.707. In fact, the three characteristic equation roots are at

$$s = -10.605, \ -175.3 + j175.4, \text{ and } -175.3 - j175.4$$

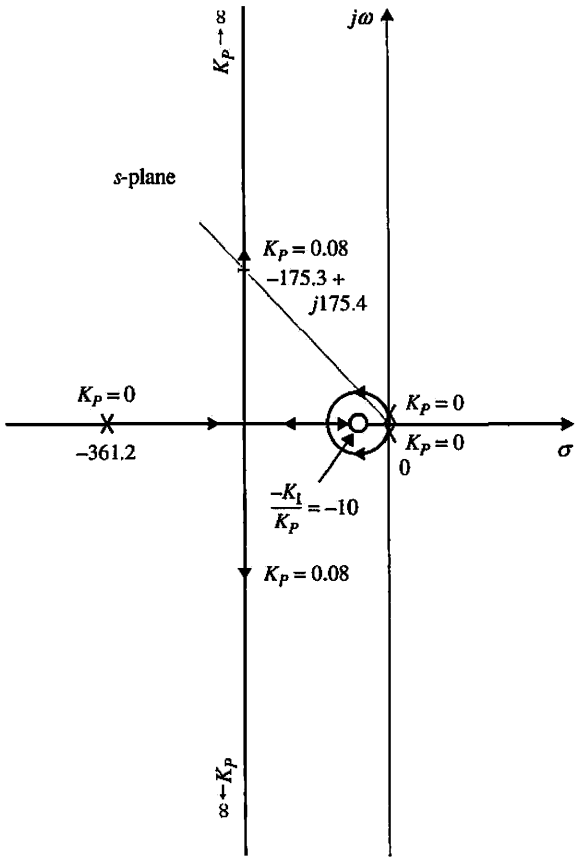


Figure 9-19 Root loci of Eq. (9-37) with $K_I/K_P = 10$; K_P varies.

The reason for this is that when we "stand" at the root at $-175.3 + j175.4$ and "look" toward the neighborhood near the origin, we see that the zero at $s = -10$ is relatively close to the origin and, thus, practically cancels one of the poles at $s = 0$. In fact, we can show that, as long as $K_P = 0.08$ and the value of K_I is chosen such that Eq. (9-38) is satisfied, the relative damping ratio of the complex roots will be very close to 0.707. For example, let us select $K_I/K_P = 5$; the three characteristic equation roots are at

$$s = -5.145, \; -178.03 + j178.03, \text{ and } -178.03 - j178.03$$

and the relative damping ratio is still 0.707. Although the real pole of the closed-loop transfer function is moved, it is very close to the zero at $s = -K_I/K_P$ so that the transient due to the real pole is negligible. For example, when $K_P = 0.08$ and $K_I = 0.4$, the closed-loop transfer function of the compensated system is

$$\frac{\Theta_y(s)}{\Theta_r(s)} = \frac{65,221.2(s+5)}{(s+5.145)(s+178.03+j178.03)(s+178.03-j178.03)} \tag{9-40}$$

Because the pole at $s = 5.145$ is very close to the zero at $s = -5$, the transient response due to this pole is negligible, and the system dynamics are essentially dominated by the two complex poles.

Toolbox 9-3-1

Root loci of Eq. (9-37) in Fig. 9-19 are obtained by the following sequence of MATLAB functions

```
KP = 0.000001; % start with a very small KP see Figure 9-19
KI=10*KP;
num = [KP KI];
den = [1 361.2 815265*KP 815265*KI];
G=tf(num,den)
rlocus(G)
```

TABLE 9-5 Attributes of the Unit-Step Responses of the System in Example 9-3-1 with PI Controller

K_I/K_P	K_I	K_P	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0	0	1.00	52.7	0.00135	0.015
20	1.60	0.08	15.16	0.0074	0.049
10	0.80	0.08	9.93	0.0078	0.0294
5	0.40	0.08	7.17	0.0080	0.023
2	0.16	0.08	5.47	0.0083	0.0194
1	0.08	0.08	4.89	0.0084	0.0114
0.5	0.04	0.08	4.61	0.0084	0.0114
0.1	0.008	0.08	4.38	0.0084	0.0115

Table 9-5 gives the attributes of the unit-step responses of the system with PI control for various values of K_I/K_P, with $K_P = 0.08$, which corresponds to a relative damping ratio of 0.707.

The results in Table 9-5 verify the fact that PI control reduces the overshoot but at the expense of longer rise time. For $K_I \leq 1$, the settling times in Table 9-5 actually show a sharp reduction, which is misleading. This is because the settling times for these cases are measured at the points where the response enters the band between 0.95 and 1.00, since the maximum overshoots are less than 5%.

The maximum overshoot of the system can still be reduced further than those shown in Table 9-5 by using smaller values of K_P than 0.08. However, the rise time and settling time will be excessive. For example, with $K_P = 0.04$ and $K_I = 0.04$, the maximum overshoot is 1.1%, but the rise time is increased to 0.0182 seconds, and the settling time is 0.024 seconds.

For the system considered, improvement on the maximum overshoot slows down for K_I less than 0.08, unless K_P is also reduced. As mentioned earlier, the value of the capacitor C_2 is inversely proportional to K_I. Thus, for practical reasons, there is a lower limit on the value of K_I.

Fig. 9-20 shows the unit-step responses of the attitude-control system with PI control, with $K_P = 0.08$ and several values of K_P. The unit-step response of the same system with the PD controller designed in Example 9-2-1, with $K_P = 1$ and $K_D = 0.00177$, is also plotted in the same figure as a comparison.

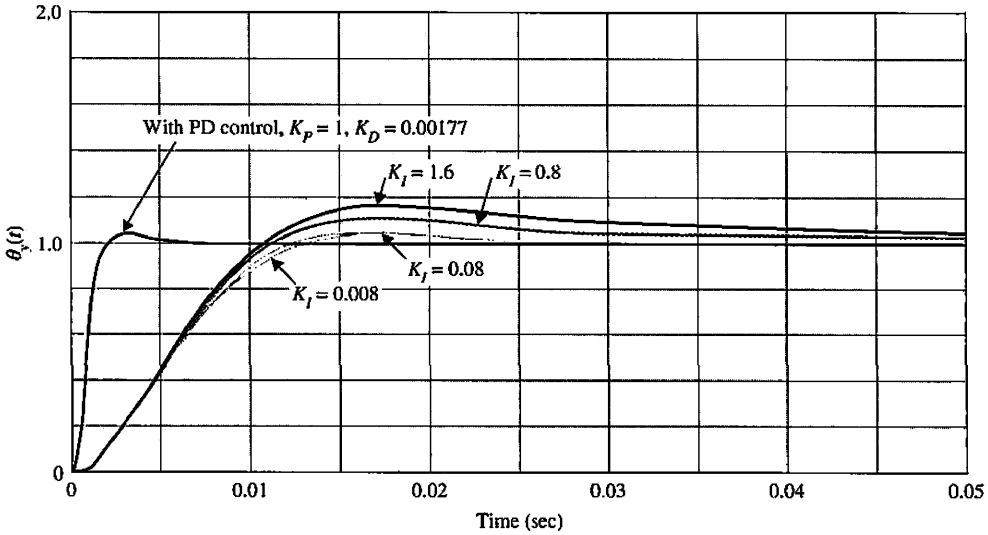


Figure 9-20 Unit-step responses of the system in Example 9-3-1 with PI control. Also, unit-step response of the system in Example 9-2-1 with PD controller.

Frequency-Domain Design

The forward-path transfer function of the uncompensated system is obtained by setting $K_P = 1$ and $K_I = 0$ in the $G(s)$ in Eq. (9-34), and the Bode plot is shown in Fig. 9-21. The phase margin is 22.68°, and the gain-crossover frequency is 868 rad/sec.

Toolbox 9-3-2

Fig. 9-20 is obtained by the following sequence of MATLAB functions

```
K=10;
num =[4500*K];
den = [1 361.2 0];
tf(num,den);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on

KI = [ 1.6 0.8 ];
KP=KI/5;
K=100;

for i = 1:length(KI)
    num = [4500*K*KP(i) 4500*K*KI(i)/KP(i)];
    den =[1 361.2 0 0];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    step(numCL,denCL)
    hold on
end
axis([0 0.05 0 2])
```

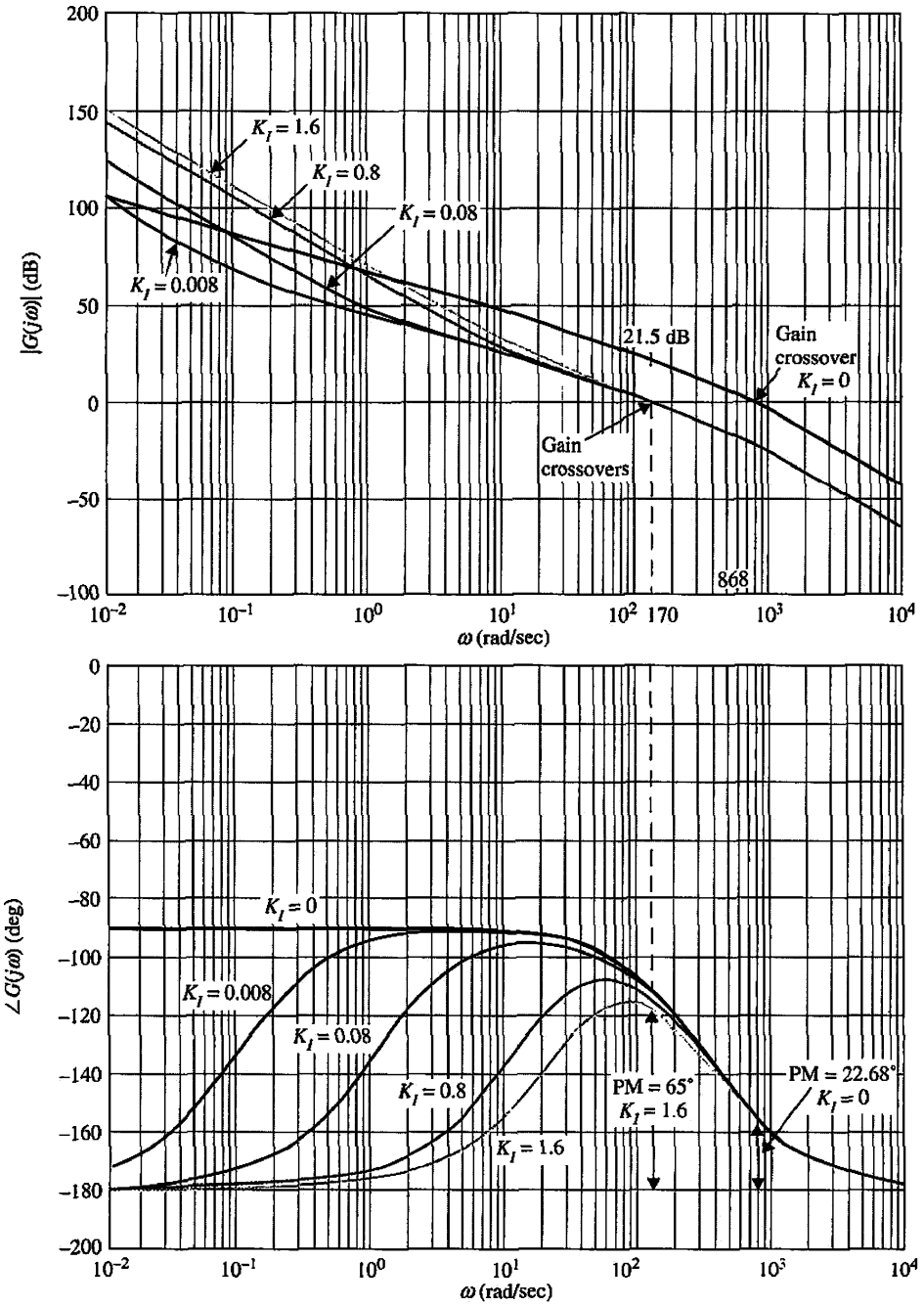


Figure 9-21 Bode plots of the control system in Example 9-3-1 with PI controller.

$$G(s) = \frac{815.265K_P(s + K_I/K_P)}{s^2(s + 361.2)}.$$

Toolbox 9-3-3

Bode plots of the control system in Example 9-3-1. Fig. 9-21 is obtained by the following sequence of MATLAB functions

```
KI = [ 0 1.6 0.8 0.08 0.008];
KP=0.08;
K=1;


for i = 1:length(KI)
    num = [815265*KP 815265*KI(i)];
    den =[1 361.2 0 0];
    bode(num,den)
    hold on
end
grid
```

Let us specify that the required phase margin should be at least 65°, and this is to be achieved with the PI controller of Eq. (9-30). Following the procedure outlined earlier in Eqs. (9-31) through (9-33) on the design of the PI controller, we conduct the following steps:

1. Look for the new gain-crossover frequency ω_g' at which the phase margin of 65° is realized. From Fig. 9-21, ω_g' is found to be 170 rad/sec. The magnitude of $G(j\omega)$ at this frequency is 21.5 dB. Thus, the PI controller should provide an attenuation of −21.5 dB at $\omega_g' = 170$ rad/sec. Substituting $\left|G\left(j\omega_g'\right)\right| = 21.5$ dB into Eq. (9-32), and solving for K_P, we get

$$K_P = 10^{-|G(j\omega_g')|_{\text{dB}}/20} = 10^{-21.5/20} = 0.084 \tag{9-41}$$

   Notice that, in the time-domain design conducted earlier, K_P was selected to be 0.08 so that the relative damping ratio of the complex characteristic equation roots will be approximately 0.707. (Perhaps we have cheated a little by selecting the desired phase margin to be 65°. This could not be just a coincidence. Can you believe that we have had no prior knowledge that, in this case, $\zeta = 0.707$ corresponds to PM = 65°?)

2. Let us choose $K_P = 0.08$, so that we can compare the design results of the frequency domain with those of the time-domain design obtained earlier. Eq. (9-33) gives the general guideline of finding K_I once K_P is determined. Thus,

$$K_I = \frac{\omega_g' K_P}{10} = \frac{170 \times 0.08}{10} = 1.36 \tag{9-42}$$

As pointed out earlier, the value of K_I is not rigid, as long as the ratio K_I/K_P is sufficiently smaller than the magnitude of the pole of $G(s)$ at −361.2. As it turns out, the value of K_I given by Eq. (9-42) is not sufficiently small for this system.

The Bode plots of the forward-path transfer function with $K_P = 0.08$ and $K_I = 0$, 0.008, 0.08, 0.8, and 1.6 are shown in Fig. 9-21. Table 9-6 shows the frequency-domain

TABLE 9-6 Frequency-Domain Performance Data of the System in Example 9-3-1 with PI Controller

K_I/K_P	K_I	K_P	GM (dB)	PM (deg)	M_r	BW (rad/sec)	Gain CO (rad/sec)	Phase CO (rad/sec)
0	0	1.00	∞	22.6	2.55	1390.87	868	∞
20	1.6	0.08	∞	58.45	1.12	268.92	165.73	∞
10	0.8	0.08	∞	61.98	1.06	262.38	164.96	∞
5	0.4	0.08	∞	63.75	1.03	258.95	164.77	∞
1	0.08	0.08	∞	65.15	1.01	256.13	164.71	∞
0.1	0.008	0.08	∞	65.47	1.00	255.49	164.70	∞

properties of the uncompensated system and the compensated system with various values of K_I. Notice that, for values of K_I/K_P that are sufficiently small, the phase margin, M_r, BW, and gain-crossover frequency all vary little.

It should be noted that the phase margin of the system can be improved further by reducing the value of K_P below 0.08. However, the bandwidth of the system will be further reduced. For example, for $K_P = 0.04$ and $K_I = 0.04$, the phase margin is increased to 75.7°, and $M_r = 1.01$, but BW is reduced to 117.3 rad/sec.

EXAMPLE 9-3-2 Now let us consider using the PI control for the third-order attitude control system described by Eq. (9-19). First, the time-domain design is carried out as follows. You may use **ACSYS** to solve this problem.

Time-Domain Design

Let the time-domain specifications bc as follows:

Steady-state error due to the parabolic input $t^2u_s(t)/2 \leq 0.2$

Maximum overshoot $\leq 5\%$

Rise time $t_r \leq 0.01$ sec

Settling time $t_s \leq 0.02$ sec

These are identical to the specifications given for the second-order system in Example 9-3-1.

Applying the PI controller of Eq. (9-24), the forward-path transfer function of the system becomes

$$G(s) = G_c(s)G_p(s) = \frac{1.5 \times 10^9 KK_P(s + K_I/K_P)}{s^2(s^2 + 3408.3s + 1,204,000)} = \frac{1.5 \times 10^9 KK_P(s + K_I/K_P)}{s^2(s + 400.26)(s + 3008)} \tag{9-43}$$

We can show that the steady-state error of the system due to the parabolic input is again given by Eq. (9-36), and, arbitrarily setting $K = 181.17$, the minimum value of K_I is 0.002215.

The characteristic equation of the closed-loop system with $K = 181.17$ is

$$s^4 + 3408.3s^3 + 1,204,000s^2 + 2.718 \times 10^9 K_P s + 2.718 \times 10^9 K_I = 0 \tag{9-44}$$

The Routh's tabulation of the last equation is performed as follows:

s^4	1	$1,204,000$	$2.718 \times 10^9 K_I$
s^3	3408.3	$2.718 \times 10^9 K_P$	0
s^2	$1,204,000 - 797465K_P$	$2.718 \times 10^9 K_I$	0
s^1	$\dfrac{1,204,000K_P - 797465K_P^2 - 3408.3K_I}{1,204,000 - 797465K_P}$	0	
s^0	$2.718 \times 10^9 K_I$	0	

The stability requirements are

$$\begin{aligned} K_I &> 0 \\ K_P &< 1.5098 \\ K_I &< 353.255K_P - 233.98K_P^2 \end{aligned} \tag{9-45}$$

The design of the PI controller calls for the selection of a small value for K_I/K_P, relative to the nearest pole of $G(s)$ to the origin, which is at -400.26. The root loci of Eq. (9-44) are plotted using the pole–zero configuration of Eq. (9-43). Fig. 9-22(a) shows the root loci as K_P

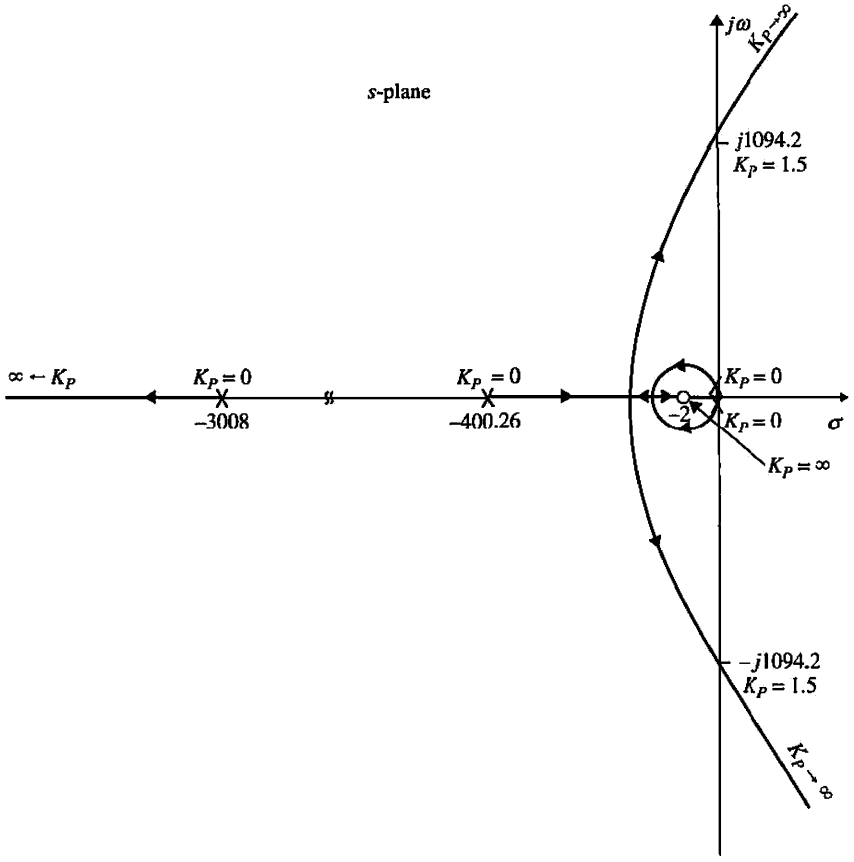
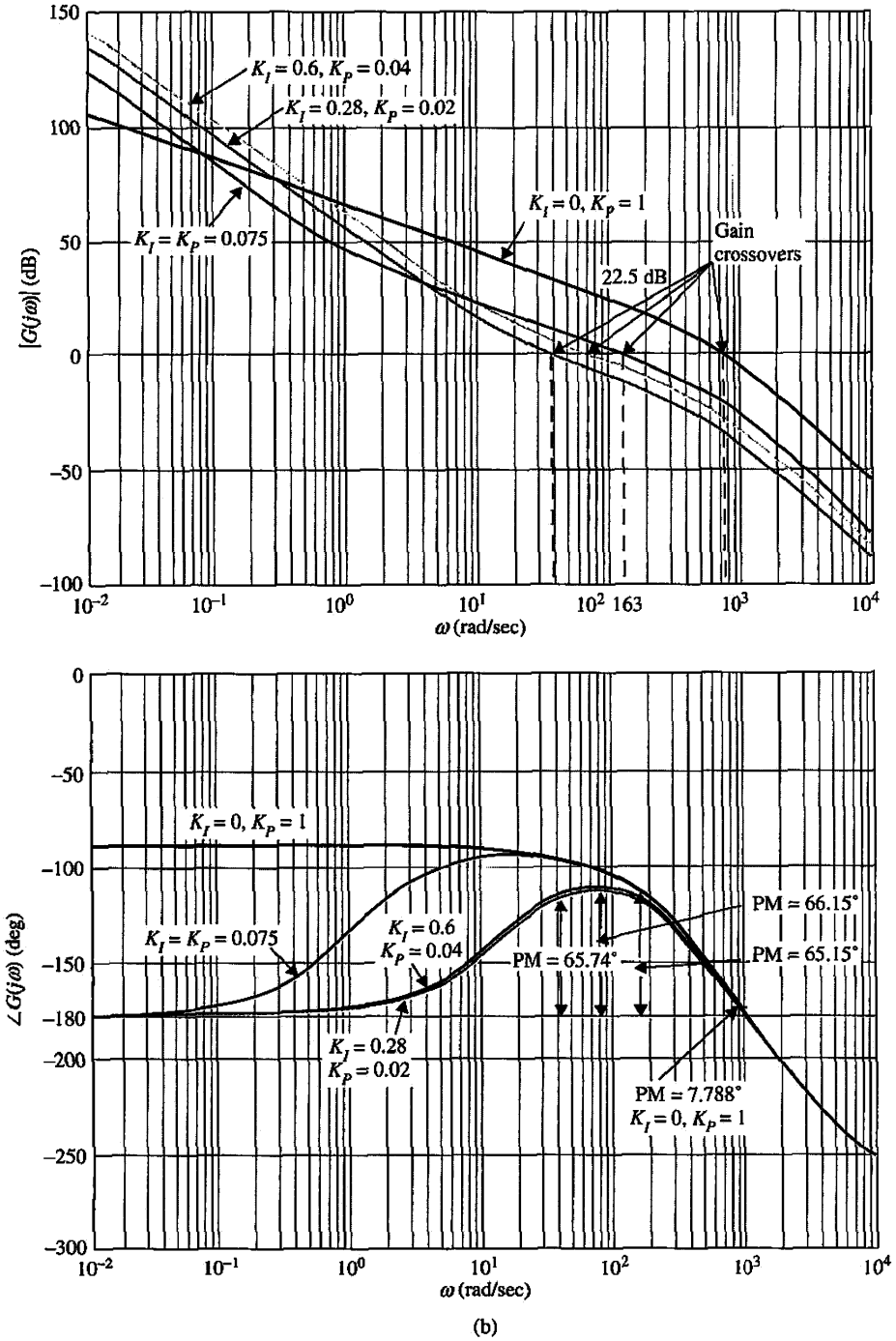


Figure 9-22 (a) Root loci of the control system in Example 9-3-2 with PI controller $K_I/K_P = 2$, $0 \leq K_P < \infty$.



(b)

Figure 9-22 (b) Bode plots of the control system in Example 9-3-2 with PI control.

varies for $K_I/K_P = 2$. The root loci near the origin due to the pole and zero of the PI controller again form a small loop, and the root loci at a distance away from the origin will be very similar to those of the uncompensated system, which are shown in Fig. 5-34. By selecting the value of K_P properly along the root loci, it may be possible to satisfy the performance specifications

TABLE 9-7 Attributes of the Unit-Step Responses of the System in Example 9-3-2 with PI Controller

K_I/K_P	K_I	K_P	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Roots of Characteristic Equation			
0	0	1	76.2	0.00158	0.0487	−3293.3	−57.5	$\pm j906.6$	
20	1.6	0.08	15.6	0.0077	0.0471	−3035	−22.7	−175.3	$\pm j180.3$
20	0.8	0.04	15.7	0.0134	0.0881	−3021.6	−259	−99	−28
5	0.4	0.08	6.3	0.00883	0.0202	−3035	−5.1	−184	$\pm j189.2$
2	0.08	0.04	2.1	0.02202	0.01515	−3021.7	−234.6	−149.9	−2
5	0.2	0.04	4.8	0.01796	0.0202	−3021.7	−240	−141.2	−5.3
2	0.16	0.08	5.8	0.00787	0.01818	−3035.2	−185.5	$\pm j190.8$	−2
1	0.08	0.08	5.2	0.00792	0.01616	−3035.2	−186	$\pm j191.4$	−1
2	0.15	0.075	4.9	0.0085	0.0101	−3033.5	−187.2	$\pm j178$	−1
2	0.14	0.070	4.0	0.00917	0.01212	−3031.8	−187.2	$\pm j164$	−1

given above. To minimize the rise time and settling time, we should select K_P so that the dominant roots are complex conjugate. Table 9-7 gives the performance attributes of several combinations of K_I/K_P and K_P. Notice that, although several combinations of these parameters correspond to systems that satisfy the performance specifications, the one with $K_P = 0.075$ and $K_I = 0.15$ gives the best rise and settling times among those shown.

Frequency-Domain Design

The Bode plot of Eq. (9-43) for $K = 181.17$, $K_P = 1$, and $K_I = 0$ is shown in Fig. 9-22(b). The performance data of the uncompensated system are as follows:

Gain margin = 3.578 dB

Phase margin = 7.788°

$M_r = 6.572$

BW = 1378 rad/sec

Let us require that the compensated system has a phase margin of at least 65°, and this is to be achieved with the PI controller of Eq. (9-30). Following the procedure outlined in Eqs. (9-31) through (9-33) on the design of the PI controller, we carry out the following steps.

1. Look for the new gain-crossover frequency ω_g' at which the phase margin of 65° is realized. From Fig. 9-20, ω_g' is found to be 163 rad/sec, and the magnitude of $G(j\omega)$ at this frequency is 22.5 dB. Thus, the PI controller should provide an attenuation of −22.5 dB at $\omega_g' = 163$ rad/sec. Substituting $\left|G\left(j\omega_g'\right)\right| = 22.5$ dB into Eq. (9-32), and solving for K_P, we get

$$K_P = 10^{-|G(j\omega_g')|_{dB}/20} = 10^{-22.5/20} = 0.075 \tag{9-46}$$

This is exactly the same result that was selected for the time-domain design that resulted in a system with a maximum overshoot of 4.9% when $K_I = 0.15$, or $K_I/K_P = 2$.

2. The suggested value of K_I is found from Eq. (9-33):

$$K_I = \frac{\omega_g' K_P}{10} = \frac{163 \times 0.075}{10} = 1.222 \tag{9-47}$$

Toolbox 9-3-4

Bode plots of the control system in Example 9-3-2. Fig. 9-22(b) is obtained by the following sequence of MATLAB functions

```
KI = [ 0.6 0.28 0.075 0];
KP = [0.04 0.02 0.075 1];
K=1;


for i = 1:length(KI)
    num = [1.5e9*KP(i) 1.5e9*KI(i)];
    den =[1 3408.3 1204000 0 0];
    bode(num,den)
    hold on
end
grid
axis([0.01 10000 -270 0]);
```

Thus, $K_I/K_P = 16.3$. However, the phase margin of the system with these design parameters is only 59.52.

To realize the desired PM of 65°, we can reduce the value of K_P or K_I. Table 9-8 gives the results of several designs with various combinations of K_P and K_I. Notice that the last three designs in the table all satisfy the PM requirements. However, the design ramifications show the following:

Reducing K_P would reduce BW and increase M_r.

Reducing K_I would increase the capacitor value in the implementing circuit.

In fact, only the $K_I = K_P = 0.075$ case gives the best all-around performance in both the frequency domain and the time domain. In attempting to increase K_I, the maximum overshoot becomes excessive. This is one example that shows the inadequacy of specifying phase margin only. The purpose of this example is to bring out the properties of the PI controller and the important considerations in its design. No details are explored further.

Fig. 9-23 shows the unit-step responses of the uncompensated system and several systems with PI control.

TABLE 9-8 Performance Summary of the System in Example 9-3-2 with PI Controller

K_I/K_P	K_I	K_P	GM (dB)	PM (deg)	M_r	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0	0	1	3.578	7.788	6.572	1378	77.2	0.0015	0.0490
16.3	1.222	0.075	25.67	59.52	1.098	264.4	13.1	0.0086	0.0478
1	0.075	0.075	26.06	65.15	1.006	253.4	4.3	0.0085	0.0116
15	0.600	0.040	31.16	66.15	1.133	134.6	12.4	0.0142	0.0970
14	0.280	0.020	37.20	65.74	1.209	66.34	17.4	0.0268	0.1616

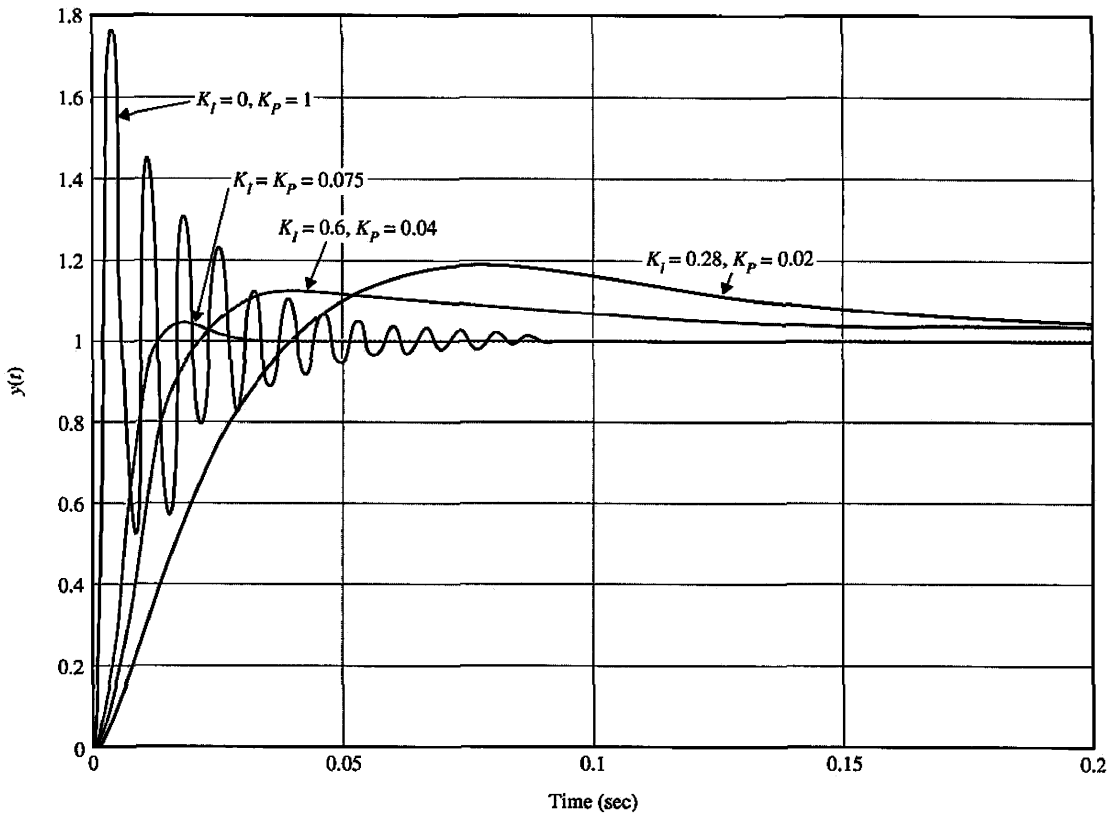


Figure 9-23 Unit-step response of system with PI controller in Example 9-3-2.

▶ 9-4 DESIGN WITH THE PID CONTROLLER

From the preceding discussions, we see that the PD controller could add damping to a system, but the steady-state response is not affected. The PI controller could improve the relative stability and improve the steady-state error at the same time, but the rise time is increased. This leads to the motivation of using a PID controller so that the best features of each of the PI and PD controllers are utilized. We can outline the following procedure for the design of the PID controller.

1. Consider that the PID controller consists of a PI portion connected in cascade with a PD portion. The transfer function of the PID controller is written as

$$G_c(s) = K_P + K_D s + \frac{K_I}{s} = (1 + K_{D1}s)\left(K_{P2} + \frac{K_{I2}}{s}\right) \tag{9-48}$$

The proportional constant of the PD portion is set to unity, since we need only three parameters in the PID controller. Equating both sides of Eq. (9-48), we have

$$K_P = K_{P2} + K_{D1}K_{I2} \tag{9-49}$$

$$K_D = K_{D1}K_{P2} \tag{9-50}$$

$$K_I = K_{I2} \tag{9-51}$$

2. Consider that the PD portion only is in effect. Select the value of K_{D1} so that a portion of the desired relative stability is achieved. In the time domain, this relative stability may be measured by the maximum overshoot, and in the frequency domain it is the phase margin.
3. Select the parameters K_{I2} and K_{P2} so that the total requirement on relative stability is satisfied.

Toolbox 9-4-1

Fig. 9-23 is obtained by the following sequence of MATLAB functions

```
KI = [0 0.6 0.28 0.075];
KP = [1 0.04 0.02 0.075];
K=1;
t = 0:0.0001:0.2;
for i = 1:length(KI)
    num = [1.5e9*KP(i) 1.5e9*KI(i)];
    den =[1 3408.3 1204000 0 0];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    step(numCL,denCL,t)
    hold on
end
grid
axis([0 0.2 0 1.8])
```

As an alternative, the PI portion of the controller can be designed first for a portion of the requirement on relative stability, and, finally, the PD portion is designed.

The following example illustrates how the PID controller is designed in the time domain and the frequency domain.

EXAMPLE 9-4-1 Consider the third-order attitude control system represented by the forward-path transfer function given in Eq. (9-19). With $K = 181.17$, the transfer function is

$$G_p(s) = \frac{2.718 \times 10^9}{s(s + 400.26)(s + 3008)} \tag{9-52}$$

You may use **ACSYS** to solve this problem; see Section 9-19. ◂

Time-Domain Design

Let the time-domain performance specifications be as follows:

Steady-state error due to a ramp input $t^2 u_s(t)/2 \leq 0.2$

Maximum overshoot $\leq 5\%$

Rise time $t_r \leq 0.005$ sec

Settling time $t_s \leq 0.005$ sec

We realize from the previous examples that these requirements cannot be fulfilled by either the PI or PD control acting alone. Let us apply the PD control with the transfer function $(1 + K_{D1}s)$. The forward-path transfer function becomes

$$G(s) = \frac{2.718 \times 10^9(1 + K_{D1}s)}{s(s + 400.26)(s + 3008)} \tag{9-53}$$

TABLE 9-9 Time-Domain Performance Characteristics of Third-Order Attitude Control System with PID Controller Designed in Example 9-4-1

K_{P2}	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Roots of Characteristic Equation			
1.0	11.1	0.00088	0.0025	−15.1	−533.2	$-1430 \pm j$	1717.5
0.9	10.8	0.00111	0.00202	−15.1	−538.7	$-1427 \pm j$	1571.8
0.8	9.3	0.00127	0.00303	−15.1	−546.5	$-1423 \pm j$	1385.6
0.7	8.2	0.00130	0.00303	−15.1	−558.4	$-1417 \pm j$	1168.7
0.6	6.9	0.00155	0.00303	−15.2	−579.3	$-1406 + j$	897.1
0.5	5.6	0.00172	0.00404	−15.2	−629	$-1382 \pm j$	470.9
0.4	5.1	0.00214	0.00505	−15.3	−1993	$-700 \pm j$	215.4
0.3	4.8	0.00271	0.00303	−15.3	−2355	$-519 \pm j$	263.1
0.2	4.5	0.00400	0.00404	−15.5	−2613	$-390 \pm j$	221.3
0.1	5.6	0.00747	0.00747	−16.1	−284	$-284 \pm j$	94.2
0.08	6.5	0.00895	0.04545	−16.5	−286.3	$-266 \pm j$	4.1

Table 9-3 shows that the best PD controller that can be obtained from the maximum overshoot standpoint is with $K_{D1} = 0.002$, and the maximum overshoot is 11.37%. The rise time and settling time are well within the required values. Next, we add the PI controller, and the forward-path transfer function becomes

$$G(s) = \frac{5.436 \times 10^6 K_{P2}(s + 500)(s + K_{I2}/K_{P2})}{s^2(s + 400.26)(s + 3008)} \tag{9-54}$$

Following the guideline of choosing a relatively small value for K_{I2}/K_{P2}, we let $K_{I2}/K_{P2} = 15$. Eq. (9-54) becomes

$$G(s) = \frac{5.436 \times 10^6 K_{P2}(s + 500)(s + 15)}{s^2(s + 400.26)(s + 3008)} \tag{9-55}$$

Table 9-9 gives the time-domain performance characteristics along with the roots of the characteristic equation for various values of K_{P2}. Apparently, the optimal value of K_{P2} is in the neighborhood of between 0.2 and 0.4.

Selecting $K_{P2} = 0.3$, and with $K_{D1} = 0.002$ and $K_{I2} = 15K_{P2} = 4.5$, the following results are obtained for the parameters of the PID controller using Eqs. (9-49) through (9-51):

$$\begin{aligned} K_I &= K_{I2} = 4.5 \\ K_P &= K_{P2} + K_{D1}K_{I2} = 0.3 + 0.002 \times 4.5 = 0.309 \\ K_D &= K_{D1}K_{P2} = 0.002 \times 0.3 = 0.0006 \end{aligned} \tag{9-56}$$

Notice that the PID design resulted in a smaller K_D and a larger K_I, which correspond to smaller capacitors in the implementing circuit.

Fig. 9-24 shows the unit-step responses of the system with the PID controller, as well as those with PD and PI controls designed in Examples 9-2-2 and 9-3-2, respectively. Notice that the PID control, when designed properly, captures the advantages of both the PD and the PI controls.

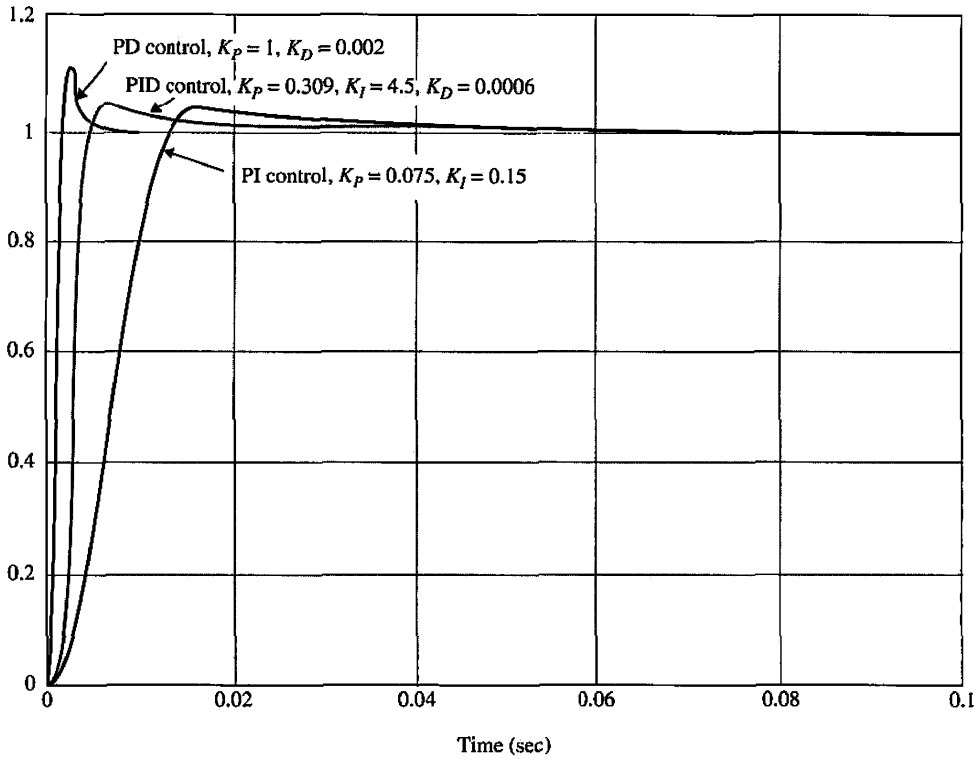


Figure 9-24 Step responses of the system in Example 9-4-1 with PD, PI, and PID controllers.

Frequency-Domain Design

The PD control of the third-order attitude control systems was already carried out in Example 9-2-2, and the results were tabulated in Table 9-3. When $K_P = 1$ and $K_D = 0.002$, the maximum overshoot is 11.37%, but this is the best that the PD control could offer. Using this PD controller, the forward-path transfer function of the system is

$$G(s) = \frac{2.718 \times 10^9(1 + 0.002s)}{s(s + 400.26)(s + 3008)} \tag{9-57}$$

and its Bode plot is shown in Fig. 9-25. Let us estimate that the following set of frequency-domain criteria corresponds to the time-domain specifications given in this problem.

Phase margin $\geq 70°$

$M_r \leq 1.1$

BW ≥ 1.000 rad/sec

From the Bode diagram in Fig. 9-25, we see that, to achieve a phase margin of 70°, the new phase-crossover frequency should be $\omega_g' = 811$ rad/sec, at which the magnitude of $G(j\omega)$ is 7 dB. Thus, using Eq. (9-32), the value of K_{P2} is calculated to be

$$K_{P2} = 10^{-7/20} = 0.45 \tag{9-58}$$

Notice that the desirable range of K_{P2} found from the time-domain design with $K_{I2}/K_{P2} = 15$ is from 0.2 to 0.4. The result given in Eq. (9-58) is slightly out of the range. Table 9-10 shows the frequency-domain performance results with $K_D = 0.002$, $K_{I2}/K_{P2} = 15$, and

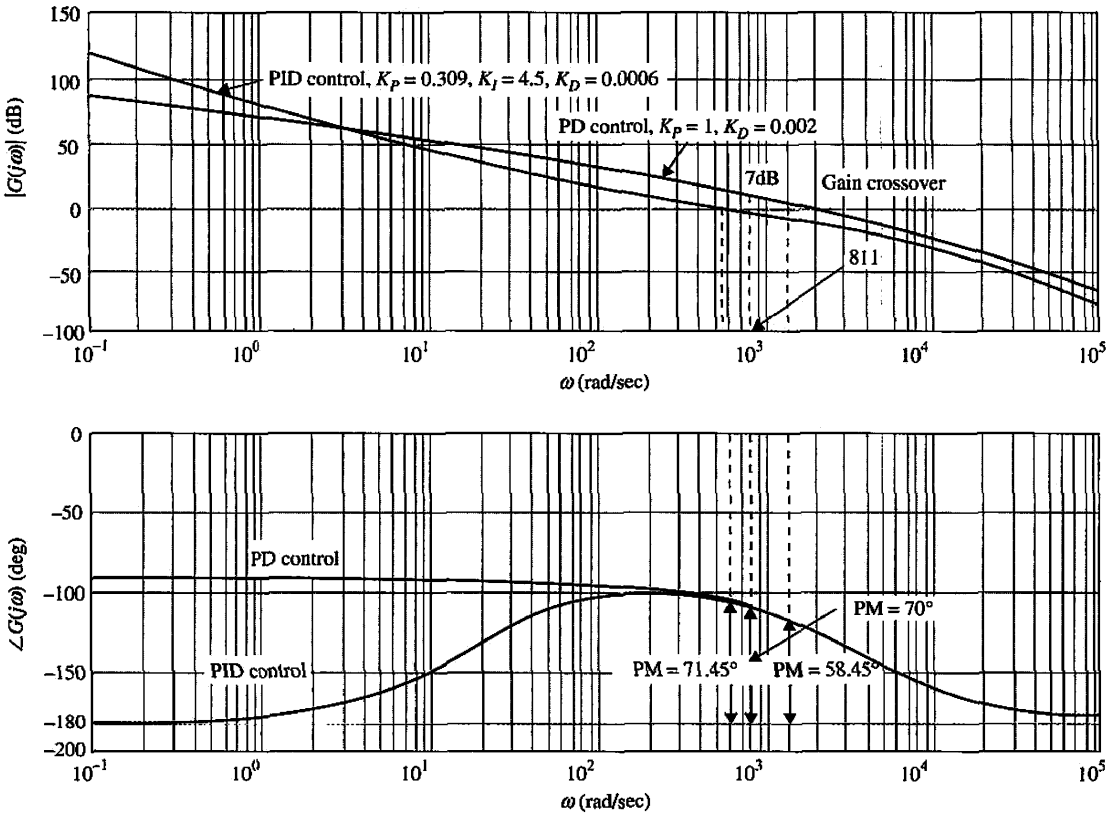


Figure 9-25 Bode plot of the system in Example 9-4-1 with PD and PID controllers.

TABLE 9-10 Frequency-Domain Performance of System in Example 9-4-1 with PID Controller

K_{P2}	K_{I2}	GM (dB)	PM (deg)	M_r	BW (rad/sec)	t_r (sec)	t_s (sec)	Maximum Overshoot (%)
1.00	0	∞	58.45	1.07	2607	0.0008	0.00255	11.37
0.45	6.75	∞	68.5	1.03	1180	0.0019	0.0040	5.6
0.40	6.00	∞	69.3	1.027	1061	0.0021	0.0050	5.0
0.30	4.50	∞	71.45	1.024	1024	0.0027	0.00303	4.8
0.20	3.00	∞	73.88	1.031	528.8	0.0040	0.00404	4.5
0.10	1.5	∞	76.91	1.054	269.5	0.0076	0.0303	5.6
0.08	1.2	∞	77.44	1.065	216.9	0.0092	0.00469	6.5

several values of K_{P2} starting with 0.45. It is interesting to note that, as K_{P2} continues to decrease, the phase margin increases monotonically, but below $K_{P2} = 0.2$, the maximum overshoot actually increases. In this case, the phase margin results are misleading, but the resonant peak M_r is a more accurate indication of this.

▶ 9-5 DESIGN WITH PHASE-LEAD CONTROLLER

The PID controller and its components in the form of PD and PI controls represent simple forms of controllers that utilize derivative and integration operations in the compensation of control systems. In general, we can regard the design of controllers of control systems as

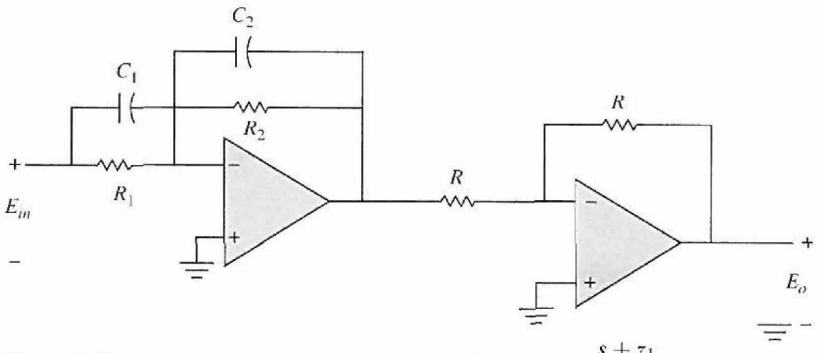


Figure 9-26 Op-amp circuit implementation of $G(s) = K_c \dfrac{s + z_1}{s + P_1}$.

a filter design problem; then there are a large number of possible schemes. From the filtering standpoint, the PD controller is a high-pass filter, the PI controller is a low-pass filter, and the PID controller is a band-pass or band-attenuate filter, depending on the values of the controller parameters. The high-pass filter is often referred to as a **phase-lead controller**, because positive phase is introduced to the system over some frequency range. The low-pass filter is also known as a **phase-lag controller**, because the corresponding phase introduced is negative. These ideas related to filtering and phase shifts are useful if designs are carried out in the frequency domain.

The transfer function of a simple lead or lag controller is expressed as

$$G_c(s) = K_c \frac{s + z_1}{s + p_1} \tag{9-59}$$

where the controller is high-pass or phase-lead if $p_1 > z_1$, and low-pass or phase-lag if $p_1 < z_1$.

The op-amp circuit implementation of Eq. (9-59) is given in Table 4-4(g) of Chapter 4 and is repeated in Fig. 9-26 with an inverting amplifier. The transfer function of the circuit is

$$G_c(s) = \frac{E_o(s)}{E_{in}(s)} = \frac{C_1}{C_2} \frac{s + \dfrac{1}{R_1 C_1}}{s + \dfrac{1}{R_2 C_2}} \tag{9-60}$$

Comparing the last two equations, we have

$$\begin{aligned} K_c &= C_1/C_2 \\ z_1 &= 1/R_1 C_1 \\ p_1 &= 1/R_2 C_2 \end{aligned} \tag{9-61}$$

We can reduce the number of design parameters from four to three by setting $C = C_1 = C_2$. Then Eq. (9-60) is written as

$$\begin{aligned} G_c(s) &= \frac{R_2}{R_1}\left(\frac{1 + R_1 Cs}{1 + R_2 Cs}\right) \\ &= \frac{1}{a}\left(\frac{1 + aTs}{1 + Ts}\right) \end{aligned} \tag{9-62}$$

where

$$a = \frac{R_1}{R_2} \tag{9-63}$$

$$T = R_2 C \tag{9-64}$$

9-5-1 Time-Domain Interpretation and Design of Phase-Lead Control

In this section, we shall first consider that Eqs. (9-60) and (9-62) represent a phase-lead controller ($z_1 < p_1$ or $a > 1$). In order that the phase-lead controller will not degrade the steady-state error, the factor a in Eq. (9-62) should be absorbed by the forward-path gain K. Then, for design purposes, $G_c(s)$ can be written as

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad (a > 1) \tag{9-65}$$

The pole–zero configuration of Eq. (9-65) is shown in Fig. 9-27. Based on the discussions given in Chapter 7 on the effects of adding a pole–zero pair (with the zero closer to the origin) to the forward-path transfer function, the phase-lead controller can improve the stability of the closed-loop system if its parameters are chosen properly. The design of phase-lead control is essentially that of placing the pole and zero of $G_c(s)$ so that the design specifications are satisfied. The root-contour method can be used to indicate the proper ranges of the parameters. The **ACSYS** MATLAB tool can be used to speed up the cut-and-try procedure considerably. The following guidelines can be made with regard to the selection of the parameters a and T.

1. Moving the zero $-1/aT$ toward the origin should improve rise time and settling time. If the zero is moved too close to the origin, the maximum overshoot may again increase, because $-1/aT$ also appears as a zero of the closed-loop transfer function.
2. Moving the pole at $-1/T$ farther away from the zero and the origin should reduce the maximum overshoot, but if the value of T is too small, rise time and settling time will again increase.

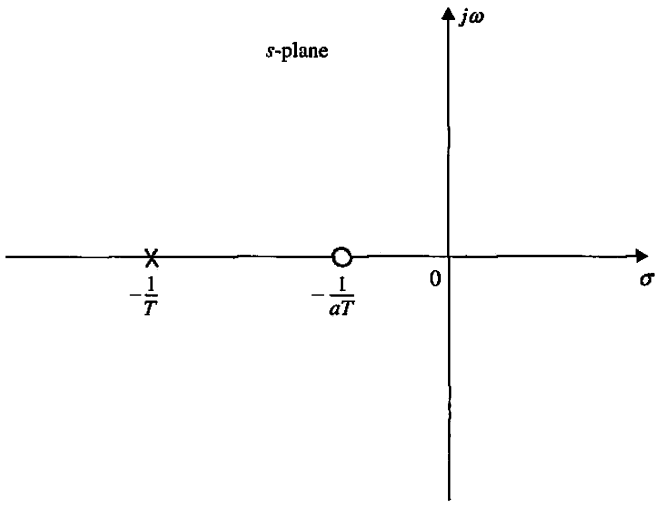


Figure 9-27 Pole–zero configuration of the phase-lead controller.

We can make the following general statements with respect to the effects of phase-lead control on the time-domain performance of a control system:

1. When used properly, it can increase damping of the system.
2. It improves rise time and settling time.
3. In the form of Eq. (9-65), phase-lead control does not affect the steady-state error, because $G_c(0) = 1$.

9-5-2 Frequency-Domain Interpretation and Design of Phase-Lead Control

The Bode plot of the phase-lead controller of Eq. (9-65) is shown in Fig. 9-28. The two corner frequencies are at $\omega = 1/aT$ and $\omega = 1/T$. The maximum value of the phase, ϕ_m, and the frequency at which it occurs, ω_m, are derived as follows. Because ω_m is the geometric mean of the two corner frequencies, we write

$$\log_{10}\omega_m = \frac{1}{2}\left(\log_{10}\frac{1}{aT} + \log_{10}\frac{1}{T}\right) \tag{9-66}$$

Thus,

$$\omega_m = \frac{1}{\sqrt{aT}} \tag{9-67}$$

To determine the maximum phase ϕ_m, the phase of $G_c(j\omega)$ is written

$$\angle G_c(j\omega) = \phi(j\omega) = \tan^{-1}\omega aT - \tan^{-1}\omega T \tag{9-68}$$

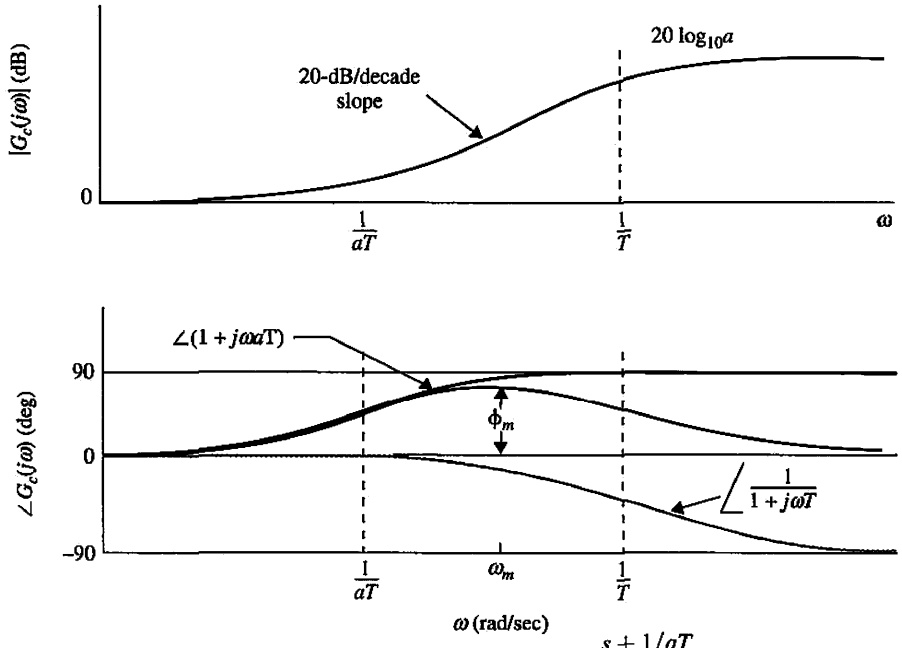


Figure 9-28 Bode plot of phase-lead controller $G_c(s) = a\dfrac{s + 1/aT}{s + 1/T}$ $a > 1$.

from which we get

$$\tan\phi(j\omega) = \frac{\omega aT - \omega T}{1 + (\omega aT)(\omega T)} \tag{9-69}$$

Substituting Eq. (9-67) into Eq. (9-69), we have

$$\tan\phi_m = \frac{a-1}{2\sqrt{a}} \tag{9-70}$$

or

$$\sin\phi_m = \frac{a-1}{a+1} \tag{9-71}$$

Thus, by knowing ϕ_m, the value of a is determined from

$$a = \frac{1+\sin\phi_m}{1-\sin\phi_m} \tag{9-72}$$

The relationship between the phase ϕ_m and a and the general properties of the Bode plot of the phase-lead controller provide an advantage of designing in the frequency domain. The difficulty is, of course, in the correlation between the time-domain and frequency-domain specifications. The general outline of phase-lead controller design in the frequency domain is given as follows. It is assumed that the design specifications simply include steady-state error and phase-margin requirements.

1. The Bode diagram of the uncompensated process $G_p(j\omega)$ is constructed with the gain constant K set according to the steady-state error requirement. The value of K has to be adjusted upward once the value of a is determined.
2. The phase margin and the gain margin of the uncompensated system are determined, and the additional amount of phase lead needed to realize the phase margin is determined. From the additional phase lead required, the desired value of ϕ_m is estimated accordingly, and the value of a is calculated from Eq. (9-72).
3. Once a is determined, it is necessary only to determine the value of T, and the design is in principle completed. This is accomplished by placing the corner frequencies of the phase-lead controller, $1/aT$ and $1/T$, such that ϕ_m is located at the new gain-crossover frequency ω_g', so the phase margin of the compensated system is benefited by ϕ_m. It is known that the high-frequency gain of the phase-lead controller is $20\log_{10}a$ dB. Thus, to have the new gain crossover at ω_m, which is the geometric mean of $1/aT$ and $1/T$, we need to place ω_m at the frequency where the magnitude of the uncompensated $G_p(j\omega)$ is $-10\log_{10}a$ dB so that adding the controller gain of $10\log_{10}a$ dB to this makes the magnitude curve go through 0 dB at ω_m.
4. The Bode diagram of the forward-path transfer function of the compensated system is investigated to check that all performance specifications are met; if not, a new value of ϕ_m must be chosen and the steps repeated.
5. If the design specifications are all satisfied, the transfer function of the phase-lead controller is established from the values of a and T.

If the design specifications also include M_r and/or BW, then these must be checked using either the Nichols chart or the output data from a computer program.

We use the following example to illustrate the design of the phase-lead controller in the time domain and frequency domain.

EXAMPLE 9-5-1 The block diagram of the sun-seeker control system described in Section 4-11 is again shown in Fig. 9-29. The system may be mounted on a space vehicle so that it will track the sun with high accuracy. The variable θ_r represents the reference angle of the solar ray, and θ_o denotes the vehicle axis. The objective of the sun-seeker system is to maintain the error α between θ_r and θ_o near zero. The parameters of the system are as follows:

$R_F = 10,000\ \Omega$	$K_b = 0.0125$ V/rad/sec
$K_i = 0.0125$ N-m/A	$R_a = 6.25\ \Omega$
$J = 10^{-6}$ kg-m^2	$K_s = 0.1$ A/rad
$K =$ to be determined	$B = 0$
$n = 800$	

The forward-path transfer function of the uncompensated system is

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{K_s R_F K K_i / n}{R_a J s^2 + K_i K_b s} \tag{9-73}$$

where $\Theta_o(s)$ and A(s) are the Laplace transforms of $\theta_o(t)$ and $\alpha(t)$, respectively.

Substituting the numerical values of the system parameters in Eq. (9-73), we get

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{2500K}{s(s + 25)} \tag{9-74}$$

You may use **ACSYS** to solve this problem after reducing the block diagram in Fig. 9-29 to a standard form. See Section 9-15.

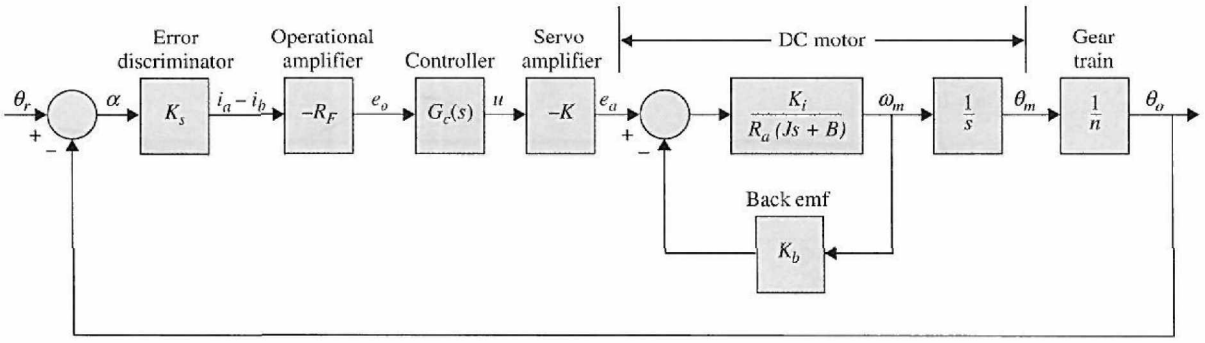


Figure 9-29 Block diagram of sun-seeker control system.

Time-Domain Design

The time-domain specifications of the system are as follows:

1. The steady-state error of $\alpha(t)$ due to a unit-ramp function input for $\theta_r(t)$ should be ≤ 0.01 rad per rad/sec of the final steady-state output velocity. In other words, the steady-state error due to a ramp input should be $\leq 1\%$.
2. The maximum overshoot of the step response should be less than 5% or as small as possible.
3. Rise time $t_r \leq 0.02$ sec.
4. Settling time $t_s \leq 0.02$ sec.

The minimum value of the amplifier gain, K, is determined initially from the steady-state requirement. Applying the final-value theorem to $\alpha(t)$, we have

$$\lim_{t\to\infty} \alpha(t) = \lim_{s\to 0} sA(s) = \lim_{s\to 0} \frac{s\Theta_r(s)}{1 + G_p(s)} \tag{9-75}$$

For a unit-ramp input, $\Theta_r(s) = 1/s^2$. By using Eq. (9-74), Eq. (9-75) leads to

$$\lim_{t\to\infty} \alpha(t) = \frac{0.01}{K} \tag{9-76}$$

Thus, for the steady-state value of $\alpha(t)$ to be ≤ 0.01, K must be ≥ 1. Let us set $K = 1$, the worst case from the steady-state error standpoint. The characteristic equation of the uncompensated system is

$$s^2 + 25s + 2500 = 0 \tag{9-77}$$

We can show that the damping ratio of the uncompensated system with $K = 1$ is only 0.25, which corresponds to a maximum overshoot of 44.4%. Fig. 9-30 shows the unit-step response of the system with $K = 1$.

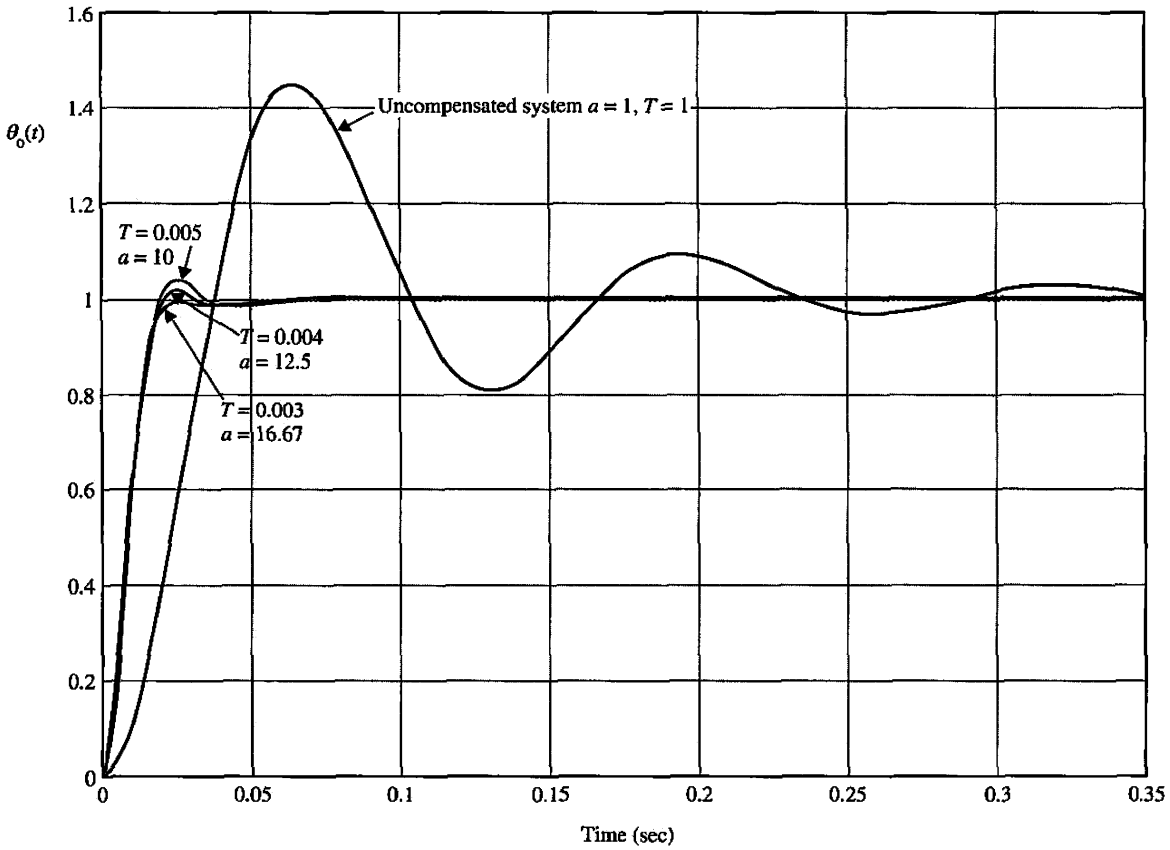


Figure 9-30 Unit-step response of sun-seeker system in Example 9-5-1.

Toolbox 9-5-1

Unit-step response for Example 9-6 in Fig. 9-30 is obtained by the following sequence of MATLAB functions

```
a = [1 10 12.5 16.67];
T = [1 0.005 0.004 0.003];

for i = 1:length(T)
    num = [2500*a(i)*T(i) 2500];
    den =[T(i) 25*T(i)+1 25 0];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    step(numCL,denCL)
    hold on
end
grid
axis([0 0.35 0 1.8])
```

A space has been reserved in the forward path of the block diagram of Fig. 9-29 for a controller with transfer function $G_c(s)$. Let us consider using the phase-lead controller of Eq. (9-62), although in the present case, a PD controller or one of the other types of phase-lead controllers may also be effective in satisfying the performance criteria given. The forward-path transfer function of the compensated system is written

$$G(s) = \frac{2500K(1 + aTs)}{as(s + 25)(1 + Ts)} \tag{9-78}$$

For the compensated system to satisfy the steady-state error requirement, K must satisfy

$$K \geq a \tag{9-79}$$

Let us set $K = a$. The characteristic equation of the system is

$$(s^2 + 25s + 2500) + Ts^2(s + 25) + 2500aTs = 0 \tag{9-80}$$

We can use the root-contour method to show the effects of varying a and T of the phase-lead controller. Let us first set $a = 0$. The characteristic equation of Eq. (9-80) becomes

$$s^2 + 25s + 2500 + Ts^2(s + 25) = 0 \tag{9-81}$$

Dividing both sides of the last equation by the terms that do not contain T, we get

$$1 + G_{eq1}(s) = 1 + \frac{Ts^2(s + 25)}{s^2 + 25s + 2500} = 0 \tag{9-82}$$

Thus, the root contours of Eq. (9-81) when T varies are determined using the pole–zero configuration of $G_{eq1}(s)$ in Eq. (9-82). These root contours are drawn as shown in Fig. 9-31. Notice that the poles of $G_{eq1}(s)$ are the roots of the characteristic equation when $a = 0$ and $T = 0$. The root contours in Fig. 9-31 clearly show that adding the factor $(1 + Ts)$ to the denominator of Eq. (9-74) alone would not improve the system performance, since the characteristic equation roots are pushed toward the right-half plane. In fact, the system becomes unstable when T is greater than 0.0133. To achieve the full effect of the phase-lead controller, we must restore the value of a in Eq. (9-80). To prepare for the root contours

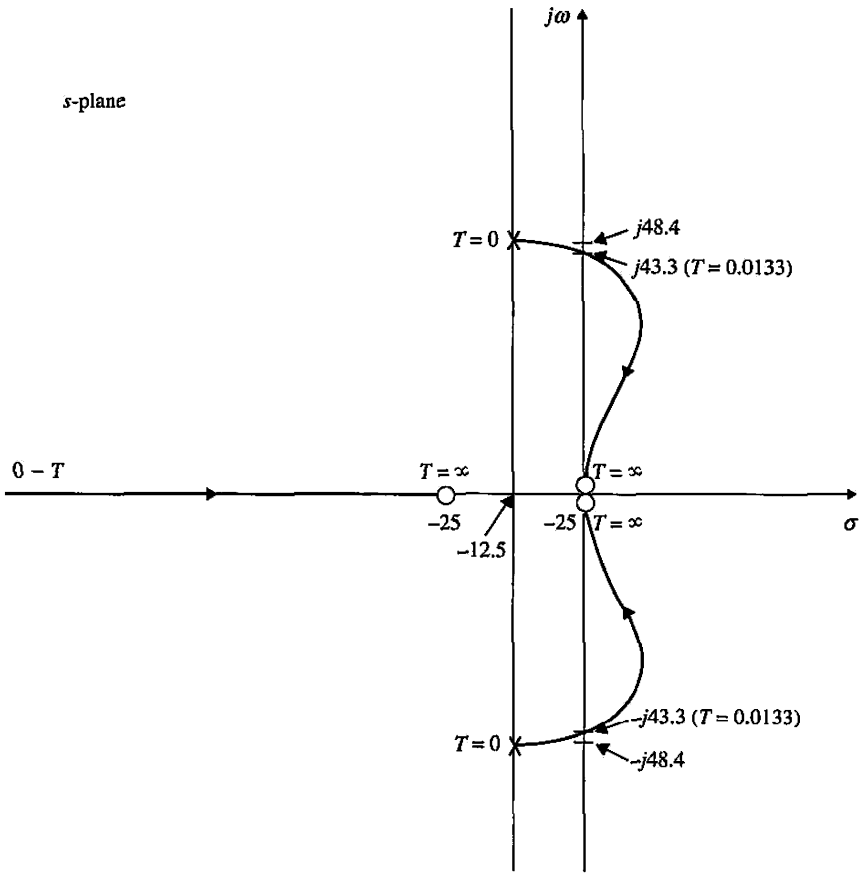


Figure 9-31 Root contours of the sun-seeker system with $a = 0$, and T varies from 0 to ∞.

with a as the variable parameter, we divide both sides of Eq. (9-80) by the terms that do not contain a, and the following equation results:

$$1 + aG_{eq2}(s) = 1 + \frac{2500aTs}{s^2 + 25s + 2500 + Ts^2(s + 25)} = 0 \tag{9-83}$$

Toolbox 9-5-2

Root contours for Fig. 9-31 are obtained by the following sequence of MATLAB functions

```
for T = 1:1:260
    num = [T 325*T 0 0];
    den = [1 25 2500];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    F = tf(num,den);
    PoleData(:,T)=pole(F);
end

plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));
```

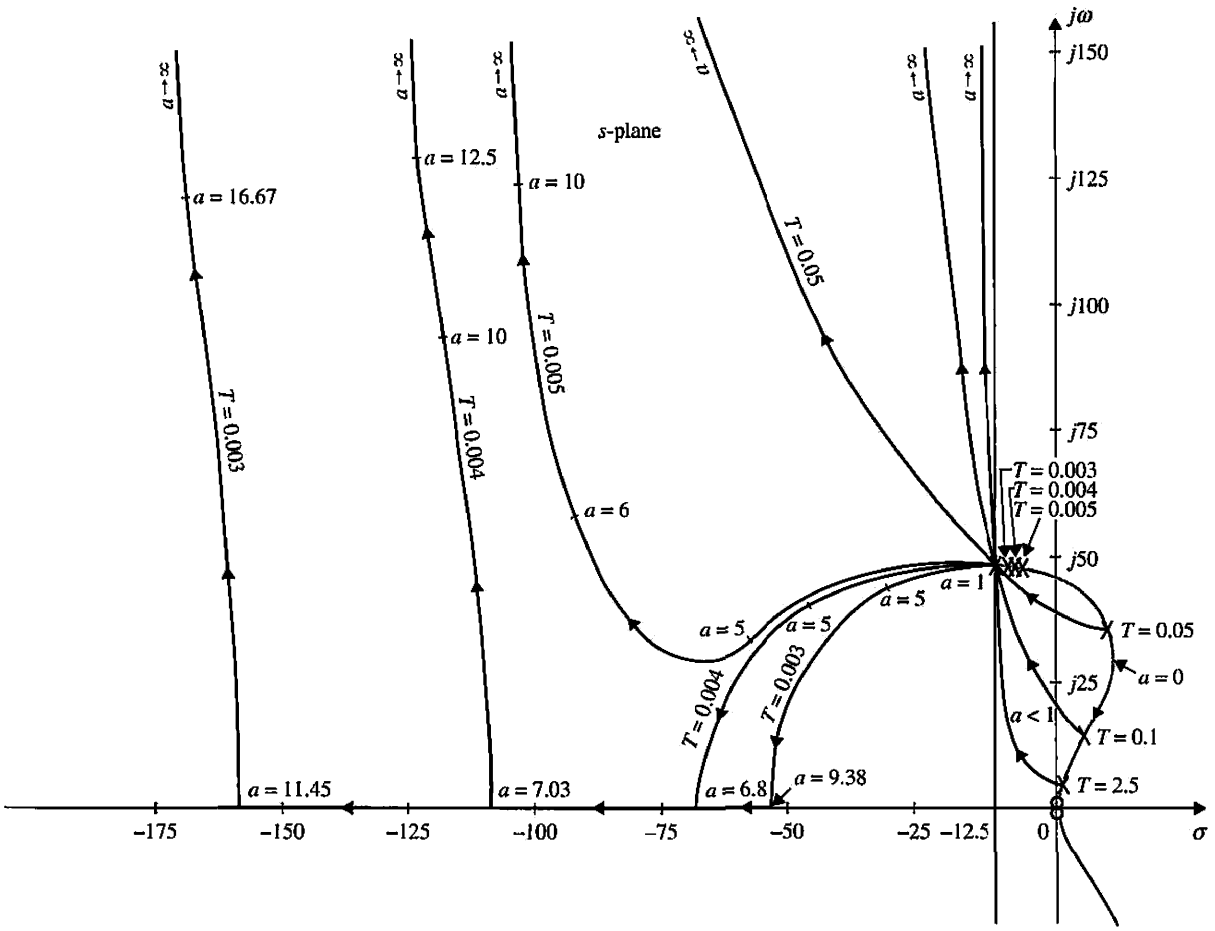


Figure 9-32 Root contours of the sun-seeker system with a phase-lead controller.

For a given T, the root contours of Eq. (9-80) when a varies are obtained based on the poles and zeros of $G_{eq2}(s)$. Notice that the poles of $G_{eq2}(s)$ are the same as the roots of Eq. (9-81). Thus, for a given T, the root contours of Eq. (9-80) when a varies must start ($a = 0$) at points on the root contours of Fig. 9-31. These root contours end ($a = \infty$) at $s = 0, \infty, \infty$, which are the zeros of $G_{eq2}(s)$. The complete root contours of Eq. (9-80) are now shown in Fig. 9-32 for several values of T, and a varies from 0 to ∞.

From the root contours of Fig. 9-32, we see that, for effective phase-lead control, the value of T should be small. For large values of T, the natural frequency of the system increases rapidly as a increases, and very little improvement is made on the damping of the system.

Let us choose $T = 0.01$ arbitrarily. Table 9-11 shows the attributes of the unit-step response when the value of aT is varied from 0.02 to 0.1. The **ACSYS** MATLAB tool was used for the calculations of the time responses. The results show that the smallest maximum overshoot is obtained when $aT = 0.05$, although the rise and settling times decrease continuously as aT increases. However, the smallest value of the maximum overshoot is 16.2%, which exceeds the design specification.

TABLE 9-11 Attributes of Unit-Step Response of System with Phase-Lead Controller in Example 9-5-1: T=0.01

aT	a	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.02	2	26.6	0.0222	0.0830
0.03	3	18.9	0.0191	0.0665
0.04	4	16.3	0.0164	0.0520
0.05	5	16.2	0.0146	0.0415
0.06	6	17.3	0.0129	0.0606
0.08	8	20.5	0.0112	0.0566
0.10	10	23.9	0.0097	0.0485

Toolbox 9-5-3

Root contours for Fig. 9-32 are obtained by the following sequence of MATLAB functions

```
T = [0.003 0.004 0.005 0.05 0.1 2.5];
for j=1:length(T)
    i=1;
    for a = 0:0.005:30
        num = [T(j) 25*T(j)+1 25+2500*a*T(j) 2500];
        den = [T(j) 25*T(j)+1 25 2500];[numCL,denCL]=cloop(num,den);
        F = tf(numCL,denCL); PoleData(:,i)=pole(F);
        i=i+1;
    end
    Count = i-1; %% for graph continuation
     for i=1:Count
       if imag(PoleData(1,i))~=0
           break;
    end         end
    count = i; %% for graph continuation
    for i=1:count
        PoleData(1,i)=PoleData(1,count);
    end

    for i=1:Count
        if imag(PoleData(2,i))~=0
            break;
    end         end
    count = i; %% for graph continuation
    for i=1:count
        PoleData(2,i)=PoleData(2,count);
    end
    PositivePos = 0;
    for i=1:Count
        if imag(PoleData(1,i)) < 0
            if PositivePos == 0
                 PositivePos = i-1;
            end
            PoleData(1,i) = PoleData(1,PositivePos);
        End         end
    PositivePos = 0;
    for i=1:Count
        if imag(PoleData(2,i)) < 0
```

```
            if PositivePos == 0
                 PositivePos = i-1;
            end
            PoleData(2,i) = PoleData(2,PositivePos);
    End             end
plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));
    hold on
end
axis([-175 0 0 150]); sgrid
```

Next, we set $aT = 0.05$ and vary T from 0.01 to 0.001, as shown in Table 9-12. Table 9-12 shows the attributes of the unit-step responses. As the value of T decreases, the maximum overshoot decreases, but the rise time and settling time increase. The cases that satisfy the design requirements are indicated in Table 9-12 for $aT = 0.05$. Fig. 9-30 shows the unit-step responses of the phase-lead-compensated system with three sets of controller parameters.

Choosing $T = 0.004$, $a = 12.5$, the transfer function of the phase-lead controller is

$$G_c(s) = a\frac{s + 1/aT}{s + 1/T} = 12.5\frac{s + 20}{s + 250} \tag{9-84}$$

The transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{31250(s + 20)}{s(s + 25)(s + 250)} \tag{9-85}$$

To find the op-amp-circuit realization of the phase-lead controller, we arbitrarily set $C = 0.1\ \mu$f, and the resistors of the circuit are found using Eqs. (9-63) and (9-64) as $R_1 = 500,000\ \Omega$ and $R_2 = 40,000\ \Omega$.

Frequency-Domain Design

Let us specify that the steady-state error requirement is the same as that given earlier. For frequency-domain design, the phase margin is to be greater than 45°. The following design steps are taken:

1. The Bode diagram of Eq. (9-74) with $K = 1$ is plotted as shown in Fig. 9-33.
2. The phase margin of the uncompensated system, read at the gain-crossover frequency, $\omega_c = 47$ rad/sec, is 28°. Because the minimum desired phase margin

TABLE 9-12 Attributes of Unit-Step Responses of System with Phase-Lead Controller in Example 9-5-1: aT=0.05

T	a	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.01	5.0	16.2	0.0146	0.0415
0.005	10.0	4.1	0.0133	0.0174
0.004	12.5	1.1	0.0135	0.0174
0.003	16.67	0	0.0141	0.0174
0.002	25.0	0	0.0154	0.0209
0.001	50.0	0	0.0179	0.0244

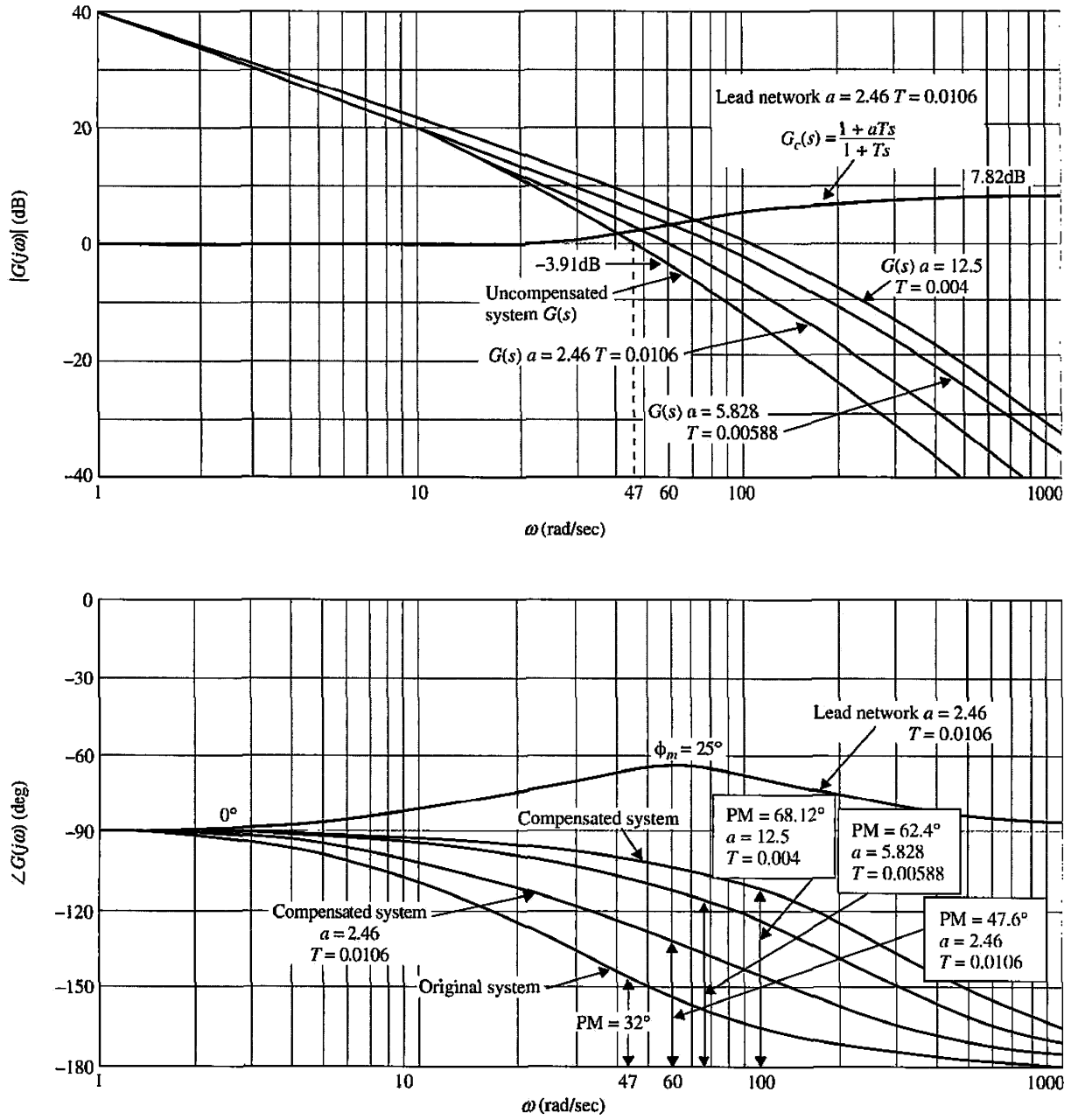


Figure 9-33 Bode diagram of the phase-lead compensation and uncompensated systems in Example 9-5-1, $G(s) = \dfrac{2500(1 + aTs)}{s(s + 25)(s + Ts)}$.

is 45°, at least 17° more phase lead should be added to the loop at the gain-crossover frequency.

3. The phase-lead controller of Eq. (9-65) must provide the additional 17° at the gain-crossover frequency of the compensated system. However, by applying the phase-lead controller, the magnitude curve of the Bode plot is also affected in such a way that the gain-crossover frequency is shifted to a higher frequency. Although it is a

simple matter to adjust the corner frequencies, $1/aT$ and $1/T$, of the controller so that the maximum phase of the controller ϕ_m falls exactly at the new gain-crossover frequency, the original phase curve at this point is no longer 28° (and could be considerably less) because the phase of most control processes decreases with the increase in frequency. In fact, if the phase of the uncompensated process decreases rapidly with increasing frequency near the gain-crossover frequency, the single-stage phase-lead controller will no longer be effective.

In view of the difficulty estimating the necessary amount of phase lead, it is essential to include some safety margin to account for the inevitable phase drop-off. Therefore, in the present case, instead of selecting a ϕ_m of a mere 17°, let ϕ_m be 25°. Using Eq. (9-72), we have

$$a = \frac{1 + \sin 25°}{1 - \sin 25°} = 2.46 \tag{9-86}$$

4. To determine the proper location of the two corner frequencies ($1/aT$ and $1/T$) of the controller, it is known from Eq. (9-67) that the maximum phase lead ϕ_m occurs at the geometric mean of the two corner frequencies. To achieve the maximum phase margin with the value of a determined, ϕ_m should occur at the new gain-crossover frequency ω_g', which is not known. The following steps are taken to ensure that ϕ_m occurs at ω_g'.

   a. The high-frequency gain of the phase-lead controller of Eq. (9-65) is

$$20 \log_{10} a = 20 \log_{10} 2.46 = 7.82 \text{ dB} \tag{9-87}$$

   b. The geometric mean ω_m of the two corner frequencies, $1/aT$ and $1/T$, should be located at the frequency at which the magnitude of the uncompensated process transfer function $G_p(j\omega)$ in dB is equal to the negative value in dB of one-half of this gain. This way, the magnitude curve of the compensated transfer function will pass through the 0-dB-axis at $\omega = \omega_m$. Thus, ω_m should be located at the frequency where

$$|G_p(j\omega)|_{\text{dB}} = -10 \log_{10} 2.46 = -3.91 \text{ dB} \tag{9-88}$$

Toolbox 9-5-4

Bode diagram for Fig. 9-33 is obtained by the following sequence of MATLAB functions

```
a = [2.46 12.5 5.828];
T = [0.0106 0.004 0.00588];

for i = 1:length(T)
    num = [2500*a(i)*T(i) 2500];
    den =[T(i) 1+25*T(i) 25 0];
    bode(num,den);
    hold on;
end
% axis([1 10000 -180 -90]);
  grid
```

Toolbox 9-5-5

Plot of G(s) for Fig. 9-34 is obtained by the following sequence of MATLAB functions

```
a = [2.46 12.5 5.828];
T = [0.0106 0.004 0.00588];
for i = 1:length(T)
    num = [2500*a(i)*T(i) 2500];
    den =[T(i) 1+25*T(i) 25 0];
    t = tf(num,den)
    nichols(t); ngrid;
    hold on;
end
```

From Fig. 9-33, this frequency is found to be $\omega_m = 60$ rad/sec. Now using Eq. (9-67), we have

$$\frac{1}{T} = \sqrt{a}\omega_m = \sqrt{2.46} \times 60 = 94.1 \text{ rad/sec} \tag{9-89}$$

Then, $1/aT = 94.1/2.46 = 38.21$ rad/sec. The transfer function of the phase-lead controller is

$$G_c(s) = a\frac{s + 1/aT}{s + 1/T} = 2.46\frac{s + 38.21}{s + 94.1} \tag{9-90}$$

The forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{6150(s + 38.21)}{s(s + 25)(s + 94.1)} \tag{9-91}$$

Fig. 9-33 shows that the phase margin of the compensated system is actually 47.6°.

In Fig. 9-34, the magnitude and phase of the original and the compensated systems are plotted on the Nichols chart for display only. These plots can be made by taking the data directly from the Bode plots of Fig. 9-33. The values of M_r, ω_r, and BW can all be determined from the Nichols chart. However, the performance data are more easily obtained with **ACSYS**.

Checking the time-domain performance of the compensated system, we have the following results:

$$\text{Maximum overshoot} = 22.3\% \quad t_r = 0.02045 \text{ sec} \quad t_s = 0.07439 \text{ sec}$$

which fall short of the time-domain specifications listed earlier. Fig. 9-33 also shows the Bode plot of the system compensated with a phase-lead controller with $a = 5.828$ and $T = 0.00588$. The phase margin is improved to 62.4°. Using Eq. (9-71), we can show that the result of $a = 12.5$ obtained in the time-domain design actually corresponds to $\phi_m = 58.41$. Adding this to the original phase of 28°, the corresponding phase margin would be 86.41°. The time-domain and frequency-domain attributes of the system with the three phase-lead controllers are summarized in Table 9-13. The results show that, with

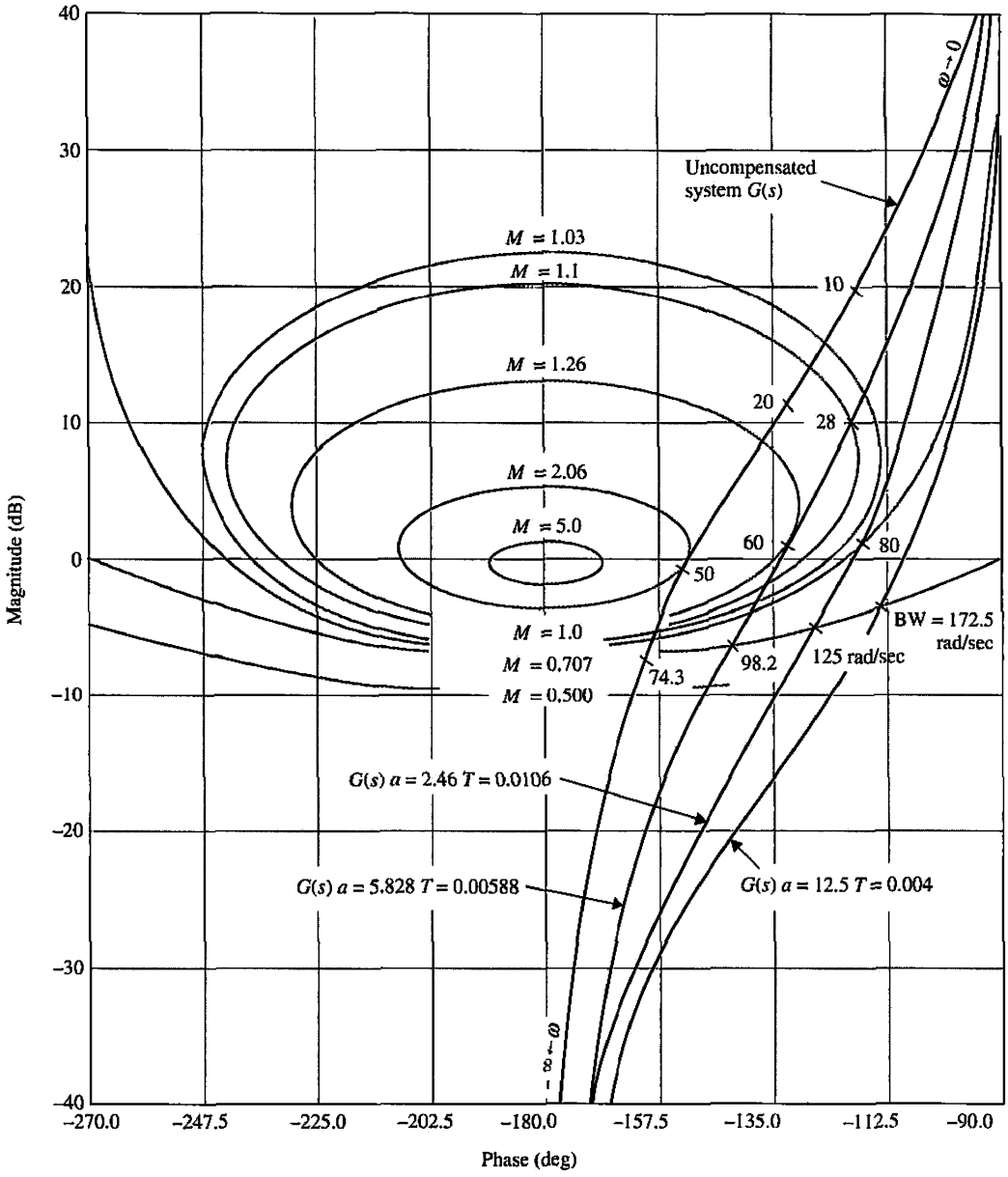


Figure 9-34 Plots of $G(s)$ in the Nichols chart for the system in Example 9-5-1.

$$G(s) = \frac{2500(1 + aTs)}{s(s + 25)(1 + Ts)}.$$

TABLE 9-13 Attributes of System with Phase-Lead Controller in Example 9-5-1

a	T	PM (deg)	M_r	Gain CO (rad/sec)	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
1	1	28.03	2.06	47.0	74.3	44.4	0.0255	0.2133
2.46	0.0106	47.53	1.26	60.2	98.2	22.3	0.0204	0.0744
5.828	0.00588	62.36	1.03	79.1	124.7	7.7	0.0169	0.0474
12.5	0.0040	68.12	1.00	113.1	172.5	1.1	0.0135	0.0174

$a = 12.5$ and $T = 0.004$, even the projected phase margin is 86.41°; the actual value is 68.12 due to the fall-off of the phase curve at the new gain crossover.

EXAMPLE 9-5-2 In this example we illustrate the application of a phase-lead controller to a third-order system with relatively high loop gain.

Let us consider that the inductance of the dc motor of the sun-seeker system described in Fig. 9-29 is not zero. The following set of system parameters is given:

$$\begin{aligned} R_F &= 10,000\,\Omega & K_b &= 0.0125\ \text{V/rad/sec} \\ K_i &= 0.0125\ \text{N-m/A} & R_a &= 6.25\,\Omega \\ J &= 10^{-6}\ \text{kg-m}^2 & K_s &= 0.3\ \text{A/rad} \\ K &= \text{to be determined} & B &= 0 \\ n &= 800 & L_a &= 10^{-3}\ \text{H} \end{aligned}$$

The transfer function of the dc motor is written

$$\frac{\Omega_m(s)}{E_a(s)} = \frac{K_i}{s(L_aJs^2 + JR_as + K_iK_b)} \tag{9-92}$$

The forward-path transfer function of the system is

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{K_sR_FKK_i}{s(L_aJs^2 + JR_as + K_iK_b)} \tag{9-93}$$

Substituting the values of the system parameters in Eq. (9-92), we get

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{4.6875 \times 10^7\,K}{s(s^2 + 625s + 156,250)} \tag{9-94}$$

You may use **ACSYS** to solve this problem.

Time-Domain Design

The time-domain specifications of the system are given as follows:

1. The steady-state error of $\alpha(t)$ due to a unit-ramp function input for $\theta_r(t)$ should be $\leq 1/300$ rad/rad/sec of the final steady-state output velocity.
2. The maximum overshoot of the step response should be less than 5% or as small as possible.
3. Rise time $t_r \leq 0.004$ sec.
4. Settling time $t_s \leq 0.02$ sec.

The minimum value of the amplifier gain K is determined initially from the steady-state requirement. Applying the final-value theorem to $\alpha(t)$, we get

$$\lim_{t\to\infty} \alpha(t) = \lim_{s\to 0} sA(s) = \lim_{s\to 0} \frac{s\Theta_r(s)}{1 + G_p(s)} \tag{9-95}$$

Substituting Eq. (9-94) into Eq. (9-95), and $\Theta_r(s) = 1/s^2$, we have

$$\lim_{t\to\infty} \alpha(t) = \frac{1}{300\,K} \tag{9-96}$$

Thus, for the steady-state value of $\alpha(t)$ to be $\leq 1/300$, K must be ≥ 1. Let us set $K = 1$; the forward-path transfer function in Eq. (9-94) becomes

$$G_p(s) = \frac{4.6875 \times 10^7}{s(s^2 + 625s + 156,250)} \tag{9-97}$$

We can show that the closed-loop sun-seeker system with $K = 1$ has the following attributes for the unit-step response.

Maximum overshoot = 43% Rise time $t_r = 0.004797$ sec Settling time $t_s = 0.04587$ sec

To improve the system response, let us select the phase-lead controller described by Eq. (9-62). The forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{4.6875 \times 10^7 K(1 + aTs)}{as(s^2 + 625s + 156,250)(1 + Ts)} \tag{9-98}$$

Now to satisfy the steady-state requirement, K must be readjusted so that $K \geq a$. Let us set $K = a$. The characteristic equation of the phase-lead compensated system becomes

$$\begin{aligned}(s^3 + 625s^2 + 156,250s + 4.6875 \times 10^7) + Ts^2(s^2 + 625s + 156,250) \\ + 4.6875 \times 10^7 aTs = 0\end{aligned} \tag{9-99}$$

We can use the root-contour method to examine the effects of varying a and T of the phase-lead controller. Let us first set a to zero. The characteristic equation of Eq. (9-99) becomes

$$(s^3 + 625s^2 + 156,250s + 4.6875 \times 10^7) + Ts^2(s^2 + 625s + 156,250) = 0 \tag{9-100}$$

Dividing both sides of the last equation by the terms that do not contain T, we get

$$1 + G_{eq1}(s) = 1 + \frac{Ts^2(s^2 + 625s + 156,250)}{s^3 + 625s^2 + 156,250s + 4.6875 \times 10^7} = 0 \tag{9-101}$$

The root contours of Eq. (9-100) when T varies are determined from the pole–zero configuration of $G_{eq1}(s)$ in Eq. (9-101) and are drawn as shown in Fig. 9-35. When a varies from 0 to ∞, we divide both sides of Eq. (9-99) by the terms that do not contain a, and we have

$$\begin{aligned}1 + G_{eq2}(s) &= 1 + \frac{4.6875 \times 10^7 aTs}{s^3 + 625s^2 + 156,250s + 4.6875 \times 10^7 + Ts^2(s^2 + 625s + 156,250)} \\ &= 0\end{aligned} \tag{9-102}$$

For a given T, the root contours of Eq. (9-99) when a varies are obtained based on the poles and zeros of $G_{eq2}(s)$. The poles of $G_{eq2}(s)$ are the same as the roots of Eq. (9-100). Thus, the root contours when a varies start ($a = 0$) at the root contours for variable T. Fig. 9-34 shows the dominant portions of the root contours when a varies for $T = 0.01$, 0.0045, 0.001, 0.0005, 0.0001, and 0.00001. Notice that, because the uncompensated system is lightly damped, for the phase-lead controller to be effective, the value of T should be very small. Even for very small values of T, there is only a small range of a that could bring increased damping, but the natural frequency of the system increases with the increase in a. The root contours in Fig. 9-35 show the approximate locations of the dominant characteristic equation roots where maximum damping occurs. Table 9-14 gives the roots of the characteristic equation and the unit-step-response attributes for the cases that correspond to near-smallest maximum overshoot for the T selected. Fig. 9-36 shows the unit-step response when $a = 500$ and $T = 0.00001$. Although the maximum overshoot is only 3.8%, the undershoot in this case is greater than the overshoot.

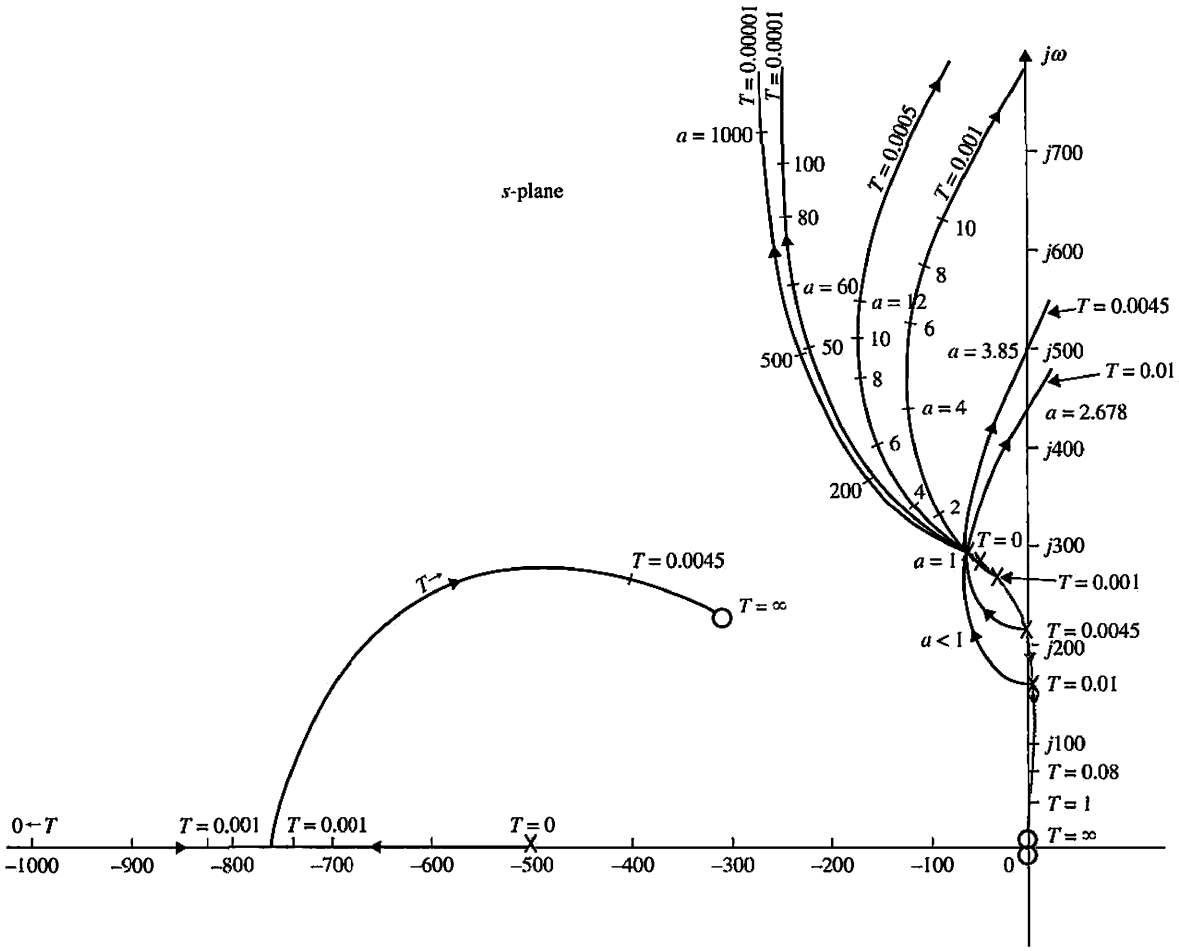


Figure 9-35 Root contours of sun-seeker system in Example 9-5-2 with phase-lead controller. $G_c(s) = \dfrac{1 + aTs}{1 + Ts}$.

Toolbox 9-5-6

Step responses for Fig. 9-36 are obtained by the following sequence of MATLAB functions

```
a = [50 100 500];
T = [0.0001 0.00005 0.00001];

for i = 1:length(T)
    num = 4.6875e7 * [a(i)*T(i) 1];
    den = conv([1 625 156250 0],[T(i) 1]);
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    step(numCL,denCL)
    hold on
end
axis([0 0.04 0 1.2])
grid
```

TABLE 9-14 Roots of Characteristic Equation and Time Response Attributes of System with Phase-Lead Controller in Example 9-5-2

T	a	Roots of Characteristic Equation			Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.001	4	−189.6	−1181.6	$-126.9 \pm j439.5$	21.7	0.0037	0.0184
0.0005	9	−164.6	−2114.2	$-173.1 \pm j489.3$	13.2	0.00345	0.0162
0.0001	50	−147	−10024	$-227 \pm j517$	5.4	0.00348	0.0150
0.00005	100	−147	−20012	$-233 \pm j515$	4.5	0.00353	0.0150
0.00001	500	−146.3	-10^5	$-238 \pm j513.55$	3.8	0.00357	0.0146

Frequency-Domain Design

The Bode plot of $G_p(s)$ in Eq. (9-97) is shown in Fig. 9-37. The performance attributes of the uncompensated system are

$$\text{PM} = 29.74°$$

$$M_r = 2.156$$

$$\text{BW} = 426.5 \text{ rad/sec}$$

We would like to show that the frequency-domain design procedure outlined earlier does not work effectively here, because the phase curve of $G_p(j\omega)$ shown in Fig. 9-37 has a very steep slope near the gain crossover. For example, if we wish to realize a phase margin

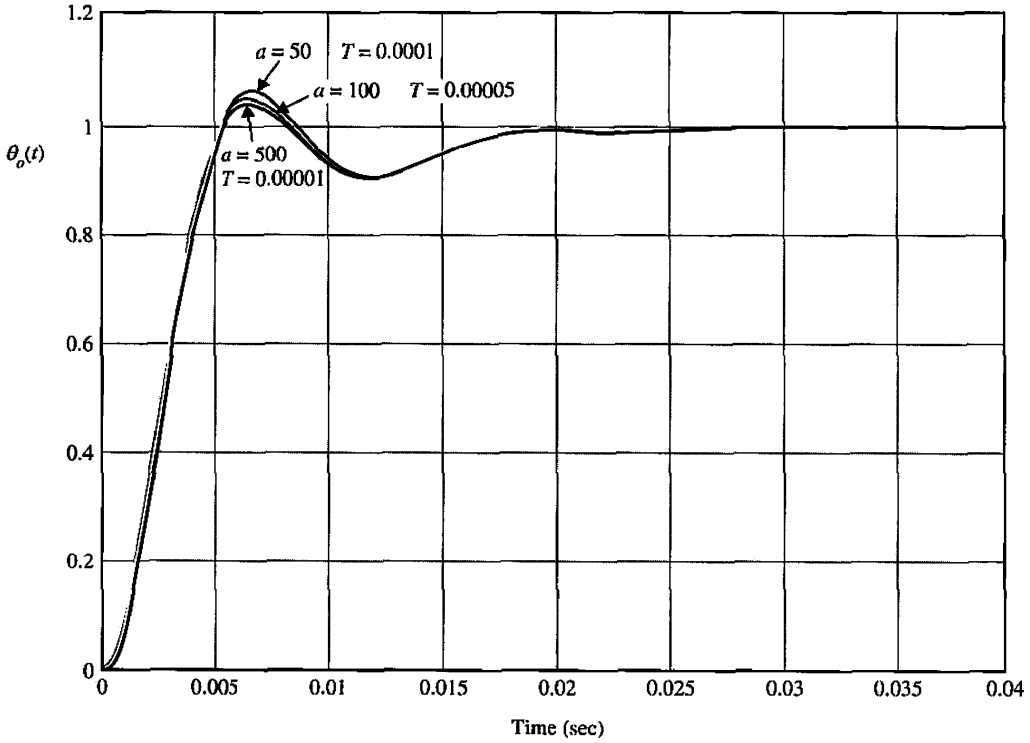


Figure 9-36 Unit-step responses of sun-seeker system in Example 9-5-2 with phase-lead controller. $G_c(s) = \dfrac{1 + aTs}{1 + Ts}$.

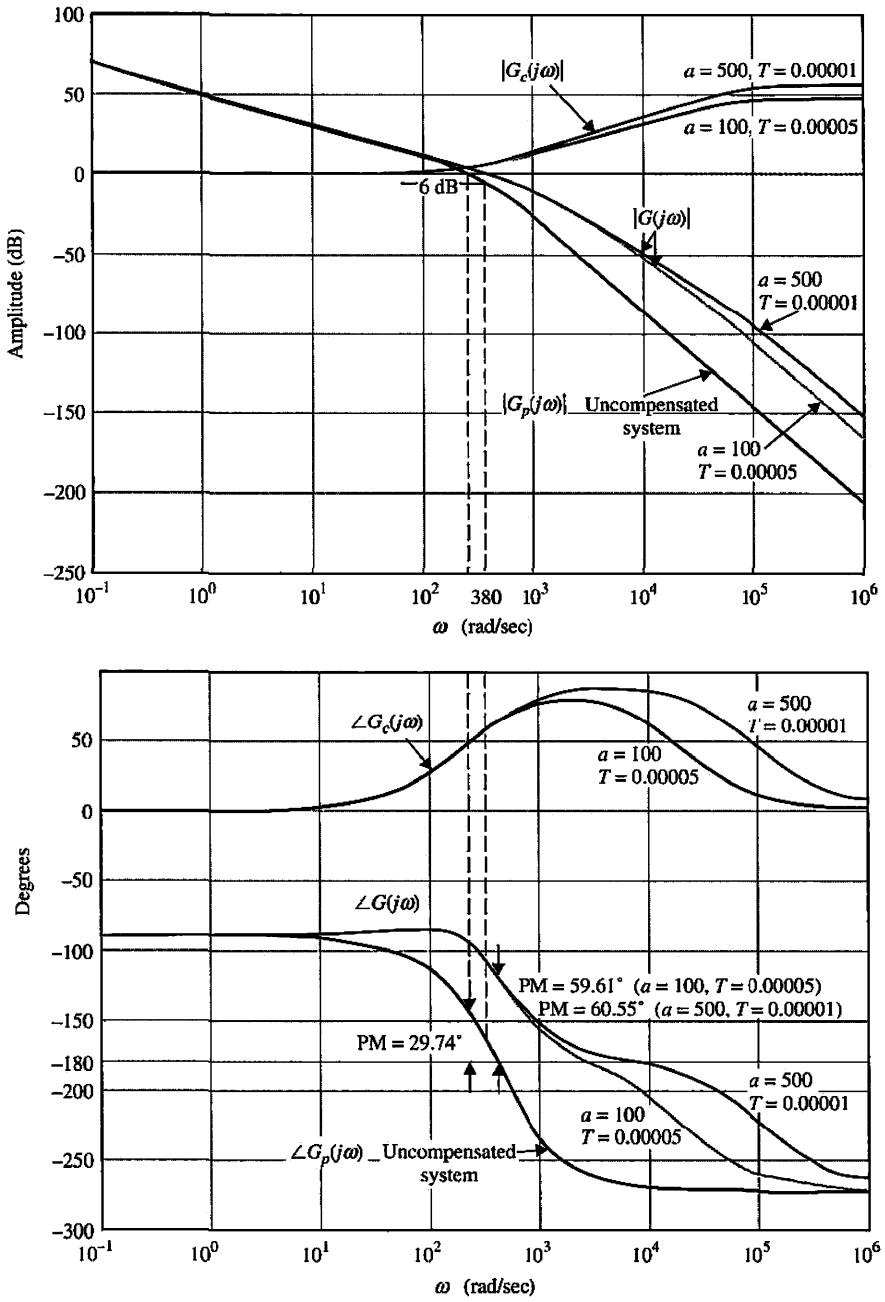


Figure 9-37 Bode plots of phase-lead controller and forward-path transfer function of sun-seeker system in Example 9-5-2. $G_c(s) = \dfrac{1 + aTs}{1 + Ts}$.

of 65°, we need at least $65 - 29.74 = 35.26°$ of phase lead. Or, $\phi_m = 35.26°$. Using Eq. (9-72), the value of a is calculated to be

$$a = \frac{1 + \sin \phi_m}{1 - \sin \phi_m} = \frac{1 + \sin 35.26°}{1 - \sin 35.26°} = 3.732 \tag{9-103}$$

Toolbox 9-5-7

Bode plots shown in Fig. 9-37 are obtained by the following sequence of MATLAB functions

```
a = [100 500];
T = [0.00005 0.00001];

for i = 1:length(T)
    num = [a(i)*T(i) 1];
    den =[T(i) 1];
    bode(num,den);
    hold on;
end

for i = 1:length(T)
    num = 4.6875e7 * [a(i)*T(i) 1];
    den = conv([1 625 156250 0],[T(i) 1]);
    bode(num,den);
    hold on;
end
axis([1 1e6 -300 90]);
grid
```

Let us choose $a = 4$. Theoretically, to maximize the utilization of ϕ_m, ω_m should be placed at the new gain crossover, which is located at the frequency where the magnitude of $G_p(j\omega)$ is $-10\log_{10}a$ dB $= -10\log_{10}4 = -6$ dB. From the Bode plot in Fig. 9-37, this frequency is found to be 380 rad/sec. Thus, we let $\omega_m = 380$ rad/sec. The value of T is found by using Eq. (9-67):

$$T = \frac{1}{\omega_m\sqrt{a}} = \frac{1}{380\sqrt{4}} = 0.0013 \tag{9-104}$$

However, checking the frequency response of the phase-lead compensated system with $a = 4$ and $T = 0.0013$, we found that the phase margin is only improved to 38.27°, and $M_r = 1.69$. The reason is the steep negative slope of the phase curve of $G_p(j\omega)$. The fact is that, at the new gain-crossover frequency of 380 rad/sec, the phase of $G_p(j\omega)$ is $-170°$, as against $-150.26°$ at the original gain crossover—a drop of almost 20°! From the time-domain design, the first line in Table 9-14 shows that, when $a = 4$ and $T = 0.001$, the maximum overshoot is 21.7%.

Checking the frequency response of the phase-lead compensated system with $a = 500$ and $T = 0.00001$, the following performance data are obtained:

$$\text{PM} = 60.55 \text{ degrees} \quad M_r = 1 \quad \text{BW} = 664.2 \text{ rad/sec}$$

This shows that the value of a has to be increased substantially just to overcome the steep drop of the phase characteristics when the gain crossover is moved upward.

Fig. 9-37 shows the Bode plots of the phase-lead controller and the forward-path transfer functions of the compensated system with $a = 100$, $T = 0.0005$ and $a = 500$, $T = 0.00001$. A summary of performance data is given in Table 9-15.

TABLE 9-15 Attributes of System with Phase-Lead Controller in Example 9-5-2

T	a	PM (deg)	GM (dB)	M_r	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_r (sec)
1	1	29.74	6.39	2.16	430.4	43.0	0.00478	0.0459
0.00005	100	59.61	31.41	1.009	670.6	4.5	0.00353	0.015
0.00001	500	60.55	45.21	1.000	664.2	3.8	0.00357	0.0146

Selecting $a = 100$ and $T = 0.00005$, the phase-lead controller is described by the transfer function

$$G_c(s) = \frac{1}{a}\frac{1+aTs}{1+Ts} = \frac{1}{100}\frac{1+0.005s}{1+0.00005s} \tag{9-105}$$

Using Eqs. (9-63) and (9-64), and letting $C = 0.01\ \mu$F, the circuit parameters of the phase-lead controller are found to be

$$R_2 = \frac{T}{C} = \frac{5 \times 10^{-5}}{10^{-8}} = 5000\,\Omega \tag{9-106}$$

$$R_1 = aR_2 = 500,000\,\Omega \tag{9-107}$$

The forward-path transfer function of the compensated system is

$$\frac{\Theta_o(s)}{A(s)} = \frac{4.6875 \times 10^7(1+0.005s)}{s(s^2+625s+156,250)(1+0.00005s)} \tag{9-108}$$

where the amplifier gain K has been set to 100 to satisfy the steady-state requirement.

From the results of the last two illustrative examples, we can summarize the effects and limitations of the single-stage phase-lead controller as follows.

9-5-3 Effects of Phase-Lead Compensation

1. The phase-lead controller adds a zero and a pole, with the zero to the right of the pole, to the forward-path transfer function. The general effect is to add more damping to the closed-loop system. The rise time and settling time are reduced in general.
2. The phase of the forward-path transfer function in the vicinity of the gain-crossover frequency is increased. This improves the phase margin of the closed-loop system.
3. The slope of the magnitude curve of the Bode plot of the forward-path transfer function is reduced at the gain-crossover frequency. This usually corresponds to an improvement in the relative stability of the system in the form of improved gain and phase margins.
4. The bandwidth of the closed-loop system is increased. This corresponds to faster time response.
5. The steady-state error of the system is not affected.

9-5-4 Limitations of Single-Stage Phase-Lead Control

In general, phase-lead control is not suitable for all systems. Successful application of single-stage phase-lead compensation to improve the stability of a control system is hinged on the following conditions:

1. Bandwidth considerations: If the original system is unstable or with a low stability margin, the additional phase lead required to realize a certain desired phase margin may be excessive. This may require a relatively large value of a for the controller, which, as a result, will give rise to a large bandwidth for the compensated system, and the transmission of high-frequency noise entering the system at the input may become objectionable. However, if the noise enters the system near the output, then the increased bandwidth may be beneficial to noise rejection. The larger bandwidth also has the advantage of robustness; that is, the system is insensitive to parameter variations and noise rejection as described before.
2. If the original system is unstable, or with low stability margin, the phase curve of the Bode plot of the forward-path transfer function has a steep negative slope near the gain-crossover frequency. Under this condition, the single-stage phase-lead controller may not be effective because the additional phase lead at the new gain crossover is added to a much smaller phase angle than that at the old gain crossover. The desired phase margin can be realized only by using a very large value of a for the controller. The amplifier gain K must be set to compensate a, so a large value for a requires a high-gain amplifier, which could be costly.

   As shown in Example 9-5-2, the compensated system may have a larger undershoot than overshoot. Often, a portion of the phase curve may still dip below the 180°-axis, resulting in a **conditionally stable** system, even though the desired phase margin is satisfied.
3. The maximum phase lead available from a single-stage phase-lead controller is less than 90°. Thus, if a phase lead of more than 90° is required, a multistage controller should be used.

9-5-5 Multistage Phase-Lead Controller

When the design with a phase-lead controller requires an additional phase of more than 90°, a multistage controller should be used. Fig. 9-38 shows an op-amp-circuit

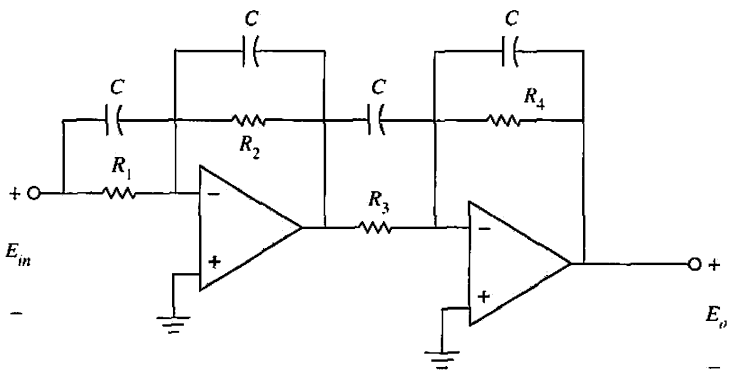


Figure 9-38 Two-stage phase-lead (phase-lag) controller.

realization of a two-stage phase-lead controller. The input–output transfer function of the circuit is

$$G_c(s) = \frac{E_o(s)}{E_{in}(s)} = \left(\frac{s + \dfrac{1}{R_1C}}{s + \dfrac{1}{R_2C}}\right)\left(\frac{s + \dfrac{1}{R_3C}}{s + \dfrac{1}{R_4C}}\right) = \frac{R_2R_4}{R_1R_3}\left(\frac{1 + R_1Cs}{1 + R_2Cs}\right)\left(\frac{1 + R_3Cs}{1 + R_4Cs}\right) \tag{9-109}$$

or

$$G_c(s) = \frac{1}{a_1a_2}\left(\frac{1 + a_1T_1s}{1 + T_1s}\right)\left(\frac{1 + a_2T_2s}{1 + T_2s}\right) \tag{9-110}$$

where $a_1 = R_1/R_2$, $a_2 = R_3/R_4$, $T_1 = R_2C$, and $T_2 = R_4C$.

The design of a multistage phase-lead controller in the time domain becomes more cumbersome, since now there are more poles and zeros to be placed. The root-contour method also becomes impractical, since there are more variable parameters. The frequency-domain design in this case does represent a better choice of the design method. For example, for a two-stage controller, we can choose the parameters of the first stage of a two-stage controller so that a portion of the phase margin requirement is satisfied, and then the second stage fulfills the remaining requirement. In general, there is no reason why the two stages cannot be identical. The following example illustrates the design of a system with a two-stage phase-lead controller.

▸ EXAMPLE 9-5-3 For the sun-seeker system designed in Example 9-5-2, let us alter the rise time and settling time requirements to be

Rise time $t_r \leq 0.001$ sec

Settling time $t_s \leq 0.005$ sec

The other requirements are not altered. One way to meet faster rise time and settling time requirements is to increase the forward-path gain of the system. Let us consider that the forward-path transfer function is

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{156,250,000}{s(s^2 + 625s + 156,250)} \tag{9-111}$$

Another way of interpreting the change in the forward-path gain is that the ramp-error constant is increased to 1000 (up from 300 in Example 9-5-1). The Bode plot of $G_p(s)$ is shown in Fig. 9-39. The closed-loop system is unstable, with a phase margin of $-15.43°$. ◂

Toolbox 9-5-8

Bode plots shown in Fig. 9-39 are obtained by the following sequence of MATLAB functions

```
num = 156250000 * [0.0087 1];
den = conv([0.000087 1],[1 625 156250]);
bode(num,den);
```

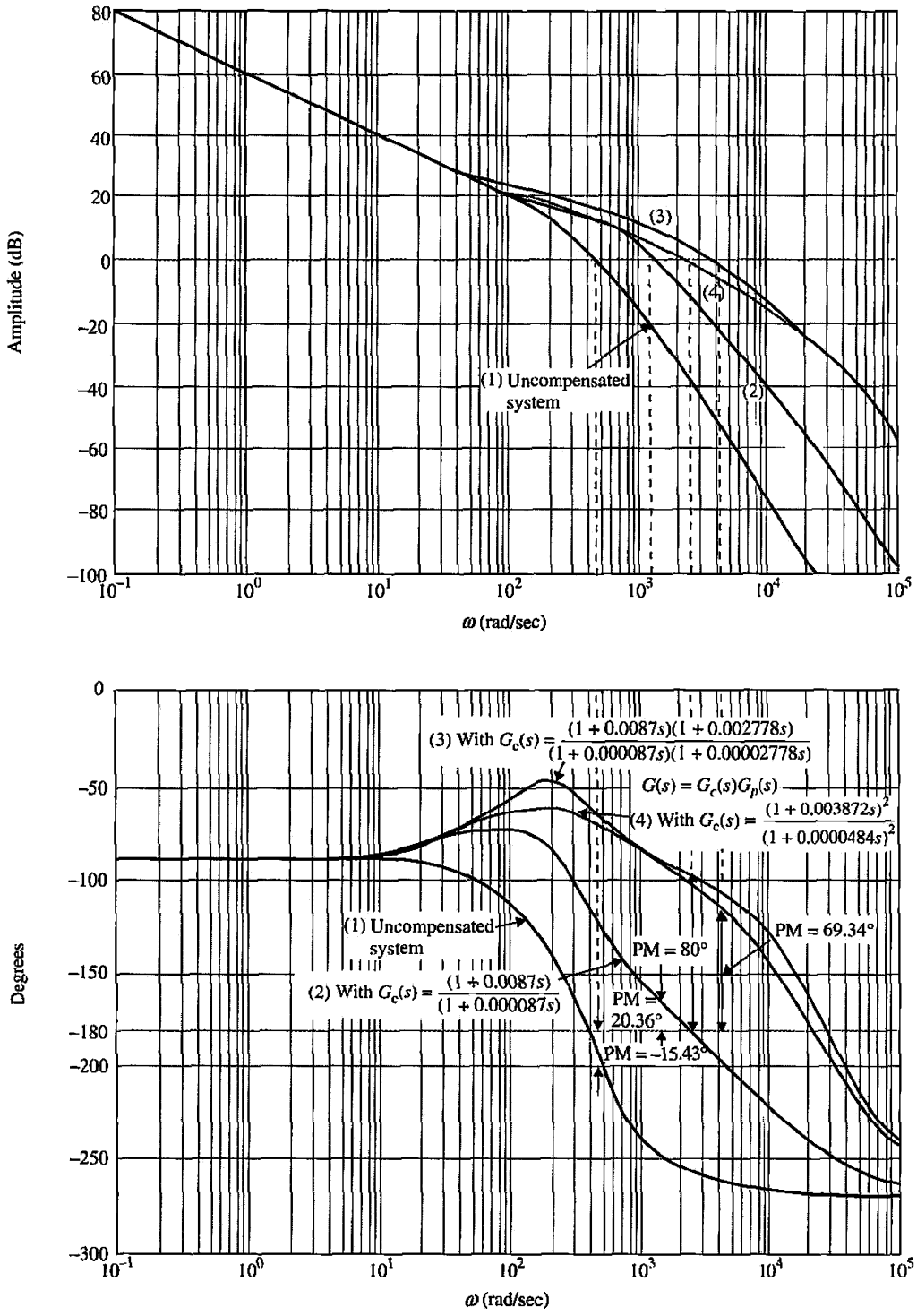


Figure 9-39 Bode plots of uncompensated and compensated sun-seeker systems in Example 9-5-2 with two-state phase-lead controller. $G_p(s) = \dfrac{156,250,000}{s(s^2 + 625s + 156,250)}$.

```
hold on;


num = 156250000 * conv([0.0087 1],[0.002778 1]);
den =conv(conv([0.000087 1],[0.00002778 1]),[1 625 156250 0]);
bode(num,den);
hold on;


num = 156250000 * conv([0.003872 1],[0.003872 1]);
den =conv(conv([0.0000484 1],[0.0000484 1]),[1 625 156250 0]);
bode(num,den);
axis([1 1e5 -300 20]);
grid
```

Because the compensated system in Example 9-5-2 had a phase margin of 60.55°, we would expect that, to satisfy the more stringent time response requirements in this example, the corresponding phase margin would have to be greater. Apparently, this increased phase margin cannot be realized with a single-stage phase-lead controller. It appears that a two-stage controller would be adequate.

The design involves some trial-and-error steps in arriving at a satisfied controller. Because we have two stages of controllers at our disposal, the design has a multitude of flexibility. We can set out by arbitrarily setting $a_1 = 100$ for the first stage of the phase-lead controller. The phase lead provided by the controller is obtained from Eq. (9-71),

$$\phi_m = \sin^{-1}\left(\frac{a_1 - 1}{a_1 + 1}\right) = \sin^{-1}\left(\frac{99}{101}\right) = 78.58° \tag{9-112}$$

To maximize the effect of ϕ_m, the new gain crossover should be at

$$-10\log_{10} a_1 = -10\log_{10} 100 = -20\,\text{dB} \tag{9-113}$$

From Fig. 9-39 the frequency that corresponds to this gain on the amplitude curve is approximately 1150 rad/sec. Substituting $\omega_{m1} = 1150$ rad/sec and $a_1 = 100$ in Eq. (9-67), we get

$$T_1 = \frac{1}{\omega_{m1}\sqrt{a_1}} = \frac{1}{1150\sqrt{100}} = 0.000087 \tag{9-114}$$

The forward-path transfer function with the one-stage phase-lead controller is

$$G(s) = \frac{156,250,000(1 + 0.0087s)}{s(s^2 + 625s + 156,250)(1 + 0.000087s)} \tag{9-115}$$

The Bode plot of the last equation is drawn as curve (2) in Fig. 9-39. We see that the phase margin of the interim design is only 20.36°. Next, we arbitrarily set the value of a_2 of the second stage at 100. From the Bode plot of the transfer function of Eq. (9-115) in Fig. 9-39, we find that the frequency at which the magnitude of $G(j\omega)$ is −20 dB is approximately 3600 rad/sec. Thus,

$$T_2 = \frac{1}{\omega_{m2}\sqrt{a_2}} = \frac{1}{3600\sqrt{100}} = 0.00002778 \tag{9-116}$$

TABLE 9-16 Attributes of Sun-Seeker System in Example 9-5-3 with Two-Stage Phase-Lead Controller

$a_1 = a_2$	T_1	T_2	PM (deg)	M_r	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
80	0.0000484	0.0000484	80	1	5686	0	0.00095	0.00475
100	0.000087	0.0000278	69.34	1	5686	0	0.000597	0.00404
70	0.0001117	0.000039	66.13	1	5198	0	0.00063	0.00404

The forward-path transfer function of the sun-seeker system with the two-stage phase-lead controller is ($a_1 = a_2 = 100$)

$$G(s) = \frac{156,250,000(1 + 0.0087s)(1 + 0.002778s)}{s(s^2 + 625s + 156,250)(1 + 0.000087s)(1 + 0.00002778s)} \tag{9-117}$$

Fig. 9-39 shows the Bode plot of the sun-seeker system with the two-stage phase-lead controller designed above [curve (3)]. As seen from Fig. 9-39, the phase margin of the system with $G(s)$ given in Eq. (9-117) is 69.34°. As shown by the system attributes in Table 9-16, the system satisfies all the time-domain specifications. In fact, the selection of $a_1 = a_2 = 100$ appears to be overly stringent. To show that the design is not critical, we can select $a_1 = a_2 = 80$, and then 70 and the time-domain specifications are still satisfied. Following similar design steps, we arrived at $T_1 = 0.0001117$ and $T_2 = 0.000039$ for $a_1 = a_2 = 70$, and $T_1 = T_2 = 0.0000484$ for $a_1 = a_2 = 80$. Curve (4) of Fig. 9-39 shows the Bode plot of the compensated system with $a_1 = a_2 = 80$. Table 9-16 summarizes all the attributes of the system performance with these three controllers.

The unit-step responses of the system with the two-stage phase-lead controller for $a_1 = a_2 = 80$ and 100 are shown in Fig. 9-40.

Toolbox 9-5-9

Fig. 9-40 is obtained by the following sequence of MATLAB functions

```
num = 156250000 * conv([100*0.000087 1],[80*0.00002778 1]);
den =conv(conv([0.000087 1],[0.00002778 1]),[1 625 156250 0]);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on

num = 156250000 * conv([80*0.0000484 1],[80*0.0000484 1]);
den =conv(conv([0.0000484 1],[0.0000484 1]),[1 625 156250 0]);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
grid
```

9-5-6 Sensitivity Considerations

The sensitivity function defined in Section 8-16, Eq. (8-122), can be used as a design specification to indicate the robustness of the system. In Eq. (9-122), the sensitivity of the

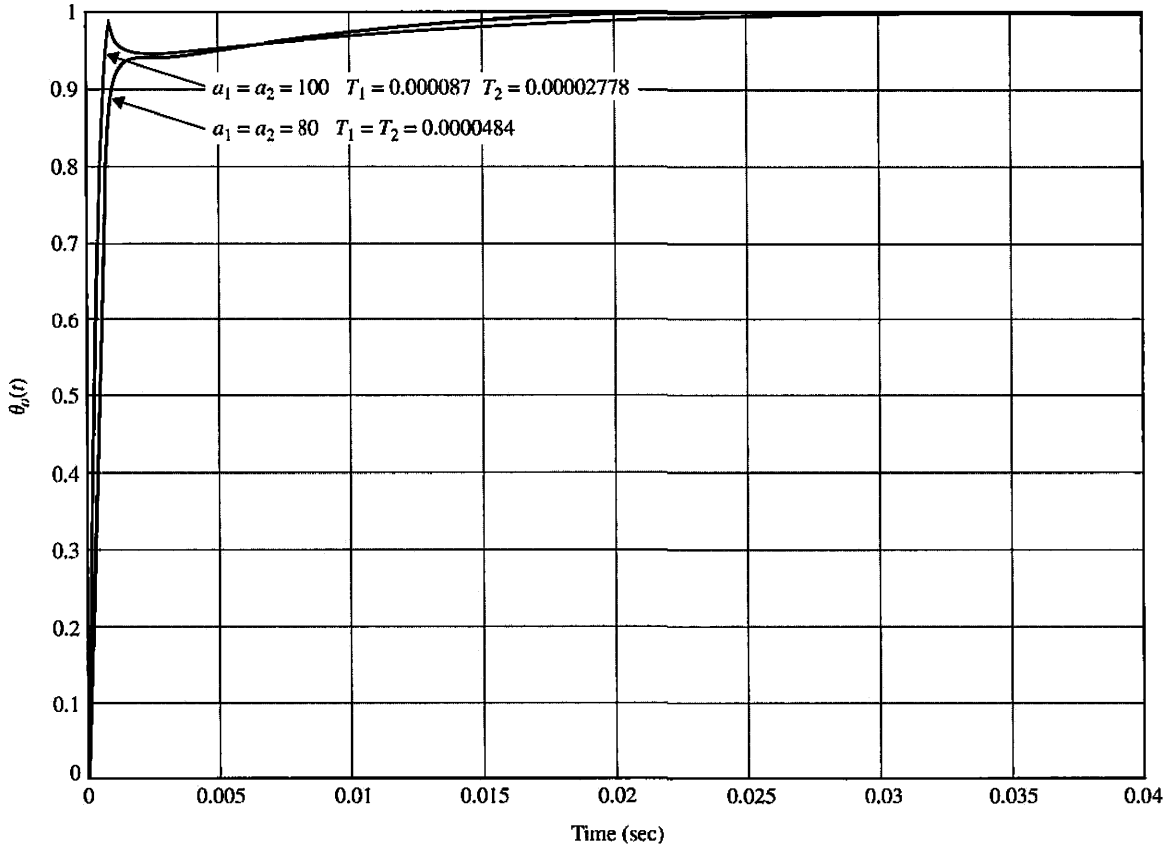


Figure 9-40 Unit-step responses of sun-seeker system in Example 9-5-2 with two-stage phase-lead controller. $G_c(s) = \left(\frac{1 + a_1 T_1 s}{1 + T_1 s}\right)\left(\frac{1 + a_2 T_2 s}{1 + T_2 s}\right)$ $G_p(s) = \frac{156,250,000}{s(s^2 + 625s + 156,250)}$.

closed-loop transfer function with respect to the variations of the forward-path transfer function is defined as

$$S_G^M(s) = \frac{\partial M(s)/M(s)}{\partial G(s)/G(s)} = \frac{G^{-1}(s)}{1 + G^{-1}(s)} = \frac{1}{1 + G(s)} \tag{9-118}$$

The plot of $|S_G^M(j\omega)|$ versus frequency gives an indication of the sensitivity of the system as a function of frequency. The ideal robust situation is for $|S_G^M(j\omega)|$ to assume a small value ($\ll 1$) over a wide range of frequencies. As an example, the sensitivity function of the sun-seeker system designed in Example 9-5-2 with the one-stage phase-lead controller with $a = 100$ and $T = 0.00005$ is plotted as shown in Fig. 9-41. Note that the sensitivity function is low at low frequencies and is less than unity for $\omega < 400$ rad/sec. Although the sun-seeker system in Example 9-5-2 does not need a multistage phase-lead controller, we shall show that, if a two-stage phase-lead controller is used, not only the value of a will be substantially reduced, resulting in lower gains for the op-amps, but the system will be more robust. Following the design procedure outlined in Example 9-5-3, a two-stage phase-lead controller is designed for the sun-seeker system with the process transfer function described by Eq. (9-96).

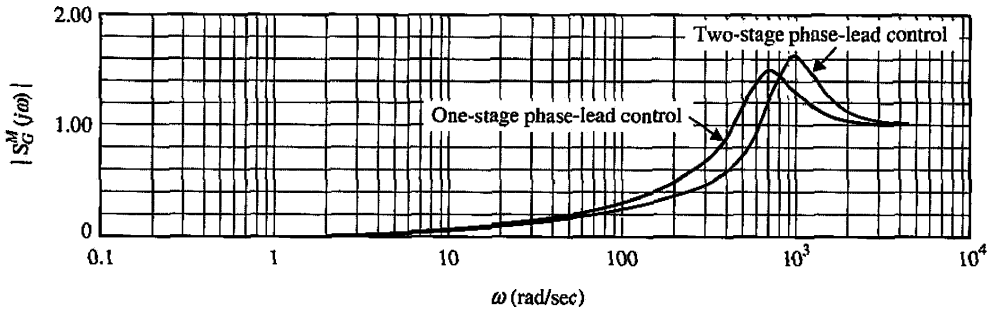


Figure 9-41 Sensitivity functions of sun-seeker system in Example 9-5-2.

The parameters of the controller are $a_1 = a_2 = 5.83$ and $T_1 = T_2 = 0.000673$. The forward-path transfer function of the compensated system is

$$G(s) = \frac{4.6875 \times 10^7 (1 + 0.0039236s)^2}{s(s^2 + 625s + 156,250)(1 + 0.000673s)^2} \tag{9-119}$$

Fig. 9-41 shows that the sensitivity function of the system with the two-stage phase-lead controller is less than unity for $\omega < 600$ rad/sec. Thus, the system with the two-stage phase-lead controller is more robust than the system with the single-stage controller. The reason for this is that the more robust system has a higher bandwidth. In general, systems with phase-lead control will be more robust due to the higher bandwidth. However, Fig. 9-41 shows that the system with the two-stage phase-lead controller has a higher sensitivity at high frequencies.

▶ 9-6 DESIGN WITH PHASE-LAG CONTROLLER

The transfer function in Eq. (9-62) represents a phase-lag controller or low-pass filter when $a < 1$. The transfer function is repeated as follows.

$$G_c(s) = \frac{1}{a}\left(\frac{1 + aTs}{1 + Ts}\right) \quad a < 1 \tag{9-120}$$

9-6-1 Time-Domain Interpretation and Design of Phase-Lag Control

The pole–zero configuration of $G_c(s)$ is shown in Fig. 9-42. Unlike the PI controller, which provides a pole at $s = 0$, the phase-lag controller affects the steady-state error only in the

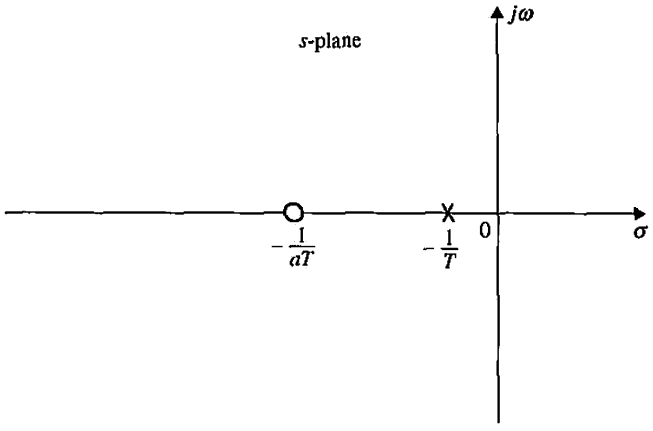


Figure 9-42 Pole–zero configuration of phase-lag controller.

sense that the zero-frequency gain of $G_c(s)$ is greater than unity. Thus, any error constant that is finite and nonzero will be increased by the factor $1/a$ from the phase-lag controller.

Because the pole at $s = -1/T$ is to the right of the zero at $-1/aT$, effective use of the phase-lag controller to improve damping would have to follow the same design principle of the PI control presented in Section 9-3. Thus, *the proper way of applying the phase-lag control is to place the pole and zero close together. For type 0 and type 1 systems, the combination should be located near the origin in the* s*-plane.* Fig. 9-43 illustrates the design strategies in the s-plane for type 0 and type 1 systems. Phase-lag control should not be applied to a type 2 system.

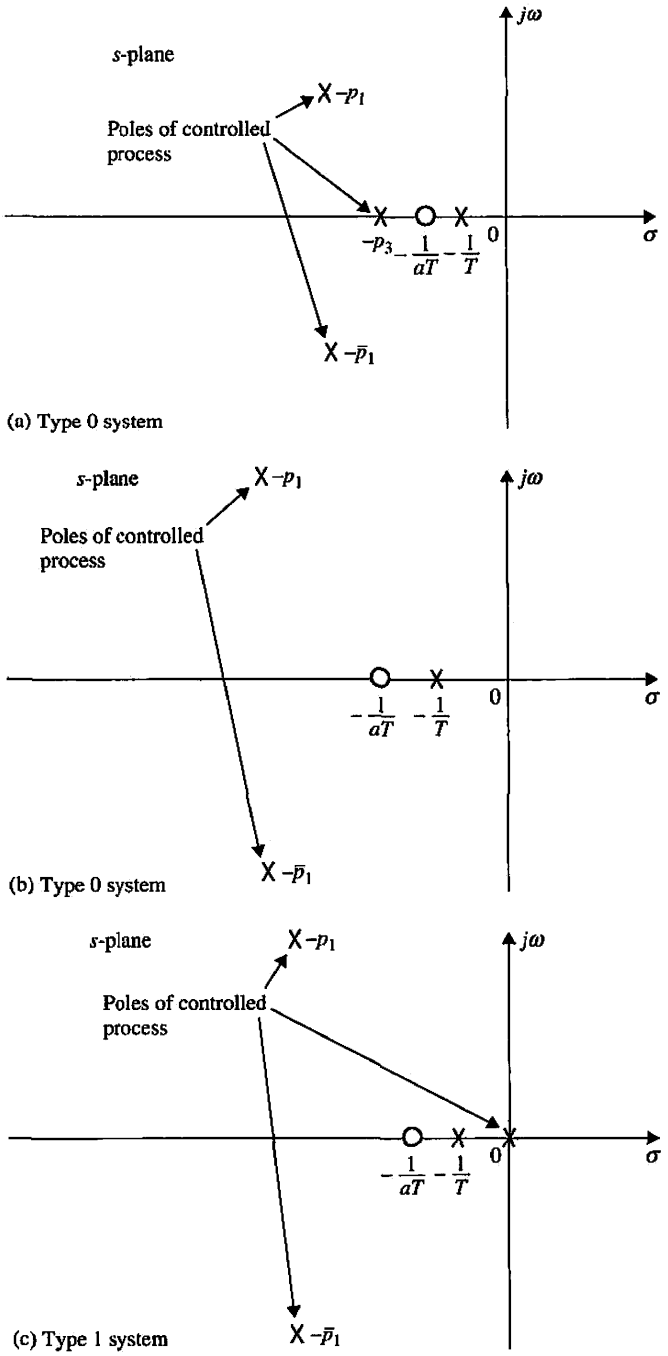


Figure 9-43 Design strategies for phase-lag control for type 0 and type 1 systems.

The design principle described above can be explained by considering that the controlled process of a type 0 control system is

$$G_p(s) = \frac{K}{(s + p_1)(s + \bar{p}_1)(s + p_3)} \tag{9-121}$$

where p_1 and $\bar{p}_1$ are complex-conjugate poles, such as the situation shown in Fig. 9-43.

Just as in the case of the phase-lead controller, we can drop the gain factor $1/a$ in Eq. (9-121), because whatever the value of a is, the value of K can be adjusted to compensate for it. Applying the phase-lag controller of (9-121), without the factor $1/a$, to the system, the forward-path transfer function becomes

$$G(s) = G_c(s)G_p(s) = \frac{K(1 + aTs)}{(s + p_1)(s + \bar{p}_1)(s + p_3)(1 + Ts)} \quad (a < 1) \tag{9-122}$$

Let us assume that the value of K is set to meet the steady-state-error requirement. Also assume that, with the selected value of K, the system damping is low or even unstable. Now let $1/T \cong 1/aT$, and place the pole–zero pair near the pole at $-1/p_3$, as shown in Fig. 9-43. Fig. 9-44 shows the root loci of the system with and without the phase-lag controller. Because the pole–zero combination of the controller is very close to the pole at $-1/p_3$, the shape of the loci of the dominant roots with and without the phase-lag control will be very similar. This is easily explained by writing Eq. (9-122) as

$$\begin{aligned} G(s) &= \frac{Ka(s + 1/aT)}{(s + p_1)(s + \bar{p}_1)(s + p_3)(s + 1/T)} \\ &\cong \frac{Ka}{(s + p_1)(s + \bar{p}_1)(s + p_3)} \end{aligned} \tag{9-123}$$

Because a is less than 1, the application of phase-lag control is equivalent to reducing the forward-path gain from K to Ka, *while not affecting the steady-state performance of the system.* Fig. 9-44 shows that the value of a can be chosen so that the damping of the compensated system is satisfactory. Apparently, the amount of damping that can be added is limited if the poles $-p_1$ and $-\bar{p}_1$ are very close to the imaginary axis. Thus, we can select a using the following equation:

$$a = \frac{K \text{ to realize the desired damping}}{K \text{ to realize the steady-state performance}} \tag{9-124}$$

The value of T should be so chosen that the pole and zero of the controller are very close together and close to $-1/p_3$.

In the time domain, phase-lag control generally has the effect of increasing the rise time and settling time.

9-6-2 Frequency-Domain Interpretation and Design of Phase-Lag Control

The transfer function of the phase-lag controller can again be written as

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad (a < 1) \tag{9-125}$$

by assuming that the gain factor $-1/a$ is eventually absorbed by the forward gain K. The Bode diagram of Eq. (9-125) is shown in Fig. 9-45. The magnitude curve has corner frequencies at $\omega = 1/aT$ and $1/T$. Because the transfer functions of the phase-lead and

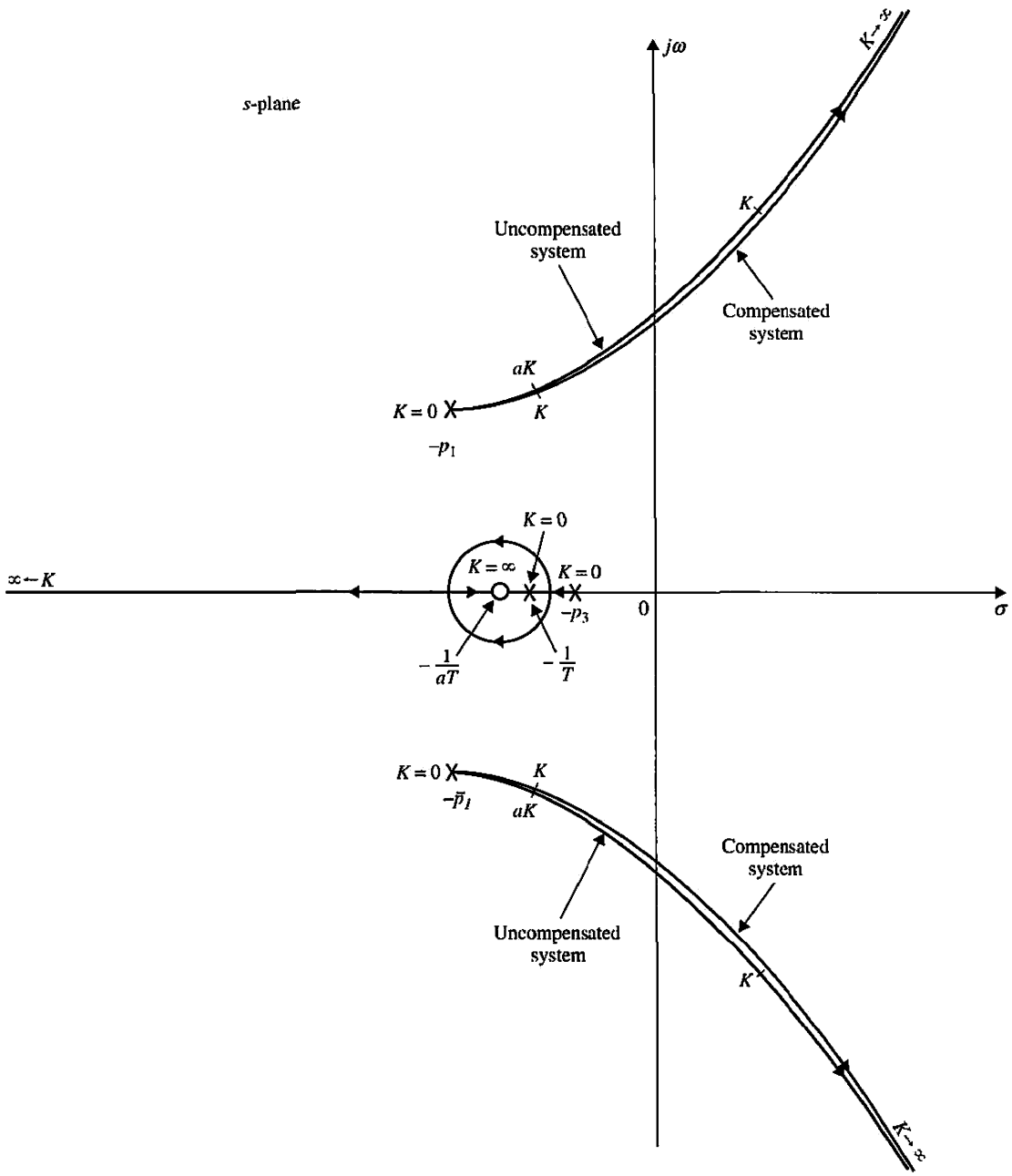


Figure 9-44 Root loci of uncompensated and phase-lag-compensated systems.

phase-lag controllers are identical in form, except for the value of a, the maximum phase lag ϕ_m of the phase curve of Fig. 9-45 is given by

$$\phi_m = \sin^{-1}\left(\frac{a-1}{a+1}\right) \quad (a<1) \tag{9-126}$$

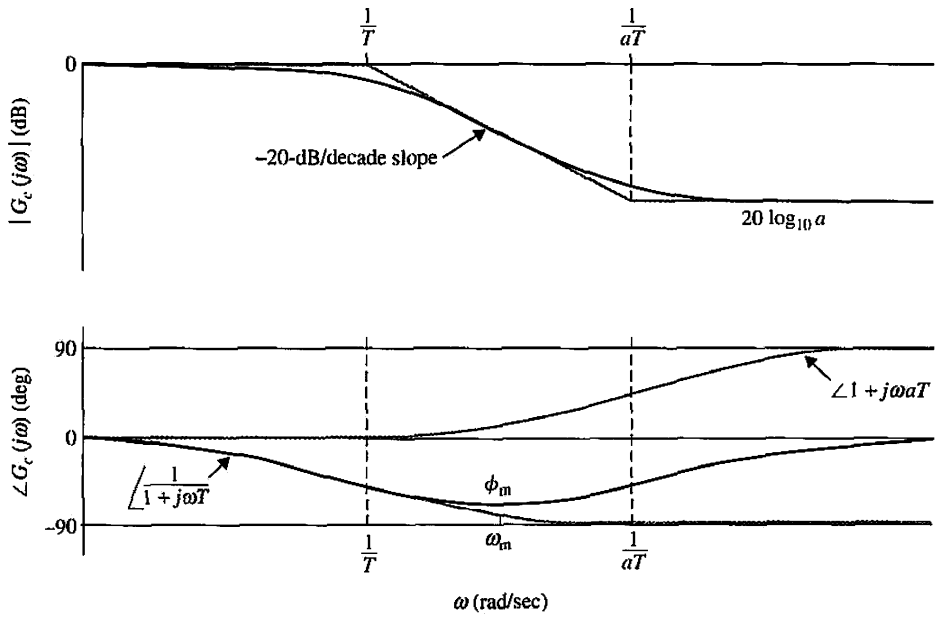


Figure 9-45 Bode diagram of the phase-lag controller. $G_c(s) = \dfrac{(1 + aTs)}{(1 + Ts)} \quad a < 1$.

Fig. 9-45 shows that the phase-lag controller essentially provides an attenuation of 20 $\log_{10} a$ at high frequencies. Thus, unlike the phase-lead control that utilizes the maximum phase lead of the controller, phase-lag control utilizes the attenuation of the controller at high frequencies. This is parallel to the situation of introducing an attenuation of a to the forward-path gain in the root-locus design. For phase-lead control, the objective of the controller is to increase the phase of the open-loop system in the vicinity of the gain crossover while attempting to locate the maximum phase lead at the new gain crossover. In phase-lag control, the objective is to move the gain crossover to a lower frequency where the desired phase margin is realized, while keeping the phase curve of the Bode plot relatively unchanged at the new gain crossover.

The design procedure for phase-lag control using the Bode plot is outlined as follows:

1. The Bode plot of the forward-path transfer function of the uncompensated system is drawn. The forward-path gain K is set according to the steady-state performance requirement.
2. The phase and gain margins of the uncompensated system are determined from the Bode plot.
3. Assuming that the phase margin is to be increased, the frequency at which the desired phase margin is obtained is located on the Bode plot. This frequency is also the new gain crossover frequency ω_g', where the compensated magnitude curve crosses the 0-dB-axis.
4. To bring the magnitude curve down to 0 dB at the new gain-crossover frequency ω_g', the phase-lag controller must provide the amount of attenuation equal to the value of the magnitude curve at ω_g'. In other words,

$$\left|G_p\left(j\omega_g'\right)\right| = -20 \log_{10} a \text{ dB} \quad (a < 1) \tag{9-127}$$

Solving for a from the last equation, we get

$$a = 10^{-|G_p(j\omega_g')|/20} \quad (a < 1) \tag{9-128}$$

Once the value of a is determined, it is necessary only to select the proper value of T to complete the design. Using the phase characteristics shown in Fig. 9-45, if the corner frequency $1/aT$ is placed far below the new gain-crossover frequency ω_g', the phase lag of the controller will not appreciably affect the phase of the compensated system near ω_g'. On the other hand, the value of $1/aT$ should not be too small because the bandwidth of the system will be too low, causing the system to be too sluggish and less robust. Usually, as a general guideline, the frequency $1/aT$ should be approximately one decade below ω_g'; that is,

$$\frac{1}{aT} = \frac{\omega_g'}{10} \text{ rad/sec} \tag{9-129}$$

Then,

$$\frac{1}{T} = \frac{a\omega_g'}{10} \text{ rad/sec} \tag{9-130}$$

5. The Bode plot of the compensated system is investigated to see if the phase margin requirement is met; if not, the values of a and T are readjusted, and the procedure is repeated. If design specifications involve gain margin, M_r, or BW, then these values should be checked and satisfied.

Because the phase-lag control brings in more attenuation to a system, then if the design is proper, the stability margins will be improved but at the expense of lower bandwidth. The only benefit of lower bandwidth is reduced sensitivity to high-frequency noise and disturbances.

The following example illustrates the design of the phase-lag controller and all its ramifications.

▶ EXAMPLE 9-6-1 In this example, we shall use the second-order sun-seeker system described in Example 9-5-1 to illustrate the principle of design of phase-lag control. The forward-path transfer function of the uncompensated system is

$$G_p(s) = \frac{\Theta_o(s)}{A(s)} = \frac{2500K}{s(s+25)} \tag{9-131}$$

You may use **ACSYS** to solve this problem. ◀

Time-Domain Design

The time-domain specifications of the system are as follows:

1. The steady-state error of $\alpha(t)$ due to a unit-ramp function input for $\theta_r(t)$ should be $\leq 1\%$.
2. The maximum overshoot of the step response should be less than 5% or as small as possible.
3. Rise time $t_r \leq 0.5$ sec.

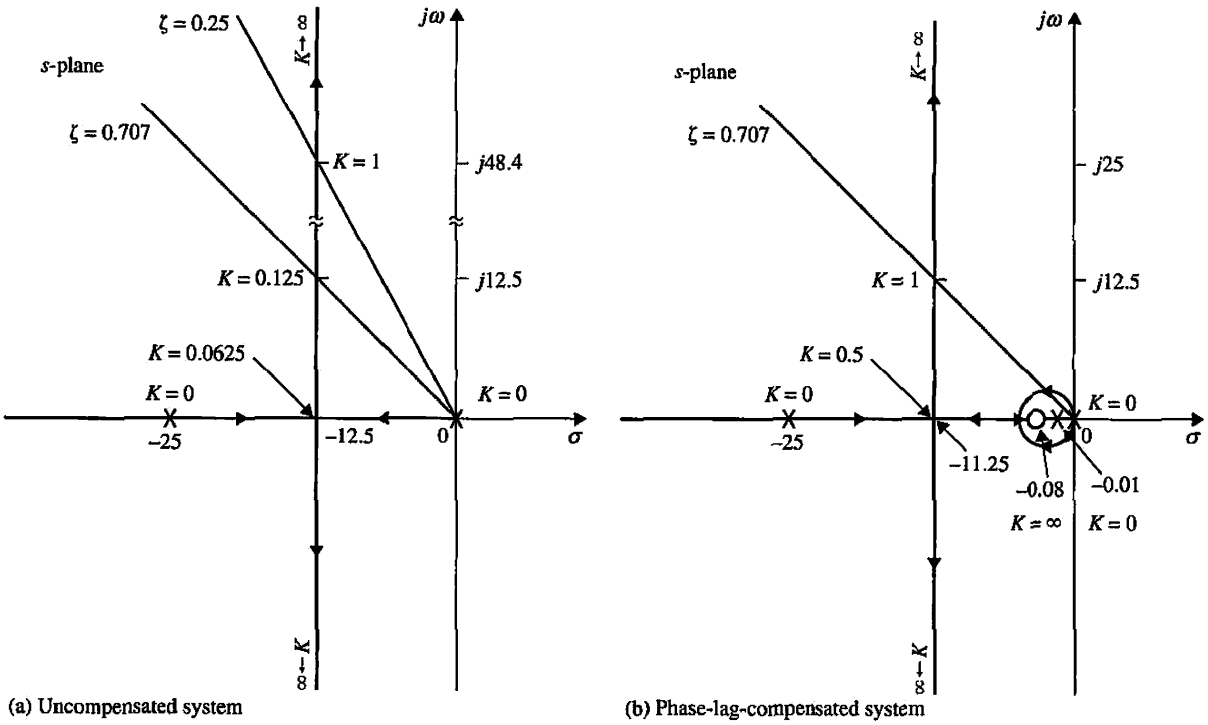


Figure 9-46 Root loci of sun-seeker system in Example 9-6-1.
$G_p(s) = \dfrac{2500K}{s(s+25)}$ $G_c(s) = \dfrac{1+aTs}{1+Ts}$ $a = 0.125$ $T = 100$.

4. Settling time $t_s \leq 0.5$ sec.
5. Due to noise problems, the bandwidth of the system must be < 50 rad/sec.

Notice that the rise-time and settling-time requirements have been relaxed considerably from the phase-lead design in Example 9-5-1. The root loci of the uncompensated system are shown in Fig. 9-46(a).

As in Example 9-5-1, we set $K = 1$ initially. The damping ratio of the uncompensated system is 0.25, and the maximum overshoot is 44.4%. Fig. 9-47 shows the unit-step response of the system with $K = 1$.

Let us select the phase-lag controller with the transfer function given in Eq. (9-121). The forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{2500K(s + 1/aT)}{s(s+25)(s+1/T)} \tag{9-132}$$

If the value of K is maintained at 1, the steady-state error will be a percent, which is better than that of the uncompensated system, since $a < 1$. *For effective phase-lag control, the pole and zero of the controller transfer function should be placed close together, and then for the type 1 system, the combination should be located relatively close to the origin of the s-plane.* From the root loci of the uncompensated system in Fig. 9-46(a), we see that, if K could be set to 0.125, the damping ratio would be 0.707, and the maximum overshoot of the system would be 4.32%. By setting the pole and zero of the controller close to $s = 0$, the shape of the loci of the dominant roots of the

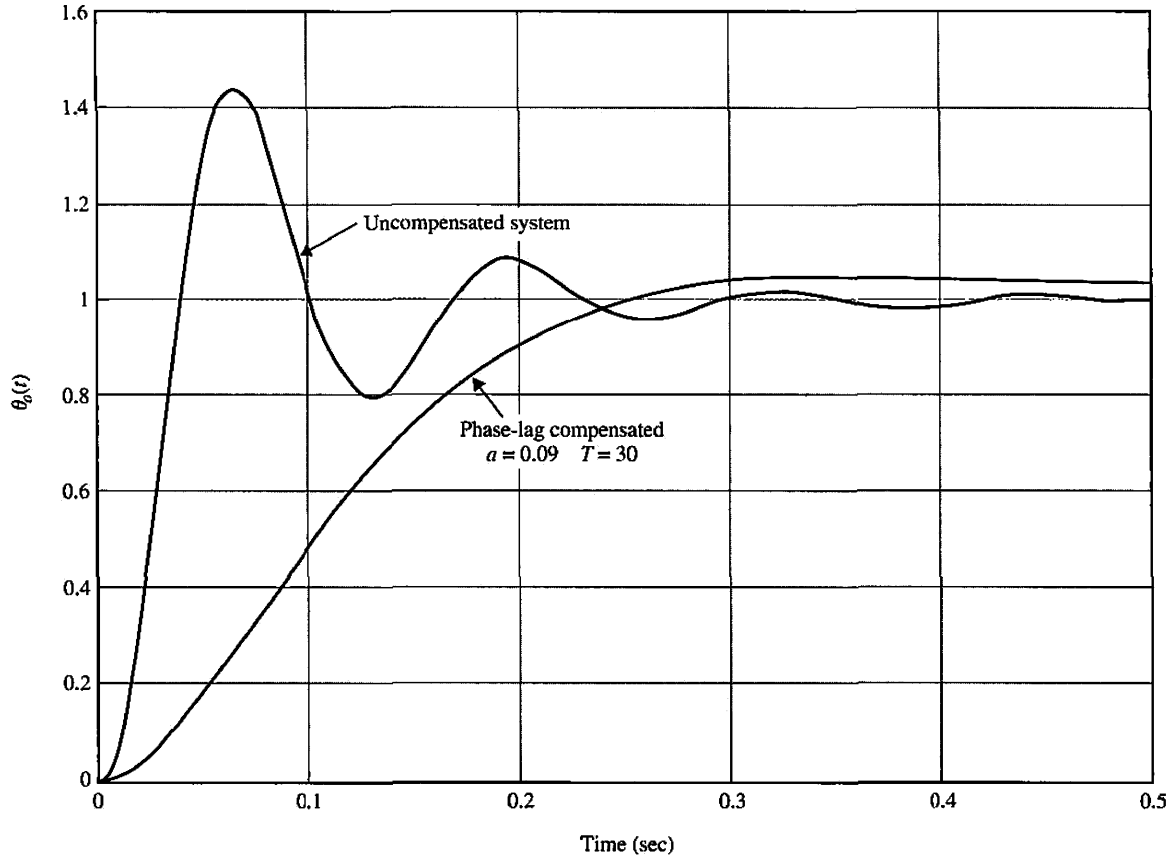


Figure 9-47 Unit-step responses of uncompensated and compensated sun-seeker systems with phase-lag controller in Example 9-6-1. $G_p(s) = \dfrac{2500K}{s(s+25)}$ $G_c(s) = \dfrac{1+aTs}{1+Ts}$ $a = 0.09$ $T = 30$.

compensated system will be very similar to those of the uncompensated system. We can find the value of a using Eq. (9-124); that is,

$$a = \frac{K \text{ to realize the desired damping}}{K \text{ to realize the steady-state performance}} = \frac{0.125}{1} = 0.125 \tag{9-133}$$

Thus, if the value of T is sufficiently large, when $K = 1$, the dominant roots of the characteristic equation will correspond to a damping ratio of approximately 0.707. Let us arbitrarily select $T = 100$. The root loci of the compensated system are shown in Fig. 9-46(b). The roots of the characteristic equation when $K = 1$, $a = 0.125$, and $T = 100$ are

$$s = -0.0805, \quad -12.465 + j12.465, \quad \text{and} \quad -12.465 - j12.465$$

which corresponds to a damping ratio of exactly 0.707. If we had chosen a smaller value for T, then the damping ratio would be slightly off 0.707. From a practical standpoint, the value of T cannot be too large, since from Eq. (9-64), $T = R_2C$, a large T would correspond to either a large capacitor or an unrealistically large resistor. To reduce the value of T and simultaneously satisfy the maximum overshoot requirement, a should also be reduced. However, a cannot be reduced indefinitely, or the zero of the controller at $-1/aT$ would be too far to the left on the real axis. Table 9-17 gives the attributes of the time-domain

TABLE 9-17 Attributes of Performance of Sun-Seeker System in Example 9-6-1 with Phase-Lag Controller

a	T	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	BW (rad/sec)	Roots of Characteristic Equation	
1.000	1	44.4	0.0255	0.2133	75.00		$-12.500 \pm j48.412$
0.125	100	4.9	0.1302	0.1515	17.67	-0.0805	$-12.465 \pm j12.465$
0.100	100	2.5	0.1517	0.2020	13.97	-0.1009	$-12.455 \pm j9.624$
0.100	50	3.4	0.1618	0.2020	14.06	-0.2037	$-12.408 \pm j9.565$
0.100	30	4.5	0.1594	0.1515	14.19	-0.3439	$-12.345 \pm j9.484$
0.100	20	5.9	0.1565	0.4040	14.33	-0.5244	$-12.263 \pm j9.382$
0.090	50	3.0	0.1746	0.2020	12.53	-0.2274	$-12.396 \pm j8.136$
0.090	30	4.4	0.1719	0.2020	12.68	-0.3852	$-12.324 \pm j8.029$
0.090	20	6.1	0.1686	0.5560	12.84	-0.5901	$-12.230 \pm j7.890$

performance of the phase-lag compensated sun-seeker system with various values for a and T. The ramifications of the various design parameters are clearly displayed.

Thus, a suitable set of controller parameters would be $a = 0.09$ and $T = 30$. With $T = 30$, selecting $C = 1\ \mu$F would require R_2 to be 30 MΩ. A smaller value for T can be realized by using a two-stage phase-lag controller. The unit-step response of the compensated system with $a = 0.09$ and $T = 30$ is shown in Fig. 9-47. Notice that the maximum overshoot is reduced at the expense of rise time and settling time. Although the settling time of the compensated system is shorter than that of the uncompensated system, it actually takes much longer for the phase-lag-compensated system to reach steady state.

It would be enlightening to explain the design of the phase-lag controller by means of the root contours. The root-contour design conducted earlier in Example 9-5-1 using Eqs. (9-80) through (9-83) for phase-lead control and Figs. 9-31 and 9-32 is still valid for phase-lag control, except that in the present case, $a < 1$. Thus, in Fig. 9-32 only the portions of the root contours that correspond to $a < 1$ are applicable for phase-lag control. These root contours clearly show that, for effective phase-lag control, the value of T should be relatively large. In Fig. 9-48 we illustrate further that the complex poles of the closed-loop transfer function are rather insensitive to the value of T when the latter is relatively large.

Frequency-Domain Design

The Bode plot of $G_p(j\omega)$ of Eq. (9-131) is shown in Fig. 9-49 for $K = 1$. The Bode plot shows that the phase margin of the uncompensated system is only 28°. Not knowing what phase margin will correspond to a maximum overshoot of less than 5%, we conduct the following series of designs using the Bode plot in Fig. 9-49. Starting with a phase margin of 45°, we observe that this phase margin can be realized if the gain-crossover frequency ω_g' is at 25 rad/sec. This means that the phase-lag controller must reduce the magnitude curve of $G_p(j\omega)$ to 0 dB at $\omega = 25$ rad/sec while it does not appreciably affect the phase curve near this frequency. Because the phase-lag controller still contributes a small negative phase when the corner frequency $1/aT$ is placed at $1/10$ of the value of ω_g', it is a safe measure to choose ω_g' at somewhat less than 25 rad/sec, say, 20 rad/sec.

From the Bode plot, the value of $\left|G_p\left(j\omega_g'\right)\right|_{\text{dB}}$ at $\omega_g' = 20$ rad/sec is 11.7 dB. Thus, using Eq. (9-128), we have

$$a = 10^{-|G_p(j\omega_g')|/20} = 10^{-11.7/20} = 0.26 \tag{9-134}$$

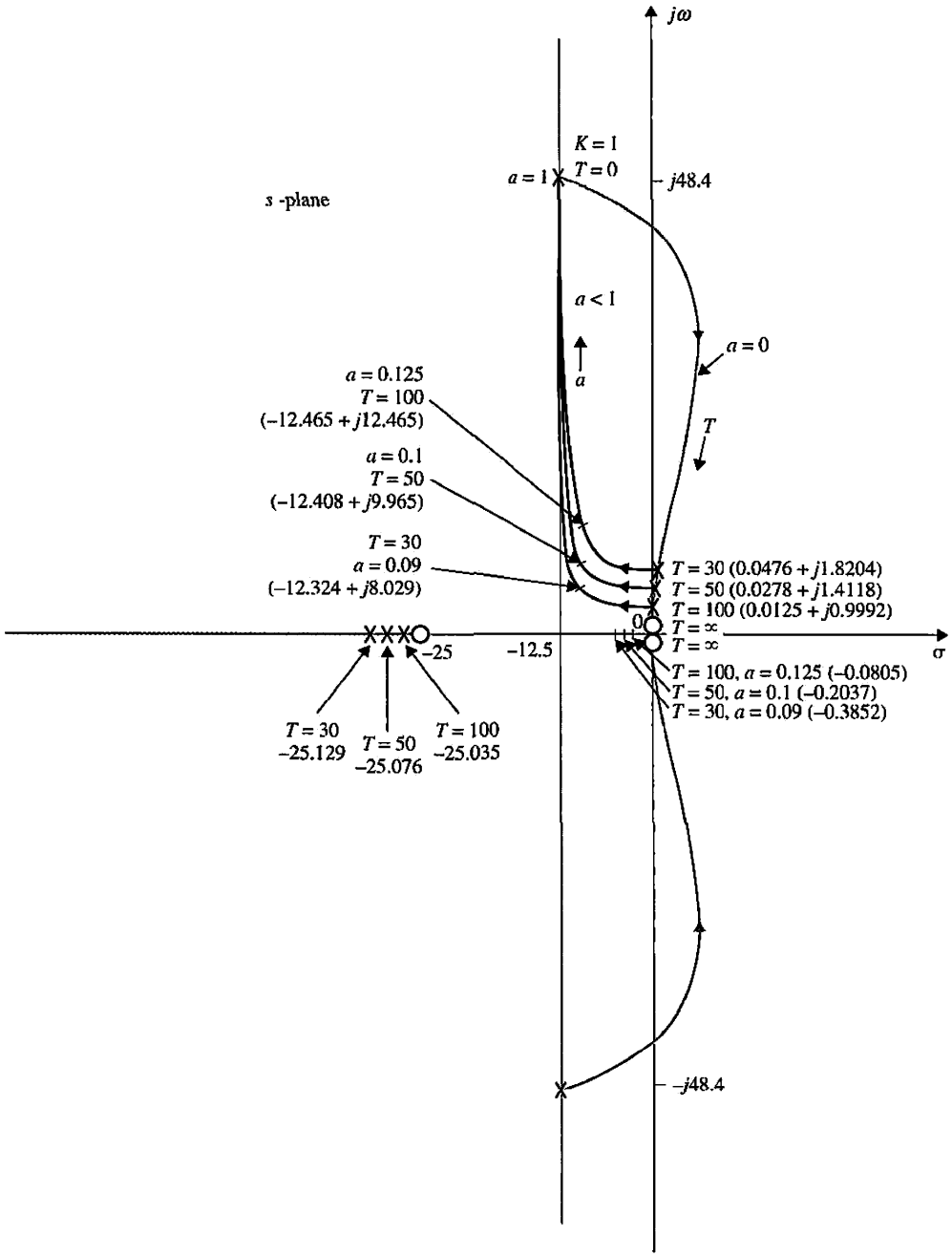


Figure 9-48 Root contours of sun-seeker system in Example 9-6-1 with phase-lag controller.

The value of $1/aT$ is chosen to be at $1/10$ the value of $\omega_g' = 20$ rad/sec. Thus,

$$\frac{1}{aT} = \frac{\omega_g'}{10} = \frac{20}{10} = 2 \text{ rad/sec} \tag{9-135}$$

and

$$T = \frac{1}{2a} = \frac{1}{0.52} = 1.923 \tag{9-136}$$

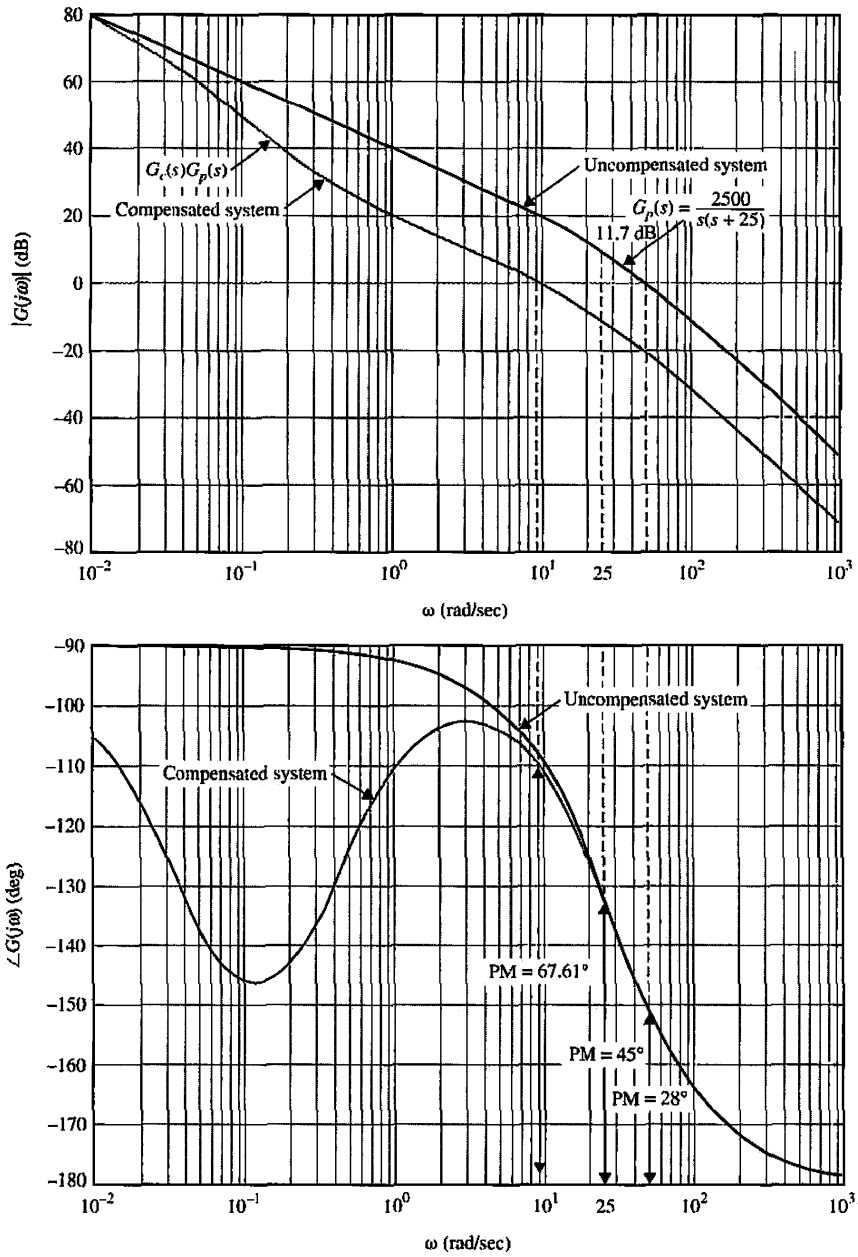


Figure 9-49 Bode plot of uncompensated and compensated systems with phase-lag controller in Example 9-6-1. $G_c(s) = \dfrac{1+3s}{1+30s}$ $G_p(s) = \dfrac{2500}{s(s+25)}$.

Checking out the unit-step response of the system with the designed phase-lag control, we found that the maximum overshoot is 24.5%. The next step is to try aiming at a higher phase margin. Table 9-18 gives the various design results by using various desired phase margins up to 80°.

Examining the results in Table 9-18, we see that none of the cases satisfies the maximum overshoot requirement of $\leq 5\%$. The $a = 0.044$ and $T = 52.5$ case yields the best maximum overshoot, but the value of T is too large to be practical. Thus, we single out the case with $a = 0.1$ and $T = 10$ and refine the design by increasing the value of T. As shown in Table 9-17,

TABLE 9-18 Performance Attributes of Sun-Seeker System in Example 9-6-1 with Phase-Lag Controller

Desired PM (deg)	a	T	Actual PM (deg)	M_r	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
45	0.26	1.923	46.78	1.27	33.37	24.5	0.0605	0.2222
60	0.178	3.75	54.0	1.19	25.07	17.5	0.0823	0.303
70	0.1	10	63.87	1.08	14.72	10.0	0.1369	0.7778
80	0.044	52.5	74.68	1.07	5.7	7.1	0.3635	1.933

when $a = 0.1$ and $T = 30$, the maximum overshoot is reduced to 4.5%. The Bode plot of the compensated system is shown in Fig. 9-49. The phase margin is 67.61°.

The unit-step response of the phase-lag-compensated system shown in Fig. 9-47 points out a major disadvantage of the phase-lag control. Because the phase-lag controller is essentially a low-pass filter, the rise time and settling time of the compensated system are usually increased. However, we shall show by the following example that phase-lag control can be more versatile and has a wider range of effectiveness in improving stability than the single-stage phase-lead controller, especially if the system has low or negative damping.

EXAMPLE 9-6-2 Consider the sun-seeker system designed in Example 9-5-3, with the forward-path transfer function given in Eq. (9-111). Let us restore the gain K, so that a root-locus plot can be made for the system. Then, Eq. (9-111) is written

$$G_p(s) = \frac{156,250,000K}{s(s^2 + 625s + 156,250)} \tag{9-137}$$

The root loci of the closed-loop system are shown in Fig. 9-50. When $K = 1$, the system is unstable, and the characteristic equation roots are at -713.14, $44.07 + j466.01$, and $44.07 - j466.01$.

Example 9-5-3 shows that the performance specification on stability cannot be achieved with a single-stage phase-lead controller. Let the performance criteria be as follows:

Maximum overshoot $\leq 5\%$

Rise time $t_r \leq 0.02$ sec

Settling time $t_s \leq 0.02$ sec

Let us assume that the desired relative damping ratio is 0.707. Fig. 9-50 shows that, when $K = 0.10675$, the dominant characteristic equation roots of the uncompensated system are at $-172.77 \pm j172.73$, which correspond to a damping ratio of 0.707. Thus, the value of a is determined from Eq. (9-124),

$$a = \frac{K \text{ to realize the desired damping}}{K \text{ to realize the steady-state performance}} = \frac{0.10675}{1} = 0.10675 \tag{9-138}$$

Let $a = 0.1$. Because the loci of the dominant roots are far away from the origin in the s-plane, the value of T has a wide range of flexibility. Table 9-19 shows the performance results when $a = 0.1$ and for various values of T.

TABLE 9-19 Performance Attributes of Sun-Seeker System in Example 9-6-2 with Phase-Lag Controller

a	T	BW (rad/sec)	PM (deg)	% Max Overshoot	t_r (sec)	t_s (sec)
0.1	20	173.5	66.94	1.2	0.01273	0.01616
0.1	10	174	66.68	1.6	0.01262	0.01616
0.1	5	174.8	66.15	2.5	0.01241	0.01616
0.1	2	177.2	64.56	4.9	0.01601	0.0101

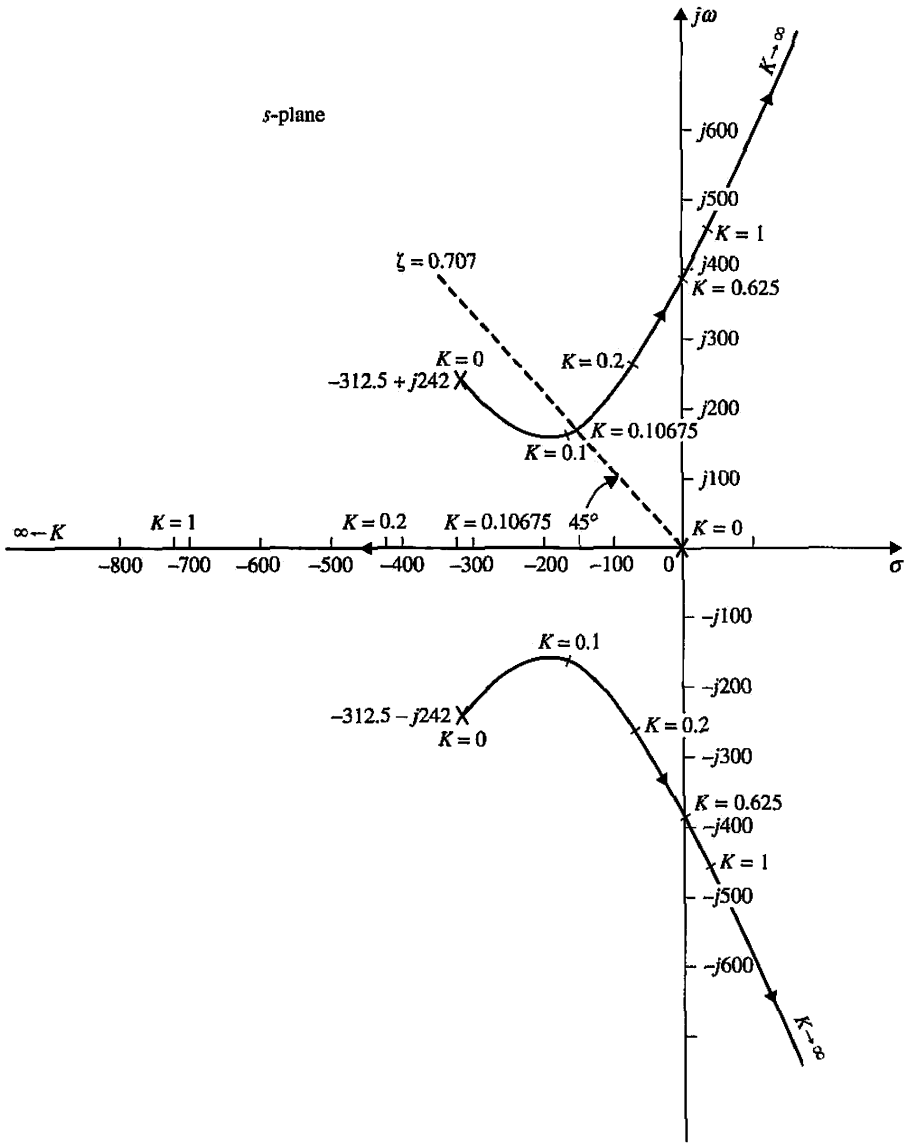


Figure 9-50 Root loci of uncompensated system in Example 9-6-2.

$$G_p(s) = \frac{156,250,000K}{s(s^2 + 625s + 156,250)}.$$

Therefore, the conclusion is that only one stage of the phase-lag controller is needed to satisfy the stability requirement, whereas two stages of the phase-lead controller are needed, as shown in Example 9-5-3. ◀

Sensitivity Function

The sensitivity function $|S_G^M(j\omega)|$ of the phase-lag compensated system with $a = 0.1$ and $T = 20$ is shown in Fig. 9-51. Notice that the sensitivity function is less than unity for frequencies up to only 102 rad/sec. This is due to the low bandwidth of the system as a result of phase-lag control.

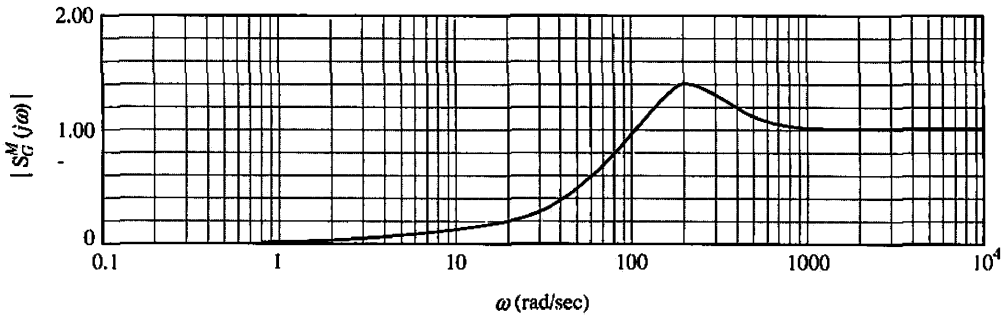


Figure 9-51 Sensitivity function of phase-lag-compensated system in Example 9-6-2.

9-6-3 Effects and Limitations of Phase-Lag Control

From the results of the preceding illustrative examples, the effects and limitations of phase-lag control on the performance of linear control systems can be summarized as follows.

1. For a given forward-path gain K, the magnitude of the forward-path transfer function is attenuated near the gain-crossover frequency, thus improving the relative stability of the system.
2. The gain-crossover frequency is decreased, and thus the bandwidth of the system is reduced.
3. The rise time and settling time of the system are usually longer, because the bandwidth is usually decreased.
4. The system is more sensitive to parameter variations because the sensitivity function is greater than unity for all frequencies approximately greater than the bandwidth of the system.

▶ 9-7 DESIGN WITH LEAD–LAG CONTROLLER

We have learned from preceding sections that phase-lead control generally improves rise time and damping but increases the natural frequency of the closed-loop system. However, phase-lag control when applied properly improves damping but usually results in a longer rise time and settling time. Therefore, each of these control schemes has its advantages, disadvantages, and limitations, and there are many systems that cannot be satisfactorily compensated by either scheme acting alone. It is natural, therefore, whenever necessary, to consider using a combination of the lead and lag controllers, so that the advantages of both schemes are utilized.

The transfer function of a simple lag–lead (or lead–lag) controller can be written

$$G_c(s) = G_{c1}(s)G_{c2}(s) = \underbrace{\left(\frac{1 + a_1T_1s}{1 + T_1s}\right)}_{|\leftarrow \text{lead} \rightarrow|}\underbrace{\left(\frac{1 + a_2T_2s}{1 + T_2s}\right)}_{|\leftarrow \text{lag} \rightarrow|} \quad (a_1 > 1,\ a_2 < 1) \tag{9-139}$$

The gain factors of the lead and lag controllers are not included because, as shown previously, these gain and attenuation are compensated eventually by the adjustment of the forward gain K.

Because the lead–lag controller transfer function in Eq. (9-139) now has four unknown parameters, its design is not as straightforward as the single-stage phase-lead or phase-lag controller. *In general, the phase-lead portion of the controller is used mainly to achieve a shorter rise time and higher bandwidth, and the phase-lag portion is brought in to provide major damping of the system.* Either the phase-lead or the phase-lag control can be designed first. We shall use Example 9-7-1 to illustrate the design steps.

▶ **EXAMPLE 9-7-1** As an illustrative example of designing a lead–lag controller, let us consider the sun-seeker system of Example 9-5-3. The uncompensated system with $K = 1$ was shown to be unstable. A two-stage phase-lead controller was designed in Example 9-6-1, and a single-stage phase-lag controller was designed in Example 9-6-2.

Based on the design in Example 9-5-3, we can first select a phase-lead control with $a = 70$ and $T_1 = 0.00004$. The remaining phase-lag control can be designed using either the root-locus method or the Bode plot method. Table 9-20 gives the results by letting $T_2 = 2$, which is an insensitive parameter, and various values of a. The results in Table 9-20 show that the optimal value of a_2, from the standpoint of minimizing the maximum overshoot, for $a_1 = 70$ and $T_2 = 0.00004$, is approximately 0.2. Compared with the single-stage phase-lag control designed in Example 9-6-1, the BW is increased to 351.4 rad/sec from 66.94 rad/sec, and the rise time is reduced to 0.00668 sec from 0.01273 sec. The system with the lead–lag controller should be more robust, because the magnitude of the sensitivity function should not increase to unity until near the BW of 351.4 rad/sec. As a comparison, the unit-step responses of the system with the two-stage phase-lead control, the single-stage phase-lag control, and the lead–lag control are shown in Fig. 9-52.

TABLE 9-20 Performance Attributes of Sun-Seeker System in Example 9-7-1 with Lead–Lag Controller: $a_1 = 70$, $T_1 = 0.00004$

a_2	T_2	PM (deg)	M_r	BW (rad/sec)	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.1	20	81.81	1.004	122.2	0.4	0.01843	0.02626
0.15	20	76.62	1.002	225.5	0.2	0.00985	0.01515
0.20	20	70.39	1.001	351.4	0.1	0.00668	0.00909
0.25	20	63.87	1.001	443.0	4.9	0.00530	0.00707

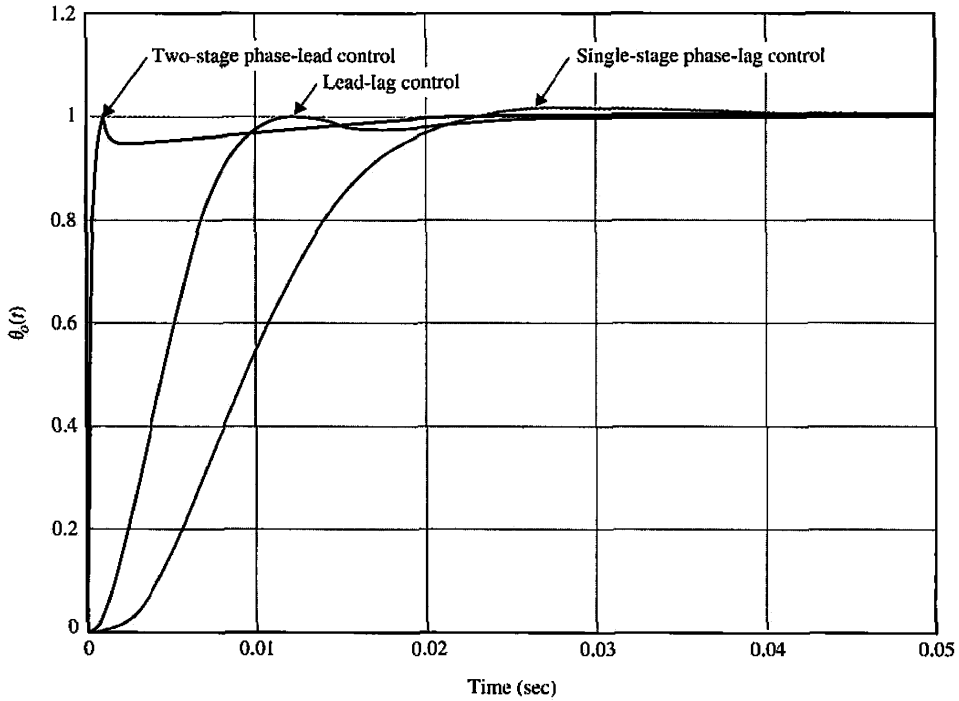


Figure 9-52 Sun-seeker system in Example 9-7-1 with single-stage phase-lag controller, lead–lag controller, and two-stage phase-lead controller.

It should be noted that the bandwidth and rise time of the sun-seeker system can be further increased and reduced, respectively, by using a larger value of a_1 for the phase-lead portion of the controller. However, the resulting step response will have a large undershoot, although the maximum overshoot can be kept small. ◀

▶ 9-8 POLE-ZERO-CANCELLATION DESIGN: NOTCH FILTER

The transfer functions of many controlled processes contain one or more pairs of complex-conjugate poles that are very close to the imaginary axis of the s-plane. These complex poles usually cause the closed-loop system to be lightly damped or unstable. One immediate solution is to use a controller that has a transfer function with zeros selected, which would cancel the undesirable poles of the controlled process, and to place the poles of the controller at more desirable locations in the s-plane to achieve the desired dynamic performance. For example, if the transfer function of a process is

$$G_p(s) = \frac{K}{s(s^2 + s + 10)} \tag{9-140}$$

in which the complex-conjugate poles may cause stability problems in the closed-loop system when the value of K is large, the suggested series controller may be of the form

$$G_c(s) = \frac{s^2 + s + 10}{s^2 + as + b} \tag{9-141}$$

The constants a and b may be selected according to the performance specifications of the closed-loop system.

There are practical difficulties with the pole-zero-cancellation design scheme that should prevent the method from being used indiscriminately. The problem is that in practice *exact* cancellation of poles and zeros of transfer functions is rarely possible. In practice, the transfer function of the process, $G_p(s)$, is usually determined through testing and physical modeling; linearization of a nonlinear process and approximation of a complex process are unavoidable. Thus, the *true* poles and zeros of the transfer function of the process may not be accurately modeled. In fact, the true order of the system may even be higher than that represented by the transfer function used for modeling purposes. Another difficulty is that the dynamic properties of the process may vary, even very slowly, due to aging of the system components or changes in the operating environment, so the poles and zeros of the transfer function may move during the operation of the system. The parameters of the controller are constrained by the actual physical components available and cannot be assigned arbitrarily. For these and other reasons, even if we could precisely design the poles and zeros of the transfer function of the controller, exact pole–zero cancellation is almost never possible in practice. We will now show that, in most cases, exact cancellation is *not* really necessary to effectively negate the influence of the undesirable poles using pole-zero-cancellation compensation schemes.

Let us assume that a controlled process is represented by

$$G_p(s) = \frac{K}{s(s + p_1)(s + \bar{p}_1)} \tag{9-142}$$

where p_1 and $\bar{p}_1$ are the two complex-conjugate poles that are to be canceled. Let the transfer function of the series controller be

$$G_c(s) = \frac{(s + p_1 + \varepsilon_1)(s + \bar{p}_1 + \bar{\varepsilon}_1)}{s^2 + as + b} \tag{9-143}$$

where ε_1 is a complex number whose magnitude is very small and $\bar{\varepsilon}_1$ is its complex conjugate. The open-loop transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{K(s + p_1 - \varepsilon_1)(s + \bar{p}_1 + \bar{\varepsilon}_1)}{s(s + p_1)(s + \bar{p}_1)(s^2 + as + b)} \tag{9-144}$$

Because of inexact cancellation, we cannot discard the terms $(s + p_1)(s + \bar{p}_1)$ in the denominator of Eq. (9-144). The closed-loop transfer function is

$$\frac{Y(s)}{R(s)} = \frac{K(s + p_1 + \varepsilon_1)(s + \bar{p}_1 + \bar{\varepsilon}_1)}{s(s + p_1)(s + \bar{p}_1)(s^2 + as + b) + K(s - p_1 + \varepsilon_1)(s + \bar{p}_1 + \varepsilon_1)} \tag{9-145}$$

The root-locus diagram in Fig. 9-53 explains the effect of inexact pole–zero cancellation. Notice that the two closed-loop poles as a result of inexact cancellation lie between the pairs of poles and zeros at $s = -p_1$, $-\bar{p}_1$ and $-p_1 - \varepsilon_1$, $-\bar{p}_1 - \bar{\varepsilon}_1$, respectively. Thus,

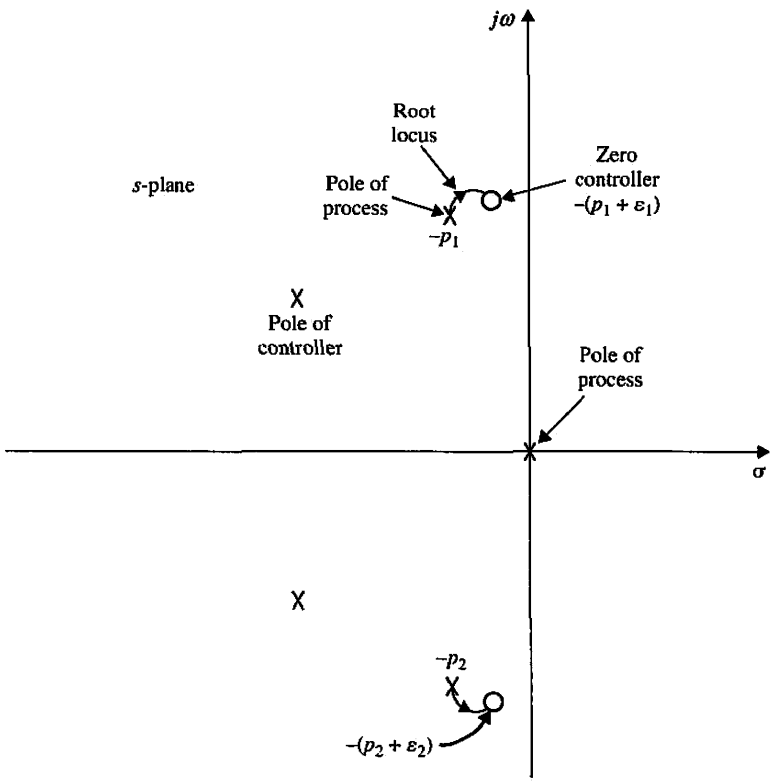


Figure 9-53 Pole–zero configuration and root loci of inexact cancellation.

these closed-loop poles are very close to the open-loop poles and zeros that are meant to be canceled. Eq. (9-145) can be approximated as

$$\frac{Y(s)}{R(s)} \cong \frac{K(s + p_1 + \varepsilon_1)(s + \bar{p}_1 + \bar{\varepsilon}_2)}{(s + p_1 + \delta_1)(s + \bar{p}_1 + \delta_1)(s^3 + as + b + K)} \tag{9-146}$$

where δ_1 and δ_1 are a pair of very small complex-conjugate numbers that depend on ε_1, $\bar{\varepsilon}_1$, and all the other parameters. The partial-fraction expansion of Eq. (9-146) is

$$\frac{Y(s)}{R(s)} \cong \frac{K_1}{s + p_1 + \delta_1} + \frac{K_2}{s + \bar{p}_1 + \delta_1} + \text{terms due to the remaining poles} \tag{9-147}$$

We can show that K_1 is proportional to $\varepsilon_1 - \delta_1$, which is a very small number. Similarly, K_2 is also very small. *This exercise simply shows that, although the poles at $-p_1$ and $-p_2$ cannot be canceled precisely, the resulting transient-response terms due to inexact cancellation will have insignificant amplitudes, so unless the controller zeros earmarked for cancellation are too far off target, the effect can be neglected for all practical purposes.* Another way of viewing this problem is that the zeros of $G(s)$ are retained as the zeros of closed-loop transfer function $Y(s)/R(s)$, so from Eq. (9-146), we see that the two pairs of poles and zeros are close enough to be canceled from the transient-response standpoint.

Keep in mind that we should never attempt to cancel poles that are in the right-half s*-plane, because any inexact cancellation will result in an unstable system. Inexact cancellation of poles could cause difficulties if the unwanted poles of the process transfer function are very close to or right on the imaginary axis of the* s*-plane. In this case, inexact cancellation may also result in an unstable system.* Fig. 9-54(a) illustrates a situation in which

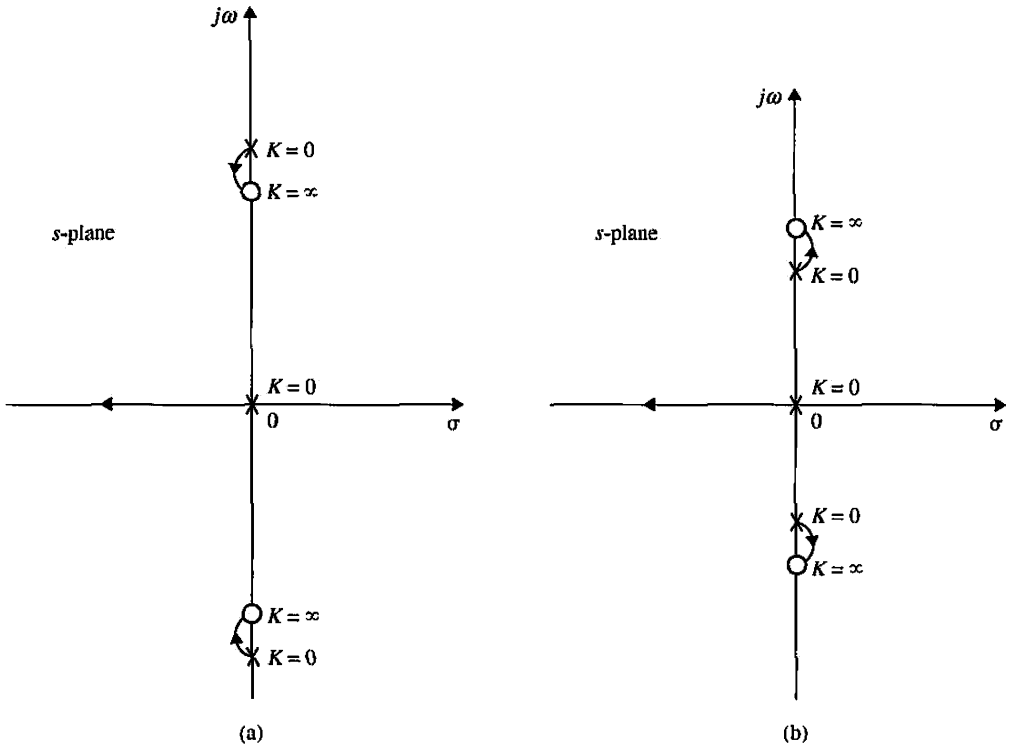


Figure 9-54 Root loci showing the effects of inexact pole–zero cancellations.

the relative positions of the poles and zeros intended for cancellation result in a stable system, whereas in Fig. 9-54(b), the inexact cancellation is unacceptable. The relative distance between the poles and zeros intended for cancellation is small, which results in residual terms in the time response solution. Although these terms have very small amplitudes, they tend to grow without bound as time increases. Hence the system response becomes unstable.

9-8-1 Second-Order Active Filter

Transfer functions with complex poles and/or zeros can be realized by electric circuits with op-amps. Consider the transfer function

$$G_c(s) = \frac{E_2(s)}{E_1(s)} = K\frac{s^2 + b_1 s + b_2}{s^2 + a_1 s + a_2} \tag{9-148}$$

where a_1, a_2, b_1, and b_2 are real constants. The active-filter realization of Eq. (9-148) can be accomplished by using the direct decomposition scheme of state variables discussed in Section 10-10. A typical op-amp circuit is shown in Fig. 9-55. The parameters of the transfer function in Eq. (9-148) are related to the circuit parameters as follows:

$$K = -\frac{R_6}{R_7} \tag{9-149}$$

$$a_1 = \frac{1}{R_1 C_1} \tag{9-150}$$

$$a_2 = \frac{1}{R_2 R_4 C_1 C_2} \tag{9-151}$$

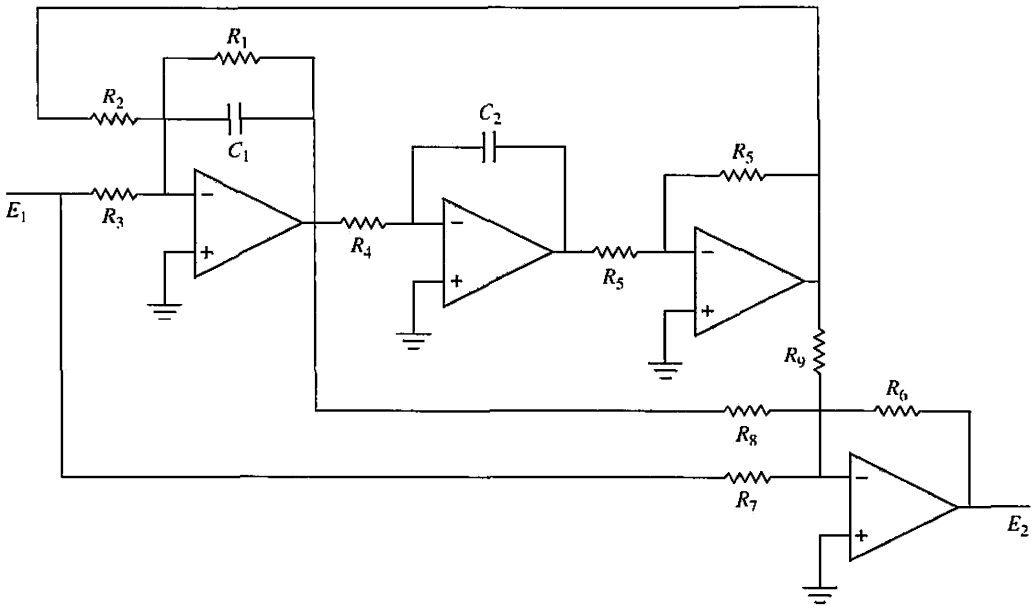


Figure 9-55 Op-amp circuit realization of the second-order transfer function. $\frac{E_2(s)}{E_1(s)} = K\frac{s^2 + b_1 s + b_2}{s^2 + a_1 s + a_2}$.

$$b_1 = \left(1 - \frac{R_1 R_7}{R_3 R_8}\right) a_1 \quad (b_1 < a_1) \tag{9-152}$$

$$b_2 = \left(1 - \frac{R_2 R_7}{R_3 R_9}\right) a_2 \quad (b_2 < a_2) \tag{9-153}$$

Because $b_1 < a_1$, the zeros of $G_c(s)$ in Eq. (9-148) are less damped and are closer to the origin in the s-plane than the poles. By setting various combinations of R_7 and R_8, and R_9 to infinity, a variety of second-order transfer functions can be realized. Note that all the parameters can be adjusted independently of one another. For example, R_1 can be adjusted to set a_1; R_4 can be adjusted to set a_2; and b_1 and b_2 are set by adjusting R_8 and R_9, respectively. The gain factor K is controlled independently by R_6.

9-8-2 Frequency-Domain Interpretation and Design

While it is simple to grasp the idea of pole-zero-cancellation design in the s-domain, the frequency-domain provides added perspective to the design principles. Fig. 9-56

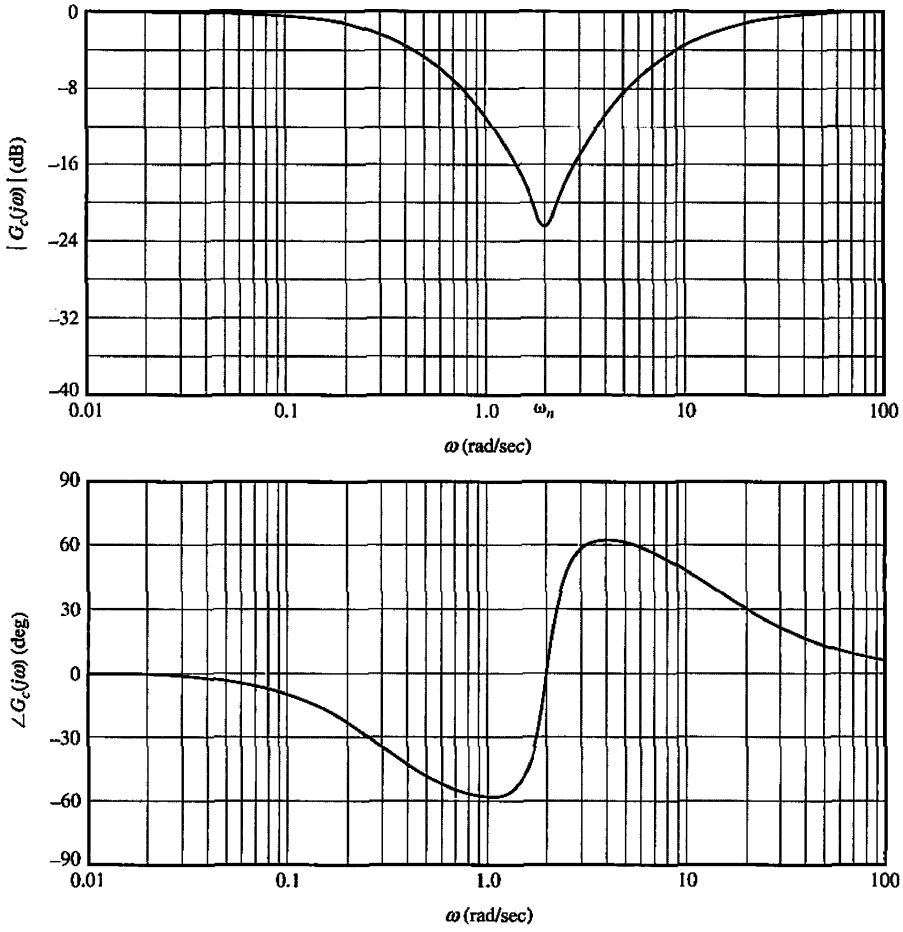


Figure 9-56 Bode plot of a notch controller with the transfer function.
$G(s) = \dfrac{(s^2 + 0.8s + 4)}{(s + 0.384)(s + 10.42)}$.

illustrates the Bode plot of the transfer function of a typical second-order controller with complex zeros. The magnitude plot of the controller typically has a "notch" at the resonant frequency ω_n. The phase plot is negative below and positive above the resonant frequency, while passing through zero degrees at the resonant frequency. The attenuation of the magnitude curve and the positive-phase characteristics can be used effectively to improve the stability of a linear system. Because of the "notch" characteristic in the magnitude curve, the controller is also referred to in the industry as a **notch filter** or **notch controller**.

From the frequency-domain standpoint, the notch controller has advantages over the phase-lead and phase-lag controllers in certain design conditions, because the magnitude and phase characteristics do not affect the high- and low-frequency properties of the system. Without using the pole-zero-cancellation principle, the design of the notch controller for compensation in the frequency domain involves the determination of the amount of attenuation required and the resonant frequency of the controller.

Let us express the transfer function of the notch controller in Eq. (9-148) as

$$G_c(s) = \frac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2} \tag{9-154}$$

where we have made the simplification by assuming that $a_2 = b_2$.

The attenuation provided by the magnitude of $G_c(j\omega)$ at the resonant frequency ω_n is

$$|G_c(j\omega_n)| = \frac{\zeta_z}{\zeta_p} \tag{9-155}$$

Thus, knowing the maximum attenuation required at ω_n, the ratio of ζ_z/ζ_p is known.

The following example illustrates the design of the notch controller based on pole–zero cancellation and required attenuation at the resonant frequency.

EXAMPLE 9-8-1 Complex-conjugate poles in system transfer functions are often due to compliances in the coupling between mechanical elements. For instance, if the shaft between the motor and load is nonrigid, the shaft is modeled as a torsional spring, which could lead to complex-conjugate poles in the process transfer function. Fig. 9-57 shows a speed-control system in which the coupling between the motor and the load is modeled as a torsional spring. The system equations are

$$T_m(t) = J_m\frac{d\omega_m(t)}{dt} + B_m\omega_m(t) + J_L\frac{d\omega_L(t)}{dt} \tag{9-156}$$

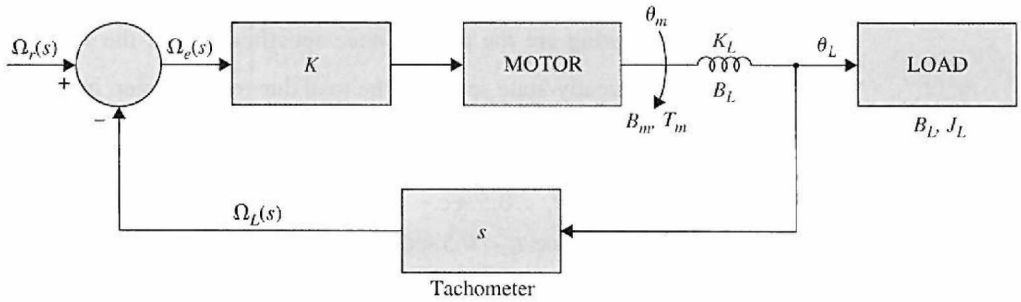


Figure 9-57 Block diagram of speed-control system in Example 9-8-1.

$$K_L[\theta_m(t) - \theta_L(t)] + B_L[\omega_m(t) - \omega_L(t)] = J_L \frac{d\omega_L(t)}{dt} \tag{9-157}$$

$$T_m(t) = K\omega_e(t) \tag{9-158}$$

$$\omega_e(t) = \omega_r(t) - \omega_L(t) \tag{9-159}$$

$T_m(t)$ = motor torque
$\omega_m(t)$ = motor angular velocity
$\omega_L(t)$ = load angular velocity
$\theta_L(t)$ = load angular displacement
$\theta_m(t)$ = motor angular displacement
J_m = motor inertia = 0.0001 oz-in.-sec^2
J_L = load inertia = 0.0005 oz-in.-sec^2
B_m = viscous-friction coefficient of motor = 0.01 oz-in.-sec
B_L = viscous-friction coefficient of shaft = 0.001 oz-in.-sec
K_L = spring constant of shaft = 100 on-in./rad
K = amplifier gain = 1

The loop transfer function of the system is

$$\begin{aligned} G_p(s) &= \frac{\Omega_L(s)}{\Omega_e(s)} \\ &= \frac{B_L s + K_L}{J_m J_L s^3 + (B_m J_L + B_L J_m + B_L J_L)s^2 + (K_L J_L + B_m B_L + K_L J_m)s + B_m K_L} \end{aligned} \tag{9-160}$$

By substituting the system parameters in the last equation, $G_p(s)$ becomes

$$\begin{aligned} G_p(s) &= \frac{20,000(s + 100,000)}{s^3 + 112s^2 + 1,200,200s + 20,000,000} \\ &= \frac{20,000(s + 100,000)}{(s + 16.69)(s + 47.66 + j1094)(s + 47.66 - j1094)} \end{aligned} \tag{9-161}$$

Thus, the shaft compliance between the motor and the load creates two complex-conjugate poles in $G_p(s)$ that are lightly damped. The resonant frequency is approximately 1095 rad/sec, and the closed-loop system is unstable. The complex poles of $G_p(s)$ would cause the speed response to oscillate even if the system were stable.

Pole-Zero-Cancellation Design with Notch Controller

The following are the performance specifications of the system:

The steady-state speed of the load due to a unit-step input should have an error of not more than 1%.

Maximum overshoot of output speed $\leq$ 5%.

Rise time $t_r < 0.5$ sec.

Settling time $t_s < 0.5$ sec.

To compensate the system, we need to get rid, or, perhaps more realistically, minimize the effect, of the complex poles of $G_p(s)$ at $s = -47.66 + j1094$ and $-47.66 - j1094$. Let

us select a notch controller with the transfer function given in Eq. (9-154) to improve the performance of the system. The complex-conjugate zeros of the controller should be so placed that they will cancel the undesirable poles of the process. Therefore, the transfer function of the notch controller should be

$$G_c(s) = \frac{s^2 + 95.3s + 1,198,606.6}{s^2 + 2\zeta_p\omega_n s + \omega_n^2} \tag{9-162}$$

The forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{20,000(s + 100,000)}{(s + 16.69)(s^2 + 2\zeta_p\omega_n s + \omega_n^2)} \tag{9-163}$$

Because the system is type 0, the step-error constant is

$$K_P = \lim_{s \to 0} G(s) = \frac{2 \times 10^9}{16.69 \times \omega_n^2} = \frac{1.198 \times 10^8}{\omega_n^2} \tag{9-164}$$

For a unit-step input, the steady-state error of the system is written

$$e_{ss} = \lim_{t \to \infty} \omega_e(t) = \lim_{s \to 0} s\Omega_e(s) = \frac{1}{1 + K_P} \tag{9-165}$$

Thus, for the steady-state error to be less than or equal to 1%, $K_P \geq 99$. The corresponding requirement on ω_n is found from Eq. (9-164),

$$\omega_n \leq 1210 \tag{9-166}$$

We can show that, from the stability standpoint, it is better to select a large value for ω_n. Thus, let $\omega_n = 1200$ rad/sec, which is at the high end of the allowable range from the steady-state error standpoint. However, the design specifications given above can only be achieved by using a very large value for ζ_p. For example, when $\zeta_p = 15,000$, the time response has the following performance attributes:

Maximum overshoot = 3.7%

Rise time $t_r = 0.1897$ sec

Settling time $t_s = 0.256$ sec

Although the performance requirements are satisfied, the solution is unrealistic, because the extremely large value for ζ_p cannot be realized by physically available controller components.

Let us choose $\zeta_p = 10$ and $\omega_n = 1000$ rad/sec. The forward-path transfer function of the system with the notch controller is

$$G(s) = G_c(s)G_p(s) = \frac{20,000(s + 100,000)}{(s + 16.69)(s + 50)(s + 19,950)} \tag{9-167}$$

We can show that the system is stable, but the maximum overshoot is 71.6%. Now we can regard the transfer function in Eq. (9-167) as a new design problem. There are a number of

TABLE 9-21 Time-Domain Performance Attributes of System in Example 9-8-1 with Notch-Phase-Lag Controller

a	T	aT	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.001	10	0.01	14.8	0.1244	0.3836
0.002	10	0.02	10.0	0.1290	0.3655
0.004	10	0.04	3.2	0.1348	0.1785
0.005	10	0.05	1.0	0.1375	0.1818
0.0055	10	0.055	0.3	0.1386	0.1889
0.006	10	0.06	0	0.1400	0.1948

possible solutions to the problem of meeting the design specifications given. We can introduce a phase-lag controller or a PI controller, among other possibilities.

Second-Stage Phase-Lag Controller Design

Let us design a phase-lag controller as the second-stage controller for the system. The roots of the characteristic equation of the system with the notch controller are at $s = -19954$, $-31.328 + j316.36$, and $-31.328 - j316.36$. The transfer function of the phase-lag controller is

$$G_{c1}(s) = \frac{1 + aTs}{1 + Ts} \quad (a < 1) \tag{9-168}$$

where for design purposes we have omitted the gain factor $1/a$ in Eq. (9-168).

Let us select $T = 10$ for the phase-lag controller. Table 9-21 gives time-domain performance attributes for various values of a. The best value of a from the overall performance standpoint appears to be 0.005. Thus, the transfer function of the phase-lag controller is

$$G_{c1}(s) = \frac{1 + aTs}{1 + Ts} = \frac{1 + 0.05s}{1 + 10s} \tag{9-169}$$

The forward-path transfer function of the compensated system with the notch-phase-lag controller is

$$G(s) = G_c(s)G_{c1}(s)G_p(s) = \frac{20,000(s + 100,000)(1 + 0.05s)}{(s + 16.69)(s + 50)(s + 19,950)(1 + 10s)} \tag{9-170}$$

The unit-step response of the system is shown in Fig. 9-58. Because the step-error constant is 120.13, the steady-state speed error due to a step input is 1/120.13, or 0.83%.

Second-Stage PI Controller Design

A PI controller can be applied to the system to improve the steady-state error and the stability simultaneously. The transfer function of the PI control is written

$$G_{c2}(s) = K_P + \frac{K_I}{s} = K_P\left(\frac{s + K_I/K_P}{s}\right) \tag{9-171}$$

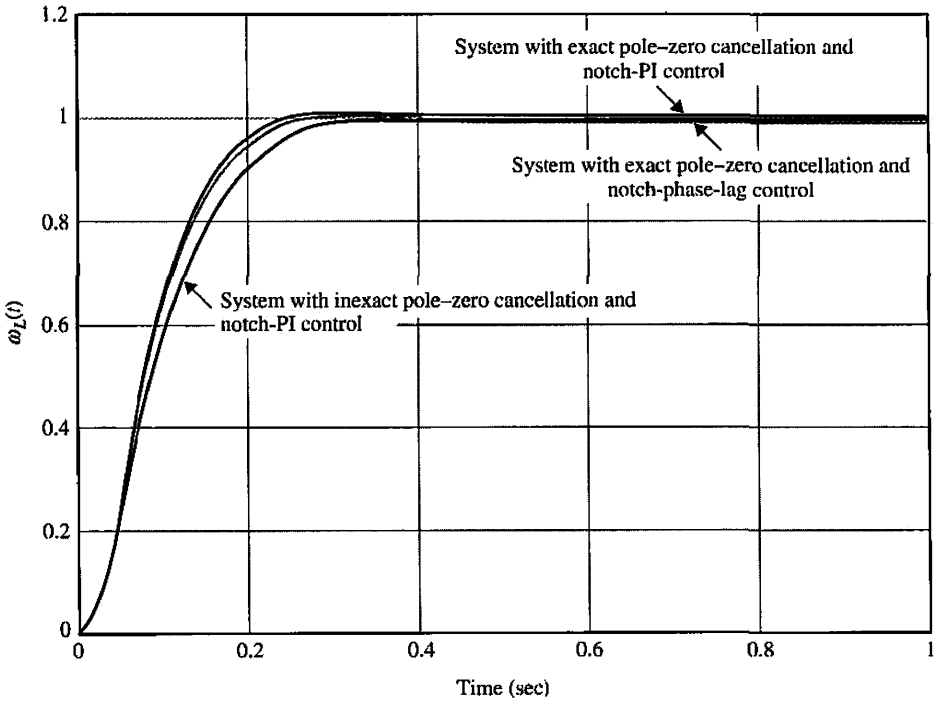


Figure 9-58 Unit-step responses of speed-control system in Example 9-8-1.

We can design the PI controller based on the phase-lag controller by writing Eq. (9-169) as

$$G_{c1}(s) = 0.005\left(\frac{s+20}{s+0.1}\right) \tag{9-172}$$

Thus, we can set $K_P = 0.005$ and $K_I/K_P = 20$. Then, $K_I = 0.1$. Fig. 9-58 shows the unit-step response of the system with the notch-PI controller. The attributes of the step response are as follows:

% Maximum overshoot = 1 %

Rise time $t_r = 0.1380$ sec

Settling time $t_s = 0.1818$ sec

which are extremely close to those with the notch-phase-lag controller, except that in the notch-PI case the steady-state velocity error is zero when the input is a step function.

Sensitivity Due to Imperfect Pole–Zero Cancellation

As mentioned earlier, exact cancellation of poles and zeros is almost never possible in real life. Let us consider that the numerator polynomial of the notch controller in Eq. (9-162) cannot be exactly realized by physical resistor and capacitor components. Rather, the transfer function of the notch controller is more realistically chosen as

$$G_c(s) = \frac{s^2 + 100s + 1,000,000}{s^2 + 20,000s + 1,000,000} \tag{9-173}$$

Fig. 9-58 shows the unit-step response of the system with the notch controller in Eq. (9-173). The attributes of the unit-step response are as follows:

% Maximum overshoot = 0.4%

Rise time $t_r = 0.17$ sec

Settling time $t_s = 0.2323$ sec

Frequency-Domain Design

To carry out the design of the notch controller, we refer to the Bode plot of Eq. (9-161) shown in Fig. 9-59. Due to the complex-conjugate poles of $G_p(s)$, the magnitude plot has a peak of 24.86 dB at 1095 rad/sec. From the Bode plot in Fig. 9-59, we see that we may want to bring the magnitude plot down to −20 dB at the resonant frequency of 1095 rad/sec so that the resonance is smoothed out. This requires an attenuation of −44.86 dB. Thus, from Eq. (9-155),

$$|G_c(j\omega_c)| = -44.86\,\text{dB} = \frac{\zeta_z}{\zeta_p} = \frac{0.0435}{\zeta_p} \tag{9-174}$$

where ζ_z is found from Eq. (9-62). Solving for ζ_p from the last equation, we get $\zeta_p = 7.612$. The attenuation should be placed at the resonant frequency of 1095 rad/sec; thus, $\omega_n = 1095$ rad/sec. The notch controller of Eq. (9-162) becomes

$$G_c(s) = \frac{s^2 + 95.3s + 1,198,606.6}{s^2 + 16,670.28s + 1,199,025} \tag{9-175}$$

The Bode plot of the system with the notch controller in Eq. (9-175) is shown in Fig. 9-59. We can see that the system with the notch controller has a phase margin of only 13.7°, and M_r is 3.92.

To complete the design, we can use a PI controller as a second-stage controller. Following the guideline given in Section 9-3 on the design of a PI controller, we assume that the desired phase margin is 80°. From the Bode plot in Fig. 9-59, we see that, to realize a phase margin of 80°, the new gain-crossover frequency should be $\omega_g' = 43$ rad/sec, and the magnitude of $G(j\omega_g')$ is 30 dB. Thus, from Eq. (9-32),

$$K_P = 10^{-|G(j\omega_g')|_{\text{dB}}^{/20}} = 10^{-30/20} = 0.0316 \tag{9-176}$$

The value of K_I is determined using the guideline given by Eq. (9-25),

$$K_I = \frac{\omega_g' K_P}{10} = \frac{43 \times 0.0316}{10} = 0.135 \tag{9-177}$$

Because the original system is type 0, the final design needs to be refined by adjusting the value of K_I. Table 9-22 gives the performance attributes when $K_P = 0.0316$ and K_I is varied from 0.135. From the best maximum overshoot, rise time, and settling time measures, the best value of K_I appears to be 0.35. The forward-path transfer function of the compensated system with the notch-PI controller is

$$G(s) = \frac{20,000(s + 100,000)(0.0316s + 0.35)}{s(s + 16.69)(s^2 + 16,670.28s + 1,199,025)} \tag{9-178}$$

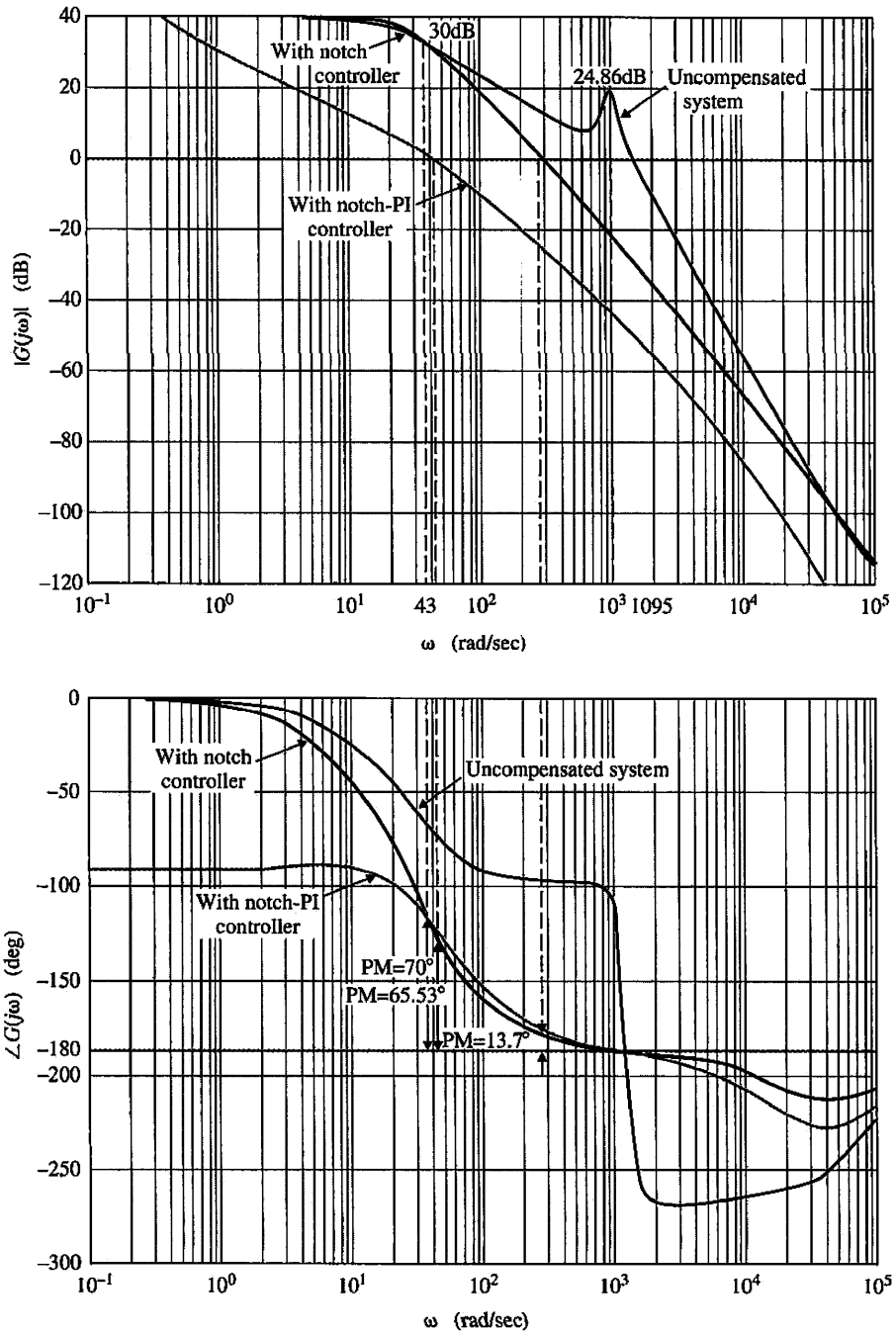


Figure 9-59 Bode plots of the uncompensated speed-control system in Example 9-8-1, with notch controller and with notch-PI controller.

Figure 9-59 shows the Bode plot of the system with the notch-PI controller, with $K_P = 0.0316$ and $K_I = 0.35$. The unit-step responses of the compensated system with $K_P = 0.0316$ and $K_I = 0.135$, 0.35, and 0.40 are shown in Fig. 9-60.

TABLE 9-22 Performance Attributes of System in Example 9-8-1 with Notch-PI Controller Designed in Frequency Domain

K_P	K_I	PM (deg)	M_r	Maximum Overshoot (%)	t_r (sec)	t_s (sec)
0.0316	0.1	76.71	1.00	0	0.2986	0.5758
0.0316	0.135	75.15	1.00	0	0.2036	0.4061
0.0316	0.200	72.22	1.00	0	0.0430	0.2403
0.0316	0.300	67.74	1.00	0	0.0350	0.1361
0.0316	0.350	65.53	1.00	1.6	0.0337	0.0401
0.0316	0.400	63.36	1.00	4.3	0.0323	0.0398

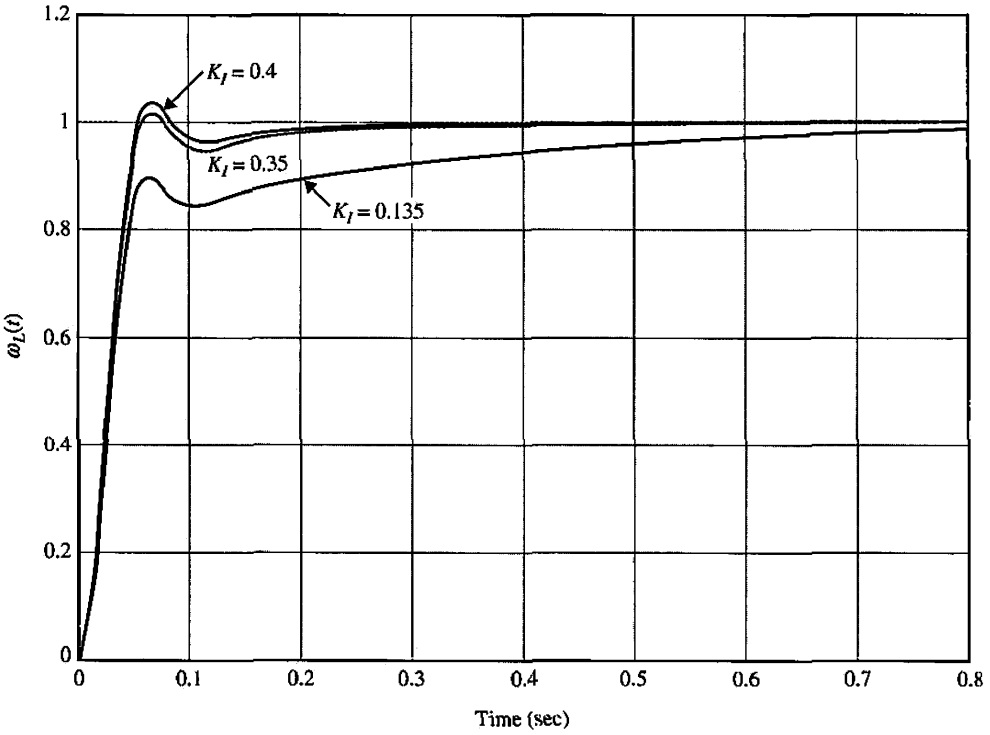


Figure 9-60 Unit-step responses of speed-control system in Example 9-8-1 with notch-PI controller, $G_c(s) = \dfrac{s^2 + 95.3s + 1,198,606.6}{s^2 + 16,670.28s + 1,199,025}$ $G_{c2}(s) = 0.0316 + \dfrac{0.35}{s}$.

▶ 9-9 FORWARD AND FEEDFORWARD CONTROLLERS

The compensation schemes discussed in the preceding sections all have one degree of freedom in that there is essentially one controller in the system, although the controller can contain several stages connected in series or in parallel. The limitations of a one-degree-of-freedom controller were discussed in Section 9-1. The two-degree-of-freedom compensation scheme shown in Fig. 9-2(d) through Fig. 9-2(f) offers design flexibility when a multiple number of design criteria have to be satisfied simultaneously.

From Fig. 9-2(e), the closed-loop transfer function of the system is

$$\frac{Y(s)}{R(s)} = \frac{G_{cf}(s)G_c(s)G_p(s)}{1 + G_c(s)G_p(s)} \tag{9-179}$$

and the error transfer function is

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_c(s)G_p(s)} \tag{9-180}$$

Thus, the controller $G_c(s)$ can be designed so that the error transfer function will have certain desirable characteristics, and the controller $G_{cf}(s)$ can be selected to satisfy performance requirements with reference to the input–output relationship. Another way of describing the flexibility of a two-degree-of-freedom design is that the controller $G_c(s)$ is usually designed to provide a certain degree of system stability and performance, but because the zeros of $G_c(s)$ always become the zeros of the closed-loop transfer function, unless some of the zeros are canceled by the poles of the process transfer function, $G_p(s)$, these zeros may cause a large overshoot in the system output even when the relative damping as determined by the characteristic equation is satisfactory. In this case and for other reasons, the transfer function $G_{cf}(s)$ may be used for the control or cancellation of the undesirable zeros of the closed-loop transfer function, while keeping the characteristic equation intact. Of course, we can also introduce zeros in $G_{cf}(s)$ to cancel some of the undesirable poles of the closed-loop transfer function that could not be otherwise affected by the controller $G_c(s)$. The feedforward compensation scheme shown in Fig. 9-2(f) serves the same purpose as the forward compensation, and the difference between the two configurations depends on system and hardware implementation considerations.

It should be kept in mind that, while the forward and feedforward compensations may seem powerful because they can be used directly for the addition or deletion of poles and zeros of the closed-loop transfer function, there is a fundamental question involving the basic characteristics of feedback. If the forward or feedforward controller is so powerful, then why do we need feedback at all? Because $G_{cf}(s)$ in the systems of Figs. 9-2(e) and 9-2(f) are outside the feedback loop, the system is susceptible to parameter variations in $G_{cf}(s)$. Therefore, in reality, these types of compensation cannot be satisfactorily applied to all situations.

EXAMPLE 9-9-1 As an illustration of the design of the forward and feedforward compensators, consider the second-order sun-seeker system with phase-lag control designed in Example 9-6-1. One of the disadvantages of phase-lag compensation is that the rise time is usually quite long. Let us consider that the phase-lag-compensated sun-seeker system has the forward-path transfer function

$$G(s) = G_c(s)G_p(s) = \frac{2500(1 + 10s)}{s(s + 25)(1 + 100s)} \tag{9-181}$$

The time-response attributes are as follows:

Maximum overshoot = 2.5%

$t_r = 0.1637$ sec

$t_s = 0.2020$ sec

We can improve the rise time and the settling time while not appreciably increasing the overshoot by adding a PD controller $G_{cf}(s)$ to the system, as shown in Fig. 9-61(a). This effectively adds a zero to the closed-loop transfer function while not affecting the characteristic equation. Selecting the PD controller as

$$G_{cf}(s) = 1 + 0.05s \tag{9-182}$$

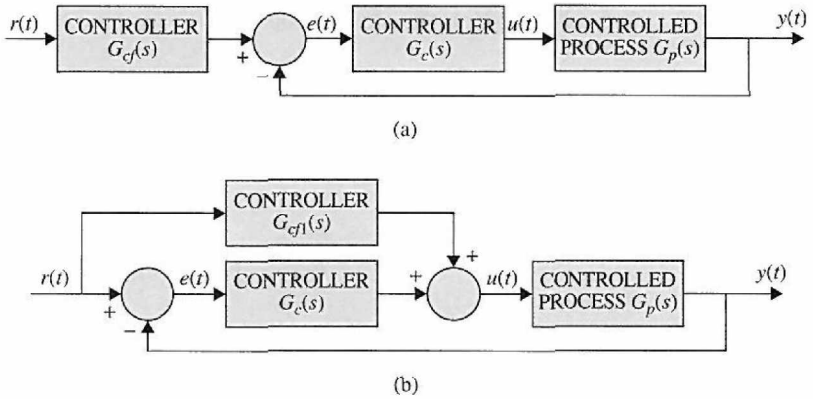


Figure 9-61 (a) Forward compensation with series compensation. (b) Feedforward compensation with series compensations.

the time-domain performance attributes are as follows:

$$\text{Maximum overshoot} = 4.3\%$$

$$t_r = 0.1069 \text{ sec}$$

$$t_s = 0.1313 \text{ sec}$$

If instead the feedforward configuration of Fig. 9-61(b) is chosen, the transfer function of $G_{cf1}(s)$ is directly related to $G_{cf}(s)$; that is, equating the closed-loop transfer functions of the two systems in Figs. 9-61(a) and 9-61(b), we have

$$\frac{[G_{cf1}(s) + G_c(s)]G_p(s)}{1 + G_c(s)G_p(s)} = \frac{G_{cf}G_c(s)G_p(s)}{1 + G_c(s)G_p(s)} \tag{9-183}$$

Solving for $G_{cf1}(s)$ from Eq. (9-183) yields

$$G_{cf1}(s) = [G_{cf}(s) - 1]G_c(s) \tag{9-184}$$

Thus, with $G_{cf}(s)$ as given in Eq. (9-179), we have the transfer function of the feedforward controller:

$$G_{cf1}(s) = 0.05s\left(\frac{1 + 10s}{1 + 100s}\right) \tag{9-185}$$

9-10 DESIGN OF ROBUST CONTROL SYSTEMS

In many control-system applications, not only must the system satisfy the damping and accuracy specifications, but the control must also yield performance that is **robust** (insensitive) to external disturbance and parameter variations. We have shown that feedback in conventional control systems has the inherent ability to reduce the effects of external disturbance and parameter variations. Unfortunately, robustness with the conventional feedback configuration is achieved only with a high loop gain, which is normally detrimental to stability. Let us consider the control system shown in Fig. 9-62.

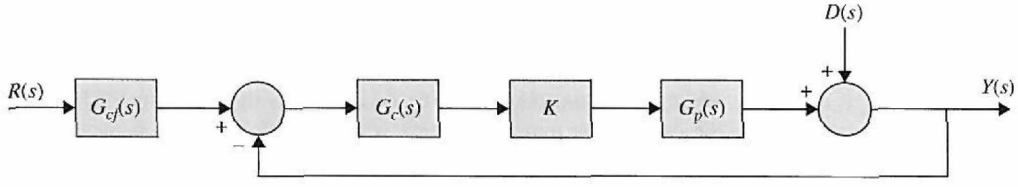


Figure 9-62 Control system with disturbance.

The external disturbance is denoted by the signal $d(t)$, and we assume that the amplifier gain K is subject to variation during operation. The input–output transfer function of the system when $d(t) = 0$ is

$$M(s) = \frac{Y(s)}{R(s)} = \frac{KG_{cf}(s)G_c(s)G_p(s)}{1 + KG_c(s)G_p(s)} \tag{9-186}$$

and the disturbance-output transfer function when $r(t) = 0$ is

$$T(s) = \frac{Y(s)}{D(s)} = \frac{1}{1 + KG_c(s)G_p(s)} \tag{9-187}$$

In general, the design strategy is to select the controller $G_c(s)$ so that the output $y(t)$ is insensitive to the disturbance over the frequency range in which the latter is dominant and to design the feedforward controller $G_{cf}(s)$ to achieve the desired transfer function between the input $r(t)$ and the output $y(t)$.

Let us define the sensitivity of $M(s)$ due to the variation of K as

$$S_K^M = \frac{\text{percent change in } M(s)}{\text{percent change in } K} = \frac{dM(s)/M(s)}{dK/K} \tag{9-188}$$

Then, for the system in Fig. 9-62,

$$S_K^M = \frac{1}{1 + KG_c(s)G_p(s)} \tag{9-189}$$

which is identical to Eq. (9-187). Thus, the sensitivity function and the disturbance-output transfer function are identical, which means that disturbance suppression and robustness with respect to variations of K can be designed with the same control schemes.

The following example shows how the two-degree-of-freedom control system of Fig. 9-62 can be used to achieve a high-gain system that will satisfy the performance and robustness requirements, as well as noise rejection.

EXAMPLE 9-10-1 Let us consider the second-order sun-seeker system in Example 9-6-1, which is compensated with phase-lag control. The forward-path transfer function is

$$G_p(s) = \frac{2500K}{s(s + 25)} \tag{9-190}$$

where $K = 1$. The forward-path transfer function of the phase-lag-compensated system with $a = 0.1$ and $T = 100$ is

$$G(s) = G_c(s)G_p(s) = \frac{2500K(1 + 10s)}{s(s + 25)(1 + 100s)} \quad (K = 1) \tag{9-191}$$

Because the phase-lag controller is a low-pass filter, the sensitivity of the closed-loop transfer function $M(s)$ with respect to K is poor. The bandwidth of the system is only 13.97 rad/sec, but it is expected that $|S_K^M(j\omega)|$ will be greater than unity at frequencies beyond 13.97 rad/sec. Fig. 9-63 shows the unit-step responses of the system when $K = 1$, the nominal value, and $K = 0.5$ and 2.0. Notice that, if for some reason, the forward gain K is changed from its nominal value, the system response of the phase-lag-compensated system would vary substantially. The attributes of the step responses and the characteristic equation roots are shown in Table 9-23 for the three values of K. Fig. 9-64 shows the root loci of the system with the phase-lag controller. The two complex roots of the characteristic equation vary substantially as K varies from 0.5 to 2.0.

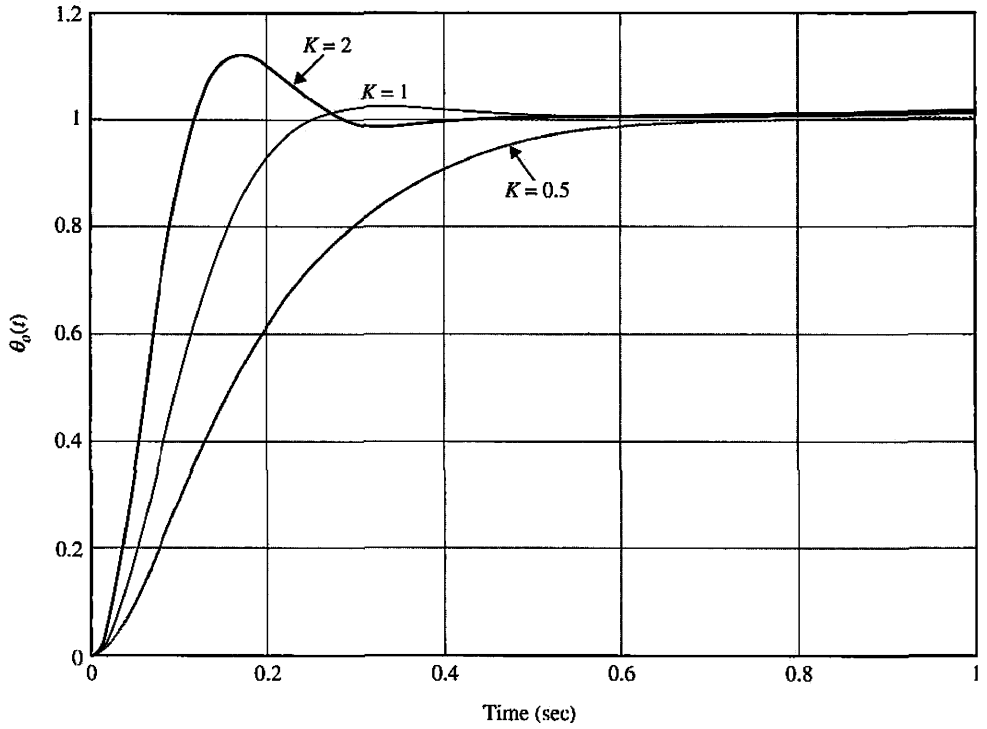


Figure 9-63 Unit-step responses of the second-order sun-seeker system with phase-lag controller, $G(s) = \dfrac{2500(1 + 10s)}{s(s + 25)(1 + 100s)}$.

TABLE 9-23 Attributes of Unit-Step Response of Second-Order Sun-Seeker System with Phase-Lag Controller in Example 9-10-1

K	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Roots of Characteristic Equation
2.0	12.6	0.07854	0.2323	$-0.1005 - 12.4548 \pm j18.51$
1.0	2.6	0.1519	0.2020	$-0.1009 - 12.4545 \pm j9.624$
0.5	1.5	0.3383	0.4646	$-0.1019 - 6.7628 - 18.1454$

Toolbox 9-10-1

Fig. 9-63 is obtained by the following sequence of MATLAB functions

```
K = 1;
num = K *2500 * [10 1]
den = conv([1 25 0], [100 1]);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)

hold on;

K = 2;
num = K*2500 * [10 1]
```

```
den = conv([1 25 0], [100 1]);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)

hold on;

K = 0.5;
num = K*2500 * [10 1]
den = conv([1 25 0], [100 1]);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)

hold on;
axis([0 1 0 1.2]);
grid
```

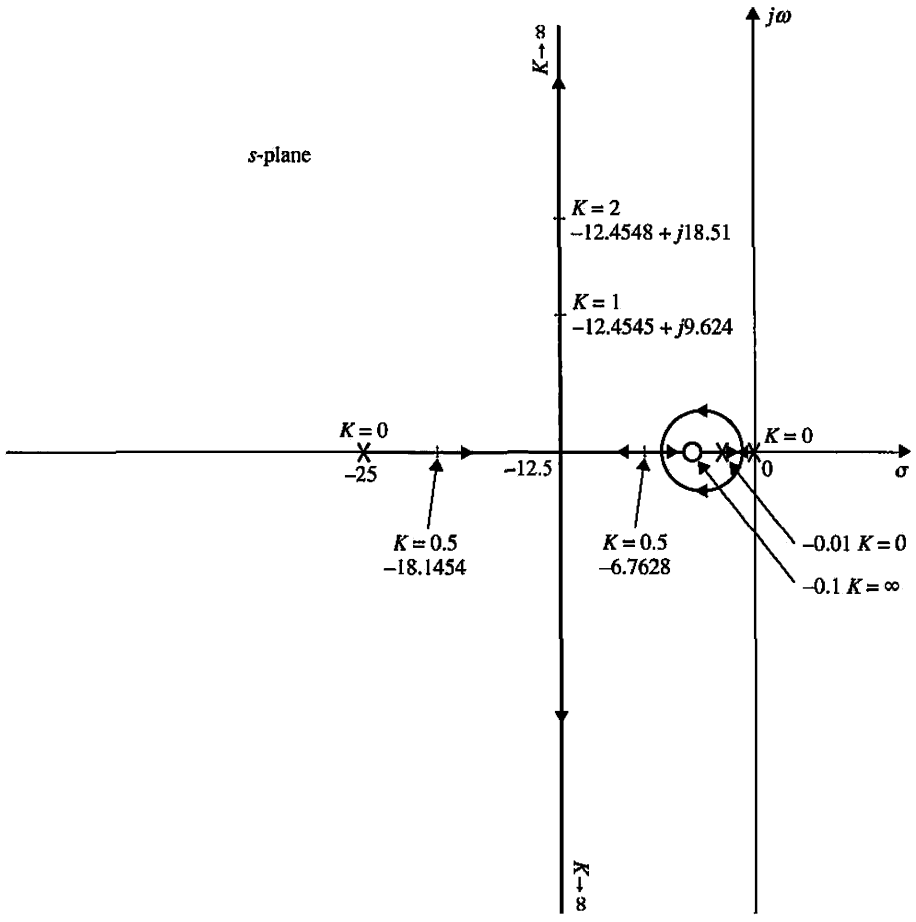


Figure 9-64 Root loci of the second-order sun-seeker system with phase-lag controller, $G(s) = \dfrac{2500(1 + 10s)}{s(s + 25)(1 + 100s)}$.

The design strategy of the robust controller is to place two zeros of the controller near the desired closed-loop poles, which according to the phase-lag-compensated system are at $s = -12.455 \pm j9.624$. Thus, we let the controller transfer function be

$$G_c(s) = \frac{(s + 13 + j10)(s + 13 - j10)}{269} = \frac{(s^2 + 26s + 269)}{269} \tag{9-192}$$

The forward-path transfer function of the system with the robust controller is

$$G(s) = \frac{9.2937K(s^2 + 26s + 269)}{s(s + 25)} \tag{9-193}$$

Toolbox 9-10-2

Fig. 9-64 is obtained by the following sequence of MATLAB functions

```
K = 1;
num = K *2500 * [10 1]
den = conv([1 25 0], [100 1]);
[numCL, denCL] = cloop(num, den)
rlocus(numCL,denCL);

hold on

K = 2;
num = K *2500 * [10 1]
den = conv([1 25 0], [100 1]);
[numCL, denCL] = cloop(num, den)
rlocus(numCL,denCL);

hold on

K = 1;
num = K *2500 * [10 1]
den = conv([1 25 0], [100 1]);
rlocus(num,den);

hold on

K = 2;
num = K *2500 * [10 1]
den = conv([1 25 0], [100 1]);
rlocus(num,den);

hold on
```

Fig. 9-65 shows the root loci of the system with the robust controller. By placing the two zeros of $G_c(s)$ near the desired characteristic equation roots, the sensitivity of the system is greatly improved. In fact, the root sensitivity near the two complex zeros at which the root loci terminate is very low. Fig. 9-65 shows that, when K approaches infinity, the two characteristic equation roots approach $-13 \pm j10$.

Toolbox 9-10-3

Fig. 9-65 is obtained by the following sequence of MATLAB functions

```
K = 1;
num = 9.2937*K *[1 26 269]
den = [1 25 0];
[numCL, denCL] = cloop(num, den)
rlocus(numCL,denCL);
```

```
hold on

K = 2;
num = 9.2937*K *[1 26 269]
den = [1 25 0];
[numCL, denCL] = cloop(num, den)
rlocus(numCL,denCL);
hold on

K = 1;
num = 9.2937*K *[1 26 269]
den = [1 25 0];
rlocus(num,den);

hold on

K = 2;
num = 9.2937*K *[1 26 269]
den = [1 25 0];
rlocus(num,den);
axis([-35 2 -15 15])
% grid
```

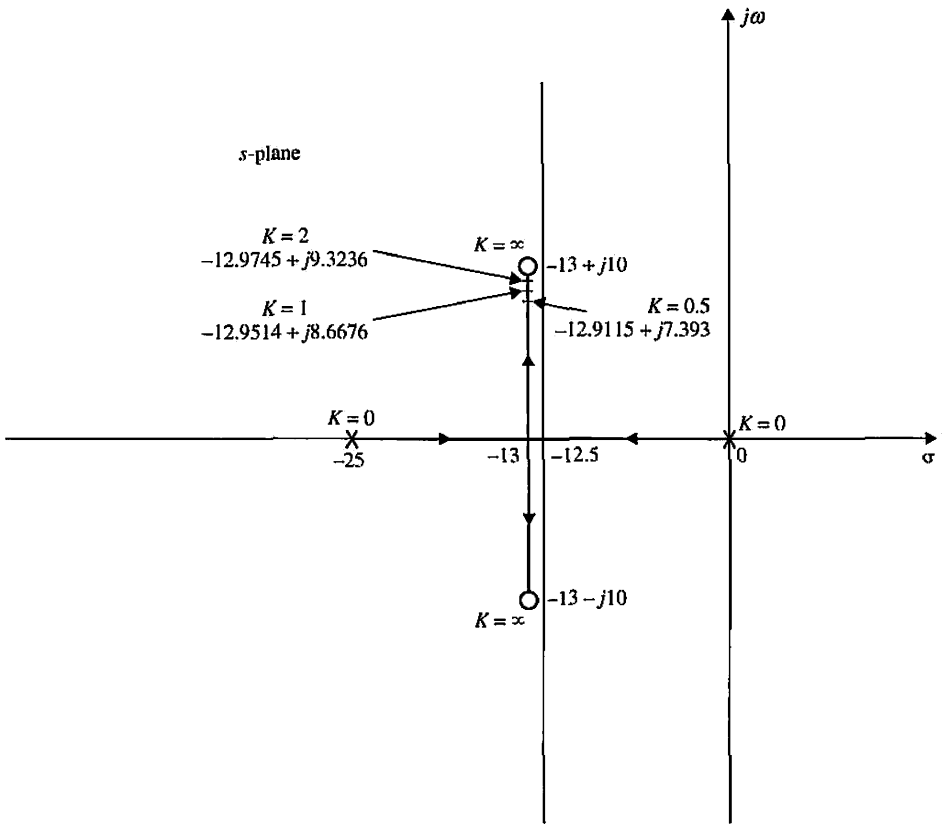


Figure 9-65 Root loci of the second-order sun-seeker system with robust controller, $G(s) = \dfrac{9.2937K(s^2 + 26s + 269)}{s(s + 25)}$.

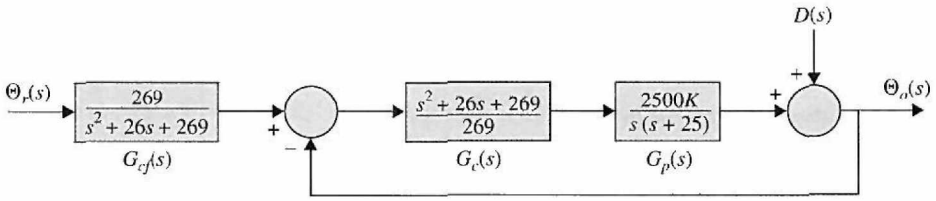


Figure 9-66 Second-order sun-seeker system with robust controller and forward controller.

Because the zeros of the forward-path transfer function are identical to the zeros of the closed-loop transfer function, the design is not complete by using only the series controller $G_c(s)$, because the closed-loop zeros will essentially cancel the closed-loop poles. This means we must add the forward controller, as shown in Fig. 9-62, where $G_{cf}(s)$ should contain poles to cancel the zeros of $s^2 + 26s + 269$ of the closed-loop transfer function. Thus, the transfer function of the forward controller is

$$G_{cf}(s) = \frac{269}{s^2 + 26s + 269} \tag{9-194}$$

The block diagram of the overall system is shown in Fig. 9-66. The closed-loop transfer function of the compensated system with $K = 1$ is

$$\frac{\Theta_o(s)}{\Theta_r(s)} = \frac{242.88}{s^2 + 25.903s + 242.88} \tag{9-195}$$

The unit-step responses of the system for $K = 0.5$, 1.0, and 2.0 are shown in Fig. 9-67, and their attributes are given in Table 9-24. As shown, the system is now very insensitive to the variation of K.

Because the system in Fig. 9-66 is now more robust, it is expected that the disturbance effect will be reduced. However, we cannot evaluate the effect of the controllers in the system of Fig. 9-66 by

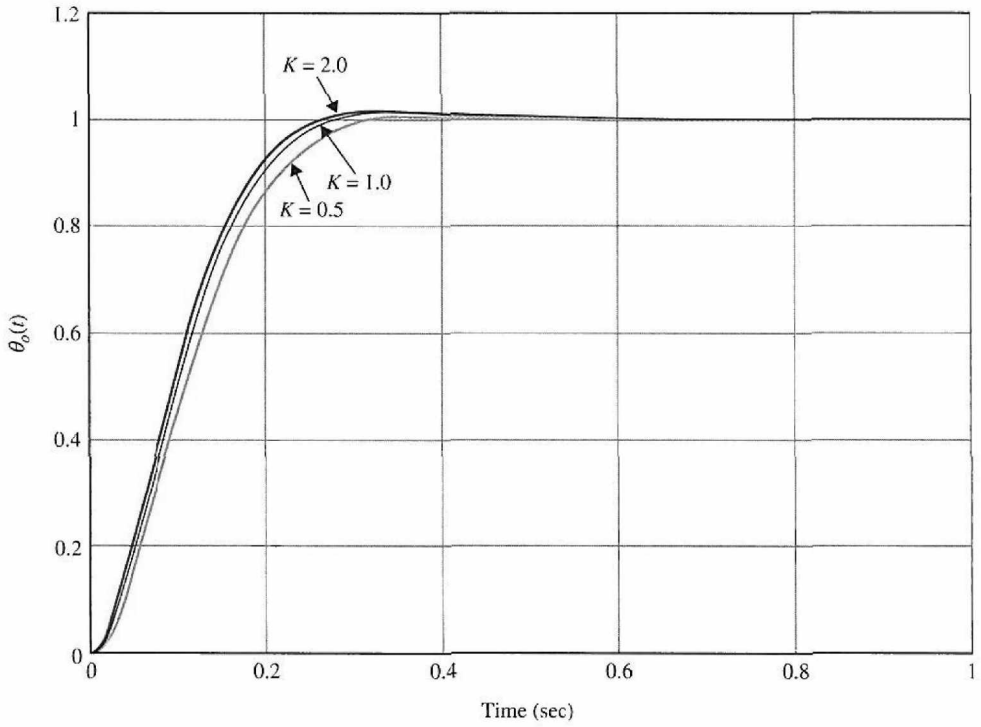


Figure 9-67 Unit-step responses of the second-order sun-seeker system with robust controller and forward controller.

TABLE 9-24 Attributes of Unit-Step Response of Second-Order Sun-Seeker System with Robust Controller in Example 9-10-1

K	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Roots of Characteristic Equation
2.0	1.3	0.1576	0.2121	$-12.9745 \pm j9.3236$
1.0	0.9	0.1664	0.2222	$-12.9514 \pm j8.6676$
0.5	0.5	0.1846	0.2525	$-12.9115 \pm j7.3930$

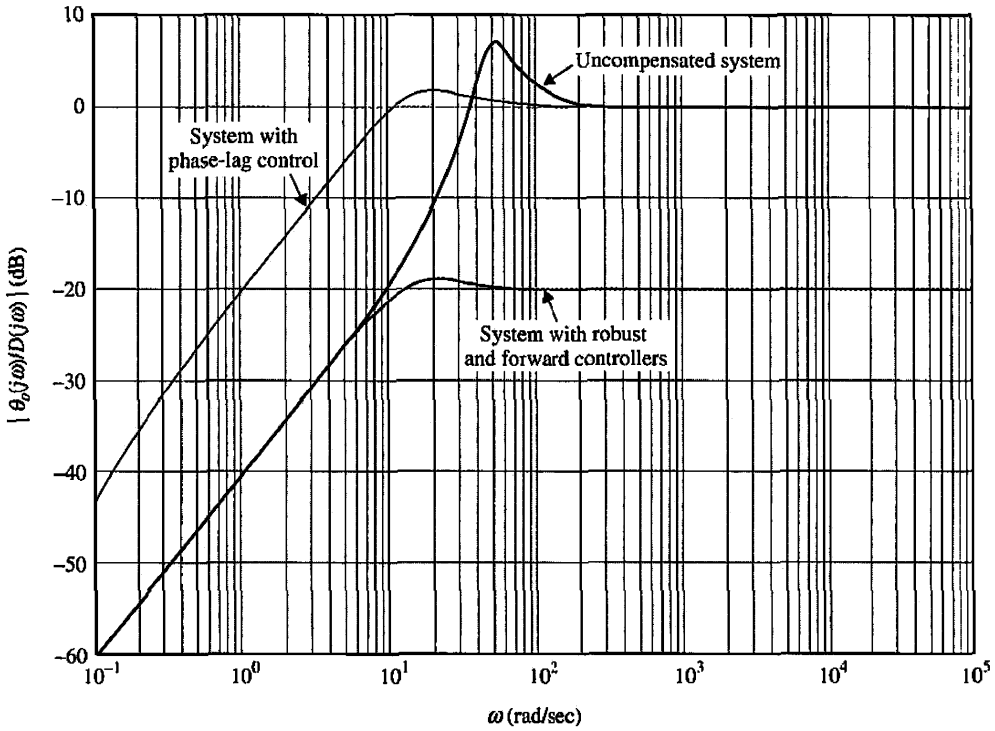


Figure 9-68 Amplitude Bode plot of response due to noise of second-order sun-seeker system.

applying a unit-step function as $d(t)$. The true improvement on the noise rejection properties is more appropriately analyzed by investigating the frequency response of $\Theta_o(s)/D(s)$. The noise-to-output transfer function, written from Fig. 9-66, is

$$\frac{\Theta_o(s)}{D(s)} = \frac{1}{1 + G_c(s)G_p(s)} = \frac{s(s+25)}{10.2937s^2 + 266.636s + 2500} \tag{9-196}$$

The amplitude Bode plot of Eq. (9-196) is shown in Fig. 9-68, along with those of the uncompensated system and the system with phase-lag control. Notice that the magnitude of the frequency response between $D(s)$ and $\Theta_o(s)$ is much smaller than those of the system without compensation and with phase-lag control. The phase-lag control also accentuates the noise for frequencies up to approximately 40 rad/sec, adding more stability to the system. ◄

► **EXAMPLE 9-10-2** In this example, a robust controller with forward compensation is designed for the third-order sun-seeker system in Example 9-6-2 with phase-lag control. The forward-path transfer function of the uncompensated system is

$$G_p(s) = \frac{156,250,000K}{s(s^2 + 625s + 156,250)} \tag{9-197}$$

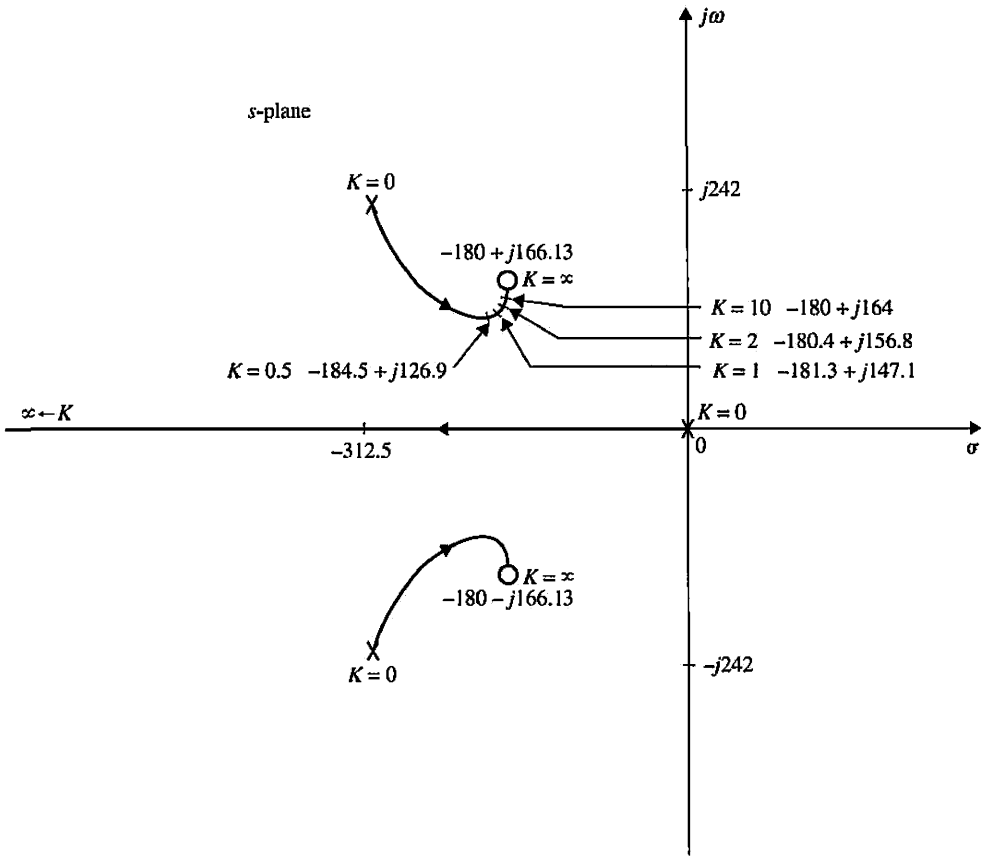


Figure 9-69 Root loci of the third-order sun-seeker system with robust and forward controllers.

where $K = 1$. The root loci of the closed-loop system are shown in Fig. 9-50, which lead to the phase-lag control with results shown in Table 9-19. Let us select the parameters of the phase-lag controller as $a = 0.1$ and $T = 20$. The dominant roots of the characteristic equation are $s = -187.73 \pm j164.93$.

Let us place the two zeros of the second-order robust controller at $-180 \pm j166.13$, so that the controller transfer function is

$$G_c(s) = \frac{s^2 + 360s + 60,000}{60,000} \tag{9-198}$$

To ease the high-frequency realization problem of the controller, we may add two nondominant poles to $G_c(s)$. The following analysis is carried out with $G_c(s)$ given in Eq. (9-198), however. The root loci of the compensated system are shown in Fig. 9-69. Thus, by placing the zeros of the controller very

TABLE 9-25 Attributes of Unit-Step Response and Characteristic Equation Roots of Third-Order Sun-Seeker System with Robust and Forward Controllers in Example 9-10-2

K	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Characteristic Equation Roots
0.5	1.0	0.01115	0.01616	$-1558.1 - 184.5 \pm j126.9$
1.0	2.1	0.01023	0.01414	$-2866.6 - 181.3 \pm j147.1$
2.0	2.7	0.00966	0.01313	$-5472.6 - 180.4 \pm j156.8$
10.0	3.2	0.00924	0.01263	$-26307 - 180.0 \pm j164.0$

close to the desired dominant roots, the system is very insensitive to changing values of K near and beyond the nominal value of K. The forward controller has the transfer function

$$G_{cf}(s) = \frac{60,000}{s^2 + 360s + 60,000} \tag{9-199}$$

The attributes of the unit-step response for $K = 0.5$, 1.0, 2.0, and 10.0 and the corresponding characteristic equation roots are given in Table 9-25.

EXAMPLE 9-10-3 In this example, we consider the design of a position-control system that has a variable load inertia. This type of situation is quite common in control systems. For example, the load inertia seen by the motor in an electronic printer will change when different printwheels are used. The system should have satisfactory performance for all the printwheels intended to be used with the system.

To illustrate the design of a robust system that is insensitive to the variation of load inertia, consider that the forward-path transfer function of a unity-feedback control system is

$$G_p(s) = \frac{KK_i}{s[(Js + B)(Ls + B) + K_iK_b]} \tag{9-200}$$

The system parameters are as follows:

K_i = motor torque constant = 1 N-m/A

K_b = motor back-emf constant = 1 V/rad/sec

R = motor resistance = 1 Ω

L = motor inductance = 0.01 H

B = motor and load viscous-friction coefficient ≅ 0

J = motor and load inertia, varies between 0.01 and 0.02 N-m/rad/sec^2

K = amplifier again

Substituting these system parameters into Eq. (9-200), we get

$$\text{For } J = 0.01 \quad G_p(s) = \frac{10,000K}{s(s^2 + 100s + 10,000)} \tag{9-201}$$

$$\text{For } J = 0.02 \quad G_p(s) = \frac{5000K}{s(s^2 + 100s + 5000)} \tag{9-202}$$

The performance specifications are as follows:

Ramp error constant $K_v \geq 200$

Maximum overshoot $\leq 5\%$ or as small as possible

Rise time $t_r < 0.05$ sec

Settling time $t_s \leq 0.05$ sec

These specifications are to be maintained for $0.01 \leq J \leq 0.02$.

To satisfy the ramp-error constant requirement, the value of K must be at least 200. Fig. 9-70 shows the root loci of the uncompensated system for $J = 0.01$ and $J = 0.02$. We see that, regardless of the value of J, the uncompensated system is unstable for $K > 100$.

To achieve robust control, let us choose the system configuration of Fig. 9-61(a). We introduce a second-order series controller with the zeros placed near the desired dominant characteristic equation of the compensated system. The zeros should be so placed that the dominant characteristic equation roots would be insensitive to the variation in J. This is done by placing the two zeros at $-55 \pm j45$, although the exact location is unimportant. By choosing the two controller zeros as designated, the root loci of the compensated system show that the two complex roots of the characteristic equation will be very close to these zeros for various values of J, especially when the value of K is large. The transfer function of the robust controller is

$$G_c(s) = \frac{(s^2 + 110s + 5050)}{5050} \tag{9-203}$$

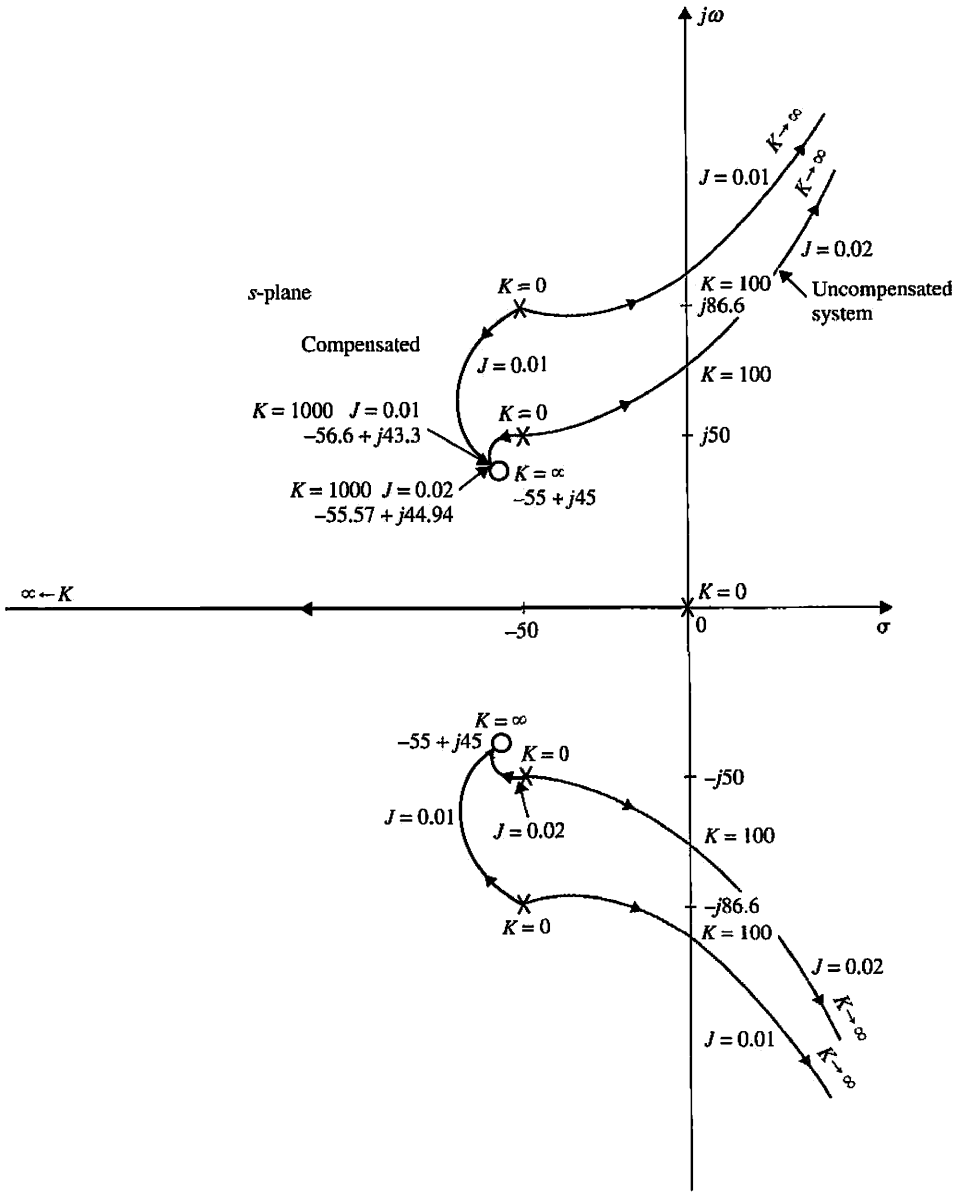


Figure 9-70 Root loci of the position-control system in Example 9-10-3 with robust and forward controllers.

As in the last example, we may add two nondominant poles to $G_c(s)$ to ease the high-frequency realization problem of the controller. The analysis is carried out with $G_c(s)$ given in Eq. (9-203).

Let $K = 1000$, although 200 would have been adequate to satisfy the K_v requirement. Then, for $J = 0.01$, the forward-path transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{1980.198(s^2 + 110s + 5050)}{s(s^2 + 100s + 10,000)} \tag{9-204}$$

and for $J = 0.02$,

$$G(s) = \frac{990.99(s^2 + 110s + 5050)}{s(s^2 + 100s + 5000)} \tag{9-205}$$

TABLE 9-26 Attributes of Unit-Step Response and Characteristic Equation Roots of System with Robust and Forward Controllers in Example 9-10-3

J N-m/rad/sec^2	Maximum Overshoot (%)	t_r (sec)	t_s (sec)	Roots of Characteristic Equation
0.01	1.6	0.03453	0.04444	$-1967 - 56.60 \pm j43.3$
0.02	2.0	0.03357	0.04444	$-978.96 - 55.57 \pm j44.94$

To cancel the two zeros of the closed-loop transfer function, the transfer function of the forward controller is

$$G_{cf}(s) = \frac{5050}{s^2 + 110s + 5050} \tag{9-206}$$

The attributes of the unit-step response and the characteristic equation roots of the compensated system with $K = 1000$, $J = 0.01$, and $J = 0.02$ are given in Table 9-26.

9-11 MINOR-LOOP FEEDBACK CONTROL

The control schemes discussed in the preceding sections have all utilized series controllers in the forward path of the main loop or feedforward path of the control system. Although series controllers are the most common because of their simplicity in implementation, depending on the nature of the system, sometimes there are advantages in placing the controller in a minor feedback loop, as shown in Fig. 9-2(b). For example, a tachometer may be coupled directly to a dc motor not only for the purpose of speed indication but more often for improving the stability of the closed-loop system by feeding back the output signal of the tachometer. The motor speed can also be generated by processing the back emf of the motor electronically. In principle, the PID controller or phase-lead and phase-lag controllers can all, with varying degrees of effectiveness, be applied as minor-loop feedback controllers. Under certain conditions, minor-loop control can yield systems that are more robust, that is, less sensitive to external disturbance or internal parameter variations.

9-11-1 Rate-Feedback or Tachometer-Feedback Control

The principle of using the derivative of the actuating signal to improve the damping of a closed-loop system can be applied to the output signal to achieve a similar effect. In other words, the derivative of the input signal is fed back and added algebraically to the actuating signal of the system. In practice, if the output variable is mechanical displacement, a tachometer may be used to convert mechanical displacement into an electrical signal that is proportional to the derivative of the displacement. Fig. 9-71 shows the block diagram of a control system with a secondary path that feeds back the derivative of the output. The

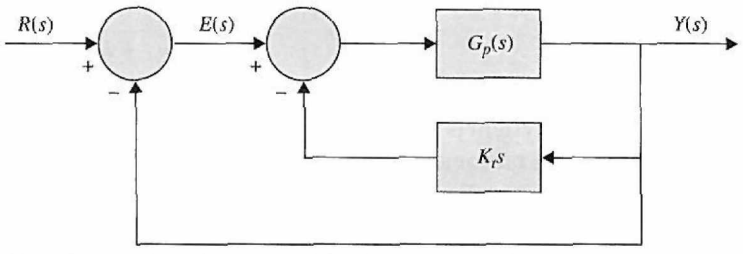


Figure 9-71 Control system with tachometer feedback.

transfer function of the tachometer is denoted by $K_t s$, where K_t is the tachometer constant, usually expressed in volts per radian per second for analytical purposes. Commercially, K_t is given in the data sheet of the tachometer, typically in volts per 1000 rpm. The effects of rate or tachometer feedback can be illustrated by applying it to a second-order prototype system. Consider that the controlled process of the system shown in Fig. 9-71 has the transfer function

$$G_p(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \tag{9-207}$$

The closed-loop transfer function of the system is

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2} \tag{9-208}$$

and the characteristic equation is

$$s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2 = 0 \tag{9-209}$$

From the characteristic equation, *it is apparent that the effect of the tachometer feedback is the increase of the damping of the system, since* K_t *appears in the same term as the damping ratio* ζ.

In this respect, tachometer-feedback control has exactly the same effect as the PD control. However, the closed-loop transfer function of the system with PD control in Fig. 9-3 is

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2(K_P + K_D s)}{s^2 + (2\zeta\omega + K_D\omega_n^2)s + \omega_n^2 K_P} \tag{9-210}$$

Comparing the two transfer functions in Eqs. (9-208) and (9-210), we see that the two characteristic equations are identical if $K_P = 1$ and $K_D = K_t$. However, Eq. (9-210) has a zero at $s = -K_P/K_D$, whereas Eq. (9-208) does not. Thus, the response of the system with tachometer feedback is uniquely defined by the characteristic equation, whereas the response of the system with the PD control also depends on the zero at $s = -K_P/K_D$, which could have a significant effect on the overshoot of the step response.

With reference to the steady-state analysis, the forward-path transfer function of the system with tachometer feedback is

$$\frac{Y(s)}{E(s)} = \frac{\omega_n^2}{s(s + 2\zeta\omega_n + K_t\omega_n^2)} \tag{9-211}$$

Because the system is still type 1, the basic characteristics of the steady-state error are not altered by the tachometer feedback; that is, when the input is a step function, the steady-state error is zero. For a unit-ramp function input, the steady-state error of the system is $(2\zeta + K_t\omega_n)/\omega_n$, whereas that of the system with PD control in Fig. 9-3 is $2\zeta/\omega_n$. Thus, *for a type 1 system, tachometer feedback decreases the ramp-error constant* K_v but does not affect the step-error constant K_P.

9-11-2 Minor-Loop Feedback Control with Active Filter

Instead of using a tachometer, an active filter with RC elements and op-amps can be used to reduce cost and save space in the minor feedback loop for compensation. We illustrate this approach with the following example.

EXAMPLE 9-11-1 Consider that, for the second-order sun-seeker system in Example 9-6-1, instead of using a series controller in the forward path, we adopt the minor-loop feedback control, as shown in Fig. 9-72(a), with

$$G_p(s) = \frac{2500}{s(s + 25)} \tag{9-212}$$

and

$$H(s) = \frac{K_t s}{1 + Ts} \tag{9-213}$$

To maintain the system as type 1, it is necessary that $H(s)$ contain a zero at $s = 0$. Eq. (9-213) can be realized by the op-amp circuit shown in Fig. 9-72(b). This circuit cannot be applied as a series controller in the forward path, because it acts as an open circuit in the steady state when the frequency is zero. As a minor-loop controller, the zero-transmission property to dc signals does not pose any problems.

The forward-path transfer function of the system in Fig. 9-72(a) is

$$\begin{aligned} \frac{\Theta_o(s)}{\Theta_e(s)} = G(s) &= \frac{G_p(s)}{1 + G_p(s)H(s)} \\ &= \frac{2500(1 + Ts)}{s[(s + 25)(1 + Ts) + 2500K_t]} \end{aligned} \tag{9-214}$$

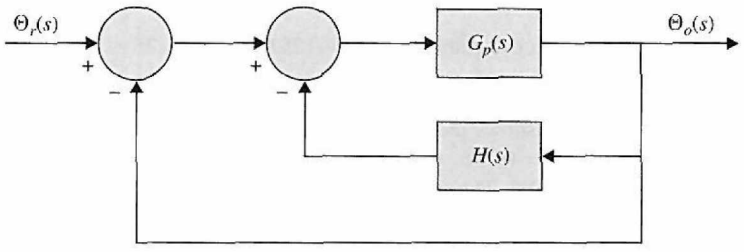
The characteristic equation of the system is

$$Ts^3 + (25T + 1)s^2 + (25 + 2500T + 2500K_t)s + 2500 = 0 \tag{9-215}$$

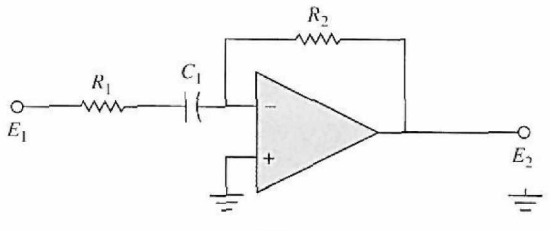
To show the effects of the parameters K_t and T, we construct the root contours of Eq. (9-215) by first considering that K_t is fixed and T is variable. Dividing both sides of Eq. (9-215) by the terms that do not contain T, we get

$$1 + \frac{Ts(s^2 + 25s + 2500)}{s^2 + (25 + 2500K_t)s + 5000} = 0 \tag{9-216}$$

When the value of K_t is relatively large, the two poles of the last equation are real with one very close to the origin. It is more effective to choose K_t so that the poles of Eq. (9-216) are complex.



(a)



(b)

Figure 9-72 (a) Sun-seeker control system with minor-loop control. (b) Op-amp circuit realization of $\frac{K_t s}{1 + Ts}$.

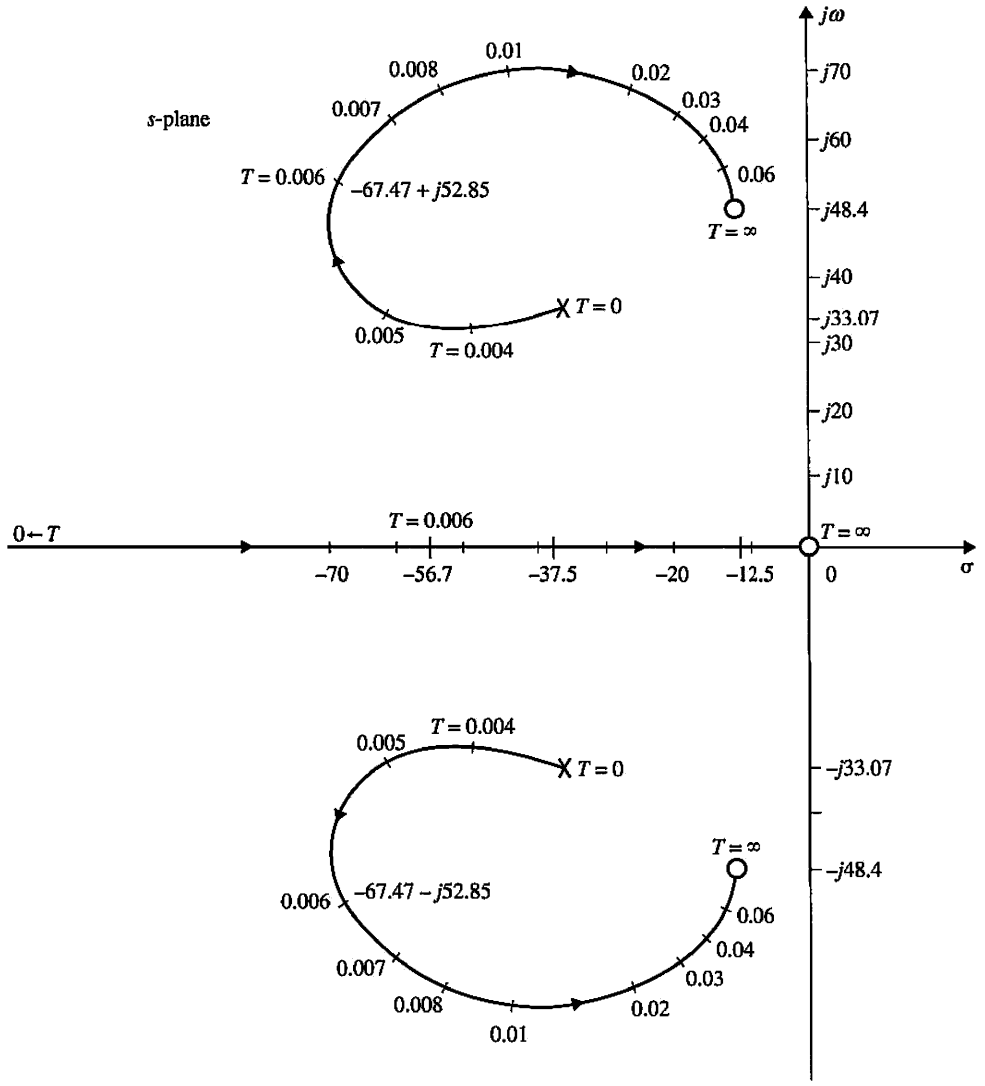


Figure 9-73 Root contours of $Ts^3 + (25T + 1)s^2 + (25 + 2500K_t + 2500T)s + 2500 = 0$, $K_t = 0.02$.

Fig. 9-73 shows the root contours of Eq. (9-215) with $K_t = 0.02$, and T varies from 0 to ∞.

When $T = 0.006$, the characteristic equation roots are at -56.72, $-67.47 + j52.85$, and $-67.47 - j52.85$. The attributes of the unit-step response are as follows:

Maximum overshoot = 0

$t_r = 0.04485$ sec

$t_s = 0.06061$ sec

$t_{\max} = 0.4$ sec

The ramp-error constant of the system is

$$K_v = \lim_{s \to 0} sG(s) = \frac{100}{1 + 100K_t} \tag{9-217}$$

Thus, just as with tachometer feedback, the minor-loop feedback controller of Eq. (9-213) reduces the ramp-error constant K_v, although the system is still type 1. ◀

► 9-12 A HYDRAULIC CONTROL SYSTEM

In this case study, we model a four-way electro-hydraulic-valve control of a linear actuator. After deriving the mathematical model for the system, we apply the model to do position control for three different applications: 1) robot arm joint (translational system), 2) dropping a disk into a hole (rotational system), and 3) variable load. At the end, we design P, PD, PI, and PID controllers for the robot-arm-joint system.

A schematic diagram of the system for the first application (robot arm joint) is shown in Fig. 9-74. The diagram is the same for all the applications except for the load. This system consists of a double-acting single rod linear actuator, a two-stage electro-hydraulic valve, controller circuitry, and potentiometer. The input to the system is the voltage corresponding to the desired output position of the load. The output position is fed back through the potentiometer. The output voltage of the potentiometer gets subtracted from the input voltage to produce the error signal. This error signal is used to control the position of the main valve displacement, which controls the pressure level and flow rate entering and leaving the linear actuator. The output force exerted by the linear actuator rod is directly proportional to the pressure difference between two sides of the piston.

9-12-1 Modeling Linear Actuator

A double-acting single rod actuator is shown in Fig. 9-75. We call it double acting since its piston can be forced in both right and left directions. The direction of the piston movement depends on the pressure difference on the two sides of the piston: port A and port B. We call this actuator single rod since the piston is connected to only one rod.

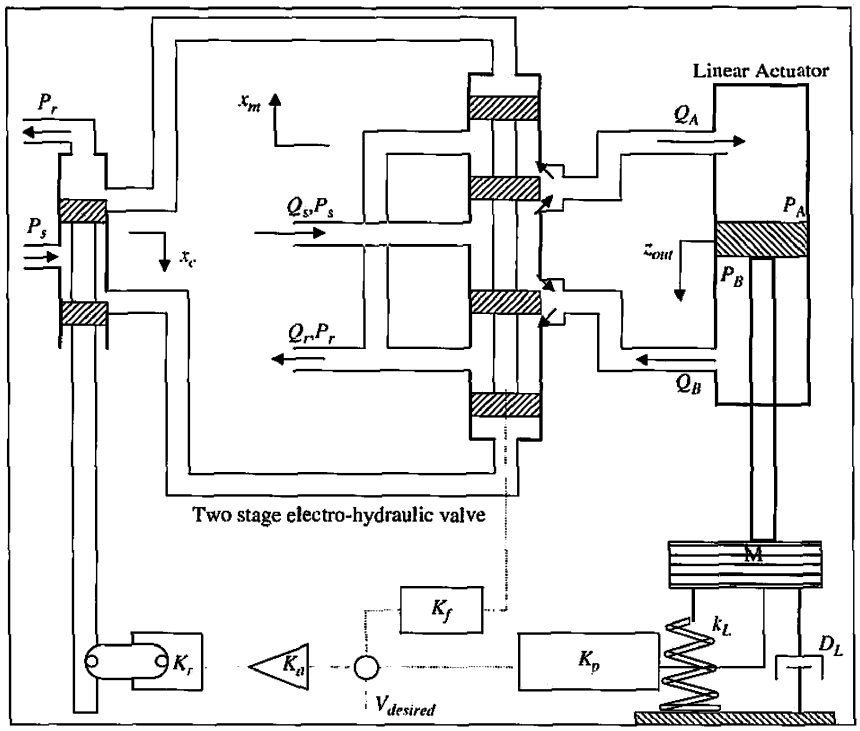


Figure 9-74 Schematic diagram of the system for the robot arm joint.

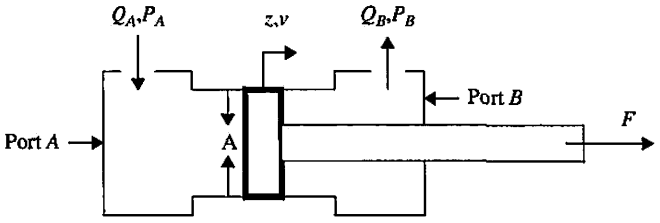


Figure 9-75 Schematic of double-acting single rod actuator.

The hydraulic fluid can enter and exit the actuator from either port A or port B. The magnitude of the flow and pressure of ports A and B are controlled by the four-way valve explained in the following section. For now, we want to develop a relationship between the fluid pressures in port A and port B, the area of the piston (A), and the displacement of the piston rod.

We can take sides A and B of the piston to have the same area A. Ignoring the pressure transient effect of the actuator, we can express the applied force F to the actuator as

$$F = (AP_A - AP_B)\eta_f \tag{9-218}$$

Note that in Eq. (9-218) η_f is the force efficiency of the actuator, which is due to the friction of the piston and the viscous shear of the hydraulic fluid. The force efficiency is always less than one. For ideal actuators, we can consider the force efficiency to be one.

Volumetric efficiency, η_v, is another source of non-ideality in linear actuators. The volumetric efficiency will always be less than one due to hydraulic fluid compression and leakage past the piston. We can express volumetric efficiency as

$$\eta_v = \frac{Av}{Q_A} \tag{9-219}$$

where A is the area of the piston, and v is the velocity of the piston. For an ideal case, we can neglect the leakage and consider the hydraulic fluid to be incompressible. For an ideal case, we can write

$$Q_A = Av = A\dot{z} \tag{9-220}$$

This equation is based on the law of conservation of mass in fluid mechanics shown in Eq. 4-124.

9-12-2 Four-Way Electro-Hydraulic Valve

The four-way electro-hydraulic valve is a two-stage control valve that takes voltage as input and provides the required fluid flow and pressure to the actuator. The first stage of the valve is an electrically actuated hydraulic valve, which controls the displacement of the spool of the second stage of the valve. The second stage is a four-way spool valve, which controls the fluid flow and pressure into and out of ports A and B of the actuator. The first stage itself is not capable of overcoming opposing spring and flow forces; therefore, the second stage is required. Fig. 9-76 shows the schematic of the electro-hydraulic valve.

From Figs. 9-75 and 9-76, we can see that the displacement of the main spool is controlled by the left and right flow rates, which are controlled by the displacement of the

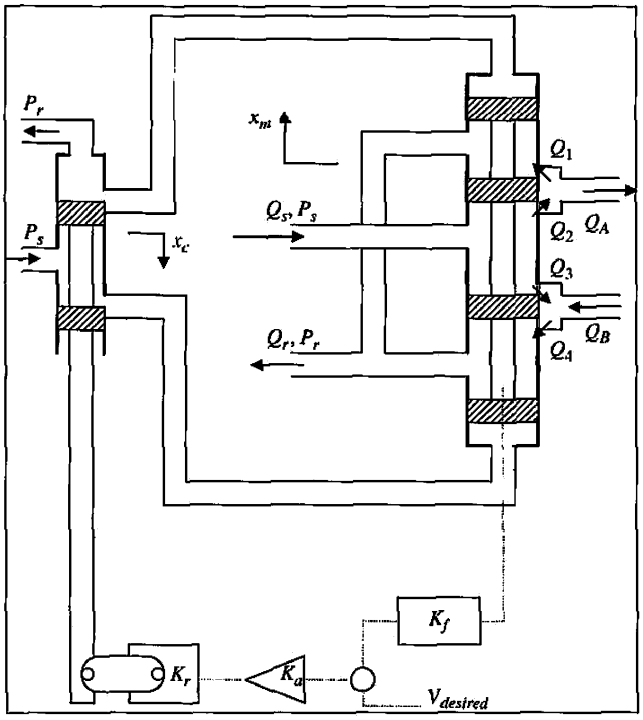


Figure 9-76 Electro-hydraulic two-stage four-way valve.

pilot spool. The control valve consists of a two-way critically centered valve, which has a pilot spool attached to a torque motor. The output from the servo amplifier actuates the torque motor and controls the displacement of the pilot spool.

The displacement of the pilot spool x_c is directly proportional to the output voltage of the servo amplifier. When the main spool is centered, there are no flow rates and both sides of the main spool are zero. We look at the operation of the valve around an operating point to be able to use the linearized flow equations derived in the following sections. At the nominal operating conditions,

$$x_m = x_c = 0,\ P_0 = P_{s0} - P_{B0} = P_{s0} - P_{A0} = \frac{1}{2}P_{s0},\ \text{and}\ P_r = 0$$

where x_m is the displacement of the main valve, x_c is the displacement of the pilot spool, P_{A0} and P_{B0} are pressure in the lines going from the main valve to the actuator, and P_r is the pressure of the return line to the reservoir. Note that, during the valve operation, the supply pressure coming from the pump is kept constant and equal for both the main valve and the control valve.

Orifice Equation

First, let's see what happens as the fluid passes through an orifice. Consider Fig. 9-77, which shows a flow passage with a sharp-edged orifice separating two sides of the flow passage. The classic orifice equation for the fluid flow can be written as

$$Q = A_0 C_d \sqrt{\frac{2}{\rho}(P_s - P_A)} \tag{9-221}$$

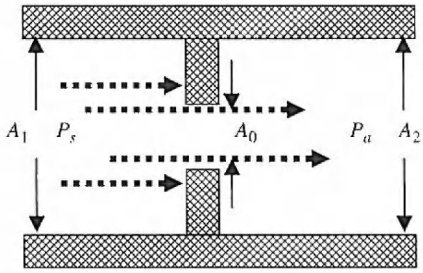


Figure 9-77 Fluid passing through an orifice.

where Q is the volumetric flow rate through the orifice, A_0 is the cross-sectional area of the orifice, ρ is the fluid density, P_s is the fluid pressure before passing through the orifice, P_A is the fluid pressure after passing through the orifice, and C_d is the discharge coefficient and is given by

$$C_d = \frac{\frac{A_2}{A_0}}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \tag{9-222}$$

where A_0, A_1 and A_2 are shown in Fig. 9-77.

Liberalized Flow Equations for the Four-Way Valve

In the orifice equation, we note that A_o and P_a are both variables. A, the opening area of the valve, is a function of the displacement of the spool. Using Taylor series, we can write Eq. (9-221) as

$$Q = Q_0 + \frac{\partial Q}{\partial A_o}\frac{\partial A_o}{\partial x}\bigg|_{x_0}(x - x_0) + \frac{\partial Q}{\partial P}\bigg|_{P_0}(P_a - P_0) \tag{9-223}$$

For Eq. (9-223) to be valid, the operating conditions of the valve should not deviate too much from the nominal operating conditions. If we take $x_0 = 0$ and $P_0 = P_{s0} - P_{B0} = P_{s0} - P_{A0} = \frac{1}{2}P_{s0}$, we can write Eq. (9-223) as

$$Q = \frac{1}{2}Q_0 + K_q x + K_c(P_s - P_A) \tag{9-224}$$

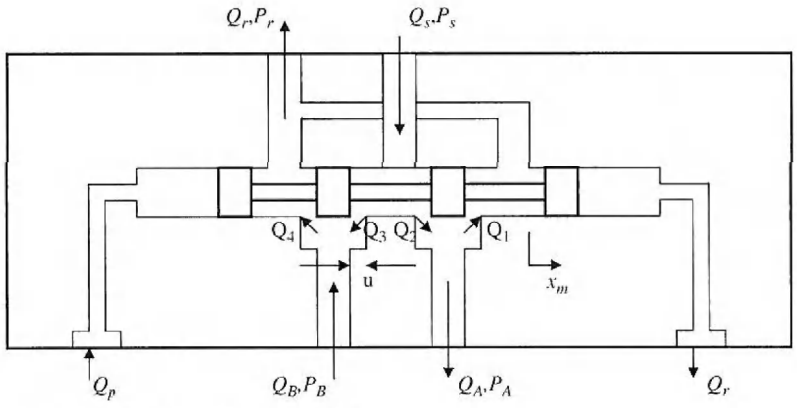


Figure 9-78 Main valve schematic.

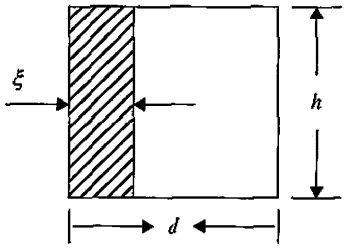


Figure 9-79 Rectangular valve-port geometry.

where K_q is called flow gain and is given by

$$K_q = C_d \sqrt{\frac{2}{\rho}(P_{s0} - P_{A0})} \left.\frac{\partial A}{\partial x}\right|_{x_0} \tag{9-225}$$

and K_c is the pressure-flow coefficient and is given by

$$K_c = \frac{AC_d}{\sqrt{2\rho(P_{s0} - P_{A0})}} \tag{9-226}$$

The term $\left.\frac{\partial A}{\partial x}\right|_{x_0}$ can be evaluated based on the valve-porting geometry. For the rectangular geometry shown in Fig. 9-79, we can write

$$A = h\xi \tag{9-227}$$

where ξ is the open distance of the valve port and h is the width of the valve port. For the open-centered valve shown in Fig. 9-79,

$$\xi = x + u \tag{9-228}$$

where x is the displacement of the spool and u is the fixed underlapped dimension.

Therefore,

$$\frac{\partial A}{\partial x} = h \tag{9-229}$$

Now we can write the following:

$$K_q = hC_d \sqrt{\frac{2}{\rho} P_0} \tag{9-230}$$

$$K_c = \frac{uhC_d}{\sqrt{2\rho P_0}} \tag{9-231}$$

where

$$P_0 = P_{s0} - P_{B0} = P_{s0} - P_{A0} = \frac{1}{2} P_{s0} \tag{9-232}$$

Let's now apply the linearized flow equation to the open-centered four-way valve shown in Fig. 9-76. For the four-way valve, ports A and B are used for directing flow to and

from the linear actuator. To direct flow into port A and receive flow from port B, the main spool moves to the right, thus allowing fluid flow from the supply line to port A and, at the same time, allowing fluid flow from port B to the return line. The flow rate and pressure levels at ports A and B are the function of the valve displacement x_m. When modeling the whole system, we can neglect the pressure drop in the pipelines connecting ports A and B of the linear actuator to ports A and B of the four-way valve. Hence, we can assume pressure levels and flow rates at ports A and B of the four-way valve are the same as pressure levels and flow rates at ports A and B of the linear actuator.

Now consider the main spool is moved to the right by a displacement x_m. Using Eq. (9-232), we can write

$$Q_1 = \frac{1}{2}Q_0 - K_q x_m + K_c(P_A - P_r) \tag{9-233}$$

$$Q_2 = \frac{1}{2}Q_0 + K_q x_m + K_c(P_s - P_A) \tag{9-234}$$

$$Q_1 = \frac{1}{2}Q_0 - K_q x_m + K_c(P_s - P_B) \tag{9-235}$$

$$Q_1 = \frac{1}{2}Q_0 + K_q x_m + K_c(P_B - P_r) \tag{9-236}$$

Note that, in the above equations, we can use the same flow gain, K_q, and the same pressure-flow coefficient, K_c, for all the metering lands because the nominal pressure across each metering land is equal to half the supply pressure $P_s/2$ and $P_r = 0$.

The volumetric flow rates into and out of the actuator can now be expressed as

$$Q_A = Q_2 - Q_1 = 2K_q x_m - 2K_c(P_A - P_s/2) \tag{9-237}$$

$$Q_B = Q_4 - Q_3 = 2K_q x_m + 2K_c(P_B - P_s/2) \tag{9-238}$$

Note that the above flow equations are valid for open-centered four-way valves. For other valve designs such as critically centered or closed-centered designs, the flow equations are different. For critically centered valves, such as the control valve in Fig. 9-76, the flow equation can be expressed as

$$Q_A = Q_B = K_q x_c \tag{9-239}$$

The flow-rate equations into and out of the actuator are very important in control analysis, which will follow. The reason is that these equations enable us to relate the flow rate and the pressure level, which delivers power to the actuator. Using these equations, we are able to control the displacement and velocity of the linear actuator (which depends on pressure level and the flow rate delivered to ports A and B of the actuator) based on the main spool displacement x_m. In other words, for this open-centered four-way valve control, we can change flow rate and pressure level simultaneously using a single variable of the main spool displacement x_m.

Relationship between Input Voltage and Main Spool Displacement x_m

Because the control valve is critically centered, the flow equation of the pilot spool can be expressed as

$$Q_P = K_q x_c \tag{9-240}$$

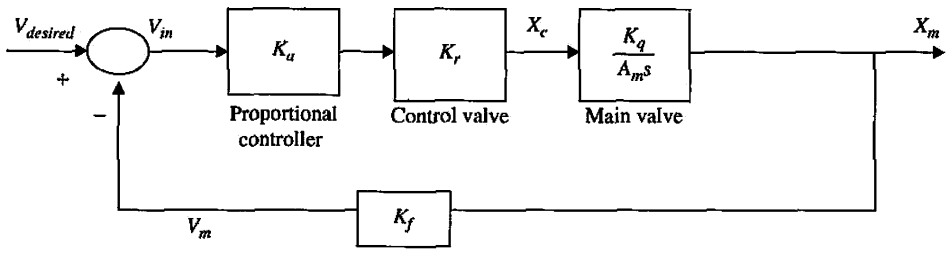


Figure 9-80 Two-stage valve block diagram.

Assuming an incompressible fluid, the interaction between the two stages can be described by

$$Q_p = A_m \frac{dx_m}{dt} \tag{9-241}$$

where A_m is the area of the main spool. The displacement of the pilot spool can be expressed as

$$x_c = V_{in} K_a K_r \tag{9-242}$$

Equating Eq. (9-241) and (9-242), we get

$$K_a K_r V_{in} = \frac{A_m}{K_q} \frac{dx_m}{dt} \tag{9-243}$$

For the two-stage valve alone (ignoring the output position feedback for the moment), using Laplace transforms and assuming zero initial conditions, we have

$$X_m(s) = \frac{K_a K_r K_q}{A_m s} V_{in}(s) \tag{9-244}$$

and

$$V_{in}(s) = V_{desired} - K_f X_m(s) \tag{9-245}$$

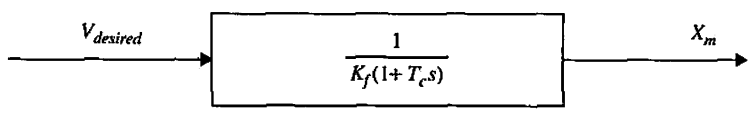
where X_m is the Laplace transform of x_m.
The transfer function of the two-stage valve can be expressed as

$$\frac{X_m(s)}{V_{desired}} = \frac{1}{K_f(1 + T_c s)} \tag{9-246}$$

where

$$T_c = \frac{A_m}{K_q K_a K_f K_r} \tag{9-247}$$

We can represent the two-stage valve with the block diagrams in Figs. 9-80 and 9-81.



Two-stage valve characteristics

Figure 9-81 Two-stage valve block diagram incorporating Eq. (9-247).

9-12-3 Modeling the Hydraulic System

Fig. 9-82 shows the schematic diagram of the whole system. In the following sections, let us rewrite the mathematical equations we derived for each component.

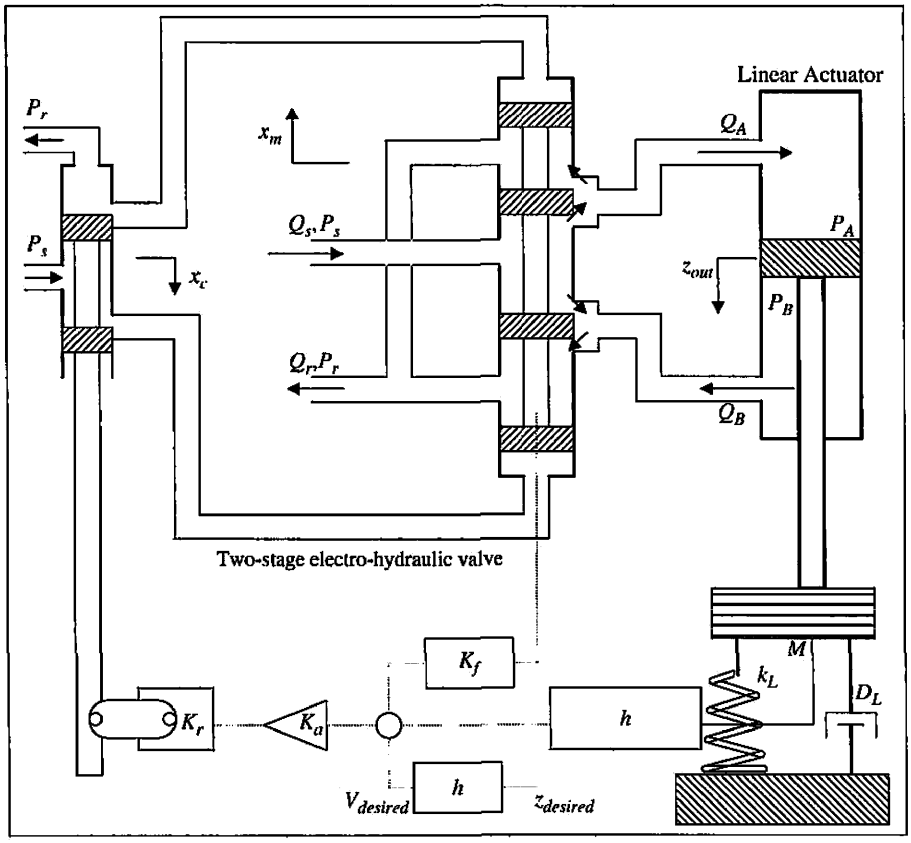


Figure 9-82 Schematic diagram of the whole translational system.

Force Balance Equation for an Ideal Linear Actuator

If the rod of the linear actuator is used for opposing a load force F_L in the z direction, using Eq. (9-247), we can express the force balance equation for the rod as

$$F_L = A(P_A - P_B) \tag{9-248}$$

where we assume the force efficiency to be one.

Expressing the Pressure Level P_A and the Pressure Level P_B

Substituting Eq. (9-220) into Eq. (9-237) and rearranging the terms, we can express P_A as

$$P_A = \frac{K_q}{K_c} x_m - \frac{A}{2K_c} \dot{z}_{out} + P_{s/2} \tag{9-249}$$

Substituting Eq. (9-220) into Eq. (9-239) and rearranging the terms, we can express P_B as

$$P_B = -\frac{K_q}{K_c}x_m + \frac{A}{2K_c}\dot{z}_{out} + P_{s/2} \tag{9-250}$$

Expressing the Pressure Difference Between Two Sides of the Linear Actuator
We can express the pressure difference, $P_A - P_B$, by subtracting Eq. (9-249) from Eq. (9-250):

$$P_A - P_B = 2\frac{K_q}{K_c}x_m - \frac{A}{K_c}\dot{z}_{out} \tag{9-251}$$

General Equation
Assuming an ideal actuator where force and volume efficiencies are one, and by substituting Eq. (9-248) into Eq. (9-251), we get

$$F_L = 2\frac{AK_q}{K_c}x_m - \frac{2A^2}{K_c}\dot{z}_{out} \tag{9-252}$$

Eq. (9-252) is the general equation that will be used to find the transfer functions for different applications in the following section.

9-12-4 Applications

In this section we apply the system model in three different applications: translational motion, rotational system, and variable load.

Position Control of a Hydraulic System (Translational Motion)
In this application, the linear actuator rod is used to move a mass, M; a spring, k_L; and a damper, D_L in the z direction, as shown in Figure 9-82.

For the two-stage valve alone,

$$X_m(s) = \frac{K_a K_r K_q}{A_m s} V_{in}(s) \tag{9-253}$$

$$V_{in}(s) = V_{desired} - K_f x_m(s) \tag{9-254}$$

The transfer function of the two-stage valve can be expressed as

$$\frac{X_m(s)}{V_{desired}} = \frac{1}{K_f(1 + T_c s)} \tag{9-255}$$

where

$$T_c = \frac{A_m}{K_q K_a K_f K_r} \tag{9-256}$$

We can represent the two-stage valve with the block diagram in Fig. 9-81.

To maintain position control of the actuator, a voltage V_0 is fed back from the actuator displacement z:

$$V_0 = hz_{out} \tag{9-257}$$

where h is the gain of the transducer. The input voltage to the two-stage valve V_{error} can be expressed as

$$V_{error} = V_{desired} - V_0 = V_{desired} - hz_{out} \tag{9-258}$$

It is desired to have position as input instead of voltage. As a result, a potentiometer is used to convert the desired input $z_{desired}$ to the desired voltage $V_{desired}$. The relationship can be expressed as

$$V_{desired} = hz_{desired} \tag{9-259}$$

Using general Eq. (9-252), the equation-relating input (main valve displacement x_m) and output (actuator displacement z_{out}) can be expressed as

$$z_{out}\left(Ms^2 + \left(D + \frac{2A^2}{K_c}\right)s + k_L\right) = 2\frac{AK_q}{K_c}X_m(s) \tag{9-260}$$

From the valve equation derived above, we know that main valve displacement x_m can be expressed as

$$X_m(s) = \left(\frac{1}{K_f(1 + T_c s)}\right)(V_{desired} - hz_{out}) \tag{9-261}$$

So the transfer function is

$$\frac{z_{out}}{z_{desired}} = \frac{2\dfrac{AK_q}{K_c}h}{K_f(1 + T_c s)\left(Ms^2 + \left(D + \dfrac{2A^2}{K_c}\right)s + k_L\right) + 2h\dfrac{AK_q}{K_c}} \tag{9-262}$$

where $T_c = \dfrac{A_m}{K_q K_a K_f K_r}$.

In the above equation,

K_q = Flow gain of the control valve
K_c = Pressure-flow coefficient
K_a = Gain of the proportional controller
K_f = Gain of the main spool feedback transducer
K_r = Ratio between the input voltage and displacement of the control spool
A_m = Area of the main spool
M = Mass of the load
D = Damping of the load
k_L = Load spring stiffness
h = Transducer gain
A = Area of the actuator piston

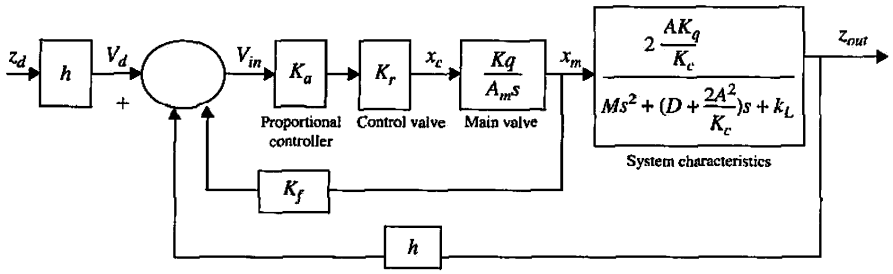


Figure 9-83 Translational system block diagram.

We can represent the system with the block diagrams in Figs. 9-83 and 9-84.

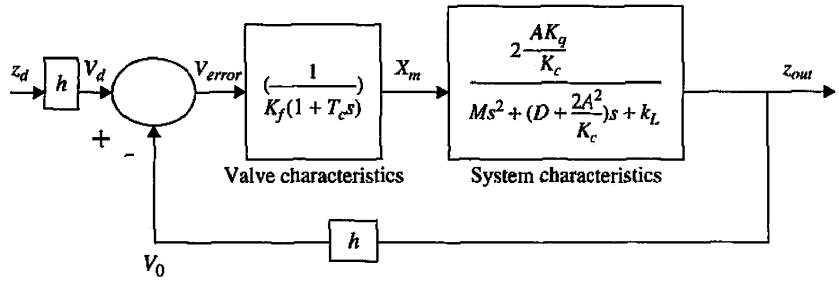
Position Control (Rotational System)

In this application we refer to Fig. 9-85, where the linear actuator rod is used to move a mass, M, θ radians. We start by writing the translational displacement of the rod in terms of the angular displacement θ:

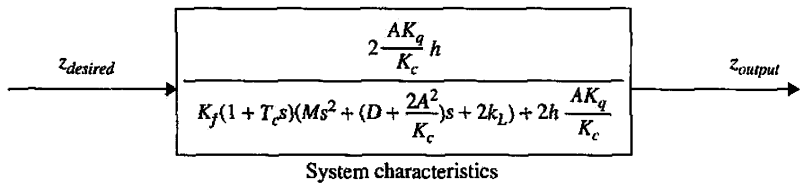
$$z_{out} = \theta_{out} L \tag{9-263}$$

From the valve equation derived in the previous section, we know that main valve displacement x_m can be expressed as

$$X_m(s) = \left(\frac{1}{K_f(1 + T_c s)}\right)(V_{desired} - h\theta_{out} L) \tag{9-264}$$



(a)



(b)

Figure 9-84 Translational reduced system block diagram.

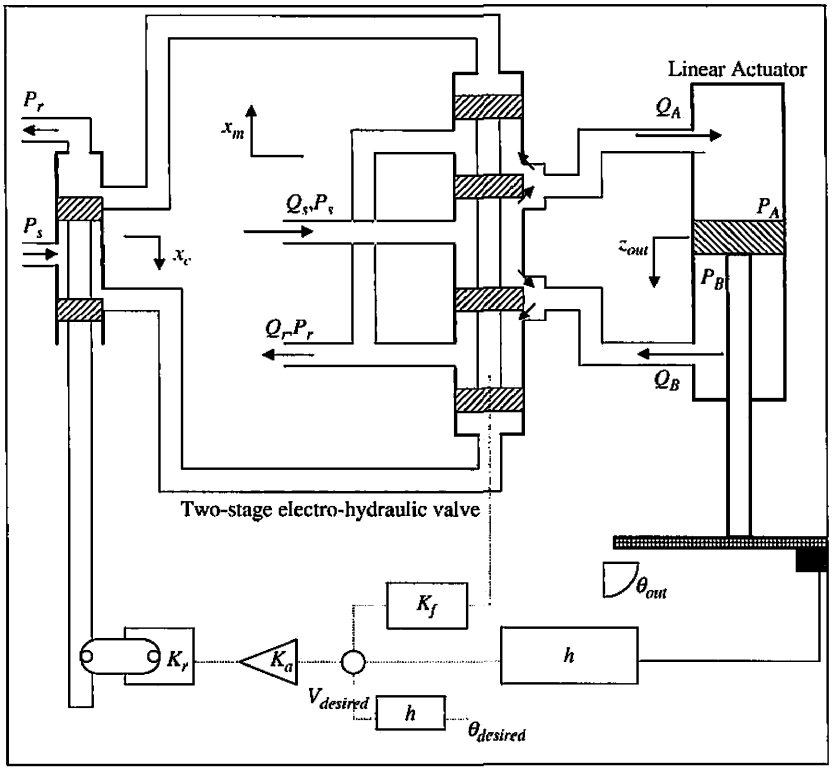


Figure 9-85 Schematic diagram of a linear actuator rod used for rotational motion.

Using Eq. (9-260) and Eq. (9-263) for the system shown, the equation-relating input (main valve displacement x_m) and output (actuator displacement θ_{out}) can be expressed as

$$\theta_{out}\left(Js^2 + \frac{2A^2}{K_c}Ls\right) = 2\frac{AK_q}{K_c}x_m \tag{9-265}$$

where J is the moment of inertia. Substituting Eq. (9-264) into Eq. (9-265), we can express the system transfer function as

$$\frac{\theta_{out}}{\theta_{desired}} = \frac{2\frac{AK_q}{K_c}h}{K_f(1+T_cs)\left(Js^2 + \frac{2A^2}{K_c}s\right) + 2hL\frac{AK_q}{K_c}} \tag{9-266}$$

Variable Load

In this application, as shown in Figure 9-86, the linear actuator rod is used to exert force on a variable load. From the valve equation derived in the previous section, we know that main valve displacement x_m can be expressed in Laplace domain as

$$X_m(s) = \left(\frac{1}{K_f(1+T_cs)}\right)(V_{desired} - hz_{out}) \tag{9-267}$$

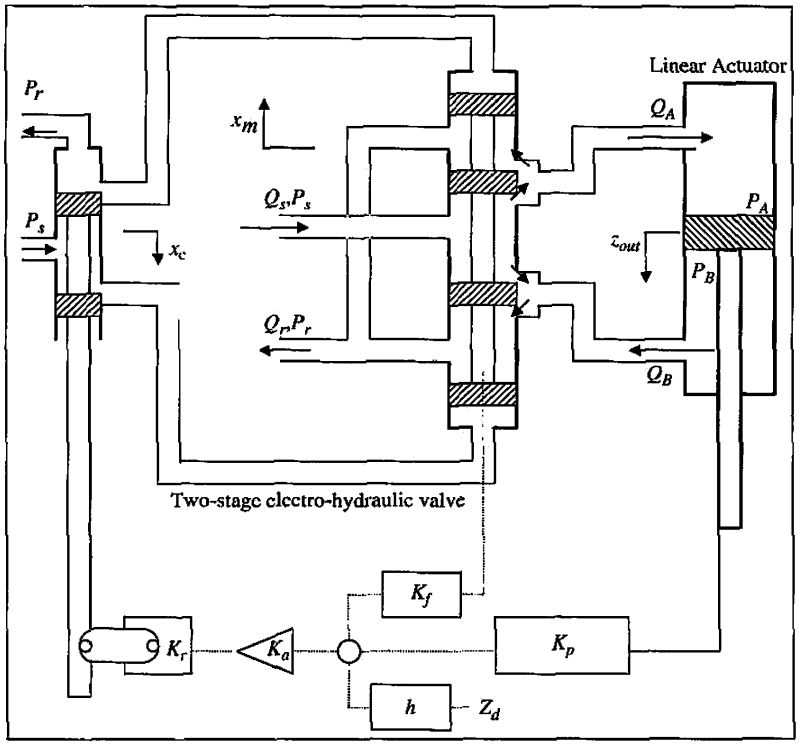


Figure 9-86 Schematic diagram of a variable load application.

Using general Eq. (9-252), the general equation for the system shown in Fig. 9-88 can be expressed as

$$F_L = 2\frac{AK_q}{K_c}\left(\frac{1}{K_f(1+T_c s)}\right)(hz_{desired} - hz_{out}) \tag{9-268}$$

▶ 9-13 CONTROLLER DESIGN

9-13-1 P Control

Consider the model of the hydraulic system shown in Fig. 9-88, where we added a proportional controller in the forward path. The new system block diagram is shown in Figure 9-87.

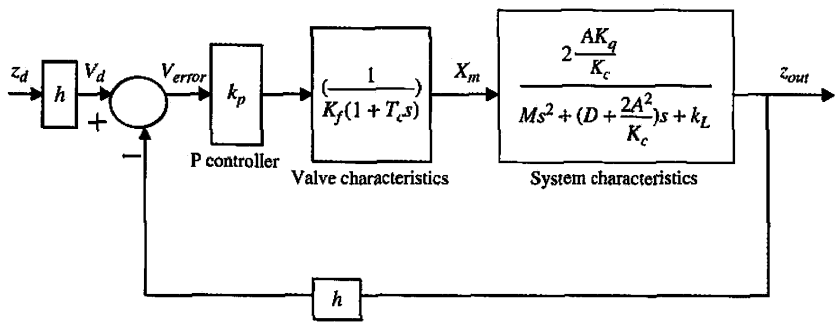


Figure 9-87 Block diagram of the hydraulic system with a proportional controller.

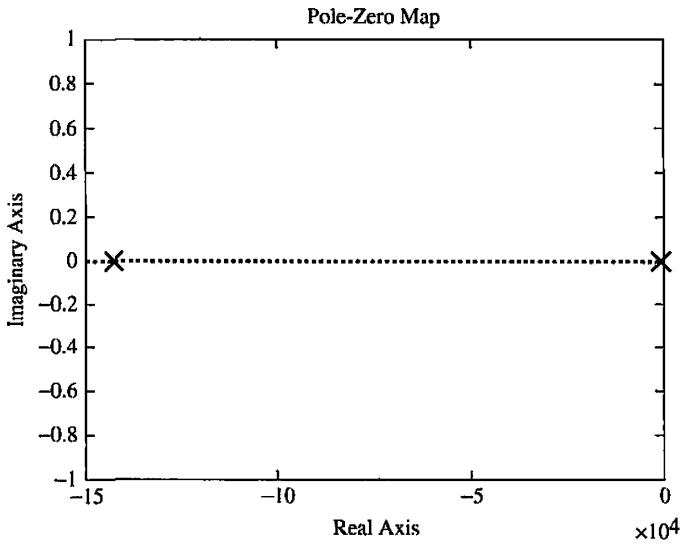


Figure 9-88 Pole–zero map of Eq. (9-269).

This system assumes an ideal actuator where force and volume efficiencies are one.

$K_q = 0.359\,m^2/s$ Flow gain of the control valve

$K_c = 1.70{*}10^{-11}$ m^3/Pa/s Pressure-flow coefficient

$K_a = 1$ Gain of the proportional controller

$K_f = 1$ Gain of the main spool feedback transducer

$K_r = 1$ Ratio between the input voltage and displacement of the control spool

$A_m = 550 \times 10^{-6}$ m^2 Area of the main spool

$M = 4$ kg Mass of the load

$D = 1$ N-s/m Damping of the load

$k_L = 2$ N/m Load spring stiffness

$h = 1$ Transducer gain

$A = 1.1 \times 10^{-3}$ m^2 Area of the actuator piston

Substituting the preceding values into the hydraulic system transfer function [Eq. (9-262)], we get

$$G(s) = \frac{3.033{*}10^{10}}{(s + 1.4235{*}10^5)(s + 653)(s + 7.02{*}10^{-6})} \tag{9-269}$$

The poles of Eq. (9-269) are shown in Fig. 9-88. Poles are at –142350, –653, and $-7.02{*}10^{-6}$.

We simplify the forward-path transfer function in Eq. (9-269) by neglecting the pole at –142350. The simplified hydraulic system transfer function is

$$G(s) = \frac{213041}{s^2 + 653s + 0.004584} \tag{9-270}$$

We note that one of the dominant poles is very close to the origin; therefore, we can write the forward-path transfer function as

$$G(s) = \frac{213041}{s(s + 653)} \tag{9-271}$$

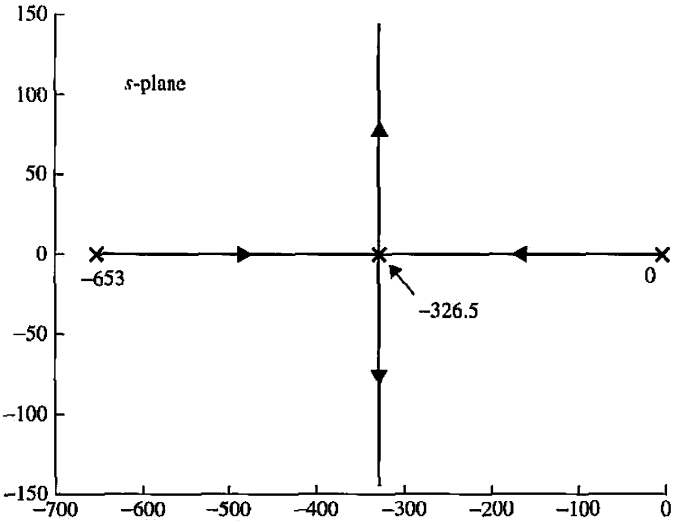


Figure 9-89 Root loci of Eq. (9-274).

Applying a proportional controller, the forward-path transfer function becomes

$$G(s) = \frac{213041(K_P)}{s^2 + 653s} \tag{9-272}$$

The closed-loop transfer function is

$$T(s) = \frac{213041(K_P)}{s^2 + 653s + 213041K_P} \tag{9-273}$$

The characteristic equation is written

$$s^2 + 653s + 213041K_P = 0 \tag{9-274}$$

The root loci of Eq. (9-274) are shown in Fig. 9-89.

By looking at the root loci in Fig. 9-90, we see that, depending on the value of K_p, we can get two real or two complex-conjugate poles. When $K_p = 0.500$, we get a damping

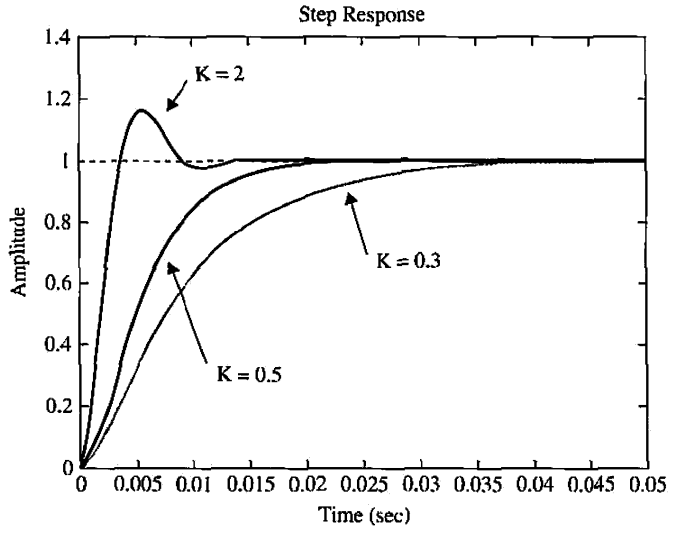


Figure 9-90 Unit-step responses with P control.

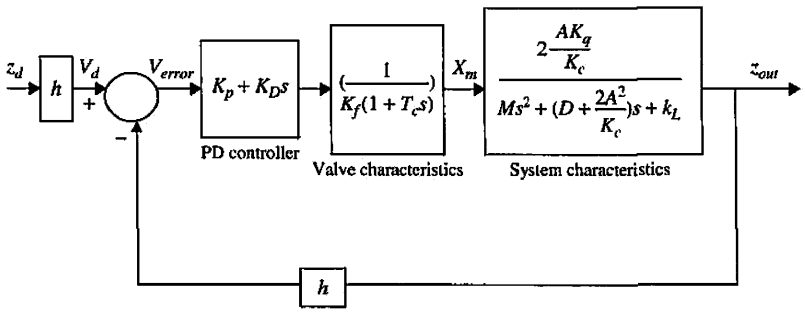


Figure 9-91 System block diagram.

ratio of 1, and the system is critically damped. For values of K_p greater than 0.500, the system is underdamped, and for values of K_p less than 0.500, the system is overdamped. We also note that the system is stable for all the values of $K_p > 0$. The step responses of the system for three different values of K_p are shown in Fig. 9-91.

Toolbox 9-13-1

Fig. 9-89 is obtained by the following sequence of MATLAB functions

```
num = [3.0333e+10];
den = conv(conv([1 1.4235e5],[1 653]),[1 7.02e-6])
pzmap(num,den);
```

Fig. 9-90 is obtained by the following sequence of MATLAB functions

```
KP=1;
KD=0;
num = [1 653+213041*KD 213041*KP];
den = [1 653 213041*KP];
tf(num,den)
rlocus(num,den)
axis([-700 0 -1 1])
```

Fig. 9-91 is obtained by the following sequence of MATLAB functions

```
KP=0.3;
num = [0 213041*KP];
den = [1 653 213041*KP];
step(num,den)
hold on;
KP=0.5;
num = [0 213041*KP];
den = [1 653 213041*KP];
step(num,den)
hold on;
KP=2;
num = [0 213041*KP];
den = [1 653 213041*KP];
step(num,den)
hold on;
```

TABLE 9-27 Attributes of the Unit-Step Response with P control

K_p	Maximum Overshoot (%)	t_s(sec)	t_r(sec)	Steady-State Error Due to Unit Step
2	16.3	0.0124	0.00253	0
0.5	—	0.0179	0.0103	0
0.3	—	0.0348	0.0191	0

Table 9-27 summarizes the attributes of the system's unit-step response for three different values of K_p.

When poles are complex conjugates, as we increase K_p, the overshoot of the system increases, but the settling time and rise time of the system remain unchanged. Also note that steady-state error due to unit-step input decreases as K_p increases.

9-13-2 PD Control

Consider the second-order forward-path transfer function of the hydraulic system discussed in Section 9-13-1. Applying a PD controller, the forward-path transfer function becomes

$$G(s) = \frac{213041(K_P + K_D s)}{s(s + 653)} \tag{9-275}$$

The system block diagram is shown in Fig. 9-91.

Now, let us set the performance specifications as follows:

Settling time $t_s \leq 0.005$ sec

Maximum overshoot $\leq 5\%$

Steady-state error due to unit-ramp input ≤ 0.00061

The closed-loop transfer function is

$$T(s) = \frac{213041(K_P + K_D s)}{s^2 + (653 + 213041K_D)s + 213041K_P} \tag{9-276}$$

The characteristic equation is written

$$s^2 + (653 + 213041K_D)s + 213041K_P = 0 \tag{9-277}$$

We start by finding the steady-state error for a unit-ramp input:

$$e_{ss}|_{ramp} = \lim_{s \to 0} \frac{1}{1 + sG(s)} = \frac{1}{1 + \frac{213041K_P}{653}} \tag{9-278}$$

Therefore, for the system to have steady-state error due to unit ramp ≤ 0.00061, we need $K_P \geq 5$. The damping ratio of the system for $K_P = 5$ can be expressed as

$$\zeta = \frac{653 + K_D 213041}{2064.2} = 0.316 + 103.209K_D \tag{9-279}$$

If we wish to have critical damping, $\zeta = 1$, the above equation gives $K_D = 0.0066$. One thing we need to check is that this value satisfies the settling-time requirement. Settling time can be expressed as

$$t_s = \frac{8}{(653 + 213041K_D)} \tag{9-280}$$

We see that, for $K_D \geq 0.0044$, we have $t_s \leq 0.005$ sec. Therefore, with $K_D = 0.006624$, we can satisfy the settling-time requirement. We note that Eq. (9-280) is an approximation and, as damping ratio comes closer to one, the actual settling time will be higher; therefore, K_D must be higher than 0.0044 to satisfy the settling time for damping ratio of one. Nevertheless, we can still use the approximation and verify our answer by simulation once a value for K_D is found. Also, we need to make sure that, for the values found for K_P and K_D, the system is stable. Applying the stability requirement, we find that for system stability

$$K_P > 0 \quad \text{and} \quad K_D > -0.00307$$

Alternatively, we can use the system's root contours to find K_P and K_D. We can apply the root-contour method to the characteristic equation in Eq. (9-277) to examine the effect of varying K_P and K_D. First, by setting K_D to zero, Eq. (9-277) becomes

$$s^2 + 653s + 213041K_P = 0 \tag{9-281}$$

The root loci of Eq. (9-281) are shown in Fig. 9-92.

The code used to plot the root locus is given in Toolbox 9-13-2 for $K_D = 0$. Alternatively you can use Toolbox 5-8-1. When $K_D \neq 0$, the characteristic equation in Eq. (9-281) becomes

$$1 + G_{eq}(s) = 1 + \frac{213041K_D s}{s^2 + 653s + 213041K_P} = 0 \tag{9-282}$$

The root contours for $K_P = 1$ and $K_P = 5$ based on the pole–zero configuration of $G_{eq}(s)$ are shown in Fig. 9-94. Note that we chose $K_P = 5$ to satisfy the steady-state error

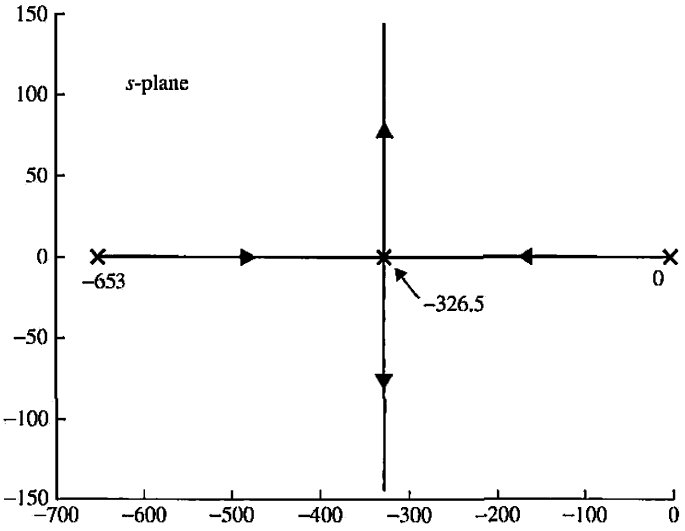


Figure 9-92 Root loci of Eq. (9-281).

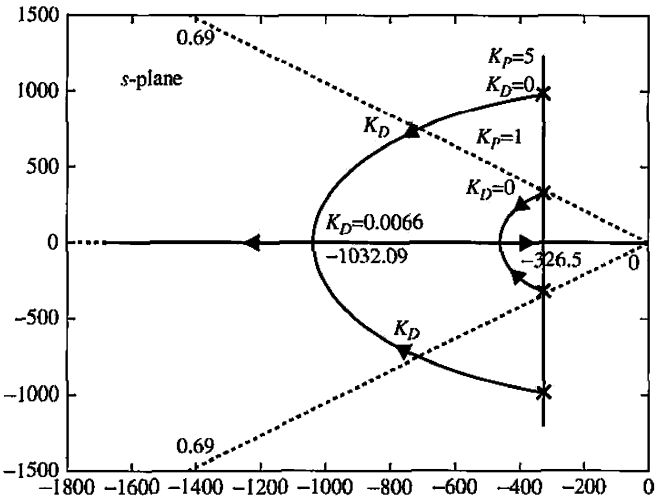


Figure 9-93 The root contours of Eq. (9-281) or Eq. (9-277) when $K_D = 0$.

requirement. For $K_P = 5$ and $K_D = 0$, the characteristic equation roots are at $-326.5 \pm j979.08$ and the damping ratio of the closed-loop system is 0.316. When the value of K_D is increased, the two characteristic equation roots move toward the real axis along a circular arc. The dashed line shows the points on the s-plane with the constant damping ratio of 0.69, which corresponds to 5% overshoot. We see that this line intersects the root contour for $K_P = 5$ at $K_D = 0.00362$. Because this value of K_D is not big enough to satisfy the settling-time requirement of less than 0.002 sec, we need a larger K_D. When K_D is increased to 0.006624, the roots are real and equal at –1032.09, and the damping is critical. At this point $K_D \geq 0.0044$; therefore, our settling-time requirement is met. When $K_p = 1$ and $K_D = 0$, the two characteristic equation roots are at $-326.5 \pm j326.3$. As K_D increases in value, the root contours again show the improved damping due to the PD controller.

Fig. 9-95 shows the unit-step responses of the closed-loop system without PD control and with $K_p = 5$ and $K_D = 0.006624$. With the PD control, although K_D is chosen for critical damping, the maximum overshoot is 0.888%. This is because of the zero at $s = -\dfrac{K_P}{K_D}$ for the

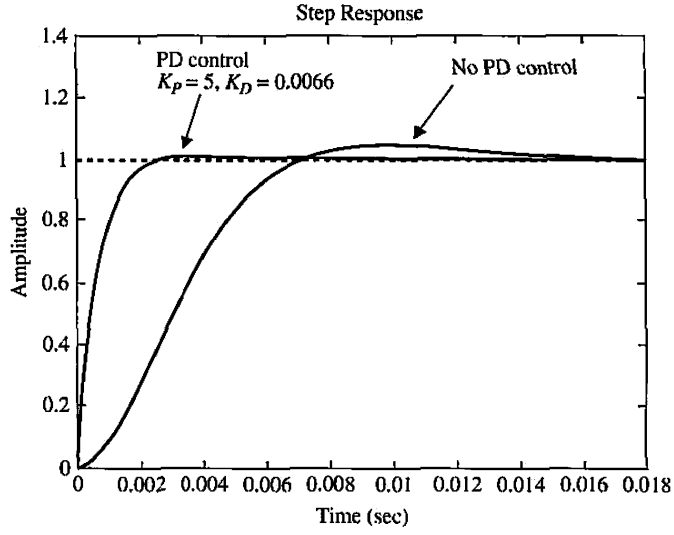


Figure 9-94 Unit-step responses with and without PD control using Toolbox 9-13-3.

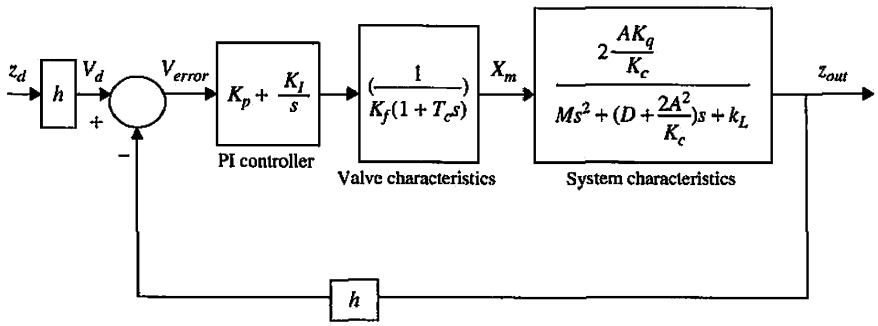


Figure 9-95 System block diagram with PI controller.

closed-loop transfer function. When $K_p = 5$ and $K_D = 0.0066$, the zero is at –754.5, which is close to the dominant poles of the system at –1032.09; therefore, it has a significant effect on the transient response of the system. Note that the zero makes the response faster by lowering the rise time and the settling time of the system.

Toolbox 9-13-2

Fig. 9-94 is obtained by the following sequence of MATLAB functions

```
%Closed loop transfer function for Kp=5
sgrid(0.69);
Kp = 5;Kd = 1e-6;
for i = 1:1:260
    if (i == 222)
        Kd = (((Kp*213041)^(0.5)*2)-653)/213041; % if this statement is not used
then the graph will not continuous
    end
    num =[213041*Kd 213041*Kp];den = [1 653 0];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    T = tf(numCL,denCL);
    PoleData(:,i)=pole(T);
    Kd = Kd +3e-5;
end
plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));

%Closed loop transfer function for Kp=1
Kp = 1; Kd = 1e-6;
for i = 1:1:260
    if (i == 44)
        Kd = (((Kp*213041)^(0.5)*2)-653)/213041;
    end
    num =[213041*Kd 213041*Kp];den = [1 653 0];
    tf(num,den);
    [numCL,denCL]=cloop(num,den);
    T = tf(numCL,denCL);
    PoleData(:,i)=pole(T);
    Kd = Kd +3e-5;
end
hold on
plot(real(PoleData(1,:)),imag(PoleData(1,:)),real(PoleData(2,:)),imag(PoleData(2,:)));
```

```
%open loop transfer function
Kp = 1; Kd = 0;
num =[213041*Kd 213041*Kp];
den = [1 653 0];
hold on
rlocus(num,den);
axis([-1800 0 -1500 1500])
step(num,den)
hold on;

KP=2;num = [0 213041*KP];den = [1 653 213041*KP];
step(num,den)
hold on;
```

Table 9-28 summarizes the attributes of the unit-step response for $K_p = 5$ and different values of K_D.

TABLE 9-28 Attributes of the Unit-Step Response with PD Control

K_D	Maximum Overshoot (%)	t_s(sec)	t_r(sec)
0.001	23.9	0.00794	0.00141
0.006624	0.888	0.00215	0.00138
0.01	—	0.00363	0.00126

Toolbox 9-13-3

Fig. 9-94 is obtained by the following sequence of MATLAB functions

```
Kp = 5;Kd = 0.0066;
num =[213041*Kd 213041*Kp];
den = [1 653 0];
tf(num,den);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on
Kp = 1;Kd = 0;
num =[213041*Kd 213041*Kp];
den = [1 653 0];
tf(num,den);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
```

At this point we need to go back to the original hydraulic transfer function expressed in Eq. (9-269), which has three poles. We need to make sure that the third pole is still far away from the origin so that the second-order approximation is valid. In the root locus of the pure proportional control system, the third pole moves to the left as K increases; therefore, the second-order approximation is valid for any value of $K \geq 0$. For the PD control design system, we introduce a zero in the forward path, which changes the behavior of the third pole. In this case, the third pole starts to move toward the origin as we increase K_D. For $K_p = 5$ and $K_D = 0.0066$, the poles of the closed-loop third-order system are at –14093, –1090, and –980.

Because the third pole is still far away from the origin relative to the other poles, it can be safely neglected, and our second-order approximation holds.

9-13-3 PI Control

Consider the second-order forward-path transfer function of the hydraulic system discussed in Section 9-13-1. Applying a PI controller, the forward-path transfer function becomes

$$G(s) = \frac{213041K_P(s + K_I/K_P)}{s^2(s + 653)} \tag{9-283}$$

The system's block diagram is shown in Fig. 9-95.

Let the time-domain performance requirement be as follows:

Settling time $t_s \leq 0.06$ sec

Rise time $t_r \leq 0.01$ sec

Maximum overshoot $\leq 1.52\%$

Steady-state error due to parabolic input ≤ 0.2

We start by finding the steady-state error for a ramp input:

$$e_{ss}|_{parabolic} = \lim_{s \to 0} \frac{1}{s^2 G(s)} = \frac{1}{213041K_I/653} \tag{9-284}$$

Therefore, for the system to have steady-state error due to parabolic input ≤ 0.2, we need $K_I \geq 0.015$. Later we need to make sure that the value of K_I used is above 0.015.

The characteristic equation of the closed-loop system is

$$s^3 + 653s^2 + 213041K_P s + 213041K_I = 0 \tag{9-285}$$

Applying Routh's test to Eq. (9-277) yields the result that the system is stable for $0 < \frac{K_I}{K_P} < 653$. This means that, if the zero of $G(s)$ be placed too far to the left in the left-half s-plane, the system will be unstable.

Let us place the zero at $-K_I/K_p$ relatively close to the origin. For this case, the most significant poles of the forward-path transfer function without the PI controller are at –653 and 0. Thus K_I/K_p should be chosen so that the following condition is satisfied

$$\frac{K_I}{K_P} \ll 653 \tag{9-286}$$

The root loci of Eq. (9-283) with $K_I/K_P = 5$ are shown in Fig. 9-96. Notice that one of the poles always has a value close to zero while the other two poles behave the same as those shown in Fig. 9-93, which is for Eq. (9-281).

With the condition in Eq. (9-286) satisfied, the pole near zero is effectively cancelled by the zero at $-K_I/K_p$, and we are left with the poles of Eq. (9-283). Therefore, Eq. (9-284) can be approximated by

$$G(s) \cong \frac{213041K_P}{(s^2 + 653s + 213041)} \tag{9-287}$$

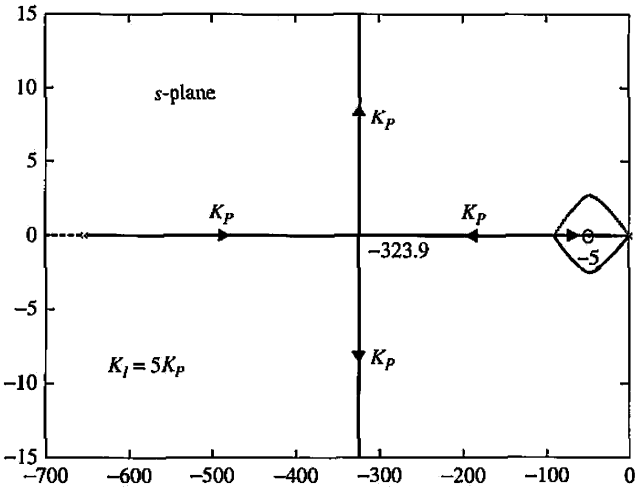


Figure 9-96 Root loci of Eq. (9-283).

Let us assume we wish to have a relative damping ratio of 0.9. From Eq. (9-287), the required value of K_P for this damping ratio is 0.62. Thus with $K_P = 0.62$ and $K_I = 3.09$, we find the roots of the characteristic equation of Eq. (9-285):

$$s = -5.13 \quad -323.94 + j152.8 \quad \text{and} \quad -323.9.3 - j152.8$$

In this case we see that the real pole of the closed-loop system is very close to the zero at $-K_I/K_p$ so that the transient due to the real pole is negligible, and the system dynamics are essentially dominated by the two complex poles; therefore, $K_P = 0.62$ will give us a damping ratio that is close to 0.9. In general, when s takes on values along the operating points on the complex portion of the root loci, we can neglect the effect of the real pole of the closed-loop system and use Eq. (9-287) to find the system characteristics.

Table 9-29 summarizes the attributes of the unit-step response for various values of K_I/K_P with $K_P = 0.62$, which corresponds to a relative damping ratio of 0.9. The results verify that PI control reduces overshoot at the expense of longer rise time.

Fig. 9-97 shows the unit-step response of the hydraulic system with PI control with $K_P = 0.62$ and $K_I = 3.09$. The unit-step response is the same for system with P control designed in Section 9-13-1.

TABLE 9-29 Attributes of the Unit-Step Response with PD Control

K_I/K_P	K_I	K_P	Maximum Overshoot (%)	t_s(sec)	t_r(sec)
0	0	5	35.1	0.0108	0.00131
20	12.4	0.62	8.78	0.0856	0.00688
12	7.44	0.62	5.59	0.099	0.00723
5	3.09	0.62	2.52	0.0555	0.00778
3	1.86	0.62	1.6	0.0121	0.0078
1	0.62	0.62	—	0.0126	0.00787
0.5	0.31	0.62	—	0.0128	0.00791

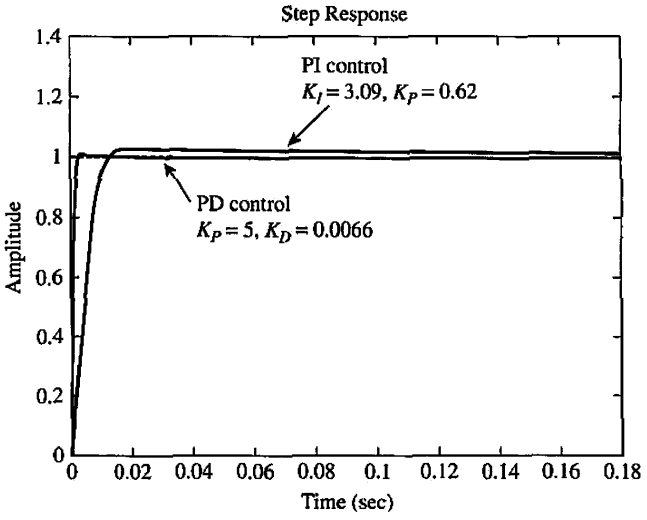


Figure 9-97 Unit-step responses of the system with PD and PI control using Toolbox 9-13-1.

Toolbox 9-13-4

Fig. 9-97 is obtained by the following sequence of MATLAB functions

```
Ki =3.09
Kp = 0.62
num = [213041*Kp 213041*Ki/Kp]
den = [1 653 0 0];
tf(num,den);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on
Kp = 5;
Kd = 0.0066;
num =[213041*Kd 213041*Kp];
den = [1 653 0];
tf(num,den);
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on
```

9-13-4 PID Control

Consider the second-order forward-path transfer function of the hydraulic system discussed previously. Applying a PID control, the forward-path transfer function of the plant is

$$G(s) = \frac{213041(K_{P1} + K_{D1}s)(K_{P2} + K_{I2}/s)}{s(s + 653)} \tag{9-288}$$

The system's block diagram is shown in Fig. 9-98.

Let the time-domain performance requirements be as follows:

Rise time $t_r \leq 0.003$ sec

Settling time $t_s \leq 0.004$ sec

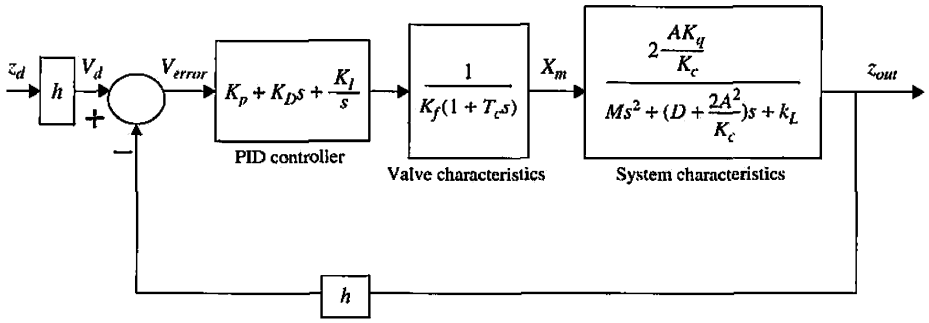


Figure 9-98 System block diagram with PID controller.

Maximum overshoot $\leq 1.2\%$

Steady-state error due to parabolic input ≤ 0.2

We realize that we need a PID controller to fulfill the above requirements. First, we apply the PD control with the transfer function $K_{P1} + k_{D1}s$. The forward-path transfer function becomes

$$G(s) = \frac{213041(K_{P1} + K_{D1}s)}{s(s + 653)} \tag{9-289}$$

Table 9-28 shows that the PD controller that can be obtained from the settling time standpoint is with $K_{D1} = 0.0066$ and $K_{P1} = 5$, and the maximum overshoot is 0.888%. The rise time and settling time are well within required values. Next we add the PI controller, and the forward-path transfer function becomes

$$G(s) = \frac{1406.07K_{P2}(s + 757.58)(s + K_{I2}/K_{P2})}{s(s^2 + 653s)} \tag{9-290}$$

Following the guidelines of choosing a relatively small value for K_{I2}/K_{P2}, we let $K_{I2}/K_{P2} = 5$. Eq. (9-290) becomes

$$G(s) = \frac{1406.07K_{P2}(s + 757.58)(s + 5)}{s(s^2 + 653s)} \tag{9-291}$$

Table 9-30 gives the time domain performance characteristics along with the roots of the characteristic equation for various values of K_{P2}.

TABLE 9-30 Attributes of the Unit-Step Response with PID Control

K_{P2}	Maximum Overshoot (%)	t_s(sec)	t_r(sec)	Roots of Characteristic Equation
2	1.23	0.00113	0.000718	−5 −799.5 −2660.6
1.2	1.27	0.00178	0.00116	−5.0 −14464.5 −870.3
0.9	1.2	0.00232	0.00151	−5.02 −956.7 $\pm j$200.2
0.62	1.05	0.00327	0.00213	−5.02 −759.9 $\pm j$282.4
0.40	0.91	0.00505	0.00321	−5.04 −605.9 $\pm j$237.8
0.20	—	0.00983	0.00622	−5.08 −542.6 −386.5

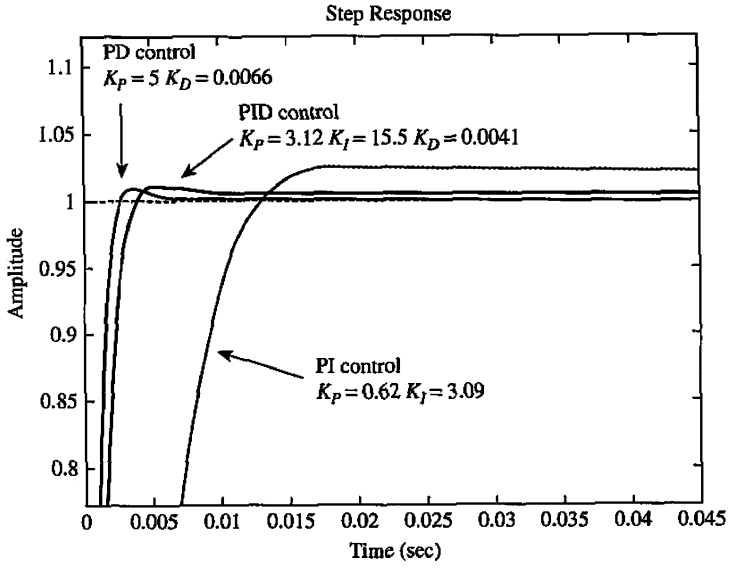


Figure 9-99 Unit-step responses with PD, PI, and PID control using Toolbox 9-13-5.

Setting $K_{P1} = 5, K_{D1} = 0.0.0066, K_{P2} = 0.62$, and $K_{I2} = 5K_{P2} = 3.09$, the following results are obtained for the parameters of the PID controller.

$$K_P = 3.12$$
$$K_I = 15.5$$
$$K_D = 0.0041$$

Fig. 9-99 shows the unit-step responses of the system with the PID controller, as well as those with PD and PI controls designed before.

Toolbox 9-13-5

Fig. 9-99 is obtained by the following sequence of MATLAB functions

```
Kp1=5;kd1=0.0066;
num = [21304*kd1 21304*Kp1];den = conv([1 0],[1 653]);
tf(num,den)
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on
Kp2=0.62;KI2=5*Kp2;
num = conv(conv([0 1406.07*Kp2],[1 757.58]),[1 KI2/Kp2])
den = conv([1 0],[1 653 0]);
tf(num,den)
[numCL,denCL]=cloop(num,den);
step(numCL,denCL)
hold on
Kp2=2;num = conv(conv([0 1406.07*Kp2],[1 755.58]),[1 5]);
den = [1 653 0 0];
tf(num,den)
[numCL,denCL]=cloop(num,den);
tf(numCL,denCL)
step(numCL,denCL)
```

9-14 MATLAB TOOLS AND CASE STUDIES

In this section we will go through the steps involved in finding some of the results and displaying many of the graphics from the examples in this chapter using the **ACSYS** software. The tools involved are all contained in the Controller Design Tool. This tool allows the user to conduct the following tasks:

- Enter the transfer function values in polynomial form. (User must use the tftool as discussed in Chapter 2 to convert the transfer function from pole-zero-gain form into polynomial form.)
- Obtain the step, impulse, parabolic, ramp, or other type input time responses.
- Obtain the closed-loop frequency plots.
- Obtain the phase and gain margin Bode plots and the polar plot of the loop transfer functions (in a single feedback loop configuration).
- Understand the effect of adding zeros and poles to the closed-loop or open-loop transfer functions.
- Design and compare various controllers including PID, lead, and lag compensators.

To run the Controller Design Tool, type "Acsys" at the MATLAB command line and click the appropriate button on the menu. Fig. 9-100 will appear on the screen.

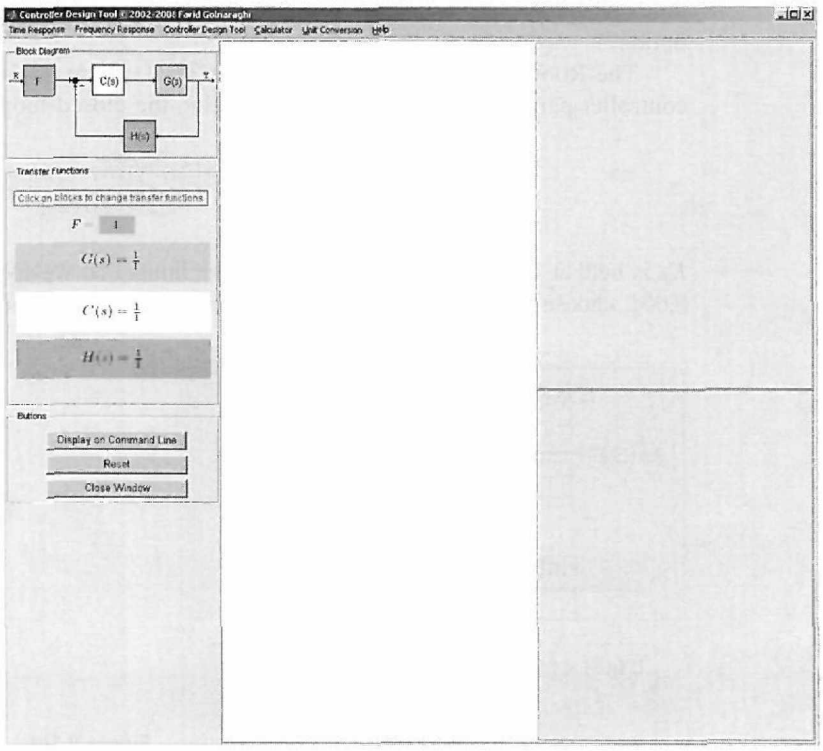


Figure 9-100 Main Controller Design Tool window.

EXAMPLE 9-2-1 Revisited Recall from Example 9-2-1 the forward-path transfer function for the altitude-control system was

$$G(s) = \frac{4500K}{s(s + 361.2)} \tag{9-292}$$

The design constraints for this problem were as follows:

Steady-state error due to unit-ramp input = 0.000443
Maximum overshoot $\leq 5\%$
Rise time $t_r \leq 0.005$ sec
Settling time $t_s \leq 0.005$ sec

To enter the transfer function into the CONTROLS tool, click on the G(s) box in the transfer function input panel. For the moment we will leave the value of K as 1, so in the numerator text box, enter [4500], and in the denominator text box, enter [1,361.2,0], as is pictured in Fig. 9-101.

Pressing the enter button will display the current closed-loop transfer function. For a more accurate representation of the transfer function, press the "Display on Command Line" button and refer to the MATLAB command window.

Now click on the C(s) box and enter the value [181.2]. Click on the "Step Response" option from the "Time Response" menu to see the step response of this system. The main window should now look like Fig. 9-102.

Press the "Print To Figure" button, , and right-click on the resulting plot to see more information about the response.

As in Example 9-2-1, we shall use a PD controller to improve the response of the system. We will use the root contour tool to see the effect that the PD controller has on the poles of the system. From the "Controller Design Tool" menu, choose the "PD design option."

The Root Contour tool plots the poles of a system as functions of certain varying controller parameters. With the PD controller, the closed-loop transfer function is

$$G(s) = \frac{\Theta_y(s)}{\Theta_e(s)} = \frac{815,265(K_P + K_D s)}{s(s + 361.2)} \tag{9-293}$$

K_P is held at 1, and K_D will vary between the limits that we specify. Choose K_D min to be 0.001, choose K_D max to be 0.005, and choose 2000 steps. Press enter and you will see the

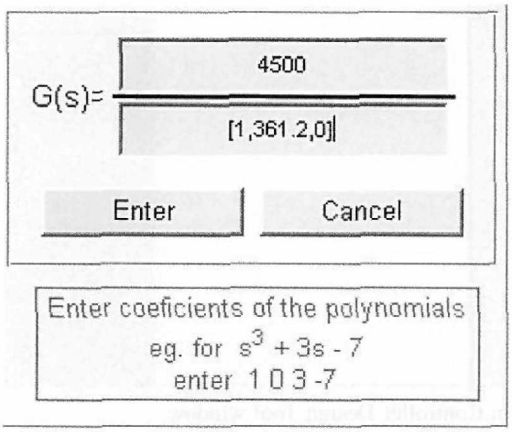


Figure 9-101 Transfer function input module.

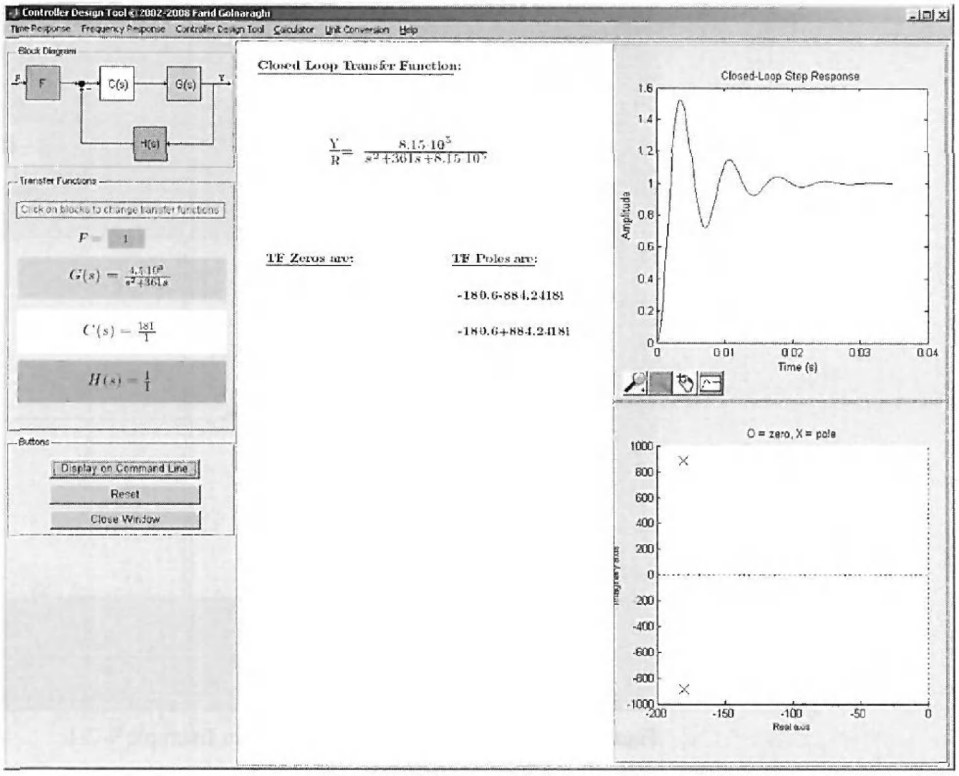


Figure 9-102 Controller Design window showing closed-loop transfer function, step response, and poles of the system.

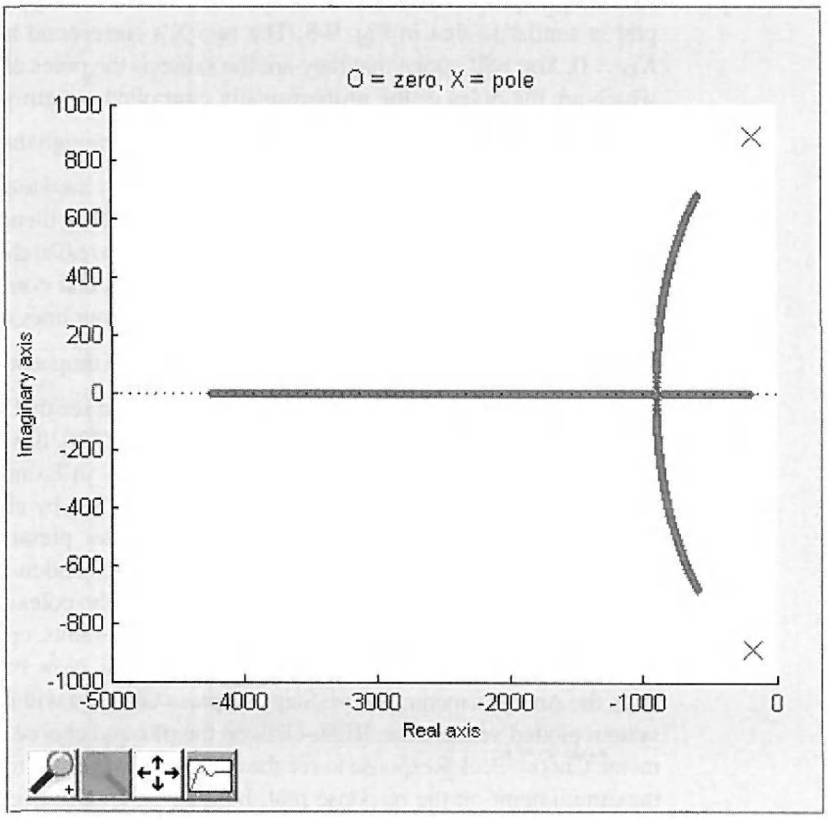


Figure 9-103 Root contours of PD-controlled system from Example 9-2-1.

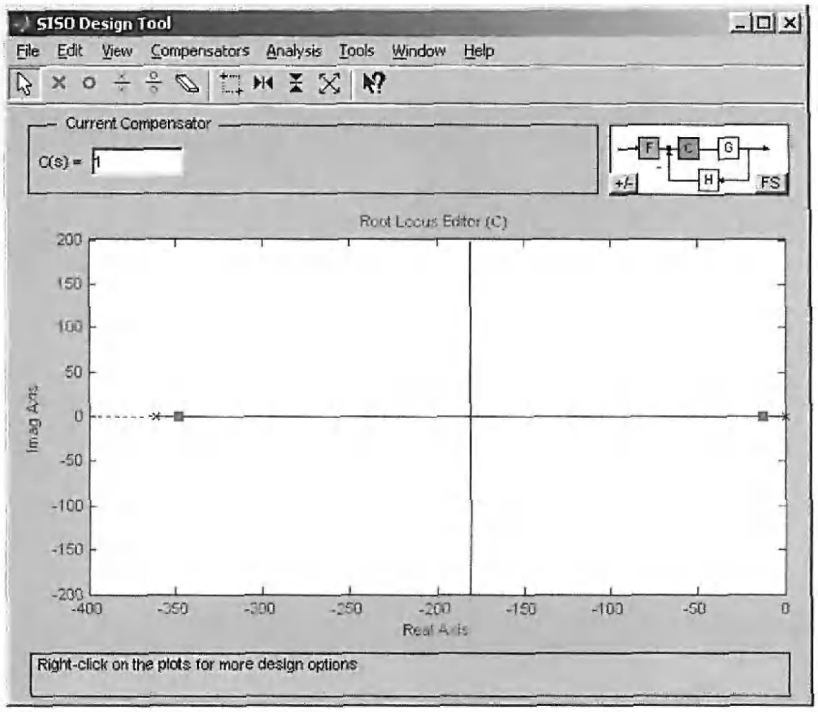


Figure 9-104 Root loci of the system from Example 9-2-1.

contour plot pictured in Fig. 9-104 in the axis in the top right of the window. Notice that this plot is similar to that in Fig. 9-8. The two X's correspond to the poles of the system with $K_D = 0$. You will notice that they are the same as the poles displayed in the lower right axis, which are the poles of the proportionally controlled system we entered earlier. Pressing the button below the axis will enable you to scroll through the different values of K_D with the arrow keys and see the corresponding poles. As K_D increases, the two poles of the system move together until their imaginary parts disappear. They then move apart on the real axis. We know that, for a second-order system such as this, two real and equal poles is a characteristic of a critically damped system. To find the value of K_D that corresponds to a critically damped system, zoom in on the point where the two root contour lines meet on the real axis. To do this, click on the zoom-in button, , and then click and drag a box on the area you would like to enlarge. By scrolling through the values of K_D, we can see that the two poles are real and equal for some value of K_D between 0.0017712 and 0.0017726. If we choose $K_D = 0.001772$, then we have found exactly what was found analytically in Example 9-2-1.

Now enter the PD controller values in the C(s) box by clicking on the box and entering 181.2*[0.001772,1] in the numerator text box. After pressing enter, the new closed-loop transfer function will be displayed in the middle of the window and the poles of the new system will be plotted in the bottom axis. Notice that while the poles are not equal they are relatively close. To check whether we have met our design constraints, open the SISO tool by clicking on the Root Locus option in the Controller Design menu. Now from the Other Responses option from the Analysis menu, choose Step and press OK. You will now see the step response of the system plotted versus time. Right-click on the plot and choose characteristics from the pop-up menu. Choose Peak Response to see the overshoot of the step response. A dot will appear at the maximum point on the response plot; holding the mouse over it will display the percentage

overshoot as 4.17%, which is within our constraint. Checking the other characteristics will show you that all the design constraints have been met.

You will notice that there are many examples in this book where multiple time-responses are plotted on the same axes as in Fig. 9-107. The capability to do this is not built into the **ACSYS** software. However, there is a short tutorial in Section 9-11 that shows you the steps involved in plotting multiple responses on the same axes.

Now let us repeat this same analysis using the SISO design tool, which is built into the Control System Toolbox, which is a component of MATLAB. Click on the C(s) block and change the value in the numerator text box back to [1].

To examine the performance of the proportional controller, we need to find the system root locus. The root locus of the system may be obtained by clicking the "Root Locus" option from the "Controller Design Tool" menu, which will activate the MATLAB SISO design tool. Fig. 9-104 shows the root locus of system. To see the poles and zeros of G and H, go to the View menu and select System Data or, alternatively, double-click the blocks G or H in the top right corner of the block diagram in the top right corner of the figure.

The squares in Fig. 9-104 correspond to the closed-loop system poles for $K = 1$. To see the closed-loop system time response to a unit-step input, select the Closed-Loop Step option from the Other Loop Responses category within the Analysis menu, which is located at the top of the screen shown in Fig. 9-105. Fig. 9-106 shows the unit-step response of the closed-loop system for $K = 1$. You may also obtain the closed-loop system poles by selecting Closed-Loop Poles from the View menu in the SISO Design Tool window. Recall the poles of the closed-loop system are

$$s_{1,2} = -180.6 \pm \sqrt{32616 - 4500K} \tag{9-294}$$

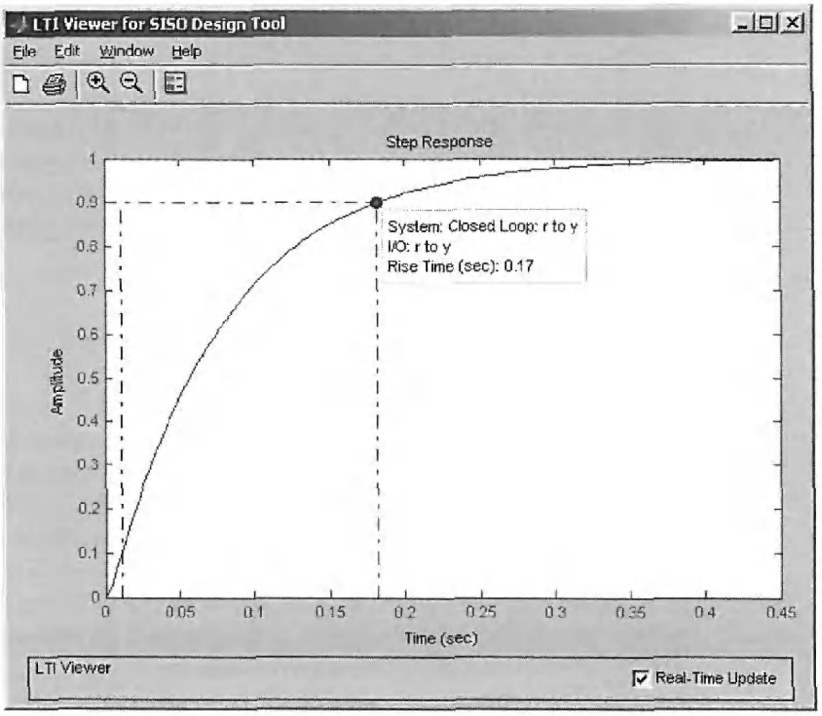


Figure 9-105 Unit-step response of the system from Example 9-2-1.

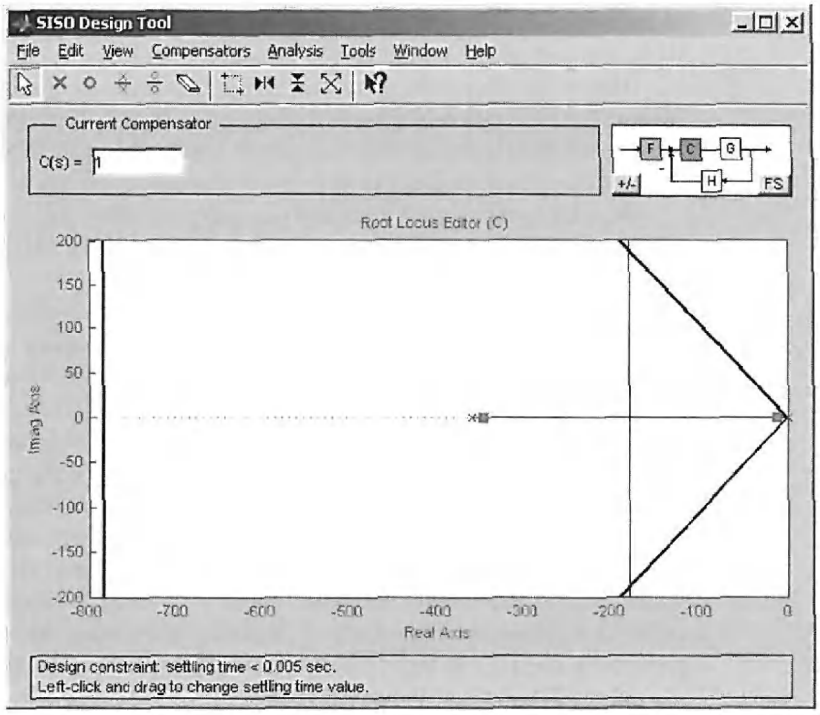


Figure 9-106 The root-locus diagram for Example 9-2-1, after incorporating the percent overshoot and the settling time as design constraints.

Changing K, therefore, affects the pole locations. In the Root Locus windows, "C(s)" represents the controller transfer function; in this case, $C(s) = K = 1$. Hence, if $C(s)$ is increased, the effective value of K increases, forcing the closed-loop poles to move together on the real axis and then ultimately to move apart to become complex.

Incorporation of the Design Criteria: As a first step to design a controller, we use the built-in design criteria option within the SISO Design Tool to establish the desired pole regions on the root locus. To add the design constraints, choose the Root Locus option from the Edit menu. "New" must then be selected within the Design Constraints option. The Design Constraints option allows the user to investigate the effect of the following:

- Settling time
- Percent overshoot
- Damping ratio
- Natural frequency

In this case we have included settling time and percent overshoot as design constraints. Enter the two constraints as they are listed at the beginning of this example, one at a time. In order also to enter the rise time as a constraint, the user must first establish a relation between the damping ratio and the natural frequency using an equation for rise time. Recall that the approximate equations for rise time for a second-order prototype system were provided in Chapter 5. Because the settling time and the percent overshoot are more important criteria in this example, we will use them as primary constraints. After designing a controller based on these constraints, we will determine whether the system complies with the rise-time constraint.

Fig. 9-106 shows the desired closed-loop system pole locations on the root locus after inclusion of the design constraints. Obviously, the poles of the system for $K = 1$ are not in

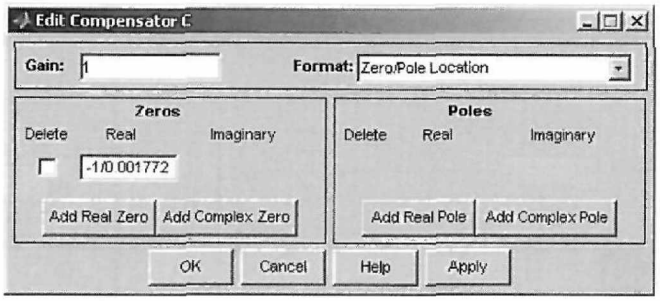


Figure 9-107 Addition of a zero to the controller $C(s)$ to create a PD controller.

the desired area. The desired poles of the system must lie to the left of the boundary imposed by the settling time between –700 and –800 markers in Fig. 9-106. Obviously, it is impossible to use the proportional controller (for any value of K) to move the poles of the closed-loop system farther to the left-half plane. However, a PD controller may be used to accomplish this task. As proposed earlier in the solution to Example 9-2-1, a PD controller with a zero at $z = -1/0.001772$ may be used for this purpose. Recall from the solution of Example 9-2-1 that this number was obtained from the steady-state error criterion by examining Eq. (9-13). In this case, if the proportional component of a PD controller is set to $K_P = 1$, the steady-state error due to a unit-ramp input is $e_{ss} = 0.000443$ for $K_D = 0.001772$.

To enter a zero to the controller, click the C(s) block in the block diagram in the top right corner of Fig. 9-106, or simply follow the instructions on the bottom of the screen shown in Fig. 9-106 and right-click the mouse to edit the controller transfer function. Fig. 9-107 shows the Edit Compensator window and how the PD controller is added.

The new root-locus diagram for the system appears in Fig. 9-108. It is now very easy to establish the required gain to push the poles to the desired region. Using the value of $K = 181.2$, which was proposed in the original solution of Example 9-2-1, would force the

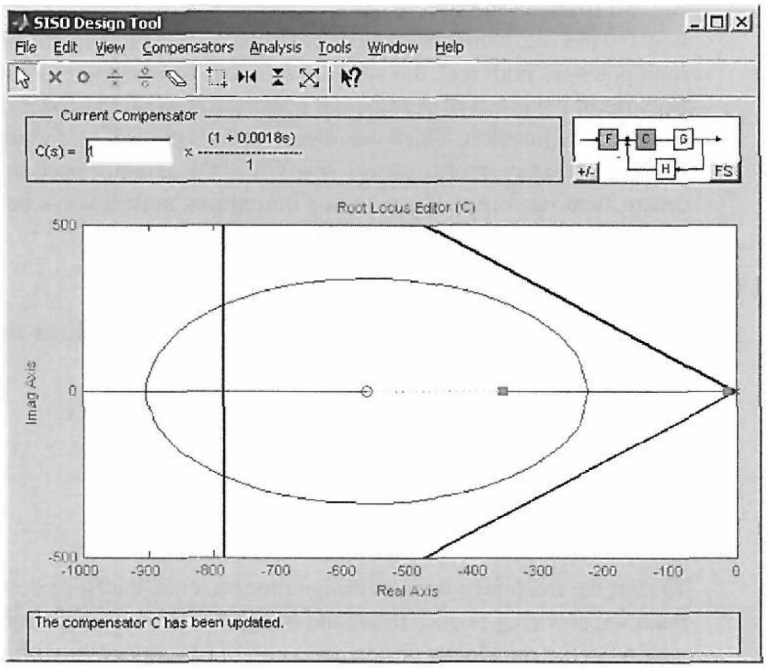


Figure 9-108 The root-locus diagram for Example 9-2-1, after incorporating a zero in the PD controller at $-1/0.001772$.

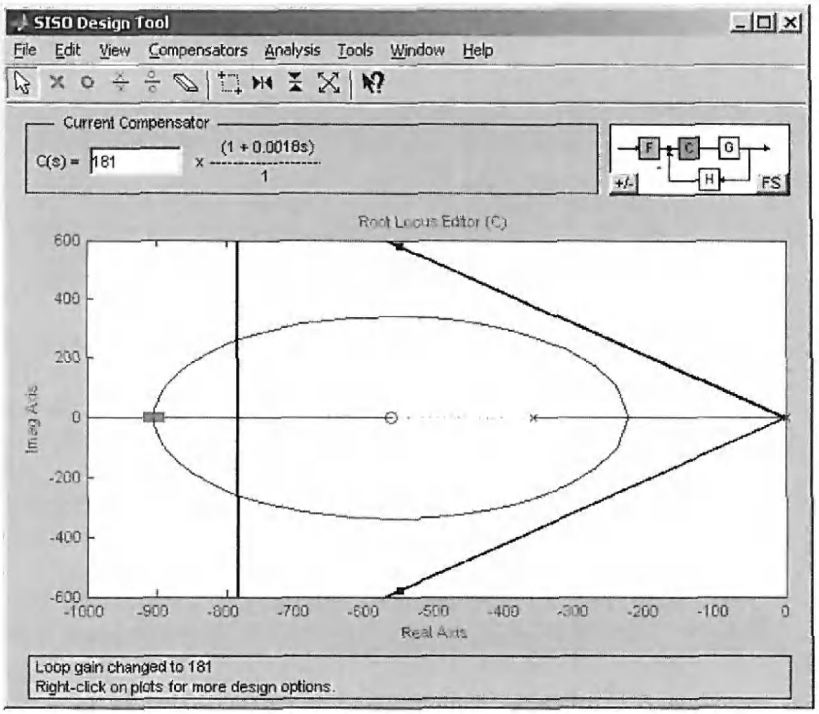


Figure 9-109 The root-locus diagram for Example 9-2-1 after incorporating a zero in the PD controller at $-1/0.001772$ and using a gain of 181.

closed-loop poles to the right region, as shown in Fig. 9-109. After dragging the poles, you can view their current location from the View menu. The step response of the controlled system in Fig. 9-110 shows the system has now complied with all design criteria. The 2% settling time is now 0.0488 sec, while the percent overshoot is 4.17. It is interesting that, although the closed-loop poles are both real, the system has a non-oscillatory response with an overshoot. This is because of the effect of zero on the response. Review the effect of adding a zero to a closed-loop transfer function, which was discussed in Section 5-11, to further appreciate this behavior.

Finally, in practice, always verify that the actuator used has enough torque or load to create such response. The actuator limitations must always be included because they are some of the most important design constraints.

Frequency-Domain Design

Now let us carry out the design of the PD controller in the frequency domain using the following performance criteria:

Steady-state error due to a unit-ramp input ≤ 0.00443

Phase margin $\geq 80°$

Resonant peak $M_r \leq 1.05$

BW ≤ 2000 rad/sec

To start the frequency-domain design process, click the Bode button in the Controller Design main window (Fig. 9-100) to get the MATLAB SISO Design Tool, as shown in Fig. 9-112.

As in the root-locus design approach, a PD controller with a zero at $z = -1/0.001772$ may be used for this purpose. Recall that this number was obtained from the steady-state error criterion. Similar to the root-locus approach, in order to enter zero for the controller,

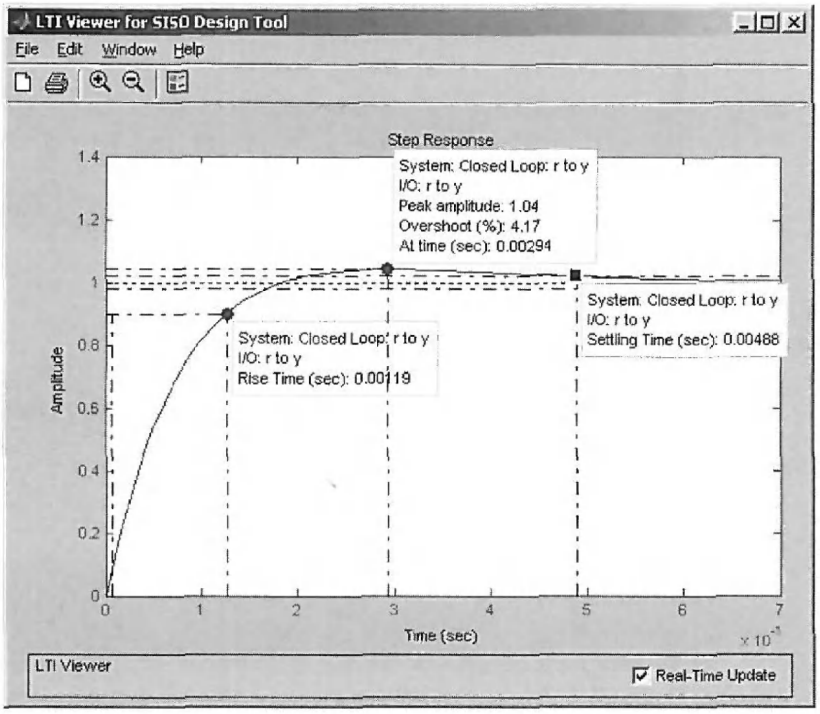


Figure 9-110 Step response of the system in Example 9-2-1 with a PD controller, $C(s) = 181(s + 1/0.001772)$.

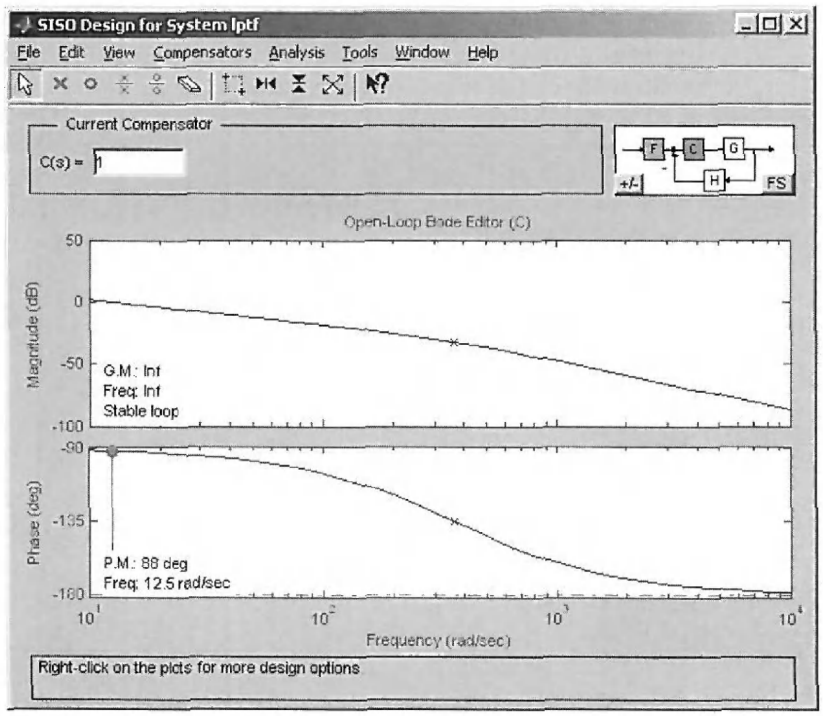


Figure 9-111 The loop magnitude and phase diagrams in the frequency-domain SISO Design Tool for Example 9-2-1.

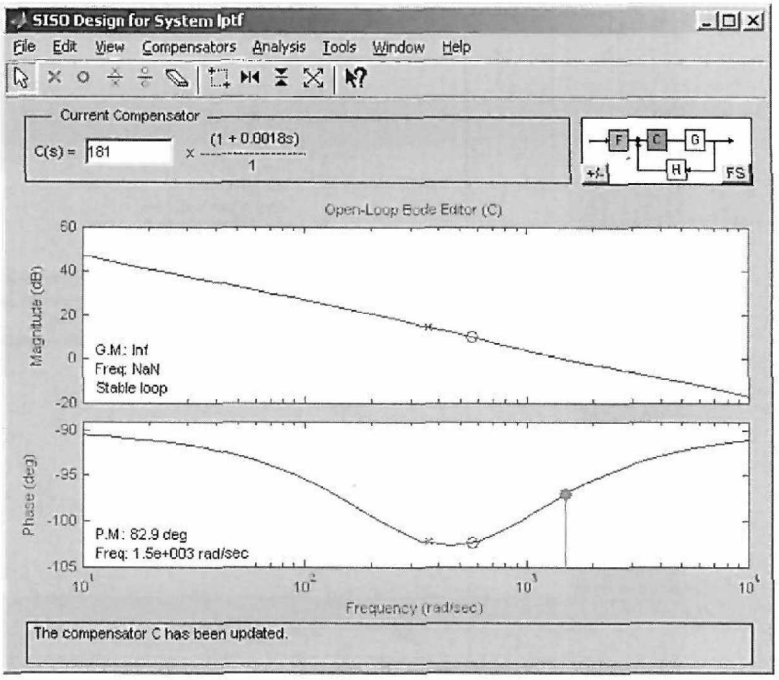


Figure 9-112 The loop magnitude and phase diagrams for Example 9-2-1, after incorporating a zero in the PD controller at $-1/0.001772$ and a gain of $K = 181.2$.

click the C(s) block in the block diagram in the top right corner of the screen shown in Fig. 9-112. The new open-loop Bode diagram of the system appears in Fig. 9-113 for the system with a PD controller $G_c(s) = C(s) = 181.2(1 + 0.001772s)$. The gain margin in this case is infinity, while the phase margin is 83°. As a result, the system has also fully complied with all the design criteria in the frequency domain. Review the effect of adding a zero to a

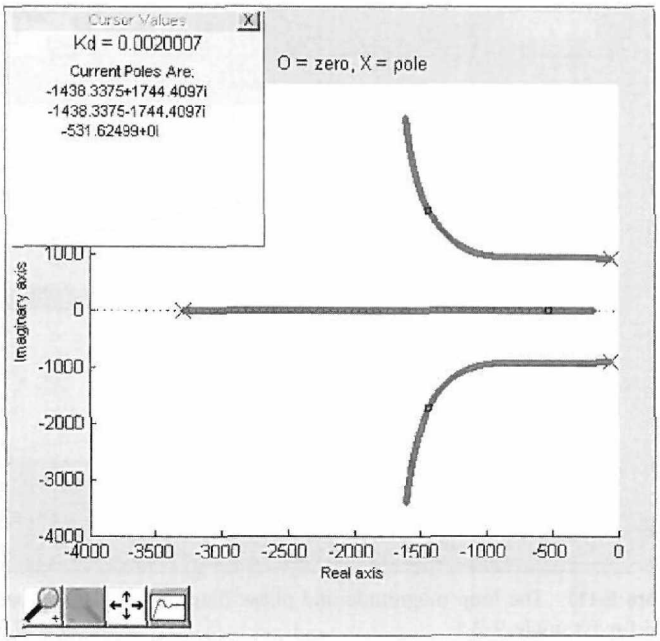


Figure 9-113 Root contours of the PD-controlled systems from Example 9-2-2.

closed-loop transfer function, which was discussed in Section 9-3 (see also Section 9-4 for addition of a pole) to appreciate this behavior. Alternatively, you may select the Open-Loop Bode option from the View menu in the root-locus diagram in Fig. 9-108 to obtain the root-locus and frequency-response representations of the system together on the same figure.

EXAMPLE 9-2-2 Revisited We will now use the **ACSYS** tool to solve the problem from Example 9-2-2. Recall that the forward-path transfer function for the system was

$$G(s) = \frac{1.5 \times 10^7 K}{s(s^2 + 3408.3s + 1,204,000)} \tag{9-295}$$

And the design contraints remain the same as in Example 9-2-2.

Click on the G(s) block to enter the plant transfer function. In the numerator text box, enter [1.5e7], and in the denominator text box, enter [1,3408.3,1204000,0]. K will be chosen to be 181.17, which we will enter in the C(s) block. Click on the C(s) block and enter [181.17] in the numerator text box. Let us again use the Root Contour tool. Choose the "PD design: Root Contour" option from the Controller Design menu. Choose K_D min to be 0.00001, choose K_D max to be 0.005, and choose 3000 steps. Pressing enter will display the plot shown in Fig. 9-113, which is similar to Fig. 9-12. Press the "arrow cursor" button and scroll through the values of P_D. You will see the behavior described in Example 9-2-2; as the value of K_D increases, the real pole moves toward the origin and the two complex poles approach asymptotically to the line at $s = -1704$.

In Example 9-2-2 the value of 0.002 is chosen for K_D; let us apply a PD controller with $K_D = 0.002$ to see its effect. K_P will be left at 1. Click on the C(s) block and enter 181.17*[.002,1] in the numerator text box. After pressing enter, click on the Step Response option from the Time Response menu and then press the Print to Figure button. Right-click on the resulting plot and choose "Characteristics" and "Peak Response" from the pop-up menu. Holding the pointer over the marker that appears at the point of maximum amplitude will display the value of the overshoot for this system. Repeating this for the rise time and the settling time will show you all the data listed in Table 9-3.

To see the Bode plots of this PD-controlled system, choose the Bode option from the Controller Design menu and you will see the window shown in Fig. 9-114.

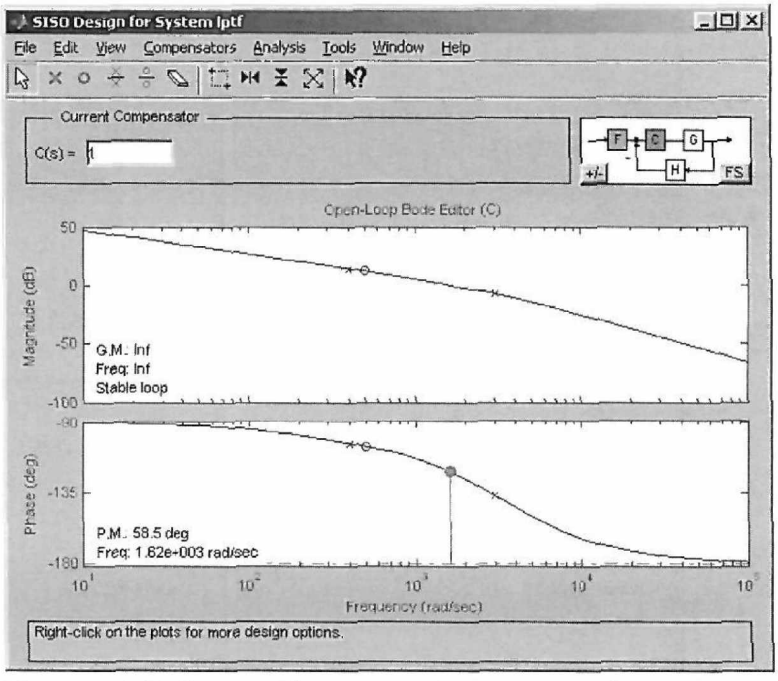


Figure 9-114 Bode plots of the PD-controlled system from Example 9-2-2.

EXAMPLE 9-3 Revisited We will now go through the steps involved in finding some of the results of Example 9-3-1. First input the plant for the system. Click on the G(s) block and enter [4500] in the numerator text box and [1,361.2,0] in the denominator text box. Adding the PI controller is accomplished by setting C(s) equal to a function of the form

$$C(s) = \frac{KK_P\left(s + \dfrac{K_I}{K_P}\right)}{s} \tag{9-296}$$

Adding the PI controller effectively adds a zero at $s = K_I/K_p$ and a pole at $s = 0$. We can use MATLAB's SISO tool to do this. As is done in Example 9-3-1 let's have $K = 181.17$. Click on the C(s) block and enter [181.17] in the numerator text box. From the Controller Design menu, choose the Root Locus option. This will open the SISO Design Tool pictured in Fig. 9-115. The plot shows the effect on the poles of the system of varying the value of a constant multiplied by $C(s)$.

Now to add the PI controller, we need to add in the pole and zero. Click on the C block in the block diagram located in the top right of the SISO window. Now, add a pole at $s = 0$, and as is done in Example 9-3-1, add a zero at $s = -10$. Click OK and you will see the plot pictured in Fig. 9-116.

Notice that this plot is similar to that in Fig. 9-19. As it is now, the value of C(s) is equal to 181.17 times the function pictured just above the Root Locus plot. Let us set the controller parameters equal to those in the third row of Table 9-5. We already have the ratio of K_I/K_P that we want. To change the value of K_P, grab one of the complex poles and drag it toward the real axis. You will see that the value of the gain in front of the controller function starts to decrease. Move the pole until the gain is equal to 0.8. Now from the Analysis menu, choose the Step option from the Other Loop Responses menu. Right-click on the resulting graph and choose "Characteristics" and "Peak Response." Hold the pointer over the dot marking the maximum response to see the percentage overshoot, as pictured in Fig. 9-117. Repeat this for the rise time and the settling time and you will see values close to those listed in Table 9-5. Note that the settling time referred to in Table 9-5 is the 5% settling time, so you may need to change the definition of settling time in the Properties menu, located in the same pop-up menu as the characteristics option. If you put both the main SISO window and the Step Response

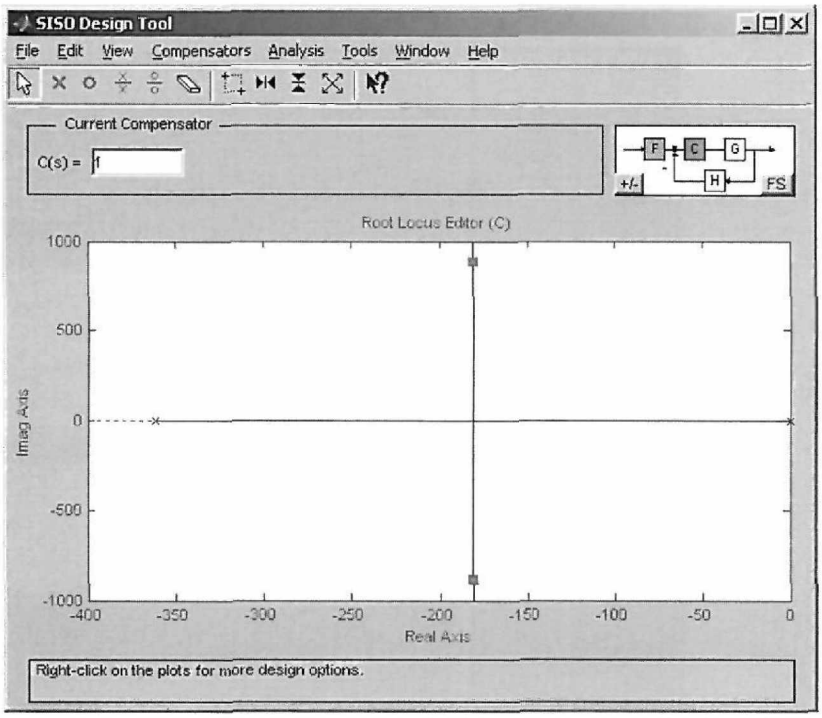


Figure 9-115 Root loci of the system from Example 9-3-1.

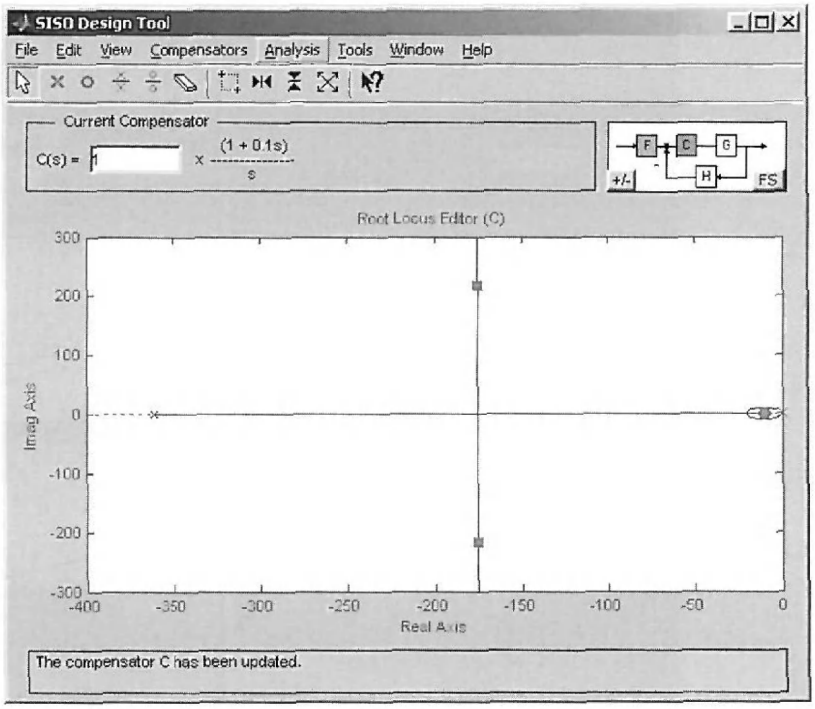


Figure 9-116 Root loci of the PI-controlled system from Example 9-3-1.

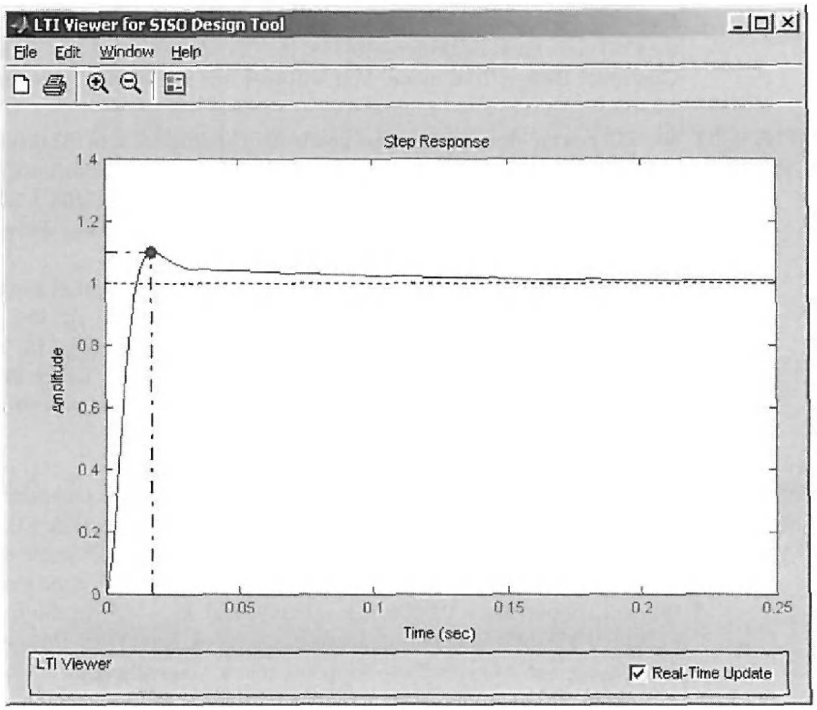


Figure 9-117 Step response of the PI-controlled system from Example 9-3-1.

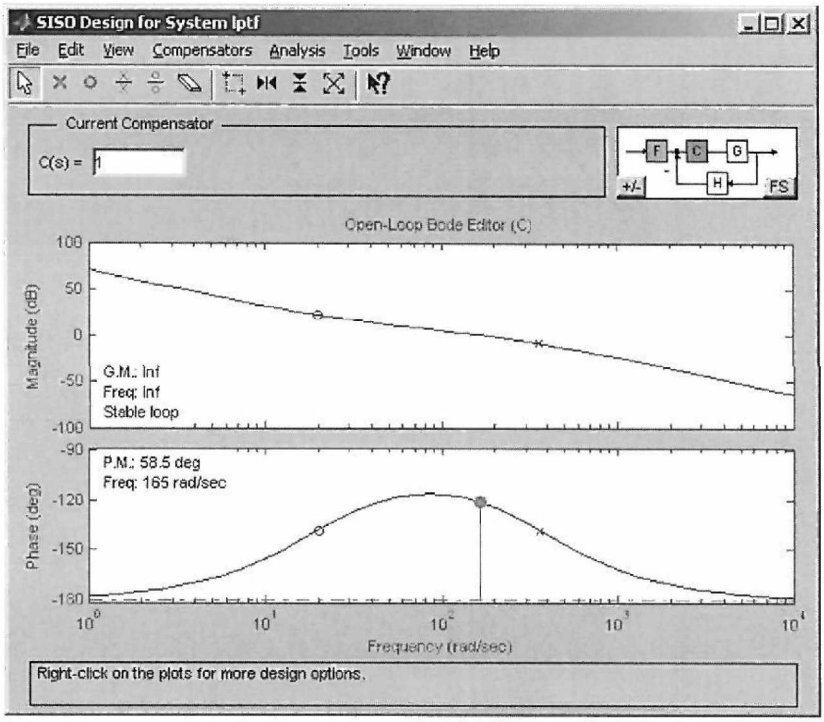


Figure 9-118 Bode plots of the PI-controlled system from Example 9-3-1.

window in view at the same time, you can see the effect that moving the poles has on the response in real time.

Now return to the Controller Design Tool window. To obtain the Bode plots shown at the end of Example 9-3-1, using the parameters from the second row in Table 9-6, enter 181.17*[0.08,1.6] into the C(s) numerator, and enter [1,0] into the C(s) denominator. Then choose the Bode option from the Controller Design Tool menu. You will then see the window pictured in Fig. 9-118. ◀

▶ **EXAMPLE 9-3-2 Revisited** We will now go through the steps involved in finding some of the results of Example 9-3-2 using the **ACSYS** software. First enter the plant transfer function. Click on the C(s) block, and in the numerator text box, enter [1.5e7]. In the denominator text box, enter [1,3408.3,1.204e6,0]. Press enter, and then click the C(s) block, and enter 181.17. From the Time Response menu, choose the Step option. You will obtain one of the plots shown in Fig. 9-23.

To improve the system, we will apply a PI controller. Let us use the parameter values from the fourth row in Table 9-8. Click on the C(s) block. In the numerator text box, enter 181.17*0.075*[1,1], and in the denominator text box, enter [1,0]. Again plot the step response and you will see one of the other plots shown in Fig. 9-23. To see the frequency response, choose the Bode option from the Controller Design Tool menu and you will see some of the information shown in Figure 9-22. ◀

▶ **EXAMPLE 9-4-1 Revisited** We will now go through the steps involved in inputting the parameters that correspond to the PID controller of Example 9-4-1. First enter the plant transfer function. Click on the G(s) block, and enter [2.718e9] in the numerator text box and [1,3408.3,1204000,0] in the denominator text box. To enter the PID controller in the C(s) text boxes, it needs to be written in the form of the left side of Eq. (9-48). We will choose $K_D = 0.0006$, $K_P = 0.309$, and $K_I = 4.5$. In the C(s) numerator text box, enter [0.0006,0.309,4.5], and in the denominator text box, enter [1,0]. Press enter, and then choose the Step Response option from the Time Response menu; you will obtain one of the plots shown in Fig. 9-24.

From the Controller Design Tool menu, choose the Bode option, and you will see the Bode plot shown in Fig. 9-25. ◀

EXAMPLE 9-5-1 Revisited We will now use **ACSYS** to obtain some of the results from Example 9-5-1. First, enter the plant of the system. In the C(s) numerator text box, enter [2500], and in the denominator text box, enter [1,25,0]. We are going to use a phase-lead controller to improve the performance of the system. Let us use the root contour tool to examine the effect that varying T has on the poles of the system. From the Controller Design Tool menu, choose the "Lead Lag Design: Root Contour" option. Set a to be 0. Set *Tmin* to be 0.0001, set *Tmax* to be 100, and choose 400 steps. After pressing enter, you will notice that the x scale on the resulting plot is much wider than that in Fig. 9-31. To get a better view, click on the zoom-in button below the plot. Then click and drag the box around the area you would like to be displayed. You should be able to generate something that looks similar to Fig. 9-119.

Now we will apply the Phase-Lead controller to see its effect on the response. Choosing the parameter values from the fourth row in Table 9-12, we will have $a = 16.67$ and $T = 0.003$. Click on the C(s) block, and in the numerator text box, enter [0.05,1]; in the denominator text box, enter [0.003,1]. Press enter, and then plot the step response from the Time Response menu. Press the Print to Figure button, and right-click on the plot to see details about the response.

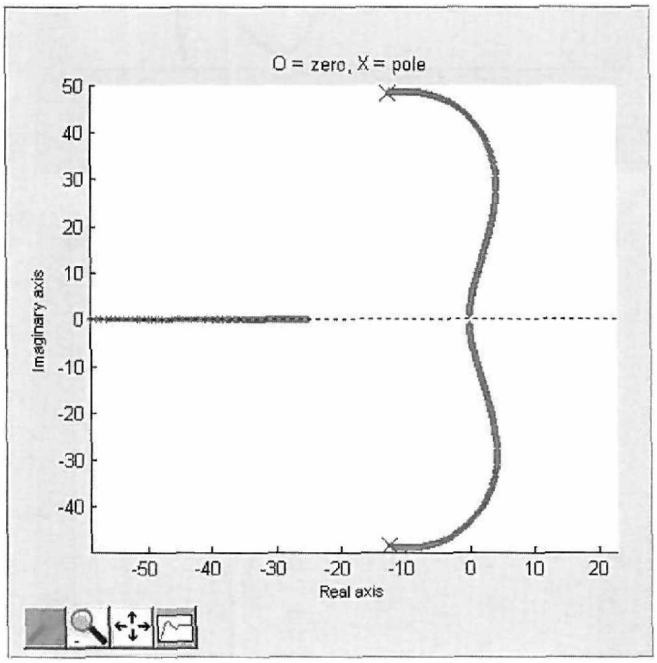


Figure 9-119 Root contours of the lead–lag controlled system from Example 9-5-1.

EXAMPLE 9-5-2 Revisited Now we will use some of the **ACSYS** tools to examine the application of a phase-lead controller to the third-order system of Example 9-5-2. First input the plant transfer function. In the G(s) numerator text box, enter [4.6875e7], and in the denominator text box, enter [1,625,156250,0]. As in Example 9-5-2, we will use a phase-lead controller to improve the performance of the system. First let's use the root contour tool to examine the effect that the controller has on the poles of the system. From the Controller Design Tool menu, choose the "Lead Lag Design: Root Contour" option. Choose $a = 0$, *Tmin* $= 0.0005$, and *Tmax* $= 1$, and choose 500 steps. After pressing enter, you will see the plot shown in Fig. 9-121, which is similar to one of the lines plotted in Fig. 9-35.

Now implement the phase-lead controller with the parameters from the last row in Table 9-14. In the C(s) numerator text box, enter [0.005,1], and in the denominator text box, enter [0.00001,1]. From the Time Response menu, choose the Step Response option, press the Print to Figure button, and right-click on the resulting plot to see the characteristics of the system, which are listed in Table 9-14. From the Controller Design menu, choose the Bode option to see the Bode plots for

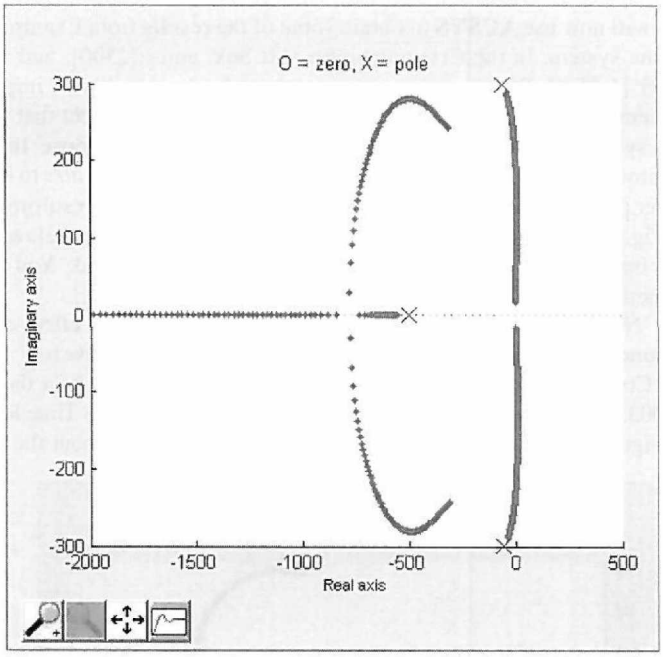


Figure 9-120 Root contours of the system from Example 9-5-2.

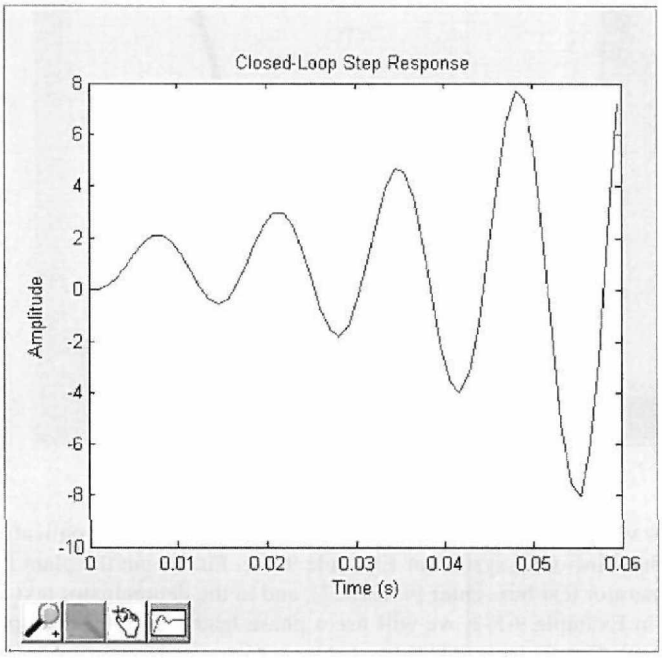


Figure 9-121 Step Response of the uncompensated system from Example 9-5-3.

this system. Right-click on the plot and change the properties to see the full range of frequencies shown in Fig. 9-37. ◀

▶ **EXAMPLE 9-5-3 Revisited** We will now go through some of the steps of Example 9-5-3 using the **ACSYS** tools. First input the plant transfer function. Click on the G(s) block, and in the numerator text box, enter [1.5625e8]. In the denominator text box, enter [1,625,156250,0]. After pressing enter, click the Bode option from the Controller Design Tool menu to see the Bode plot of the uncompensated system, which is shown in Fig. 9-39. From the Time Response menu, choose the Step command and you will see that the system is indeed unstable. The step response of the uncompensated system is shown in Fig. 9-121.

To improve the performance we will use a two-stage phase-lead controller. Let us use the parameter values from the first row of Table 9-16. To input the controller transfer function, we need to rewrite the numerator and denominator in polynomial form. In the C(s) numerator text box, enter [1.499e-5,7.744e-3,1], and in the denominator text box, enter [2.343e-9,9.68e-5,1]. Press enter and plot the step response from the Time Response menu to see one of the plots shown in Fig. 9-40.

EXAMPLE 9-6-1 Revisited Now we will go through some of the steps involved in applying the phase-lag controller to a second-order system, as in Example 9-6-1. Input the plant transfer function. Click on the G(s) block. K will be set equal to 1, so in the G(s) denominator text box, enter [2500]; in the numerator text box, enter [1,25,0]. Choose the Step option from the Time Response menu to see the step response of the uncompensated system, which is shown in Fig. 9-47.

Let us use the controller parameters from the second-to-last row in Table 9-17. Click on the C(s) block, and in the numerator text box, enter [2.7,1]. In the denominator text box, enter [30,1]. Press enter, and again plot the step response. Press the Print to Figure button, and right-click on the resulting plot. By clicking the Properties option, you can set the x-limits of the response plot to get a better view of the response.

EXAMPLE 9-6-2 Revisited We will now go through some of the steps involved in Example 9-6-2 with the **ACSYS** tools. First input the plant transfer function. Click on the G(s) block, and enter [1.5625e8] in the numerator text box, and enter [1,625,156250,0] in the denominator text box. After pressing enter, select the Root Locus option from the Controller Design menu. You will see the root loci plot pictured in Fig. 9-50.

We will implement a phase-lag controller to improve the performance of the system. Let's use the parameter values from the second row of Table 9-19. Click on the C(s) block, enter [1,1] in the numerator text box, and enter [10,1] in the denominator text box. Click the enter button, and choose the Step option from the Time Response menu. Right-click on the resulting plot to check the characteristics of the current system.

9-15 PLOTTING TUTORIAL

You will notice that in many of the examples in this book there are figures containing multiple response plots, each corresponding to a certain parameter value. Arranging plots in this way is obviously very helpful in discovering the effect that a certain parameter has on a system. The functionality to overlay the response plots on the same figure is not built into the **ACSYS** software. This short tutorial will go through the steps involved. This tutorial can also serve as an introduction to manipulating graphics and data at the MATLAB command line.

The plots we wish to display are the step responses of the system in Example 9-2-1. One plot will be the response of the system with a proportional controller, and the other will be the system with a PD controller. What we will do is generate the step response plots for each system and then plot the data from each on another figure; this will leave us with a graph similar to that in Fig. 9-110.

Press the G(s) block, enter [4500] in the numerator text box, and enter [1,361.2,0] in the denominator text box. Press the C(s) block, and enter [181.17] in the numerator text box. After pressing enter, choose the Step Response option from the Time Response menu, and press the Print to Figure button under the resulting plot. Now change the value in the C (s) numerator to 181.17*[.00177,1]. Repeat the procedure to generate a second figure. We now have two separate figures, each with its own set of axes and its own set of data plotted.

We now want to extract the plotted data from each figure so we can plot them again on new axes on a new figure. Choose "Data Cursor" from the toolbar on one of the figures, and click somewhere on the line so a marker appears on it. Now, at the MATLAB command line, type the following:

```
t1 = get(gco, 'xdata'); y1 = get(gco, 'ydata');
```

Now repeat the procedure for the second figure, this time using the variable names "t2" and "y2."

The line is a graphics object that has a list of properties that define it. GCO stands for "get current object." We make the line the current object by clicking on it with the cursor tool; the "get" command puts the value of the specified property in a variable of our choice. Xdata is a property of the line, which is a row vector of values. We now have two vectors, a time vector $t1$ and an amplitude vector $y1$. When we plot, we will be plotting each $y1$ value versus its corresponding $t1$ value. Repeat the procedure for the second figure and place the xdata and ydata in the variables $t2$ and $y2$, respectively.

Now we are ready to plot the data on a single axis. At the command line, type the following:

```
figure; axes; plot(t1, y1); hold on; plot(t2, y2);
plot([0, 0.05], [1, 1], 'color', [0, 0, 0], 'linestyle', ':');
```

The "hold on" command stops MATLAB from erasing the first plot when the second plot command is given. The third plot command just draws a black dotted line between the points (0,1) and (.05,1).

There is one final adjustment to make the new graph complete. It would be much better to display the two plots in different colors. Again, choose "Data Cursor" from the toolbar, and click on one of the lines. Now type the following:

```
set(gco, 'color', [1, 0, 0]);
```

The "set" command gives the specified value to the specified property of the specified object. You should now have a window that looks like Fig. 9-122.

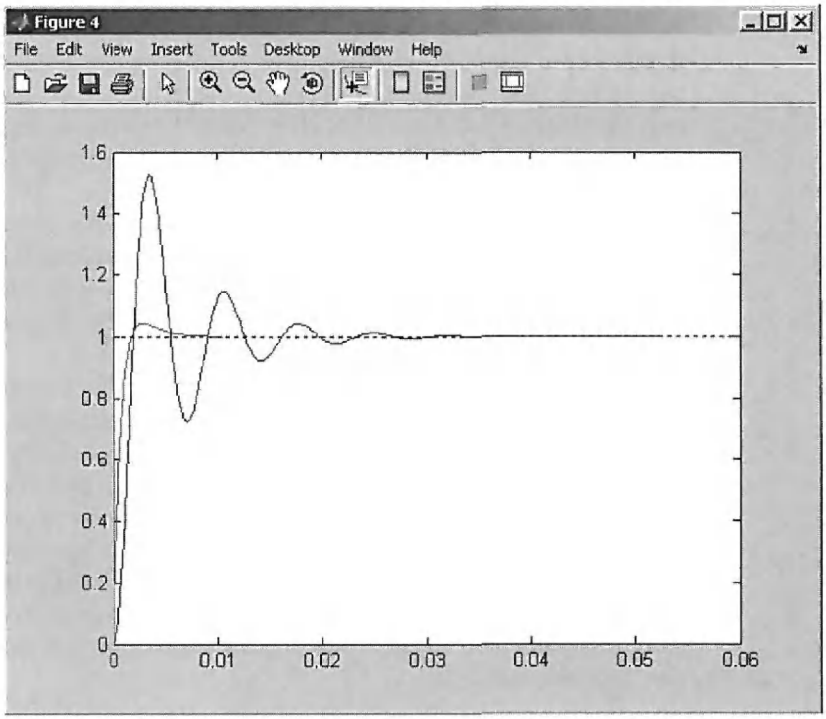


Figure 9-122 Step responses of system.

▶ 9-16 SUMMARY

This chapter was devoted to the design of linear continuous-data control systems. It began by giving some fundamental considerations on system design and then reviewed the specifications in the time domain and frequency domain. Fixed configurations of compensation schemes used in practice, such as series, forward and feedforward, feedback and minor loop, and state feedback were illustrated. The types of controllers considered were the PD, PI, PID, phase-lead, phase-lag, lead–lag, pole–zero cancellation, and notch filters. Designs were carried out in the time-domain (s-domain) as well as the frequency domain. The time-domain design was characterized by specifications such as the relative damping ratio, maximum overshoot, rise time, delay time, settling time, or simply the location of the characteristic-equation roots, keeping in mind that the zeros of the system transfer function also affect the transient response. The performance was generally measured by the step response and the steady-state error. Frequency-domain designs were generally carried out using Bode diagrams or gain-phase plots. The performance specifications in the frequency domain were the phase margin, gain margin, M_r, BW, and the like.

The effect of feedforward control on noise and disturbance reduction was demonstrated. A section was devoted to the design of robust control systems.

While the design techniques covered in this chapter were outlined with analytical procedures, the text promotes the use of MATLAB Toolboxes and specifically ACSYS. The Controller Design Tool has been developed by the authors for this purpose. Through the GUI approach, ACSYS creates a user-friendly environment to reduce the complexity of control systems design.

▶ REVIEW QUESTIONS

1. What is a PD controller? Write its input–output transfer function.

2. A PD controller has the constants K_D and K_P. Give the effects of these constants on the steady-state error of the system. Does the PD control change the type of a system?

3. Give the effects of the PD control on rise time and settling time of a control system.

4. How does the PD controller affect the bandwidth of a control system?

5. Once the value of K_D of a PD controller is fixed, increasing the value of K_P will increase the phase margin monotonically. **(T) (F)**

6. If a PD controller is designed so that the characteristic-equation roots have better damping than the original system, then the maximum overshoot of the system is always reduced. **(T) (F)**

7. What does it mean when a control system is described as being robust?

8. A system compensated with a PD controller is usually more robust than the system compensated with a PI controller. **(T) (F)**

9. What is a PI controller? Write its input–output transfer function.

10. A PI controller has the constants K_P and K_I. Give the effects of the PI controller on the steady-state error of the system. Does the PI control change the system type?

11. Give the effects of the PI control on the rise time and settling time of a control system.

12. How does the PI controller affect the bandwidth of a control system?

13. What is a PID controller? Write its input–output transfer function.

14. Give the limitations of the phase-lead controller.

15. How does the phase-lead controller affect the bandwidth of a control system?

16. Give the general effects of the phase-lead controller on rise time and settling time.

17. For the phase-lead controller, $G_c(s) = (1 + aTs)/(1 + Ts)$, $a > 1$, what is the effect of the controller on the steady-state performance of the system?

18. The phase-lead controller is generally less effective if the uncompensated system is very unstable to begin with. **(T) (F)**

19. The maximum phase that is available from a single-stage phase-lead controller is 90°. **(T) (F)**

20. The design objective of the phase-lead controller is to place the maximum phase lead at the new gain-crossover frequency. **(T) (F)**

21. The design objective of the phase-lead controller is to place the maximum phase lead at the frequency where the magnitude of the uncompensated $G_p(j\omega)$ is $-10 \log_{10} a$, where a is the gain of the phase-lead controller. **(T) (F)**

22. The phase-lead controller may not be effective if the negative slope of the un-compensated process transfer function is too steep near the gain-crossover frequency. **(T) (F)**

23. For the phase-lag controller, $G_c(s) = (1 + aTs)/(1 + Ts)$, $a < 1$, what is the effect of the controller on the steady-state performance of the system?

24. Give the general effects of the phase-lag controller on rise time and settling time.

25. How does the phase-lag controller affect the bandwidth?

26. For a phase-lag controller, if the value of T is large and the value of a is small, it is equivalent to adding a pure attenuation of a to the original uncompensated system at low frequencies. **(T) (F)**

27. The principle of design of the phase-lag controller is to utilize the zero-frequency attenuation property of the controller. **(T) (F)**

28. The corner frequencies of the phase-lag controller should not be too low or else the bandwidth of the system will be too low. **(T) (F)**

29. Give the limitations of the pole-zero-cancellation control scheme.

30. How does the sensitivity function relate to the bandwidth of a system?

Answers to these review questions can be found on this book's companion Web site: www.wiley.com/college/golnaraghi.

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► PROBLEMS

Most of the following problems can be solved using a computer program. This is highly recommended if the reader has access to such a program.

9-1. The block diagram of a control system with a series controller is shown in Fig. 9P-1. Find the transfer function of the controller $G_c(s)$ so that the following specifications are satisfied:

The ramp-error constant K_v is 5.

The closed-loop transfer function is of the form

$$M(s) = \frac{Y(s)}{R(s)} = \frac{K}{(s^2 + 20s + 200)(s + a)}$$

where K and a are real constants. Find the values of K and a.

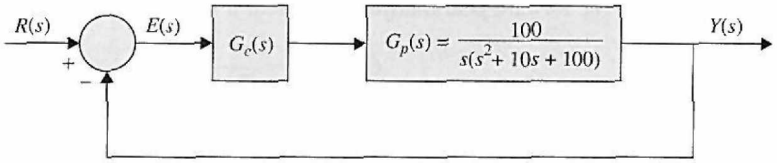


Figure 9P-1

The design strategy is to place the closed-loop poles at $-10 + j10$ and $-10 - j10$, and then adjust the values of K and a to satisfy the steady-state requirement. The value of a is large so that it will not affect the transient response appreciably. Find the maximum overshoot of the designed system.

9-2. Repeat Problem 9-1 if the ramp-error constant is to be 9. What is the maximum value of K_v that can be realized? Comment on the difficulties that may arise in attempting to realize a very large K_v.

9-3. The forward-path transfer function of a unity-feedback control system is

$$G(s) = \frac{K}{s(\tau s + 1)}$$

Find the value of K and τ so that the overshoot of the system is 25.4% at the $\zeta = 0.4$.

9-4. The forward-path transfer function of a system is

$$G(s)H(s) = \frac{24}{s(s + 1)(s + 6)}$$

Design a PD controller that satisfies the following factors:

(a) The steady-state error is less than $\pi/10$ when the input is a ramp with a slope of 2π rad/sec.

(b) The phase margin is between 40 and 50 degrees.

(c) Gain-crossover frequency is greater than 1 rad/sec.

9-5. A control system with a PD controller is shown in Fig. 9P-5.

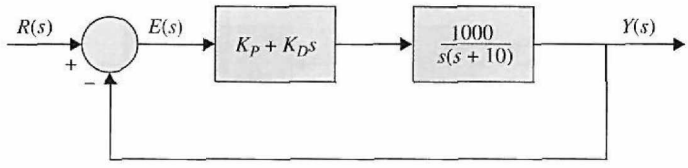


Figure 9P-5

(a) Find the values of K_P and K_D so that the ramp-error constant K_v is 1000 and the damping ratio is 0.5.

(b) Find the values of K_P and K_D so that the ramp-error constant K_v is 1000 and the damping ratio is 0.707.

(c) Find the values of K_P and K_D so that the ramp-error constant K_v is 1000 and the damping ratio is 1.0.

9-6. For the control system shown in Fig. 9P-5, set the value of K_P so that the ramp-error constant is 1000.

(a) Vary the value of K_D from 0.2 to 1.0 in increments of 0.2 and determine the values of phase margin, gain margin, M_r, and BW of the system. Find the value of K_D so that the phase margin is maximum.

(b) Vary the value of K_D from 0.2 to 1.0 in increments of 0.2 and find the value of K_D so that the maximum overshoot is minimum.

9-7. The loop transfer function of a system is

$$G(s)H(s) = \frac{1}{(2s+1)(s+1)(0.5s+1)}$$

Design a PD controller such that the $K_P = 9$ and the phase margin is greater than 25 degrees.

9-8. The loop transfer function of a system is

$$G(s)H(s) = \frac{60}{s(0.4s+1)(s+1)(s+6)}$$

(a) Design a PD controller to statisfy the following specifications:
(i) $K_v = 10$.
(ii) The phase margin is 45 degrees.

(b) Use MATLAB to plot the Bode diagram of the compensated system.

9-9. Consider the second-order model of the aircraft attitude control system shown in Fig. 5-31. The transfer function of the process is

$$G_p(s) = \frac{4500\,K}{s(s+361.2)}$$

(a) Design a series PD controller with the transfer function $G_c(s) = K_D + K_P s$ so that the following performance specifications are satisfied:

Steady-state error due to a unit-ramp input ≤ 0.001
Maximum overshoot $\leq 5\%$
Rise time $t_r \leq 0.005$ sec
Settling time $t_s \leq 0.005$ sec

(b) Repeat part (a) for all the specifications listed, and, in addition, the bandwidth of the system must be less than 850 rad/sec.

9-10. Fig. 9P-10 shows the block diagram of the liquid-level control system described in Problem 5-42. The number of inlets is denoted by N. Set $N = 20$. Design the PD controller so that with a unit-step input the tank is filled to within 5% of the reference level in less than 3 sec without overshoot.

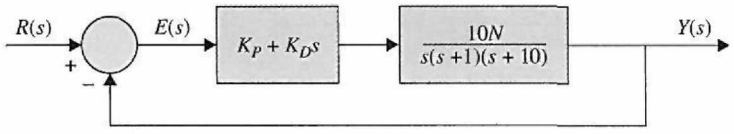


Figure 9P-10

9-11. For the liquid-level control system described in Problem 9-10,
(a) Set K_P so that the ramp-error constant is 1. Vary K_D from 0 to 0.5 and find the value of K_D that gives the maximum phase margin. Record the gain margin, M_r, and BW.

(b) Plot the sensitivity functions $|S_G^M(j\omega)|$ of the uncompensated system and the compensated system with the values of K_D and K_P determined in part (a). How does the PD controller affect the sensitivity?

9-12. The block diagram of a servo system is shown in Fig. 9P-12.

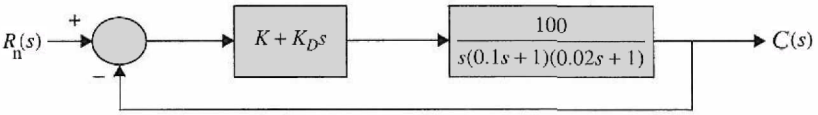


Figure 9P-12

Design the PD controller so that the phase margin is greater than 50 degrees and the BW is greater than 20 rad/sec. Use MATLAB to verify your answer.

9-13. The loop transfer function of a unity-feedback system is

$$G(s)H(s) = \frac{1000K}{s(0.2s + 1)(0.005s + 1)}$$

Design a compensator such that the steady-state error to the unit-step input is less than 0.01 and the closed-loop damping ratio $\zeta > 0.4$.
Use MATLAB to plot the Bode diagram of the compensated system.

9-14. The open-loop transfer function of a dc motor is

$$G(s) = \frac{250}{s(0.2s + 1)}$$

Design a PD controller so that the steady-state error to the input ramp is less than 0.005, the maximum overshoot is 20% for the unit-step input, and the BW must be maintained at a value approximately the same as that of the uncompensated system.

9-15. The open-loop plant model of a plastic extrusion is given by

$$G(s) = \frac{40}{(s + 1)(0.25s + 1)}$$

Design a series of lead compensator, which is described by

$$G_c(s) = \frac{r(\tau s + 1)}{(r\tau s + 1)}$$

so that the phase margin is 45 degrees and the BW must be maintained at a value approximately the same as that of the uncompensated system.

9-16. Repeat Problem 9-15 assuming that the $r < 0.1$.

9-17. The loop transfer function of a unity-feedback control system is

$$G(s)H(s) = \frac{1000K}{s(0.2s + 1)(0.05s + 1)}$$

(a) Design a compensator such that
- (i) The steady-state error is less than 0.01 for a unit-ramp input.
- (ii) The phase margin is greater than 45 degrees.
- (iii) The steady-state error is less than 0.004 for a sinusoidal input with $\omega < 0.2$.
- (iv) The noise for the frequencies greater than 200 rad/sec reduced to 100 at the output.

(b) Use MATLAB to plot the Bode diagram of the compensated system and verify or refine your design in part (a).

9-18. A control system with a type 0 process $G_p(s)$ and a PI controller is shown in Fig. 9P-18.

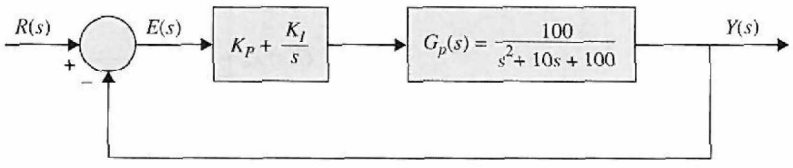


Figure 9P-18

(a) Find the value of K_I so that the ramp-error constant K_v is 10.

(b) Find the value of K_P so that the magnitude of the imaginary parts of the complex roots of the characteristic equation of the system is 15 rad/sec. Find the roots of the characteristic equation.

(c) Sketch the root contours of the characteristic equation with the value of K_I as determined in part (a) and for $0 \leq K_P < \infty$.

9-19. For the control system described in Problem 9-18,

(a) Set K_I so that the ramp-error constant is 10. Find the value of K_P so that the phase margin is minimum. Record the values of the phase margin, gain margin, M_r, and BW.

(b) Plot the sensitivity functions $|S_G^M(j\omega)|$ of the uncompensated and the compensated systems with the values of K_I and K_P selected in part (a). Comment on the effect of the PI control on sensitivity.

9-20. For the control system shown in Fig. 9P-18, perform the following:

(a) Find the value of K_I so that the ramp-error constant K_v is 100.

(b) With the value of K_I found in part (a), find the critical value of K_P so that the system is stable. Sketch the root contours of the characteristic equation for $0 \leq K_P < \infty$.

(c) Show that the maximum overshoot is high for both large and small values of K_P. Use the value of K_I found in part (a). Find the value of K_P when the maximum overshoot is a minimum. What is the value of this maximum overshoot?

9-21. Repeat Problem 9-20 for $K_v = 10$.

9-22. The transfer function of a system with unity feedback is

$$G(s) = \frac{24}{s(s+1)(s+6)}$$

(a) Design a PI controller that satisfies the following factors:
   (i) The ramp error constant $K_v > 20$.
   (ii) The phase margin is between 40 and 50 degrees.
   (iii) Gain-crossover frequency is greater than 1 rad/sec.

(b) Use MATLAB to plot the Bode diagram of the closed-loop system.

9-23. The transfer function of a robot arm–positioning system with unity feedback is represented by

$$G(s) = \frac{40}{s(s+2)(s+20)}$$

(a) Design a PI controller such that:
   (i) The steady-state error is less than 5% of the slope for a ramp input.
   (ii) The phase margin is between 32.5 and 37.5 degrees.
   (iii) Gain-crossover frequency is 1 rad/sec.

(b) Use MATLAB to plot the Bode diagram of the closed-loop system and verify your design in part (a).

9-24. The transfer function of a system with unity feedback is

$$G(s) = \frac{210}{s(5s+7)(s+3)}$$

Design a PI controller with a unity dc gain so that the phase margin of the system is greater than 40 degrees, and then find the BW of the system.

9-25. The transfer function of the steering of a ship is given by

$$G(s) = \frac{2353K(71 - 500s)}{71s(40s+13)(5000s+181)}$$

Design a PI controller such that:

(a) The ramp error constant $K_v = 2$.

(b) The phase margin is greater than 50 degrees.

(c) For all frequencies greater than crossover frequency, PM > 0. This means the system is always stable without any condition.

(d) Show the closed-loop poles in the root locus with respect to values of K.

9-26. The transfer function of a unity-feedback system is

$$G(s) = \frac{2 \times 10^5}{s(s+20)(s^2+50s+10000)}$$

(a) A PD controller with the transfer function of $H(s) = \dfrac{(\tau s+1)}{(r\tau s+1)}$ is designed with $r = 0.2$ and $\tau = 0.05$. It is desired to find the gain so that the crossover frequency is 31.6 rad/sec.

(b) Find the ramp error constant Kv by applying the controller designed in part (a).

(c) Consider the PD controller designed in part (a) is applied to the system. Find the value of K for a PI controller so that the ramp error constant $K_v = 100$.

(d) If the PI controller pole is at 3.16 rad/sec and the crossover frequency maintains at 31.6 rad/sec, what is the zero of the PI controller? [Consider the transfer function of the PI controller is $H(s) = \dfrac{r(\tau s+1)}{(r\tau s+1)}$.]

(e) Use MATLAB to plot the Bode diagram of the compensated system and find the phase margin.

9-27. A control system with a type 0 process and a PID controller is shown in Fig. 9P-27. Design the controller parameters so that the following specifications are satisfied:

Ramp-error constant $K_v = 100$

Rise time $t_r < 0.01$ sec

Maximum overshoot $< 2\%$

Plot the unit-step response of the designed system.

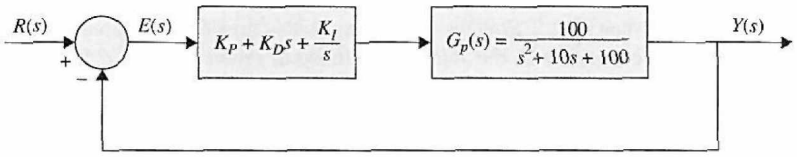


Figure 9P-27

9-28. A considerable amount of effort is being spent by automobile manufacturers to meet the exhaust-emission-performance standards set by the government. Modern automotive-power-plant systems consist of an internal combustion engine that has an internal cleanup device called the catalytic converter. Such a system requires control of the engine air–fuel ratio (A/F), the ignition-spark timing, exhaust-gas recirculation, and injection air. The control system problem considered in this exercise deals with the control of the air–fuel ratio. In general, depending on fuel composition and other factors, a typical stoichiometric A/F is 14.7:1, that is, 14.7 grams of air to each gram of fuel. An A/F greater or less than stoichiometry will cause high hydrocarbons, carbon monoxide, and nitrous oxide in the tailpipe emission. The control system whose block diagram is shown in Fig. 9P-28 is devised to control the air–fuel ratio so that a desired output variable is maintained for a given command signal. Fig. 9P-28 shows that the sensor senses the composition of the exhaust-gas mixture entering the catalytic converter. The electronic controller detects the difference or the error between the command and the sensor signals and computes the control signal necessary to achieve the desired exhaust-gas composition. The output variable $y(t)$ denotes the effective air–fuel ratio. The transfer function of the engine is given by

$$\frac{Y(s)}{U(s)} = G_p(s) = \frac{e^{-T_d s}}{1+\tau s}$$

where T_d is the time delay and is 0.2 sec. The time constant τ is 0.25 sec. Approximate the time delay by a power series:

$$e^{-T_d s} \cong \frac{1}{1 + T_d s + T_d^2 s^2/2}$$

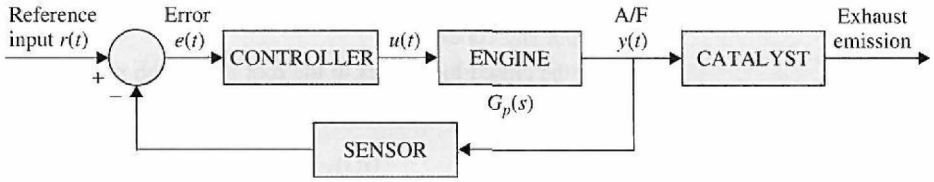


Figure 9P-28

(a) Let the controller be a PI controller so that

$$G_c(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s}$$

Find the value of K_I so that the ramp-error constant K_v is 2. Determine the value of K_P so that the maximum overshoot of the unit-step response is a minimum and the settling time is a minimum. Give the values of the maximum overshoot and the settling time. Plot the unit-step response of $y(t)$. Find the marginal value of K_P for system stability.

(b) Can the system performance be further improved by using a PID controller?

$$G_c(s) = \frac{U(s)}{E(s)} = K_P + K_D s + \frac{K_I}{s}$$

9-29. One of the advantages of the frequency-domain analysis and design methods is that systems with pure time delays can be treated without approximation. Consider the automobile-engine control system treated in Problem 9-28.

The process has the transfer function

$$G_p(s) = \frac{e^{-0.2s}}{1 + 0.25s}$$

Let the controller be of the PI type so that $G_c(s) = K_P + K_I/s$. Set the value of K_I so that the ramp-error constant K_v is 2. Find the value of K_P so that the phase margin is a maximum. How does this "optimal" K_P compare with the value of K_P found in Problem 9-28(a)? Find the critical value of K_P for system stability. How does this value of K_P compare with the critical value of K_P found in Problem 9-28?

9-30. Fig. 9P-30 shows a simplified design of an airplane attitude controller.

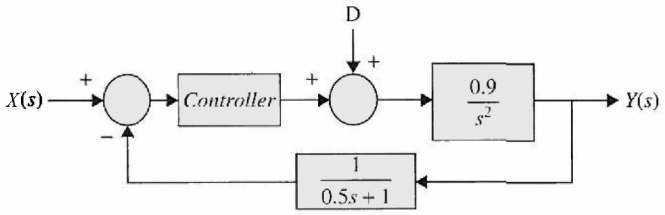


Figure 9P-30

where D is the disturbance torque. Design a PID controller with the following satisfactions:

(a) Zero steady-state error

(b) PM = 65°

(c) High bandwidth (as high as possible)

9-31. Consider the open-loop plant model of a plastic extrusion given in Problem 9-15.

Design a series of lead–lag compensator that is described by

$$H(s) = \frac{(\tau_1 s + 1)(\tau_2 s + 1)}{\tau_1 \tau_1 s^2 + \left(\tau_1 + \frac{\tau_2}{r}\right)s + 1}$$

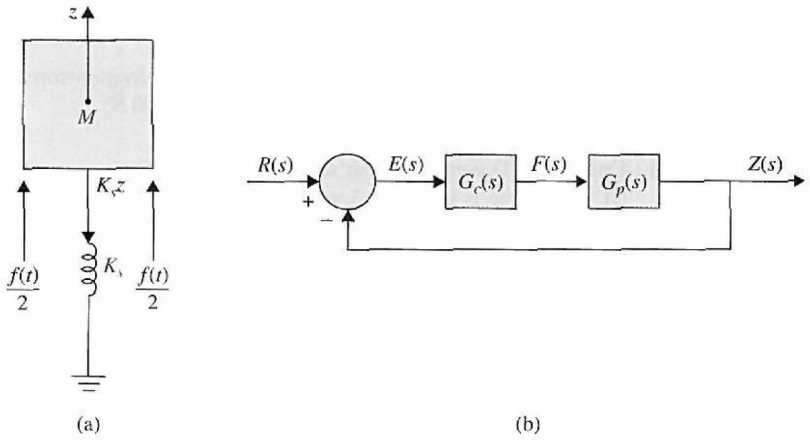
and satisfies the following:

(a) The phase margin is 45 degrees.

(b) The steady-state error of a closed-loop system to the unit-step input is less than 1%.

(c) The gain-crossover frequency is 5 rad/sec.

9-32. A telescope for tracking stars and asteroids on a space shuttle may be modeled as a pure mass M. It is suspended by magnetic bearings so that there is no friction, and its attitude is controlled by magnetic actuators located at the base of the payload. The dynamic model for the control of the z-axis motion is shown in Fig. 9P-32(a). Because there are electrical components on the telescope, electric power must be brought to the telescope through a cable. The spring shown is used to model the wire-cable attachment, which exerts a spring force on the mass. The force produced by the magnetic actuators is denoted by $f(t)$. The force equation of motion in the z direction is



(a) (b)

Figure 9P-32

$$f(t) - K_s z(t) = M\frac{d^2 z(t)}{dt^2}$$

where $K_s = 1$ lb/ft, and $M = 150$ lb (mass); $f(t)$ is in pounds, and $z(t)$ is measured in feet.

(a) Show that the natural response of the system output $z(t)$ is oscillatory without damping. Find the natural undamped frequency of the open-loop space-shuttle system.

(b) Design the PID controller

$$G_c(s) = K_P + K_D s + \frac{K_I}{s}$$

shown in Fig. 9P-32(b) so that the following performance specifications are satisfied:

Ramp-error constant $K_v = 100$.

The complex charactreristic equation roots correspond to a relative damping ratio of 0.707 and a natural undamped frequency of 1 rad/sec.

Compute and plot the unit-step response of the designed system. Find the maximum overshoot. Comment on the design results.

(c) Design the PID controller so that the following specifications are satisfied:

Ramp-error constant $K_v = 100$

Maximum overshoot $< 5\%$

Compute and plot the unit-step response of the designed system. Find the roots of the characteristic equation of the designed system.

9-33. Repeat Problem 9-32(b) with the following specifications:

Ramp-error constant $K_v = 100$.

The complex charactreristic equation roots correspond to a relative damping ratio of 1.0 and a natural undamped frequency of 1 rad/sec.

9-34. Consider a cruise control system shown in Fig. 9P-34.

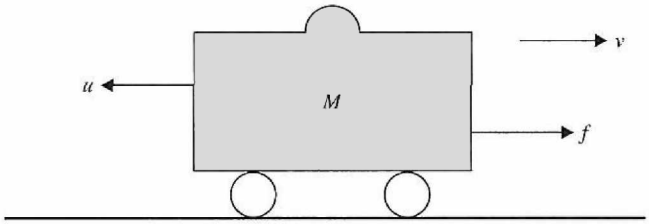


Figure 9P-34

where f is the engine force, v is the velocity, u is the friction force, and $u = \mu v$. Assuming $M = 1000\,\text{kg}$, $\mu = 50\,\text{Nsec/m}$, and $f = 500\,\text{N}$:

(a) Find the transfer function of the system.

(b) Design a PID controller that satisfies the following:

(i) Rise time is less than 5 sec.

(ii) Maximum overshoot is less than 10%.

(iii) Steady-state error is less than 2%.

9-35. An inventory control system is modeled by the following state equations:

$$\frac{dx_1(t)}{dt} = -2x_2(t)$$
$$\frac{dx_2(t)}{dt} = -2u(t)$$

where $x_1(t)$ = level of inventory, $x_2(t)$ = rate of sales of product, and $u(t)$ = production rate. The output equation is $y(t) = x_1(t)$. One unit of time is one day. Fig. 9P-35 shows the block diagram of the closed-loop inventory control system with a series controller. Let the controller be a PD controller, $G_c(s) = K_P + K_D s$.

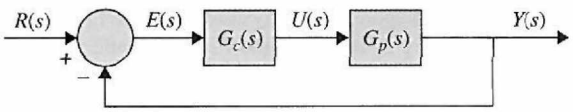


Figure 9P-35

(a) Find the parameters of the PD controller, K_P and K_D, so that the roots of the characteristic equation correspond to a relative damping ratio of 0.707 and $\omega_n = 1$ rad/sec. Plot the unit-step response of $y(t)$ and find the maximum overshoot.

(b) Find the values of K_P and K_D so that the overshoot is zero and the rise time is less than 0.06 sec.

(c) Design the PD controller so that $M_r = 1$ and $\text{BW} \leq 40$ rad/sec.

9-36. The block diagram of a type 2 control system with a series controller $G_c(s)$ is shown in Fig. 9P-36.

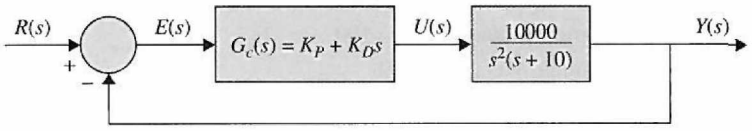


Figure 9P-36

The objective is to design a PD controller so that the following specifications are satisfied:

Maximum overshoot $< 10\%$

Rise time < 0.5 sec

(a) Obtain the characteristic equation of the closed-loop system, and determine the ranges of the values of K_P and K_D for stability. Show the region of stability in the K_D-versus-K_P plane.

(b) Construct the root loci of the characteristic equation with $K_D = 0$ and $0 \leq K_P < \infty$. Then construct the root contours for $0 \leq K_D < \infty$ and several fixed values of K_P ranging from 0.001 to 0.01.

(c) Design the PD controller to satisfy the performance specifications given. Use the information on the root contours to help your design. Plot the unit-step response of $y(t)$.

(d) Check the design results obtained in part (c) in the frequency domain. Determine the phase margin, gain margin, M_r, and BW of the designed system.

9-37. Consider a dc motor shown in Fig. 9P-37 and described in Section 4-7-3.

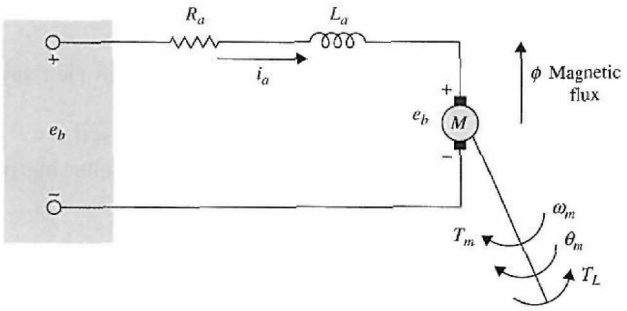


Figure 9P-37

Assuming the following:

The rotor inertia $(J) = 0.01\ \text{kg.m}^2/\text{s}^2$

Damping ratio of the mechanical system $(\zeta) = 0.1$ Nms

Back-emf constant $(K_b) = 0.01$ Nm/Amp

Torque constant $(K_t) = 0.01$ Nm/Amp

Armature resistance $(R_a) = 1\ \Omega$

Armature inductance $(L_a) = 0.5$ H

Design a PID controller that satisfies the following:

(a) Settling time is less than 2 sec.

(b) Maximum overshoot is less than 5%.

(c) Steady-state error is less than 1%.

9-38. For the dc motor described in Problem 9-37, assuming the following:

The rotor inertia $(J) = 3.2284\text{E-}6\text{kg.m}^2/\text{s}^2$

Damping ratio of the mechanical system $(\zeta) = 3.5077\text{E-}6$ Nms

Back-emf constant $(K_b) = 0.0274$ Nm/Amp

Torque constant $(K_t) = 0.0274$ Nm/Amp

Armature resistance $(R_a) = 4\ \Omega$

Armature inductance $(L_a) = 2.75\text{E-}6$ H

Design a PID controller that satisfies the following:

(a) Settling time is less than 40 milliseconds.

(b) Maximum overshoot is less than 16%.

(c) Zero steady-state error is less than 1%.

(d) Zero steady-state error due to a disturbance.

9-39. Consider the broom-balancing control system described in Problems 4-21 and 10-51. The $\mathbf{A}*$ and $\mathbf{B}*$ matrices are given in Problem 10-51 for the small-signal linearized model.

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{A}^*\Delta\mathbf{x}(t) + \mathbf{B}^*\Delta r(t)$$
$$\Delta y(t) = \mathbf{C}\Delta\mathbf{x}(t)$$
$$\mathbf{D}* = [0 \quad 0 \quad 1 \quad 0]$$

Fig. 9P-39 shows the block diagram of the system with a series PD controller. Determine if the PD controller can stabilize the system; if so, find the values of K_P and K_D. If the PD controller cannot stabilize the system, explain why not.

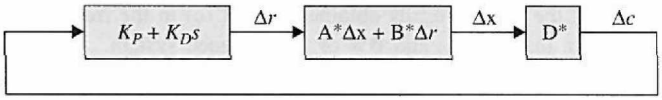


Figure 9P-39

9-40. The process of a unity-feedback control system has the transfer function

$$G_p(s) = \frac{100}{s^2 + 10s + 100}$$

Design a series controller (PD, PI, or PID) so that the following performance specifications are satisfied:

Steady-state error due to a step input = 0
Maximum overshoot < 2%
Rise time < 0.02 sec

Carry out the design in the frequency domain and check the design in the time domain.

9-41. The forward path of a unity-feedback control system that includes a disturbance signal $D(s)$ is given by

$$G(s) = \frac{1}{(s^2 + 3.6s + 9)}$$

(a) Design a PID controller with the transfer function of $H(s) = \dfrac{K(\tau_1 s + 1)(\tau_2 s + 1)}{s}$ so that the response to any step disturbance is damped in less than 3 sec at the 2% settling time.

(b) Use MATLAB to plot the response of the closed-loop system to various step disturbance inputs and verify your design in part (a).

9-42. For the inventory control system shown in Fig. 9P-35, let the controller be of the phase-lead type:

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a > 1$$

Determine the values of a and T so that the following performance specifications are satisfied:

Steady-state error due to a step input = 0
Maximum overshoot < 5%

(a) Design the controller using the root contours with T and a as variable parameters. Plot the unit-step response of the designed system. Plot the Bode diagram of $G(s) = G_c(s)G_p(s)$, and find PM, GM, M_r, and BW of the designed system.

(b) Design the phase-lead controller so that the following performance specifications are satisfied:

Steady-state error due to a step input = 0
Phase margin > 75°
$M_r < 1.1$

Construct the Bode diagram of $G(s)$ and carry out the design in the frequency domain. Find the attributes of the time response of the designed system.

9-43. Consider that the process of a unity-feedback control system is

$$G_p(s) = \frac{1000}{s(s+10)}$$

Let the series controller be a single-stage phase-lead controller:

$$G_c(s) = \frac{1+aTs}{1+Ts} \quad a > 1$$

(a) Determine the values of a and T so that the zero of $G_c(s)$ cancels the pole of $G_p(s)$ at $s = -10$. The damping ratio of the designed system should be unity. Find the attributes of the unit-step response of the designed system.

(b) Carry out the design in the frequency domain using the Bode plot. The design specifications are as follows:

Phase margin $> 75°$

$M_r < 1.1$

Find the attributes of the unit-step response of the designed system.

9-44. Fig. 9P-44 shows the quarter-car model realization with 2 degrees of freedom.

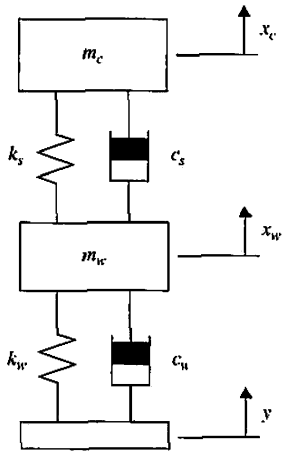


Figure 9P-44

Assuming:

Body mass $(m_c) = 2500\,\text{kg}$

Suspension mass $(m_w) = 320\,\text{kg}$

Spring constant of suspension system $(k_c) = 80,000\,\text{N/m}$

Spring constant of wheel and tire $(k_w) = 500,000\,\text{N/m}$

Damping constant of suspension system $(c_s) = 350\,\text{Ns/m}$

Damping constant of wheel and tire $(c_w) = 15,020\,\text{Ns/m}$

When the vehicle is experiencing any road disturbance, the vehicle body should not have large oscillations, and the oscillations should be damped quickly. If the deformation tire is negligible, and the road disturbance **(D)** is considered a step input,

(a) Design a PID controller that satisfies the following requirements:

(i) Overshoot less than 5%

(ii) Settling time shorter than 5 seconds

(b) Use MATLAB to plot the response of the closed-loop system to various step disturbance inputs and verify your design in part (a).

9-45. Consider that the controller in the liquid-level control system shown in Fig. 9P-10 is a phase-lead controller:

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a > 1$$

(a) For $N = 20$, select the values of a and T so that the maximum overshoot is barely 0%. The value of a must not exceed 1000. Find the attributes of the unit-step response of the designed system. Plot the unit-step response.

(b) For $N = 20$, design the phase-lead controller in the frequency domain. Find the values of a and T so that the phase margin is maximized subject to the condition that BW > 100. The value of a must not exceed 1000.

9-46. The transfer function of the process of a unity-feedback control system is

$$G_p(s) = \frac{6}{s(1 + 0.2s)(1 + 0.5s)}$$

(a) Construct the Bode diagram of $G_p(j\omega)$ and determine the PM, GM, M_r, and BW of the system.

(b) Design a series single-stage series phase-lead controller with the transfer function

$$G_c(s) = \left(\frac{1 + aTs}{1 + Ts}\right) \quad a > 1$$

so that the phase margin is maximum. The value of a must not be greater than 1000. Determine PM and M_r of the designed system. Determine the attributes of the unit-step response.

(c) Using the system designed in part (b) as a basis, design a two-stage phase-lead controller so that the phase margin is at least 85°. The transfer function of the two-stage phase-lead controller is

$$G_c(s) = \left(\frac{1 + aT_1 s}{1 + T_1 s}\right)\left(\frac{1 + bT_2 s}{1 + T_2 s}\right) \quad a > 1 \quad b > 1$$

where a and T_1 are determined in part (b). The value of T_2 should not exceed 1000. Find the values of PM and M_r of the designed system. Find the attributes of the unit-step response.

(d) Plot the unit-step responses of the output in parts (a), (b), and (c).

9-47. Fig. 9P-47 shows an inverted pendulum on a cart.

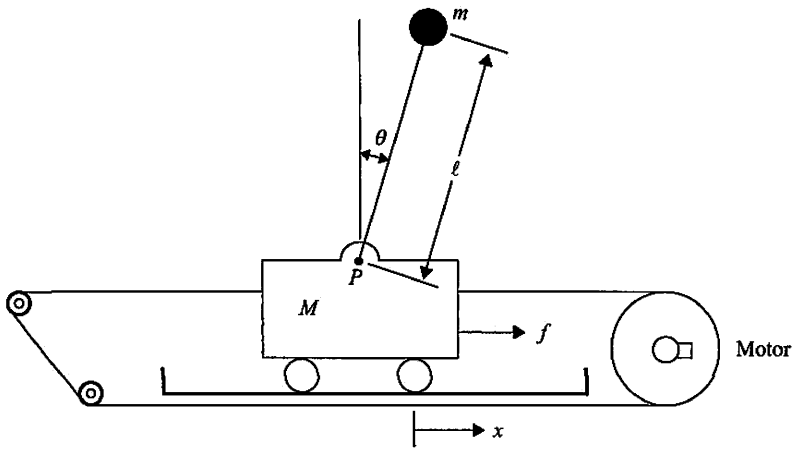


Figure 9P-47

Assuming:

M	mass of the cart	0.5 kg
m	mass of the pendulum	0.2 kg
μ	friction of the cart	0.1 N/m/sec
l	length to pendulum center of mass	0.3 m
I	inertia of the pendulum	0.006 kg*m^2

(a) Design a PID controller so that the settling time is less than 5 seconds and the pendulum angle is never more than 0.05 radians from the vertical position.

(b) If the step input is applied to the cart, design a PID controller so that the settling time for x and θ is less than 5 seconds, the rise time for x is less than 0.5 seconds, and the overshoot of theta is less than 20 degrees (0.35 radians).

9-48. A phase-lock-loop, dc-motor-speed-control system is described in Problem 4-46. The block diagram of the system is shown in Fig. 9P-48. The system parameters and transfer functions are given as follows:

Reference speed command, $\omega_r = 120$ pulse/sec

Phase-detector gain, $K_p = 0.06$ V/pulse/sec

Amplifier gain, $K_a = 20$

Encoder gain, $K_e = 5.73$ pulse/rad

Counter gain, $N = 1$

Motor transfer function,

$$\frac{\Omega_m(s)}{E_a(s)} = \frac{10}{s(1 + 0.05s)}$$

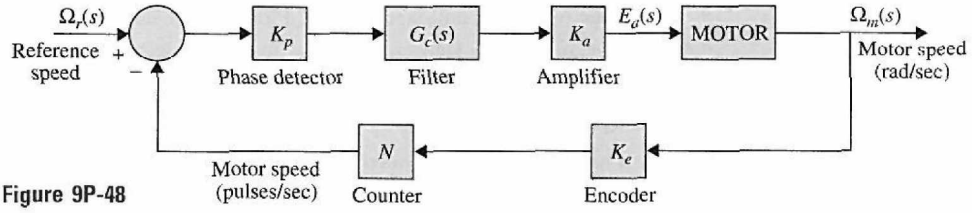


Figure 9P-48

(a) Let the filter (controller) transfer function be of the form

$$G_c(s) = \frac{E_o(s)}{E_i(s)} = \frac{1 + R_2Cs}{R_1Cs}$$

where $R_1 = 2 \times 10^6\ \Omega$ and $C = 1\ \mu F$. Determine the value of R_2 so that the complex roots of the closed-loop characteristic equation have a maximum relative damping ratio. Sketch the root loci of the characteristic equation for $0 \leq R_2 < \infty$. Compute and plot the unit-step responses of the motor speed $f_\omega(t)$ (pulse/sec) with the values of R_2 found, when the input is 120 pulse/sec. Convert the speed in pulse/sec to rpm.

(b) Let the filter transfer function be

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a > 1$$

where $T = 0.01$. Find a so that the complex roots of the characteristic equation have a maximum relative damping ratio. Compute and plot the unit-step response of the motor speed $f_\omega(t)$ (pulse/sec) when the input is 120 pulse/sec.

(c) Design the phase-lead controller in the frequency domain so that the phase margin is at least 60°.

9-49. Consider that the controller in the liquid-level control system shown in Fig. 9P-10 is a single-stage phase-lag controller:

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a < 1$$

(a) For $N = 20$, select the values of a and T so that the two complex roots of the characteristic equation correspond to a relative damping ratio of approximately 0.707. Plot the unit-step response of the output $y(t)$. Find the attributes of the unit-step response. Plot the Bode plot of $G_c(s)G_p(s)$ and determine the phase margin of the designed system.

(b) For $N = 20$, design the phase-lag controller in the frequency domain so that the phase margin is approximately 60°. Plot the unit-step response of the output $y(t)$, and find the attributes of the unit-step response.

9-50. The controlled process of a unity-feedback control system is

$$G_p(s) = \frac{K}{s(s+5)^2}$$

The series controller has the transfer function

$$G_c(s) = \frac{1 + aTs}{1 + Ts}$$

(a) Design a phase-lead controller ($a > 1$) so that the following performance specifications are satisfied:

Ramp-error constant $K_v = 10$

Maximum overshoot is near minimum

The value of a must not exceed 1000. Plot the unit-step response and give its attributes.

(b) Design a phase-lead controller in the frequency domain so that the following performance specifications are satisfied:

Ramp-error constant $K_v = 10$

Phase margin is near maximum

The value of a must not exceed 1000

(c) Design a phase-lag controller ($a < 1$) so that the following performance specifications are satisfied:

Ramp-error constant $K_v = 10$

Maximum overshoot $< 1\%$

Rise time $t_r < 2$ sec

Settling time $t_s < 2.5$ sec

Find the PM, GM, M_r, and BW of the designed system.

(d) Design the phase-lag controller in the frequency domain so that the following performance specifications are satisfied:

Ramp-error constant $K_v = 10$

Phase margin $\geq 70°$

Check the unit-step response attributes of the designed system and compare with those obtained in part (c).

9-51. Fig. 9P-51 shows the "beam and ball" system that is described in Problem 4-11.

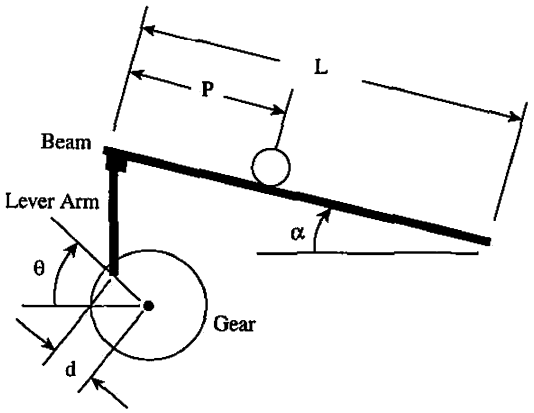


Figure 9P-51

Assuming:

$m = 0.11\,\text{kg}$	mass of the ball	$I = 9.99\text{e-}6\,\text{kg m}^2$	ball's moment of inertia
$r = 0.015$	radius of the ball	P	ball position coordinate
$d = 0.03\,\text{m}$	lever arm offset	α	beam angle coordinate
$g = 9.8\,\text{m/s}^2$	gravitational acceleration	θ	servo gear angle
$L = 1.0\,\text{m}$	length of the beam		

Design a PID controller so that the settling time is less than 3 seconds and the maximum overshoot is no more than 5%.

9-52. The controlled process of a dc-motor control system with unity feedback has the transfer function

$$G_p(s) = \frac{6.087 \times 10^{10}}{s(s^3 + 423.42s^2 + 2.6667 \times 10^6 s + 4.2342 \times 10^8)}$$

Due to the compliance in the motor shaft, the process transfer function contains two lightly damped poles, which will cause oscillations in the output response. The following performance criteria are to be satisfied:

Maximum overshoot $< 1\%$

Rise time $t_r < 0.15$ sec

Settling time $t_s < 0.15$ sec

Output response should not have oscillations

Ramp-error constant is not affected

(a) Design a series phase-lead controller,

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a > 1$$

so that all the step-response attributes (except for the oscillations) are satisfied.

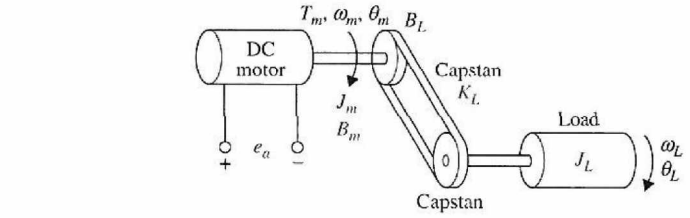
(b) To eliminate the oscillations due to the motor shaft compliance, add another stage to the controller with the transfer function

$$G_{c1}(s) = \frac{s^2 + 2\zeta_z\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\omega_n s + \omega_n^2}$$

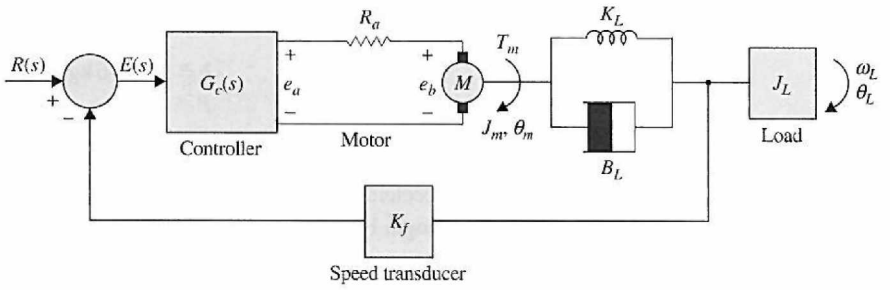
so that the zeros of $G_{c1}(s)$ will cancel the two complex poles of $G_p(s)$. Set the value of ζ_p so that the two poles of $G_{c1}(s)$ will not have an appreciable affect on the system response. Determine the

attributes of the unit-step response to see if all the requirements are satisfied. Plot the unit-step responses of the uncompensated system and the compensated system with the phase-lead controller designed in part (b).

9-53. A computer-tape-drive system utilizing a permanent-magnet dc-motor is shown in Fig. 9P-53(a). The closed-loop system is modeled by the block diagram in Fig. 9P-53(b). The constant K_L represents the spring constant of the elastic tape, and B_L denotes the viscous-friction coefficient between the tape and the capstans. The system parameters are as follows:



(a)



(b)

Figure 9P-53

K_i = motor torque constant = 10 oz-in./A
K_b = motor back-emf constant = 0.0706 V/rad/sec
B_m = motor friction coefficent = 3 oz-in./rad/sec
$R_a = 0.25\,\Omega$ $L_a \cong 0$ H
K_L = 3000 oz-in./rad B_L = 10 oz-in./rad/sec
J_L = 6 oz-in./rad/sec^2 K_f = 1 V/rad/sec
J_m = 0.05 oz-in./rad/sec^2

(a) Write the state equations of the system between e_a and θ_L using θ_L, ω_L, θ_m, and ω_m as state variables and e_a as input. Draw a state diagram using the state equations. Derive the transfer functions:

$$\frac{\Omega_m(s)}{E_a(s)} \quad \text{and} \quad \frac{\Omega_L(s)}{E_a(s)}$$

(b) The objective of the system is to control the speed of the tape, ω_L, accurately. Consider that a PI controller with the transfer function $G_c(s) = K_P + K_I/s$ is to be used. Find the values of K_P and K_I so that the following specifications are satisfied:

Ramp-error constant $K_v = 100$
Rise time < 0.02 sec
Settling time < 0.02 sec
Maximum overshoot $< 1\%$ or at minimum

Plot the unit-step response of $\omega_L(t)$ of the system.

(c) Design the PI controller in the frequency domain. The value of K_I is to be selected as in part (b). Vary the value of K_P and compute the values of PM, GM, M_r, and BW. Find the value of K_P so that PM is maximum. How does this value of K_P compare with the result obtained in part (b)?

9-54. Fig. 9P-54 shows the block diagram of a motor-control system that has a flexible shaft between the motor and the load. The transfer function between the motor torque and motor displacement is

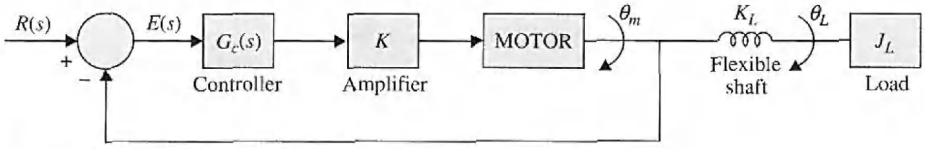


Figure 9P-54

$$G_p(s) = \frac{\Theta_m(s)}{T_m(s)} = \frac{J_L s^2 + B_L s + K_L}{s[J_m J_L s^3 + (B_m J_L + B_L J_m)s^2 + (K_L J_m + K_L J_L + B_m B_L)s + B_m K_L]}$$

where $J_L = 0.01$, $B_L = 0.1$, $K_L = 10$, $J_m = 0.01$, $B_m = 0.1$, and $K = 100$.

(a) Compute and plot the unit-step response of $\theta_m(t)$. Find the attributes of the unit-step response.

(b) Design a second-order notch controller with the transfer function

$$G_c(s) = \frac{s^2 + 2\zeta_z \omega_n s + \omega_n^2}{s^2 + 2\zeta_p \omega_n s + \omega_n^2}$$

so that its zeros cancel the complex poles of $G_p(s)$. The two poles of $G_c(s)$ should be selected so that they do not affect the steady-state response of the system, and the maximum overshoot is a minimum. Compute the attributes of the unit-step response and plot the response.

(c) Carry out design of the second-order controller in the frequency domain. Plot the Bode diagram of the uncompensated $G_p(s)$, and find the values of PM, GM, M_r, and BW. Set the two zeros of $G_c(s)$ to cancel the two complex poles of $G_p(s)$. Determine the value of ζ_p by determining the amount of attenuation required from the second-order notch controller and using Eq. (9-155). Find the PM, GM, M_r, and BW of the compensated system. How do the frequency-domain design results compare with the results in part (b)?

9-55. The transfer function of the process of a unity-feedback control system is

$$G_p(s) = \frac{500(s + 10)}{s(s^2 + 10s + 1000)}$$

(a) Plot the Bode diagram of $G_p(s)$ and determine the PM, GM, M_r, and BW of the uncompensated system. Compute and plot the unit-step response of the system.

(b) Design a series second-order notch controller with the transfer function

$$G_c(s) = \frac{s^2 + 2\zeta_z \omega_n s + \omega_n^2}{s^2 + 2\zeta_p \omega_n s + \omega_n^2}$$

so that its zeros cancel the complex poles of $G_p(s)$. Determine the value of ζ_p using the method outlined in Section 9-8-2. Find the PM, GM, M_r, and BW of the designed system. Compute and plot the unit-step response.

(c) Design the series second-order notch controller so that its zeros cancel the complex poles of $G_p(s)$. Determine the value of ζ_p so that the following specifications are satisfied:

Maximum overshoot $< 1\%$

Rise time < 0.4 sec

Settling time < 0.5 sec

9-56. Design the controllers $G_{cf}(s)$ and $G_c(s)$ for the system shown in Fig. 9P-56 so that the following specifications are satisfied:

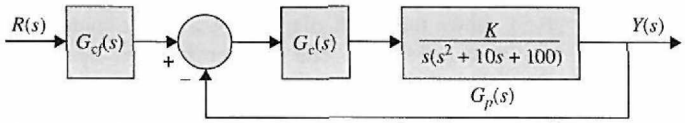


Figure 9P-56

Ramp-error constant $K_v = 50$

Dominant roots of the characteristic equation at $-5 \pm j5$ approximately

Rise time < 0.1 sec

System must be robust when K varies $\pm 20\%$ from the nominal value, with the rise time and overshoot staying within specifications

Compute and plot the unit-step responses to check the design.

9-57. Fig. 9P-57 shows the block diagram of a motor-control system. The transfer function of the controlled process is

$$G_p(s) = \frac{1000K}{s(s + a)}$$

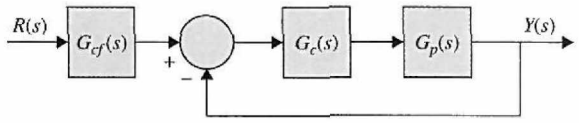


Figure 9P-57

where K denotes the aggregate of the amplifier gain and motor torque constant, and a is the inverse of the motor time constant. Design the controllers $G_{cf}(s)$ and $G_c(s)$ so that the following performance specifications are satisfied.

Ramp-error constant $K_v = 100$ when $a = 10$

Rise time < 0.3 sec

Maximum overshoot $< 8\%$

Dominant characteristic equation roots $= -5 \pm j5$

System must be robust when a varies between 8 and 12

Compute and plot the unit-step responses to verify the design.

9-58. Fig. 9P-58 shows the block diagram of a dc-motor control system with tachometer feedback. Find the values of K and K_t so that the following specifications are satisfied:

Ramp-error constant $K_v = 1$

Dominant characteristic equation roots correspond to a damping ratio of approximately 0.707; if there are two solutions, select the larger value of K

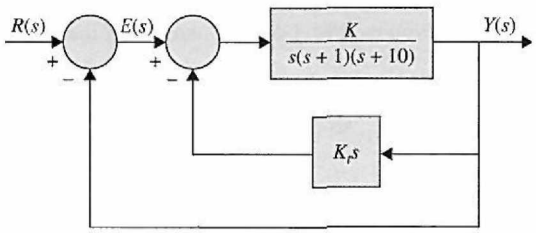


Figure 9P-58

9-59. Carry out the design with the specifications given in Problem 9-58 for the system shown in Fig. 9P-59.

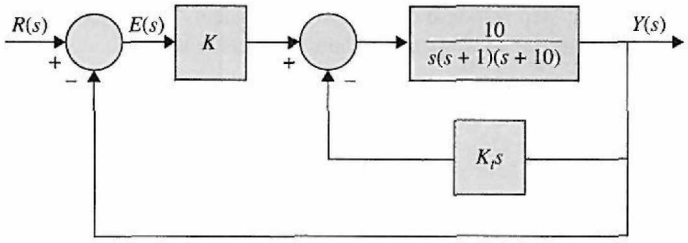


Figure 9P-59

9-60. The block diagram of a control system with a type 2 process is shown in Fig. 9P-60. The system is to be compensated by tachometer feedback and a series controller. Find the values of a, T, K, and K_t so that the following performance specifications are satisfied:

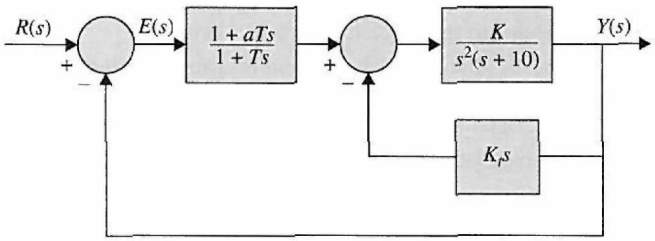


Figure 9P-60

Ramp-error constant $K_v = 100$

Dominant characteristic equation roots correspond to a damping ratio of 0.707

9-61. The aircraft-attitude control system described in Section 5-8 is modeled by the block diagram shown in Fig. 9P-61. The system parameters are as follows:

K = variable	$K_s = 1$	$K_1 = 10$	$K_2 = 0.5$	K_t = variable	$R_a = 5$
$L_a = 0.003$	$K_i = 9.0$	$K_b = 0.0636$	$J_m = 0.0001$	$J_L = 0.01$	
$B_m = 0.005$	$B_L = 1.0$	$N = 0.1$			

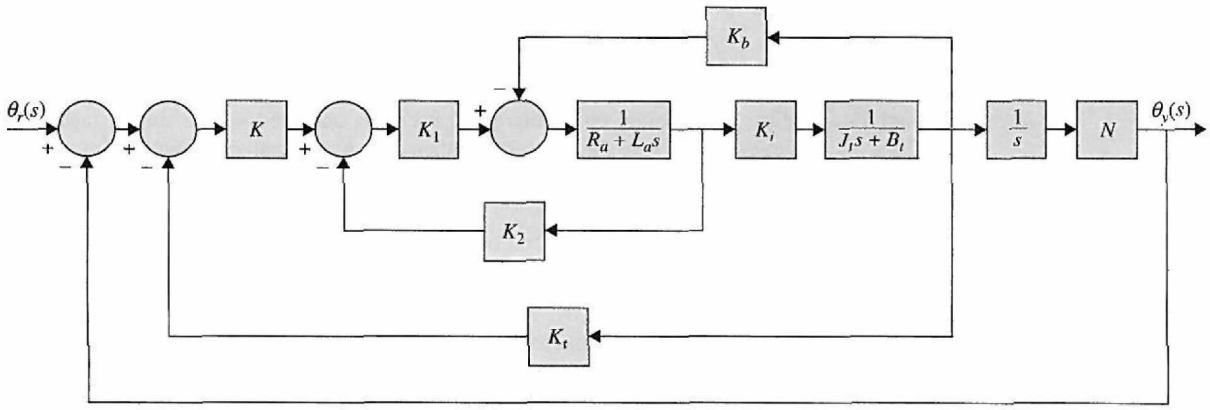


Figure 9P-61

Find the values of K and K_t so that the following specifications are satisfied:

Ramp-error constant $K_v = 100$

Relative damping ratio of the complex roots of the characteristic equation is approximately 0.707

Plot the unit-step response of the designed system. Show that the system performance is extremely insensitive to the value of K. Explain why this is so.

9-62. Fig. 9P-62 shows the block diagram of a position-control system with a series controller $G_c(s)$.

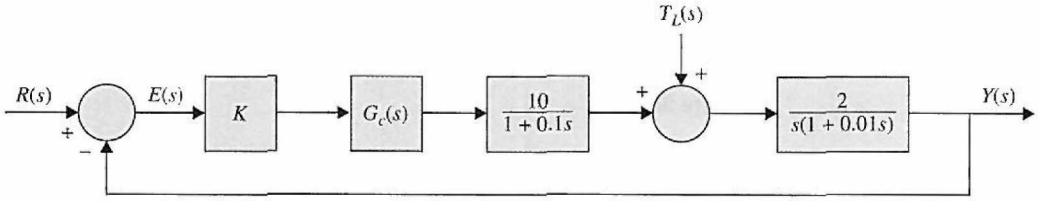


Figure 9P-62

(a) Determine the minimum value of the amplifier gain K so that the steady-state value of the output $y(t)$ due to a unit-step torque disturbance is ≤ 0.01.

(b) Show that the uncompensated system is unstable with the minimum value of K determined in part (a). Construct the Bode diagram for the open-loop transfer function $G(s) = Y(s)/E(s)$, and find the values of PM and GM.

(c) Design a single-stage phase-lead controller with the transfer function

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a > 1$$

so that the phase margin is 30°. Show that this is nearly the highest phase margin that can be achieved with a single-stage phase-lead controller. Find GM, M_r, and BW of the compensated system.

(d) Design a two-stage phase-lead controller using the system arrived at in part (c) as a basis so that the phase margin is 55°. Show that this is the best PM that can be obtained for this system with a two-stage phase-lead controller. Find GM, M_r, and BW of the compensated system.

9-63. The transfer function of the process of a unity-feedback control system is

$$G_c(s) = \frac{60}{s(1 + 0.2s)(1 + 0.5s)}$$

Show that, due to the relative high gain, the uncompensated system is unstable.

(a) Design a two-stage phase-lead controller with

$$G_c(s) = \left(\frac{1 + aT_1s}{1 + T_1s}\right)\left(\frac{1 + bT_2s}{1 + T_2s}\right) \quad a > 1, \quad b > 1$$

so that the phase margin is greater than 60°. Conduct the design by first determining the values of a and T_1 to realize a maximum phase margin that can be achieved with a single-stage phase-lead controller. The second stage of the controller is then designed to realize the balance of the 60° phase margin. Determine GM, M_r, and BW of the compensated system. Compute and plot the unit-step response of the compensated system.

(b) Design a single-stage phase-lag controller with

$$G_c(s) = \frac{1 + aTs}{1 + Ts} \quad a < 1$$

so that the phase margin of the compensated system is greater than 60°. Determine GM, M_r, and BW of the compensated system. Compute and plot the unit-step response of the compensated system.

(c) Design a lag–lead controller with $G_c(s)$ as in the equation in part (a). Design the phase-lag portion first by setting the phase margin at 40°. The resulting system is then compensated by the

phase-lead portion to achieve a total of 60° of phase margin. Determine GM, M_r, and BW of the compensated system. Compute and plot the unit-step response of the compensated system.

9-64. The block diagram of the steel-rolling system described in Problem 4-18 is shown in Fig. 9P-64. The transfer function of the process is

$$G_p(s) = \frac{Y(s)}{E(s)} = \frac{5e^{-0.1s}}{s(1 + 0.1s)(1 + 0.5s)} \quad K_s = 1$$

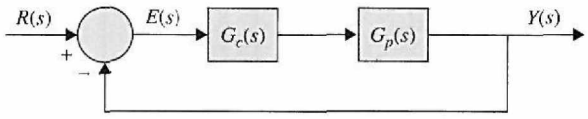


Figure 9P-64

(a) Approximate the time delay by

$$e^{-0.1s} \cong \frac{1 - 0.05s}{1 + 0.05s}$$

Design a series controller of your choice so that the phase margin of the compensated system is at least 60°. Determine GM, M_r, and BW of the compensated system. Compute and plot the unit-step responses of the compensated and the uncompensated systems.

(b) Repeat part (a) without using the approximation of the time delay.

9-65. Human beings breathe in order to provide for gas exchange for the entire body. A respiratory control system is needed to ensure that the body's needs for this gas exchange are adequately met. The criterion of control is adequate ventilation, which ensures satisfactory levels of both oxygen and carbon dioxide in the arterial blood. Respiration is controlled by neural impulses that originate within the lower brain and are transmitted to the chest cavity and diaphragm to govern the rate and tidal volume. One source of signals consists of the chemoreceptors located near the respiratory center, which are sensitive to carbon dioxide and oxygen concentrations. Fig. 9P-65 shows the block diagram of a simplified model of the human respiratory control system. The objective is to control the effective ventilation of the lungs so that a satisfactory balance of concentrations of carbon dioxide and oxygen is maintained in the blood circulated at the chemoreceptor.

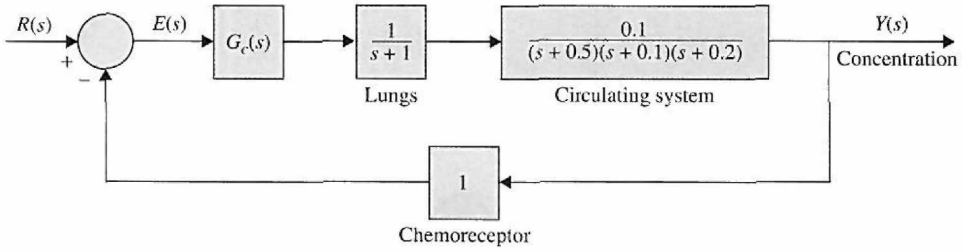


Figure 9P-65

(a) Plot the Bode diagram of the transfer function $G(s) = Y(s)/E(s)$ when $G_c(s) = 1$. Find the PM and GM. Determine the stability of the system.

(b) Design a PI controller, $G_c(s) = K_P + K_I/s$, so that the following specifications are satisfied:

Ramp-error constant $K_v = 1$

Phase margin is maximized

Plot the unit-step response of the system. Find the attributes of the unit-step response.

(c) Design a PI controller so that the following specifications are satisfied:

Ramp-error constant $K_v = 1$

Maximum overshoot is minimized

Plot the unit-step response of the system. Find the attributes of the unit-step response. Compare the design results in parts (b) and (c).

9-66. The block diagram of a control system with state feedback is shown in Fig. 9P-66. Find the real feedback gains k_1, k_2, and k_3 so that:

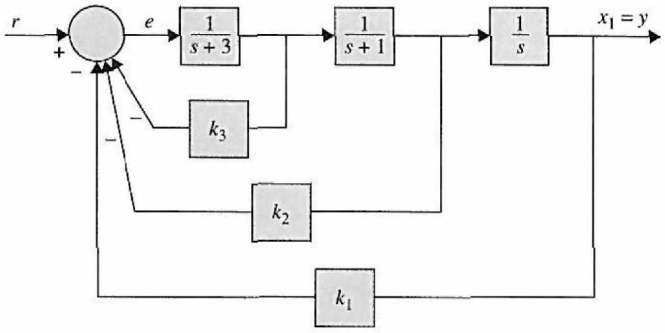


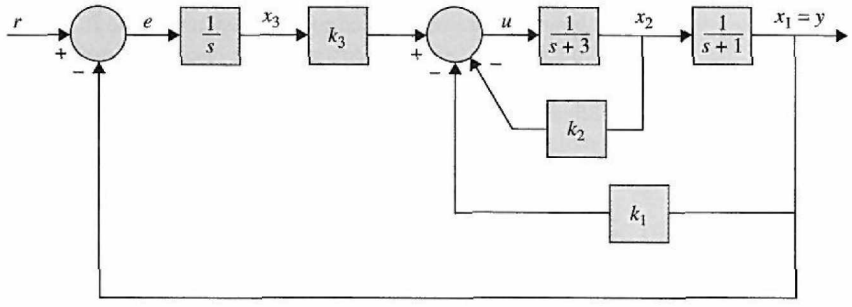
Figure 9P-66

The steady-state error e_{ss} [$e(t)$ is the error signal] due to a step input is zero.

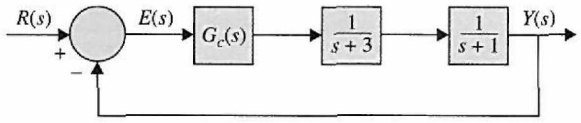
The complex roots of the characteristic equation are at $-1 + j$ and $-1 - j$.

Find the third root. Can all three roots be arbitrarily assigned while still meeting the steady-state requirement?

9-67. The block diagram of a control system with state feedback is shown in Fig. 9P-67(a). The feedback gains k_1, k_2, and k_3 are real constants.



(a)



(b)

Figure 9P-67

(a) Find the values of the feedback gains so that:

The steady-state error e_{ss} [$e(t)$ is the error signal] due to a step input is zero.

The characteristic equation roots are at $-1 + j$, $-1 - j$, and -10.

(b) Instead of using state feedback, a series controller is implemented, as shown in Fig. 9P-67(b). Find the transfer function of the controller $G_c(s)$ in terms of k_1, k_2, and k_3 found in part (a) and the other system parameters.

CHAPTER 10

State Variable Analysis

10-1 INTRODUCTION

In Chapter 2 we presented the concept and definition of state variables and state equations for linear continuous-data and discrete-data dynamic systems. In Chapter 3 we used block-diagram and signal-flow-graph (SFG) methods to obtain the transfer function of linear systems. In this chapter, the SFG concept is extended to the modeling of the state equations, and the result is the **state diagram**. In contrast to the transfer-function approach to the analysis and design of linear control systems, the state-variable method is regarded as modern, since it uses underlying force for optimal control. The basic characteristic of the state-variable formulation is that linear and nonlinear systems, time-invariant and time-varying systems, and single-variable and multivariable systems can all be modeled in a unified manner. Transfer functions, on the other hand, are defined only for linear time-invariant systems.

The objective of this chapter is to introduce the basic methods of state variables and state equations so that the reader can gain a working knowledge of the subject for further studies when the state-space approach is used for modern and optimal control design. Specifically, the closed-form solutions of linear time-invariant state equations are presented. Various transformations that may be used to facilitate the analysis and design of linear control systems in the state-variable domain are introduced. The relationship between the conventional transfer-function approach and the state-variable approach is established so that the analyst will be able to investigate a system problem with various alternative methods. Finally, the controllability and observability of linear systems are defined and their applications investigated. Some state-space controller design problems appear in the end. At the end of the chapter, we also present MATLAB tools to solve most state-space problems.

10-2 BLOCK DIAGRAMS, TRANSFER FUNCTIONS, AND STATE DIAGRAMS

10-2-1 Transfer Functions (Multivariable Systems)

The definition of a transfer function is easily extended to a system with multiple inputs and outputs. A system of this type is often referred to as a multivariable system. In a multivariable system, a differential equation of the form of Eq. (2-217) may be used to describe the relationship between a pair of input and output variables, when all other inputs are set to zero. This equation is restated as

$$\begin{aligned}&\frac{d^n y(t)}{dt^n} + a_{n-1}\frac{d^{n-1}y(t)}{dt^{n-1}} + \cdots + a_1\frac{dy(t)}{dt} + a_0 y(t)\\ &\quad = b_m\frac{d^m u(t)}{dt^m} + b_{m-1}\frac{d^{m-1}u(t)}{dt^{m-1}} + \cdots + b_1\frac{du(t)}{dt} + b_0 u(t)\end{aligned} \tag{10-1}$$

The coefficients $a_0, a_1, \ldots, a_{n-1}$ and $b_0, b_1, \ldots, b_m$ are real constants. Because the principle of superposition is valid for linear systems, the total effect on any output due to all the inputs acting simultaneously is obtained by adding up the outputs due to each input acting alone.

In general, if a linear system has p inputs and q outputs, the transfer function between the jth input and the ith output is defined as

$$G_{ij}(s) = \frac{Y_i(s)}{R_j(s)} \tag{10-2}$$

with $R_k(s) = 0, k = 1, 2, \ldots, p, k \neq j$. Note that Eq. (10-2) is defined with only the jth input in effect, whereas the other inputs are set to zero. When all the p inputs are in action, the ith output transform is written

$$Y_i(s) = G_{i1}(s)R_1(s) + G_{i2}(s)R_2(s) + \cdots + G_{ip}(s)R_p(s) \tag{10-3}$$

It is convenient to express Eq. (10-3) in matrix-vector form:

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{R}(s) \tag{10-4}$$

where

$$\mathbf{Y}(s) = \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_q(s) \end{bmatrix} \tag{10-5}$$

is the $q \times 1$ transformed output vector,

$$\mathbf{R}(s) = \begin{bmatrix} R_1(s) \\ R_2(s) \\ \vdots \\ R_p(s) \end{bmatrix} \tag{10-6}$$

is the $p \times 1$ transformed input vector, and

$$\mathbf{G}(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \cdots & G_{1p}(s) \\ G_{21}(s) & G_{22}(s) & \cdots & G_{2p}(s) \\ \cdot & \cdot & \cdots & \cdot \\ G_{q1}(s) & G_{q2}(s) & \cdots & G_{qp}(s) \end{bmatrix} \tag{10-7}$$

is the $q \times p$ transfer-function matrix.

10-2-2 Block Diagrams and Transfer Functions of Multivariable Systems

In this section, we shall illustrate the block diagram and matrix representations of multivariable systems. Two block-diagram representations of a multivariable system with p inputs and q outputs are shown in Fig. 10-1(a) and (b). In Fig. 10-1(a), the

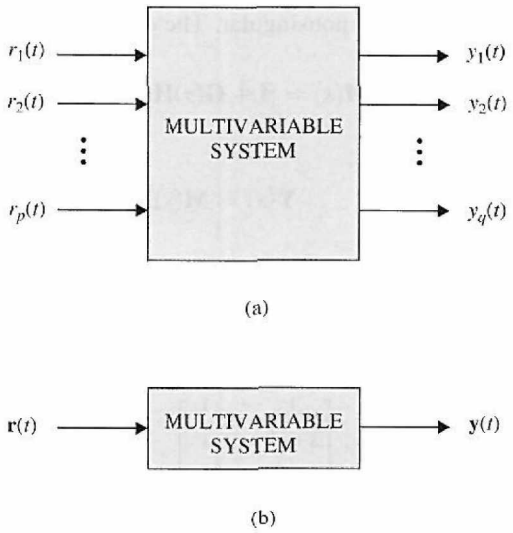


Figure 10-1 Block diagram representations of a multivariable system.

individual input and output signals are designated, whereas in the block diagram of Fig. 10-1(b), the multiplicity of the inputs and outputs is denoted by vectors. The case of Fig. 10-1(b) is preferable in practice because of its simplicity.

Fig. 10-2 shows the block diagram of a multivariable feedback control system. The transfer function relationships of the system are expressed in vector-matrix form (see Section 10-3 for more detail):

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s) \tag{10-8}$$

$$\mathbf{U}(s) = \mathbf{R}(s) - \mathbf{B}(s) \tag{10-9}$$

$$\mathbf{B}(s) = \mathbf{H}(s)\mathbf{Y}(s) \tag{10-10}$$

where $\mathbf{Y}(s)$ is the $q \times 1$ output vector; $\mathbf{U}(s)$, $\mathbf{R}(s)$, and $\mathbf{B}(s)$ are all $p \times 1$ vectors; and $\mathbf{G}(s)$ and $\mathbf{H}(s)$ are $q \times p$ and $p \times q$ transfer-function matrices, respectively. Substituting Eq. (10-9) into Eq. (10-8) and then from Eq. (10-8) to Eq. (10-10), we get

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{R}(s) - \mathbf{G}(s)\mathbf{H}(s)\mathbf{Y}(s) \tag{10-11}$$

Solving for $\mathbf{Y}(s)$ from Eq. (10-11) gives

$$\mathbf{Y}(s) = [\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]^{-1}\,\mathbf{G}(s)\mathbf{R}(s) \tag{10-12}$$

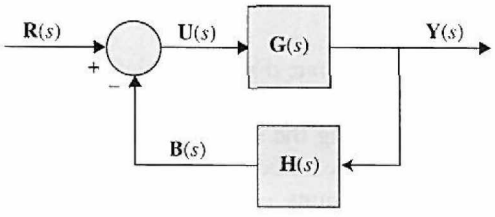


Figure 10-2 Block diagram of a multivariable feedback control system.

provided that $\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)$ is nonsingular. The closed-loop transfer matrix is defined as

$$\mathbf{M}(s) = [\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]^{-1}\,\mathbf{G}(s) \tag{10-13}$$

Then Eq. (10-12) is written

$$\mathbf{Y}(s) = \mathbf{M}(s)\mathbf{R}(s) \tag{10-14}$$

▶ EXAMPLE 10-2-1 Consider that the forward-path transfer function matrix and the feedback-path transfer function matrix of the system shown in Fig. 10-2 are

$$\mathbf{G}(s) = \begin{bmatrix} \dfrac{1}{s+1} & -\dfrac{1}{s} \\ 2 & \dfrac{1}{s+2} \end{bmatrix} \qquad \mathbf{H}(s) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{10-15}$$

respectively. The closed-loop transfer function matrix of the system is given by Eq. (10-14) and is evaluated as follows:

$$\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s) = \begin{bmatrix} 1 + \dfrac{1}{s+1} & -\dfrac{1}{s} \\ 2 & 1 + \dfrac{1}{s+2} \end{bmatrix} = \begin{bmatrix} \dfrac{s+2}{s+1} & -\dfrac{1}{s} \\ 2 & \dfrac{s+3}{s+2} \end{bmatrix} \tag{10-16}$$

The closed-loop transfer function matrix is

$$\mathbf{M}(s) = [\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]^{-1}\,\mathbf{G}(s) = \frac{1}{\Delta}\begin{bmatrix} \dfrac{s+3}{s+2} & \dfrac{1}{s} \\ -2 & \dfrac{s+2}{s+1} \end{bmatrix}\begin{bmatrix} \dfrac{1}{s+1} & -\dfrac{1}{s} \\ 2 & \dfrac{1}{s+2} \end{bmatrix} \tag{10-17}$$

where

$$\Delta = \frac{s+2}{s+1}\frac{s+3}{s+2} + \frac{2}{s} = \frac{s^2+5s+2}{s(s+1)} \tag{10-18}$$

Thus,

$$\mathbf{M}(s) = \frac{s(s+1)}{s^2+5s+2}\begin{bmatrix} \dfrac{3s^2+9s+4}{s(s+1)(s+2)} & -\dfrac{1}{s} \\ 2 & \dfrac{3s+2}{s(s+1)} \end{bmatrix} \tag{10-19}$$

◀

10-2-3 State Diagram

In this section, we introduce the state diagram, which is an extension of the SFG to portray state equations and differential equations. The significance of the state diagram is that it forms a close relationship among the state equations, computer simulation, and transfer functions. A state diagram is constructed following all the rules of the SFG using the Laplace-transformed state equations.

The basic elements of a state diagram are similar to the conventional SFG, except for the **integration** operation. Let the variables $x_1(t)$ and $x_2(t)$ be related by the first-order differentiation:

$$\frac{dx_1(t)}{dt} = x_2(t) \tag{10-20}$$

Integrating both sides of the last equation with respect to t from the initial time t_0, we get

$$x_1(t) = \int_{t_0}^{t} x_2(\tau)d\tau + x_1(t_0) \tag{10-21}$$

Because the SFG algebra does not handle integration in the time domain, we must take the Laplace transform on both sides of Eq. (10-20). We have

$$\begin{aligned} X_1(s) &= \mathcal{L}\left[\int_{t_0}^{t} x_2(\tau)d\tau\right] + \frac{x_1(t_0)}{s} = \mathcal{L}\left[\int_0^t x_2(\tau)d\tau - \int_0^{t_0} x_2(\tau)d\tau\right] + \frac{x_1(t_0)}{s} \\ &= \frac{X_2(s)}{s} - \mathcal{L}\left[\int_0^{t_0} x_2(\tau)d\tau\right] + \frac{x_1(t_0)}{s} \end{aligned} \tag{10-22}$$

Because the past history of the integrator is represented by $x_1(t_0)$, and the state transition is assumed to start at $\tau = t_0$, $x_2(\tau) = 0$ for $0 < \tau < t_0$. Thus, Eq. (10-22) becomes

$$X_1(s) = \frac{X_2(s)}{s} + \frac{x_1(t_0)}{s} \quad \tau \geq t_0 \tag{10-23}$$

Eq. (10-23) is now algebraic and can be represented by an SFG, as shown in Fig. 10-3. Fig. 10-3 shows that *the output of the integrator is equal to s^{-1} times the input, plus the initial condition $x_1(t_0)/s$.* An alternative SFG with fewer elements for Eq. (10-23) is shown in Fig. 10-4.

Before embarking on several illustrative examples on the construction of state diagrams, let us point out the important uses of the state diagram.

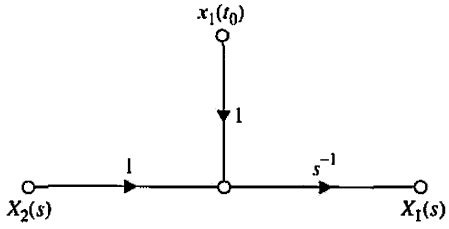


Figure 10-3 Signal-flow graph representation of $X_1(s) = [X_2(s)/s] + [x_1(t_0)/s]$.

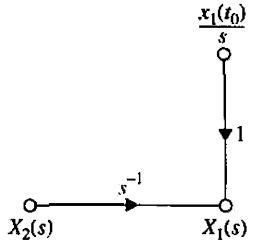


Figure 10-4 Signal-flow graph representation of $X_1(s) = [X_2(s)/s] + [x_1(t_0)/s]$.

1. A state diagram can be constructed directly from the system's differential equation. This allows the determination of the state variables and the state equations.
2. A state diagram can be constructed from the system's transfer function. This step is defined as the **decomposition** of transfer functions (Section 10-10).
3. The state diagram can be used to program the system on an analog computer or for simulation on a digital computer.
4. The state-transition equation in the Laplace transform domain may be obtained from the state diagram by using the SFG gain formula.
5. The transfer functions of a system can be determined from the state diagram.
6. The state equations and the output equations can be determined from the state diagram.

The details of these techniques will follow.

10-2-4 From Differential Equations to State Diagrams

When a linear system is described by a high-order differential equation, a state diagram can be constructed from these equations, although a direct approach is not always the most convenient. Consider the following differential equation:

$$\frac{d^n y(t)}{dt^n} + a_n \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_2 \frac{dy(t)}{dt} + a_1 y(t) = r(t) \tag{10-24}$$

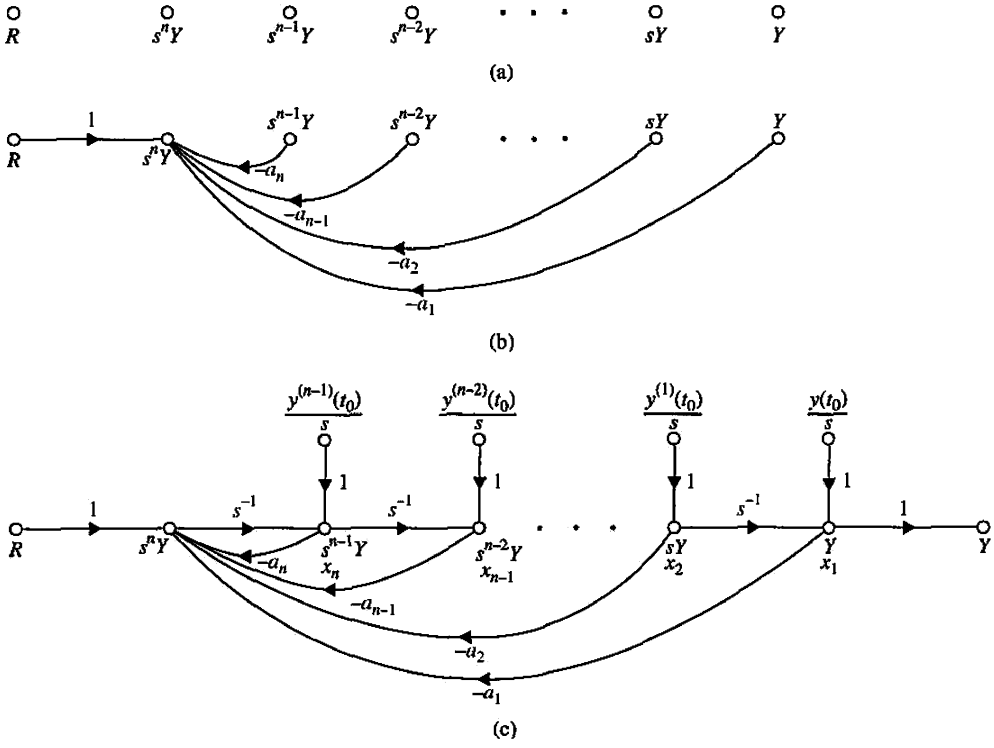


Figure 10-5 State-diagram representation of the differential equation of Eq. (10-24).

To construct a state diagram using this equation, we rearrange the equation as

$$\frac{d^n y(t)}{dt^n} - -a_n \frac{d^{n-1} y(t)}{dt^{n-1}} - \cdots - a_2 \frac{dy(t)}{dt} - a_1 y(t) + r(t) \tag{10-25}$$

As a first step, the nodes representing $R(s), s^n Y(s), s^{n-1} Y(s), \ldots, sY(s)$, and $Y(s)$ are arranged from left to right, as shown in Fig. 10-5(a). Because $s^i Y(s)$ corresponds to $d^i y(t)/dt^i, i = 0, 1, 2, \ldots, n$, in the Laplace domain, as the next step, the nodes in Fig. 10-5(a) are connected by branches to portray Eq. (10-25), resulting in Fig. 10-5(b). Finally, the integrator branches with gains of s^{-1} are inserted, and the initial conditions are added to the outputs of the integrators, according to the basic scheme in Fig. 10-3. The complete state diagram is drawn as shown in Fig. 10-5(c). *The outputs of the integrators are defined as the state variables*, $x_1, x_2, \ldots, x_n$. This is usually the natural choice of state variables once the state diagram is drawn.

When the differential equation has derivatives of the input on the right side, the problem of drawing the state diagram directly is not as straightforward as just illustrated. We will show that, in general, it is more convenient to obtain the transfer function from the differential equation first and then arrive at the state diagram through decomposition (Section 10-10).

EXAMPLE 10-2-2 Consider the differential equation

$$\frac{d^2 y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = r(t) \tag{10-26}$$

Equating the highest-ordered term of the last equation to the rest of the terms, we have

$$\frac{d^2 y(t)}{dt^2} = -3\frac{dy(t)}{dt} - 2y(t) + r(t) \tag{10-27}$$

Following the procedure just outlined, the state diagram of the system is drawn as shown in Fig. 10-6. The state variables x_1 and x_2 are assigned as shown.

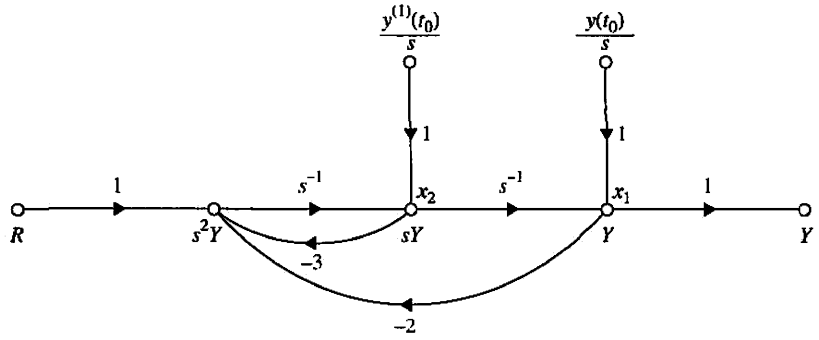


Figure 10-6 State diagram for Eq. (10-26).

10-2-5 From State Diagrams to Transfer Functions

The transfer function between an input and an output is obtained from the state diagram by using the gain formula and setting all other inputs and initial states to zero. The following example shows how the transfer function is obtained directly from a state diagram.

EXAMPLE 10-2-3 Consider the state diagram of Fig. 10-6. The transfer function between $R(s)$ and $Y(s)$ is obtained by applying the gain formula between these two nodes and setting the initial states to zero. We have

$$\frac{Y(s)}{R(s)} = \frac{1}{s^2 + 3s + 2} \tag{10-28}$$

10-2-6 From State Diagrams to State and Output Equations

The state equations and the output equations can be obtained directly from the state diagram by using the SFG gain formula. The general form of a state equation and the output equation for a linear system is described in Chapter 2 and presented here.

State equation:

$$\frac{dx(t)}{dt} = ax(t) + br(t) \tag{10-29}$$

Output equation:

$$y(t) = cx(t) + dr(t) \tag{10-30}$$

where $x(t)$ is the state variable; $r(t)$ is the input; $y(t)$ is the output; and a, b, c, and d are constant coefficients. Based on the general form of the state and output equations, the following procedure of deriving the state and output equations from the state diagram are outlined:

1. Delete the initial states and the integrator branches with gains s^{-1} from the state diagram, since the state and output equations do not contain the Laplace operator s or the initial states.
2. For the state equations, regard the nodes that represent the derivatives of the state variables as output nodes, since these variables appear on the left-hand side of the state equations. The output $y(t)$ in the output equation is naturally an output node variable.
3. Regard the state variables and the inputs as input variables on the state diagram, since these variables are found on the right-hand side of the state and output equations.
4. Apply the SFG gain formula to the state diagram.

EXAMPLE 10-2-4 Fig. 10-7 shows the state diagram of Fig. 10-6 with the integrator branches and the initial states eliminated. Using $dx_1(t)/dt$ and $dx_2(t)/dt$ as the output nodes and $x_1(t)$, $x_2(t)$, and $r(t)$ as input nodes, and applying the gain formula between these nodes, the state equations are obtained as

$$\frac{dx_1(t)}{dt} = x_2(t) \tag{10-31}$$

$$\frac{dx_2(t)}{dt} = -2x_1(t) - 3x_2(t) + r(t) \tag{10-32}$$

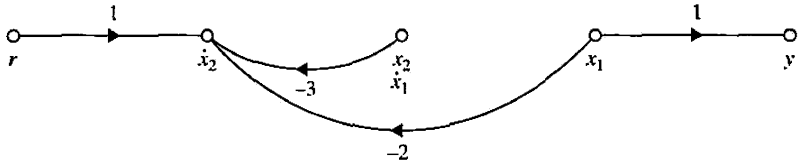


Figure 10-7 State diagram of Fig. 10-6 with the initial states and the integrator branches left out.

Applying the gain formula with $x_1(t)$, $x_2(t)$, and $r(t)$ as input nodes and $y(t)$ as the output node, the output equation is written

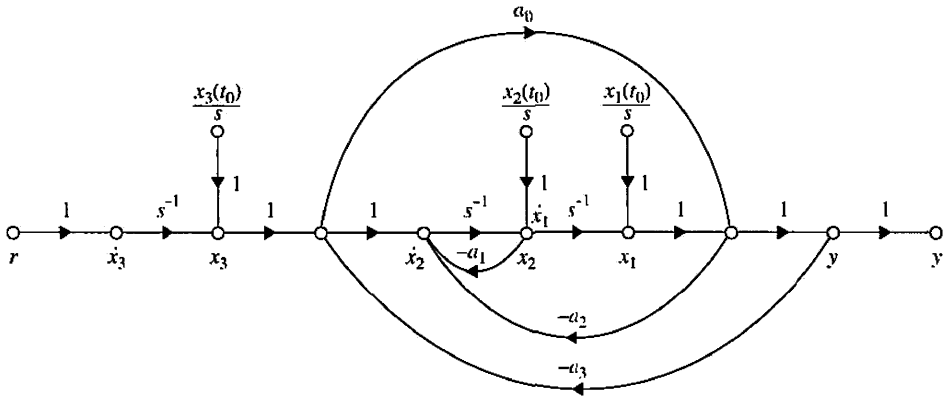
$$y(t) = x_1(t) \tag{10-33}$$

EXAMPLE 10-2-5 As another example on the determination of the state equations from the state diagram, consider the state diagram shown in Fig. 10-8(a). This example will also emphasize the importance of applying the gain formula. Fig. 10-8(b) shows the state diagram with the initial states and the integrator branches deleted. Notice that, in this case, the state diagram in Fig. 10-8(b) still contains a loop. By applying the gain formula to the state diagram in Fig. 10-8(b) with $\dot{x}_1(t)$, $\dot{x}_2(t)$, and $\dot{x}_3(t)$ as output-node variables and $r(t)$, $x_1(t)$, $x_2(t)$, and $x_3(t)$ as input nodes, the state equations are obtained as follows in vector-matrix form:

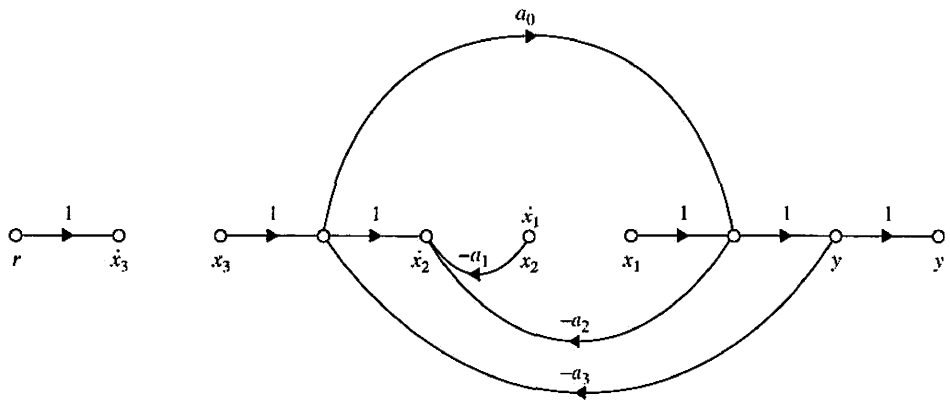
$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \\ \dfrac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ \dfrac{-(a_2 + a_3)}{1 + a_0 a_3} & -a_1 & \dfrac{1 - a_0 a_2}{1 + a_0 a_3} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r(t) \tag{10-34}$$

The output equation is

$$y(t) = \frac{1}{1 + a_0 a_3} x_1(t) + \frac{a_0}{1 + a_0 a_3} x_3(t) \tag{10-35}$$



(a)



(b)

Figure 10-8 (a) State diagram. (b) State diagram in part (a) with all initial states and integrators left out.

▶ 10-3 VECTOR-MATRIX REPRESENTATION OF STATE EQUATIONS

Let the n state equations of an nth-order dynamic system be represented as

$$\frac{dx_1(t)}{dt} = f_i[x_1(t), x_2(t), \ldots, x_n(t), u_1(t), u_2(t), \ldots, u_p(t), w_1(t), w_2(t), \ldots, w_v(t)] \tag{10-36}$$

where $i = 1, 2, \ldots, n$. The ith state variable is represented by $x_i(t)$; $u_j(t)$ denotes the jth input for $j = 1, 2, \ldots, p$; and $w_k(t)$ denotes the kth disturbance input, with $k = 1, 2, \ldots, v$.

Let the variables $y_1(t), y_2(t), \ldots, y_q(t)$ be the q output variables of the system. In general, the output variables are functions of the state variables and the input variables. The **output equations** can be expressed as

$$y_j(t) = g_j[x_1(t), x_2(t), \ldots, x_n(t), u_1(t), u_2(t), \ldots, u_p(t), w_1(t), w_2(t), \ldots, w_v(t)] \tag{10-37}$$

where $j = 1, 2, \ldots, q$.

The set of n state equations in Eq. (10-36) and q output equations in Eq. (10-37) together form the **dynamic equations**. For ease of expression and manipulation, it is convenient to represent the dynamic equations in vector-matrix form. Let us define the following vectors:

State vector:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \quad (n \times 1) \tag{10-38}$$

Input vector:

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_p(t) \end{bmatrix} \quad (p \times 1) \tag{10-39}$$

Output vector:

$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_q(t) \end{bmatrix} \quad (q \times 1) \tag{10-40}$$

Disturbance vector:

$$\mathbf{w}(t) = \begin{bmatrix} w_1(t) \\ w_2(t) \\ \vdots \\ w_v(t) \end{bmatrix} \quad (v \times 1) \tag{10-41}$$

By using these vectors, the n state equations of Eq. (10-36) can be written

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t), \mathbf{w}(t)] \tag{10-42}$$

where $\mathbf{f}$ denotes an $n \times 1$ column matrix that contains the functions $f_1, f_2, \ldots, f_n$ as elements. Similarly, the q output equations in Eq. (10-37) become

$$\mathbf{y}(t) = \mathbf{g}[\mathbf{x}(t), \mathbf{u}(t), \mathbf{w}(t)] \tag{10-43}$$

where $\mathbf{g}$ denotes a $q \times 1$ column matrix that contains the functions $g_1, g_2, \ldots, g_q$ as elements.

For a linear time-invariant system, the dynamic equations are written as

State equations:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{E}\mathbf{w}(t) \tag{10-44}$$

Output equations:

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) + \mathbf{H}\mathbf{w}(t) \tag{10-45}$$

where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad (n \times n) \tag{10-46}$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} \quad (n \times p) \tag{10-47}$$

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{q1} & c_{q2} & \cdots & c_{qn} \end{bmatrix} \quad (q \times n) \tag{10-48}$$

$$\mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1p} \\ d_{21} & d_{22} & \cdots & d_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ d_{q1} & d_{q2} & \cdots & d_{qp} \end{bmatrix} \quad (q \times p) \tag{10-49}$$

$$\mathbf{E} = \begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1v} \\ e_{21} & e_{22} & \cdots & e_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ e_{n1} & e_{n2} & \cdots & e_{nv} \end{bmatrix} \quad (n \times v) \tag{10-50}$$

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1v} \\ h_{12} & h_{22} & \cdots & h_{2v} \\ \vdots & \vdots & \ddots & \vdots \\ h_{q1} & h_{q2} & \cdots & h_{qv} \end{bmatrix} \quad (q \times v) \tag{10-51}$$

10-4 STATE-TRANSITION MATRIX

Once the state equations of a linear time-invariant system are expressed in the form of Eq. (10-44), the next step often involves the solutions of these equations given the initial state vector $\mathbf{x}(t_0)$, the input vector $\mathbf{u}(t)$, and the disturbance vector $\mathbf{w}(t)$, for $t \geq t_0$. The first term on the right-hand side of Eq. (10-44) is known as the homogeneous part of the state equation, and the last two terms represent the forcing functions $\mathbf{u}(t)$ and $\mathbf{w}(t)$.

The **state-transition matrix** is defined as a matrix that satisfies the linear homogeneous state equation:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) \tag{10-52}$$

Let $\boldsymbol{\phi}(t)$ be the $n \times n$ matrix that represents the state-transition matrix; then it must satisfy the equation

$$\frac{d\boldsymbol{\phi}(t)}{dt} = \mathbf{A}\boldsymbol{\phi}(t) \tag{10-53}$$

Furthermore, let $\mathbf{x}(0)$ denote the initial state at $t = 0$; then $\boldsymbol{\phi}(t)$ is also defined by the matrix equation

$$\mathbf{x}(t) = \boldsymbol{\phi}(t)\mathbf{x}(0) \tag{10-54}$$

which is the solution of the homogeneous state equation for $t \geq 0$.

One way of determining $\boldsymbol{\phi}(t)$ is by taking the Laplace transform on both sides of Eq. (10-52); we have

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) \tag{10-55}$$

Solving for $\mathbf{X}(s)$ from Eq. (10-55), we get

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) \tag{10-56}$$

where it is assumed that the matrix $(s\mathbf{I} - \mathbf{A})$ is nonsingular. Taking the inverse Laplace transform on both sides of Eq. (10-56) yields

$$\mathbf{x}(t) = \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A})^{-1}\right]\mathbf{x}(0) \quad t \geq 0 \tag{10-57}$$

By comparing Eq. (10-54) with Eq. (10-57), the state-transition matrix is identified to be

$$\boldsymbol{\phi}(t) = \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A})^{-1}\right] \tag{10-58}$$

An alternative way of solving the homogeneous state equation is to assume a solution, as in the classical method of solving linear differential equations. We let the solution to Eq. (10-52) be

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) \tag{10-59}$$

for $t \geq 0$, where $e^{\mathbf{A}t}$ represents the following power series of the matrix $\mathbf{A}t$, and

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{A}^2t^2 + \frac{1}{3!}\mathbf{A}^3t^3 + \cdots \tag{10-60}$$

It is easy to show that Eq. (10-59) is a solution of the homogeneous state equation, since, from Eq. (10-60),

$$\frac{de^{\mathbf{A}t}}{dt} = \mathbf{A}e^{\mathbf{A}t} \tag{10-61}$$

Therefore, in addition to Eq. (10-58), we have obtained another expression for the state-transition matrix:

$$\boldsymbol{\phi}(t) = e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{A}^2t^2 + \frac{1}{3!}\mathbf{A}^3t^3 + \cdots \tag{10-62}$$

Eq. (10-62) can also be obtained directly from Eq. (10-58). This is left as an exercise for the reader (Problem 10-5).

10-4-1 Significance of the State-Transition Matrix

Because the state-transition matrix satisfies the homogeneous state equation, it represents the **free response** of the system. In other words, it governs the response that is excited by the initial conditions only. In view of Eqs. (10-58) and (10-62), the state-transition matrix is dependent only upon the matrix **A** and, therefore, is sometimes referred to as the **state-transition matrix of A**. As the name implies, the state-transition matrix $\boldsymbol{\phi}(t)$ completely defines the transition of the states from the initial time $t = 0$ to any time t when the inputs are zero.

10-4-2 Properties of the State-Transition Matrix

The state-transition matrix $\boldsymbol{\phi}(t)$ possesses the following properties:

1. $\boldsymbol{\phi}(0) = \mathbf{I}$ (the identity matrix) (10-63)

   Proof: Eq. (10-63) follows directly from Eq. (10-62) by setting $t = 0$.

2. $\boldsymbol{\phi}^{-1}(t) = \boldsymbol{\phi}(-t)$ (10-64)

   Proof: Post-multiplying both sides of Eq. (10-62) by $e^{-\mathbf{A}t}$, we get

$$\boldsymbol{\phi}(t)e^{-\mathbf{A}t} = \mathbf{e}^{\mathbf{A}t}\mathbf{e}^{-\mathbf{A}t} = \mathbf{I} \tag{10-65}$$

Then, pre-multiplying both sides of Eq. (10-65) by $\phi^{-1}(t)$, we get

$$e^{-\mathbf{A}t} = \phi^{-1}(t) \tag{10-66}$$

Thus,

$$\phi(-t) = \phi^{-1}(t) = e^{-\mathbf{A}t} \tag{10-67}$$

An interesting result from this property of $\phi(t)$ is that Eq. (10-59) can be rearranged to read

$$\mathbf{x}(0) = \phi(-t)\mathbf{x}(t) \tag{10-68}$$

which means that the state-transition process can be considered as bilateral in time. That is, the transition in time can take place in either direction.

3. $\phi(t_2 - t_1)\phi(t_1 - t_0) = \phi(t_2 - t_0)$ for any t_0, t_1, t_2 (10-69)

Proof:

$$\begin{aligned}\phi(t_2 - t_1)\phi(t_1 - t_0) &= e^{\mathbf{A}(t_2 - t_1)}e^{\mathbf{A}(t_1 - t_0)} \\ &= e^{\mathbf{A}(t_2 - t_0)} = \phi(t_2 - t_0)\end{aligned} \tag{10-70}$$

This property of the state-transition matrix is important because it implies that a state-transition process can be divided into a number of sequential transitions. Fig. 10-9 illustrates that the transition from $t = t_0$ to $t = t_2$ is equal to the transition from t_0 to t_1 and then from t_1 to t_2. In general, of course, the state-transition process can be divided into any number of parts.

4. $[\phi(t)]^k = \phi(kt)$ for k = positive integer (10-71)

Proof:

$$\begin{aligned}[\phi(t)]^k &= e^{\mathbf{A}t}e^{\mathbf{A}t}\ldots e^{\mathbf{A}t} \\ &= e^{k\mathbf{A}t} = \phi(kt)\end{aligned} \tag{10-72}$$

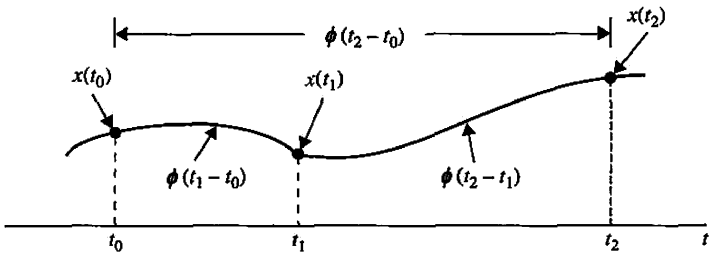


Figure 10-9 Property of the state-transition matrix.

10-5 STATE-TRANSITION EQUATION

The **state-transition equation** *is defined as the solution of the linear homogeneous state equation.* The linear time-invariant state equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{E}\mathbf{w}(t) \tag{10-73}$$

can be solved using either the classical method of solving linear differential equations or the Laplace transform method. The Laplace transform solution is presented in the following equations.

Taking the Laplace transform on both sides of Eq. (10-73), we have

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s) \tag{10-74}$$

where $\mathbf{x}(0)$ denotes the initial-state vector evaluated at $t = 0$. Solving for $\mathbf{X}(s)$ in Eq. (10-74) yields

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)] \tag{10-75}$$

The state-transition equation of Eq. (10-73) is obtained by taking the inverse Laplace transform on both sides of Eq. (10-75):

$$\begin{aligned}\mathbf{x}(t) &= \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A})^{-1}\right]\mathbf{x}(0) + \mathcal{L}^{-1}\left\{(s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)]\right\} \\ &= \boldsymbol{\phi}(t)\mathbf{x}(0) + \int_0^t \boldsymbol{\phi}(t-\tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau \quad t \geq 0\end{aligned} \tag{10-76}$$

The state-transition equation in Eq. (10-76) is useful only when the initial time is defined to be at $t = 0$. In the study of control systems, especially discrete-data control systems, it is often desirable to break up a state-transition process into a sequence of transitions, so a more flexible initial time must be chosen. Let the initial time be represented by t_0 and the corresponding initial state by $\mathbf{x}(t_0)$, and assume that the input $\mathbf{u}(t)$ and the disturbance $\mathbf{w}(t)$ are applied at $t \geq 0$. We start with Eq. (10-76) by setting $t = t_0$, and solving for $\mathbf{x}(0)$, we get

$$\mathbf{x}(0) = \boldsymbol{\phi}(-t_0)\mathbf{x}(t_0) - \boldsymbol{\phi}(-t_0)\int_0^{t_0} \boldsymbol{\phi}(t_0 - \tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau \tag{10-77}$$

where the property on $\boldsymbol{\phi}(t)$ of Eq. (10-64) has been applied.

Substituting Eq. (10-77) into Eq. (10-76) yields

$$\begin{aligned}\mathbf{x}(t) = \boldsymbol{\phi}(t)\boldsymbol{\phi}(-t_0)\mathbf{x}(t_0) - \boldsymbol{\phi}(t)\boldsymbol{\phi}(-t_0)\int_0^{t_0} \boldsymbol{\phi}(t_0 - \tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau \\ + \int_0^t \boldsymbol{\phi}(t-\tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau\end{aligned} \tag{10-78}$$

Now by using the property of Eq. (10-69) and combining the last two integrals, Eq. (10-78) becomes

$$\mathbf{x}(t) = \boldsymbol{\phi}(t - t_0)\mathbf{x}(t_0) + \int_0^t \boldsymbol{\phi}(t-\tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau \quad t \geq t_0 \tag{10-79}$$

It is apparent that Eq. (10-79) reverts to Eq. (10-77) when $t_0 = 0$.

Once the state-transition equation is determined, the output vector can be expressed as a function of the initial state and the input vector simply by substituting $\mathbf{x}(t)$ from Eq. (10-79) into Eq. (10-45). Thus, the output vector is

$$\mathbf{y}(t) = \mathbf{C}\boldsymbol{\phi}(t-t_0)\mathbf{x}(t_0) + \int_{t_0}^{t} \mathbf{C}\boldsymbol{\phi}(t-\tau)[\mathbf{B}\mathbf{u}(\tau) + \mathbf{E}\mathbf{w}(\tau)]d\tau + \mathbf{D}\mathbf{u}(t) + \mathbf{H}\mathbf{w}(t) \quad t \geq t_0 \tag{10-80}$$

The following example illustrates the determination of the state-transition matrix and equation.

► **EXAMPLE 10-5-1** Consider the state equation

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \tag{10-81}$$

The problem is to determine the state-transition matrix $\boldsymbol{\phi}(t)$ and the state vector $\mathbf{x}(t)$ for $t \geq 0$ when the input is $u(t) = 1$ for $t \geq 0$. The coefficient matrices are identified to be

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{E} = 0 \tag{10-82}$$

Therefore,

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \tag{10-83}$$

The inverse matrix of $(s\mathbf{I} - \mathbf{A})$ is

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \tag{10-84}$$

The state-transition matrix of $\mathbf{A}$ is found by taking the inverse Laplace transform of Eq. (10-84). Thus,

$$\boldsymbol{\phi}(t) = \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A})^{-1}\right] = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \tag{10-85}$$

The state-transition equation for $t \geq 0$ is obtained by substituting Eq. (10-85), $\mathbf{B}$, and $u(t)$ into Eq. (10-76). We have

$$\mathbf{x}(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \mathbf{x}(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau \tag{10-86}$$

or

$$\mathbf{x}(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \mathbf{x}(0) + \begin{bmatrix} 0.5 - e^{-t} + 0.5e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix} \quad t \geq 0 \tag{10-87}$$

As an alternative, the second term of the state-transition equation can be obtained by taking the inverse Laplace transform of $(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$. Thus, we have

$$\begin{aligned} \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A})^{-1}\right]\mathbf{B}\mathbf{U}(s) &= \mathcal{L}^{-1}\left(\frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{s}\right) \\ &= \mathcal{L}^{-1}\left(\frac{1}{s^2 + 3s + 2} \begin{bmatrix} \frac{1}{s} \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0.5 - e^{-t} + 0.5e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix} \quad t \geq 0 \end{aligned} \tag{10-88}$$

◄

10-5-1 State-Transition Equation Determined from the State Diagram

Eqs. (10-75) and (10-76) show that the Laplace transform method of solving the state equations requires obtaining the inverse of matrix $(s\mathbf{I} - \mathbf{A})$. We shall now show that the state diagram described earlier in Section 10-2-3 and the SFG gain formula (Chapter 3) can be used to solve for the state-transition equation in the Laplace domain of Eq. (10-75). Let the initial time be t_0; then Eq. (10-75) is rewritten as

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(t_0) + (s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{BU}(s) + \mathbf{EW}(s)] \quad t \geq t_0 \tag{10-89}$$

The last equation can be written directly from the state diagram using the gain formula, with $X_i(s)$, $i = 1, 2, \ldots, n$ *as the output nodes.* The following example illustrates the state-diagram method of finding the state-transition equations for the system described in Example 10-2-1.

▶ EXAMPLE 10-5-2 The state diagram for the system described by Eq. (10-81) is shown in Fig. 10-10 with t_0 as the initial time. The outputs of the integrators are assigned as state variables. Applying the gain formula to the state diagram in Fig. 10-10, with $X_1(s)$ and $X_2(s)$ as output nodes and $x_1(t_0)$, $x_2(t_0)$, and $U(s)$ as input nodes, we have

$$X_1(s) = \frac{s^{-1}(1 + 3s^{-1})}{\Delta}x_1(t_0) + \frac{s^{-2}}{\Delta}x_2(t_0) + \frac{s^{-2}}{\Delta}U(s) \tag{10-90}$$

$$X_2(s) = \frac{-2s^{-2}}{\Delta}x_1(t_0) + \frac{s^{-1}}{\Delta}x_2(t_0) + \frac{s^{-1}}{\Delta}U(s) \tag{10-91}$$

where

$$\Delta = 1 + 3s^{-1} + 2s^{-2} \tag{10-92}$$

After simplification, Eqs. (10-90) and (10-91) are presented in vector-matrix form:

$$\begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \frac{1}{(s+1)(s+2)}\begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}\begin{bmatrix} x_1(t_0) \\ x_2(t_0) \end{bmatrix} + \frac{1}{(s+1)(s+2)}\begin{bmatrix} 1 \\ s \end{bmatrix}U(s) \tag{10-93}$$

The state-transition equation for $t \geq t_0$ is obtained by taking the inverse Laplace transform on both sides of Eq. (10-93).

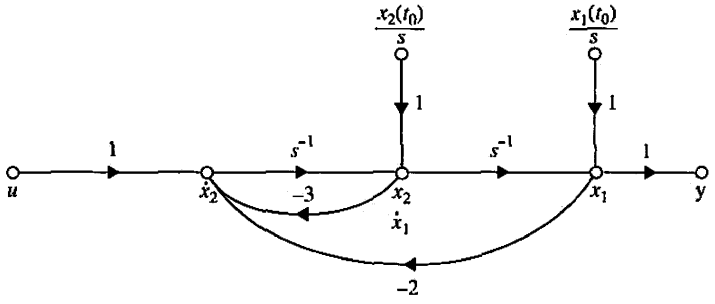


Figure 10-10 State diagram for Eq. (10-81).

Consider that the input $u(t)$ is a unit-step function applied at $t = t_0$. Then the following inverse Laplace transform relationships are identified:

$$\mathcal{L}^{-1}\left(\frac{1}{s}\right) = u_s(t - t_0) \quad t \geq t_0 \tag{10-94}$$

$$\mathcal{L}^{-1}\left(\frac{1}{s+a}\right) = e^{-a(t-t_0)}u_s(t - t_0) \quad t \geq t_0 \tag{10-95}$$

Because the initial time is defined to be t_0, the Laplace transform expressions here do not have the delay factor $e^{-t_0 s}$. The inverse Laplace transform of Eq. (10-93) is

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-(t-t_0)} - e^{-2(t-t_0)} & e^{-(t-t_0)} - e^{-2(t-t_0)} \\ -2e^{-(t-t_0)} + 2e^{-2(t-t_0)} & -e^{(t-t_0)} + 2e^{-2(t-t_0)} \end{bmatrix} \begin{bmatrix} x_1(t_0) \\ x_2(t_0) \end{bmatrix} + \begin{bmatrix} 0.5u_s(t - t_0) - e^{-(t-t_0)} + 0.5e^{-2(t-t_0)} \\ e^{-(t-t_0)} - e^{-2(t-t_0)} \end{bmatrix} t \geq t_0 \tag{10-96}$$

The reader should compare this result with that in Eq. (10-87), which is obtained for $t \geq 0$. ◀

EXAMPLE 10-5-3 In this example, we illustrate the utilization of the state-transition method to a system with input discontinuity. An *RL* network is shown in Fig. 10-11. The history of the network is completely specified by the initial current of the inductance, $i(0)$ at $t = 0$. At time $t = 0$, the voltage $e_{in}(t)$ with the profile shown in Fig. 10-12 is applied to the network. The state equation of the network for $t \geq 0$ is

$$\frac{di(t)}{dt} = -\frac{R}{L}i(t) + \frac{1}{L}e_{in}(t) \tag{10-97}$$

Comparing the last equation with Eq. (10-44), the scalar coefficients of the state equation are identified to be

$$A = -\frac{R}{L} \quad B = \frac{1}{L} \quad E = 0 \tag{10-98}$$

The state-transition matrix is

$$\phi(t) = e^{-At} = e^{-Rt/L} \tag{10-99}$$

The conventional approach of solving for $i(t)$ for $t \geq 0$ is to express the input voltage as

$$e(t) = E_{in}u_s(t) + E_{in}u_s(t - t_1) \tag{10-100}$$

where $u_s(t)$ is the unit-step function. The Laplace transform of $e(t)$ is

$$E_{in}(s) = \frac{E_{in}}{s}(1 + e^{-t_1 s}) \tag{10-101}$$

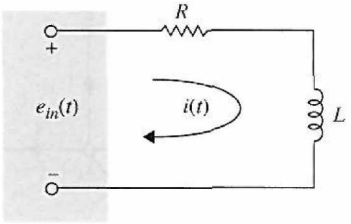


Figure 10-11 *RL* network.

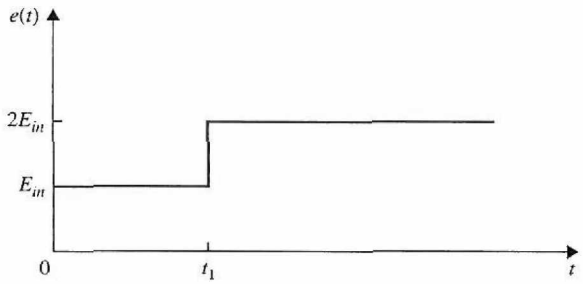


Figure 10-12 Input voltage waveform for the network in Fig. 10-3.

Then

$$(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{BU}(s) = \frac{E_{in}}{Ls(s + R/L)}(1 + e^{-t_1 s}) \tag{10-102}$$

By substituting Eq. (10-102) into Eq. (10-76), the state-transition equation, the current for $t \geq 0$ is obtained:

$$i(t) = e^{-Rt/L}i(0)u_s(t) + \frac{E_{in}}{R}\left(1 - e^{-Rt/L}\right)u_s(t) + \frac{E_{in}}{R}\left(1 - e^{-R(t-t_1)/L}\right)u_s(t - t_1) \tag{10-103}$$

Using the state-transition approach, we can divide the transition period into two parts: $t = 0$ to $t = t_1$, and $t = t_1$ to $t = \infty$. First, for the time interval $0 \leq t \leq t_1$, the input is

$$e(t) = E_{in}u_s(t) \quad 0 \leq t < t_1 \tag{10-104}$$

Then

$$(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{BU}(s) = \frac{E_{in}}{Ls(s + R/L)} = \frac{E_{in}}{Rs[1 + (L/R)s]} \tag{10-105}$$

Thus, the state-transition equation for the time interval $0 \leq t \leq t_1$ is

$$i(t) = \left[e^{-Rt/L}i(0) + \frac{E_{in}}{R}\left(1 - e^{-Rt/L}\right)\right]u_s(t) \tag{10-106}$$

Substituting $t = t_1$ into Eq. (10-106), we get

$$i(t_1) = e^{-Rt_1/L}i(0) + \frac{E_{in}}{R}\left(1 - e^{-Rt_1/L}\right) \tag{10-107}$$

The value of $i(t)$ at $t = t_1$ is now used as the initial state for the next transition period of $t_1 \leq t < \infty$. The amplitude of the input for the interval is $2E_{in}$. The state-transition equation for the second transition period is

$$i(t) = e^{-R(t-t_1)/L}i(t_1) + \frac{2E_{in}}{R}\left(1 - e^{-R(t-t_1)/L}\right) \quad t \geq t_1 \tag{10-108}$$

where $i(t_1)$ is given by Eq. (10-107).

This example illustrates two possible ways of solving a state-transition problem. In the first approach, the transition is treated as one continuous process, whereas in the second, the transition period is divided into parts over which the input can be more easily presented. Although the first approach requires only one operation, the second method yields relatively simple results to the state-transition equation, and it often presents computational advantages. Notice that, in the second method, the state at $t = t_1$ is used as the initial state for the next transition period, which begins at t_1.

10-6 RELATIONSHIP BETWEEN STATE EQUATIONS AND HIGH-ORDER DIFFERENTIAL EQUATIONS

In the preceding sections, we defined the state equations and their solutions for linear time-invariant systems. Although it is usually possible to write the state equations directly from the schematic diagram of a system, in practice the system may have been described by a high-order differential equation or transfer function. It becomes necessary to investigate how state equations can be written directly from the high-order differential equation or the transfer function. In Chapter 2 we illustrated how the state variables of an nth-order differential equation in Eq. (2-97) are intuitively defined, as shown in Eq. (2-105). The results are the n state equations in Eq. (2-106).

The state equations are written in vector-matrix form:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \tag{10-109}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \quad (n \times n) \tag{10-110}$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad (n \times 1) \tag{10-111}$$

Notice that the last row of **A** contains the negative values of the coefficients of the homogeneous part of the differential equation in ascending order, except for the coefficient of the highest-order term, which is unity. **B** is a column matrix with the last row equal to one, and the rest of the elements are all zeros. The state equations in Eq. (10-109) with **A** and **B** given in Eqs. (10-110) and (10-111) are known as the **phase-variable canonical form (PVCF)**, or the **controllability canonical form (CCF)**.

The output equation of the system is written

$$y(t) = \mathbf{C}\mathbf{x}(t) = x_1(t) \tag{10-112}$$

where

$$\mathbf{C} = [1 \quad 0 \quad 0 \quad \cdots \quad 0] \tag{10-113}$$

We have shown earlier that the state variables of a given system are not unique. In general, we seek the most convenient way of assigning the state variables as long as the definition of state variables is satisfied. In Section 10-9 we shall show that, by first writing the transfer function and then drawing the state diagram of the system by decomposition of the transfer function, the state variables and state equations of any system can be found very easily.

EXAMPLE 10-6-1 Consider the differential equation

$$\frac{d^3y(t)}{dt^3} + 5\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + 2y(t) = u(t) \tag{10-114}$$

Rearranging the last equation so that the highest-order derivative term is set equal to the rest of the terms, we have

$$\frac{d^3y(t)}{dt^3} = -5\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) + u(t) \tag{10-115}$$

The state variables are defined as

$$\begin{aligned} x_1(t) &= y(t) \\ x_2(t) &= \frac{dy(t)}{dt} \\ x_3(t) &= \frac{d^2y(t)}{dt^2} \end{aligned} \tag{10-116}$$

Then the state equations are represented by the vector-matrix equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \tag{10-117}$$

where $\mathbf{x}(t)$ is the 2×1 state vector, $u(t)$ is the scalar input, and

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -1 & -5 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{10-118}$$

The output equation is

$$y(t) = x_1(t) = [1 \quad 0]\mathbf{x}(t) \tag{10-119}$$

10-7 RELATIONSHIP BETWEEN STATE EQUATIONS AND TRANSFER FUNCTIONS

We have presented the methods of modeling a linear time-invariant system by transfer functions and dynamic equations. We now investigate the relationship between these two representations.

Consider a linear time-invariant system described by the following dynamic equations:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{E}\mathbf{w}(t) \tag{10-120}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) + \mathbf{H}\mathbf{w}(t) \tag{10-121}$$

where

$\mathbf{x}(t) = n \times 1$ state vector
$\mathbf{u}(t) = p \times 1$ input vector
$\mathbf{y}(t) = q \times 1$ output vector
$\mathbf{w}(t) = v \times 1$ disturbance vector

and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{H}$ are coefficient matrices of appropriate dimensions.

Taking the Laplace transform on both sides of Eq. (10-120) and solving for $\mathbf{X}(s)$, we have

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)] \tag{10-122}$$

The Laplace transform of Eq. (10-121) is

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s) + \mathbf{H}\mathbf{W}(s) \tag{10-123}$$

Substituting Eq. (10-122) into Eq. (10-123), we have

$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + \mathbf{C}(s\mathbf{I} - \mathbf{A})[\mathbf{B}\mathbf{U}(s) + \mathbf{E}\mathbf{W}(s)] + \mathbf{D}\mathbf{U}(s) + \mathbf{H}\mathbf{W}(s) \tag{10-124}$$

Because the definition of a transfer function requires that the initial conditions be set to zero, $\mathbf{x}(0) = \mathbf{0}$; thus, Eq. (10-124) becomes

$$\mathbf{Y}(s) = \left[\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}\right]\mathbf{U}(s) + [\mathbf{C}(s\mathbf{I} - \mathbf{A})\mathbf{E} + \mathbf{H}]\mathbf{W}(s) \tag{10-125}$$

Let us define

$$\mathbf{G}_u(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \tag{10-126}$$

$$\mathbf{G}_w(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{E} + \mathbf{H} \tag{10-127}$$

where $\mathbf{G}_u(s)$ is a $q \times p$ transfer-function matrix between $\mathbf{u}(t)$ and $\mathbf{y}(t)$ when $\mathbf{w}(t) = \mathbf{0}$, and $\mathbf{G}_w(s)$ is a $q \times v$ transfer-function matrix between $\mathbf{w}(t)$ and $\mathbf{y}(t)$ when $\mathbf{u}(t) = 0$.

Then, Eq. (10-125) becomes

$$\mathbf{Y}(s) = \mathbf{G}_u(s)\mathbf{U}(s) + \mathbf{G}_w(s)\mathbf{W}(s) \tag{10-128}$$

EXAMPLE 10-7-1 Consider that a multivariable system is described by the differential equations

$$\frac{d^2y_1(t)}{dt^2} + 4\frac{dy_1(t)}{dt} - 3y_2(t) = u_1(t) + 2w(t) \tag{10-129}$$

$$\frac{dy_1(t)}{dt} + \frac{dy_2(t)}{dt} + y_1(t) + 2y_2(t) = u_2(t) \tag{10-130}$$

The state variables of the system are assigned as:

$$\begin{aligned} x_1(t) &= y_1(t) \\ x_2(t) &= \frac{dy_1(t)}{dt} \\ x_3(t) &= y_2(t) \end{aligned} \tag{10-131}$$

These state variables are defined by mere inspection of the two differential equations, because no particular reasons for the definitions are given other than that these are the most convenient. Now equating the first term of each of the equations of Eqs. (10-129) and (10-130) to the rest of the terms and using the state-variable relations of Eq. (10-131), we arrive at the following state equations and output equations in vector-matrix form:

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \\ \dfrac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -4 & 3 \\ -1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} w(t) \tag{10-132}$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \mathbf{C}\mathbf{x}(t) \tag{10-133}$$

To determine the transfer-function matrix of the system using the state-variable formulation, we substitute the **A**, **B**, **C**, **D**, and **E** matrices into Eq. (10-125). First, we form the matrix $(s\mathbf{I} - \mathbf{A})$:

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & -1 & 0 \\ 0 & s+4 & -3 \\ 1 & 1 & s+2 \end{bmatrix} \tag{10-134}$$

The determinant of $(s\mathbf{I} - \mathbf{A})$ is

$$|s\mathbf{I} - \mathbf{A}| = s^3 + 6s^2 + 11s + 3 \tag{10-135}$$

Thus,

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{|s\mathbf{I} - \mathbf{A}|} \begin{bmatrix} s^2 + 6s + 11 & s+2 & 3 \\ -3 & s(s+2) & 3s \\ -(s+4) & -(s+1) & s(s+4) \end{bmatrix} \tag{10-136}$$

The transfer-function matrix between $\mathbf{u}(t)$ and $\mathbf{y}(t)$ is

$$\mathbf{G}_u(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{1}{s^3 + 6s^2 + 11s + 3}\begin{bmatrix} s+2 & 3 \\ -(s+1) & s(s+4) \end{bmatrix} \tag{10-137}$$

and that between $\mathbf{w}(t)$ and $\mathbf{y}(t)$ is

$$\mathbf{G}_w(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{E} = \frac{1}{s^3 + 6s + 11s + 3}\begin{bmatrix} 2(s+2) \\ -2(s+1) \end{bmatrix} \tag{10-138}$$

Using the conventional approach, we take the Laplace transform on both sides of Eqs. (10-129) and (10-130) and assume zero initial conditions. The resulting transformed equations are written in vector-matrix form as

$$\begin{bmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{bmatrix}\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} W(s) \tag{10-139}$$

Solving for $\mathbf{Y}(s)$ from Eq. (10-139), we obtain

$$\mathbf{Y}(s) = \mathbf{G}_u(s)\mathbf{U}(s) + \mathbf{G}_w(s)\mathbf{W}(s) \tag{10-140}$$

where

$$\mathbf{G}_u(s) = \begin{bmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{bmatrix}^{-1} \tag{10-141}$$

$$\mathbf{G}_w(s) = \begin{bmatrix} s(s+4) & -3 \\ s+1 & s+2 \end{bmatrix}^{-1}\begin{bmatrix} 2 \\ 0 \end{bmatrix} \tag{10-142}$$

which will give the same results as in Eqs. (10-137) and (10-138), respectively, when the matrix inverses are carried out.

10-8 CHARACTERISTIC EQUATIONS, EIGENVALUES, AND EIGENVECTORS

Characteristic equations play an important role in the study of linear systems. They can be defined with respect to differential equations, transfer functions, or state equations.

10-8-1 Characteristic Equation from a Differential Equation

Consider that a linear time-invariant system is described by the differential equation

$$\begin{aligned} &\frac{d^n y(t)}{dt^n} + a_{n-1}\frac{d^{n-1}y(t)}{dt^{n-1}} + \cdots + a_1\frac{dy(t)}{dt} + a_0 y(t) \\ &\quad = b_m\frac{d^m u(t)}{dt^m} + b_{m-1}\frac{d^{m-1}u(t)}{dt^{m-1}} + \cdots + b_1\frac{du(t)}{dt} + b_0 u(t) \end{aligned} \tag{10-143}$$

where $n > m$. By defining the operator s as

$$s^k = \frac{d^k}{dt^k} \qquad k = 1, 2, \ldots, n \tag{10-144}$$

Eq. (10-143) is written

$$(s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0)y(t) = (b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0)u(t) \tag{10-145}$$

The **characteristic equation** of the system is defined as

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0 \tag{10-146}$$

which is obtained by setting the homogeneous part of Eq. (10-145) to zero.

EXAMPLE 10-8-1 Consider the differential equation in Eq. (10-114). The characteristic equation is obtained by inspection,

$$s^3 + 5s^2 + s + 2 = 0 \tag{10-147}$$

10-8-2 Characteristic Equation from a Transfer Function

The transfer function of the system described by Eq. (10-143) is

$$G(s) = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0} \tag{10-148}$$

The characteristic equation is obtained by equating the denominator polynomial of the transfer function to zero.

EXAMPLE 10-8-2 The transfer function of the system described by the differential equation in Eq. (10-114) is

$$\frac{Y(s)}{U(s)} = \frac{1}{s^3 + 5s^2 + s + 2} \tag{10-149}$$

The same characteristic equation as in Eq. (10-147) is obtained by setting the denominator polynomial of Eq. (10-149) to zero.

10-8-3 Characteristic Equation from State Equations

From the state-variable approach, we can write Eq. (10-126) as

$$\begin{aligned} \mathbf{G}_u(s) &= \mathbf{C}\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{(s\mathbf{I} - \mathbf{A})}\mathbf{B} + \mathbf{D} \\ &= \frac{\mathbf{C}[\text{adj}(s\mathbf{I} - \mathbf{A})]\mathbf{B} + |s\mathbf{I} - \mathbf{A}|\mathbf{D}}{|s\mathbf{I} - \mathbf{A}|} \end{aligned} \tag{10-150}$$

Setting the denominator of the transfer-function matrix $\mathbf{G}_u(s)$ to zero, we get the characteristic equation

$$|s\mathbf{I} - \mathbf{A}| = 0 \tag{10-151}$$

which is an alternative form of the characteristic equation but should lead to the same equation as in Eq. (10-146). *An important property of the characteristic equation is that, if the coefficients of* $\mathbf{A}$ *are real, then the coefficients of* $|s\mathbf{I} - \mathbf{A}|$ *are also real.*

EXAMPLE 10-8-3 The matrix $\mathbf{A}$ for the state equations of the differential equation in Eq. (10-114) is given in Eq. (10-128). The characteristic equation of $\mathbf{A}$ is

$$|s\mathbf{I} - \mathbf{A}| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 2 & 1 & s+5 \end{vmatrix} = s^3 + 5s^2 + s + 2 - 0 \tag{10-152}$$

10-8-4 Eigenvalues

The roots of the characteristic equation are often referred to as the eigenvalues of the matrix **A**.

Some of the important properties of eigenvalues are given as follows.

1. If the coefficients of **A** are all real, then its eigenvalues are either real or in complex-conjugate pairs.
2. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of **A**, then

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^{n} \lambda_i \tag{10-153}$$

That is, the trace of **A** is the sum of all the eigenvalues of **A**.

3. If λ_i, $i = 1, 2, \ldots, n$, is an eigenvalue of **A**, then it is an eigenvalue of $\mathbf{A}'$.
4. If **A** is nonsingular, with eigenvalues λ_i, $i = 1, 2, \ldots, n$, then $1/\lambda_i$, $i = 1, 2, \ldots, n$, are the eigenvalues of $\mathbf{A}^{-1}$.

EXAMPLE 10-8-4 The eigenvalues or the roots of the characteristic equation of the matrix **A** in Eq. (10-118) are obtained by solving for the roots of Eq. (10-152). The results are

$$s = -0.06047 + j0.63738 \quad s = -0.06047 - j0.63738 \quad s = -4.87906 \tag{10-154}$$

10-8-5 Eigenvectors

Eigenvectors are useful in modern control methods, one of which is the similarity transformation, which will be discussed in a later section.

Any nonzero vector $\mathbf{p}_i$ *that satisfies the matrix equation*

$$(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{p}_i = \mathbf{0} \tag{10-155}$$

where λ_i, $i = 1, 2, \ldots, n$, *denotes the ith eigenvalue of* **A**, *called the* eigenvector *of* **A** *associated with the eigenvalue* λ_i. If **A** has distinct eigenvalues, the eigenvectors can be solved directly from Eq. (10-155).

EXAMPLE 10-8-5 Consider that the state equation of Eq. (10-44) has the coefficient matrices

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{E} = \mathbf{0} \tag{10-156}$$

The characteristic equation of **A** is

$$|s\mathbf{I} - \mathbf{A}| = s^2 - 1 \tag{10-157}$$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = -1$. Let the eigenvectors be written as

$$\mathbf{p}_1 = \begin{bmatrix} p_{11} \\ p_{21} \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} p_{12} \\ p_{22} \end{bmatrix} \tag{10-158}$$

Substituting $\lambda_1 = 1$ and $\mathbf{p}_1$ into Eq. (10-155), we get

$$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{10-159}$$

Thus, $p_{21} = 0$, and p_{11} is arbitrary, which in this case can be set equal to 1.

Similarly, for $\lambda_2 = -1$, Eq. (10-155) becomes

$$\begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_{12} \\ p_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{10-160}$$

which leads to

$$-2p_{12} + p_{22} = 0 \tag{10-161}$$

The last equation has two unknowns, which means that one can be set arbitrarily. Let $p_{12} = 1$, then $p_{22} = 2$. The eigenvectors are

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \tag{10-162}$$

10-8-6 Generalized Eigenvectors

It should be pointed out that if **A** has multiple-order eigenvalues and is nonsymmetric, not all the **eigenvectors** can be found using Eq. (10-155). Let us assume that there are $q(< n)$ distinct eigenvalues among the n eigenvalues of **A**. The eigenvectors that correspond to the q distinct eigenvalues can be determined in the usual manner from

$$(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{p}_i = 0 \tag{10-163}$$

where λ_i denotes the ith distinct eigenvalue, $i = 1, 2, \ldots, q$. Among the remaining high-order eigenvalues, let λ_j be of the mth order ($m \leq n - q$). The corresponding eigenvectors are called the **generalized eigenvectors** and can be determined from the following m vector equations:

$$\begin{aligned} (\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+1} &= \mathbf{0} \\ (\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+2} &= -\mathbf{p}_{n-q+1} \\ (\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+3} &= -\mathbf{p}_{n-q+2} \\ &\vdots \\ (\lambda_j \mathbf{I} - \mathbf{A})\mathbf{p}_{n-q+m} &= -\mathbf{p}_{n-q+m-1} \end{aligned} \tag{10-164}$$

EXAMPLE 10-8-6 Given the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 6 & -5 \\ 1 & 0 & 2 \\ 3 & 2 & 4 \end{bmatrix} \tag{10-165}$$

The eigenvalues of **A** are $\lambda_1 = 2$, $\lambda_2 = \lambda_3 = 1$. Thus, **A** is a second-order eigenvalue at 1. The eigenvector that is associated with $\lambda_1 = 2$ is determined using Eq. (10-163). Thus,

$$(\lambda_1 \mathbf{I} - \mathbf{A})\mathbf{p}_1 = \begin{bmatrix} 2 & -6 & 5 \\ -1 & 2 & -2 \\ -3 & -2 & -2 \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{21} \\ p_{31} \end{bmatrix} = \mathbf{0} \tag{10-166}$$

Because there are only two independent equations in Eq. (10-166), we arbitrarily set $p_{11} = 2$, and we have $p_{21} = -1$ and $p_{31} = -2$. Thus,

$$\mathbf{p}_1 = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix} \tag{10-167}$$

For the generalized eigenvectors that are associated with the second-order eigenvalues, we substitute $\lambda_2 = 1$ into the first equation of Eq. (10-164). We have

$$(\lambda_2\mathbf{I} - \mathbf{A})\mathbf{p}_2 = \begin{bmatrix} 1 & -6 & 5 \\ -1 & 1 & -2 \\ -3 & -2 & -3 \end{bmatrix} \begin{bmatrix} p_{12} \\ p_{22} \\ p_{32} \end{bmatrix} = \mathbf{0} \tag{10-168}$$

Setting $p_{12} = 1$ arbitrarily, we have $p_{22} = -\frac{3}{7}$ and $p_{32} = -\frac{5}{7}$. Thus,

$$\mathbf{p}_2 = \begin{bmatrix} 1 \\ -\frac{3}{7} \\ -\frac{5}{7} \end{bmatrix} \tag{10-169}$$

Substituting $\lambda_3 = 1$ into the second equation of Eq. (10-164), we have

$$(\lambda_3\mathbf{I} - \mathbf{A})\mathbf{p}_3 = \begin{bmatrix} 1 & -6 & -5 \\ -1 & 1 & -2 \\ -3 & -2 & -3 \end{bmatrix} \begin{bmatrix} p_{13} \\ p_{23} \\ p_{33} \end{bmatrix} = -\mathbf{p}_2 = \begin{bmatrix} -1 \\ \frac{3}{7} \\ \frac{5}{7} \end{bmatrix} \tag{10-170}$$

Setting p_{13} arbitrarily to 1, we have the generalized eigenvector

$$\mathbf{p}_3 = \begin{bmatrix} 1 \\ -\frac{22}{49} \\ \frac{46}{49} \end{bmatrix} \tag{10-171}$$

10-9 SIMILARITY TRANSFORMATION

The dynamic equations of a single-input, single-output (SISO) system are

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \tag{10-172}$$

$$y(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t) \tag{10-173}$$

where $\mathbf{x}(t)$ is the $n \times 1$ state vector, and $u(t)$ and $y(t)$ are the scalar input and output, respectively. When carrying out analysis and design in the state domain, it is often advantageous to transform these equations into particular forms. For example, as we will show later, the controllability canonical form (CCF) has many interesting properties that make it convenient for controllability tests and state-feedback design.

Let us consider that the dynamic equations of Eqs. (10-172) and (10-173) are transformed into another set of equations of the same dimension by the following transformation:

$$\mathbf{x}(t) = \mathbf{P}\bar{\mathbf{x}}(t) \tag{10-174}$$

where $\mathbf{P}$ is an $n \times n$ nonsingular matrix, so

$$\bar{\mathbf{x}}(t) = \mathbf{P}^{-1}\mathbf{x}(t) \tag{10-175}$$

The transformed dynamic equations are written

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \bar{\mathbf{A}}\bar{\mathbf{x}}(t) + \bar{\mathbf{B}}u(t) \tag{10-176}$$

$$\bar{y}(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t) + \bar{\mathbf{D}}u(t) \tag{10-177}$$

Taking the derivative on both sides of Eq. (10-175) with respect to t, we have

$$\begin{aligned}\frac{d\bar{\mathbf{x}}(t)}{dt} &= \mathbf{P}^{-1}\frac{d\mathbf{x}(t)}{dt} = \mathbf{P}^{-1}\mathbf{A}\mathbf{x}(t) + \mathbf{P}^{-1}\mathbf{B}u(t) \\ &= \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\bar{\mathbf{x}}(t) + \mathbf{P}^{-1}\mathbf{B}u(t)\end{aligned} \tag{10-178}$$

Comparing Eq. (10-178) with Eq. (10-176), we get

$$\bar{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \tag{10-179}$$

and

$$\bar{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B} \tag{10-180}$$

Using Eq. (10-174), Eq. (10-177) is written

$$\bar{y}(t) = \mathbf{C}\mathbf{P}\mathbf{x}(t) + \bar{\mathbf{D}}u(t) \tag{10-181}$$

Comparing Eq. (10-181) with Eq. (10-173), we see that

$$\bar{\mathbf{C}} = \mathbf{C}\mathbf{P} \quad \bar{\mathbf{D}} = \mathbf{D} \tag{10-182}$$

The transformation just described is called a **similarity transformation**, because in the transformed system such properties as the characteristic equation, eigenvectors, eigenvalues, and transfer function are all preserved by the transformation. We shall describe the controllability canonical form (CCF), the observability canonical form (OCF), and the diagonal canonical form (DCF) transformations in the following sections. The transformation equations are given without proofs.

10-9-1 Invariance Properties of the Similarity Transformations

One of the important properties of the similarity transformations is that the characteristic equation, eigenvalues, eigenvectors, and transfer functions are invariant under the transformations.

Characteristic Equations, Eigenvalues, and Eigenvectors

The characteristic equation of the system described by Eq. (10-176) is $|s\mathbf{I} - \bar{\mathbf{A}}| = 0$ and is written

$$|s\mathbf{I} - \bar{\mathbf{A}}| = |s\mathbf{I} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}| = |s\mathbf{P}^{-1}\mathbf{P} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}| \tag{10-183}$$

Because the determinant of a product matrix is equal to the product of the determinants of the matrices, the last equation becomes

$$|s\mathbf{I} - \overline{\mathbf{A}}| = |\mathbf{P}^{-1}||s\mathbf{I} - \mathbf{A}||\mathbf{P}| = |s\mathbf{I} - \mathbf{A}| \tag{10-184}$$

Thus, the characteristic equation is preserved, which naturally leads to the same eigenvalues and eigenvectors.

Transfer-Function Matrix

From Eq. (10-126), the transfer-function matrix of the system of Eqs. (10-176) and (10-177) is

$$\begin{aligned}\overline{\mathbf{G}}(s) &= \overline{\mathbf{C}}(s\mathbf{I} - \overline{\mathbf{A}})\overline{\mathbf{B}} + \overline{\mathbf{D}} \\ &= \mathbf{CP}(s\mathbf{I} - \mathbf{P}^{-1}\mathbf{AP})\mathbf{P}^{-1}\mathbf{B} + \mathbf{D}\end{aligned} \tag{10-185}$$

which is simplified to

$$\overline{\mathbf{G}}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})\mathbf{B} + \mathbf{D} = \mathbf{G}(s) \tag{10-186}$$

10-9-2 Controllability Canonical Form (CCF)

Consider the dynamic equations given in Eqs. (10-172) and (10-173). The characteristic equation of $\mathbf{A}$ is

$$|s\mathbf{I} - \mathbf{A}| = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0 \tag{10-187}$$

The dynamic equations in Eqs. (10-172) and (10-173) are transformed into CCF of the form of Eqs. (10-176) and (10-177) by the transformation of Eq. (10-174), with

$$\mathbf{P} = \mathbf{SM} \tag{10-188}$$

where

$$\mathbf{S} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \ldots \mathbf{A}^{n-1}\mathbf{B}] \tag{10-189}$$

and

$$\mathbf{M} = \begin{bmatrix} a_1 & a_2 & \cdots & a_{n-1} & 1 \\ a_2 & a_3 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n-1} & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix} \tag{10-190}$$

Then,

$$\overline{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \tag{10-191}$$

$$\overline{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{10-192}$$

The matrices $\overline{\mathbf{C}}$ and $\overline{\mathbf{D}}$ are given by Eq. (10-182) and do not follow any particular pattern. The CCF transformation requires that $\mathbf{P}^{-1}$ exists, which implies that the matrix $\mathbf{S}$ must have an inverse, because the inverse of $\mathbf{M}$ always exists because its determinant is $(-1)^{n-1}$, which is nonzero. The $n \times n$ matrix $\mathbf{S}$ in Eq. (10-189) is later defined as the **controllability matrix**.

EXAMPLE 10-9-1 Consider the coefficient matrices of the state equations in Eq. (10-172):

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \tag{10-193}$$

The state equations are to be transformed to CCF.

The characteristic equation of $\mathbf{A}$ is

$$|s\mathbf{I} - \mathbf{A}| = \begin{vmatrix} s-1 & -2 & -1 \\ 0 & s-1 & -3 \\ -1 & -1 & s-1 \end{vmatrix} = s^3 - 3s^2 - s - 3 = 0 \tag{10-194}$$

Thus, the coefficients of the characteristic equation are identified as $a_0 = -3, a_1 = -1$, and $a_2 = -3$. From Eq. (10-190),

$$\mathbf{M} = \begin{bmatrix} a_1 & a_2 & 1 \\ a_2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -3 & 1 \\ -3 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \tag{10-195}$$

The controllability matrix is

$$\mathbf{S} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 10 \\ 0 & 3 & 9 \\ 1 & 2 & 7 \end{bmatrix} \tag{10-196}$$

We can show that $\mathbf{S}$ is nonsingular, so the system can be transformed into the CCF. Substituting $\mathbf{S}$ and $\mathbf{M}$ into Eq. (10-188), we get

$$\mathbf{P} = \mathbf{SM} = \begin{bmatrix} 3 & -1 & 1 \\ 0 & 3 & 0 \\ 0 & -1 & 1 \end{bmatrix} \tag{10-197}$$

Thus, from Eqs. (10-191) and (10-192), the CCF model is given by

$$\overline{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & 1 & 3 \end{bmatrix} \quad \overline{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{10-198}$$

which could have been determined once the coefficients of the characteristic equation are known; however, the exercise is to show how the CCF transformation matrix $\mathbf{P}$ is obtained.

10-9-3 Observability Canonical Form (OCF)

A dual form of transformation of the CCF is the **observability canonical form** (OCF). The system described by Eqs. (10-172) and (10-173) is transformed to the OCF by the transformation

$$\mathbf{x}(t) = \mathbf{Q}\overline{\mathbf{x}}(t) \tag{10-199}$$

The transformed equations are as given in Eqs. (10-176) and (10-177). Thus,

$$\overline{\mathbf{A}} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} \quad \overline{\mathbf{B}} = \mathbf{Q}^{-1}\mathbf{B} \quad \overline{\mathbf{C}} = \mathbf{C}\mathbf{Q} \quad \overline{\mathbf{D}} = \mathbf{D} \tag{10-200}$$

where

$$\overline{\mathbf{A}} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix} \tag{10-201}$$

$$\overline{\mathbf{C}} = \mathbf{C}\mathbf{Q} = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \tag{10-202}$$

The elements of the matrices $\overline{\mathbf{B}}$ and $\overline{\mathbf{D}}$ are not restricted to any form. Notice that $\overline{\mathbf{A}}$ and $\overline{\mathbf{C}}$ are the transpose of the $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ in Eqs. (10-191) and (10-192), respectively.

The OCF transformation matrix $\mathbf{Q}$ is given by

$$\mathbf{Q} = (\mathbf{M}\mathbf{V})^{-1} \tag{10-203}$$

where $\mathbf{M}$ is as given in Eq. (10-190), and

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \mathbf{C}\mathbf{A}^2 \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix} \quad (n \times n) \tag{10-204}$$

The matrix $\mathbf{V}$ is often defined as the **observability matrix**, and $\mathbf{V}^{-1}$ must exist in order for the OCF transformation to be possible.

EXAMPLE 10-9-2 Consider that the coefficient matrices of the system described by Eqs. (10-172) and (10-138) are

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = [1 \quad 1 \quad 0] \quad \mathbf{D} = \mathbf{0} \tag{10-205}$$

Because the matrix **A** is identical to that of the system in Example 10-9-1, the matrix **M** is the same as that in Eq. (10-195). The observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 4 \\ 5 & 9 & 14 \end{bmatrix} \tag{10-206}$$

We can show that **V** is nonsingular, so the system can be transformed into the OCF. Substituting **V** and **M** into Eq. (10-203), we have the OCF transformation matrix,

$$\mathbf{Q} = (\mathbf{MV})^{-1} = \begin{bmatrix} 0.3333 & -0.1667 & 0.3333 \\ -0.3333 & 0.1667 & 0.6667 \\ 0.1667 & 0.1667 & 0.1667 \end{bmatrix} \tag{10-207}$$

From Eq. (10-191), the OCF model of the system is described by

$$\overline{\mathbf{A}} = \mathbf{Q}^{-1}\mathbf{AQ} = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix} \quad \overline{\mathbf{C}} = \mathbf{CQ} = [0 \quad 0 \quad 1] \quad \overline{\mathbf{B}} = \mathbf{Q}^{-1}\mathbf{B} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \tag{10-208}$$

Thus, $\overline{\mathbf{A}}$ and $\overline{\mathbf{C}}$ are of the OCF form given in Eqs. (10-201) and (10-202), respectively, and $\overline{\mathbf{B}}$ does not conform to any particular form.

10-9-4 Diagonal Canonical Form (DCF)

Given the dynamic equations in Eqs. (10-172) and (10-173), if **A** has distinct eigenvalues, there is a nonsingular transformation

$$\mathbf{x}(t) = \mathbf{T}\overline{\mathbf{x}}(t) \tag{10-209}$$

which transforms these equations to the dynamic equations of Eqs. (10-176) and (10-177), where

$$\overline{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{AT} \quad \overline{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B} \quad \overline{\mathbf{C}} = \mathbf{CT} \quad \overline{\mathbf{D}} = \mathbf{D} \tag{10-210}$$

The matrix $\overline{\mathbf{A}}$ is a diagonal matrix,

$$\overline{\mathbf{A}} = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix} \quad (n \times n) \tag{10-211}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the n distinct eigenvalues of **A**. The coefficient matrices $\overline{\mathbf{B}}$, $\overline{\mathbf{C}}$, and $\overline{\mathbf{D}}$ are given in Eq. (10-210) and do not follow any particular form.

It is apparent that one of the advantages of the DCF is that the transformed state equations are *decoupled* from each other and, therefore, can be solved individually.

We show in the following that the DCF transformation matrix **T** can be formed by use of the eigenvectors of **A** as its columns; that is,

$$\mathbf{T} = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \cdots \quad \mathbf{p}_n] \tag{10-212}$$

where $\mathbf{p}_i$, $i = 1, 2, \ldots, n$, denotes the eigenvector associated with the eigenvalue λ_i. This is proved by use of Eq. (10-155), which is written as

$$\lambda_i \mathbf{p}_i = \mathbf{A}\mathbf{p}_i \quad i = 1, 2, \ldots, n \tag{10-213}$$

Now, forming the $n \times n$ matrix,

$$\begin{aligned}[\lambda_1\mathbf{p}_1 \quad \lambda_2\mathbf{p}_2 \quad \cdots \quad \lambda_n\mathbf{p}_n] &= [\mathbf{A}\mathbf{p}_1 \quad \mathbf{A}\mathbf{p}_2 \quad \cdots \quad \mathbf{A}\mathbf{p}_n] \\ &= \mathbf{A}[\mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_n]\end{aligned} \tag{10-214}$$

The last equation is written

$$[\mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_n]\overline{\mathbf{A}} = \mathbf{A}[\mathbf{p}_1 \quad \mathbf{p}_2 \quad \cdots \quad \mathbf{p}_n] \tag{10-215}$$

where $\overline{\mathbf{A}}$ is as given in Eq. (10-211). Thus, if we let

$$\mathbf{T} = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad) \quad \mathbf{p}_n] \tag{10-216}$$

Eq. (10-215) is written

$$\overline{\mathbf{A}} = \mathbf{T}^{21}\mathbf{A}\mathbf{T} \tag{10-217}$$

If the matrix $\mathbf{A}$ is of the CCF and $\mathbf{A}$ has distinct eigenvalues, then the DCF transformation matrix is the Vandermonde matrix,

$$\mathbf{T} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \lambda_3^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix} \tag{10-218}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of $\mathbf{A}$. This can be proven by substituting the CCF of $\mathbf{A}$ in Eq. (10-110) into Eq. (10-155). The result is that the ith eigenvector $\mathbf{p}_i$ is equal to the ith column of $\mathbf{T}$ in Eq. (10-218).

▶ **EXAMPLE 10.9.3** Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \tag{10-219}$$

which has eigenvalues $\lambda_1 = -1$, $\lambda_2 = -2$, and $\lambda_3 = -3$. Because $\mathbf{A}$ is CCF, to transform it into DCF, the transformation matrix can be the Vandermonde matrix in Eq. (10-218). Thus,

$$\mathbf{T} = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} \tag{10-220}$$

Thus, the DCF of $\mathbf{A}$ is written

$$\overline{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \tag{10-221}$$

◀

10-9-5 Jordan Canonical Form (JCF)

In general, when the matrix **A** has multiple-order eigenvalues, unless the matrix is symmetric with real elements, it cannot be transformed into a diagonal matrix. However, there exists a similarity transformation in the form of Eq. (10-217) such that the matrix $\overline{\mathbf{A}}$ is almost diagonal. The matrix $\overline{\mathbf{A}}$ is called the **Jordan canonical form (JCF)**. A typical JCF is shown below.

$$\overline{\mathbf{A}} = \begin{bmatrix} \lambda_1 & 1 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & 0 & \lambda_2 & 0 \\ 0 & 0 & 0 & 0 & \lambda_3 \end{bmatrix} \tag{10-222}$$

where it is assumed that **A** has a third-order eigenvalue λ_1 and distinct eigenvalues λ_2 and λ_3.

The JCF generally has the following properties:

1. The elements on the main diagonal are the eigenvalues.
2. All the elements below the main diagonal are zero.
3. Some of the elements immediately above the multiple-order eigenvalues on the main diagonal are 1s, as shown in Eq. (10-222).
4. The 1s together with the eigenvalues form typical blocks called the **Jordan blocks**. As shown in Eq. (10-222), the Jordan blocks are enclosed by dashed lines.
5. When the nonsymmetrical matrix **A** has multiple-order eigenvalues, its eigenvectors are not linearly independent. For an **A** that is $n \times n$, there are only r (where r is an integer that is less than n and is dependent on the number of multiple-order eigenvalues) linearly independent eigenvectors.
6. The number of Jordan blocks is equal to the number of independent eigenvectors r. There is one and only one linearly independent eigenvector associated with each Jordan block.
7. The number of 1s above the main diagonal is equal to $n - r$.

To perform the JCF transformation, the transformation matrix **T** *is again formed by using the eigenvectors and generalized eigenvectors as its columns.*

EXAMPLE 10-9-4 Consider the matrix given in Eq. (10-165). We have shown that the matrix has eigenvalues 2, 1, and 1. Thus, the DCF transformation matrix can be formed by using the eigenvector and generalized eigenvector given in Eqs. (10-167), (10-169), and (10-171), respectively. That is,

$$\mathbf{T} = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3] = \begin{bmatrix} 2 & 1 & 1 \\ -1 & -\frac{3}{7} & -\frac{22}{49} \\ -2 & -\frac{5}{7} & -\frac{46}{49} \end{bmatrix} \tag{10-223}$$

Thus, the DCF is

$$\overline{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \tag{10-224}$$

Note that in this case there are two Jordan blocks, and there is one element of 1 above the main diagonal. ◀

10-10 DECOMPOSITIONS OF TRANSFER FUNCTIONS

Up to this point, various methods of characterizing linear systems have been presented. To summarize, it has been shown that the starting point of modeling a linear system may be the system's differential equation, transfer function, or dynamic equations; all these methods are closely related. Furthermore, the state diagram is also a useful tool that can not only lead to the solutions of state equations but also serve as a vehicle of transformation from one form of description to the others. The block diagram of Fig. 10-13 shows the relationships among the various ways of describing a linear system. For example, the block diagram shows that, starting with the differential equation of a system, one can find the solution by the transfer-function or state-equation method. The block diagram also shows that the majority of the relationships are bilateral, so a great deal of flexibility exists between the methods.

One subject remains to be discussed, which involves the construction of the state diagram from the transfer function between the input and the output. The process of going from the transfer function to the state diagram is called **decomposition**. In general, there are three basic ways to decompose transfer functions. These are **direct decomposition**, **cascade decomposition**, and **parallel decomposition**. Each of these three schemes of decomposition has its own merits and is best suited for a particular purpose.

10-10-1 Direct Decomposition

Direct decomposition is applied to an input–output transfer function that is not in factored form. Consider the transfer function of an nth-order SISO system between the input $U(s)$

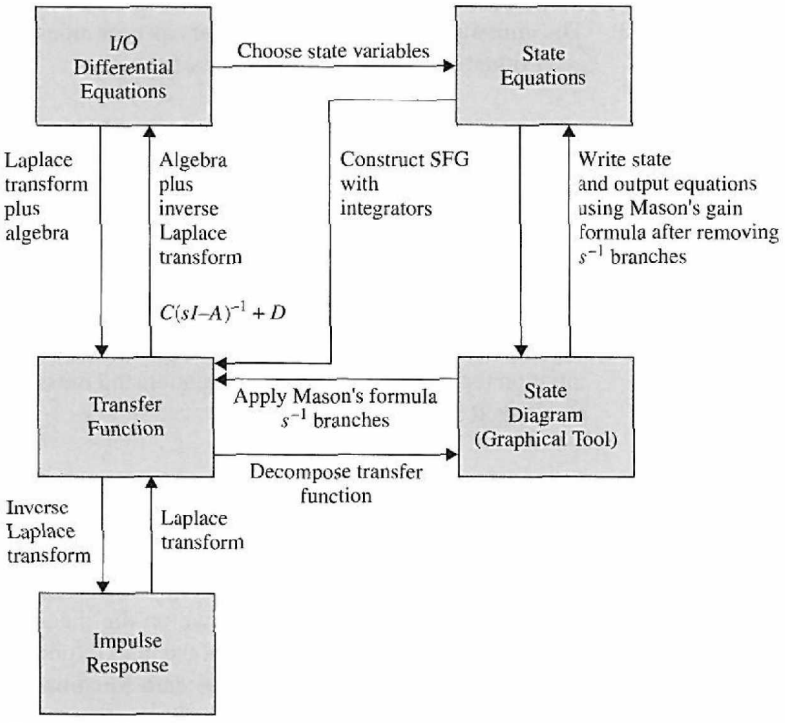


Figure 10-13 Block diagram showing the relationships among various methods of describing linear systems.

and output $Y(s)$:

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{b-2}s^{n-2} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0} \tag{10-225}$$

where we have assumed that the order of the denominator is at least one degree higher than that of the numerator.

We next show that the direct decomposition can be conducted in at least two ways, one leading to a state diagram that corresponds to the CCF and the other to the OCF.

Direct Decomposition to CCF

The objective is to construct a state diagram from the transfer function of Eq. (10-225). The following steps are outlined:

1. Express the transfer function in negative powers of s. This is done by multiplying the numerator and the denominator of the transfer function by s^{-n}.
2. Multiply the numerator and the denominator of the transfer function by a dummy variable $X(s)$. By implementing the last two steps, Eq. (10-225) becomes

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{-1} + b_{n-2}s^{-2} + \cdots + b_1 s^{-n+1} + b_0 s^{-n}}{1 + a_{n-1}s^{-1} + \cdots + a_1 s^{-n+1} + a_0 s^{-n}} \frac{X(s)}{X(s)} \tag{10-226}$$

3. The numerators and the denominators on both sides of Eq. (10-226) are equated to each other, respectively. The results are:

$$Y(s) = (b_{n-1}s^{-1} + b_{n-2}s^{-2} + \cdots + b_1 s^{-n+1} + b_0 s^{-n})X(s) \tag{10-227}$$

$$U(s) = (1 + a_{n-1}s^{-1} + \cdots + a_1 s^{-n+1} + a_0 s^{-n})X(s) \tag{10-228}$$

4. To construct a state diagram using the two equations in Eqs. (10-227) and (10–228), they must first be in the proper cause-and-effect relation. It is apparent that Eq. (10-227) already satisfies this prerequisite. However, Eq. (10-228) has the input on the left-hand side of the equation and must be rearranged. Eq. (10-228) is rearranged as

$$X(s) = U(s) - (a_{n-1}s^{-1} + a_{n-2}s^{-2} + \cdots + a_1 s^{-n+1} + a_0 s^{-n})X(s) \tag{10-229}$$

The state diagram is drawn as shown in Fig. 10-14 using Eqs. (10-227) and (10-228). For simplicity, the initial states are not drawn on the diagram. The state variables $x_1(t)$, $x_2(t)$, . . . , $x_n(t)$ are defined as the outputs of the integrators and are arranged in order from the right to the left on the state diagram. The state equations are obtained by applying the SFG gain formula to Fig. 10-14 with the derivatives of the state variables as the outputs and the state variables and $u(t)$ as the inputs, and overlooking the integrator branches. The

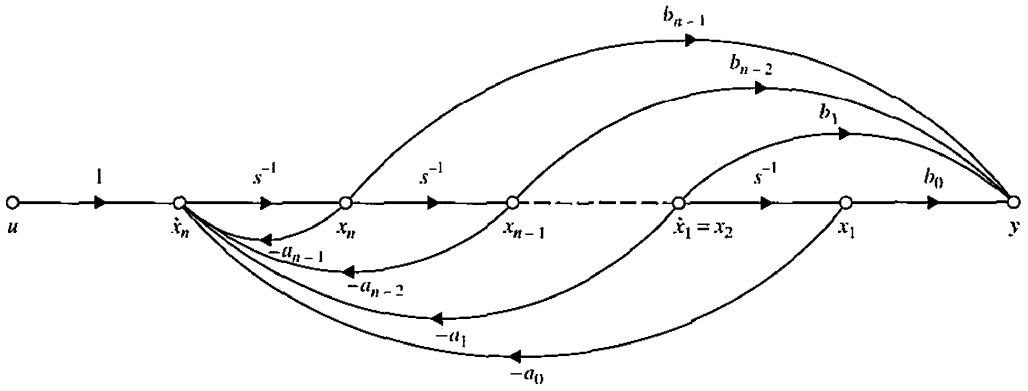


Figure 10-14 CCF state diagram of the transfer function in Eq. (10-225) by direct decomposition.

output equation is determined by applying the gain formula among the state variables, the input, and the output $y(t)$. The dynamic equations are written

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \tag{10-230}$$

$$y(t) = \mathbf{C}\mathbf{x}(t) + Du(t) \tag{10-231}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{10-232}$$

$$\mathbf{C} = [b_0 \quad b_1 \quad \cdots \quad b_{n-2} \quad b_{n-1}] \qquad \mathbf{D} = 0 \tag{10-233}$$

Apparently, **A** and **B** in Eq. (10-232) are of the CCF.

Direct Decomposition to OCF

Multiplying the numerator and the denominator of Eq. (10-225) by s^{-n}, the equation is expanded as

$$\begin{aligned}(1 + a_{n-1}s^{-1} + \cdots + a_1 s^{-n+1} + a_0 s^{-n})Y(s) \\ = (b_{n-1}s^{-1} + b_{n-2}s^{-2} + \cdots + b_1 s^{-n+1} + b_0 s^{-n})U(s)\end{aligned} \tag{10-234}$$

or

$$\begin{aligned}Y(s) = -(a_{n-1}s^{-1} + \cdots + a_1 s^{-n+1} + a_0 s^{-n})Y(s) \\ +(b_{n-1}s^{-1} + b_{n-2}s^{-2} + \cdots + b_1 s^{-n+1} + b_0 s^{-n})U(s)\end{aligned} \tag{10-235}$$

Fig. 10-15 shows the state diagram that results from using Eq. (10-235). The outputs of the integrators are designated as the state variables. However, unlike the usual convention, the state variables are assigned in *descending order* from right to left. Applying the SFG gain

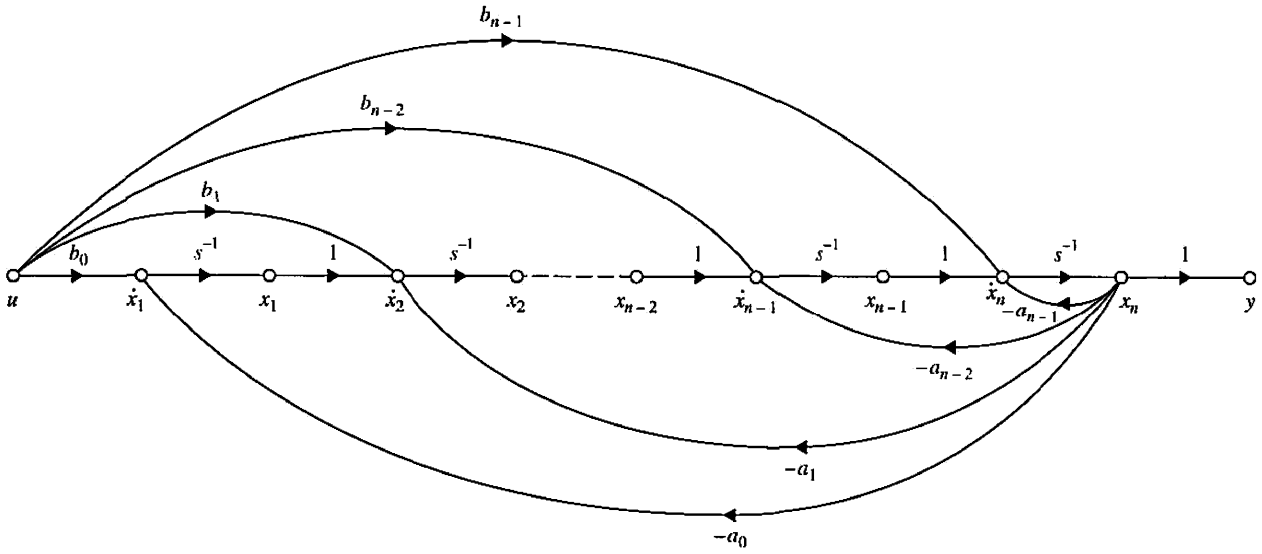


Figure 10-15 CCF state diagram of the transfer function in Eq. (10-225) by direct decomposition.

formula to the state diagram, the dynamic equations are written as in Eqs. (10-230) and (10-231), with

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_{n-1} \end{bmatrix} \tag{10-236}$$

and

$$\mathbf{C} = [0 \quad 0 \quad \cdots \quad 0 \quad 1] \quad \mathbf{D} = 0 \tag{10-237}$$

The matrices **A** and **C** are in OCF.

It should be pointed out that, given the dynamic equations of a system, the input–output transfer function is unique. However, given the transfer function, the state model is not unique, as shown by the CCF, OCF, and DCF, and many other possibilities. In fact, even for any one of these canonical forms (for example, CCF), while matrices **A** and **B** are defined, the elements of **C** and **D** could still be different depending on how the state diagram is drawn, that is, how the transfer function is decomposed. In other words, referring to Fig. 10-14, whereas the feedback branches are fixed, the feedforward branches that contain the coefficients of the numerator of the transfer function can still be manipulated to change the contents of **C**.

▶ **EXAMPLE 10-10-1** Consider the following input–output transfer function:

$$\frac{Y(s)}{U(s)} = \frac{2s^2 + s + 5}{s^3 + 6s^2 + 11s + 4} \tag{10-238}$$

The CCF state diagram of the system is shown in Fig. 10-16, which is drawn from the following equations:

$$Y(s) = (2s^{-1} + s^{-2} + 5s^{-3})X(s) \tag{10-239}$$

$$X(s) = U(s) - (6s^{-1} + 11s^{-2} + 4s^{-3})X(s) \tag{10-240}$$

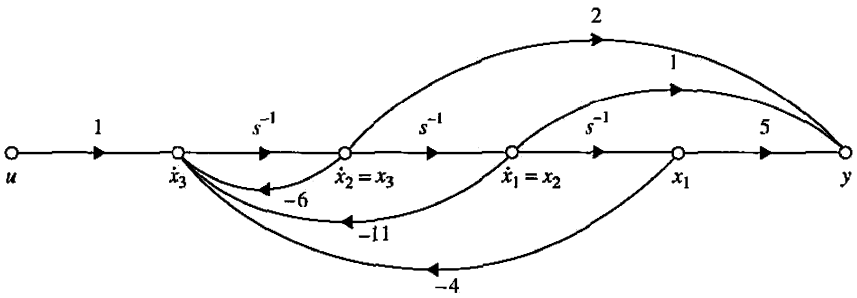


Figure 10-16 CCF state diagram of the transfer function in Eq. (10-238).

The dynamic equations of the system in CCF are

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \\ \frac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) \tag{10-241}$$

$$y(t) = [5 \quad 1 \quad 2]\mathbf{x}(t) \tag{10-242}$$

For the OCF, Eq. (10-238) is expanded to

$$Y(s) = (2s^{-1} + s^{-2} + 5s^{-3})U(s) - (6s^{-1} + 11s^{-2} + 4s^{-3})Y(s) \tag{10-243}$$

which leads to the OCF state diagram shown in Fig. 10-17. The OCF dynamic equations are written

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \\ \frac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -4 \\ 1 & 0 & -11 \\ 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} u(t) \tag{10-244}$$

$$y(t) = [0 \quad 0 \quad 1]\mathbf{x}(t) \tag{10-245}$$

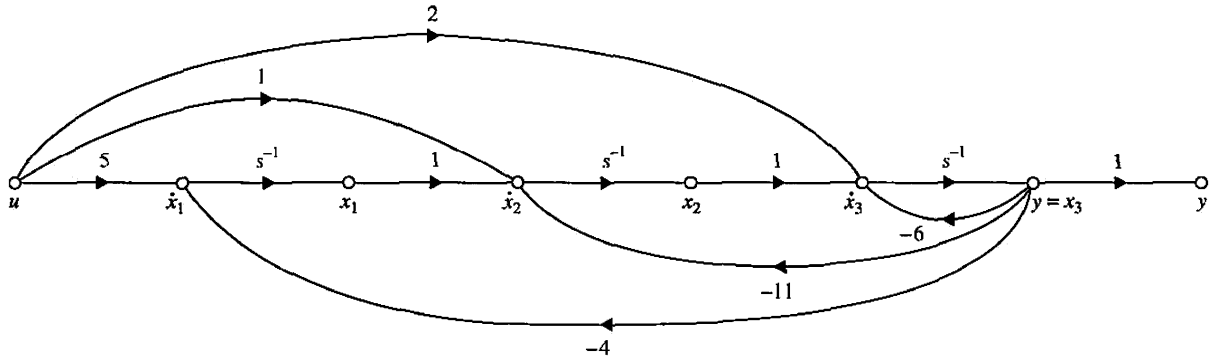


Figure 10-17 OCF state diagram of the transfer function in Eq. (10-238).

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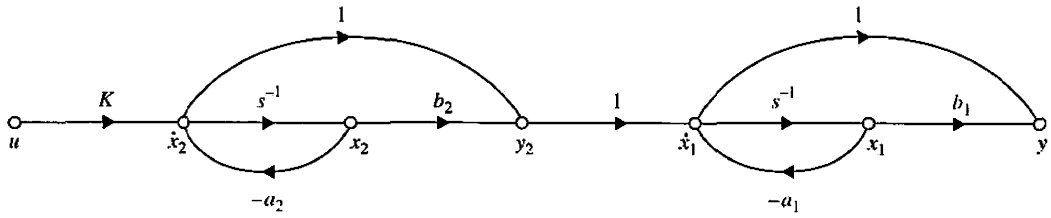

Figure 10-18 State diagram of the transfer function in Eq. (10-246) by cascade decomposition.

10-10-2 Cascade Decomposition

Cascade compensation refers to transfer functions that are written as products of simple first-order or second-order components. Consider the following transfer function, which is the product of two first-order transfer functions.

$$\frac{Y(s)}{U(s)} = K\left(\frac{s+b_1}{s+a_1}\right)\left(\frac{s+b_2}{s+a_2}\right) \tag{10-246}$$

where a_1, a_2, b_1, and b_2 are real constants. Each of the first-order transfer functions is decomposed by the direct decomposition, and the two state diagrams are connected in cascade, as shown in Fig. 10-18. The state equations are obtained by regarding the derivatives of the state variables as outputs and the state variables and $u(t)$ as inputs and then applying the SFG gain formula to the state diagram in Fig. 10-18. The integrator branches are neglected when applying the gain formula. The results are

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} -a_1 & b_2 - a_2 \\ 0 & -a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} K \\ K \end{bmatrix} u(t) \tag{10-247}$$

The output equation is obtained by regarding the state variables and $u(t)$ as inputs and $y(t)$ as the output and applying the gain formula to Fig. 10-18. Thus,

$$y(t) = [b_1 - a_1 \quad b_2 - a_2]\mathbf{x}(t) + Ku(t) \tag{10-248}$$

When the overall transfer function has complex poles or zeros, the individual factors related to these poles or zeros should be in second-order form. As an example, consider the following transfer function:

$$\frac{Y(s)}{U(s)} = \left(\frac{s+5}{s+2}\right)\left(\frac{s+1.5}{s^2+3s+4}\right) \tag{10-249}$$

where the poles of the second term are complex. The state diagram of the system with the two subsystems connected in cascade is shown in Fig. 10–19. The dynamic equations of the system are

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \\ \dfrac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -3 & 3 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u(t) \tag{10-250}$$

$$y(t) = [1.5 \quad 1 \quad 0]\mathbf{x}(t) \tag{10-251}$$

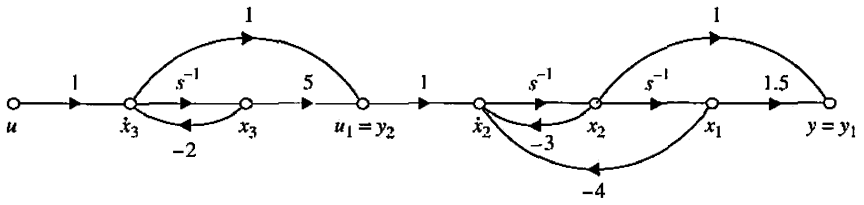

Figure 10-19 State diagram of the transfer function in Eq. (10-249) by cascade decomposition.

10-10-3 Parallel Decomposition

When the denominator of the transfer function is in factored form, the transfer function may be expanded by partial-fraction expansion. The resulting state diagram will consist of simple first- or second-order systems connected in parallel, which leads to the state equations in DCF or JCF, the latter in the case of multiple-order eigenvalues.

Consider that a second-order system is represented by the transfer function

$$\frac{Y(s)}{U(s)} = \frac{Q(s)}{(s+a_1)(s+a_2)} \tag{10-252}$$

where $Q(s)$ is a polynomial of order less than 2, and a_1 and a_2 are real and distinct. Although, analytically, a_1 and a_2 may be complex, in practice, complex numbers are difficult to implement on a computer. Eq. (10-253) is expansion by partial fractions:

$$\frac{Y(s)}{U(s)} = \frac{K_1}{s+a_1} + \frac{K_2}{s+a_2} \tag{10-253}$$

where K_1 and K_2 are real constants.

The state diagram of the system is drawn by the parallel combination of the state diagrams of each of the first-order terms in Eq. (10-253), as shown in Fig. 10-20. The dynamic equations of the system are

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} -a_1 & 0 \\ 0 & -a_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t) \tag{10-254}$$

$$y(t) = [K_1 \quad K_2]\mathbf{x}(t) \tag{10-255}$$

Thus, the state equations are of the DCF.

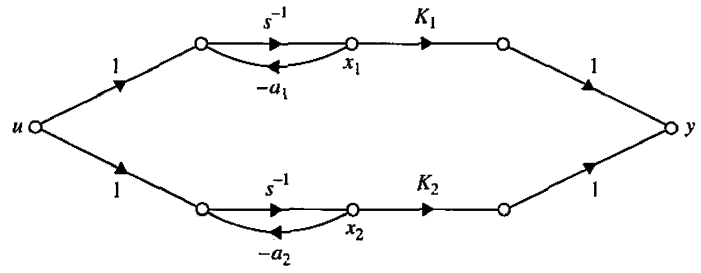

Figure 10-20 State diagram of the transfer function of Eq. (10-252) by parallel decomposition.

The conclusion is that, for transfer functions with distinct poles, parallel decomposition will lead to the DCF for the state equations. For transfer functions with multiple-order eigenvalues, parallel decomposition to a state diagram with a minimum number of integrators will lead to the JCF state equations. The following example will clarify this point.

EXAMPLE 10-10-2 Consider the following transfer function and its partial-fraction expansion:

$$\frac{Y(s)}{U(s)} = \frac{2s^2 + 6s + 5}{(s+1)^2(s+2)} = \frac{1}{(s+1)^2} + \frac{1}{s+1} + \frac{1}{s+2} \tag{10-256}$$

Note that the transfer function is of the third order, and, although the total order of the terms on the right-hand side of Eq. (10-256) is four, only three integrators should be used in the state diagram, which is drawn as shown in Fig. 10-21. The minimum number of three integrators is used, with one integrator being shared by two channels. The state equations of the system are written directly from Fig. 10-21.

$$\begin{bmatrix} \dfrac{dx_1(t)}{dt} \\ \dfrac{dx_2(t)}{dt} \\ \dfrac{dx_3(t)}{dt} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u(t) \tag{10-257}$$

which is recognized to be the JCF.

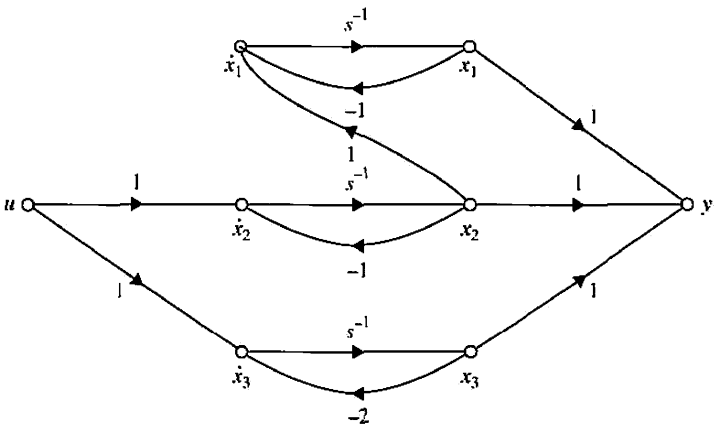

Figure 10-21 State diagram of the transfer function of Eq. (10-256) by parallel decomposition.

10-11 CONTROLLABILITY OF CONTROL SYSTEMS

The concepts of **controllability** and **observability**, introduced first by Kalman [3], play an important role in both theoretical and practical aspects of modern control. The conditions on controllability and observability essentially govern the existence of a solution to an optimal control problem. This seems to be the basic difference between optimal control theory and classical control theory. In the classical control theory, the design techniques are dominated by trial-and-error methods so that given a set of design specifications the designer at the outset does not know if any solution exists. Optimal control theory, on the

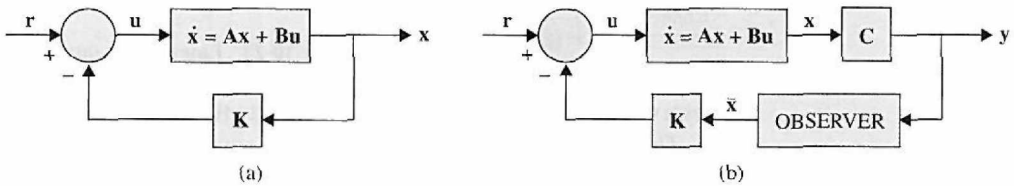

Figure 10-22 (a) Control system with state feedback. (b) Control system with observer and state feedback.

other hand, has criteria for determining at the outset if the design solution exists for the system parameters and design objectives.

We shall show that the condition of controllability of a system is closely related to the existence of solutions of state feedback for assigning the values of the eigenvalues of the system arbitrarily. The concept of *observability* relates to the condition of observing or estimating the state variables from the output variables, which are generally measurable.

The block diagram shown in Fig. 10-22 illustrates the motivation behind investigating controllability and observability. Figure 10-22(a) shows a system with the process dynamics described by

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \tag{10-258}$$

The closed-loop system is formed by feeding back the state variables through a constant feedback gain matrix $\mathbf{K}$. Thus, from Fig. 10-22,

$$\mathbf{u}(t) = -\mathbf{K}\mathbf{x}(t) + \mathbf{r}(t) \tag{10-259}$$

where $\mathbf{K}$ is a $p \times n$ feedback matrix with constant elements. The closed-loop system is thus described by

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}\mathbf{r}(t) \tag{10-260}$$

This problem is also known as the **pole-placement design** through state feedback. The design objective in this case is to find the feedback matrix $\mathbf{K}$ such that the eigenvalues of $(\mathbf{A} - \mathbf{B}\mathbf{K})$, or of the closed-loop system, are of certain prescribed values. The word *pole* refers here to the poles of the closed-loop transfer function, which are the same as the eigenvalues of $(\mathbf{A} - \mathbf{B}\mathbf{K})$.

We shall show later that the existence of the solution to the pole-placement design with arbitrarily assigned pole values through state feedback is directly based on the controllability of the states of the system. The result is that *if the system of Eq. (10-225) is controllable, then there exists a constant feedback matrix* $\mathbf{K}$ *that allows the eigenvalues of* $(\mathbf{A} - \mathbf{B}\mathbf{K})$ *to be arbitrarily assigned.*

Once the closed-loop system is designed, the practical problems of implementing the feeding back of the state variables must be considered. There are two problems with implementing state feedback control: First, the number of state variables may be excessive, which will make the cost of sensing each of these state variables for feedback prohibitive. Second, not all the state variables are physically accessible, and so it may be necessary to design and construct an **observer** that will estimate the state vector from the output vector $\mathbf{y}(t)$. Fig. 10-22(b) shows the block diagram of a closed-loop system with an observer. The

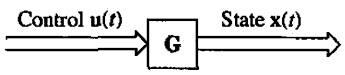

Figure 10-23 Linear time-invariant system.

observed state vector $\bar{\mathbf{x}}(t)$ is used to generate the control $\mathbf{u}(t)$ through the feedback matrix $\mathbf{K}$. *The condition that such an observer can be designed for the system is called the observability of the system.*

10-11-1 General Concept of Controllability

The concept of controllability can be stated with reference to the block diagram of Fig. 10-22(a). *The process is said to be* ***completely controllable*** *if every state variable of the process can be controlled to reach a certain objective in finite time by some unconstrained control* $\mathbf{u}(t)$, as shown in Fig. 10-23. Intuitively, it is simple to understand that, if any one of the state variables is independent of the control $\mathbf{u}(t)$, there would be no way of driving this particular state variable to a desired state in finite time by means of a control effort. Therefore, this particular state is said to be uncontrollable, and, as long as there is at least one uncontrollable state, the system is said to be not completely controllable or, simply, uncontrollable.

As a simple example of an uncontrollable system, Fig. 10-24 illustrates the state diagram of a linear system with two state variables. Because the control $\mathbf{U}(t)$ affects only the state $x_1(t)$, the state $x_2(t)$ is uncontrollable. In other words, it would be impossible to drive $x_2(t)$ from an initial state $x_2(t_0)$ to a desired state $x_2(t_f)$ in finite time interval $t_f - t_0$ by the control $\mathbf{U}(t)$. Therefore, the entire system is said to be uncontrollable.

The concept of controllability given here refers to the states and is sometimes referred to as **state controllability**. Controllability can also be defined for the outputs of the system, so there is a difference between state controllability and output controllability.

10-11-2 Definition of State Controllability

Consider that a linear time-invariant system is described by the following dynamic equations:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \tag{10-261}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \tag{10-262}$$

where $\mathbf{x}(t)$ is the $n \times 1$ state vector, $\mathbf{u}(t)$ is the $r \times 1$ input vector, and $\mathbf{y}(t)$ is the $p \times 1$ output vector. $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are coefficients of appropriate dimensions.

The state $\mathbf{x}(t)$ *is said to be controllable at* $t = t_0$ *if there exists a piecewise continuous input* $\mathbf{u}(t)$ *that will drive the state to any final state* $\mathbf{x}(t_f)$ *for a finite time* $(t_f - t_0) \geq 0$. *If*

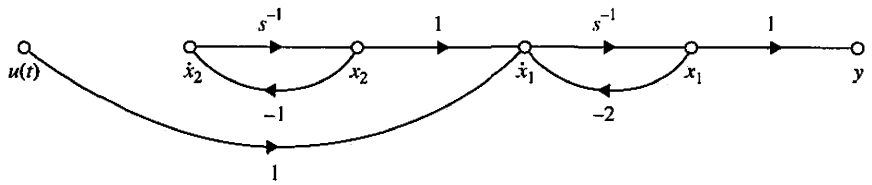

Figure 10-24 State diagram of the system that is not state controllable.

every state $\mathbf{x}(t_0)$ *of the system is controllable in a finite time interval, the system is said to be completely state controllable or, simply, controllable.*

The following theorem shows that the condition of controllability depends on the coefficient matrices **A** and **B** of the system. The theorem also gives one method of testing for state controllability.

Theorem 10-1. *For the system described by the state equation of Eq. (10-261) to be* ***completely state controllable****, it is necessary and sufficient that the following* $n \times nr$ ***controllability matrix*** *has a rank of* n:

$$\mathbf{S} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B} \quad \cdots \quad \mathbf{A}^{n-1}\mathbf{B}] \tag{10-263}$$

Because the matrices **A** *and* **B** *are involved, sometimes we say that the pair* [**A**, **B**] *is controllable, which implies that* **S** *is of rank* n.

The proof of this theorem is given in any standard textbook on optimal control systems. The idea is to start with the state-transition equation of Eq. (10-79) and then proceed to show that Eq. (10-263) must be satisfied in order that all the states are accessible by the input.

Although the criterion of state controllability given in Theorem 10-1 is quite straightforward, manually, it is not very easy to test for high-order systems and/or systems with many inputs. If **S** is nonsquare, we can form the matrix $\mathbf{SS}'$, which is $n \times n$; then, if $\mathbf{SS}'$ is nonsingular, **S** has rank n.

10-11-3 Alternate Tests on Controllability

There are several alternate methods of testing controllability, and some of these may be more convenient to apply than the condition in Eq. (10-263).

Theorem 10-2. *For a single-input, single-output (SISO) system described by the state equation of Eq. (10-261) with* $r = 1$*, the pair* [**A**, **B**] *is completely controllable if* **A** *and* **B** *are in CCF or transformable into CCF by a similarity transformation.*

The proof of this theorem is straightforward, since it was established in Section 10-9 that the CCF transformation requires that the controllability matrix **S** be nonsingular. Because the CCF transformation in Section 10-9 was defined only for SISO systems, the theorem applies only to this type of system.

Theorem 10-3. *For a system described by the state equation of Eq. (10-261), if* **A** *is in DCF or JCF, the pair* [**A**, **B**] *is completely controllable if all the elements in the rows of* **B** *that correspond to the last row of each Jordan block are nonzero.*

The proof of this theorem comes directly from the definition of controllability. Let us assume that **A** is diagonal and that it has distinct eigenvalues. Then, the pair [**A**, **B**] is controllable if **B** does not have any row with all zeros. The reason is that, if **A** is diagonal, all the states are decoupled from each other, and, if any row of **B** contains all zero elements, the corresponding state would not be accessed from any of the inputs, and that state would be uncontrollable.

For a system in JCF, such as the **A** and **B** matrices illustrated in Eq. (10-264), for controllability only the elements in the row of **B** that correspond to the last row of the Jordan block cannot all be zeros. The elements in the other rows of **B** need not all be

nonzero, since the corresponding states are still coupled through the 1s in the Jordan blocks of **A**.

$$\mathbf{A} = \begin{bmatrix} \lambda_1 & 1 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 \\ 0 & 0 & \lambda_1 & 0 \\ 0 & 0 & 0 & \lambda_2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix} \tag{10-264}$$

Thus, the condition of controllability for the **A** and **B** in Eq. (10-264) is $b_{31} \neq 0$, $b_{32} \neq 0$, $b_{41} \neq 0$, and $b_{42} \neq 0$.

EXAMPLE 10-11-1 The following matrices are for a system with two identical eigenvalues, but the matrix **A** is diagonal.

$$\mathbf{A} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} \tag{10-265}$$

The system is uncontrollable, since the two state equations are dependent; that is, it would not be possible to control the states independently by the input. We can easily show that in this case $\mathbf{S} = [\mathbf{B}\ \mathbf{AB}]$ is singular.

EXAMPLE 10-11-2 Consider the system shown in Fig. 10-24, which was reasoned earlier to be uncontrollable. Let us investigate the same system using the condition of Eq. (10-263). The state equations of the system are written in the form of Eq. (10-263) with

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{10-266}$$

Thus, from Eq. (10-263), the controllability matrix is

$$\mathbf{S} = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \tag{10-267}$$

which is singular, and the system is uncontrollable.

EXAMPLE 10-11-3 Consider that a third-order system has the coefficient matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & -4 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{10-268}$$

The controllability matrix is

$$\mathbf{S} = [\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}] = \begin{bmatrix} 0 & -1 & -4 \\ 0 & 0 & 0 \\ 1 & 3 & 8 \end{bmatrix} \tag{10-269}$$

which is singular. Thus, the system is not controllable.

The eigenvalues of **A** are $\lambda_1 = 2$, $\lambda_2 = 2$, and $\lambda_3 = 1$. The JCF of **A** and **B** are obtained with the transformation $\mathbf{x}(t) = \mathbf{T}\bar{\mathbf{x}}(t)$, where

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix} \tag{10-270}$$

Then,

$$\bar{\mathbf{A}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \bar{\mathbf{B}} = \mathbf{T}^{-1}\mathbf{B} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \tag{10-271}$$

Because the last row of $\bar{\mathbf{B}}$, which corresponds to the Jordan block for the eigenvalue λ_3, is zero, the transformed state variable $\bar{x}_3(t)$ is uncontrollable. From the transformation matrix **T** in Eq. (10-235), $x_2 = \bar{x}_3$, which means that x_2 is uncontrollable in the original system. It should be noted that the minus sign in front of the 1 in the Jordan block does not alter the basic definition of the block. ◀

10-12 OBSERVABILITY OF LINEAR SYSTEMS

The concept of observability was covered earlier in Section 10-11 on controllability and observability. Essentially, *a system is completely observable if every state variable of the system affects some of the outputs.* In other words, it is often desirable to obtain information on the state variables from the measurements of the outputs and the inputs. If any one of the states cannot be observed from the measurements of the outputs, the state is said to be unobservable, and the system is not completely observable or, simply, unobservable. Fig. 10-25 shows the state diagram of a linear system in which the state x_2 is not connected to the output $y(t)$ in any way. Once we have measured $y(t)$, we can observe the state $x_1(t)$, since $x_1(t) = y(t)$. However, the state x_2 cannot be observed from the information on $y(t)$. Thus, the system is unobservable.

10-12-1 Definition of Observability

Given a linear time-invariant system that is described by the dynamic equations of Eqs. (10-261) and (10-262), the state $\mathbf{x}(t_0)$ *is said to be observable if given any input* $\mathbf{u}(t)$, *there exists a finite time* $t_f \geq t_0$ *such that the knowledge of* $\mathbf{u}(t)$ *for* $t_0 \leq t < t_f$, *matrices* **A**, **B**, **C**, *and* **D**; *and the output* $\mathbf{y}(t)$ *for* $t_0 \leq t < t_f$ *are sufficient to determine* $\mathbf{x}(t_0)$. *If every state of the system is observable for a finite* t_f, *we say that the system is completely observable, or, simply, observable.*

The following theorem shows that the condition of observability depends on the matrices **A** and **C** of the system. The theorem also gives one method of testing observability.

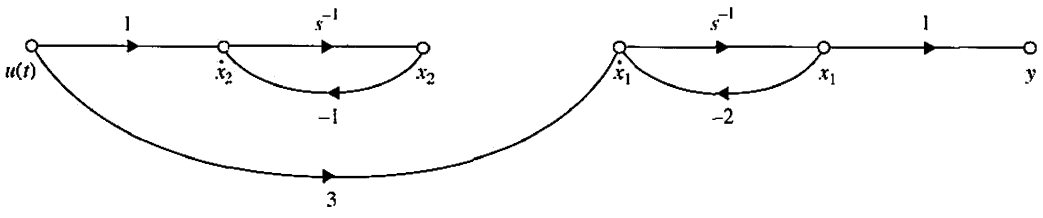

Figure 10-25 State diagram of a system that is not observable.

■ **Theorem 10-4.** *For the system described by Eqs. (10-261) and (10-262) to be completely observable, it is necessary and sufficient that the following $n \times np$* ***observability matrix*** *has a rank of n:*

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \tag{10-272}$$

The condition is also referred to as the pair [**A**, **C**] *being observable. In particular, if the system has only one output,* **C** *is a $1 \times n$ row matrix;* **V** *is an $n \times n$ square matrix. Then the system is completely observable if* **V** *is nonsingular.*

The proof of this theorem is not given here. It is based on the principle that Eq. (10-272) must be satisfied so that $\mathbf{x}(t_0)$ can be uniquely determined from the output $\mathbf{y}(t)$.

10-12-2 Alternate Tests on Observability

Just as with controllability, there are several alternate methods of testing observability. These are described in the following theorems.

■ **Theorem 10-5.** *For an SISO system, described by the dynamic equations of Eqs. (10-261) and (10-262) with $r = 1$ and $p = 1$, the pair* [**A**, **C**] *is completely observable if* **A** *and* **C** *are in OCF or transformable into OCF by a similarity transformation.*

The proof of this theorem is straightforward, since it was established in Section 10-8 that the OCF transformation requires that the observability matrix **V** be nonsingular.

■ **Theorem 10-6.** *For a system described by the dynamic equations of Eqs. (10-261) and (10-262), if* **A** *is in DCF or JCF, the pair* [**A**, **C**] *is completely observable if all the elements in the columns of* **C** *that correspond to the first row of each Jordan block are nonzero.*

Note that this theorem is a dual of the test of controllability given in Theorem 10-3. If the system has distinct eigenvalues, **A** is diagonal, then the condition on observability is that none of the columns of **C** can contain all zeros.

▶ **EXAMPLE 10-12-1** Consider the system shown in Fig. 10-25, which was earlier defined to be unobservable. The dynamic equations of the system are expressed in the form of Eqs. (10-261) and (10-262) with

$$\mathbf{A} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \mathbf{C} = [1 \quad 0] \tag{10-273}$$

Thus, the observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix} \tag{10-274}$$

which is singular. Thus, the pair [**A**, **C**] is unobservable. In fact, because **A** is of DCF and the second column of **C** is zero, this means that the state $x_2(t)$ is unobservable, as conjectured from Fig. 10-24. ◁

10-13 RELATIONSHIP AMONG CONTROLLABILITY, OBSERVABILITY, AND TRANSFER FUNCTIONS

In the classical analysis of control systems, transfer functions are used for modeling of linear time-invariant systems. Although controllability and observability are concepts and tools of modern control theory, we shall show that they are closely related to the properties of transfer functions.

Theorem 10-7. *If the input–output transfer function of a linear system has pole–zero cancellation, the system will be uncontrollable or unobservable, or both, depending on how the state variables are defined. On the other hand, if the input–output transfer function does not have pole–zero cancellation, the system can always be represented by dynamic equations as a completely controllable and observable system.*

The proof of this theorem is not given here. The importance of this theorem is that, if a linear system is modeled by a transfer function with no pole–zero cancellation, then we are assured that it is a controllable and observable system, no matter how the state-variable model is derived. Let us amplify this point further by referring to the following SISO system.

$$\mathbf{A}=\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -2 & 0 & 0\\ 0 & 0 & -3 & 0\\ 0 & 0 & 0 & -4\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}1\\1\\0\\0\end{bmatrix}\quad \mathbf{C}=[1\quad 0\quad 1\quad 0]\quad \mathbf{D}=0 \tag{10-275}$$

Because **A** is a diagonal matrix, the controllability and observability conditions of the four states are determined by inspection. They are as follows:

x_1: Controllable and observable (C and O)

x_2: Controllable but unobservable (C but UO)

x_3: Uncontrollable but observable (UC but O)

x_4: Uncontrollable and unobservable (UC and UO)

The block diagram of the system in Fig. 10-26 shows the DCF decomposition of the system. Clearly, the transfer function of the controllable and observable system should be

$$\frac{Y(s)}{U(s)}=\frac{1}{s+1} \tag{10-276}$$

whereas the transfer function that corresponds to the dynamics described in Eq. (10-275) is

$$\frac{Y(s)}{U(s)}=\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}=\frac{(s+2)(s+3)(s+4)}{(s+1)(s+2)(s+3)(s+4)} \tag{10-277}$$

which has three pole–zero cancellations. This simple-minded example illustrates that a "minimum-order" transfer function without pole–zero cancellation is the only component that corresponds to a system that is controllable and observable.

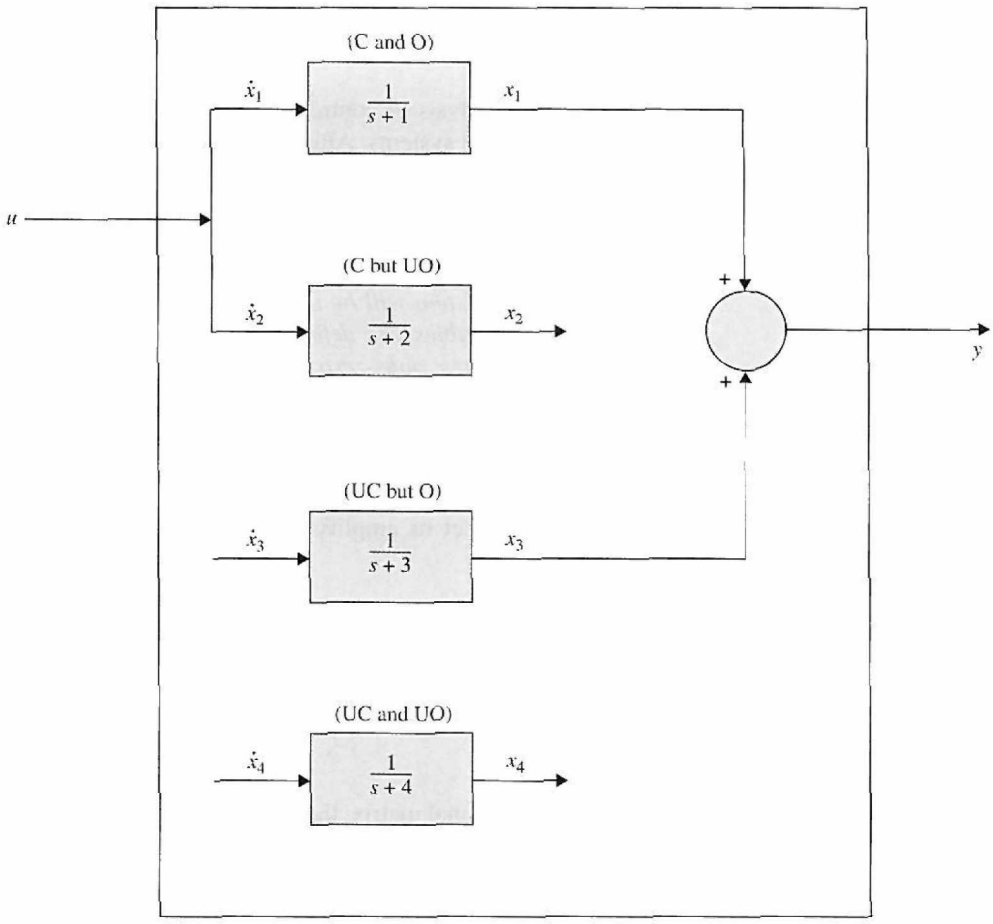

Figure 10-26 Block diagram showing the controllable, uncontrollable, observable, and unobservable components of the system described in Eq. (10-275).

▶ **EXAMPLE 10-13-1** Let us consider the transfer function

$$\frac{Y(s)}{U(s)} = \frac{s+2}{(s+1)(s+2)} \tag{10-278}$$

which is a reduced form of Eq. (10-277). Eq. (10-278) is decomposed into CCF and OCF as follows:

CCF:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = [1 \quad 1] \tag{10-279}$$

Because the CCF transformation can be made, the pair [**A**, **B**] of the CCF is controllable. The observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \tag{10-280}$$

which is singular, and the pair [**A**, **C**] of the CCF is unobservable.

OCF:

$$\mathbf{A} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{C} = [0 \quad 1] \tag{10-281}$$

Because the OCF transformation can be made, the pair [**A**, **C**] of the OCF is observable. However, the controllability matrix is

$$\mathbf{S} = [\mathbf{B} \quad \mathbf{AB}] = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \tag{10-282}$$

which is singular, and the pair [**A**, **B**] of the OCF is uncontrollable.

The conclusion that can be drawn from this example is that, given a system that is modeled by transfer function, the controllability and observability conditions of the system depend on how the state variables are defined. ◄

► 10-14 INVARIANT THEOREMS ON CONTROLLABILITY AND OBSERVABILITY

We now investigate the effects of the similarity transformations on controllability and observability. The effects of controllability and observability due to state feedback will be investigated.

■ **Theorem 10-8.** *Invariant theorem on similarity transformations: Consider the system described by the dynamic equations of Eqs. (10-261) and (10-262). The similarity transformation* $\mathbf{x}(t) = \mathbf{P}\bar{\mathbf{x}}(t)$, *where* $\mathbf{P}$ *is nonsingular, transforms the dynamic equations to*

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = \bar{\mathbf{A}}\bar{\mathbf{x}}(t) + \bar{\mathbf{B}}\mathbf{u}(t) \tag{10-283}$$

$$\bar{\mathbf{y}}(t) = \bar{\mathbf{C}}\mathbf{x}(t) + \bar{\mathbf{D}}\mathbf{u}(t) \tag{10-284}$$

where

$$\bar{\mathbf{A}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P} \qquad \bar{\mathbf{B}} = \mathbf{P}^{-1}\mathbf{B} \tag{10-285}$$

The controllability of $[\bar{\mathbf{A}}, \bar{\mathbf{B}}]$ *and the observability of* $[\bar{\mathbf{A}}, \bar{\mathbf{C}}]$ *are not affected by the transformation.*

In other words, controllability and observability are preserved through similar transformations. The theorem is easily proven by showing that the ranks of $\bar{\mathbf{S}}$ and $\mathbf{S}$ and the ranks of $\bar{\mathbf{V}}$ and $\mathbf{V}$ are identical, where $\bar{\mathbf{S}}$ and $\bar{\mathbf{V}}$ are the controllability and observability matrices, respectively, of the transformed system.

■ **Theorem 10-9.** *Theorem on controllability of closed-loop systems with state feedback: If the open-loop system*

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \tag{10-286}$$

is completely controllable, then the closed-loop system obtained through state feedback,

$$\mathbf{u}(t) = \mathbf{r}(t) - \mathbf{K}\mathbf{x}(t) \tag{10-287}$$

so that the state equation becomes

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{BK})\mathbf{x}(t) + \mathbf{Br}(t) \tag{10-288}$$

is also completely controllable. On the other hand, if **[A, B]** *is uncontrollable, then there is no* **K** *that will make the pair* **[A − BK, B]** *controllable. In other words, if an open-loop system is uncontrollable, it cannot be made controllable through state feedback.*

Proof: The controllability of **[A, B]** implies that there exists a control $\mathbf{u}(t)$ over the time interval $[t_0, t_f]$ such that **the initial state** $\mathbf{x}(t_0)$ is driven to the final state $\mathbf{x}(t_f)$ over the finite time interval $t_f - t_0$. We can write Eq. (10-252) as

$$\mathbf{r}(t) = \mathbf{u}(t) + \mathbf{Kx}(t) \tag{10-289}$$

which is the control of the closed-loop system. Thus, if $\mathbf{u}(t)$ exists that can drive $\mathbf{x}(t_0)$ to any $\mathbf{x}(t_f)$ in finite time, then we cannot find an input $\mathbf{r}(t)$ that will do the same to $\mathbf{x}(t)$, because otherwise we can set $\mathbf{u}(t)$ as in Eq. (10-287) to control the open-loop system.

■ **Theorem 10-10.** *Theorem on observability of closed-loop systems with state feedback: If an open-loop system is controllable and observable, then state feedback of the form of Eq. (10-287) could destroy observability. In other words, the observability of open-loop and closed-loop systems due to state feedback is unrelated.*

The following example illustrates the relation between observability and state feedback.

▶ **EXAMPLE 10-14-1** Let the coefficient matrices of a linear system be

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{C} = [1 \quad 2] \tag{10-290}$$

We can show that the pair **[A, B]** is controllable and **[A, C]** is observable.

Let the state feedback be defined as

$$u(t) = r(t) - \mathbf{Kx}(t) \tag{10-291}$$

where

$$\mathbf{K} = [k_1 \quad k_2] \tag{10-292}$$

Then the closed-loop system is described by the state equation

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{BK})\mathbf{x}(t) + \mathbf{B}r(t) \tag{10-293}$$

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} -k_1 & 1 - k_2 \\ -2 - k_1 & -3 - g_2 \end{bmatrix} \tag{10-294}$$

The observability matrix of the closed-loop system is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C} \\ \mathbf{C(A - BK)} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -k_1 - 4 & -3k_2 - 5 \end{bmatrix} \tag{10-295}$$

The determinant of $\mathbf{V}$ is

$$|\mathbf{V}| = 6k_1 - 3k_2 + 3 \tag{10-296}$$

Thus, if k_1 and k_2 are chosen so that $|\mathbf{V}| = 0$, the closed-loop system would be uncontrollable. ◀

10-15 CASE STUDY: MAGNETIC-BALL SUSPENSION SYSTEM

As a case study to illustrate some of the material presented in this chapter, let us consider the magnetic-ball suspension system shown in Fig. 10-27. The objective of the system is to regulate the current of the electromagnet so that the ball will be suspended at a fixed distance from the end of the magnet. The dynamic equations of the system are

$$M\frac{d^2x(t)}{dt^2} = Mg - \frac{ki^2(t)}{x(t)} \tag{10-297}$$

$$v(t) = Ri(t) + L\frac{di(t)}{dt} \tag{10-298}$$

where Eq. (10-262) is nonlinear. The system variables and parameters are as follows:

$v(t)$ = input voltage (V)
$i(t)$ = winding current (A)
R = winding resistance = 1Ω
M = ball mass = 1.0 kg
$x(t)$ = ball position (m)
k = proportional constant = 1.0
L = winding inductance = 0.01 H
g = gravitational acceleration = 32.2 m/sec^2

The state variables are defined as

$$\begin{aligned} x_1(t) &= x(t) \\ x_2(t) &= \frac{dx(t)}{dt} \\ x_3(t) &= i(t) \end{aligned} \tag{10-299}$$

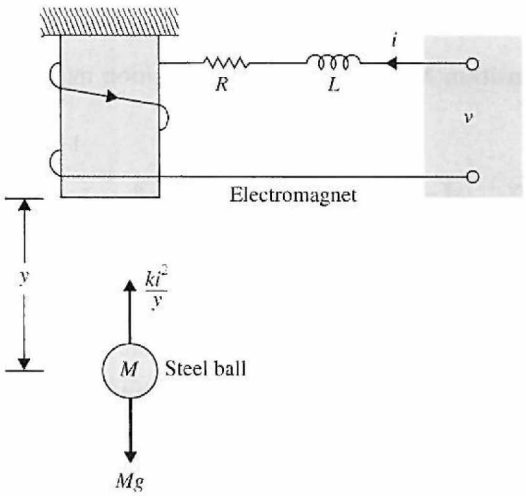

Figure 10-27 Ball-suspension system.

The state equations are

$$\frac{dx_1(t)}{dt} = x_2(t) \tag{10-300}$$

$$\frac{dx_2(t)}{dt} = g - \frac{k}{M}\frac{x_3^2(t)}{x_1(t)} \tag{10-301}$$

$$\frac{dx_3(t)}{dt} = -\frac{R}{L}x_3(t) + \frac{v(t)}{L} \tag{10-302}$$

These nonlinear state equations are linearized about the equilibrium point, $x_1(t) = x(t) = 0.5$ m, using the method described in Section 4-9. After substituting the parameter values, the linearized equations are written

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{A}^*\Delta\mathbf{x}(t) + \mathbf{B}^*\Delta v(t) \tag{10-303}$$

where $\Delta\mathbf{x}(t)$ denotes the state vector, and $\Delta v(t)$ is the input voltage of the linearized system. The coefficient matrices are

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix} \tag{10-304}$$

All the computations done in the following section can be carried out with the MATLAB Toolbox appearing later in this chapter. To show the analytical method, we carry out the steps of the derivations as follows.

The Characteristic Equation

$$|s\mathbf{I} - \mathbf{A}^*| = \begin{bmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 0 & 0 & s+100 \end{bmatrix} = s^3 + 100s^2 - 64.4s - 6440 = 0 \tag{10-305}$$

Eigenvalues: The eigenvalues of $\mathbf{A}^*$, or the roots of the characteristic equation, are

$$s = -100 \quad s = -8.025 \quad s = 8.025$$

The State-Transition Matrix: The state-transition matrix of $\mathbf{A}^*$ is

$$\phi(t) = \mathcal{L}^{-1}\left[(s\mathbf{I} - \mathbf{A}^*)^{-1}\right] = \mathcal{L}^{-1}\left(\begin{bmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 0 & 0 & s+100 \end{bmatrix}^{-1}\right) \tag{10-306}$$

or

$$\phi(t) = \mathcal{L}^{-1}\left(\frac{1}{(s+100)(s+8.025)(s-8.025)}\begin{bmatrix} s(s+100) & s+100 & -16 \\ 64.4(s+100) & s(s+100) & -16s \\ 0 & 0 & s^2-64.4 \end{bmatrix}\right) \tag{10-307}$$

By performing the partial-fraction expansion and carrying out the inverse Laplace transform, the state-transition matrix is

$$\phi(t) = \begin{bmatrix} 0 & 0 & -0.0016 \\ 0 & 0 & 0.16 \\ 0 & 0 & 1 \end{bmatrix} e^{-100t} + \begin{bmatrix} 0.5 & -0.062 & 0.0108 \\ -4.012 & 0.5 & -0.087 \\ 0 & 0 & 0 \end{bmatrix} e^{-8.025t} + \begin{bmatrix} 0.5 & 0.062 & -0.0092 \\ 4.012 & 0.5 & -0.074 \\ 0 & 0 & 0 \end{bmatrix} e^{8.025t} \tag{10-308}$$

Because the last term in Eq. (10-308) has a positive exponent, the response of $\phi(t)$ increases with time, and the system is unstable. This is expected, since without control, the steel ball would be attracted by the magnet until it hits the bottom of the magnet.

Transfer Function: Let us define the ball position $x(t)$ as the output $y(t)$; then, given the input, $v(t)$, the input–output transfer function of the system is

$$\frac{Y(s)}{V(s)} = \mathbf{C}^*(s\mathbf{I} - \mathbf{A}^*)^{-1}\mathbf{B}^* = [1 \quad 0 \quad 0](s\mathbf{I} - \mathbf{A}^*)^{-1}\mathbf{B}^* = \frac{-1600}{(s+100)(s+8.025)(s-8.025)} \tag{10-309}$$

Controllability: The controllability matrix is

$$\mathbf{S} = [\mathbf{B}^* \quad \mathbf{A}^*\mathbf{B}^* \quad \mathbf{A}^*2\mathbf{B}^*] = \begin{bmatrix} 0 & 0 & -1{,}600 \\ 0 & -1{,}600 & 160{,}000 \\ 100 & -10{,}000 & 1{,}000{,}000 \end{bmatrix} \tag{10-310}$$

Because the rank of **S** is 3, the system is completely controllable.

Observability: The observability of the system depends on which variable is defined at the output. For state-feedback control, which will be discussed later in Chapter 10, the full controller requires feeding back all three state variables, x_1, x_2, and x_3. However, for reasons of economy, we may want to feed back only one of the three state variables. To make the problem more general, we may want to investigate which state, if chosen as the output, would render the system unobservable.

1. $y(t)$ = ball position = $x(t)$: $\mathbf{C}* = [1\ 0\ 0]$
   The observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^*2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 64.4 & 0 & -16 \end{bmatrix} \tag{10-311}$$

   which has a rank of 3. Thus, the system is completely observable.

2. $y(t)$ = all velocity = $dx(t)/dt$: $\mathbf{C}^* = [0\ 1\ 0]$
   The observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^*2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 64.4 & 1600 \end{bmatrix} \tag{10-312}$$

   which has a rank of 3. Thus, the system is completely observable.

3. $y(t)$ = winding current = $i(t)$: $\mathbf{C}^* = [0\ 0\ 1]$
The observability matrix is

$$\mathbf{V} = \begin{bmatrix} \mathbf{C}^* \\ \mathbf{C}^*\mathbf{A}^* \\ \mathbf{C}^*\mathbf{A}^*2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -100 \\ 0 & 0 & -10,000 \end{bmatrix} \tag{10-313}$$

which has a rank of 1. Thus, the system is unobservable. The physical interpretation of this result is that, if we choose the current $i(t)$ as the measurable output, we would not be able to reconstruct the state variables from the measured information.

The interested reader can enter the data of this system into any available computer program and verify the results obtained.

▶ 10-16 STATE-FEEDBACK CONTROL

A majority of the design techniques in modern control theory is based on the state-feedback configuration. That is, instead of using controllers with fixed configurations in the forward or feedback path, control is achieved by feeding back the state variables through real constant gains. The block diagram of a system with state-feedback control is shown in Fig. 9-2(c). A more detailed block diagram is shown in Fig. 10-28.

We can show that the PID control and the tachometer-feedback control discussed earlier are all special cases of the state-feedback control scheme. In the case of tachometer-feedback control, let us consider the second-order prototype system described in Eq. (5-87). The process is decomposed by direct decomposition and is represented by the state diagram of Fig. 10-29(a). If the states $x_1(t)$ and $x_2(t)$ are physically accessible, these variables may be fed back through constant real gains $-k_1$ and $-k_2$, respectively, to form the control $u(t)$, as shown in Fig. 10-29(b). The transfer function of the system with state feedback is

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\zeta\omega_n + K_2)s + K_1} \tag{10-314}$$

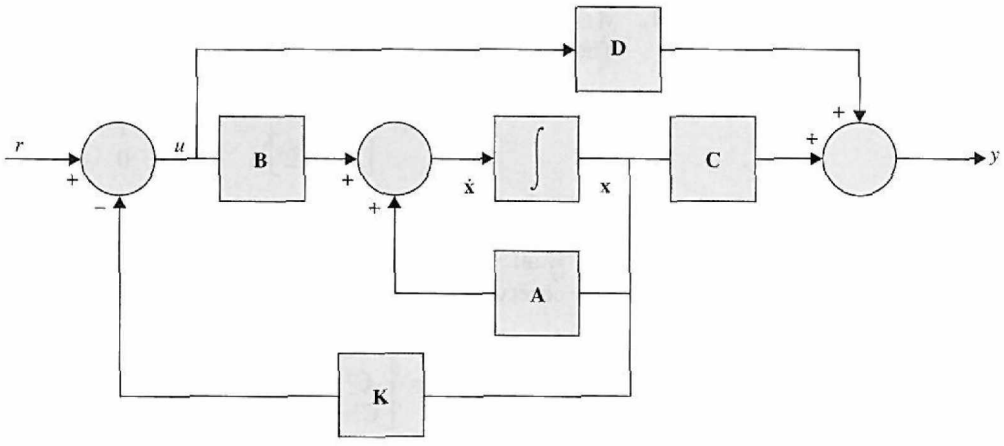

Figure 10-28 Block diagram of a control system with state feedback.

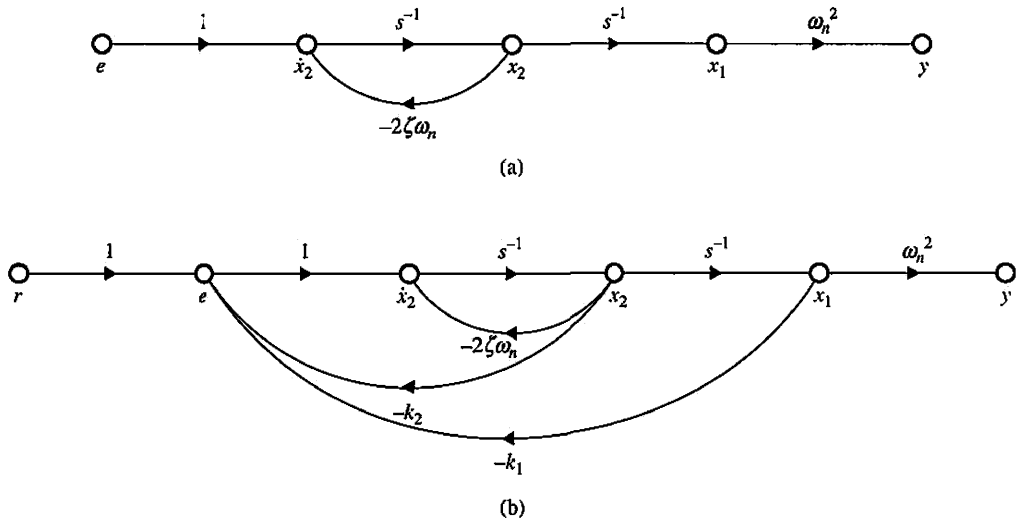

Figure 10-29 Control of a second-order system by state feedback.

For comparison purposes, we display the transfer functions of the systems with tachometer feedback and with PD control as follows:

Tachometer feedback:

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2} \tag{10-315}$$

PD control:

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2(K_P + K_D s)}{s^2 + (2\zeta\omega + K_D\omega_n^2)s + \omega_n^2 K_P} \tag{10-316}$$

Thus, tachometer feedback is equivalent to state feedback if $k_1 = \omega_n^2$ and $k_2 = K_t\omega_n^2$. Comparing Eq. (10-314) with Eq. (10-316), we see that the characteristic equation of the system with state feedback would be identical to that of the system with PD control if $k_1 = \omega_n^2 K_p$ and $k_2 = \omega_n^2 K_D$. However, the numerators of the two transfer functions are different.

The systems with zero reference input, $r(t) = 0$, are commonly known as **regulators**. When $r(t) = 0$, the control objective is to drive any arbitrary initial conditions of the system to zero in some prescribed manner, for example, "as quickly as possible." Then a second-order system with PD control is the same as state-feedback control.

It should be emphasized that the comparisons just made are all for second-order systems. For higher-order systems, the PD control and tachometer-feedback control are equivalent to feeding back only the state variables x_1 and x_2, while state-feedback control feeds back all the state variables.

Because PI control increases the order of the system by one, it cannot be made equivalent to state feedback through constant gains. We show in Section 10-18 that if we combine state feedback with integral control we can again realize PI control in the sense of state-feedback control.

10-17 POLE-PLACEMENT DESIGN THROUGH STATE FEEDBACK

When root loci are utilized for the design of control systems, the general approach may be described as that of **pole placement**; the poles here refer to that of the closed-loop transfer function, which are also the roots of the characteristic equation. Knowing the relation between the closed-loop poles and the system performance, we can effectively carry out the design by specifying the location of these poles.

The design methods discussed in the preceding sections are all characterized by the property that the poles are selected based on what can be achieved with the fixed-controller configuration and the physical range of the controller parameters. A natural question would be: *Under what condition can the poles be placed arbitrarily*? This is an entirely new design philosophy and freedom that apparently can be achieved only under certain conditions.

When we have a controlled process of the third order or higher, the PD, PI, single-stage phase-lead, and phase-lag controllers would not be able to control independently all the poles of the system, because there are only two free parameters in each of these controllers.

To investigate the condition required for arbitrary pole placement in an nth-order system, let us consider that the process is described by the following state equation:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \tag{10-317}$$

where $\mathbf{x}(t)$ is an $n \times 1$ state vector, and $u(t)$ is the scalar control. The state-feedback control is

$$u(t) = -\mathbf{K}\mathbf{x}(t) + r(t) \tag{10-318}$$

where $\mathbf{K}$ is the $1 \times n$ feedback matrix with constant-gain elements. By substituting Eq. (10-318) into Eq. (10-317), the closed-loop system is represented by the state equation

$$\frac{d\mathbf{x}(t)}{dt} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}r(t) \tag{10-319}$$

It will be shown in the following that if the pair $[\mathbf{A}, \mathbf{B}]$ is completely controllable, then a matrix $\mathbf{K}$ exists that can give an arbitrary set of eigenvalues of $(\mathbf{A} - \mathbf{B}\mathbf{K})$; that is, the n roots of the characteristic equation

$$|s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K}| = 0 \tag{10-320}$$

can be arbitrarily placed. To show that this is true, that if a system is completely controllable, it can always be represented in the controllable canonical form (CCF); that is, in Eq. (10-317),

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{10-321}$$

The feedback gain matrix $\mathbf{K}$ is expressed as

$$\mathbf{K} = [k_1 \quad k_2 \quad \cdots \quad k_n] \tag{10-322}$$

where $k_1, k_2, \ldots, k_n$ are real constants. Then,

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 - k_1 & -a_1 - k_2 & -a_2 - k_3 & \cdots & -a_{n-1} - k_n \end{bmatrix} \tag{10-323}$$

The eigenvalues of $\mathbf{A} - \mathbf{BK}$ are then found from the characteristic equation

$$|s\mathbf{I} - (\mathbf{A} - \mathbf{BK})| = s^n + (a_{n-1} + k_n)s^{n-1} + (a_{n-2} + k_{n-1})s^{n-2} + \cdots + (a_0 + k_1) = 0 \tag{10-324}$$

Clearly, the eigenvalues can be arbitrarily assigned, because the feedback gains $k_1, k_2, \ldots k_n$ are isolated in each coefficient of the characteristic equation. Intuitively, it makes sense that a system must be controllable for the poles to be placed arbitrarily. If one or more state variables are uncontrollable, then the poles associated with these state variables are also uncontrollable and cannot be moved as desired. The following example illustrates the design of a control system with state feedback.

▶ EXAMPLE 10-17-1 Consider the magnetic-ball suspension system analyzed in Section 10-15. This is a typical regulator system for which the control problem is to maintain the ball at its equilibrium position. It is shown in Section 10-15 that the system without control is unstable.

The linearized state model of the magnetic-ball system is represented by the state equation

$$\frac{d\Delta\mathbf{x}(t)}{dt} = \mathbf{A}^*\Delta\mathbf{x}(t) + \mathbf{B}^*\Delta v(t) \tag{10-325}$$

where $\Delta\mathbf{x}(t)$ denotes the linearized state vector, and $\Delta v(t)$ is the linearized input voltage. The coefficient matrices are

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix} \tag{10-326}$$

The eigenvalues of $\mathbf{A}^*$ are $s = -100, -8.025$, and 8.025. Thus, the system without feedback control is unstable.

Let us give the following design specifications:

1. The system must be stable.
2. For any initial disturbance on the position of the ball from its equilibrium position, the ball must return to the equilibrium position with zero steady-state error.
3. The time response should settle to within 5% of the initial disturbance in not more than 0.5 sec.
4. The control is to be realized by state feedback

$$\Delta v(t) = -\mathbf{K}\Delta\mathbf{x}(t) = -[k_1 \quad k_2 \quad k_3]\Delta\mathbf{x}(t) \tag{10-327}$$

where k_1, k_2, and k_3 are real constants.

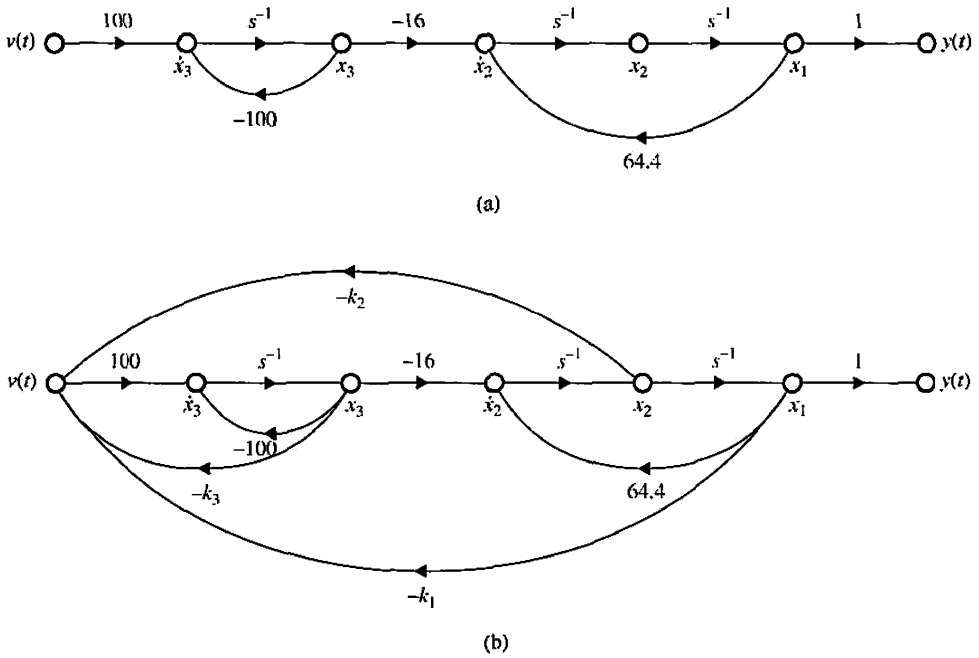

Figure 10-30 (a) State diagram of magnetic-ball-suspension system. (b) State diagram of magnetic-ball-suspension system with state feedback.

A state diagram of the "open-loop" ball-suspension system is shown in Fig. 10-30(a), and the same of the "closed-loop" system with state feedback is shown in Fig. 10-30(b).

We must select the desired location of the eigenvalues of $(s\mathbf{I} - \mathbf{A}^* + \mathbf{B}^*\mathbf{K})$ so that requirement 3 in the preceding list on the time response is satisfied. Without entirely resorting to trial and error, we can start with the following decisions:

1. The system dynamics should be controlled by two dominant roots.
2. To achieve a relatively fast response, the two dominant roots should be complex.
3. The damping that is controlled by the real parts of the complex roots should be adequate, and the imaginary parts should be high enough for the transient to die out sufficiently fast.

After a few trial-and-error runs, using the **ACSYS/MATLAB** tool, we found that the following characteristic equation roots should satisfy the design requirements:

$$s = -20 \quad s = -6 + j4.9 \quad s = -6 - j4.9$$

The corresponding characteristic equation is

$$s^3 + 32s^2 + 300s + 1200 = 0 \tag{10-328}$$

The characteristic equation of the closed-loop system with state feedback is written

$$|s\mathbf{I} - \mathbf{A}^* + \mathbf{B}^*\mathbf{K}| = \begin{vmatrix} s & -1 & 0 \\ -64.4 & s & 16 \\ 100k_1 & 100k_2 & s + 100 + 100k_3 \end{vmatrix}$$
$$= s^3 + 100(k_3 + 1)s^2 - (64.4 + 1600k_2)s - 1600k_1 - 6440(k_3 + 1) = 0 \tag{10-329}$$

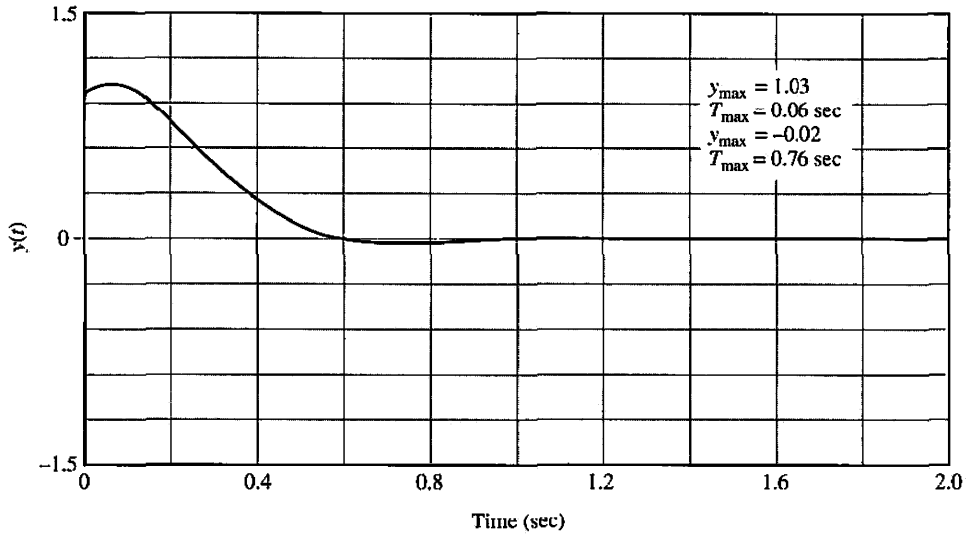

Figure 10-31 Output response of magnetic-ball-suspension system with state feedback, subject to initial condition $y(0) = x_1(0) = 1$.

which can also be obtained directly from Fig. 10-30(b) using the SFG gain formula. Equating like coefficients of Eqs. (10-328) and (10-329), we get the following simultaneous equations:

$$\begin{aligned} 100(k_3 + 1) &= 32 \\ -64.4 - 1600k_2 &= 300 \\ -1600k_1 - 6440(k_3 + 1) &= 1200 \end{aligned} \tag{10-330}$$

Solving the last three equations, and being assured that the solutions exist and are unique, we get the feedback-gain matrix

$$\mathbf{K} = [k_1 \quad k_2 \quad k_3] = [-2.038 \quad -0.22775 \quad -0.68] \tag{10-331}$$

Fig. 10.31 shows the output response $y(t)$ when the system is subject to the initial condition

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{10-331}$$

◄

► **EXAMPLE 10-17-2** In this example, we shall design a state-feedback control for the second-order sun-seeker system treated in Example 4-11-2 and throughout Chapter 9. The CCF state diagram of the process with $K = 1$ is shown in Fig. 10-32(a). The problem involves the design of a state-feedback control with

$$\theta_e(t) = -\mathbf{K}\mathbf{x}(t) = -[k_1 \quad k_2]\mathbf{x}(t) \tag{10-332}$$

The state equations are represented in vector-matrix form as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\theta_e(t) \tag{10-333}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -25 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{10-334}$$

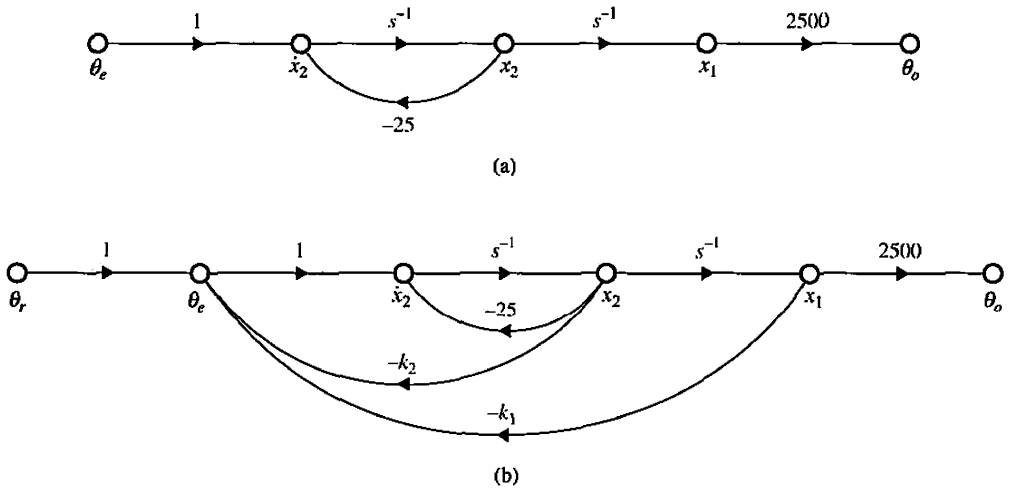

Figure 10-32 (a) State diagram of second-order sun-seeker system. (b) State diagram of second-order sun-seeker system with state feedback.

The output equation is

$$\theta_o(t) = \mathbf{C}\mathbf{x}(t) \tag{10-335}$$

where

$$\mathbf{C} = [1 \quad 0] \tag{10-336}$$

The design objectives are as follows:

1. The steady-state error due to a step function input should equal 0.
2. With the state-feedback control, the unit-step response should have minimum overshoot, rise time, and settling time.

The transfer function of the system with state feedback is written

$$\frac{\Theta_o(s)}{\Theta_r(s)} = \frac{2500}{s^2 + (25 + k_2)s + k_1} \tag{10-337}$$

Thus, for a step input, if the output has zero steady-state error, the constant terms in the numerator and denominator must be equal to each other—that is, $k_1 = 2500$. This means that, while the system is completely controllable, we cannot arbitrarily assign the two roots of the characteristic equation, which is now

$$s^2 + (25 + k_2)s + 2500 = 0 \tag{10-338}$$

In other words, only one of the roots of Eq. (10-338) can be arbitrarily assigned. The problem is solved using **ACSYS**. After a few trial-and-error runs, we found out that the maximum overshoot, rise time, and settling time are all at a minimum when $k_2 = 75$. The two roots are $s = -50$ and -50. The attributes of the unit-step response are

$$\text{maximum overshoot} = 0\% \quad t_r = 0.06717 \text{ sec} \quad t_s = 0.09467 \text{ sec}$$

The state-feedback gain matrix is

$$\mathbf{K} = [2500 \quad 75] \tag{10-339}$$

The lesson that we learned from this illustrative example is that state-feedback control generally produces a system that is type 0. For the system to track a step input without steady-state error, which requires a type 1 or higher-type system, the feedback gain k_1 of the system in the CCF state diagram

cannot be assigned arbitrarily. This means that, for an nth-order system, only $n - 1$ roots of the characteristic equation can be placed arbitrarily.

10-18 STATE FEEDBACK WITH INTEGRAL CONTROL

The state-feedback control structured in the preceding section has one deficiency in that it does not improve the type of the system. As a result, the state-feedback control with constant-gain feedback is generally useful only for regulator systems for which the system does not track inputs, if all the roots of the characteristic equation are to be placed at will.

In general, most control systems must track inputs. One solution to this problem is to introduce integral control, just as with PI controller, together with the constant-gain state feedback. The block diagram of a system with constant-gain state feedback and integral control feedback of the output is shown in Fig. 10-33. The system is also subject to a noise input $n(t)$. For a SISO system, the integral control adds one integrator to the system. As shown in Fig. 10-33, the output of the $(n + 1)$st integrator is designated as x_{n+1}. The dynamic equations of the system in Fig. 10-33 are written as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{E}n(t) \tag{10-340}$$

$$\frac{dx_{n+1}(t)}{dt} = r(t) - y(t) \tag{10-341}$$

$$y(t) = \mathbf{C}\mathbf{x}(t) + Du(t) \tag{10-342}$$

where $\mathbf{x}(t)$ is the $n \times 1$ state vector; $u(t)$ and $y(t)$ are the scalar actuating signal and output, respectively; $r(t)$ is the scalar reference input; and $n(t)$ is the scalar disturbance input. The coefficient matrices are represented by **A, B, C**, D and **E**, with appropriate dimensions. The actuating signal $u(t)$ is related to the state variables through constant-state and integral feedback,

$$u(t) = -\mathbf{K}\mathbf{x}(t) - k_{n+1}x_{n+1}(t) \tag{10-343}$$

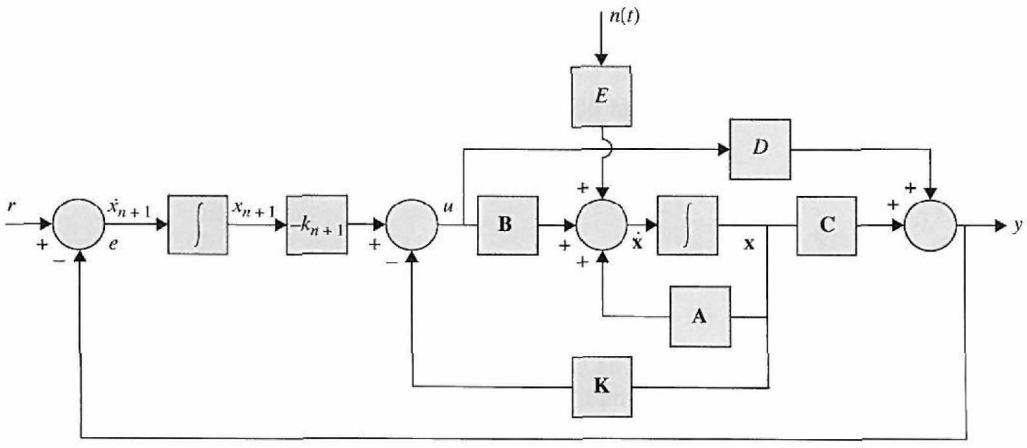

Figure 10-33 Block diagram of a control system with state feedback and integral output feedback.

where

$$\mathbf{K} = [k_1 \quad k_2 \quad k_3 \quad \cdots \quad k_n] \tag{10-344}$$

with constant real gain elements, and k_{n+1} is the scalar integral-feedback gain.

Substituting Eq. (10-343) into Eq. (10-340) and combining with Eq. (10-341), the $n + 1$ state equations of the overall system with constant-gain and integral feedback are written

$$\frac{d\bar{\mathbf{x}}(t)}{dt} = (\bar{\mathbf{A}} - \bar{\mathbf{B}}\,\bar{\mathbf{K}})\bar{\mathbf{x}}(t) + \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} r(t) + \bar{\mathbf{E}}n(t) \tag{10-345}$$

where

$$\bar{\mathbf{x}}(t) = \begin{bmatrix} \dfrac{d\mathbf{x}(t)}{dt} \\ \dfrac{dx_{n+1}(t)}{dt} \end{bmatrix} \quad (n+1) \times 1 \tag{10-346}$$

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & 0 \end{bmatrix} \quad (n+1) \times (n+1) \qquad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ D \end{bmatrix} \quad (n+1) \times 1 \tag{10-347}$$

$$\bar{\mathbf{K}} = [K \quad k_{n+1}] = [k_1 \quad k_2 \quad \cdots \quad k_n \quad k_{n+1}] \quad 1 \times (n+1) \tag{10-348}$$

$$\bar{\mathbf{E}} = \begin{bmatrix} \mathbf{E} \\ 0 \end{bmatrix} \quad [(n+1) \times 1] \tag{10-349}$$

Substituting Eq. (10-343) into Eq. (10-342), the output equation of the overall system is written

$$y(t) = \bar{\mathbf{C}}\bar{\mathbf{x}}(t) \tag{10-350}$$

where

$$\bar{\mathbf{C}} = [\mathbf{C} - D\mathbf{K} \quad D\mathbf{K}] \quad [1 \times (n+1)] \tag{10-351}$$

The design objectives are as follows:

1. The steady-state value of the output $y(t)$ follows a step-function input with zero error; that is,

$$e_{ss} = \lim_{t \to \infty} e(t) = 0 \tag{10-352}$$

2. The $n + 1$ eigenvalues of $(\bar{\mathbf{A}} - \bar{\mathbf{B}}\,\bar{\mathbf{K}})$ are placed at desirable locations. For the last condition to be possible, the pair $[\bar{\mathbf{A}}, \bar{\mathbf{B}}]$ must be completely controllable.

The following example illustrates the applications of state-feedback with integral control.

▶ EXAMPLE 10-18-1 We have shown in Example 10-17-2 that, with constant-gain state-feedback control, the second-order sun-seeker system can have only one of its two roots placed at will for the system to track a step input without steady-state error. Now let us consider the same second-order sun-seeker system in Example 10-17-2, except that an integral control is added to the forward path. The state diagram of the overall system is shown in Fig. 10-34. The coefficient matrices are

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -25 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{C} = [2500 \quad 0] \quad D = 0 \tag{10-353}$$

From Eq. (10-347),

$$\overline{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -25 & 0 \\ -2500 & 0 & 0 \end{bmatrix} \quad \overline{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ D \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \tag{10-354}$$

We can show that the pair $[\overline{\mathbf{A}}, \overline{\mathbf{B}}]$ is completely controllable. Thus, the eigenvalues of $(s\mathbf{I} - \overline{\mathbf{A}} + \overline{\mathbf{B}}\,\overline{\mathbf{K}})$ can be arbitrarily placed. Substituting $\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, and $\overline{\mathbf{K}}$ in the characteristic equation of the closed-loop system with state and integral feedback, we have

$$\begin{aligned} |s\mathbf{I} - \overline{\mathbf{A}} + \overline{\mathbf{B}}\,\overline{\mathbf{K}}| &= \begin{vmatrix} s & -1 & 0 \\ k_1 & s + 25 + k_2 & k_3 \\ -2500 & 0 & s \end{vmatrix} \\ &= s^3 + (25 + k_2)s^2 + k_1 s + 2500 k_3 = 0 \end{aligned} \tag{10-355}$$

which can also be found from Fig. 10-34 using the SFG gain formula.

The design objectives are as follows:

1. The steady-state output must follow a step function input with zero error.
2. The rise time and settling time must be less than 0.05 sec.
3. The maximum overshoot of the unit-step response must be less than 5%.

Because all three roots of the characteristic equation can be placed arbitrarily, it is not realistic to require minimum rise and settling times, as in Example 10-17-2.

Again, to realize a fast rise time and settling time, the roots of the characteristic equation should be placed far to the left in the s-plane, and the natural frequency should be high. Keep in mind that *roots with large magnitudes will lead to high gains for the state-feedback matrix.*

The **ACSYS/MATLAB** software was used to carry out the design. After a few trial-and-error runs, the design specifications can be satisfied by placing the roots at

$$s = -200 \quad -50 + j50 \quad \text{and} \quad -50 - j50$$

The desired characteristic equation is

$$s^3 + 300s^2 + 25{,}000s + 1{,}000{,}000 = 0 \tag{10-356}$$

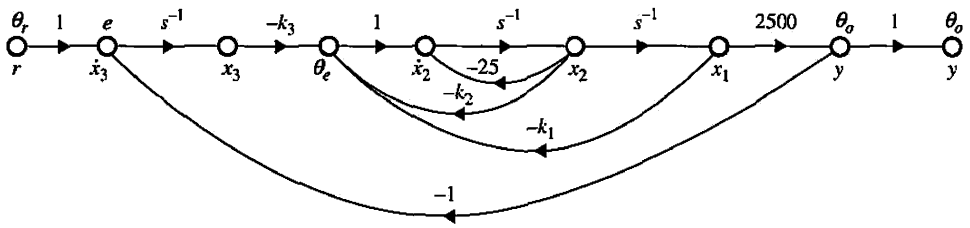

Figure 10-34 Sun-seeker system with state feedback and integral control in Example 10-18-1.

Equating like coefficients of Eqs. (10-355) and (10-356), we get

$$k_1 = 25,000 \quad k_2 = 275 \quad \text{and} \quad k_3 = 400$$

The attributes of the unit-step response are as follows:

Maximum overshoot = 4%

$t_r = 0.03247$ sec

$t_s = 0.04667$ sec

Notice that the high feedback gain of k_1, which is due to the large values of the roots selected, may pose physical problems; if so, the design specifications may have to be revised.

► **EXAMPLE 10-18-2** In this example we illustrate the application of state-feedback with integral control to a system with a disturbance input.

Consider a dc-motor control system that is described by the following state equations:

$$\frac{d\omega(t)}{dt} = \frac{-B}{J}\omega(t) + \frac{K_i}{J}i_a(t) - \frac{1}{J}T_L \tag{10-357}$$

$$\frac{di_a(t)}{dt} = \frac{-K_b}{L}\omega(t) - \frac{R}{L}i_a(t) + \frac{1}{L}e_a(t) \tag{10-358}$$

where

$i_a(t)$ = armature current, A

$e_a(t)$ = armature applied voltage, V

$\omega(t)$ = motor velocity, rad/sec

B = viscous-friction coefficient of motor and load = 0

J = moment of inertia of motor and load = 0.02 N-m/rad/sec^2

K_i = motor torque constant = 1 N-m/A

K_b = motor back-emf constant = 1 V/rad/sec

T_L = constant load torque(magnitude not known), N-m

L = armature inductance = 0.005 H

R = armature resistance = 1Ω

The output equation is

$$y(t) = \mathbf{C}\mathbf{x}(t) = [1 \quad 0]\mathbf{x}(t) \tag{10-359}$$

The design problem is to find the control $u(t) = e_a(t)$ through state feedback and integral control such that

1. $\lim_{t\to\infty} i_a(t) = 0 \quad \text{and} \quad \lim_{t\to\infty} \frac{d\omega(t)}{dt} = 0$ (10-360)
2. $\lim_{t\to\infty} \omega(t) = \text{step input } r(t) = u_s(t)$ (10-361)
3. The eigenvalues of the closed-loop system with state feedback and integral control are at $s = -300$, $-10 + j10$, and $-10 - j10$.

Let the state variables be defined as $x_1(t) = \omega(t)$ and $x_2(t) = i_a(t)$. The state equations in Eqs. (10-357) and (10-358) are written in vector-matrix form:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{E}n(t) \tag{10-362}$$

where $n(t) = T_L u_s(t)$.

$$\mathbf{A} = \begin{bmatrix} -\frac{B}{J} & \frac{K_i}{J} \\ -\frac{K_b}{L} & -\frac{R}{L} \end{bmatrix} = \begin{bmatrix} 0 & 50 \\ -200 & -200 \end{bmatrix} \tag{10-363}$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \end{bmatrix} \tag{10-364}$$

$$\mathbf{E} = \begin{bmatrix} -\frac{1}{J} \\ 0 \end{bmatrix} = \begin{bmatrix} -50 \\ 0 \end{bmatrix} \tag{10-365}$$

From Eq. (10-347),

$$\overline{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & 0 \\ -\mathbf{C} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 50 & 0 \\ -200 & -200 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad \overline{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \\ 0 \end{bmatrix} \tag{10-366}$$

$$\overline{\mathbf{C}} = [\mathbf{C} \quad 0] = [1 \quad 0 \quad 0] \quad \overline{\mathbf{E}} = \begin{bmatrix} \mathbf{E} \\ 0 \end{bmatrix} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix} \tag{10-367}$$

The control is given by

$$u(t) = -\mathbf{K}\mathbf{x}(t) - k_{n+1}x_{n+1}(t) = \overline{\mathbf{K}}\overline{\mathbf{x}}(t) \tag{10-368}$$

where

$$\overline{\mathbf{K}} = [k_1 \quad k_2 \quad k_3] \tag{10-369}$$

Fig. 10-35 shows the state diagram of the overall designed system.

The coefficient matrix of the closed-loop system is

$$\overline{\mathbf{A}} - \overline{\mathbf{B}}\,\overline{\mathbf{K}} = \begin{bmatrix} 0 & 50 & 0 \\ -200 - 200k_1 & -200 - 200k_2 & -200k_3 \\ -1 & 0 & 0 \end{bmatrix} \tag{10-370}$$

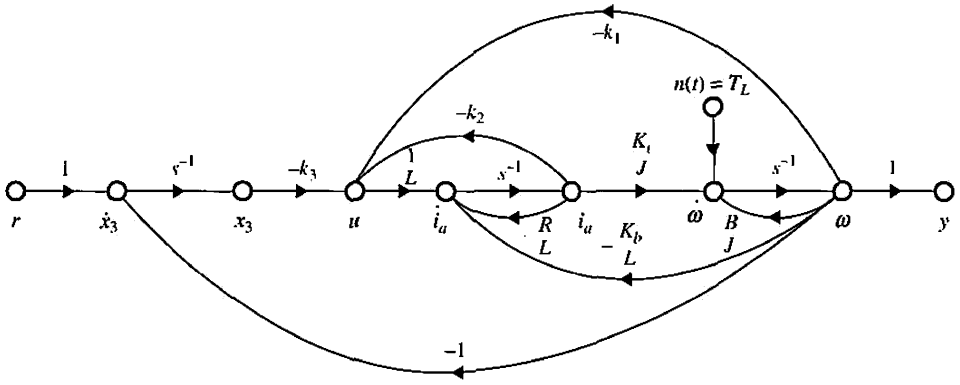

Figure 10-35 DC-motor control system with state feedback and integral control and disturbance torque in Example 10-18-2.

The characteristic equation is

$$|s\mathbf{I} - \overline{\mathbf{A}} + \overline{\mathbf{B}}\,\overline{\mathbf{K}}| = s^3 + 200(1 + k_2)s^2 + 10,000(1 + k_1)s - 10,000k_3 = 0 \tag{10-371}$$

which is more easily determined by applying the gain formula of SFG to Fig. 10-35.

For the three roots assigned, the last equation must equal

$$s^3 + 320s^2 + 6,200s + 60,000 = 0 \tag{10-372}$$

Equating the like coefficients of Eqs. (10-371) and (10-372), we get

$$k_1 = -0.38 \quad k_2 = 0.6 \quad k_3 = -6.0$$

Applying the SFG gain formula to Fig. 10-35 between the inputs $r(t)$ and $n(t)$ and the states $\omega(t)$ and $i_a(t)$, we have

$$\begin{bmatrix} \Omega(s) \\ I_a(s) \end{bmatrix} = \frac{1}{\Delta_c(s)} \begin{bmatrix} -\frac{1}{J}\left(s^2 + \frac{R}{L}s + \frac{k_2}{L}s\right) & -\frac{k_3 K_i}{JL} \\ -\frac{1}{J}\left(-\frac{K_b}{L}s - \frac{k_1}{L}s + \frac{k_3}{L}\right) & -\frac{k_3}{L}\left(s + \frac{B}{J}\right) \end{bmatrix} \begin{bmatrix} \frac{T_L}{s} \\ \frac{1}{s} \end{bmatrix} \tag{10-373}$$

where $\Delta_c(s)$ is the characteristic polynomial given in Eq. (10-372).

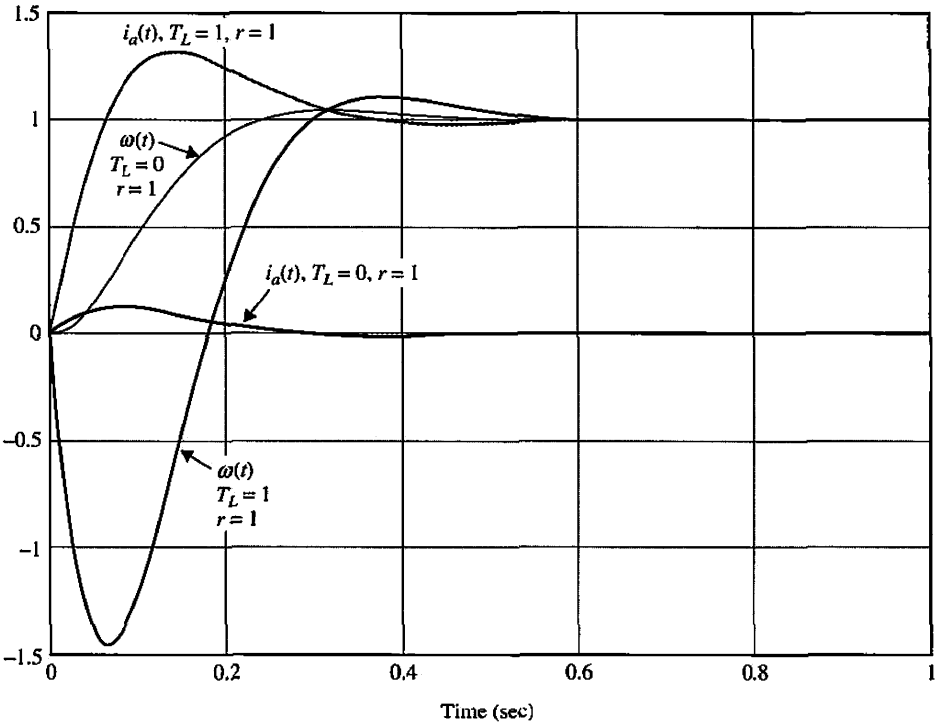

Figure 10-36 Time responses of dc-motor control system with state feedback and integral control and disturbance torque in Example 10-18-2.

Applying the final-value theorem to the last equation, the steady-state values of the state variables are found to be

$$\lim_{t\to\infty}\begin{bmatrix}\omega(t)\\ i_a(t)\end{bmatrix} = \lim_{s\to 0} s\begin{bmatrix}\Omega(s)\\ I_a(s)\end{bmatrix} = \begin{bmatrix}0 & K_i\\ 1 & B\end{bmatrix}\begin{bmatrix}T_L\\ 1\end{bmatrix} = \begin{bmatrix}1\\ T_L\end{bmatrix} \tag{10-374}$$

Thus, the motor velocity $\omega(t)$ will approach the constant reference input step function $r(t) = u_s(t)$ as t approaches infinity, independent of the disturbance torque T_L. Substituting the system parameters into Eq. (10-373), we get

$$\begin{bmatrix}\Omega(s)\\ I_a(s)\end{bmatrix} = \frac{1}{\Delta_c(s)}\begin{bmatrix}-50(s+320)s & 60,000\\ 6200s+60,000 & 1,200s\end{bmatrix}\begin{bmatrix}\frac{T_L}{s}\\ \frac{1}{s}\end{bmatrix} \tag{10-375}$$

Fig. 10-36 shows the time responses of $\omega(t)$ and $i_a(t)$ when $T_L = 1$ and $T_L = 0$. The reference input is a unit-step function.

10-19 MATLAB TOOLS AND CASE STUDIES

In this section we present a MATLAB tool to solve most problems addressed in this chapter. The reader is encouraged to apply this tool to all the problems identified by a MATLAB Toolbox in the left margin of the text throughout this chapter. We use MATLAB's Symbolic Tool to solve some of the initial problems in this chapter involving inverse Laplace transformations. We will also use MATLAB to convert from transfer functions to state-space representation. These programs allow the user to conduct the following tasks:

- Enter the state matrices.
- Find the system's characteristic polynomial, eigenvalues, and eigenvectors.
- Find the similarity transformation matrices.
- Examine the system controllability and observability properties.
- Obtain the step, impulse, and natural (response to initial conditions) responses, as well as the time response to any function of time.
- Use MATLAB Symbolic Tool to find the state-transition matrix using the inverse Laplace command.
- Convert a transfer function to state-space form or vice versa.

To better illustrate how to use the software, let us go through some of the steps involved in solving earlier examples in this chapter.

10-19-1 Description and Use of the State-Space Analysis Tool

The State-Space Analysis Tool (statetool) consists of a number of m-files and GUIs for the analysis of state-space systems. The statetool can be invoked from the Automatic Control Systems launch applet (**ACSYS**) by clicking on the appropriate button. You will then see the window pictured in Fig. 10-37. We use the example in Section 10-16 and Examples 10-5-1 and 10-5-2 to describe how to use the statetool.

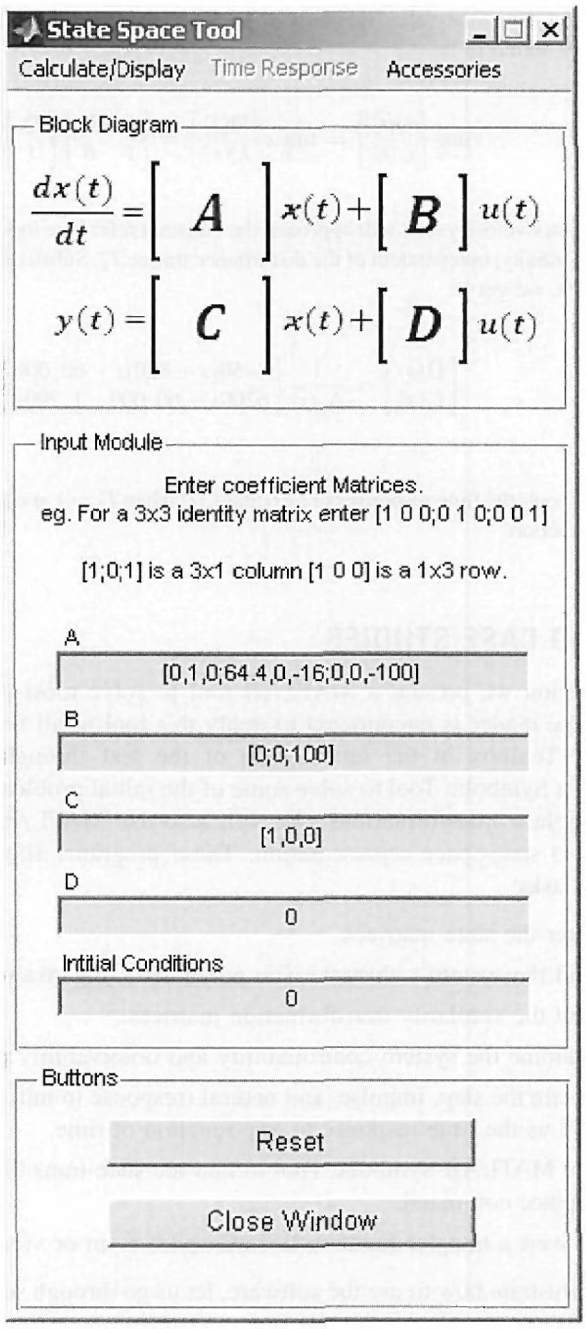

Figure 10-37 The State-Space Analysis window.

First consider the example in Section 10-14. To enter the following coefficient matrices,

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 \\ 64.4 & 0 & -16 \\ 0 & 0 & -100 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix} \quad \mathbf{C}^* = [1 \quad 0 \quad 0] \tag{10-378}$$

enter the values in the appropriate edit boxes. Note that the default value of initial conditions is set to zero and you do not have to adjust it for this example. Follow the instructions on the screen very carefully. The elements in the row of a matrix may be separated by a space or a comma, while the rows themselves must be separated by a semicolon. For example, to enter matrix **A**, enter [0,1,0;64.4,0,-16;0,0,-100] in the **A** edit box, and to enter matrix **B**, enter [0;0;100] in the **B** edit box, as shown in Fig. 10-38. In this case, the **D** matrix is set to zero (default value). To find the characteristic Eq. (10-270),

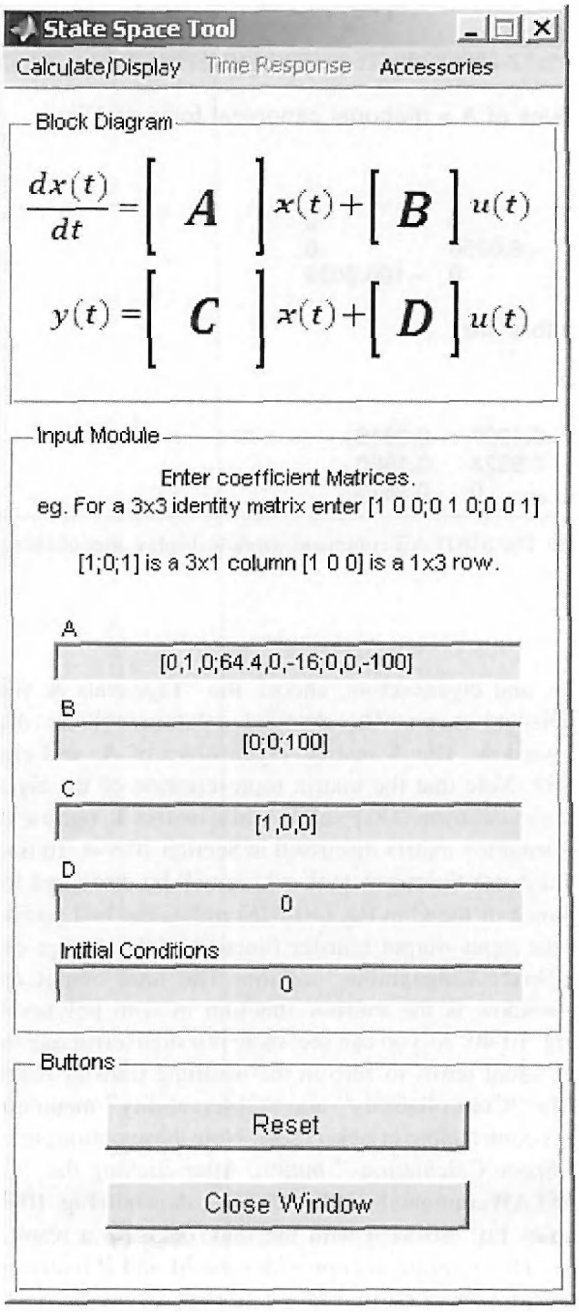

Figure 10-38 Inputing values in the State-Space window.

```
The A matrix is:

Amat =

         0    1.0000         0
   64.4000         0   -16.0000
         0         0  -100.0000

Characteristic polynomial:

ans =

s^3+100*s^2-2265873562520787/35184372088832*s-6440

Eigenvalues of A = diagonal canonical form of A is:

Abar =

    8.0250         0         0
         0   -8.0250         0
         0         0  -100.0000

Eigenvectors are

T =

    0.1237   -0.1237   -0.0016
    0.9923    0.9923    0.1590
         0         0    0.9873
```

Figure 10-39 The MATLAB command window display after clicking the "Eigenvals & vects of A" button.

eigenvalues, and eigenvectors, choose the "Eigenvals & vects of A" option from the Calculate/Display menu. The detailed solution will be displayed on the MATLAB command window. The **A** matrix, eigenvalues of **A**, and eigenvectors of **A** are shown in Fig. 10-39. Note that the matrix representation of the eigenvalues corresponds to the diagonal canonical form (DCF) of **A**, while matrix **T**, representing the eigenvectors, is the DCF transformation matrix discussed in Section 10-9-4. To find the state-transition matrix $\phi(t)$, you must use the tfsym tool, which will be discussed in Section 10-19-2.

The choice of the **C** in Eq. (10-376) makes the ball position the output $y(t)$ for input $v(t)$. Then the input–output transfer function of the system can be obtained by choosing the "State-Space Calculations" option. The final output appearing in the MATLAB command window is the transfer function in both polynomial and factored forms, as shown in Fig. 10-40. As you can see, there is a small error due to numerical simulation. You may set the small terms to zero in the resulting transfer function to get Eq. (10-309).

Click the "Controllability" and "Observability" menu options to determine whether the system is controllable or observable. Note these options are only enabled after pressing the "State-Space Calculations" button. After clicking the "Controllability" button, you get the MATLAB command window display, shown in Fig. 10-41. The **S** matrix in this case is the same as Eq. (10-310) with the rank of 3. As a result, the system is completely controllable. The program also provides the **M** and **P** matrices and the system controllability canonical form (CCF) representation as defined in Section 10-9.

```
State-space model is:

a =

       x1   x2     x3
x1      0    1      0
x2   64.4    0    -16
x3      0    0   -100


b =

       u1
x1      0
x2      0
x3    100


c =

     x1   x2   x3
y1    1    0    0


d =

     u1
y1    0

Continuous-time model.
Characteristic polynomial:

ans =

s^3+100*s^2-2265873562520787/35184372088832*s - 6440

Equivalent transfer-function model is:

Transfer function:
4.263e - 014 s^2 + 8.527e-014 s - 1600
-------------------------------------
     s^3 + 100s^2 - 64.4s - 6440

Pole, zero form:

Zero/pole/gain:
4.2633e - 014 (s+1.937e008)(s - 1.937e008)
------------------------------------------
        (s+100)(s+8.025)(s - 8.025)
```

Figure 10-40 The MATLAB command window after clicking the "State-Space Calculations" button.

```
The Controllibility matrix [B AB A^2B ...] is =

Smat =

     0         0    -1600
     0    -1600    160000
   100   -10000   1000000

The system is therefore controllable, rank of S matrix is =

rankS =

   3

Mmat =

  -64.4000   100.0000   1.0000
  100.0000     1.0000        0
    1.0000          0        0

The controllability canonical form (CCF) transformation matrix is:

Ptran =

   -1600         0     0
       0    -1600     0
   -6440         0   100

The transformed matrices using CCF are:

Abar =

1.0e+003 *

       0    0.0010         0
       0         0    0.0010
  6.4400    0.0644   -0.1000

Bbar =

  0
  0
  1

Cbar =

  -1600     0     0

Dbar =

  0
```

Figure 10-41 The MATLAB command window after clicking the "Controllability" button.

```
The observability matrix (transpose:[C CA CA^2 ...]) is =

 Vmat =

   1.0000         0         0
        0    1.0000         0
  64.4000         0  -16.0000

The system is therefore observable, rank of V matrix is =

rankV =

   3

Mmat =

 -64.4000  100.0000    1.0000
 100.0000    1.0000         0
   1.0000         0         0

The observability canonical form (OCF) transformation matrix is:

Qtran =

         0         0    1.0000
         0    1.0000 -100.0000
   -0.0625    6.2500 -625.0000

The transformed matrices using OCF are:

Abar =

1.0e+003 *

  0.0000   -0.0000    6.4400
  0.0010   -0.0000    0.0644
       0    0.0010   -0.1000

Bbar =

   -1600
       0
       0

Cbar =

  0   0   1

Dbar =

  0
```

Figure 10-42 The MATLAB command window after clicking the "Observability" button.

Once you choose the "Observability" option, the system observability is assessed in the MATLAB command window, as shown in Fig. 10-42. The system is completely observable, since the **V** matrix has a rank of 3. Note the **V** matrix in Fig. 10-42 is the same as in Eq. (10-311). The program also provides the **M** and **Q** matrices and the system

observability canonical form (OCF) representation as defined in Section 10-9. As an exercise, the user is urged to reproduce Eqs. (10-312) and (10-313) using this software.

You may obtain the output $y(t)$ natural time response (response to initial conditions only), the step response, the impulse response, or the time response to any other input function by choosing the appropriate option from the Time Response menu.

The statetool program may be used on all the examples identified by a MATLAB Toolbox in the left margin of the text throughout this chapter, except problems involving inverse Laplace transformations and closed-form solutions. To address the analytical solutions, we need to use the tfsym tool, which requires the Symbolic Tool of MATLAB.

10-19-2 Description and Use of tfsym for State-Space Applications

You may run the Transfer Function Symbolic Tool by clicking the "Transfer Function Symbolic" button in the **ACSYS** window. You should get the window in Fig. 10-43. For

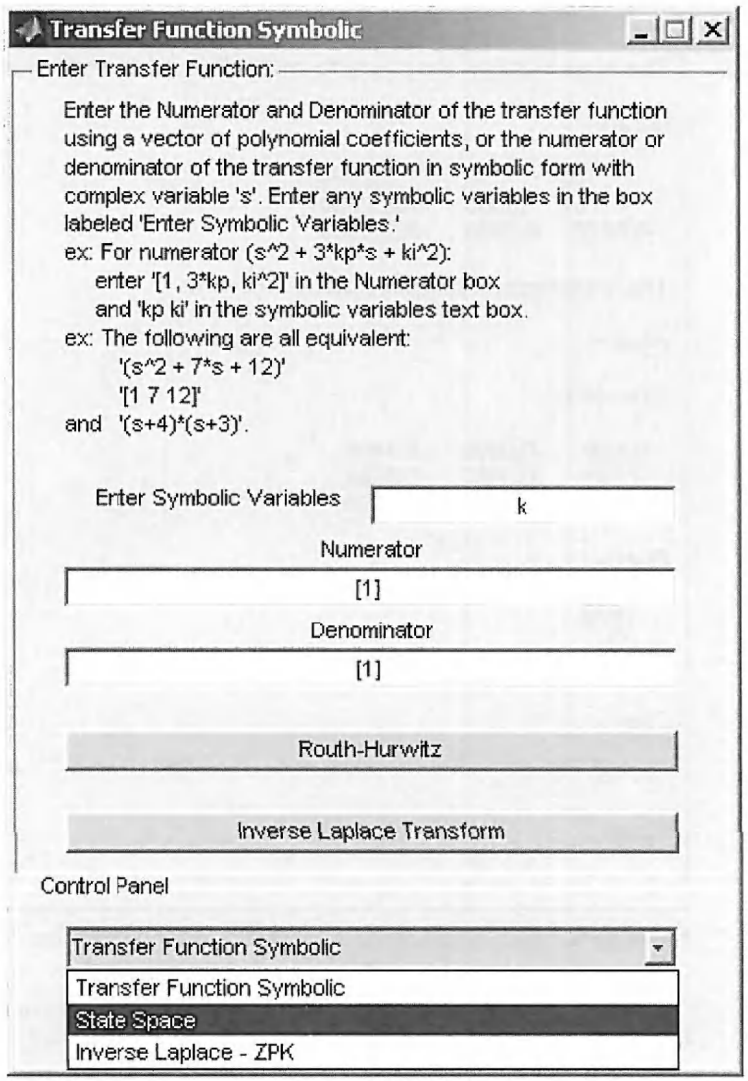

Figure 10-43 The Transfer Function Symbolic window.

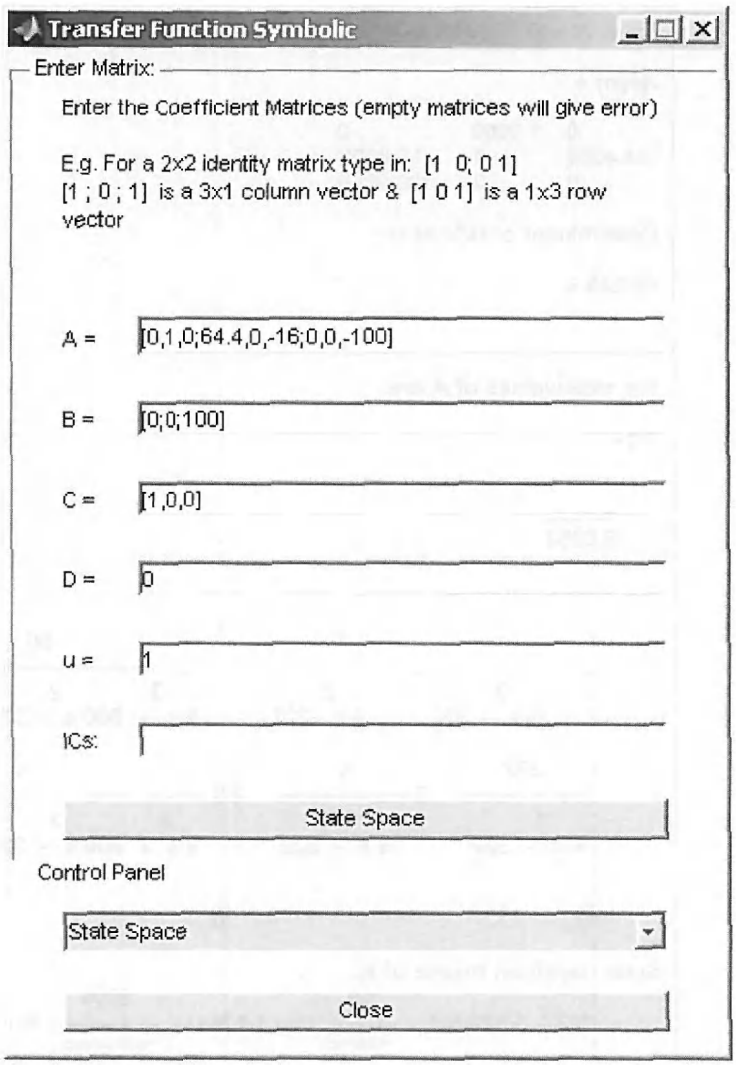

Figure 10-44 Inputting values into the Transfer Function Symbolic window.

this example we will use the State-Space mode. Choose the appropriate option from the drop-down menu as shown in Fig. 10-43.

Let us continue our solution to the example in Section 10-16. Fig. 10-44 shows the input of the matrices for this example into the state-space window. The input and output displays in the MATLAB command window are selectively shown in Fig. 10-45. Note that at first glance the $(s\mathbf{I}\text{-}\mathbf{A})^{-1}$ and $\phi(t)$ matrices may appear different from Eqs. (10-366) and (10-367). However, after minor manipulations, you may be able to verify that they are the same. This difference in representation is because of MATLAB symbolic approach. You may further simplify these matrices by using the "simple" command in the MATLAB command window. For example, to simplify $\phi(t)$, type "simple(phi)" in the MATLAB command window. If the desired format has not been achieved, you may have reached the software limit.

```
Enter A = [0 1 0;64.4 0-16;0 0 -100]

Asym =

       0   1.0000         0
 64.4000         0   -16.0000
       0         0  -100.0000

Determinant of (s*I-A) is:

detSIA =

s^3+100*s^2-322/5*s-6440

the eigenvalues of A are:

eigA =

-100.0000
    8.0250
   -8.0250

Inverse of (s*I-A) is:

[          s                5                                  80                  ]
[5 ----------------- , ----------------- , ------------------------------------------]
[          2                2                3         2                           ]
[     5 s  -  322      5 s - 322       5 s  +  500 s  -  322 s  -  32200]
[                                                                                   ]
[     322                 s                                  s                    ]
[----------------- , 5 ----------------- , -80 ------------------------------------]
[    2                   2                3         2                           ]
[ 5 s  -  322        5 s  -  322       5 s  +  500 s  -  322 s  -  32200]
[                                                                                   ]
[                                                                          1        ]
[0 ,                                  0 ,                                 ---------]
[                                                                        s + 100]

State transition matrix of A:
[                                40                                2000                 40          ]
[%2 ,  1/322 %1, - ---------- exp (-100 t) - ---------------- %1 + --------- %2]
[                                24839                             3999079              24839       ]
[                                                                                                  ]
[                                 4000                              4000                           ]
[1/5 %1 ,      %2 ,    ---------- exp (-100 t) - -------- %2 + 8/24839 %1]
[                                 24839                             24839                          ]
[                                                                                                  ]
[0 ,                                        0 ,                                exp (-100 t)]

                                        1/2                  1/2
                          %1 : = 1610    sinh (1/5 1610          t)

                                                      1/2
                                %2 : = cosh (1/5 1610    t)

Transfer function between u(t) and y(t) is:

                                  8000
                  - ---------------------------------
                      3    2
                    5 s + 500 s - 322 s - 32200
```

Figure 10-45 Selective display of the MATLAB command window for the tfsym tool.

10-20 SUMMARY

This chapter was devoted to the state-variable analysis of linear systems. The fundamentals on state variables and state equations were introduced in Chapters 2 and 3, and formal discussions on these subjects were covered in this chapter. Specifically, the state-transition matrix and state-transition equations were introduced and the relationship between the state equations and transfer functions was established. Given the transfer function of a linear system, the state equations of the system can be obtained by decomposition of the transfer function. Given the state equations and the output equations, the transfer function can be determined either analytically or directly from the state diagram.

Characteristic equations and eigenvalues were defined in terms of the state equations and the transfer function. Eigenvectors of **A** were also defined for distinct and multiple-order eigenvalues. Similarity transformations to controllability canonical form (CCF), observability canonical form (OCF), diagonal canonical form (DCF), and Jordan canonical form (JCF) were discussed. State controllability and observability of linear time-invariant systems were defined and illustrated, and a final example, on the magnetic-ball-suspension system, summarized the important elements of the state-variable analysis of linear systems.

The MATLAB software tools statetool, tfsym, and tfcal were described in the last section. The program functionality was discussed with two examples. Together these tools can solve most of the homework problems and examples in this chapter.

REVIEW QUESTIONS

1. What are the components of the dynamic equations of a linear system?
2. Given the state equations of a linear system as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

give two expressions of the state-transition matrix $\phi(t)$ in terms of **A**.

3. List the properties of the state-transition matrix $\phi(t)$.
4. Given the state equations as in Review Question 2, write the state-transition equation.
5. List the advantages of expressing a linear system in the controllability canonical form (CCF). Give an example of **A** and **B** in CCF.
6. Given the state equations as in Review Question 2, give the conditions for **A** and **B** to be transformable into CCF.
7. Express the characteristic equation in terms of the matrix **A**.
8. List the three methods of decomposition of a transfer function.
9. What special forms will the state equations be in if the transfer function is decomposed by direct decomposition?
10. What special form will the state equations be in if the transfer function is decomposed by parallel decomposition?
11. What is the advantage of using cascade decomposition?
12. State the relationship between the CCF and controllability.
13. For controllability, does the magnitude of the inputs have to be finite?
14. Give the condition of controllability in terms of the matrices **A** and **B**.
15. What is the motivation behind the concept of observability?
16. Give the condition of observability in terms of the matrices **A** and **C**.

17. What can be said about the controllability and observability conditions if the transfer function has pole–zero cancellation?

18. State the relationship between OCF and observability.

Answers to these review questions can be found on this book's companion Web site: www.wiley.com/college/golnaraghi.

▶ REFERENCES

State Variables and State Equations

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Controllability and Observability

3. R. E. Kalman, "On the General Theory of Control Systems," *Proc. IFAC*, Vol. 1, pp. 481–492, Butterworths, London, 1961.
4. W. L. Brogan, *Modern Control Theory*, 2nd Ed., Prentice Hall, Englewood Cliffs, NJ, 1985.

▶ PROBLEMS

10-1. The following differential equations represent linear time-invariant systems. Write the dynamic equations (state equations and output equations) in vector-matrix form.

(a) $\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + y(t) = 5r(t)$

(b) $2\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 2y(t) = r(t)$

(c) $\frac{d^3y(t)}{dt^3} + 5\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + y(t) + \int_0^t y(\tau)d\tau = r(\tau)$

(d) $\frac{d^4y(t)}{dt^4} + 1.5\frac{d^3y(t)}{dt^3} + 2.5\frac{dy(t)}{dt} + y(t) = 2r(t)$

10-2. The following transfer functions show linear time-invariant systems. Write the dynamic equations (state equations and output equations) in vector-matrix form.

(a) $G(s) = \frac{s+3}{s^2+3s+2}$

(b) $G(s) = \frac{6}{s^3 + +6s^2 + 11s + 6}$

(c) $G(s) = \frac{s+2}{s^2+7s+12}$

(d) $G(s) = \frac{s^3 + 11s^2 + 35s + 250}{s^2(s^3 + 4s^2 + 39s + 108)}$

10-3. Repeat Problem 10-2 by using MATLAB.

10-4. Write the state equations for the block diagrams of the systems shown in Fig. 10P-4.

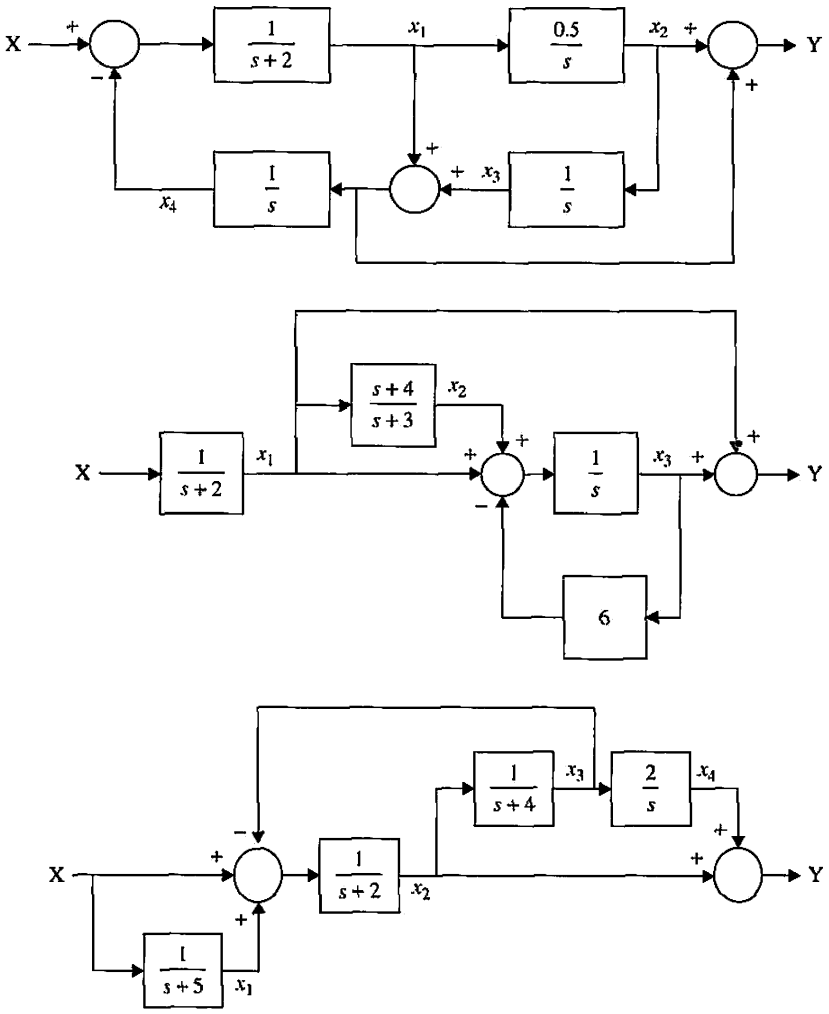


Figure 10P-4

10-5. By use of Eq. (10-58), show that

$$\phi(t) = \mathbf{I} + \mathbf{A}t + \frac{1}{2!}\mathbf{A}^2t^2 + \frac{1}{3!}\mathbf{A}^3t^3 + \cdots$$

10-6. The state equations of a linear time-invariant system are represented by

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

Find the state-transition matrix $\phi(t)$, the characteristic equation, and the eigenvalues of $\mathbf{A}$ for the following cases:

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(d) $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(e) $\mathbf{A} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(f) $\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

(g) $\mathbf{A} = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -5 & 1 \\ 0 & 0 & -5 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

10-7. Find $\phi(t)$ and the characteristic equation of the state variables in Problem 10-6 using a computer program.

10-8. Find the state-transition equation of each of the systems described in Problem 10-6 for $t \geq 0$. Assume that $\mathbf{x}(0)$ is the initial state vector, and the components of the input vector $\mathbf{u}(t)$ are all unit-step functions.

10-9. Find out if the matrices given in the following can be state-transition matrices. [Hint: check the properties of $\phi(t)$.]

(a) $\begin{bmatrix} -e^{-t} & 0 \\ 0 & 1-e^{-t} \end{bmatrix}$

(b) $\begin{bmatrix} 1-e^{-t} & 0 \\ 1 & e^{-t} \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 1-e^{-t} & e^{-t} \end{bmatrix}$

(d) $\begin{bmatrix} e^{-2t} & te^{-2t} & t^2e^{-2t}/2 \\ 0 & e^{-2t} & te^{-2t} \\ 0 & 0 & e^{-2t} \end{bmatrix}$

10-10. Find the time response of the following systems:

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} u \quad y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

10-11. Given a system described by the dynamic equations:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \quad y(t) = \mathbf{C}\mathbf{x}(t)$$

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\mathbf{C} = [1 \quad 0 \quad 0]$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\mathbf{C} = [1 \quad 1]$

(c) $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\mathbf{C} = [1 \quad 1 \quad 0]$

(1) Find the eigenvalues of **A**. Use the **ACSYS** computer program to check the answers. You may get the characteristic equation and solve for the roots using tfsym or tcal components of **ACSYS**.

(2) Find the transfer-function relation between $\mathbf{X}(s)$ and $U(s)$.

(3) Find the transfer function $Y(s)/U(s)$.

10-12. Given the dynamic equations of a time-invariant system:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \quad y(t) = \mathbf{C}\mathbf{x}(t)$$

where

$$\mathbf{A}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -2 & -3\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}\quad \mathbf{C}=[1\quad 1\quad 0]$$

Find the matrices $\mathbf{A}_1$ and $\mathbf{B}_1$ so that the state equations are written as

$$\frac{d\bar{\mathbf{x}}(t)}{dt}=\mathbf{A}_1\bar{\mathbf{x}}(t)+\mathbf{B}_1u(t)$$

where

$$\bar{\mathbf{x}}(t)=\begin{bmatrix}x_1(t)\\ y(t)\\ \dfrac{dy(t)}{dt}\end{bmatrix}$$

10-13. Given the dynamic equations

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}\mathbf{u}(t)\quad y(t)=\mathbf{C}\mathbf{x}(t)$$

(a) $\mathbf{A}=\begin{bmatrix}0 & 2 & 0\\ 1 & 2 & 0\\ -1 & 0 & 1\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}0\\ 1\\ 1\end{bmatrix}\quad \mathbf{C}=[1\quad 0\quad 1]$

(b) $\mathbf{A}=\begin{bmatrix}0 & 2 & 0\\ 1 & 2 & 0\\ -1 & 1 & 1\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}1\\ 1\\ 0\end{bmatrix}\quad \mathbf{C}=[1\quad 0\quad 1]$

(c) $\mathbf{A}=\begin{bmatrix}-2 & 1 & 0\\ 0 & -2 & 0\\ -1 & -2 & -3\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}1\\ 1\\ 1\end{bmatrix}\quad \mathbf{C}=[1\quad 0\quad 0]$

(d) $\mathbf{A}=\begin{bmatrix}-1 & 1 & 0\\ 0 & -1 & 1\\ 0 & 0 & -1\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}0\\ 1\\ 1\end{bmatrix}\quad \mathbf{C}=[1\quad 0\quad 1]$

(e) $\mathbf{A}=\begin{bmatrix}1 & 1\\ -2 & -3\end{bmatrix}\quad \mathbf{B}=\begin{bmatrix}0\\ 1\end{bmatrix}\quad \mathbf{C}=[1\quad 0]$

Find the transformation $\mathbf{x}(t)=\mathbf{P}\bar{\mathbf{x}}(t)$ that transforms the state equations into the controllability canonical form (CCF).

10-14. For the systems described in Problem 10-13, find the transformation $\mathbf{x}(t)=\mathbf{Q}\bar{\mathbf{x}}(t)$ so that the state equations are transformed into the observability canonical form (OCF).

10-15. For the systems described in Problem 10-13, find the transformation $\mathbf{x}(t)=\mathbf{T}\bar{\mathbf{x}}(t)$ so that the state equations are transformed into the diagonal canonical form (DCF) if $\mathbf{A}$ has distinct eigenvalues and Jordan canonical form (JCF) if $\mathbf{A}$ has at least one multiple-order eigenvalue.

10-16. Consider the following transfer functions. Transform the state equations into the controllability canonical form (CCF) and observibility canonical form (OCF).

(a) $\dfrac{s^2-1}{s^2(s^2-2)}$ (b) $\dfrac{2s+1}{s^2+4s+4}$

10-17. The state equation of a linear system is described by

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}\mathbf{u}(t)$$

The coefficient matrices are given as follows. Explain why the state equations cannot be transformed into the controllability canonical form (CCF).

(a) $\mathbf{A} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 2 \\ \sqrt{2} \end{bmatrix}$

(d) $\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

10-18. Check the controlibility of the following systems:

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$

(c) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 0 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

(d) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 3 \end{bmatrix} u$

(e) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -5 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 3 & 0 \\ 0 & 0 \\ 2 & \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

(f) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & -5 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 1 \\ 3 \\ 0 \end{bmatrix} u$

10-19. Check the observability of the following systems:

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mathbf{y} = [1 \quad 3] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mathbf{y} = [0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(c) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 3 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

(d) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

(e) $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\\ \dot{x}_5\end{bmatrix}=\begin{bmatrix}2&1&0&0&0\\0&2&1&0&0\\0&0&2&0&0\\0&0&0&-3&1\\0&0&0&0&-3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix}\qquad\begin{bmatrix}y_1\\y_2\end{bmatrix}=\begin{bmatrix}1&1&1&0&0\\0&1&1&1&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix}$$

(f) $$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\\ \dot{x}_5\end{bmatrix}=\begin{bmatrix}2&1&0&0&0\\0&2&1&0&0\\0&0&2&0&0\\0&0&0&-3&1\\0&0&0&0&-3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix}\qquad\begin{bmatrix}y_1\\y_2\end{bmatrix}=\begin{bmatrix}1&1&1&0&0\\0&1&1&0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix}$$

10-20. The equations that describe the dynamics of a motor control system are

$$\begin{aligned}e_a(t)&=R_a i_a(t)+L_a\frac{di_a(t)}{dt}+K_b\frac{d\theta_m(t)}{dt}\\T_m(t)&=K_i i_a(t)\\T_m(t)&=J\frac{d^2\theta_m(t)}{dt^2}+B\frac{d\theta_m(t)}{dt}+K\theta_m(t)\\e_a(t)&=K_a e(t)\\e(t)&=K_s[\theta_r(t)-\theta_m(t)]\end{aligned}$$

(a) Assign the state variables as $x_1(t)=\theta_m(t)$, $x_2(t)=d\theta_m(t)/dt$, and $x_3(t)=i_a(t)$. Write the state equations in the form of

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}\theta_r(t)$$

Write the output equation in the form $y(t)=\mathbf{C}\mathbf{x}(t)$, where $y(t)=\theta_m(t)$.

(b) Find the transfer function $G(s)=\Theta_m(s)/E(s)$ when the feedback path from $\Theta_m(s)$ to $E(s)$ is broken. Find the closed-loop transfer function $M(s)=\Theta_m(s)/\Theta_r(s)$.

10-21. Given the matrix $\mathbf{A}$ of a linear system described by the state equation

$$\frac{d\mathbf{x}(t)}{dt}=\mathbf{A}\mathbf{x}(t)+\mathbf{B}u(t)$$

(a) $\mathbf{A}=\begin{bmatrix}0&1\\-1&0\end{bmatrix}$

(b) $\mathbf{A}=\begin{bmatrix}-1&0\\0&-2\end{bmatrix}$

(c) $\mathbf{A}=\begin{bmatrix}0&1\\1&0\end{bmatrix}$

Find the state-transition matrix $\phi(t)$ using the following methods:

(1) Infinite-series expansion of $e^{\mathbf{A}t}$, expressed in closed form

(2) The inverse Laplace transform of $(s\mathbf{I}-\mathbf{A})^{-1}$

10-22. The schematic diagram of a feedback control system using a dc motor is shown in Fig. 10P-22. The torque developed by the motor is $T_m(t)=K_i i_a(t)$, where K_i is the torque constant.

The constants of the system are

$K_s=2$ $\quad R=2\,\Omega$ $\quad R_s=0.1\,\Omega$

$K_b=5$ V/rad/sec $\quad K_i=5$ N-m/A $\quad L_a\cong 0$ H

$J_m+J_L=0.1$ N-m-sec^2 $\quad B_m\cong 0$ N-m-sec

Assume that all the units are consistent so that no conversion is necessary.

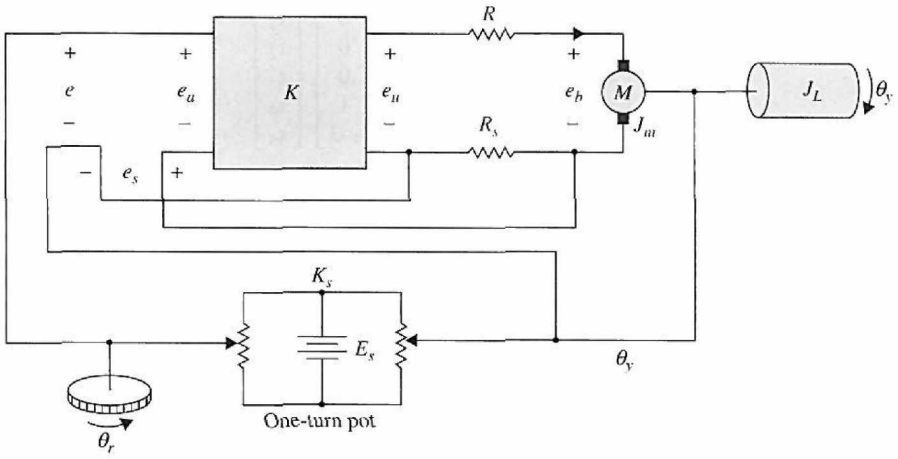


Figure 10P-22

(a) Let the state variables be assigned as $x_1 = \theta_y$ and $x_2 = d\theta_y/dt$. Let the output be $y = \theta_y$. Write the state equations in vector-matrix form. Show that the matrices **A** and **B** are in CCF.

(b) Let $\theta_r(t)$ be a unit-step function. Find $\mathbf{x}(t)$ in terms of $\mathbf{x}(0)$, the initial state. Use the Laplace transform table.

(c) Find the characteristic equation and the eigenvalues of **A**.

(d) Comment on the purpose of the feedback resistor R_s.

10-23. Repeat Problem 10-22 with the following system parameters:

$K_s = 1$	$K = 9$	$R_a = 0.1\ \Omega$
$R_s = 0.1\ \Omega$	$K_b = 1$ V/rad/s	$K_i = 1$ N-m/A
$L_a \cong 0$ H	$J_m + J_L = 0.01$ N-m-sec^2	$B_m \cong 0$ N-m-sec

10-24. Consider that matrix **A** can be diagonalized. Show that $e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}$, where **P** transforms **A** into a diagonal matrix, and $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$, where **D** is a diagonal matrix.

10-25. Consider that matrix **A** can be transformed to the Jordan canonical form, then $e^{\mathbf{A}t} = \mathbf{S}e^{\mathbf{J}t}\mathbf{S}^{-1}$, where **S** transforms **A** into a Jordan canonical form and **J** is in a Jordan canonical form.

10-26. The block diagram of a feedback control system is shown in Fig. 10P-26.

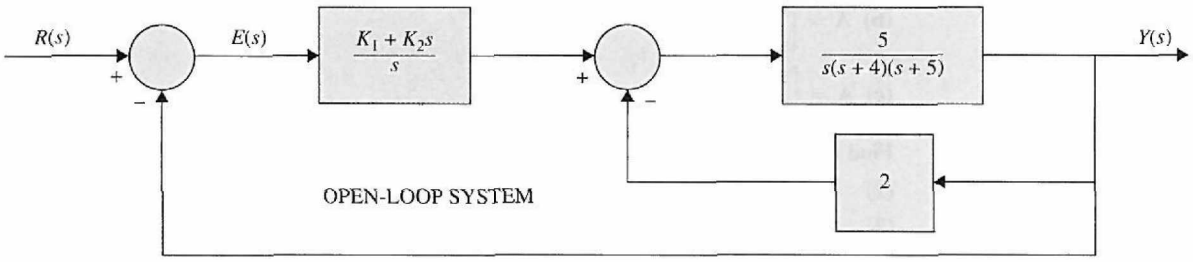


Figure 10P-26

(a) Find the forward-path transfer function $Y(s)/E(s)$ and the closed-loop transfer function $Y(s)/R(s)$.

(b) Write the dynamic equations in the form of

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}r(t) \quad y(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}r(t)$$

Find **A**, **B**, **C**, and **D** in terms of the system parameters.

(c) Apply the final-value theorem to find the steady-state value of the output $y(t)$ when the input $r(t)$ is a unit-step function. Assume that the closed-loop system is stable.

10-27. For the linear time-invariant system whose state equations have the coefficient matrices given by Eqs. (10-191) and (10-192) (CCF), show that

$$\text{adj}(s\mathbf{I} - \mathbf{A})\mathbf{B} = \begin{bmatrix} 1 \\ s \\ s^2 \\ \vdots \\ s^{n-1} \end{bmatrix}$$

and the characteristic equation of **A** is

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

10-28. A linear time-invariant system is described by the differential equation

$$\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + y(t) = r(t)$$

(a) Let the state variables be defined as $x_1 = y$, $x_2 = dy/dt$, and $x_3 = d^2y/dt^2$. Write the state equations of the system in vector-matrix form.

(b) Find the state-transition matrix $\phi(t)$ of **A**.

(c) Let $y(0) = 1$, $dy(0)/dt = 0$, $d^2y(0)/dt^2 = 0$, and $r(t) = u_s(t)$. Find the state-transition equation of the system.

(d) Find the characteristic equation and the eigenvalues of **A**.

10-29. A spring-mass-friction system is described by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = r(t)$$

(a) Define the state variables as $x_1(t) = y(t)$ and $x_2(t) = dy(t)/dt$. Write the state equations in vector-matrix form. Find the state-transition matrix $\phi(t)$ of **A**.

(b) Define the state variables as $x_1(t) = y(t)$ and $x_2(t) = y(t) + dy(t)/dt$. Write the state equations in vector-matrix form. Find the state-transition matrix $\phi(t)$ of **A**.

(c) Show that the characteristic equations, $|s\mathbf{I} - \mathbf{A}| = 0$, for parts (a) and (b) are identical.

10-30. Given the state equations $d\mathbf{x}(t)/dt = \mathbf{A}\mathbf{x}(t)$, where σ and ω are real numbers:

(a) Find the state transition matrix of **A**.

(b) Find the eigenvalues of **A**.

10-31. **(a)** Show that the input–output transfer functions of the two systems shown in Fig. 10P-31 are the same.

(b) Write the dynamic equations of the system in Fig. 10P-31(a) as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}_1\mathbf{x}(t) + \mathbf{B}_1u_1(t) \qquad y_1(t) = \mathbf{C}_1\mathbf{x}(t)$$

and those of the system in Fig. 10-31(b) as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}_2\mathbf{x}(t) + \mathbf{B}_2u_2(t) \qquad y_2(t) = \mathbf{C}_2\mathbf{x}(t)$$

10-32. Draw the state diagrams for the following systems.

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

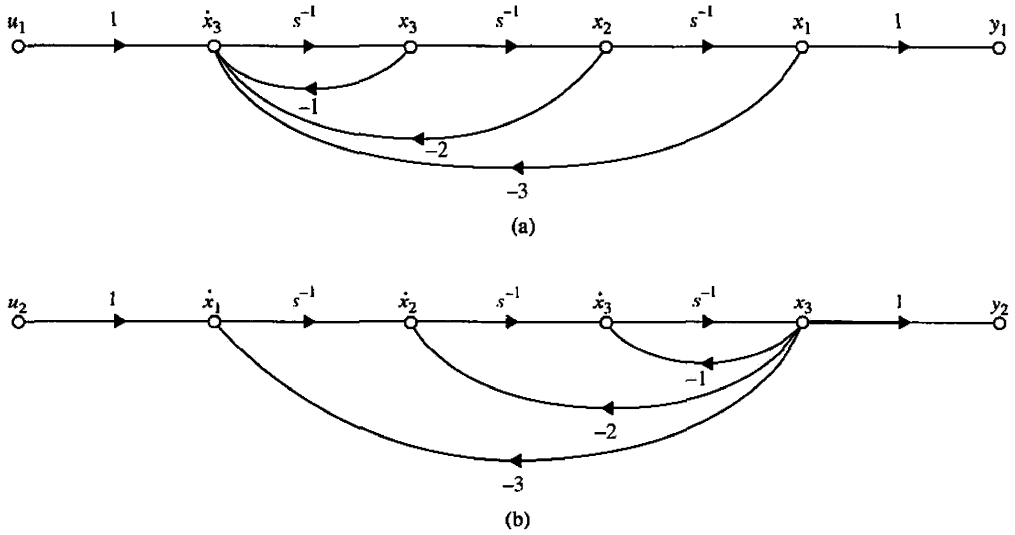


Figure 10P-31

(a) $\mathbf{A} = \begin{bmatrix} -3 & 2 & 0 \\ -1 & 0 & 1 \\ -2 & -3 & -4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(b) Same **A** as in part (a), but with

$$\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

10-33. Draw state diagrams for the following transfer functions by direct decomposition. Assign the state variables from right to left for $x_1, x_2, \ldots$. Write the state equations from the state diagram and show that the equations are in CCF.

(a) $G(s) = \dfrac{10}{s^3 + 8.5s^2 + 20.5s + 15}$

(b) $G(s) = \dfrac{10(s+2)}{s^2(s+1)(s+3.5)}$

(c) $G(s) = \dfrac{5(s+1)}{s(s+2)(s+10)}$

(d) $G(s) = \dfrac{1}{s(s+5)(s^2+2s+2)}$

10-34. Draw state diagrams for the systems described in Problem 10-33 by parallel decomposition. Make certain that the state diagrams contain a minimum number of integrators. The constant branch gains must be real. Write the state equations from the state diagram.

10-35. Draw the state diagrams for the systems described in Problem 10-33 by using cascade decomposition. Assign the state variables in ascending order from right to left. Write the state equations from the state diagram.

10-36. The block diagram of a feedback control system is shown in Fig. 10P-36.

(a) Draw a state diagram for the system by first decomposing $G(s)$ by direct decomposition. Assign the state variables in ascending order, $x_1, x_2, \ldots$, from right to left. In addition to the state-variable-related nodes, the state diagram should contain nodes for $R(s)$, $E(s)$, and $C(s)$.

(b) Write the dynamic equations of the system in vector-matrix form.

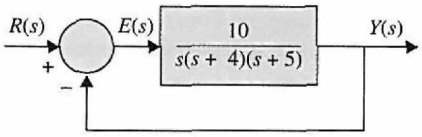


Figure 10P-36

(c) Find the state-transition equations of the system using the state equations found in part (b). The initial state vector is $\mathbf{x}(0)$, and $r(t) = u_s(t)$.

(d) Find the output $y(t)$ for $t \geq 0$ with the initial state $\mathbf{x}(0)$, and $r(t) = u_s(t)$.

10-37. **(a)** Find the closed-loop transfer function $Y(s)/R(s)$, and draw the state diagram.
(b) Perform a direct decomposition to $Y(s)/R(s)$, and draw the state diagram.
(c) Assign the state variables from right to left in ascending order, and write the state equations in vector-matrix form.
(d) Find the state-transition equations of the system using the state equations found in part (c). The initial state vector is $\mathbf{x}(0)$, and $r(t) = u_s(t)$.
(e) Find the output $y(t)$ for $t \geq 0$ with the initial state $\mathbf{x}(0)$, and $r(t) = u_s(t)$.

10-38. The block diagram of a linearized idle-speed engine-control system of an automobile is shown in Fig. 10P-38. (For a discussion on linearization of nonlinear systems, refer to Section 4-9.) The system is linearized about a nominal operating point, so all the variables represent linear-perturbed quantities. The following variables are defined: $T_m(t)$ is the engine torque; T_D, the constant load-disturbance torque; $\omega(t)$, the engine speed; $u(t)$, the input-voltage to the throttle actuator; and α, the throttle angle. The time delay in the engine model can be approximated by

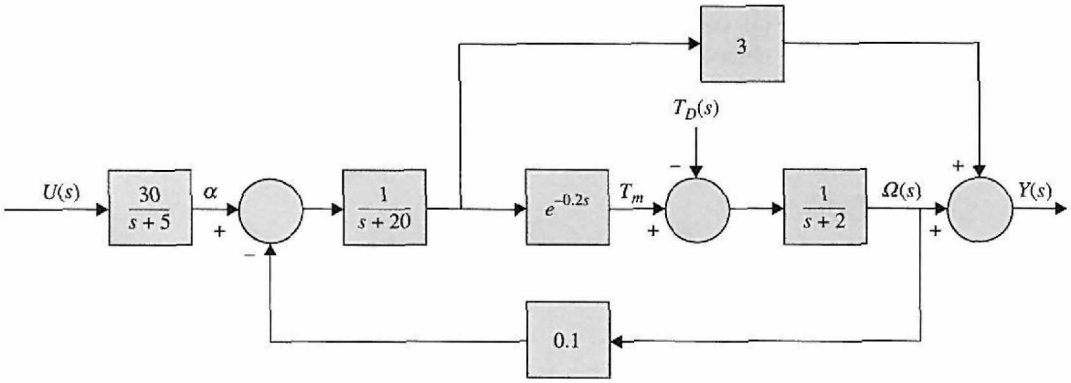


Figure 10P-38

$$e^{-0.2s} \cong \frac{1 - 0.1s}{1 + 0.1s}$$

(a) Draw a state diagram for the system by decomposing each block individually. Assign the state variables from right to left in ascending order.

(b) Write the state equations from the state diagram obtained in part (a), in the form of

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\begin{bmatrix} u(t) \\ T_D(t) \end{bmatrix}$$

(c) Write $Y(s)$ as a function of $U(s)$ and $T_D(s)$. Write $\Omega(s)$ as a function of $U(s)$ and $T_D(s)$.

10-39. The state diagram of a linear system is shown in Fig. 10P-39.

(a) Assign state variables on the state diagram from right to left in ascending order. Create additional artificial nodes if necessary so that the state-variable nodes satisfy as "input nodes" after the integrator branches are deleted.

(b) Write the dynamic equations of the system from the state diagram in part (a).

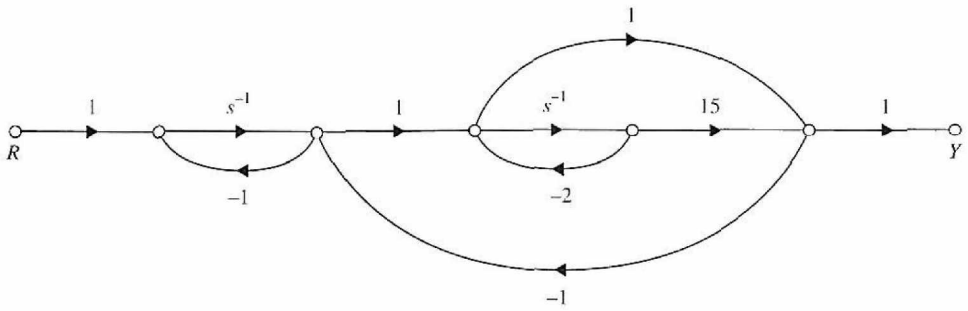


Figure 10P-39

10-40. The block diagram of a linear spacecraft-control system is shown in Fig. 10P-40.

(a) Determine the transfer function $Y(s)/R(s)$.

(b) Find the characteristic equation and its roots of the system. Show that the roots of the characteristic equation are not dependent on K.

(c) When $K = 1$, draw a state diagram for the system by decomposing $Y(s)/R(s)$, using a minimum number of integrators.

(d) Repeat part (c) when $K = 4$.

(e) Determine the values of K that must be avoided if the system is to be both state controllable and observable.

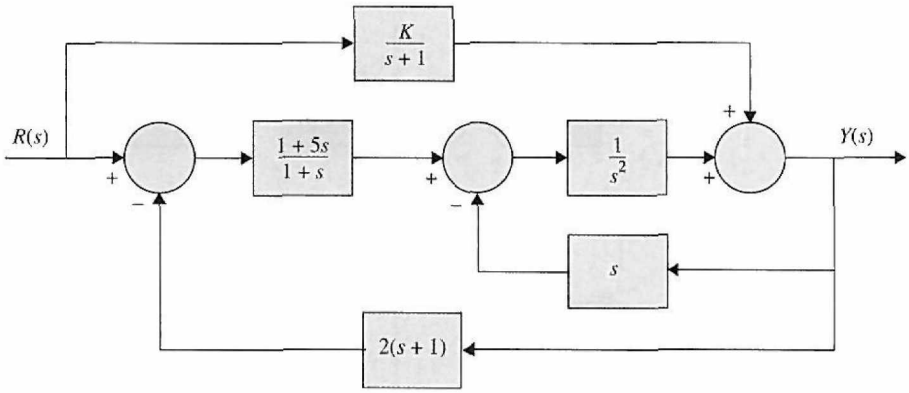


Figure 10P-40

10-41. A considerable amount of effort is being spent by automobile manufacturers to meet the exhaust-emission-performance standards set by the government. Modern automobile-power-plant systems consist of an internal combustion engine that has an internal cleanup device called a catalytic converter. Such a system requires control of such variables as the engine air–fuel (A/F) ratio, ignition-spark timing, exhaust-gas recirculation, and injection air. The control-system problem considered in this problem deals with the control of the A/F ratio. In general, depending on fuel composition and other factors, a typical stoichiometric A/F is 14.7:1, that is, 14.7 grams of air to each gram of fuel. An A/F greater or less than stoichiometry will cause high hydrocarbons, carbon monoxide, and nitrous oxides in the tailpipe emission. The control system shown in Fig. 10P-41 is devised to control the air–fuel ratio so that a desired output is achieved for a given input command.

The sensor senses the composition of the exhaust-gas mixture entering the catalytic converter. The electronic controller detects the difference or the error between the command and the error and computes the control signal necessary to achieve the desired exhaust-gas composition. The output $y(t)$ denotes the effective air–fuel ratio. The transfer function of the engine is given by

$$G_p(s) = \frac{Y(s)}{U(s)} = \frac{e^{-T_d s}}{1 + 0.5s}$$

where $T_d = 0.2$ sec is the time delay and is approximated by

$$e^{-T_d s} = \frac{1}{e^{T_d s}} = \frac{1}{1 + T_d s + T_d^2 s^2/2! + \cdots} \cong \frac{1}{1 + T_d s + T_d^2 s^2/2!}$$

The gain of the sensor is 1.0.

(a) Using the approximation for $e^{-T_d s}$ given, find the expression for $G_p(s)$. Decompose $G_p(s)$ by direct decomposition, and draw the state diagram with $u(t)$ as the input and $y(t)$ as the output. Assign state variables from right to left in ascending order, and write the state equations in vector-matrix form.

(b) Assuming that the controller is a simple amplifier with a gain of 1, i.e., $u(t) = e(t)$, find the characteristic equation and its roots of the closed-loop system.

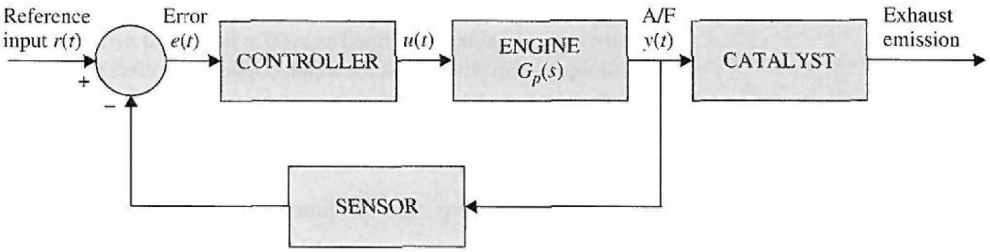


Figure 10P-41

10-42. Repeat Problem 10-41 when the time delay of the automobile engine is approximated as

$$e^{-T_d s} \cong \frac{1 - T_d s/3}{1 + \frac{2}{3}T_d s + \frac{1}{6}T_d^2 s^2} \qquad T_d = 0.2 \text{ sec}$$

10-43. The schematic diagram in Fig. 10P-43 shows a permanent-magnet dc-motor-control system with a viscous-inertia damper. The system can be used for the control of the printwheel of an electronic word processor. A mechanical damper such as the viscous-inertia type is sometimes used in practice as a simple and economical way of stabilizing a control system. The damping effect is achieved by a rotor suspended in a viscous fluid. The differential and algebraic equations that describe the dynamics of the system are as follows:

$$e(t) = K_s[\omega_r(t) - \omega_m(t)] \qquad K_s = 1 \text{ V/rad/sec}$$

$$e_a(t) = Ke(t) = R_a i_a(t) + e_b(t) \qquad K = 10$$

$$e_b(t) = K_b \omega_m(t) \qquad K_b = 0.0706 \text{ V/rad/sec}$$

$$T_m(t) = J\frac{d\omega_m(t)}{dt} + K_D[\omega_m(t) - \omega_D(t)] \qquad J = J_h + J_m = 0.1 \text{ oz-in.-sec}^2$$

$$T_m(t) = K_i i_a(t) \qquad K_i = 10 \text{ oz-in./A}$$

$$K_D[\omega_m(t) - \omega_D(t)] = J_R\frac{d\omega_D(t)}{dt} \qquad J_R = 0.05 \text{ oz-in.-sec}^2$$

$$R_a = 1\,\Omega \qquad K_D = 1 \text{ oz-in.-sec}$$

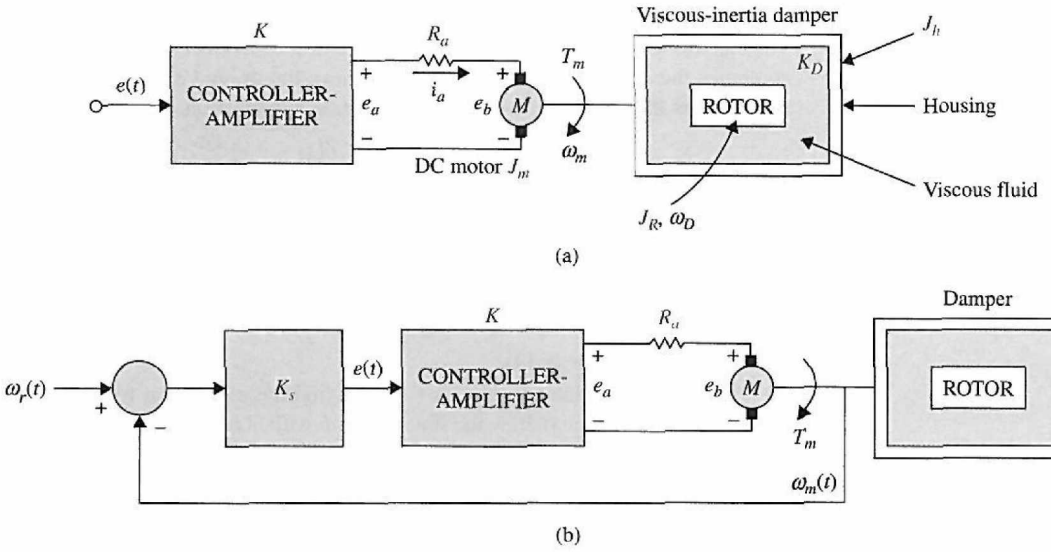


Figure 10P-43

(a) Let the state variables be defined as $x_1(t) = \omega_m(t)$ and $x_2(t) = \omega_D(t)$. Write the state equations for the open-loop system with $e(t)$ as the input. (*Open-loop* refers to the feedback path from ω_m to e being open.)

(b) Draw the state diagram for the overall system using the state equations found in part (a) and $e(t) = K_s[\omega_r(t) - \omega_m(t)]$.

(c) Derive the open-loop transfer function $\Omega_m(s)/E(s)$ and the closed-loop transfer function $\Omega_m(s)/\Omega_r(s)$.

10-44. Determine the state controllability of the system shown in Fig. 10P-44.

(a) $a = 1$, $b = 2$, $c = 2$, and $d = 1$.

(b) Are there any nonzero values for a, b, c, and d such that the system is uncontrollable?

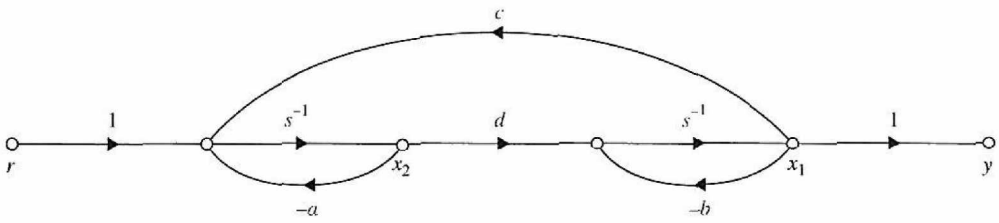


Figure 10P-44

10-45. Determine the controllability of the following systems:

(a) $\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

10-46. Determine the controllability and observability of the system shown in Fig. 10P-46 by the following methods:

(a) Conditions on the **A**, **B**, **C**, and **D** matrices

(b) Conditions on the pole–zero cancellation of the transfer functions

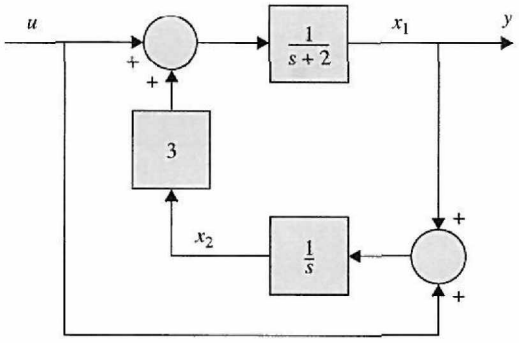


Figure 10P-46

10-47. The transfer function of a linear control system is

$$\frac{Y(s)}{R(s)} = \frac{s+\alpha}{s^3+7s^2+14s+8}$$

(a) Determine the value(s) of α so that the system is either uncontrollable or unobservable.

(b) With the value(s) of α found in part (a), define the state variables so that one of them is uncontrollable.

(c) With the value(s) of α found in part (a), define the state variables so that one of them is unobservable.

10-48. Consider the system described by the state equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & a \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ b \end{bmatrix}$$

Find the region in the a–b plane such that the system is completely controllable.

10-49. Determine the condition on b_1, b_2, c_1, and c_2 so that the following system is completely controllable and observable.

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \quad y(t) = \mathbf{C}\mathbf{x}(t)$$
$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad \mathbf{C} = [d_1 \quad d_2]$$

10-50. The schematic diagram of Fig. 10P-50 represents a control system whose purpose is to hold the level of the liquid in the tank at a desired level. The liquid level is controlled by a float whose position $h(t)$ is monitored. The input signal of the open-loop system is $e(t)$. The system parameters and equations are as follows:

Motor resistance R_a	= 10 Ω	Motor inductance L_a	= 0 H
Torque constant K_i	= 10 oz-in./A	Rotor inertia J_m	= 0.005 oz-in.-sec^2
Back-emf constant K_b	= 0.0706 V/rad/sec	Gear ratio n	= N_1/N_2 = 1/100
Load inertia J_L	= 10 oz-in.-sec^2	Load and motor friction	= negligible
Amplifier gain K_a	= 50	Area of tank A	= 50 ft^2

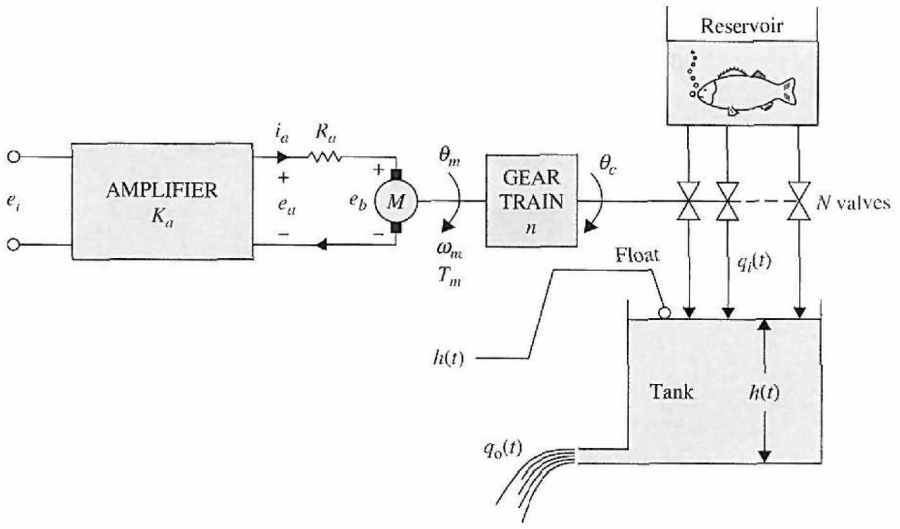


Figure 10P-50

$$e_a(t) = R_a i_a(t) + K_b \omega_m(t) \qquad \omega_m(t) = \frac{d\theta_m(t)}{dt}$$

$$T_m(t) = K_i i_a(t) = \left(J_m + n^2 J_L\right)\frac{d\omega_m(t)}{dt} \qquad \theta_y(t) = n\theta_m(t)$$

The number of valves connected to the tank from the reservoir is $N = 10$. All the valves have the same characteristics and are controlled simultaneously by θ_y. The equations that govern the volume of flow are as follows:

$$q_i(t) = K_I N \theta_y(t) \qquad K_I = 10\,\text{ft}^3/\text{sec-rad}$$

$$q_o(t) = K_o h(t) \qquad K_o = 50\,\text{ft}^2/\text{sec}$$

$$h(t) = \frac{\text{volume of tank}}{\text{area of tank}} = \frac{1}{A}\int [q_i(t) - q_o(t)]dt$$

(a) Define the state variables as $x_1(t) = h(t)$, $x_2(t) = \theta_m(t)$, and $x_3(t) = d\theta_m(t)/dt$. Write the state equations of the system in the form of $d\,\mathbf{x}(t)/dt = \mathbf{A}\mathbf{x}(t) + \mathbf{B}e_i(t)$. Draw a state diagram for the system.

(b) Find the characteristic equation and the eigenvalues of the **A** matrix found in part (a).

(c) Show that the open-loop system is completely controllable; that is, the pair [**A**, **B**] is controllable.

(d) For reasons of economy, only one of the three state variables is measured and fed back for control purposes. The output equation is $y = \mathbf{C}\mathbf{x}$, where **C** can be one of the following forms:
(1) $\mathbf{C} = [1 \quad 0 \quad 0]$ **(2)** $\mathbf{C} = [0 \quad 1 \quad 0]$ **(3)** $\mathbf{C} = [0 \quad 0 \quad 1]$
Determine which case (or cases) corresponds to a completely observable system.

10-51. The "broom-balancing" control system described in Problem 4-21 has the following parameters:

$$M_b = 1\,\text{kg} \quad M_c = 10\,\text{kg} \quad L = 1\,\text{m} \quad g = 32.2\,\text{ft/sec}^2$$

The small-signal linearized state equation model of the system is

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{A}^*\Delta\mathbf{x}(t) + \mathbf{B}^*\Delta r(t)$$

where

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 25.92 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -2.36 & 0 & 0 & 0 \end{bmatrix} \qquad \mathbf{B}^* = \begin{bmatrix} 0 \\ -0.0732 \\ 0 \\ 0.0976 \end{bmatrix}$$

(a) Find the characteristic equation of $\mathbf{A}^*$ and its roots.

(b) Determine the controllability of [$\mathbf{A}^*$, $\mathbf{B}^*$].

(c) For reason of economy, only one of the state variables is to be measured for feedback.

The output equation is written

$$\Delta y(t) = \mathbf{C}^* \Delta \mathbf{x}(t)$$

where

(1) $\mathbf{C}^* = [1 \quad 0 \quad 0 \quad 0]$ (2) $\mathbf{C}^* = [0 \quad 1 \quad 0 \quad 0]$

(3) $\mathbf{C}^* = [0 \quad 0 \quad 1 \quad 0]$ (4) $\mathbf{C}^* = [0 \quad 0 \quad 0 \quad 1]$

Determine which $\mathbf{C}^*$ corresponds to an observable system.

10-52. The double-inverted pendulum shown in Fig. 10P-52 is approximately modeled by the following linear state equation:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

where

$$\mathbf{x}(t) = \begin{bmatrix} \theta_1(t) \\ \dot{\theta}_1(t) \\ \theta_2(t) \\ \dot{\theta}_2(t) \\ x(t) \\ \dot{x}(t) \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 16 & 0 & -8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ -16 & 0 & 16 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Determine the controllability of the states.

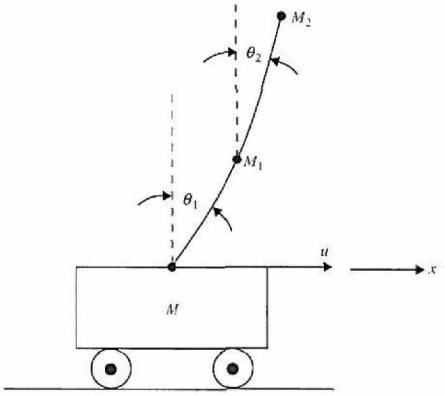


Figure 10P-52

10-53. The block diagram of a simplified control system for the large space telescope (LST) is shown in Fig. 10P-53. For simulation and control purposes, model the system by state equations and by a state diagram.

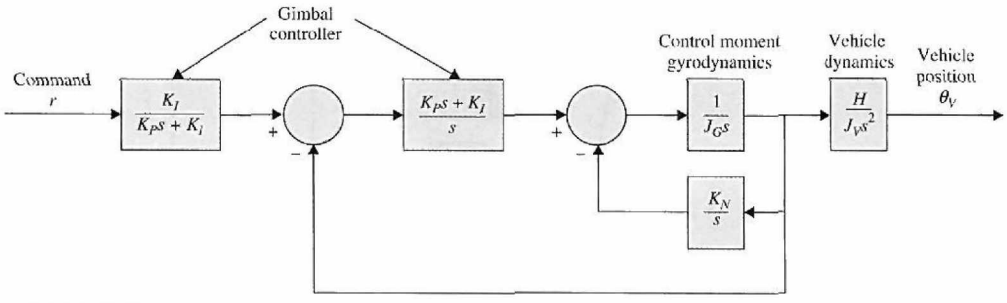


Figure 10P-53

(a) Draw a state diagram for the system and write the state equations in vector-matrix form. The state diagram should contain a minimum number of state variables, so it would be helpful if the transfer function of the system is written first.

(b) Find the characteristic equation of the system.

10-54. The state diagram shown in Fig. 10P-54 represents two subsystems connected in cascade.

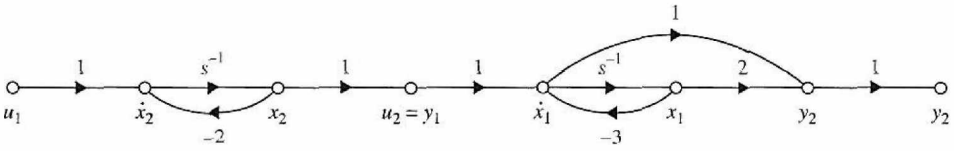


Figure 10P-54

(a) Determine the controllability and observability of the system.

(b) Consider that output feedback is applied by feeding back y_2 to u_2; that is, $u_2 = -ky_2$, where k is a real constant. Determine how the value of k affects the controllability and observability of the system.

10-55. Given the system

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \quad y(t) = \mathbf{C}\mathbf{x}(t)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{C} = [1 \quad 1]$$

(a) Determine the state controllability and observability of the system.

(b) Let $u(t) = -\mathbf{K}\mathbf{x}(t)$, where $\mathbf{K} = [k_1 k_2]$, and k_1 and k_2 are real constants. Determine if and how controllability and observability of the closed-loop system are affected by the elements of $\mathbf{K}$.

10-56. The torque equation for part (a) of Problem 10-21 is

$$J\frac{d^2\theta(t)}{dt^2} = K_F d_1 \theta(t) + T_s d_2 \delta(t)$$

where $K_F d_1 = 1$ and $J = 1$. Define the state variables as $x_1 = \theta$ and $x_2 = d\theta/dt$. Find the state-transition matrix $\phi(t)$ using any available computer program.

10-57. Starting with the state equation $d\mathbf{x}(t)/dt = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\theta_r$ obtained in Problem 10-22, use **ACSYS**/MATLAB or any other available computer program to do the following:

(a) Find the state-transition matrix of $\mathbf{A}$, $\phi(t)$.

(b) Find the characteristic equation of $\mathbf{A}$.

(c) Find the eigenvalues of $\mathbf{A}$.

(d) Compute and plot the unit-step response of $y(t) = \theta_y(t)$ for 3 seconds. Set all the initial conditions to zero.

10-58. The block diagram of a control system with state feedback is shown in Fig. 10P-58. Find the real feedback gains k_1, k_2, and k_3 so that:

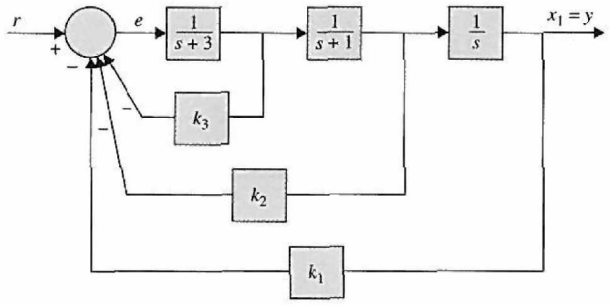
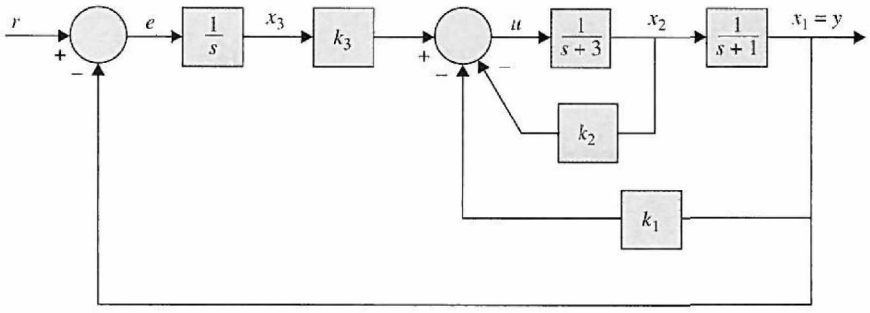


Figure 10P-58

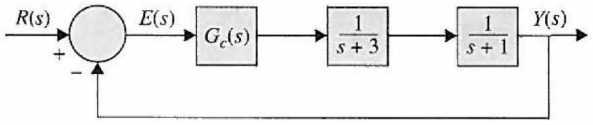
The steady-state error e_{ss} [$e(t)$ is the error signal] due to a step input is zero.
The complex roots of the characteristic equation are at $-1 + j$ and $-1 - j$.

Find the third root. Can all three roots be arbitrarily assigned while still meeting the steady-state requirement?

10-59. The block diagram of a control system with state feedback is shown in Fig. 10P-59(a). The feedback gains k_1, k_2, and k_3 are real constants.



(a)



(b)

Figure 10P-59

(a) Find the values of the feedback gains so that:

The steady-state error e_{ss} [$e(t)$ is the error signal] due to a step input is zero.
The characteristic equation roots are at $-1 + j$, $-1 - j$, and -10.

(b) Instead of using state feedback, a series controller is implemented, as shown in Fig. 10P-59(b). Find the transfer function of the controller $G_c(s)$ in terms of k_1, k_2, and k_3 found in part (a) and the other system parameters.

10-60. Problem 9-39 has revealed that it is impossible to stabilize the broom-balancing control system described in Problems 4-21 and 10-51 with a series PD controller. Consider that the system is now controlled by state feedback with $\Delta r(t) = -\mathbf{K}\mathbf{x}(t)$, where

$$\mathbf{K} = [k_1 \quad k_2 \quad k_3 \quad k_4]$$

(a) Find the feedback gains k_1, k_2, k_3, and k_4 so that the eigenvalues of $\mathbf{A}^* - \mathbf{B}^*\mathbf{K}$ are at $-1 + j$, $-1 - j$, -10, and -10. Compute and plot the responses of $\Delta x_1(t)$, $\Delta x_2(t)$, $\Delta x_3(t)$, and $\Delta x_4(t)$ for the initial condition, $\Delta x_1(0) = 0.1$, $\Delta\theta(0) = 0.1$, and all other initial conditions are zero.

(b) Repeat part (a) for the eigenvalues at $-2 + j2$, $-2 - j2$, -20, and -20. Comment on the difference between the two systems.

10-61. The linearized state equations of the ball-suspension control system described in Problem 4-57 are expressed as

$$\Delta\dot{\mathbf{x}}(t) = \mathbf{A}^*\Delta\mathbf{x}(t) + \mathbf{B}^*\Delta i(t)$$

where

$$\mathbf{A}^* = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 115.2 & -0.05 & -18.6 & 0 \\ 0 & 0 & 0 & 1 \\ -37.2 & 0 & 37.2 & -0.1 \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} 0 \\ -6.55 \\ 0 \\ -6.55 \end{bmatrix}$$

Let the control current $\Delta i(t)$ be derived from the state feedback $\Delta i(t) = -\mathbf{K}\Delta\mathbf{x}(t)$, where

$$\mathbf{K} = [k_1 \quad k_2 \quad k_3 \quad k_4]$$

(a) Find the elements of $\mathbf{K}$ so that the eigenvalues of $\mathbf{A}^* - \mathbf{B}^*\mathbf{K}$ are at $-1 + j, -1 - j, -10$, and -10.

(b) Plot the responses of $\Delta x_1(t) = \Delta y_1(t)$ (magnet displacement) and $\Delta x_3(t) = \Delta y_2(t)$ (ball displacement) with the initial condition

$$\Delta\mathbf{x}(0) = \begin{bmatrix} 0.1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(c) Repeat part (b) with the initial condition

$$\Delta\mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0.1 \\ 0 \end{bmatrix}$$

Comment on the responses of the closed-loop system with the two sets of initial conditions used in (b) and (c).

10-62. The temperature $x(t)$ in the electric furnace shown in Fig. 10P-62 is described by the differential equation

$$\frac{dx(t)}{dt} = -2x(t) + u(t) + n(t)$$

where $u(t)$ is the control signal, and $n(t)$ the constant disturbance of unknown magnitude due to heat loss. It is desired that the temperature $x(t)$ follows a reference input r that is a constant.

(a) Design a control system with state and integral control so that the following specifications are satisfied:

$$\lim_{t \to \infty} x(t) = r = \text{constant}$$

The eigenvalues of the closed-loop system are at -10 and -10.

Plot the responses of $x(t)$ for $t \geq 0$ with $r = 1$ and $n(t) = -1$, and then with $r = 1$ and $n(t) = 0$, all with $x(0) = 0$.

(b) Design a PI controller so that

$$G_c(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s}$$
$$E(s) = R(s) - X(s)$$

where $R(s) = R/s$.

Find K_P and K_I so that the characteristic equation roots are at -10 and -10. Plot the responses of $x(t)$ for $t \geq 0$ with $r = 1$ and $n(t) = -1$, and then with $r = 1$ and $n(t) = 0$, all $x(0) = 0$.

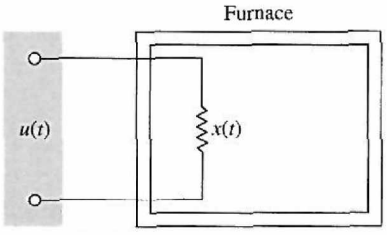


Figure 10P-62

10-63. The transfer function of a system is given by

$$G(s) = \frac{10}{(s+1)(s+2)(s+3)}$$

Find the state-space model of the system if:

$x_1 = y$
$x_2 = \dot{x}_1$
$x_3 = \dot{x}_2$

Design a state control feedback $u = -Kx$ so that the closed-loop poles are located at $s = -2 + j2\sqrt{3}$, $s = -2 - j2\sqrt{3}$, and $s = -10$

10-64. Fig. 10P-64 shows an inverted pendulum on a moving platform.
Assuming $M = 2$ kg, $m = 0.5$ kg, and $l = 1$ m.

(a) Find the state-space model of the system if $x_1 = \theta, x_2 = \dot{\theta}, x_3 = x, x_4 = \dot{x}$, $y_1 = x_1 = \theta$, and $y_2 = x_3 = x$

(b) Design a state feedback control with gain $-K$ so that the closed-loop poles are located at $s = -4 + 4j$, $s = -4 - 4j$, $s = 210$, and $s = 210$

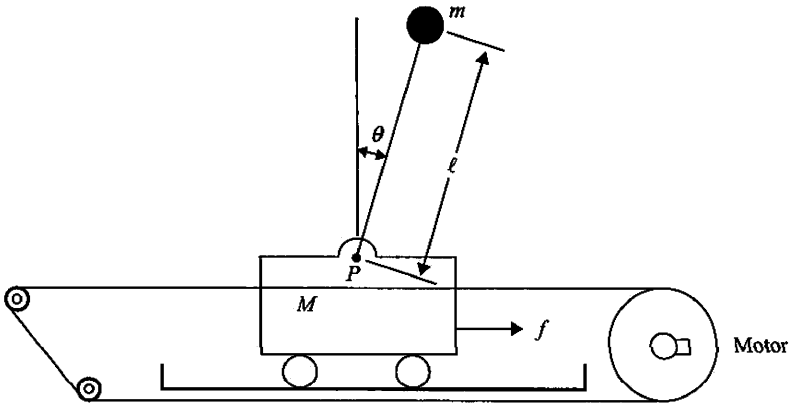


Figure 10P-64

10-65. Consider the following state-space equation of a system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

(a) Design a state feedback controller so that:

(i) The damping ratio is $\zeta = 0.707$.

(ii) Peak time of the unit-step response is 3 sec.

(b) Use MATLAB to plot the step response of the system and show how your design meets the specification in part (a).

10-66. Consider the following state-space equation of a system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -2 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} u$$

(a) Design a state feedback controller so that:

(i) Settling time is less than 5 sec (1% settling time).

(ii) Overshoot is less than 10%.

(b) Use MATLAB to verify your design.

10-67. Fig. 10P-67 shows an RLC circuit.

(a) Find the state equation for the circuit when $v(t)$ is an input, $i(t)$ is an output, and capacitor voltage and the inductor current are the state variables.

(b) Find the condition that the system is controllable.

(c) Find the condition that the system is observable.

(d) Repeat parts (a), (b), and (c) when $v(t)$ is an input, the voltage of the R_2 is output, and capacitor voltage and the inductor current are the state variables.

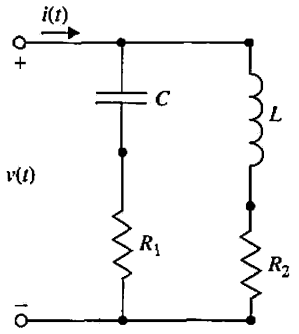


Figure 10P-67

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APPENDIX A

Elementary Matrix Theory and Algebra

A-1 ELEMENTARY MATRIX THEORY

In the study of modern control theory, it is often desirable to use matrix notation to simplify complex mathematical expressions. The matrix notation usually makes the equations much easier to handle and manipulate.

As a motivation to the reason of using matrix notation, let us consider the following set of n simultaneous algebraic equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= y_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= y_2 \\ \cdots\cdots\cdots\

which are simply bracketed arrays of coefficients and variables. These examples of matrices prompted the following definition of a matrix.

A-1-1 Definition of a Matrix

A matrix is a collection of elements arranged in a rectangular or square array. There are several ways of bracketing a matrix. In this text, the square brackets, such as those in Eqs. (A-3) through (A-5), are used to represent matrices. It is important to distinguish between a **matrix** and a **determinant**. The basic characteristics of these are listed as follows:

Matrix	Determinant
• An array of numbers or elements with n rows and m columns. • Does not have a value, although a square matrix ($n = m$) has a determinant.	• An array of numbers or elements with n rows and n columns (always square). • Has a value.

Some important definitions of matrices are given in the following paragraphs.

Matrix Elements: When a matrix is written

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{A-6}$$

where a_{ij} is defined as the element in the ith **row** and the jth **column** of the matrix. As a rule, we always refer to the row first and the column last.

Order of a Matrix: *The order of a matrix refers to the total number of rows and columns of the matrix*. For example, the matrix in Eq. (A-6) has three rows and three columns and is called a 3×3 (three-by-three) matrix. A matrix with n rows and m columns is termed $n \times m$, or n by m.

Square Matrix: *A square matrix is one that has the same number of rows as columns.*

Column Matrix: *A column matrix is one that has one column and more than one row, that is, an $m \times 1$ matrix, $m > 1$.* Quite often, a column matrix is referred to as a **column vector** or simply an **m-vector** if there are m rows and one column. The matrix in Eq. (A-4) is a typical n-vector.

Row Matrix: *A row matrix is one that has one row and more than one column, that is, a $1 \times n$ matrix, where $n > 1$.* A row matrix can also be referred as a **row-vector**.

Diagonal Matrix: *A diagonal matrix is a square matrix with $a_{ij} = 0$ for all $i \neq j$.* Examples of a diagonal matrix are

$$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \quad \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

Unity Matrix (Identity Matrix): *A unity matrix is a diagonal matrix with all the elements on the main diagonal ($i = j$) equal to 1.* A unity matrix is often designated by

I or **U**. An example of a unity matrix is

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{A-7}$$

Null Matrix: *A null matrix is one whose elements are all equal to zero.*

Symmetric Matrix: *A symmetric matrix is a square matrix that satisfies the condition* $a_{ij} = a_{ji}$ *for all i and j.* A symmetric matrix has the property that, if its rows are interchanged with its columns, the same matrix is obtained. Two examples of the symmetric matrix are

$$\mathbf{A} = \begin{bmatrix} 6 & 5 & 1 \\ 5 & 0 & 10 \\ 1 & 10 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & -4 \\ -4 & 1 \end{bmatrix} \tag{A-8}$$

Determinant of a Matrix: *With each square matrix, a determinant having the same elements and order as the matrix may be defined. The determinant of a square matrix* **A** *is designated by*

$$\det \mathbf{A} = \Delta_{\mathrm{A}} = |\mathbf{A}| \tag{A-9}$$

For example, the determinant of the matrix in Eq. (A-6) is

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \tag{A-10}$$

Cofactor of a Determinant Element: *Given any* n*th-order determinant* $|\mathbf{A}|$, *the cofactor* A_{ij} *of any element* a_{ij} *is the determinant obtained by eliminating all elements of the ith row and jth column and then multiplied by* $(-1)^{i+j}$. For example, the cofactor of the element a_{11} of $|\mathbf{A}|$ in Eq. (A-10) is

$$A_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{23}a_{32} \tag{A-11}$$

In general, the value of a determinant can be written in terms of the cofactors. Let **A** be an $n \times n$ matrix, then the determinant of **A** can be written in terms of the cofactor of any row or the cofactor of any column. That is,

$$\det \mathbf{A} = \sum_{j=1}^{n} a_{ij}A_{ij} \quad (i = 1, \text{ or } 2, \ldots, \text{or } n) \tag{A-12}$$

or

$$\det \mathbf{A} = \sum_{i=1}^{n} a_{ij}A_{ij} \quad (j = 1, \text{or } 2, \ldots, \text{or } n) \tag{A-13}$$

▶ **EXAMPLE A-1-1** The value of the determinant in Eq. (A-10) is

$$\begin{aligned} \det \mathbf{A} = |\mathbf{A}| &= a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \end{aligned} \tag{A-14}$$

Singular Matrix: *A square matrix is said to be singular if the value of its determinant is zero.* If a square matrix has a nonzero determinant, it is called a **nonsingular matrix**. When a matrix is

singular, it usually means that not all the rows or not all the columns of the matrix are independent of each other. When the matrix is used to represent a set of algebraic equations, singularity of the matrix means that these equations are not independent of each other. ◀

▶ **EXAMPLE A-1-2** Consider the following set of equations:

$$\begin{aligned} 2x_1 - 3x_2 + x_3 &= 0 \\ -x_1 + x_2 + x_3 &= 0 \\ x_1 - 2x_2 + 2x_3 &= 0 \end{aligned} \tag{A-15}$$

The third equation is equal to the sum of the first two. Thus, these three equations are not completely independent. In matrix form, these equations may be written as

$$\mathbf{Ax} = 0 \tag{A-16}$$

where

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 1 \\ -1 & 1 & 1 \\ 1 & -2 & 2 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \tag{A-17}$$

and $\mathbf{0}$ is a 3×1 null vector. The determinant of $\mathbf{A}$ is 0, and, thus, the matrix $\mathbf{A}$ is singular. In this case, the rows of $\mathbf{A}$ are dependent.

Transpose of a Matrix: *The transpose of a matrix* $\mathbf{A}$ *is defined as the matrix that is obtained by interchanging the corresponding rows and columns in* $\mathbf{A}$. Let $\mathbf{A}$ be an $n \times m$ matrix that is represented by

$$\mathbf{A} = [a_{ij}]_{n,m} \tag{A-18}$$

The transpose of $\mathbf{A}$, denoted by $\mathbf{A}'$, is given by

$$\mathbf{A}' = \text{transpose of } \mathbf{A} = [a_{ji}]_{m,n} \tag{A-19}$$

Notice that the order of $\mathbf{A}$ is $n \times m$, but the order of $\mathbf{A}'$ is $m \times n$. ◀

▶ **EXAMPLE A-1-3** Consider the 2×3 matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & -1 & 5 \end{bmatrix} \tag{A-20}$$

The transpose of $\mathbf{A}$ is obtained by interchanging the rows and the columns.

$$\mathbf{A}' = \begin{bmatrix} 3 & 0 \\ 2 & -1 \\ 1 & 5 \end{bmatrix} \tag{A-21}$$

◀

Some Properties of Matrix Transpose

1. $(\mathbf{A}')' = \mathbf{A}$ (A-22)
2. $(k\mathbf{A})' = k\mathbf{A}'$, where k is a scalar (A-23)
3. $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$ (A-24)
4. $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ (A-25)

Adjoint of a Matrix: *Let* **A** *be a square matrix of order n. The adjoint matrix of* **A**, *denoted by* adj **A**, *is defined as*

$$\text{adj}\,\mathbf{A} = \left[A_{ij} \text{ of } \det \mathbf{A}\right]'_{n,n} \tag{A-26}$$

where A_{ij} *denotes the cofactor of* a_{ij}.

▶ **EXAMPLE A-1-4** Consider the 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \tag{A-27}$$

The cofactors are $A_{11} = a_{22}$, $A_{12} = -a_{21}$, $A_{21} = -a_{12}$, and $A_{22} = a_{11}$. Thus, the adjoint matrix of **A** is

$$\text{adj}\,\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}' = \begin{bmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{bmatrix}' = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \tag{A-28}$$

Trace of a Square Matrix: *Given an* $n \times n$ *matrix with elements* a_{ij}, *the* **trace of A**, *denoted as* tr (**A**), *is defined as the sum of the elements on the main diagonal of* **A**; *that is*

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^{n} a_{ii} \tag{A-29}$$

The trace of a matrix has the following properties:

1. $\text{tr}(\mathbf{A}') = \text{tr}(\mathbf{A})$ (A-30)
2. For $n \times n$ square matrices **A** and **B**,

$$\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}) \tag{A-31}$$

◀

▶ A-2 MATRIX ALGEBRA

When carrying out matrix operations, it is necessary to define matrix algebra in the form of addition, subtraction, multiplication, and division.

A-2-1 Equality of Matrices

Two matrices **A** and **B** are said to be equal to each other if they satisfy the following conditions:

1. They are of the same order.
2. The corresponding elements are equal; that is,

$$a_{ij} = b_{ij} \quad \text{for every } i \text{ and } j \tag{A-32}$$

▶ **EXAMPLE A-2-1**

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \tag{A-33}$$

implies that $a_{11} = b_{11}$, $a_{12} = b_{12}$, $a_{21} = b_{21}$, $a_{22} = b_{22}$. ◀

A-2-2 Addition and Subtraction of Matrices

Two matrices **A** and **B** can be added or subtracted to form $\mathbf{A} \pm \mathbf{B}$ if they are of the same order. That is,

$$\mathbf{A} \pm \mathbf{B} = [a_{ij}]_{n,m} \pm [b_{ij}]_{n,m} = \mathbf{C} = [c_{ij}]_{n,m} \tag{A-34}$$

where

$$c_{ij} = a_{ij} \pm b_{ij} \quad \text{for all } i \text{ and } j. \tag{A-35}$$

The order of the matrices is preserved after addition or subtraction.

▶ **EXAMPLE A-2-2** Consider the matrices

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ -1 & 4 \\ 0 & -1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 3 \\ -1 & 2 \\ 1 & 0 \end{bmatrix} \tag{A-36}$$

which are of the same order. Then the sum of **A** and **B** is

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = \begin{bmatrix} 3+0 & 2+3 \\ -1-1 & 4+2 \\ 0+1 & -1+0 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -2 & 6 \\ 1 & -1 \end{bmatrix} \tag{A-37}$$

◀

A-2-3 Associative Law of Matrix (Addition and Subtraction)

The associative law of scalar algebra still holds for matrix addition and subtraction. That is,

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}) \tag{A-38}$$

A-2-4 Commutative Law of Matrix (Addition and Subtraction)

The commutative law for matrix addition and subtraction states that the following matrix relationship is true:

$$\mathbf{A} + \mathbf{B} + \mathbf{C} = \mathbf{B} + \mathbf{C} + \mathbf{A} = \mathbf{A} + \mathbf{C} + \mathbf{B} \tag{A-39}$$

as well as other possible commutative combinations.

A-2-5 Matrix Multiplication

The matrices **A** and **B** may be multiplied together to form the product **AB** if they are **conformable**. This means that the number of columns of **A** must equal the number of rows of **B**. In other words, let

$$\mathbf{A} = [a_{ij}]_{n,p} \quad \mathbf{B} = [b_{ij}]_{q,m} \tag{A-40}$$

Then **A** and **B** are conformable to form the product

$$\mathbf{C} = \mathbf{AB} = [a_{ij}]_{n,p}[b_{ij}]_{q,m} = [c_{ij}]_{n,m} \tag{A-41}$$

if and only if $p = q$. The matrix **C** will have the same number of rows as **A** and the same number of columns as **B**.

It is important to note that **A** and **B** may be conformable to form **AB**, but they may not be conformable for the product **BA**, unless in Eq. (A-41), n also equals m. This points out an important fact that *the commutative law is not generally valid for matrix multiplication.* It is also noteworthy that, even though **A** and **B** are conformable for both **AB** and **BA**, usually $\mathbf{AB} \neq \mathbf{BA}$. The following references are made with respect to matrix manipulation whenever they exist:

$$\mathbf{AB} = \mathbf{A} \text{ postmultiplied by } \mathbf{B} \quad \text{or} \quad \mathbf{B} \text{ premultiplied by } \mathbf{A} \tag{A-42}$$

A-2-6 Rules of Matrix Multiplication

When the matrices $\mathbf{A}$ $(n \times p)$ and $\mathbf{B}$ $(p \times m)$ are conformable to form the matrix $\mathbf{C} = \mathbf{AB}$, the ijth element of $\mathbf{C}$, c_{ij}, is given by

$$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} \tag{A-43}$$

for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots, m$.

▶ **EXAMPLE A-2-3** Given the matrices

$$\mathbf{A} = [a_{ij}]_{2,3} \quad \mathbf{B} = [b_{ij}]_{3,1} \tag{A-44}$$

the two matrices are conformable for the product **AB** but not for **BA**. Thus,

$$\mathbf{AB} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \end{bmatrix} \tag{A-45}$$

◀

▶ **EXAMPLE A-2-4** Given the matrices

$$\mathbf{A} = \begin{bmatrix} 3 & -1 \\ 0 & 1 \\ 2 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} \tag{A-46}$$

the two matrices are conformable for **AB** and **BA**.

$$\mathbf{AB} = \begin{bmatrix} 3 & -1 \\ 0 & 1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -3 \\ 2 & 1 & 0 \\ 2 & 0 & -2 \end{bmatrix} \tag{A-47}$$

$$\mathbf{BA} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 0 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 6 & -1 \end{bmatrix} \tag{A-48}$$

Therefore, even though **AB** and **BA** both exist, they are not equal. In fact, in this case the products are not of the same order.

Although the commutative law does not hold in general for matrix multiplication, the **associative** and **distributive** laws are valid. For the distributive law, we state that

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC} \tag{A-49}$$

if the products are conformable. For the associative law,

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \tag{A-50}$$

if the product is conformable. ◀

A-2-7 Multiplication by a Scalar *k*

Multiplying a matrix **A** *by any scalar k is equivalent to multiplying each element of* **A** *by k.*

A-2-8 Inverse of a Matrix (Matrix Division)

In the algebra of scalar quantities, when we write $y = ax$, it implies that $x = y/a$ is also true. In matrix algebra, if $\mathbf{Ax} = \mathbf{y}$, then it *may be possible* to write

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y} \tag{A-51}$$

where $\mathbf{A}^{-1}$ denotes the **matrix inverse** of **A**. The conditions that $\mathbf{A}^{-1}$ exists are as follows:

1. **A** is a square matrix.
2. **A** must be nonsingular.
3. If $\mathbf{A}^{-1}$ exists, it is given by

$$\mathbf{A}^{-1} = \frac{\text{adj}\,\mathbf{A}}{|\mathbf{A}|} \tag{A-52}$$

▶ **EXAMPLE A-2-5** Given the matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \tag{A-53}$$

the inverse of **A** is given by

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \frac{\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}}{a_{11}a_{22} - a_{12}a_{21}} \tag{A-54}$$

where for **A** to be nonsingular, $|\mathbf{A}| \neq 0$, or $a_{11}a_{22} - a_{12}a_{21} \neq 0$.
Eq. (A-54) shows that adj **A** of a 2×2 matrix is obtained by *interchanging the two elements on the main diagonal and changing the signs of the elements off the diagonal of* **A**. ◀

▶ **EXAMPLE A-2-6** Given the matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{A-55}$$

to find the inverse of **A**, the adjoint of **A** is

$$\text{adj}\,\mathbf{A} = \begin{bmatrix} a_{22}a_{33} - a_{23}a_{32} & (-a_{12}a_{33} - a_{13}a_{32}) & a_{12}a_{23} - a_{13}a_{22} \\ -(a_{21}a_{33} - a_{23}a_{31}) & a_{11}a_{33} - a_{13}a_{31} & -(a_{11}a_{23} - a_{21}a_{13}) \\ a_{21}a_{32} - a_{22}a_{31} & (-a_{11}a_{32} - a_{12}a_{31}) & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix} \tag{A-56}$$

The determinant of **A** is

$$|\mathbf{A}| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{32}a_{21} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} \tag{A-57}$$

◀

Some Properties of Matrix Inverse

1. $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (A-58)
2. $\left(\mathbf{A}^{-1}\right)^{-1} = \mathbf{A}$ (A-59)

3. In matrix algebra, in general, $\mathbf{AB} = \mathbf{AC}$ does not necessarily imply that $\mathbf{B} = \mathbf{C}$. The reader can easily construct an example to illustrate this property. However, if $\mathbf{A}$ is a square, nonsingular matrix, we can premultiply both sides of $\mathbf{AB} = \mathbf{AC}$ by $\mathbf{A}^{-1}$. Then,

$$\mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{AC} \tag{A-60}$$

which leads to $\mathbf{B} = \mathbf{C}$.

4. If $\mathbf{A}$ and $\mathbf{B}$ are square matrices and are nonsingular, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \tag{A-61}$$

A-2-9 Rank of a Matrix

The rank of a matrix $\mathbf{A}$ *is the maximum number of linearly independent columns of* $\mathbf{A}$; *that is, it is the order of the largest nonsingular matrix contained in* $\mathbf{A}$.

▶ **EXAMPLE A-2-7** Several examples on the rank of a matrix are as follows:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{rank} = 1 \qquad \begin{bmatrix} 0 & 5 & 1 & 4 \\ 3 & 0 & 3 & 2 \end{bmatrix} \quad \text{rank} = 2$$

$$\begin{bmatrix} 3 & 9 & 2 \\ 1 & 3 & 0 \\ 2 & 6 & 1 \end{bmatrix} \quad \text{rank} = 2 \qquad \begin{bmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{rank} = 3$$

The following properties are useful in the determination of the rank of a matrix. Given an $n \times m$ matrix $\mathbf{A}$,

1. Rank of $\mathbf{A}'$ = Rank of $\mathbf{A}$.
2. Rank of $\mathbf{A}'\mathbf{A}$ = Rank of $\mathbf{A}$.
3. Rank of $\mathbf{AA}'$ = Rank of $\mathbf{A}$.

Properties 2 and 3 are useful in the determination of rank; because, $\mathbf{A}'\mathbf{A}$ and $\mathbf{AA}'$ are always square, the rank condition can be checked by evaluating the determinant of these matrices. ◀

▶ A-3 COMPUTER-AIDED SOLUTIONS OF MATRICES

Many commercial software packages such as MATLAB, **Maple,** and **MATHCAD** contain routines for matrix manipulations.

▶ REFERENCES

1. R. Bellman, *Introduction to Matrix Analysis,* McGraw-Hill, New York 1960.
2. F. Ayres, Jr., *Theory and Problems of Matrices*, Shaum's Outline Series, McGraw-Hill, New York 1962.

APPENDIX B

Difference Equations

B-1 DIFFERENCE EQUATIONS

Because digital controllers are frequently used in control systems, it is necessary to establish equations that relate digital and discrete-time signals. Just as differential equations are used to represent systems with analog signals, difference equations are used for computers or systems utilizing discrete data. Difference equations are also used to approximate differential equations, because the former are more easily programmed on a computer and are easier to solve.

A linear nth-order difference equation with constant coefficients can be written as

$$y(k+n) + a_{n-1}y(k+n-1) + \cdots + a_1 y(k+1) + a_0 y(k) = f(k) \tag{B-1}$$

where $y(i)$, $i = k, k+1, \ldots, k+n$ denotes the discrete dependent variable y at the ith instant if the independent variable is time. In general, the independent variable can be any real quantity.

Similar to the case of the analog systems, it is convenient to use a set of first-order difference equations, or state equations, to represent a high-order difference equation. For the difference equation in Eq. (B-1), if we let

$$\begin{aligned} x_1(k) &= y(k) \\ x_2(k) &= x_1(k+1) = y(k+1) \\ &\vdots \\ x_{n-1}(k) &= x_{n-2}(k+1) = y(k+n-2) \\ x_n(k) &= x_{n-1}(k+1) = y(k+n-1) \end{aligned} \tag{B-2}$$

then by equating the highest-order term to the rest, the equation is written as

$$x_n(k+1) = -a_0 x_1(k) - a_1 x_2(k) - \cdots - a_{n-1} x_n(k) + f(k) \tag{B-3}$$

The first $n-1$ state equations are taken directly from the last $n-1$ equations in Eq. (B-2), and the last one is given by Eq. (B-3). The n state equations are written in vector-matrix form:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k) \tag{B-4}$$

where

$$\mathbf{x}(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_n(k) \end{bmatrix} \tag{B-5}$$

is the $n \times 1$ state vector, and

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{B-6}$$

and $u(k) = f(k)$.

▶ REFERENCES

1. C. R. Wylie, Jr., *Advanced Engineering Mathematics,* 2nd Ed. McGraw-Hill, New York 1960.
2. B. C. Kuo, *Linear Networks and Systems,* McGraw-Hill, New York 1967.

APPENDIX C

Laplace Transform Table

Laplace Transform $F(s)$	Time Function $f(t)$
1	Unit-impulse function $\delta(t)$
$\frac{1}{s}$	Unit-step function $u_s(t)$
$\frac{1}{s^2}$	Unit-ramp function t
$\frac{n!}{s^{n+1}}$	$t^n (n = \text{positive integer})$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}$
$\frac{1}{(s+\alpha)^2}$	$te^{-\alpha t}$
$\frac{n!}{(s+\alpha)^{n+1}}$	$t^n e^{-\alpha t} (n = \text{positive integer})$
$\frac{1}{(s+\alpha)(s+\beta)}$	$\frac{1}{\beta-\alpha}(e^{-\alpha t} - e^{-\beta t}) (\alpha \neq \beta)$
$\frac{s}{(s+\alpha)(s+\beta)}$	$\frac{1}{\beta-\alpha}(\beta e^{-\beta t} - \alpha e^{-\alpha t}) (\alpha \neq \beta)$
$\frac{1}{s(s+\alpha)}$	$\frac{1}{\alpha}(1 - e^{-\alpha t})$
$\frac{1}{s(s+\alpha)^2}$	$\frac{1}{\alpha^2}(1 - e^{-\alpha t} - \alpha t e^{-\alpha t})$
$\frac{1}{s^2(s+\alpha)}$	$\frac{1}{\alpha^2}(\alpha t - 1 + e^{-\alpha t})$
$\frac{1}{s^2(s+\alpha)^2}$	$\frac{1}{\alpha^2}\left[t - \frac{2}{\alpha} + \left(t + \frac{2}{\alpha}\right)e^{-\alpha t}\right]$
$\frac{s}{(s+\alpha)^2}$	$(1 - \alpha t)e^{-\alpha t}$

Laplace Transform $F(s)$	Time Function $f(t)$
$\frac{\omega_n}{s^2+\omega_n^2}$	$\sin\omega_n t$
$\frac{s}{s^2+\omega_n^2}$	$\cos\omega_n t$
$\frac{\omega_n^2}{s(s^2+\omega_n^2)}$	$1-\cos\omega_n t$
$\frac{\omega_n^2(s+\alpha)}{s^2+\omega_n^2}$	$\omega_n\sqrt{\alpha^2+\omega_n^2}\,\sin(\omega_n t+\theta)$ where $\theta=\tan^{-1}(\omega_n/\alpha)$
$\frac{\omega_n}{(s+\alpha)(s^2+\omega_n^2)}$	$\frac{\omega_n}{\alpha^2+\omega_n^2}e^{-\alpha t}+\frac{1}{\sqrt{\alpha^2+\omega_n^2}}\sin(\omega_n t-\theta)$ where $\theta=\tan^{-1}(\omega_n/\alpha)$
$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$	$\frac{\omega_n}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\omega_n\sqrt{1-\zeta^2}\;t\;(\zeta<1)$
$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}\;t+\theta\right)$ where $\theta=\cos^{-1}\zeta\quad(\zeta<1)$
$\frac{s\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$	$\frac{-\omega_n^2}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}\;t-\theta\right)$ where $\theta=\cos^{-1}\zeta\quad(\zeta<1)$
$\frac{\omega_n^2(s+\alpha)}{s^2+2\zeta\omega_n s+\omega_n^2}$	$\omega_n\sqrt{\frac{\alpha^2-2\alpha\zeta\omega_n+\omega_n^2}{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}\;t+\theta\right)$ where $\theta=\tan^{-1}\frac{\omega_n\sqrt{1-\zeta^2}}{\alpha-\zeta\omega_n}\quad(\zeta<1)$
$\frac{\omega_n^2}{s^2(s^2+2\zeta\omega_n s+\omega_n^2)}$	$t-\frac{2\zeta}{\omega_n}+\frac{1}{\omega_n\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}\;t+\theta\right)$ where $\theta=\cos^{-1}(2\zeta^2-1)\quad(\zeta<1)$

APPENDIX D

z-Transform Table

z-Transform Table

Laplace Transform	Time Function	*z*-Transform
1	Unit impulse $\delta(t)$	1
$\frac{1}{s}$	Unit step $u_s(t)$	$\frac{z}{z-1}$
$\frac{1}{1-e^{-Ts}}$	$\delta_T(t)=\sum_{n=0}^{\infty}\delta(t-nT)$	$\frac{z}{z-1}$
$\frac{1}{s^2}$	t	$\frac{Tz}{(z-1)^2}$
$\frac{1}{s^3}$	$\frac{t^2}{2}$	$\frac{T^2z(z+1)}{2(z-1)^3}$
$\frac{1}{s^{n+1}}$	$\frac{t^n}{n!}$	$\lim_{\alpha\to 0}\frac{(-1)^n}{n!}\frac{\partial^n}{\partial\alpha^n}\left[\frac{z}{z-e^{-\alpha T}}\right]$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}$	$\frac{z}{z-e^{-\alpha T}}$
$\frac{1}{(s+\alpha)^2}$	$te^{-\alpha t}$	$\frac{Tze^{-\alpha T}}{(z-e^{-\alpha T})^2}$
$\frac{\alpha}{s(s+\alpha)}$	$1-e^{-\alpha t}$	$\frac{(1-e^{-\alpha T})z}{(z-1)(z-e^{-\alpha T})}$
$\frac{\omega}{s^2+\omega^2}$	$\sin\omega t$	$\frac{z\sin\omega T}{z^2-2z\cos\omega T+1}$
$\frac{\omega}{(s+\alpha)^2+\omega^2}$	$e^{-\alpha t}\sin\omega t$	$\frac{ze^{-\alpha T}\sin\omega T}{z^2-ze^{-\alpha T}\cos\omega T+e^{-2\alpha T}}$
$\frac{s}{s^2+\omega^2}$	$\cos\omega t$	$\frac{z(z-\cos\omega T)}{z^2-2z\cos\omega T+1}$
$\frac{s+\alpha}{(s+\alpha)^2+\omega^2}$	$e^{-\alpha t}\cos\omega t$	$\frac{z^2-ze^{-\alpha T}\cos\omega T}{z^2-2ze^{-\alpha T}\cos\omega T+e^{-2\alpha T}}$

APPENDIX E

Properties and Construction of the Root Loci

The following properties of the root loci are useful for constructing the root loci manually and for understanding the root loci. The properties are developed based on the relationship between the poles and zeros of $G(s)H(s)$ and the zeros of $1 + G(s)H(s)$, which are the roots of the characteristic equation.

E-1 $K = 0$ AND $K = \pm\infty$ POINTS

The $K = 0$ points on the root loci are at the poles of $G(s)H(s)$.

The $K = \pm\infty$ points on the root loci are at the zeros of $G(s)H(s)$.

The poles and zeros referred to here include those at infinity, if any.

The reason for this is seen from the condition of the root loci given by Eq. (7-12)

$$G_1(s)H_1(s) = -\frac{1}{K} \tag{E-1}$$

As the magnitude of K approaches zero, $G_1(s)H_1(s)$ approaches infinity, so s must approach the poles of $G_1(s)H_1(s)$ or $G(s)H(s)$. Similarly, as the magnitude of K approaches infinity, s must approach the poles of $G(s)H(s)$.

EXAMPLE E-1-1 Consider the equation

$$s(s+2)(s+3) + K(s+1) = 0 \tag{E-2}$$

When $K = 0$, the three roots of the equation are at $s = 0, -2$, and -3. When the magnitude of K is infinite, the three roots of the equation are at $s = -1, \infty$ and ∞. It is useful to consider that infinity in the s-plane is a point concept. We can visualize that the finite s-plane is only a small portion of a sphere with an infinite radius. Then, infinity in the s-plane is a point on the opposite side of the sphere that we face.

Dividing both sides of Eq. (E-2) by the terms that do not contain K, we get

$$1 + G(s)H(s) = 1 + \frac{K(s+1)}{s(s+2)(s+3)} = 0 \tag{E-3}$$

which gives

$$G(s)H(s) = \frac{K(s+1)}{s(s+2)(s+3)} \tag{E-4}$$

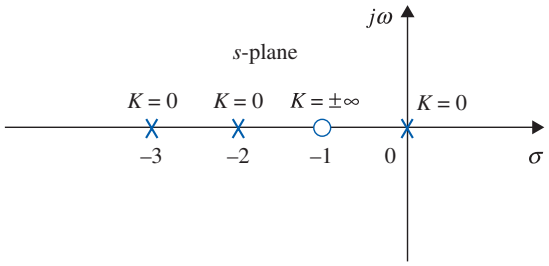

Figure E-1 Roots at which $K = 0$ on the root loci of $s(s + 2)(s + 3) + K(s + 1) = 0$.

Thus, the three roots of Eq. (E-2) when $K = 0$ are the same as the poles of the function $G(s)H(s)$. The three roots of Eq. (E-2) when $K = \pm\infty$ are at the three zeros of $G(s)H(s)$, including those at infinity. The three points on the root loci at which $K = 0$ and those at which $K = \pm\infty$ are shown in Fig. E-1. ◀

▶ E-2 NUMBER OF BRANCHES ON THE ROOT LOCI

A branch of the root loci is the locus of one root when K varies between $-\infty$ and ∞. The following property of the root loci results, since the number of branches of the root loci must equal the number of roots of the equation.

The number of branches of the root loci of $F(s) = P(s) + KQ(s) = 0$ is equal to the order of the polynomial.

• It is important to keep track of the total number of branches of the root loci.

Keeping track of the individual branches and the total number of branches of the root locus diagram is important in making certain that the plot is done correctly. This is particularly true when the root locus plot is done by a computer, since unless each root locus branch is coded by a different color, it is up to the user to make the distinctions.

▶ **EXAMPLE E-2-1** The number of branches of the root loci of

$$s(s + 2)(s + 3) + K(s + 1) = 0 \tag{E-5}$$

is three, since the equation is of the third order. In other words, the equation has three roots, and thus there should be three root loci. ◀

▶ E-3 SYMMETRY OF THE ROOT LOCI

• It is important to pay attention to the symmetry of the root loci.

The root loci are symmetrical with respect to the real axis of the s-plane. In general, the root loci are symmetrical with respect to the axes of symmetry of the pole-zero configuration of $G(s)H(s)$.

The reason behind this property is because for a polynomial with real coefficients the roots must be real or in complex-conjugate pairs. In general, if the poles and zeros of $G(s)H(s)$ are symmetrical to an axis in addition to the real axis in the s-plane, we can regard this axis of symmetry as if it were the real axis of a new complex plane obtained through a linear transformation.

▶ **EXAMPLE E-3-1** Consider the equation

$$s(s+1)(s+2)+K=0 \tag{E-6}$$

Dividing both sides of the equation by the terms that do not contain K, we get

$$G(s)H(s)=\frac{K}{s(s+1)(s+2)} \tag{E-7}$$

The root loci of Eq. (E-6) are shown in Fig. E-2 for $K=-\infty$ to $K=\infty$. Since the pole-zero configuration of $G(s)H(s)$ is symmetrical with respect to the real axis as well as the $s=-1$ axis, the root locus plot is symmetrical to the two axes.

As a review of all the properties of the root loci presented thus far, we conduct the following exercise with regard to the root loci in Fig. E-2.

The points at which $K=0$ are at the poles of $G(s)H(s)$, $s=0$, -1, and -2. The function $G(s)H(s)$ has three zeros at $s=\infty$ at which $K=\pm\infty$. The reader should try to trace out the three separate branches of the root loci by starting from one of the $K=-\infty$ points, through the $K=0$ point on the same branch, and ending at $K=\infty$ at $s=\infty$.

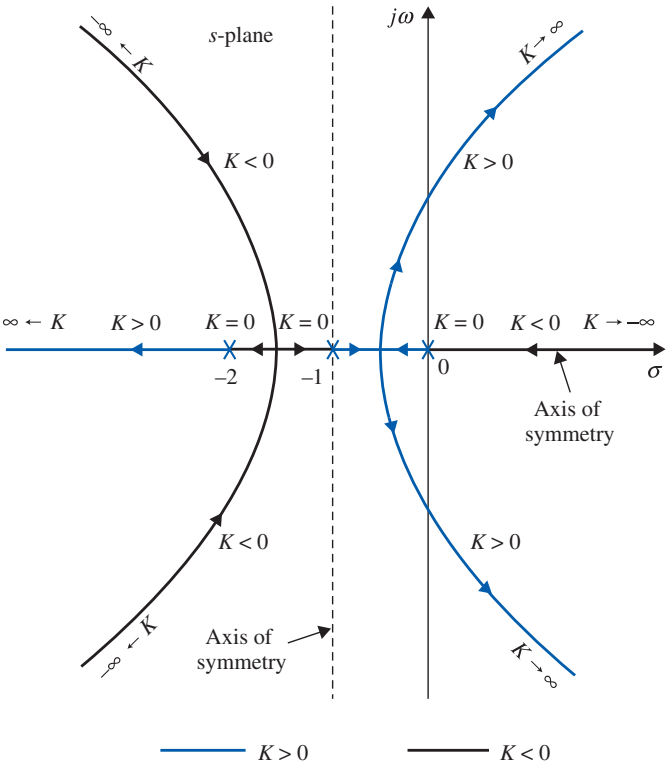

Figure E-2 Root loci of $s(s+2)(s+3)+K(s+1)=0$, showing the properties of symmetry.

▶ **EXAMPLE E-3-2** When the pole-zero configuration of $G(s)H(s)$ is symmetrical with respect to a point in the s-plane, the root loci will also be symmetrical to that point. This is illustrated by the root locus plot of

$$s(s+1)(s+1+j)(s+1-j)+K=0 \tag{E-8}$$

shown in Fig. E-3.

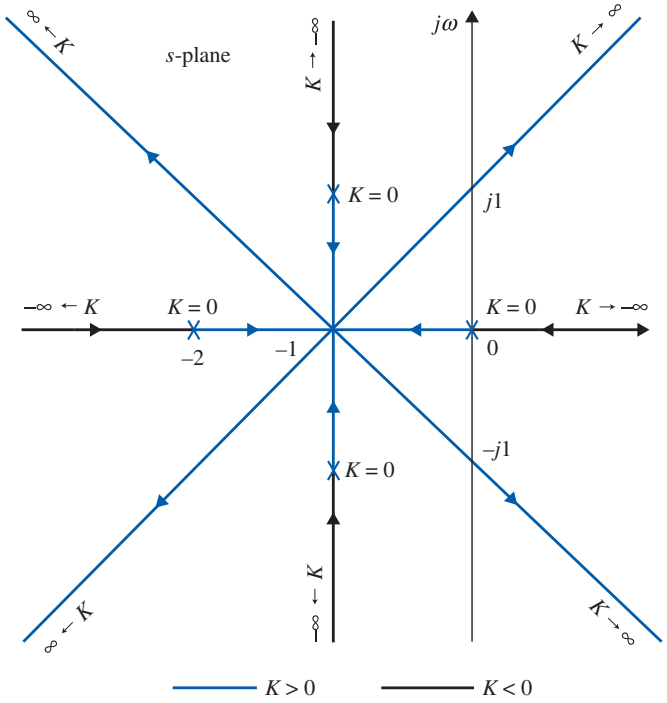

Figure E-3 Root loci of $s(s+2)(s^2+2s+2)+K=0$, showing the properties of symmetry.

▶ E-4 ANGLES OF ASYMPTOTES OF THE ROOT LOCI AND BEHAVIOR OF THE ROOT LOCI AT $|s|=\infty$

• Asymptotes of root loci refers to behavior of root loci at $|s|\to\infty$.

As shown by the root loci in Figs. E-2 and E-3, when n, the order of $P(s)$ is not equal to m, the order of $Q(s)$, $2|n-m|$ of the loci will approach infinity in the s-plane. The properties of the root loci near infinity in the s-plane are described by the **asymptotes** of the loci when $|s|\to\infty$. The angles of the asymptotes and their intersect with the real axis of the s-plane are described as follows.

For large values of s, the root loci for K ≥ 0 (RL) are asymptotic to asymptotes with angles given by

$$\theta_i=\frac{(2i+1)}{|n-m|}\times 180^\circ \qquad n\neq m \tag{E-9}$$

where i = 0, 1, 2, . . . , |n − m| − 1; and n and m are the number of finite poles and zeros of G(s)H(s), respectively.

For $K \leq 0$ (RL), the angles of the asymptotes are

$$\theta_i = \frac{2i}{|n-m|} \times 180^\circ \qquad n \neq m \tag{E-10}$$

where $\mathbf{i = 0, 1, 2, \ldots, |n - m| - 1}$.

▶ E-5 INTERSECT OF THE ASYMPTOTES (CENTROID)

The intersection of the $2|n - m|$ asymptotes of the root loci lies on the real axis of the s-plane, at

$$\sigma_1 = \frac{\Sigma \text{ finite poles of } G(s)H(s) - \Sigma \text{ finite zeros of } G(s)H(s)}{n - m} \tag{E-11}$$

where n is the number of finite poles and m is the number of finite zeros of G(s)H(s), respectively. The intersection of the asymptotes σ_1 represents the center of gravity of the root loci, and is always a real number.

Since the poles and zeros of $G(s)H(s)$ are either real or in complex-conjugate pairs, the imaginary parts in the numerator of Eq. (7-35) always cancel each other out. Thus, in Eq. (E-11), the terms in the summations may be replaced by the real parts of the poles and zeros of $G(s)H(s)$, respectively. That is,

$$\sigma_1 = \frac{\Sigma \text{ real parts of poles of } G(s)H(s) - \Sigma \text{ real parts of zeros of } G(s)H(s)}{n - m} \tag{E-12}$$

▶ EXAMPLE E-5-1 Consider the transfer function

$$G(s)H(s) = \frac{K(s+1)}{s(s+4)(s^2+2s+2)} \tag{E-13}$$

which corresponds to the characteristic equation

$$s(s+4)(s^2+2s+2) + K(s+1) = 0 \tag{E-14}$$

The pole-zero configuration of $G(s)H(s)$ is shown in Fig. E-4. From the six properties of the root loci discussed so far, the following information concerning the root loci of Eq. (E-14) when K varies from $-\infty$ to ∞ is obtained:

1. $K = 0$: The points at which $K = 0$ on the root loci are at the poles of $G(s)H(s)$: $s = 0, -4, -1 + j$, and $-1 - j$.
2. $K = \pm\infty$: The points at which $K = \pm\infty$ on the root loci are at the zeros of $G(s)H(s)$: $s = -1, \infty, \infty$, and ∞.
3. There are four root loci branches, since Eqs. (E-13) and (E-14) are of the fourth order.
4. The root loci are symmetrical to the real axis.

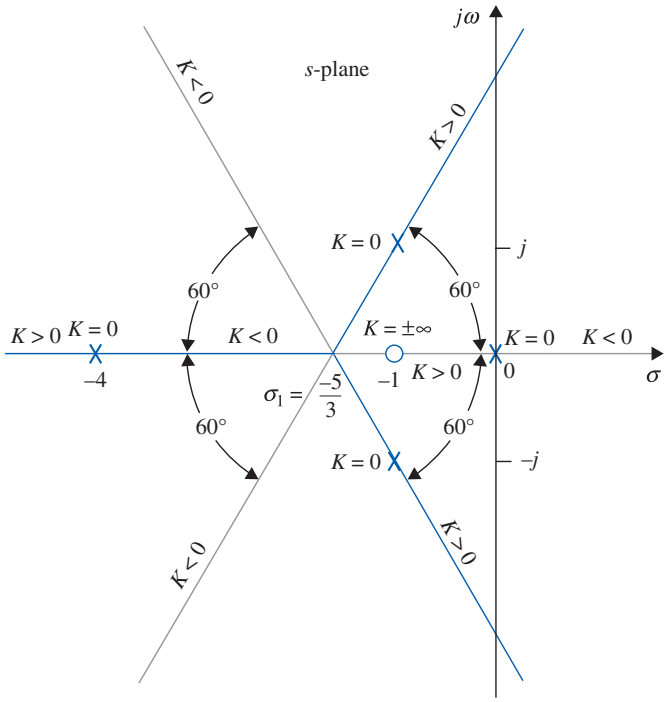

Figure E-4 Asymptotes of the root loci of $s(s + 4)(s^2 + 2s + 2) + K(s + 1) = 0$.

5. Since the number of finite poles of $G(s)H(s)$ exceeds the number of finite zeros of $G(s)H(s)$ by three ($n - m = 4 - 1 = 3$), when $K = \pm\infty$, three root loci approach $s = \infty$.

The angles of the asymptotes of the RL ($K \geq 0$) are given by Eq. (E-9):

$$j = 0: \quad \theta_0 = \frac{180^\circ}{3} = 60^\circ$$

$$j = 1: \quad \theta_1 = \frac{540^\circ}{3} = 180^\circ$$

$$j = 2: \quad \theta_2 = \frac{900^\circ}{3} = 300^\circ$$

The angles of the asymptotes of the root loci for $K \leq 0$ are given by Eq. (E-10), and are calculated to be 0°, 120°, and 240°.

6. The intersection of the asymptotes is given by Eq. (E-12):

$$\sigma_1 = \frac{(-4 - 1 - 1) - (-1)}{4 - 1} = -\frac{5}{3} \tag{E-15}$$

The asymptotes of the root loci are shown in Fig. E-4.

◀

▶ **EXAMPLE E-5-2** The asymptotes of the root loci of several equations are shown in Fig. E-5.

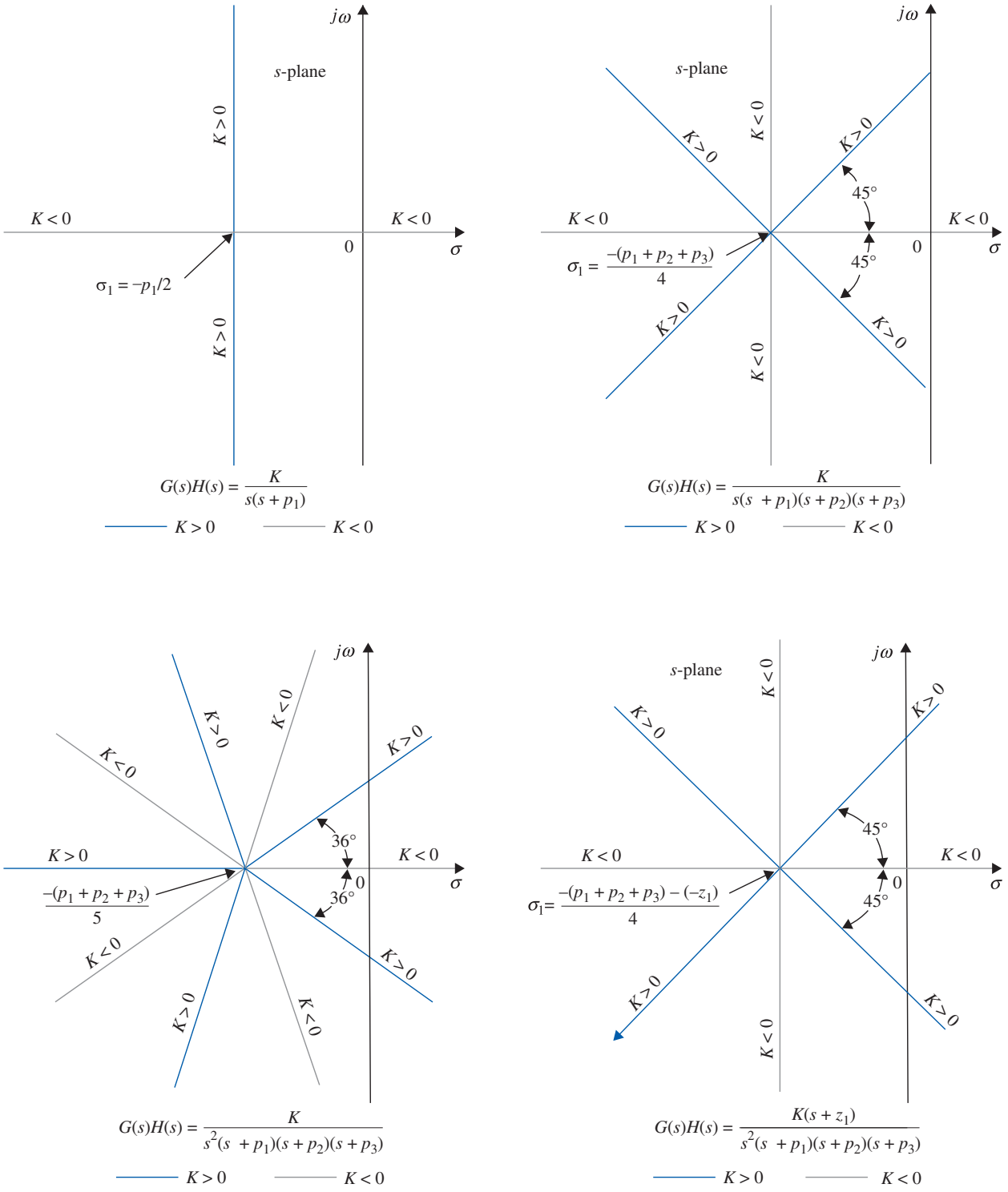

Figure E-5 Examples of the asymtotes of the root loci.

▶ E-6 ROOT LOCI ON THE REAL AXIS

The entire real axis of the s-plane is occupied by the root loci for $-\infty \leq K \leq \infty$.

• The entire real axis of the s-plane is occupied by root loci.

1. ***$K \geq 0$: On a given section of the real axis, root loci are found in the section only if the total number of poles and zeros of G(s)H(s) to the right of the section is odd.***
2. ***$K \leq 0$: On a given section of the real axis, root loci are found in the section only if the total number of real poles and zeros of G(s)H(s) to the right of the section is even. Complex poles and zeros of G(s)H(s) do not affect the type of root loci found on the real axis.***

These properties are arrived at based on the following observations:

1. At any point s_1 on the real axis, the angles of the vectors drawn from the complex-conjugate poles and zeros of $G(s)H(s)$ add up to zero. Thus, only the real zeros and poles of $G(s)H(s)$ contribute to the angular relations in Eqs. (7-18) and (7-19).
2. Only the real poles and zeros of $G(s)H(s)$ that lie to the right of the point s_1 contribute to Eqs. (7-18) and (7-19), because real poles and zeros that lie to the left of the point contribute nothing.
3. Each real pole of $G(s)H(s)$ to the right of s_1 contributes -180 degrees, and each real zero of $G(s)H(s)$ to the right of s_1 contributes $+180$ degrees to Eqs. (7-18) and (7-19).

The last observation shows that for s_1 to be a point on the root locus, there must be an **odd** number of poles and zeros of $G(s)H(s)$ to the right of the point. For s_1 to be a point on the branch of the root loci for $K \leq 0$, the total number of poles and zeros of $G(s)H(s)$ to the right of the point must be **even**. The following example illustrates the determination of the properties of the root loci on the real axis of the s-plane.

▶ EXAMPLE E-6-1 The root loci on the real axis for two pole-zero configurations of $G(s)H(s)$ are shown in Fig. E-6. Notice that the entire real axis is occupied by the root loci for all values of K.

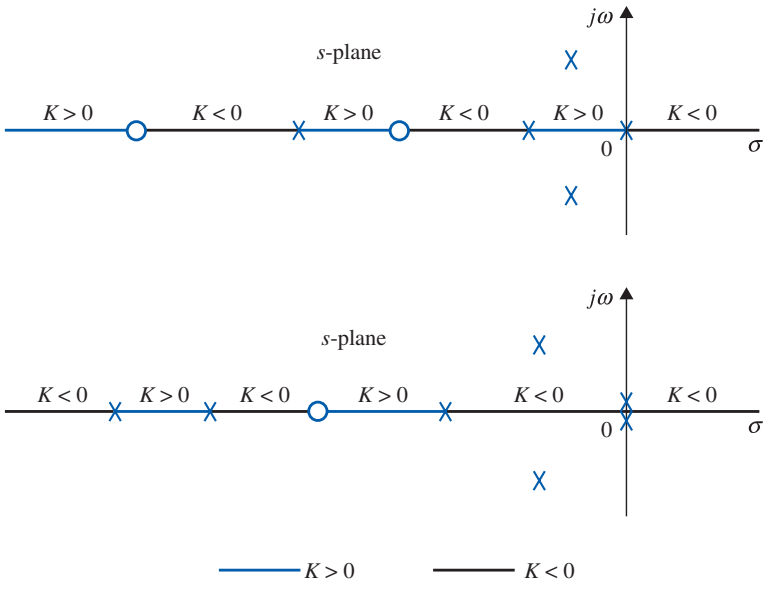

Figure E-6 Properties of root loci on the real axis. ◀

▶ E-7 ANGLES OF DEPARTURE AND ANGLES OF ARRIVAL OF THE ROOT LOCI

The angle of departure or arrival of a root locus at a pole or zero of $G(s)H(s)$ denotes the angle of the tangent to the locus near the point.

The angles of departure and arrival are determined using Eq. (7-18) for root loci for positive K and Eq. (7-19) for root loci for negative K. The details are illustrated by the following example.

▶ EXAMPLE E-7-1 For the root-locus diagram shown in Fig. E-7, the root locus near the pole $s = -1 + j$ may be more accurately sketched by knowing the angle at which the root locus leaves the pole. As shown in Fig. E-7, the angle of departure of the root locus at $s = -1 + j$ is represented by θ_2, measured with respect to the real axis. Let us assign s_1 to be a point on the RL leaving the pole at $-1 + j$ and is very close to the pole. Then, s_1 must satisfy Eq. (7-18). Thus,

$$\angle G(s_1)H(s_1) = -(\theta_1 + \theta_2 + \theta_3 + \theta_4) = (2i + 1)180^\circ \qquad \text{(E-16)}$$

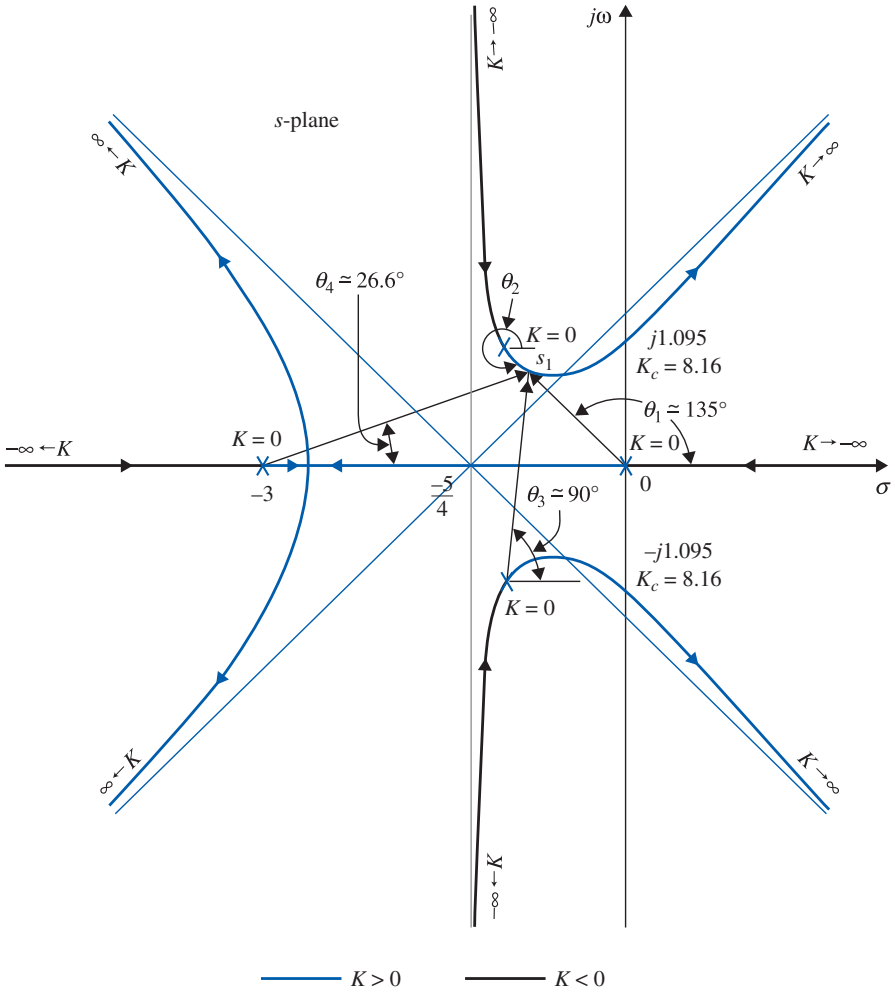

Figure E-7 Root loci of $s(s + 3)(s^2 + 2s + 2) + K = 0$ to illustrate the angles of departure or arrival.

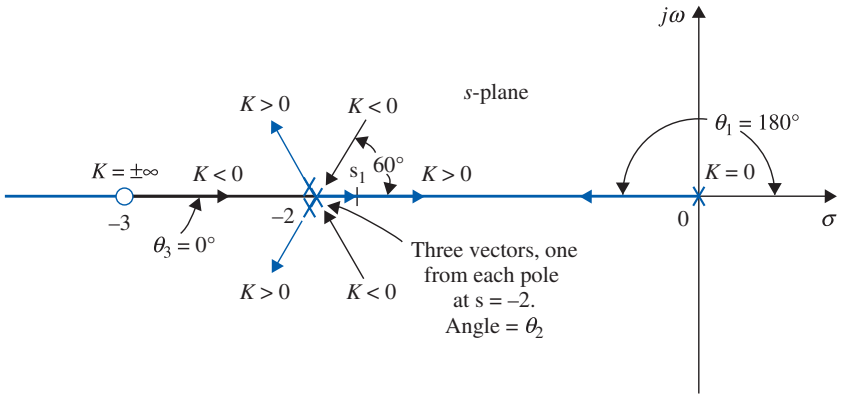

Figure E-8 Angles of departure and arrival at a third-order pole.

where i is any integer. Since s_1 is assumed to be very close to the pole at $-1 + j$, the angles of the vectors drawn from the other three poles are approximated by considering that s_1 is at $-1 + j$. From Fig. E-7, Eq. (E-16) is written

$$-(135^\circ + \theta_2 + 90^\circ + 26.6^\circ) = (2i + 1)180^\circ \tag{E-17}$$

where θ_2 is the only unknown angle. In this case, we can set i to be -1, and the result for θ_2 is -71.6°.

When the angle of departure or arrival of a root locus for positive K at a simple pole or zero of $G(s)H(s)$ is determined, the angle of arrival or departure of the root locus for negative K at the same point differs from this angle by 180°, and Eq. (7-19) is now used. Figure E-7 shows that the angle of arrival of the root locus for negative K at $-1 + j$ is 108.4°, which is 180° − 71.6°. Similarly, for the root-locus diagram in Fig. E-8, we can show that the root locus for negative K arrives at the pole $s = -3$ with an angle of 180°, and the root locus for positive K leaves the same pole at 0°. For the pole at $s = 0$, the angle of arrival of the negative-K root locus is 180°, whereas the angle of departure of the positive-K root locus is 180°. These angles are also determined from the knowledge of the type of root loci on sections of the real axis separated by the poles and zeros of $G(s)H(s)$. Since the total angles of the vectors drawn from complex-conjugate poles and zeros to any point on the real axis add up to be zero, the angles of arrival and departure of root loci on the real axis are not affected by the complex poles and zeros of $G(s)H(s)$. ◀

▶ **EXAMPLE E-7-2** In this example we examine the angles of departure and arrival of the root loci at a multiple-order pole or zero of $G(s)H(s)$. Consider that a $G(s)H(s)$ has a multiple-order (third-order) pole on the real axis, as shown in Fig. E-8. Only the real poles and zeros of $G(s)H(s)$ are shown, since the complex ones do not affect the type or the angles of arrival and departure of the root loci on the real axis. For the third-order pole at $s = -2$, there are three positive-K loci leaving and three negative-K loci arriving at the point. To find the angles of departure of the positive-K root loci, we assign a point s_1 on one of the loci near $s = -2$, and apply Eq. (7-18). The result is

$$-\theta_1 - 3\theta_2 + \theta_3 = (2i + 1)180^\circ \tag{E-18}$$

where θ_1 and θ_3 denote the angles of the vectors drawn from the pole at 0 and the zero at -3, respectively, to s_1. The angle θ_2 is multiplied by 3, since there are three poles at $s = -2$, so that there are three vectors drawn from -2 to s_1. Setting i to zero in Eq. (E-18), and since $\theta_1 = 180^\circ$, $\theta_3 = 0^\circ$ we have $\theta_2 = 0^\circ$, which is the angle of departure of the positive-K root loci that lies between $s = 0$ and $s = -2$. For the angles of departure of the other two positive-K loci, we set $i = 1$ and $i = 2$ successively in Eq. (E-18), and we have $\theta_2 = 120^\circ$ and -120°. Similarly, for the three negative-K root loci that arrive at $s = -2$, Eq. (7-19) is used, and the angles of arrivals are found to be 60°, 180°, and -60°. ◀

▶ E-8 INTERSECTION OF THE ROOT LOCI WITH THE IMAGINARY AXIS

• Routh-Hurwitz criterion may be used to find the intersection of the root loci on the imaginary axis.

The points where the root loci intersect the imaginary axis of the s-plane, and the corresponding values of K, may be determined by means of the Routh-Hurwitz criterion. For complex situations, when the root loci have multiple number of intersections on the imaginary axis, the intersects and the critical values of K can be determined with the help of the root-locus computer program. The Bode diagram method in Chapters 2 and 8 associated with the frequency response, can also be used for this purpose.

▶ EXAMPLE E-8-1 The root loci shown in Fig. E-7 is for the equation

$$s(s+3)(s^2+2s+2)+K=0 \tag{E-19}$$

Fig. E-7 shows that the root loci intersect the $j\omega$ axis at two points. Applying the Routh-Hurwitz criterion to Eq. (E-19), and by solving the auxiliary equation, we have the critical value of K for stability at $K = 8.16$, and the corresponding crossover points on the $j\omega$-axis are at $\pm$j1.095. ◀

▶ E-9 BREAKAWAY POINTS

E-9-1 (Saddle Points) on the Root Loci

Breakaway points on the root loci of an equation correspond to multiple-order roots of the equation.

Fig. E-9(a) illustrates a case in which two branches of the root loci meet at the breakaway point on the real axis and then depart from the axis in opposite directions. In this

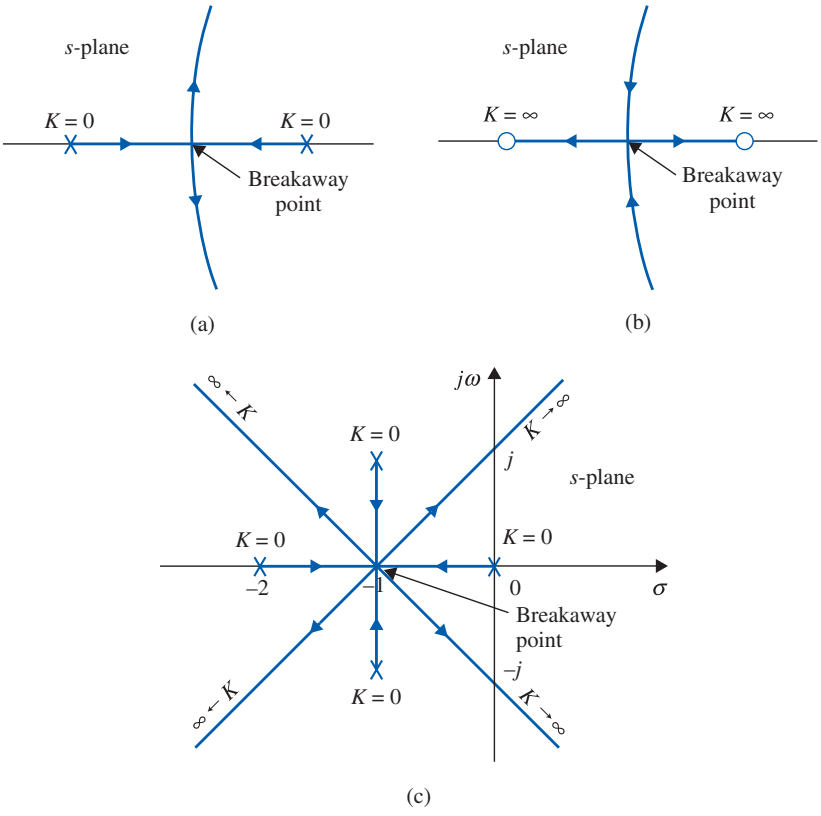

Figure E-9 Examples of breakaway points on the real axis in the s-plane.

case, the breakaway point represents a double root of the equation when the value of K is assigned the value corresponding to the point. Fig. E-9(b) shows another common situation when two complex-conjugate root loci approach the real axis, meet at the breakaway point, and then depart in opposite directions along the real axis. In general, a breakaway point may involve more than two root loci. Fig. E-9(c) illustrates a situation when the breakaway point represents a fourth-order root.

• A root-locus plot may have more than one breakaway points.

A root-locus diagram can have, of course, more than one breakaway point. Moreover, the breakaway points need not always be on the real axis. Because of the conjugate symmetry of the root loci, the breakaway points not on the real axis must be in complex-conjugate pairs. Refer to Fig. E-12 for an example of root loci with complex breakaway points. The properties of the breakaway points of root loci are given as follows:

• Breakaway points may be complex conjugates in the s-plane

The breakaway points on the root loci of $1 + KG_1(s)H_1(s) = 0$ ***must satisfy***

$$\frac{dG_1(s)H_1(s)}{ds} = 0 \tag{E-20}$$

It is important to point out that the condition for the breakaway point given in Eq. (E-20) is *necessary* but *not sufficient*. In other words, all breakaway points on the root loci for all values of K must satisfy Eq. (E-20), but not all solutions of Eq. (E-20) are breakaway points. To be a breakaway point, the solution of Eq. (E-20) must also satisfy the equation $1 + KG_1(s)H_1(s) = 0$, that is, must also be a point on the root loci for some real K. In general, the following conclusions may be made with regard to the solutions of Eq. (E-20):

1. All *real* solutions of Eq. (E-20) are breakaway points on the root loci for all values of K, since the entire real axis of the s-plane is occupied by the root loci.
2. The complex-conjugate solutions of Eq. (E-20) are breakaway points only if they satisfy the characteristic equation or are points on the root loci.
3. Since the condition of the root loci is

$$K = -\frac{1}{G_1(s)H_1(s)} \tag{E-21}$$

taking the derivative on both sides of the equation with respect to s, we have

$$\frac{dK}{ds} = \frac{dG_1(s)H_1(s)/ds}{[G_1(s)H_1(s)]^2} \tag{E-22}$$

Thus, the breakaway point condition can also be written as

$$\frac{dK}{ds} = 0 \tag{E-23}$$

where K is expressed as in Eq. (E-21).

E-9-2 The Angle of Arrival and Departure of Root Loci at the Breakaway Point

The angles at which the root loci arrive or depart from a breakaway point depend on the number of loci that are involved at the point. For example, the root loci shown in

Figs. E-9(a) and E-9(b) all arrive and break away at 180° apart, whereas in Fig. E-9(c), the four root loci arrive and depart with angles 90° apart, whereas in Fig. E-9(c), the four root loci arrive and depart with angles 90° apart. In general, ***n root loci*** $(-\infty \leq K \leq \infty)$ ***arrive or depart a breakaway point at 180/n degrees apart***.

Many root-locus computer programs have features that will obtain the breakaway points, which a rather tedious task to do manually.

▶ **EXAMPLE E-9-1** Consider the second-order equation

$$s(s+2)+K(s+4)=0 \tag{E-24}$$

Based on some of the properties of the root loci described thus far, the root loci of Eq. (E-24) are sketched as shown in Fig. E-10 for $-\infty < K < \infty$. It can be proven that the complex portion of the root loci is a circle. The two breakaway points are on the real axis, one between 0 and -2 and the other between -4 and $-\infty$. From Eq. (E-24), we have

$$G_1(s)H_1(s)=\frac{s+4}{s(s+2)} \tag{E-25}$$

Applying Eq. (E-20), the breakaway points on the root loci must satisfy

$$\frac{dG_1(s)H_1(s)}{ds}=\frac{s(s+2)-2(s+1)(s+4)}{s^2(s+2)^2}=0 \tag{E-26}$$

or

$$s^2+8s+8=0 \tag{E-27}$$

Solving Eq. (E-27), we find the two breakaway points of the root loci at $s=-1.172$ and -6.828. Fig. E-10 shows that the two breakaway points are all on the root loci for positive K.

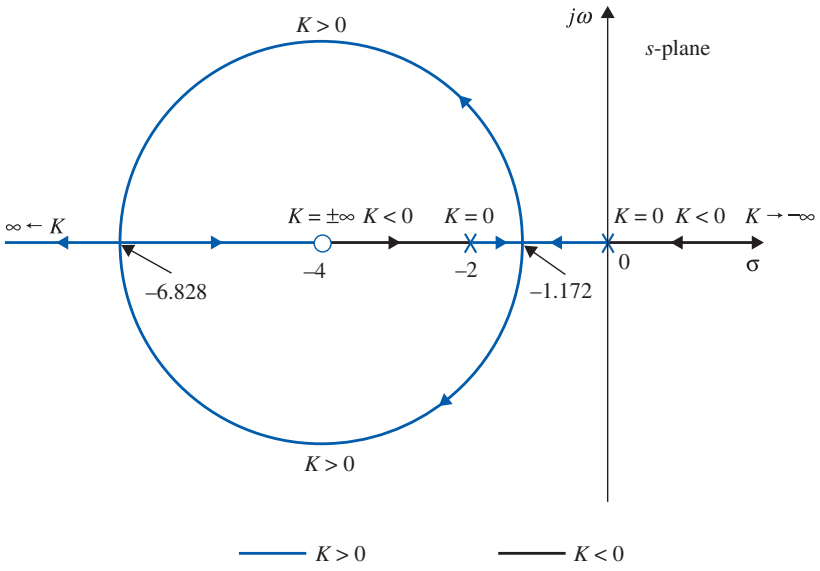

Figure E-10 Root loci of $s(s+2)+K(s+4)=0$.

◀

▶ **EXAMPLE E-9-2** Consider the equation

$$s^2+2s+2+K(s+2)=0 \tag{E-28}$$

The equivalent $G(s)H(s)$ is obtained by dividing both sides of Eq. (E-28) by the terms that do not contain K. We have

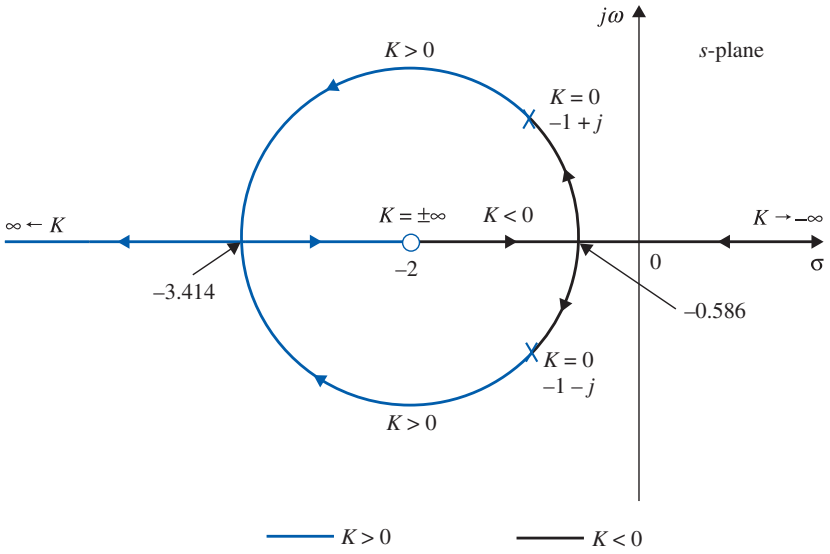

Figure E-11 Root loci of $s^2 + 2s + 2 + K(s + 2) = 0$.

$$G(s)H(s) = \frac{K(s+2)}{s^2+2s+2} \tag{E-29}$$

Based on the poles and zeros of $G(s)H(s)$, the root loci of Eq. (E-29) are plotted as shown in Fig. E-11. The plot shows that there are two breakaway points, one for $K > 0$ and one for $K < 0$. These breakaway points are determined from

$$\frac{dG_1(s)H_1(s)}{ds} = \frac{d}{ds}\left(\frac{s+2}{s^2+2s+2}\right) = \frac{s^2+2s+2-2(s+1)(s+2)}{(s^2+2s+2)^2} = 0 \tag{E-30}$$

or

$$s^2 + 4s + 2 = 0 \tag{E-31}$$

The solution of this equation gives the breakaway point as $s = -0.586$ and $s = -3.414$. ◀

▶ **EXAMPLE E-9-3** Fig. E-12 shows the root loci of the equation

$$s(s+4)(s^2+4s+20) + K = 0 \tag{E-32}$$

Dividing both sides of the last equation by the terms that do not contain K, we have

$$1 + KG_1(s)H_1(s) = 1 + \frac{K}{s(s+4)(s^2+4s+20)} = 0 \tag{E-33}$$

Since the poles of $G_1(s)H_1(s)$ are symmetrical about the axes $\sigma = -2$ *and* $\omega = 0$ *in the* s-plane, the root loci of the equation are also symmetrical with respect to these two axes. Taking the derivative of $G_1(s)H_1(s)$ with respect to s, we get

$$\frac{dG_1(s)H_1(s)}{ds} = -\frac{4s^3+24s^2+72s+80}{[s(s+4)(s^2+4s+20)]^2} = 0 \tag{E-34}$$

or

$$s^3 + 6s^2 + 18s + 20 = 0 \tag{E-35}$$

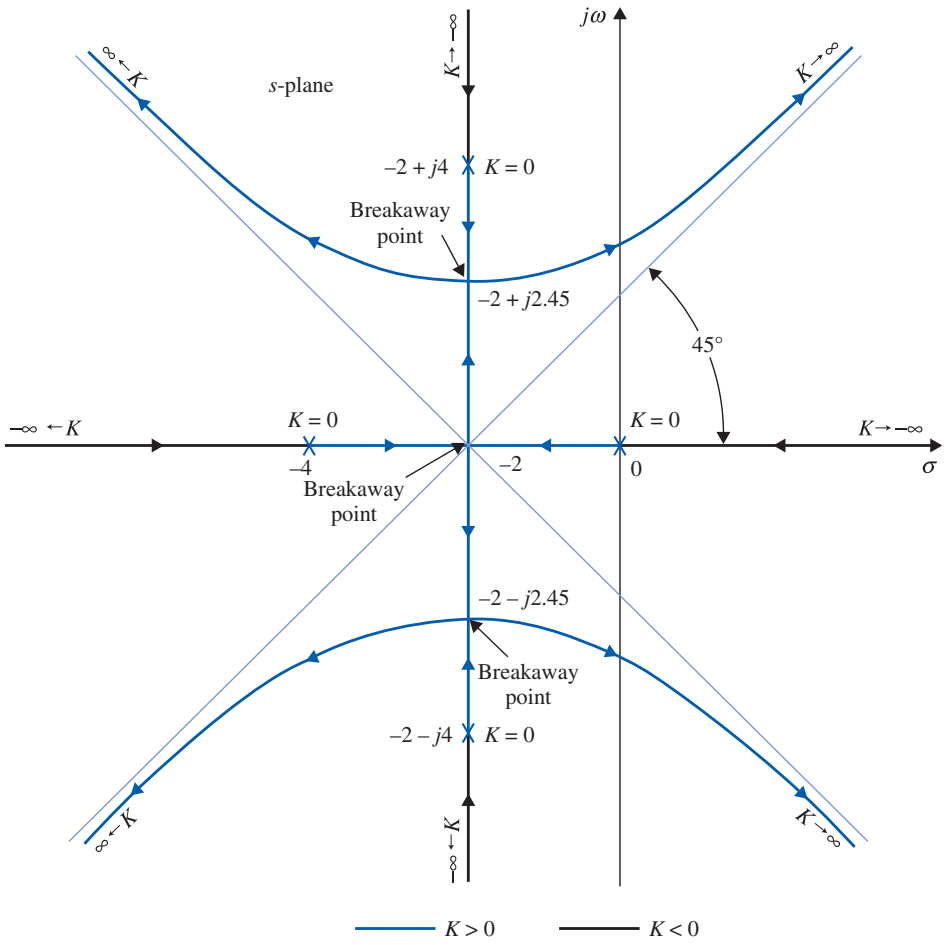

Figure E-12 Root loci of $s(s + 4)(s^2 + 4s + 20) + K = 0$.

The solutions of the last equation are $s = -2$, $-2 + j2.45$, and $-2 - j2.45$. In this case, Fig. E-12 shows that all the solutions of Eq. (E-35) are breakaway points on the root loci, and two of these points are complex. ◀

▶ **EXAMPLE E-9-4** In this example, we shall show that not all the solutions of Eq. (E-20) are breakaway points on the root loci. The root loci of the equation

$$s(s^2 + 2s + 2) + K = 0 \tag{E-36}$$

are shown in Fig. E-13. The root loci show that neither the $K \geq 0$ loci nor the $K \leq 0$ loci has any breakaway point in this case. However, writing Eq. (E-36) as

$$1 + KG_1(s)H_1(s) = 1 + \frac{K}{s(s^2 + 2s + 2)} = 0 \tag{E-37}$$

and applying Eq. (E-20), we have the equation for the breakaway points:

$$3s^2 + 4s + 2 = 0 \tag{E-38}$$

The roots of Eq. (E-38) are $s = -0.667 + j0.471$ and $-0.667 - j0.471$. These two roots are **not** breakaway points on the root loci, since they do not satisfy Eq. (E-36) for any real value of K.

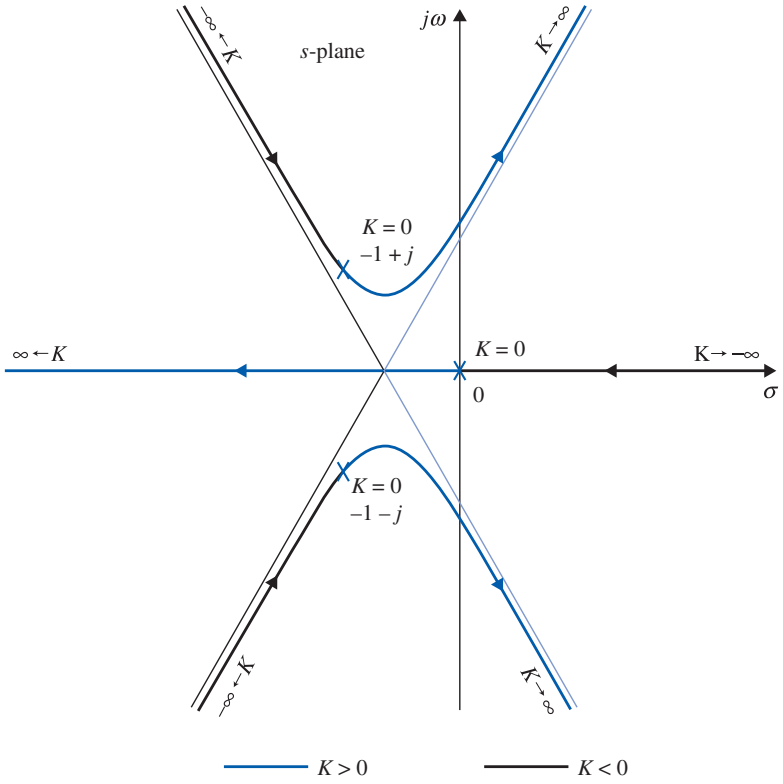

Figure E-13 Root loci of $s(s^2 + 2s + 2) + K = 0$.

▶ E-10 CALCULATION OF K ON THE ROOT LOCI

Once the root loci are constructed, the values of K at any point s_1 on the loci can be determined by use of the defining equation of Eq. (7-20). Graphically, the magnitude of K can be written as

$$|K| = \frac{\Pi \text{ lengths of vectors drawn from the poles of } G_1(s)H_1(s) \text{ to } s_1}{\Pi \text{ lengths of vectors drawn from the zeros of } G_1(s)H_1(s) \text{ to } s_1} \tag{E-39}$$

▶ EXAMPLE E-10-1 As an illustration on the determination of the value of K on the root loci, the root loci of the equation

$$s^2 + 2s + 2 + K(s + 2) = 0 \tag{E-40}$$

are shown in Fig. E-14. The value of K at the point s_1 is given by

$$K = \frac{A \times B}{C} \tag{E-41}$$

where A and B are the lengths of the vectors drawn from the poles of $G(s)H(s) = K(s + 2)/(s^2 + 2s + 2)$ to the point s_1, and C is the length of the vector drawn from the zero of $G(s)H(s)$ to s_1. In this case, s_1 is on the locus where K is positive. In general, the value of K at the point where the root loci intersect the imaginary axis can also be found by the method just described. Figure E-14 shows that the value of K at $s = 0$ is -1. The computer method and the Routh-Hurwitz criterion are other convenient alternatives of finding the critical value of K for stability.

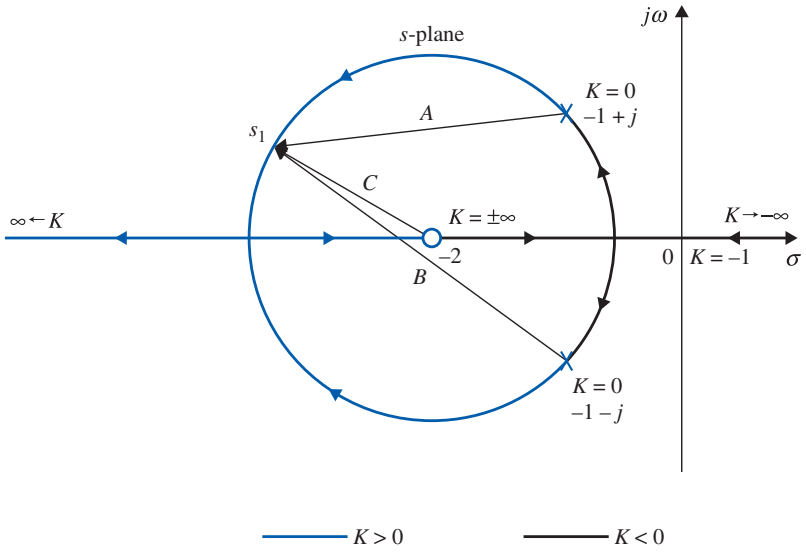

Figure E-14 Graphical method of finding the values of K on the real axis. ◀

In summary, except for extremely complex cases, the properties on the root loci presented here should be adequate for making a reasonably accurate sketch of the root-locus diagram short of plotting it point by point. The computer program can be used to solve for the exact root locations, the breakaway points, and some of the other specific details of the root loci, including the plotting of the final loci. However, one cannot rely on the computer solution completely, since the user still has to decide on the range and resolution of K so that the root-locus plot has a reasonable appearance. For quick reference, the important properties described are summarized in Table E-1.

TABLE E-1 Properties of the Root Loci of $F(s) = 1 + KG_1(s)H_1(s) = 0$

1.	$K = 0$ points	The $K = 0$ points are at the poles of $G(s)H(s)$, including those at $s = \infty$.
2.	$K = \pm\infty$ points	The $K = \pm\infty$ points are at the zeros of $G(s)H(s)$, including those at $s = \infty$.
3.	Number of separate root loci	The total number of root loci is equal to the order of the characteristic equation $F(s)$
4.	Symmetry of root loci	The root loci are symmetrical about the axes of symmetry of the pole-zero configuration of $G(s)H(s)$.
5.	Asymptotes of root loci as $\lvert s\rvert \to \infty$	For large values of s, the root loci for $K > 0$ are asymptotic to asymptotes with angles given by $\theta_i = \dfrac{2i+1}{\lvert n-m\rvert} \times 180^\circ$ For $K < 0$, the root loci are asymptotic to $\theta_i = \dfrac{2i}{\lvert n-m\rvert} \times 180^\circ$ where $i = 0, 1, 2, \ldots, \lvert n-m\rvert - 1$, n = number of finite poles of $G(s)H(s)$, and m = number of finite zeros of $G(s)H(s)$.

(Continued)

TABLE E-1 (Continued)

6.	Intersection of the asymptotes	(a) The intersection of the asymptotes lies only on the real axis in the s-plane. (b) The point of intersection of the asymptotes is given by $$\sigma_1 = \frac{\Sigma \text{ real parts of poles of } G(s)H(s) - \Sigma \text{ real parts of zeros of } G(s)H(s)}{n-m}$$
7.	Root loci on the real axis	Root loci for $K > 0$ are found in a section of the real axis only if the total number of real poles and zeros of $G(s)H(s)$ to the **right** of the section is **odd**. If the total number of real poles and zeros to the right of a given section is **even**, root loci for $K < 0$ are found.
8.	Angles of departure	The angle of departure or arrival of the root loci from a pole or a zero of $G(s)H(s)$ can be determined by assuming a point s_1 that is very close to the pole, or zero, and applying the equation $$\begin{aligned}\angle G(s_1)H(s_1) &= \sum_{k=1}^{m}\angle(s_1 + z_k) - \sum_{j=1}^{n}\angle(s_1 + p_j)\\ &= 2(i+1)180^\circ \quad K>0\\ &= 2i \times 180^\circ \quad K<0\end{aligned}$$ where $i = 0, \pm 1, \pm 2, \ldots$.
9.	Intersection of the root loci	The crossing points of the root loci on the imaginary axis and with the imaginary axis the corresponding values of K may be found by use of the Routh-Hurwitz criterion.
10.	Breakaway points	The breakaway points on the root loci are determined by finding the roots of $dK/ds = 0$, or $dG(s)H(s)/ds = 0$. These are necessary conditions only.
11.	Calculation of the values of K	The absolute value of K at any point s_1 on the root loci is on the root loci determined from the equation $$|K| = \frac{1}{|G_1(s_1)H_1(s_1)|}$$

The following example illustrates the construction of a root locus diagram manually, step by step, using the root locus properties given in Table E-1.

▶ EXAMPLE E-10-2 Consider the equation

$$s(s+5)(s+6)(s^2+2s+2) + K(s+3) = 0 \tag{E-42}$$

Dividing both sides of the last equation by the terms that do not contain K, we have

$$G(s)H(s) = \frac{K(s+3)}{s(s+5)(s+6)(s^2+2s+2)} \tag{E-43}$$

The following properties of the root loci are determined:

1. The $K = 0$ points are at the poles of $G(s)H(s)$: $s = -5, -6, -1 + j$, and $-1 - j$.
2. The $K = \pm\infty$ points are at the zeros of $G(s)H(s)$: $s = -3, \infty, \infty, \infty, \infty$.

3. There are five separate branches on the root loci.
4. The root loci are symmetrical with respect to the real axis of the s-plane.
5. Since $G(s)H(s)$ has five poles and one finite zero, four RL and CRL should approach infinity along the asymptotes. The angles of the asymptotes of the RL are given by [Eq. (E-9)]

$$\theta_i = \frac{2i+1}{|n-m|}180^\circ = \frac{2i+1}{|5-1|}180^\circ \quad 0 \le K < \infty \tag{E-44}$$

for $i = 0, 1, 2, 3$. Thus, the four root loci that approach infinity as K approaches infinity should approach asymptotes with angles of 45°, −45°, 135°, and −135°, respectively. The angles of the asymptotes of the CRL at infinity are given by Eq. (E-10):

$$\theta_i = \frac{2i}{|n-m|}180^\circ = \frac{2i}{|5-1|}180^\circ \quad -\infty < K \le 0 \tag{E-45}$$

for $i = 0, 1, 2, 3$. Thus, as K approaches $-\infty$, four root loci for $K < 0$ should approach infinity along asymptotes with angles of 0°, 90°, 180°, and 270°.

6. The intersection of the asymptotes is given by [Eq. (E-12)]

$$\sigma_1 = \frac{\Sigma(-5-6-1-1)-(-3)}{4} = -2.5 \tag{E-46}$$

The results from these six steps are illustrated in Fig. E-15. It should be pointed out that in general the properties of the asymptotes do not indicate on which side of the asymptotes the root loci lie. The asymptotes indicate nothing more than the behavior of the root loci as

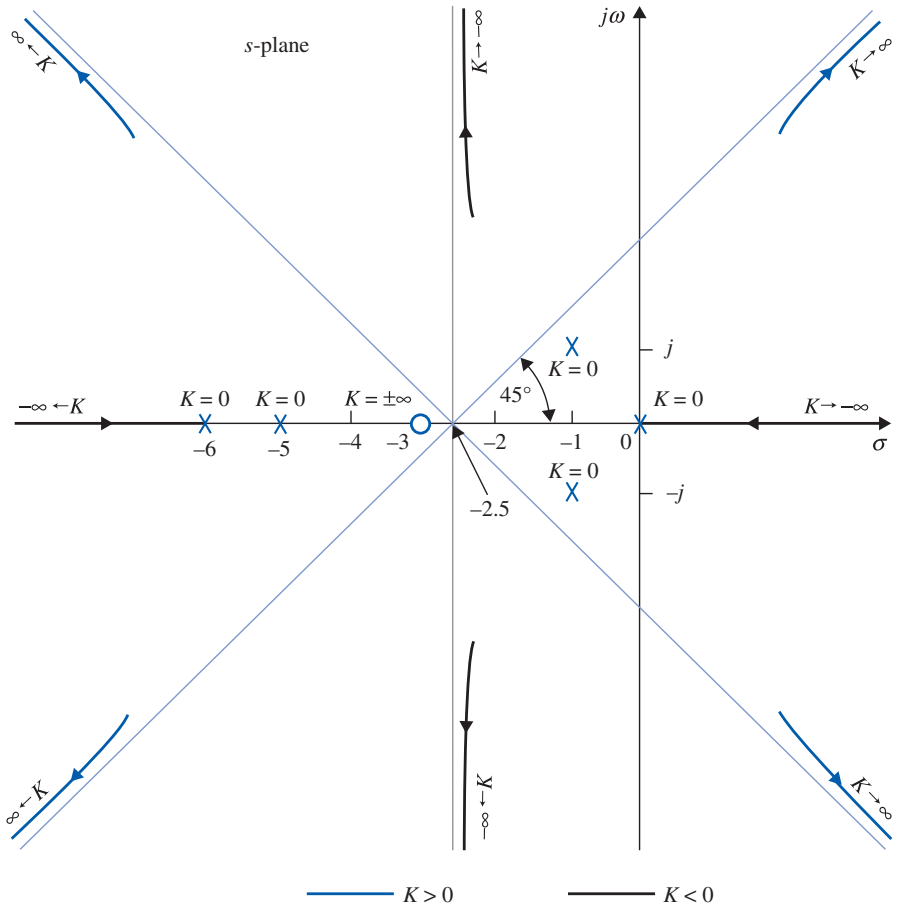

Figure E-15 Preliminary calculation of the root loci of $s(s+5)(s+6)(s^2+2s+2)+K(s+3)=0$.

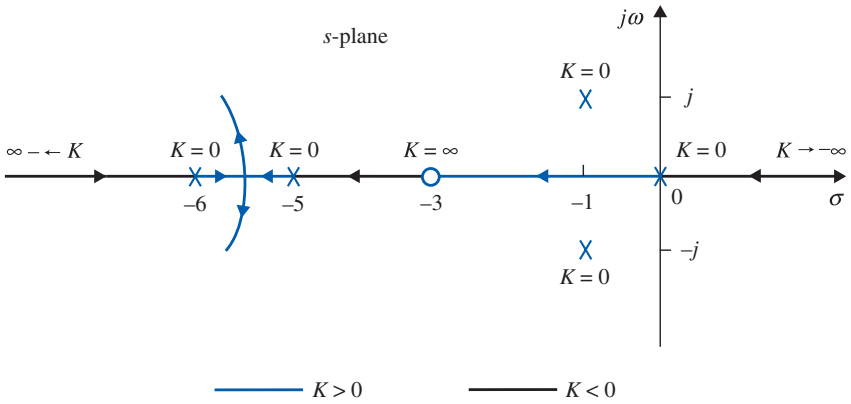

Figure E-16 Root loci of $s(s + 5)(s + 6)(s^2 + 2s + 2) + K(s + 3) = 0$ on the real axis.

$|s| \to \infty$ In fact, the root locus can even cross an asymptote in the finite s domain. The segments of the root loci shown in Fig. E-15 can be accurately plotted only if additional information is obtained.

7. Root loci on the real axis: There are $K \geq 0$ root loci on the real axis between $s = 0$ and -3, and $s = -5$ and -6. There are $K \leq 0$ root loci on the remaining portions of the real axis, that is, between $s = -3$ and -5, and $s = -6$ and $-\infty$, as shown in Fig. E-16.

8. Angles of departure: The angle of departure θ of the root loci leaving the pole at $-1 + j$ is determined using Eq. (7-18). If s_1 is a point on the root loci leaving the pole at $-1 + j$, and s_1 is very close to the pole, as shown in Fig E-17, Eq. (7-18) gives

$$\begin{aligned} &\angle(s_1 + 3) - \angle s_1 - \angle(s_1 + 1 + j) - \angle(s_1 + 5) - \angle(s_1 + 1 - j) \\ &\quad = (2i + 1)180^\circ \end{aligned} \tag{E-47}$$

or

$$26.6^\circ - 135^\circ - 90^\circ - 14^\circ - 11.4^\circ - \theta \cong (2i + 1)180^\circ \tag{E-48}$$

for $i = 0, \pm 1, \pm 2, \ldots$ Therefore, selecting $i = 2$,

$$\theta \cong -43.8^\circ \tag{E-49}$$

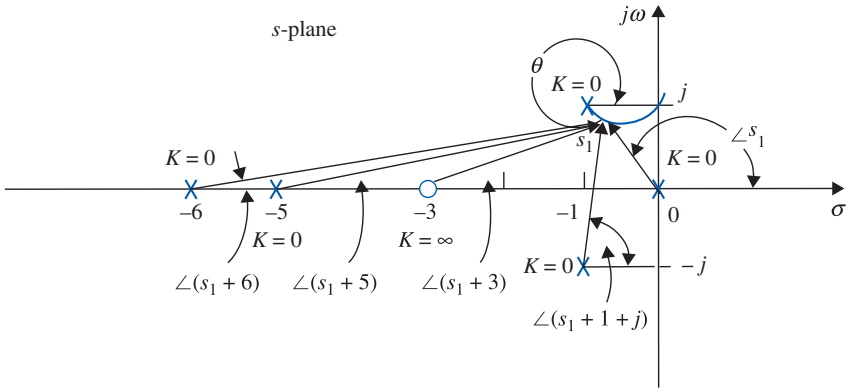

Figure E-17 Computation of angle of departure of the root loci of $s(s + 5)(s + 6)(s^2 + 2s + 2) + K(s + 3) = 0$.

Similarly, Eq. (7-19) is used to determine the angle of arrival θ' of the K ≤ 0 root loci arriving at the pole $-1 + j$. It is easy to see that θ' differs from θ by 180°; thus,

$$\theta' = 180^\circ - 43.8^\circ = 136.2^\circ \tag{E-50}$$

9. The intersection of the root loci on the imaginary axis is determined using Routh's tabulation. Equation (E-42) is written

$$s^5 + 13s^4 + 54s^3 + 82s^2 + (60 + K)s + 3K = 0 \tag{E-51}$$

Routh's tabulation is

s^5	1	54	$60 + K$
s^4	13	82	$3K$
s^3	47.7	$0.769K$	0
s^2	$65.6 - 0.212K$	$3K$	0
s^1	$\dfrac{3940 - 105K - 0.163K^2}{65.6 - 0.212K}$	0	0
s^0	$3K$	0	0

For Eq. (E-51) to have no roots on the imaginary axis or in the right-half of the s-plane, the elements in the first column of Routh's tabulation must all be of the same sign. Thus, the following inequalities must be satisfied:

$$65.6 - 0.212K > 0 \quad \text{or} \quad K < 309 \tag{E-52}$$

$$3940 - 105K - 0.163K^2 > 0 \quad \text{or} \quad K < 35 \tag{E-53}$$

$$K > 0 \tag{E-54}$$

Thus, all the roots of Eq. (E-51) will stay in the left-half s-plane if K lies between 0 and 35, which means that the root loci of Eq. (E-51) cross the imaginary axis when $K = 35$ and $K = 0$. The coordinates at the crossover points on the imaginary axis that correspond to $K = 35$ are determined from the auxiliary equation:

$$A(s) = (65.6 - 0.212K)s^2 + 3K = 0 \tag{E-55}$$

which is obtained by using the coefficients from the row just above the row of zeros in the s^1 row that would have happened when K is set to 35. Substituting $K = 35$ in Eq. (E-55), we get

$$58.2s^2 + 105 = 0 \tag{E-56}$$

The roots of Eq. (E-56) are $s = j1.34$ and $-j1.34$, which are the points at which the root loci cross the *jω-axis.*

10. Breakaway points: Based on the information gathered from the preceding nine steps, a trial sketch of the root loci indicates that there can be only one breakaway point on the entire root loci, and the point should lie between the two poles of $G(s)H(s)$ at $s = -5$ and -6. To find the breakaway point, we take the derivative on both sides of Eq. (E-43) with respect to s and set it to zero; the resulting equation is

$$s^5 + 13.5s^4 + 66s^3 + 142s^2 + 123s + 45 = 0 \tag{E-57}$$

Since there is only one breakaway expected, only one root of the last equation is the correct solution of the breakaway point. The five roots of Eq. (E-57) are:

$$s = 3.33 + j1.204 \qquad s = 3.33 - j1.204$$
$$s = -0.656 + j0.468 \qquad s = -0.656 - j0.468$$
$$s = -5.53$$

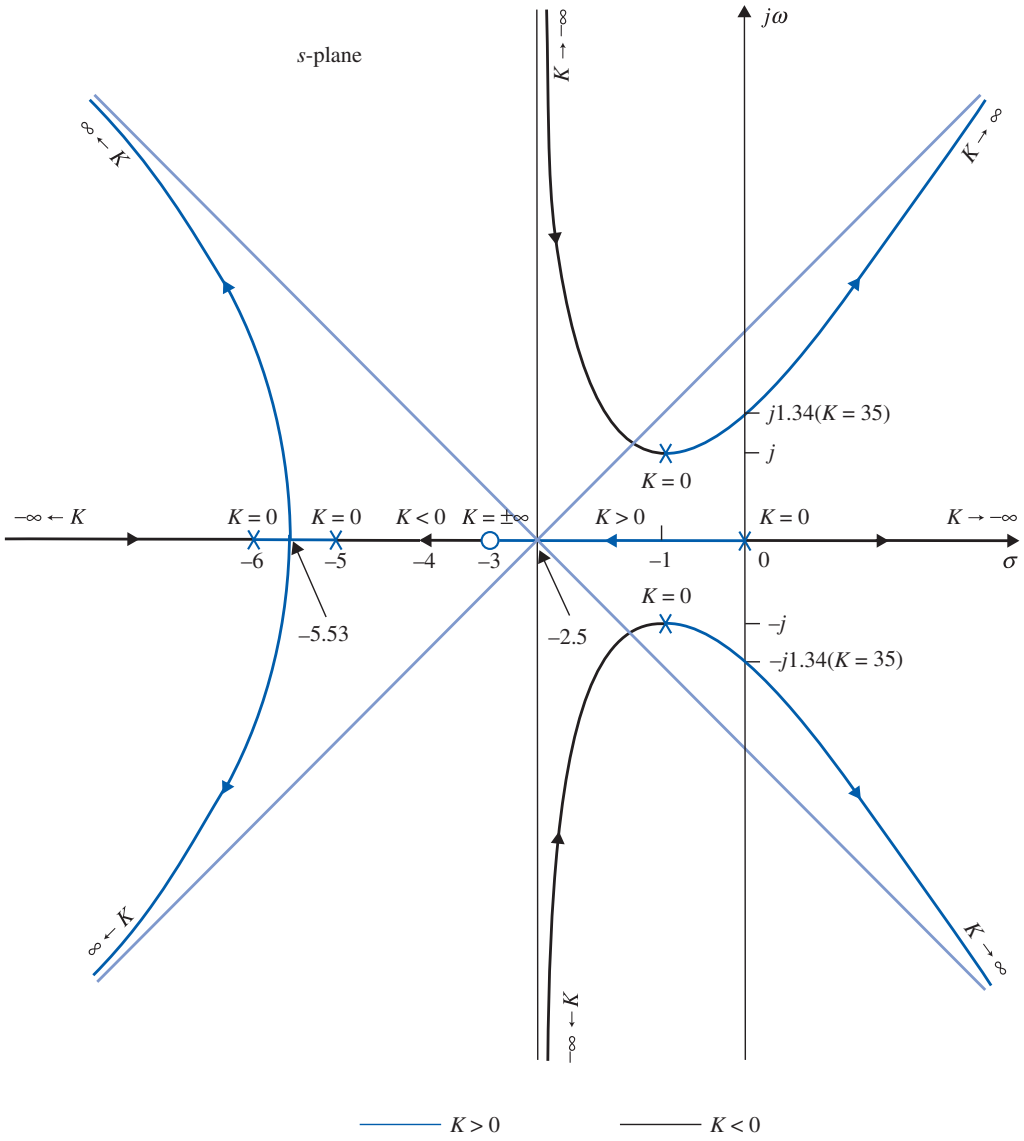

Figure E-18 Root loci of $s(s + 5)(s + 6)(s^2 + 2s + 2) + K(s + 3) = 0$.

Clearly, the breakaway point is at -5.53. The other four solutions do not satisfy Eq. (E-51) and are not breakaway points. Based on the information obtained in the last 10 steps, the root loci of Eq. (E-51) are sketched as shown in Fig. E-18. ◀

APPENDIX F

General Nyquist Criterion

F-1 FORMULATION OF NYQUIST CRITERION

The original Nyquist criterion presented in Chapter 8 is cumbersome to apply when the loop transfer function is of the nonminimum-phase type; that is, $L(s)$ has either poles and/or zeros in the right-half s-plane. We shall show that if the loop transfer function is of the nonminimum-phase type, then plotting the Nyquist plot of $L(s)$ only for $s = j\omega$ to $s = j0$ and not enclosing the $(-1, j0)$ point in the $L(s)$-plane is only a necessary, but not sufficient, condition for closed-loop stability. For a system with a nonminimum-phase loop transfer function $L(s)$, the original Nyquist criterion requires that the $L(s)$ plot that corresponds to the entire Nyquist path in Fig. 8-20 be made. If the loop transfer function $L(s)$ has poles or zeros on the $j\omega$-axis, then the Nyquist path of Fig. 8-20 must have small indentations around them on the $j\omega$-axis. This adds even more complexity to the construction of the $L(s)$ plot. Our MATLAB Toolbox (**ACSYS**) or other computer software can all be used to construct the plots of only functions that correspond to the positive $j\omega$-axis of the s-plane. The rest of the Nyquist plot that corresponds to the small indentations and the large semicircle on the Nyquist path have to be plotted manually. With modern computer facilities and software, the analyst should not be burdened with the chores of manual plotting. Therefore, we are introducing *a simplified Nyquist criterion that can be applied by using only the positive* jω*-axis of the Nyquist path and then observing its behavior with reference to the* $(-1, j0)$ *point.*

Yeung [1] introduced a general and yet simplified version of the Nyquist criterion that allows the determination of stability of closed-loop systems of minimum- as well as nonminimum-phase loop transfer functions by using only the positive part of the $j\omega$-axis of the Nyquist path. However, if the system is of the minimum-phase type, the test of whether the $(-1, j0)$ point is enclosed is still simpler to apply. We shall show that, for nonminimum-phase systems, if the $(-1, j0)$ point is enclosed, the system is still unstable. However, if the $(-1, j0)$ point is not enclosed, then an additional angle condition is all that must be satisfied by the Nyquist plot of $L(s)$ for the system to be stable.

Let us consider the two Nyquist paths shown in Figs. F-1(a) and F-1(b). Apparently, the Nyquist path Γ_{s1} in Fig. F-1(a) is the original one shown in Fig. 8-20, whereas the path Γ_{s2} in Fig. F-1(b) encircles not only the entire right-half s-plane but also all the poles and zeros of $L(s)$ on the $j\omega$-axis, if there are any. Let us define the following quantities.

Z = number of zeros of $1 + L(s)$ that are in the right-half s-plane.

P = number of poles of $L(s)$, or of $1 + L(s)$, that are in the right-half s-plane.

P_ω = number of poles of $L(s)$, or of $1 + L(s)$, that are on the $j\omega$-axis, including the origin.

N_1 = number of times the $(-1, j0)$ point of the $L(s)$-plane that is encircled by the Nyquist plot of $L(s)$ corresponding to Γ_{s1}.

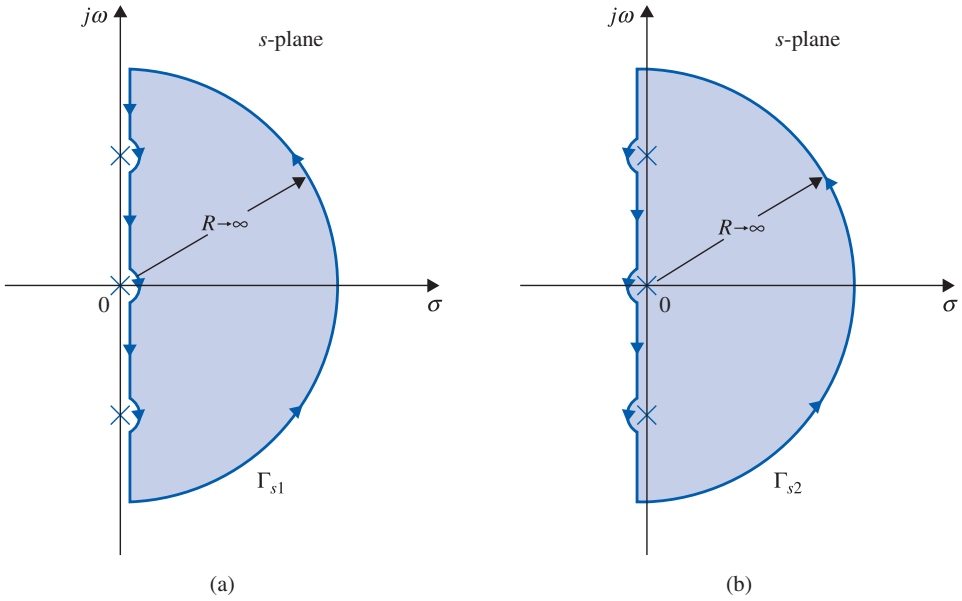

Figure F-1 (a) Nyquist path. (b) An alternative Nyquist path. (Source: K. S. Yeung, "A Reformulation of Nyquist's Criterion," *IEEE Trans. Educ.*, Vol. E-28, pp. 58–60, Feb. 1985.)

N_2 = number of times the $(-1, j0)$ point of the $L(s)$-plane that is encircled by the Nyquist plot of $L(s)$ corresponding to Γ_{s2}.

Then, with reference to the two Nyquist paths in Fig. F-1, and according to the Nyquist criterion,

$$N_1 = Z - P \tag{F-1}$$

and

$$N_2 = Z - P - P_\omega \tag{F-2}$$

Let Φ_1 and Φ_2 represent the net angles traversed by the Nyquist plot of $L(s)$ with respect to the $(-1, j0)$ point, corresponding to Γ_{s1} and Γ_{s2}, respectively. Then,

$$\Phi_1 = N_1 \times 360^\circ = (Z - P)360^\circ \tag{F-3}$$

$$\Phi_2 = N_2 \times 360^\circ = (Z - P - P_\omega)360^\circ \tag{F-4}$$

Let us consider that each of the Nyquist paths Γ_{s1} and Γ_{s2} is composed of three portions:

1. The portion from $s = -j\infty$ to $+j\infty$ along the semicircle with infinite radius
2. The portion along the $j\omega$-axis, excluding all the small indentations
3. All the small indentations on the $j\omega$-axis

Because the Nyquist paths in Fig. F-1 are symmetrical about the real axis in the s-plane, the angles traversed by the Nyquist plots are identical for positive and negative values of ω. Thus, Φ_1 and Φ_2 are written

$$\Phi_1 = 2\Phi_{11} + \Phi_{12} + \Phi_{13} \tag{F-5}$$

$$\Phi_2 = 2\Phi_{11} - \Phi_{12} + \Phi_{13} \tag{F-6}$$

where

Φ_{11} = angle traversed by the Nyquist plot of $L(s)$ with respect to the $(-1, j0)$ point, corresponding to the positive $j\omega$-axis or the $-j\omega$-axis of the s-plane, excluding the small indentations.

Φ_{12} = angle traversed by the Nyquist plot of $L(s)$ with respect to the $(-1, j0)$ point, corresponding to the small indentations on the $j\omega$-axis of Γ_{s1}. Because on Γ_{s2} the directions of the small indentations are opposite to that of Γ_{s1}, the sign of Φ_{12} in Eq. (F-5) is negative.

Φ_{13} = angle traversed by the Nyquist plot of $L(s)$ with respect to the $(-1, j0)$ point, corresponding to the semicircle with infinite radius on the Nyquist paths.

For a transfer function $L(s)$ that does not have more zeros than poles, the Nyquist plot of $L(s)$ that corresponds to the infinite semicircle must either be a point on the real axis or a trajectory around the origin of the $L(s)$-plane. Thus, the angle Φ_{13} traversed by the phasor drawn from the $(-1, j0)$ point to the Nyquist plot along the semicircle with infinite radius is always zero.

Now adding Eq. (F-5) to Eq. (F-6) and using Eqs. (F-3) and (F-4), we get

$$\begin{aligned}\Phi_1 + \Phi_2 &= 4\Phi_{11} \\ &= (2Z - 2P - P_\omega)360^\circ\end{aligned} \tag{F-7}$$

Solving for Φ_{11}, we get

$$\Phi_{11} = (Z - P - 0.5P_\omega)180^\circ \tag{F-8}$$

The equation states:

The total angle traversed by the phasor drawn from the $(-1, j0)$ point to the $L(s)$ Nyquist plot that corresponds to the portion on the positive $j\omega$-axis of the s-plane, excluding the small indentations, if any, equals

$$\begin{aligned}&\text{The number of zeros of } 1 + L(s) \text{ in the right-half } s\text{-plane} \\ &\text{The number of poles of } L(s) \text{ in the right-half } s\text{-plane} \\ &0.5(\text{the number of poles of } L(s) \text{ on the } j\omega = \text{axis})180^\circ\end{aligned} \tag{F-9}$$

Thus, *the Nyquist stability criterion can be carried out by constructing only the Nyquist plot that corresponds to the $s = j\infty$ to $s = 0$ portion on the Nyquist path.* Furthermore, if the closed-loop system is unstable, by knowing the values of Φ_{11}, P_ω, and P, Eq. (F-8) gives the number of roots of the characteristic equation that are in the right-half s-plane.

For the closed-loop system to be stable, Z must equal zero. Thus, the Nyquist criterion for stability of the closed-loop system is

$$\Phi_{11} = -(0.5P_\omega + P)180^\circ \tag{F-10}$$

Because P_ω and P cannot be negative, the last equation indicates that ***if the phase traversed by the Nyquist plot of $L(j\omega)$ as ω varies from ∞ to $\mathbf{0, \Phi_{11}}$, is positive with respect to the $\mathbf{(-1, j0)}$ point, the closed-loop system is unstable.***

However, if Φ_{11} is negative, it still has to satisfy Eq. (F-9) for the system to be stable. With reference to the Nyquist plot of $L(j\omega)$ and the $(-1, j0)$ point, we see that, when the angle variation Φ_{11} is positive, it corresponds to the $(-1, j0)$ point being enclosed. Thus, the condition that the Nyquist plot of $L(j\omega)$ not enclosing the $(-1, j0)$ point is a necessary condition for closed-loop stability for nonminimum-phase systems. However, if

the $(-1, j0)$ point is not enclosed by the Nyquist plot of $L(j\omega)$, for the nonminimum-phase system to be closed-loop stable, the angle variation Φ_{11} still has to satisfy Eq. (F-9).

F-1-1 System with Minimum-Phase Loop Transfer Functions

If $L(s)$ is of the minimum-phase type, then $P = 0$ and P_ω denotes the number of poles of $L(s)$ that are at the origin. Eq. (F-8) becomes

$$\Phi_{11} = (Z - 0.5P_\omega)180^\circ \tag{F-11}$$

For closed-loop stability, $Z = 0$; Eq. (F-11) becomes

$$\Phi_{11} = -P_\omega \times 90^\circ \tag{F-12}$$

Because P_ω denotes the number of poles of $L(s)$ that are at the origin, it is easy to see that, if the $(-1, j0)$ point is not enclosed by the Nyquist plot of $L(s)$, Φ_{11} will always be given by Eq. (F-12). Thus, when $L(s)$ is of the minimum-phase type, the condition that the $(-1, j0)$ point not be enclosed by the Nyquist plot is a necessary and sufficient condition for closed-loop stability.

F-1-2 Systems with Improper Loop Transfer Functions

Eq. (F-8) is derived based on the condition that $\Phi_{13} = 0$, which is true only if $L(s)$ is strictly proper; that is, it has more poles than zeros. For improper transfer functions, we can again use the method discussed in Section 8-6 by plotting the Nyquist plot of $1/L(s)$.

▶ F-2 ILLUSTRATIVE EXAMPLES—GENERAL NYQUIST CRITERION MINIMUM AND NONMINIMUM TRANSFER FUNCTIONS

In the following example we shall show that the Nyquist plot of a nonminimum-phase transfer function does not enclose the $(-1, j0)$ point, and yet the system is unstable.

▶ EXAMPLE F-2-1 Consider that the loop transfer function of a control system is given by

$$L(s) = \frac{s^2 - s + 1}{s(s^2 - 6s + 5)} \tag{F-13}$$

Because $L(s)$ has one pole at the origin and two poles in the right-half s-plane, $P_\omega = 1$ and $P = 2$. From Eq. (F-9), the closed-loop system is stable if the following condition is satisfied:

$$\Phi_{11} = -(0.5P_\omega + P)180^\circ = -450^\circ \tag{F-14}$$

The Nyquist plot of $L(j\omega)$ for $\omega = \infty$ to $\omega = 0$ is plotted as shown in Fig. F-2. Apparently, in this case, the $(-1, j0)$ point is not enclosed by the Nyquist plot. However, because Φ_{11} is -90°, and not -450°, the system is unstable. Substituting $\Phi_{11} = 90^\circ$ into Eq. (8-82) and solving for Z, we have $Z = 2$, which means that there are two closed-loop poles in the right-half s-plane. ◀

▶ EXAMPLE F-2-2 Consider the system described in Example 8-1-1. The loop transfer function of the system, $L(s)$, is given in Eq. (9-59) and is repeated below.

$$L(j\omega) = \frac{K}{s(s + 2)(s + 10)} \tag{F-15}$$

The Nyquist plot of $L(j\omega)$ is shown in Fig. 8-25. It is shown in Example 8-1-1 that, for closed-loop stability, the $(-1, j0)$ point in the $L(j\omega)$-plane must be to the left of the intersect of the Nyquist plot with the real axis.

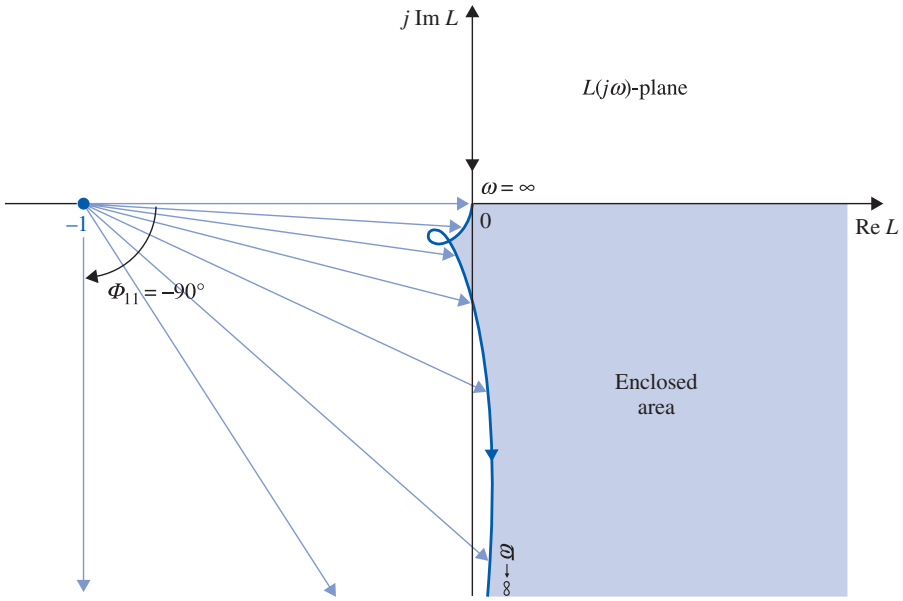

Figure F-2 Nyquist plot of $L(s) = \dfrac{s^2 - s + 1}{s(s^2 - 6s + 5)}$.

Now we shall apply the generalized Nyquist criterion to the system. Because $L(s)$ is of the minimum-phase type, $P = 0$, and it has one pole at the origin, thus, $P_\omega = 1$. Substituting these quantities into Eq. (F-11), we have

$$\Phi_{11} = (Z - 0.5)180^\circ \tag{F-16}$$

For closed-loop stability, Z must equal zero; thus, the last equation gives $\Phi_{11} = -90^\circ$. This means that the phasor drawn from the $(-1, j0)$ point to the Nyquist plot, from $\omega = \infty$ to $\omega = 0$, must equal -90°, or 90° in the CW direction. Fig. F-3(a) shows that, if the $(-1, j0)$ point is to the left of the intersect of $L(j\omega)$ with the real axis, Φ_{11} is indeed -90°. On the other hand, if the $(-1, j0)$ point is to the right of the intersect, as shown in Fig. F-3(b), when the value of K is greater than 240, then Φ_{11} is $+270^\circ$, or, (more easily observed when the critical point is enclosed), the system would be unstable. Substituting $\Phi_{11} = 270^\circ$ into Eq. (F-16), we get $Z = 2$, which means that the characteristic equation has two roots in the right-half s-plane. Thus, for systems with minimum-phase loop transfer functions, the Nyquist

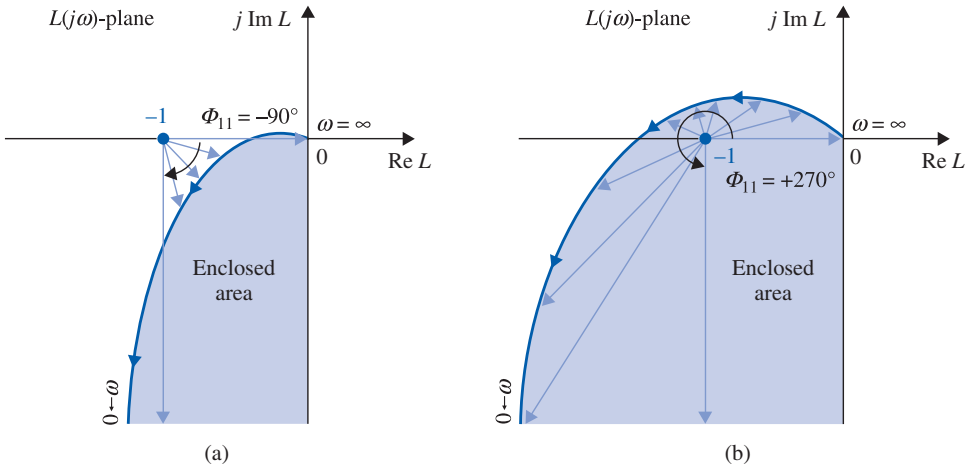

Figure F-3 Nyquist plot of $L(s) = \dfrac{K}{s(s + 2)(s + 10)}$. (a) $K < 240$. (b) $K > 240$.

criterion with the "enclosure" test is easier to observe, but when the system is unstable, it does not tell how many characteristic equation roots are in the right-half plane; the general Nyquist criterion does. ◀

▶ **EXAMPLE F-2-3** Consider that a control system has the loop transfer function

$$L(s) = \frac{K(s-1)}{s(s+1)} \tag{F-17}$$

We observed from the last equation that $P_\omega = 1$, and $P = 0$. The function $L(s)$ is of the nonminimum-phase type, because it has a zero at $s = 1$. Thus, the Nyquist criterion on enclosure cannot be used adequately in this case. From Eq. (F-8), the requirement for closed-loop stability is

$$\Phi_{11} = -(0.5P_\omega + P)180^\circ = -90^\circ \tag{F-18}$$

Thus, the stability criterion requires that the phasor drawn from the $(-1, j0)$ point to the Nyquist plot of $L(j\omega)$ should traverse -90° as ω varies from ∞ to 0.
To sketch the Nyquist plot of $L(s)$ that corresponds to the postive portion of the $j\omega$-axis of the s-plane, we set $s = j\omega$ in Eq. (F-17). We get

$$L(j\omega) = \frac{K(j\omega - 1)}{j\omega(j\omega + 1)} = \frac{K(j\omega - 1)}{-\omega^2 + j\omega} \tag{F-19}$$

When $\omega = \infty$,

$$L(j\infty) = \frac{K}{j\omega}_{\omega=\infty} = 0\angle -90^\circ \tag{F-20}$$

When $\omega = 0$,

$$L(j0) = \frac{K}{j\omega}_{\omega=0} = \infty\angle 90^\circ \tag{F-21}$$

To find the intersect of the $L(j\omega)$ plot on the real axis, we rationalize the function by multiplying the numerator and the denominator of Eq. (F-19) by $-\omega^2 - j\omega$. We have

$$L(j\omega) = \frac{K(j\omega - 1)(-\omega^2 - j\omega)}{\omega^4 + \omega^2} = \frac{K[2\omega + j(1 - \omega^2)]}{\omega(\omega^2 + 1)} \tag{F-22}$$

Setting the imaginary part of $L(j\omega)$ to zero and solving for ω^2, we have

$$\omega^2 = \pm 1 \text{ rad/sec} \tag{F-23}$$

For $\omega = 1$,

$$L(j1) = K \tag{F-24}$$

◀

Nyquist Plot for $K > 0$
Based on the preceding information, the Nyquist plot of $L(j\omega)$ that corresponds to the positive portion of the $j\omega$-axis is sketched as shown in Fig. F-4 for $K > 0$. Fig. F-4 shows that, as ω varies from ∞ to 0 along the Nyquist plot, the net angle Φ_{11} traversed by the phasor drawn from the $(-1, j0)$ point to the Nyquist plot is $+90^\circ$. Thus, the system is unstable because Φ_{11} is positive. We can also readily see that $(-1, j0)$ is enclosed by the Nyquist plot, so the same conclusion on closed-loop stability can be drawn.

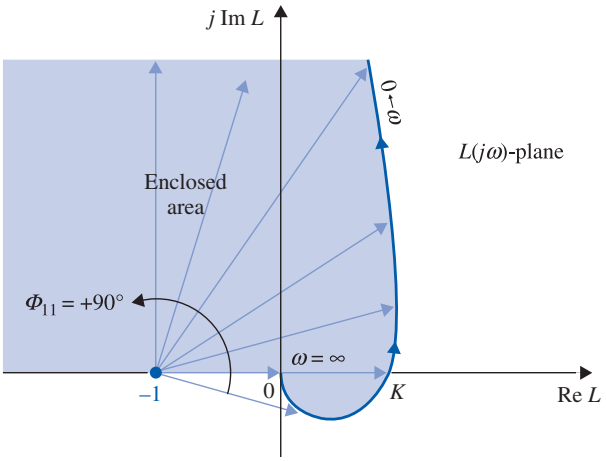

Figure F-4 Nyquist plot of the system in Example F-2-3.
$L(s) = \dfrac{K(s-1)}{s(s+1)}$.

From Eq. (F-7),

$$\Phi_{11} = (Z - 0.5P_\omega - P)180^\circ = (Z - 0.5)180^\circ = 90^\circ \tag{F-25}$$

Thus, $Z = 1$, which means that the characteristic equation of the closed-loop system has one root in the right-half s-plane. The characteristic equation of the system is

$$s^2 + (1 + K)s - K = 0 \tag{F-26}$$

We can easily verify that stability requires

$$0 > K > -1 \tag{F-27}$$

Nyquist Plot for $K < 0$

Fig. F-5(a) shows the Nyquist plot of $L(j\omega)$ when K lies between 0 and -1. Notice that the plot is obtained by *rotating* the $L(j\omega)$ plot of Fig. F-4 by 180° about the origin. As ω is varied from ∞ to 0, the angle Φ_{11} in Fig. F-5(a) has a net rotation of -90°, which agrees with the stability requirement in Eq. (F-18), and the closed-loop system is stable. It should be reminded that since $L(j\omega)$ is of the minimum-phase type, the fact that the Nyquist plot of $L(j\omega)$ of Fig. F-5(a) does not enclose the $(-1, j0)$ is *not* the reason that the system is stable.

Figure F-5(b) shows the Nyquist plot when $K < -1$. Now we see that the $(-1, j0)$ point is enclosed, so the system is unstable. Checking the value of Φ_{11}, we have $\Phi_{11} = 270^\circ$, which differs from the required -90°. Using Eq. (F-8),

$$\Phi_{11} = (Z - 0.5P_\omega - P)180^\circ = (Z - 0.5)180^\circ = 270^\circ \tag{F-28}$$

Thus, $Z = 2$, which means that the characteristic equation has two roots in the right-half s-plane.

In general, when K changes signs, it is not necessary to redraw the Nyquist plot as shown in Fig. F-5. Eq. (8-40) can be written as

$$1 + L(s) = 1 + KL_1(s) = 0 \tag{F-29}$$

where K is positive. For negative K, the last equation can be written as

$$1 - KL_1(s) = 0 \tag{F-30}$$

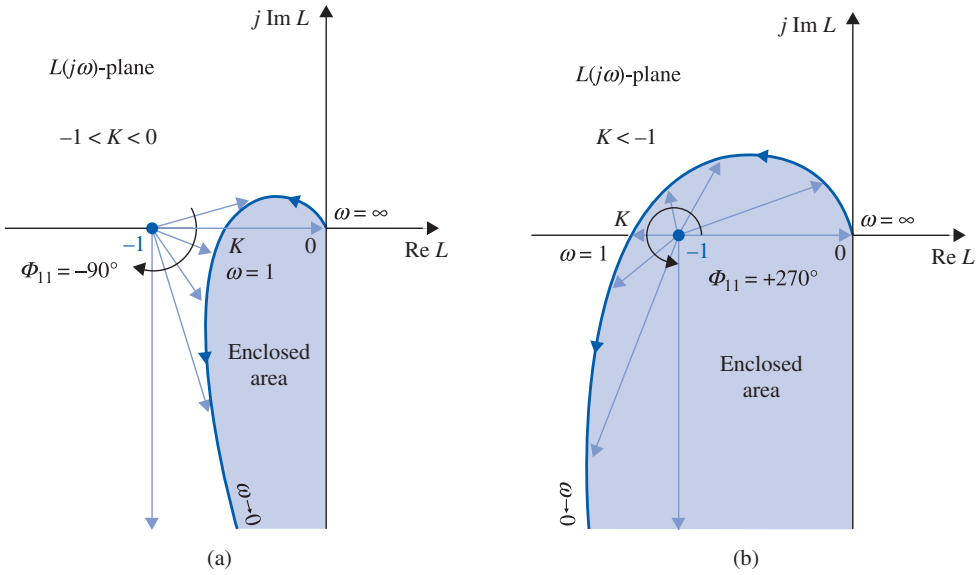

Figure F-5 Nyquist plots of the system in Example F-2-3. $L(s) = \dfrac{K(s-1)}{s(s+1)}$. (a) $-1 < K < 0$. (b) $K < -1$.

or

$$KL_1(s) = 1 \tag{F-31}$$

where K is now positive. Thus, Eq. (F-31) shows that, when K is negative, we can still use the $L(j\omega)$ plot for positive K but designate the $(+1, j0)$ point as the critical point for stability analysis.

Fig. F-6 shows the Nyquist plot of Eq. (F-17) for $K > 0$. When K is negative, the $(+1, j0)$ point is regarded as the critical point. As shown in Fig. F-6, for $-1 < K < 0$, Φ_{11} is $-90°$, which is the required value, and the system is stable. When $K < -1$, the $(+1, j0)$ point is enclosed by the Nyquist plot, and the system is unstable. These results agree with those obtained from Fig. F-5 when the Nyquist plots for $K < 0$ were actually constructed.

It is of interest to compare the Nyquist stability analysis with the root-locus analysis. Fig. F-7 shows the root loci of the characteristic equation of the system with the loop transfer function given in Eq. (F-17). The stability condition of the system as a function of

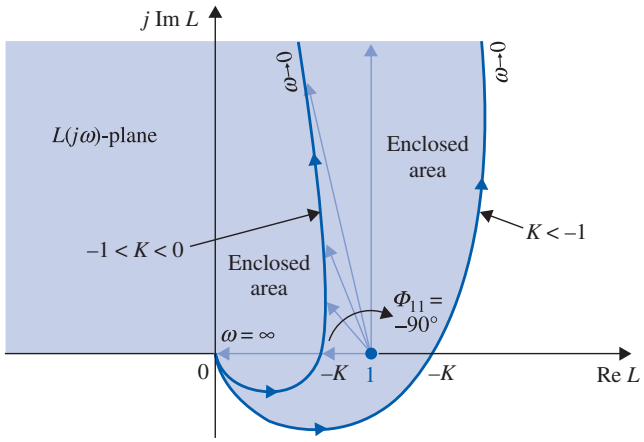

Figure F-6 Nyquist plot of $L(s) = \dfrac{K(s-1)}{s(s+1)}$ with $K < 0$. The $(+1, j0)$ point is the critical point.

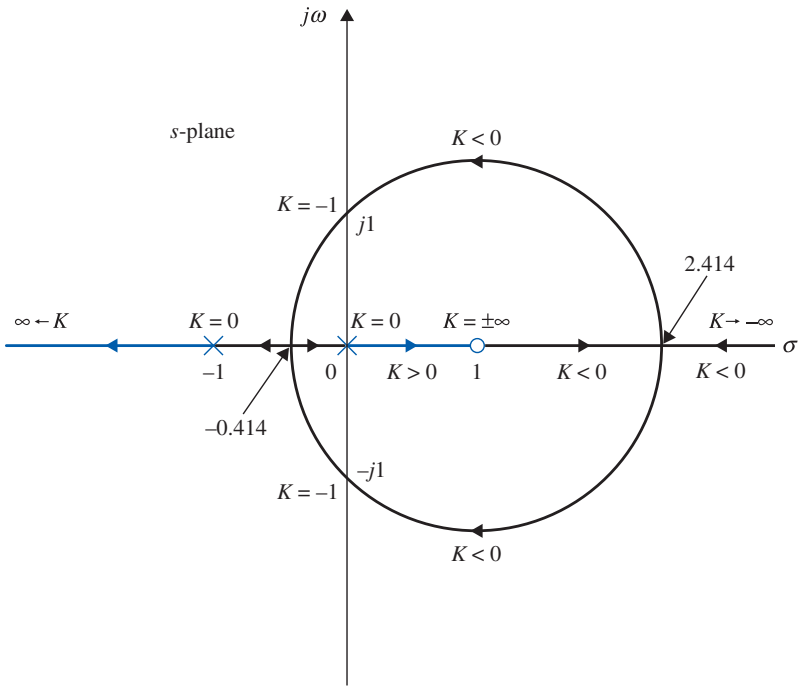

Figure F-7 Complete root loci of system in Example F-2-3.

K is clearly indicated by the root-locus diagram. The RL between 0 and +1 on the real axis indicates that the system is unstable for $0 < K < \infty$. The RL indicates that, for negative values of K, the system is unstable for $-\infty < K < -1$, and the system is stable only for the range of $-1 < K < 0$. The root loci cross the $j\omega$-axis at $\omega = \pm 1$ rad/sec, which are the values of ω at which the Nyquist plot of $L(j\omega)$ intersects the negative real axis.

▶ EXAMPLE F-2-4 Consider the control system shown in Fig. F-8. It is desired to determine the range of K for which the system is stable. The loop transfer function of the system is

$$L(s) = \frac{Y(s)}{E(s)} = \frac{10K(s+2)}{s^3 + 3s^2 + 10} \tag{F-32}$$

The poles of $L(s)$ are found to be at $s = -3.72$, $0.361 + j1.6$, and $0.361 - j1.6$. Or, we can use the Routh-Hurwitz criterion to verify that $L(s)$ has two poles in the right-half s-plane. Thus, $P = 2$, and $P_\omega = 0$. The transfer function $L(s)$ is of the nonminimum-phase type.

From Eq. (F-9), the requirement for the closed-loop system to be stable is

$$\Phi_{11} = -(0.5P_\omega + P)180^\circ = -360^\circ \tag{F-33}$$

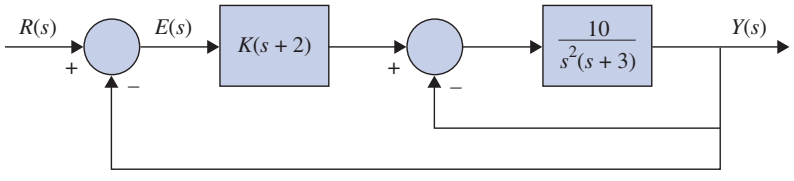

Figure F-8 Block diagram of the control system in Example F-2-4.

Setting $s = j\omega$, Eq. (F-32) becomes

$$L(j\omega) = \frac{10K(j\omega + 2)}{(10 - 3\omega^2) - j\omega^3} \tag{F-34}$$

At $\omega = \infty$,

$$L(j\infty) = 0\angle 180^\circ \tag{F-35}$$

At $\omega = 0$,

$$L(j0) = 2K \tag{F-36}$$

To find the intersect on the real axis of the $L(j\omega)$-plane, we rationalize $L(j\omega)$ as

$$L(j\omega) = \frac{10K\{2(10 - 3\omega^2) - \omega^4 + j[\omega(10 - 3\omega^2) + 2\omega^2]\}}{(10 - 3\omega^2)^2 + \omega^6} \tag{F-37}$$

Setting the imaginary part of $L(j\omega)$ to zero, we have

$$\omega(10 - 3\omega^2) + 2\omega^3 = 0 \tag{F-38}$$

The solutions of the last equation are $\omega = 0$ and $\omega = \pm\sqrt{10} = 3.16$ rad/sec, which are the frequencies at which the $L(j\omega)$ plot intersects the real axis of the $L(j\omega)$-plane. When $\omega = 0$, we already have $L(j0) = 2K$ in Eq. (F-36). When $\omega = 3.16$ rad/sec,

$$L(j3.16) = -K \tag{F-39}$$

Fig. F-9(a) shows the Nqyuist plot of $L(j\omega)$ for $0 < K < 1$. Because the $(-1, j0)$ point is enclosed by the Nyquist plot, the closed-loop system is unstable. We can also show that the angle traversed by Φ_{11} is 0°, not -360°, as required in Eq. (F-33). Fig. F-9(b) shows the Nyquist plot of $L(j\omega)$ when K is greater than unity. In this case, the angle Φ_{11} rotates a total of -360°; thus, the system is stable.

When K is negative, we can use the plots in Fig. F-9 and regard the $(+1, j0)$ point as the critical point. The following stability conditions are observed:

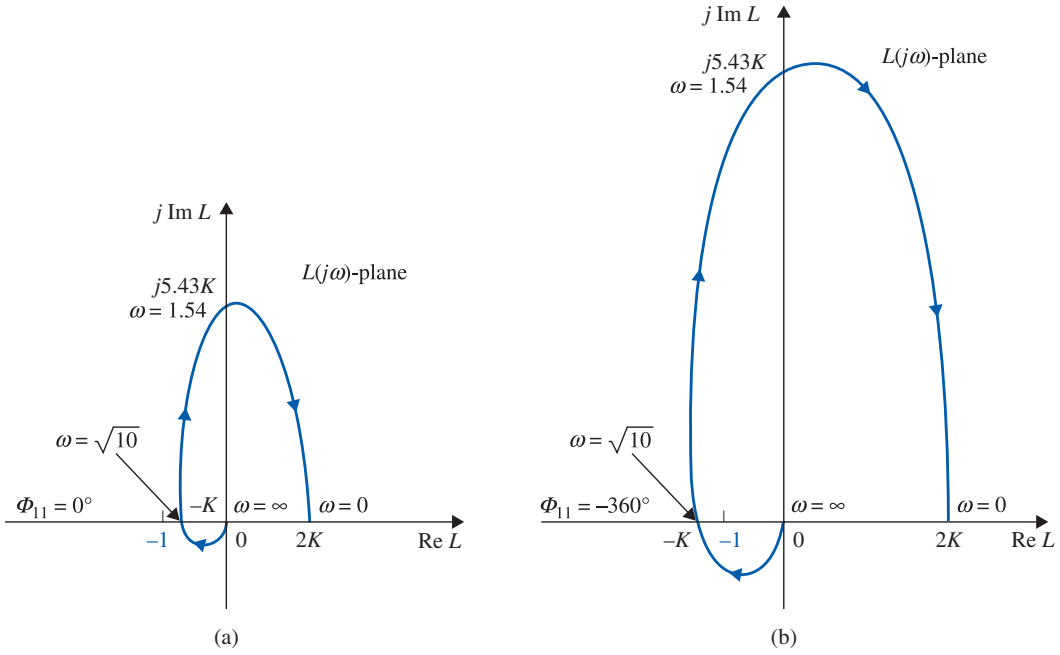

Figure F-9 Nyquist plots of the system in Example F-2-4. (a) $1 > K > 0$. (b) $K > 1$.

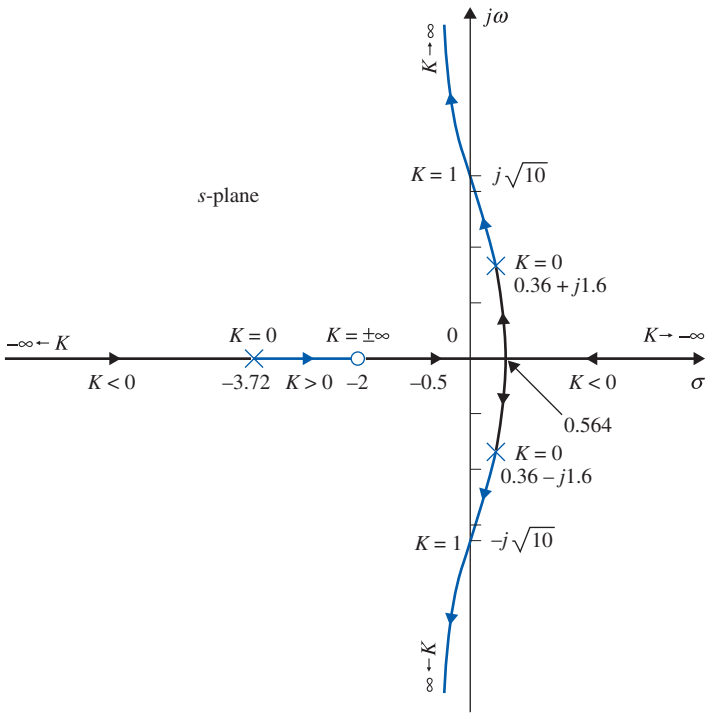

Figure F-10 Complete root loci of the system in Example F-2-4.

- $2K < -1$: The $(+1, j0)$ point lies between 0 and $2K$ and is not enclosed by $L(j\omega)$, but $\Phi_{11} = -180°$; the system is unstable. For stability, Φ_{11} must equal $-360°$.
- $-1 < 2K < 0$: The $(+1, j0)$ point is to the right of the point $2K$ and is enclosed by the Nyquist plot; the system is unstable. In this case, $\Phi_{11} = 0°$.

The conclusion is that the system is stable for $K > 1$. The root loci of the system are shown in Fig. F-10. Clearly, when K is negative, one branch of the RL will always stay in the right-half plane, and the system is unstable. The system is stable only for $K > 1$, and the root loci cross the $j\omega$-axis at $\omega = \pm 3.16$ rad/sec, which corresponds to the frequency at which the $L(j\omega)$ plot intersects the negative real axis. The value of K at the crossing point on the $j\omega$-axis is 1. ◀

▶ **EXAMPLE F-2-5** Consider a control system with the loop transfer function

$$L(s) = \frac{K}{(s+2)(s^2+4)} \tag{F-40}$$

which has a pair of imaginary poles at $s = j2$ and $-j2$. Thus, $P_\omega = 2$, and $P = 0$. To apply the Nyquist criterion in the original form, we would have to define the Nyquist path with small indentations around these poles.

Instead of constructing the entire Nyquist plot, the portion that corresponds to $s = j\infty$ to $j0$ is plotted as shown in Fig. F-11. The data for this Nyquist plot are easily obtained using any of the frequency-domain programs mentioned earlier.

From Eq. (F-9), the value of Φ_{11} required for stability is

$$\Phi_{11} = -(0.5P_\omega + P)180° = -180° \tag{F-41}$$

As seen from Fig. F-11, the magnitude of $L(j\omega)$ goes to infinity when $\omega = 2$ rad/sec. When K is positive, the critical point $(-1, j0)$ is not enclosed by the Nyquist plot, and the system is unstable. For the angle check,

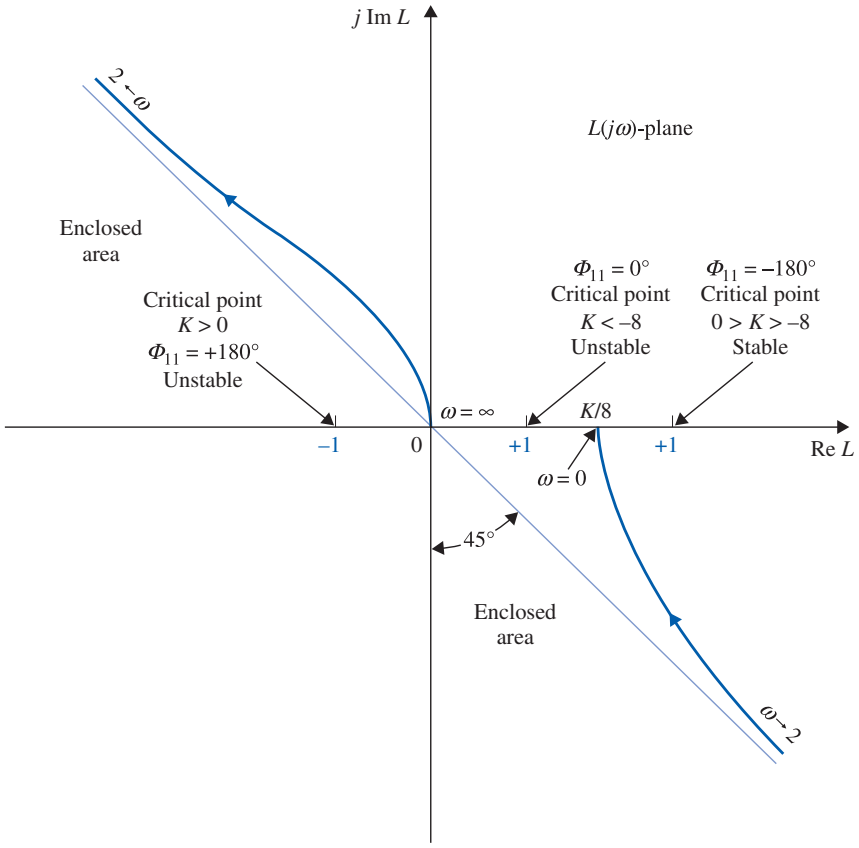

Figure F-11 Nyquist plot of the control system in Example F-2-5.

when ω varies from ∞ to 2, the angle Φ_{11} is $+135°$, and for the portion of $\omega = 2$ to 0, Φ_{11} is $+45°$. Thus, the total Φ_{11} is $+180°$, not $-180°$. The system is unstable for all positive values of K.

When K is negative, the critical point in Fig. F-11 is at $(+1, j0)$. Fig. F-11 shows that, if the $(+1, j0)$ point lies between 0 and $K/8$, it is enclosed by the Nyquist path, and the system is unstable. Thus, the system is unstable for $K < -8$. When the $(+1, j0)$ point is to the right of the $K/8$ point, Φ_{11} from $\omega = \infty$ to $\omega = 2$ is $-45°$, and from $\omega = 2$ to $\omega = 0$ is $-135°$. Thus the total Φ_{11} as ω varies from ∞ to 0 is $-180°$, which agrees with the value required in Eq. (F-41). The system is stable for $0 > K > -8$. The summary of the Nyquist criterion application to this system is as follows.

Range of K	Φ_{11} (deg) for ω = 2 to 0	Φ_{11} (deg) for ω = 2 to ∞	Total Φ_{11} (deg)	Critical Point	Stability Condition
$K > 0$	$+135$	$+45$	$+180$	-1 point enclosed	Unstable
$K < -8$	-45	$+45$	0	$+1$ point enclosed	Unstable
$-8 < K < 0$	-45	-135	-180		Stable

The complete root loci of the characteristic equation of the system are constructed in Fig. F-12 using the pole–zero configuration of Eq. (F-40). The stability condition of $-8 < K < 0$ is easily viewed from the root loci.

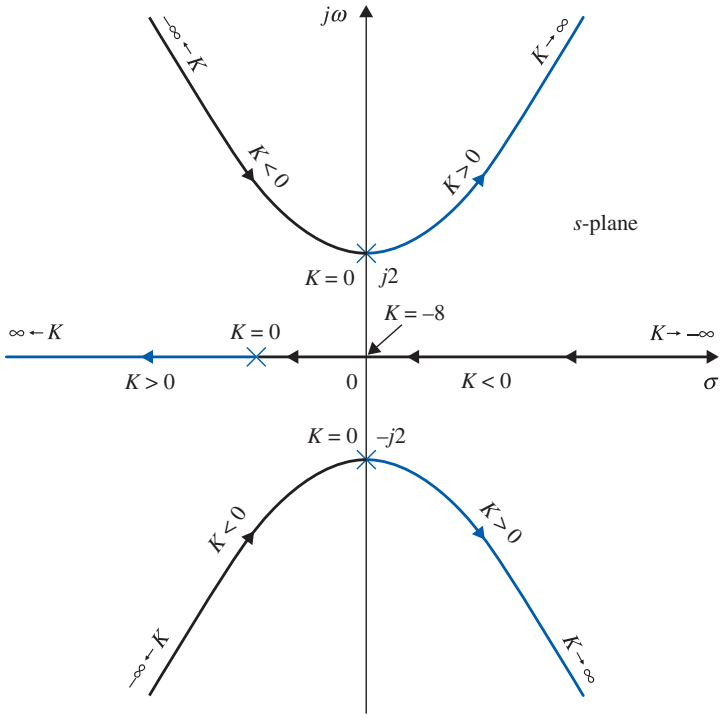

Figure F-12 Complete root loci of the system in Example F-2-5.

▶ F-3 STABILITY ANALYSIS OF MULTILOOP SYSTEMS

The Nyquist stability analyses conducted in the preceding sections are all directed toward the loop transfer function $L(s)$. It does not matter whether the system is with single loop or multiple loops, because once the loop transfer function is obtained, stability analysis can be conducted using the Routh-Hurwitz criterion, root loci, or the Nyquist criterion.

For multiloop-feedback systems, it may be advantageous to analyze the stability of the system by working from the inner loop toward the outer loop, one at a time. This way, more insight may be gained on the stability of the individual loops of the system. The following example will illustrate this approach.

▶ EXAMPLE F-3-1

Fig. F-13 shows the block diagram of a system that controls the gun turret of a tank. During servicing of the turret control system, the mechanic accidentally opened the outer loop of the system. With the power turned on, the gun turret went out of control and finally flew apart. The purpose of this example is to show that it is inadequate to investigate just the stability of the overall system. In general, for a multiloop control system, one should conduct a systematic stability analysis of all the inner loops of the system. It is admissible to have unstable inner loops, as long as the overall system is stable. However, if such a situation exists, it is important to forewarn or take precautionary measures to prevent opening the loops during operation.

The loop transfer function of the inner loop is

$$G(s) = \frac{6}{s(s+1)(s+2)} \tag{F-42}$$

Fig. F-14 shows the Nyquist plot of $G(s)$. Because the plot intersects the real axis at the -1 point at $\omega = 1.414$ rad/sec, the inner loop is marginally stable. Therefore, if the outer loop of the system is opened, the system will oscillate continuously with a frequency of 1.414 rad/sec. The loop transfer

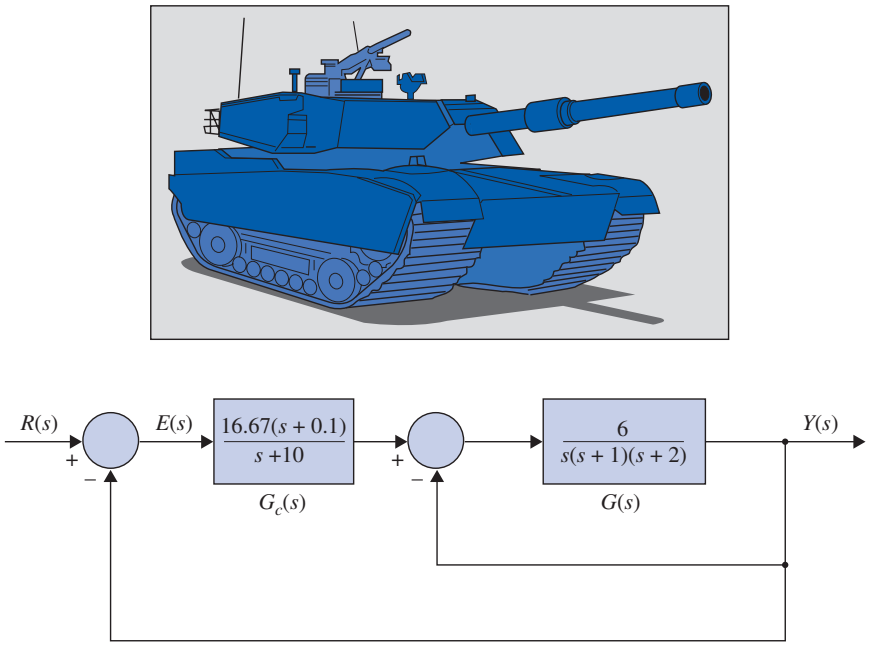

Figure F-13 Multiloop feedback control system for tank turret control.

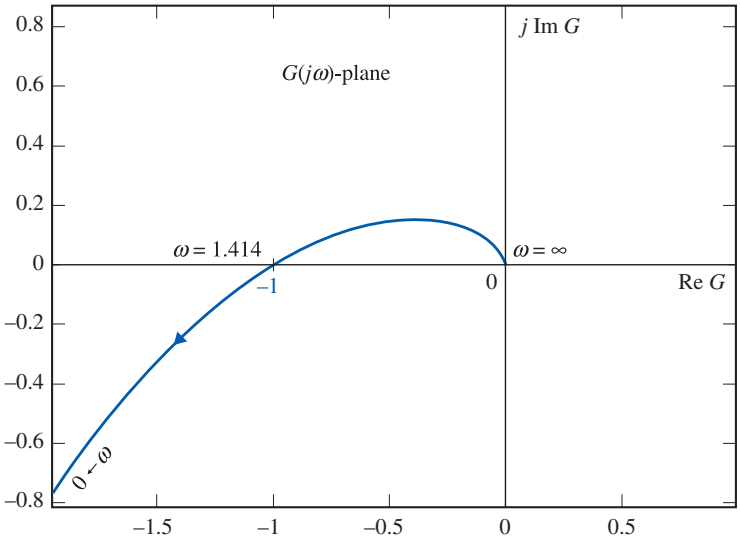

Figure F-14 Nyquist plot of $G(s) = \dfrac{6}{s(s+1)(s+2)}$.

function of the overall system is

$$L(s) = \frac{G_c(s)G(s)}{1 + G(s)} = \frac{100(s + 0.1)}{(s + 10)(s^3 + 3s^2 + 2s + 6)} \tag{F-43}$$

Because $L(s)$ has two poles on the $j\omega$-axis and the rest are in the left-half s-plane, $P_\omega = 2$ and $P = 0$. The function is also of the nonminimum-phase type, so we must use Eq. (F-9) for the stability test of the overall system. Thus,

$$\Phi_{11} = -(0.5P_\omega + P)180^\circ = -180^\circ \tag{F-44}$$

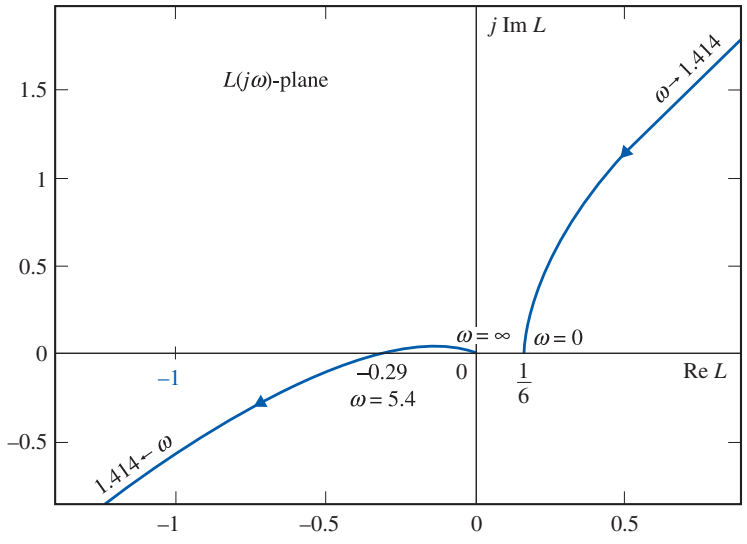

Figure F-15 Nyquist plot of $L(s) = \dfrac{100(s + 0.1)}{(s + 10)(s^3 + 3s^2 + 2s + 6)}$.

The Nyquist plot of $L(s)$ is plotted as shown in Fig. F-15. The angle Φ_{11} for $\omega = \infty$ to $\omega = 1.414$ rad/sec is $-90°$, and from $\omega = 1.414$ rad/sec to $\omega = 0$ is $-90°$. Thus, the total value of Φ_{11} for $\omega = \infty$ to $\omega = 0$ is $-180°$, and the overall system is stable.

In general, when more than two loops are involved, the proper way is to start with the stability of the innermost loop by opening all the outer loops, and then add one loop at a time, until the outermost loop is closed. ◀

▶ PROBLEMS

F-1. The loop transfer functions $L(s)$ of single-feedback-loop systems are given in the following equations. Sketch the Nyquist plot of $L(j\omega)$ for $\omega = 0$ to ∞. Determine the stability of the closed-loop system. If the system is unstable, find the number of poles of the closed-loop transfer function that are in the right-half s-plane. Solve for the intersect of $L(j\omega)$ on the negative real axis of the $L(j\omega)$-plane analytically. You may construct the Nyquist plot of $L(j\omega)$ using any computer program.

(a) $L(s) = \dfrac{5(s - 2)}{s(s + 1)(s - 1)}$

(b) $L(s) = \dfrac{50}{s(s + 5)(s - 1)}$

(c) $L(s) = \dfrac{3(s + 2)}{s(s^3 + 3s + 1)}$

(d) $L(s) = \dfrac{100}{s(s + 1)(s^2 + 2)}$

(e) $L(s) = \dfrac{s^2 - 5s + 2}{s(s^3 + 2s^2 + 2s + 10)}$

(f) $L(s) = \dfrac{-0.1(s^2 - 1)(s + 2)}{s(s^2 + s + 1)}$

F-2. The loop transfer functions of single-feedback-loop control systems are given in the following equations. Apply the Nyquist criterion and determine the values of K for the system to be stable. Sketch the Nyquist plot of $L(j\omega)$ with $K = 1$ for $\omega = 0$ to $\omega = \infty$. You may use a computer program to plot the Nyquist plots.

(a) $L(s) = \dfrac{K(s - 2)}{s(s^2 - 1)}$

(b) $L(s) = \dfrac{K}{s(s + 10)(s - 2)}$

(c) $L(s) = \dfrac{K(s + 1)}{s(s^3 + 3s + 1)}$

(d) $L(s) = \dfrac{K(s^2 - 5s + 2)}{s(s^3 + 2s^2 + 2s + 10)}$

(e) $L(s) = \dfrac{K(s^2-1)(s+2)}{s(s^2+s+1)}$ **(f)** $L(s) = \dfrac{K(s^2-5s+1)}{s(s+1)(s^2+4)}$

F-3. Fig. FP-3 shows the Nyquist plots of the loop transfer function $L(j\omega)$ for $\omega = 0$ to $\omega = \infty$ for single-feedback-loop control systems. The number of poles of $L(j\omega)$ that are on the $j\omega$-axis, P_ω, and in the right-half s-plane, P, are indicated for each case. Determine the stability of the closed-loop system by applying the Nyquist criterion. For the unstable systems, give the number of zeros of $1 + L(s)$ that are in the right-half s-plane.

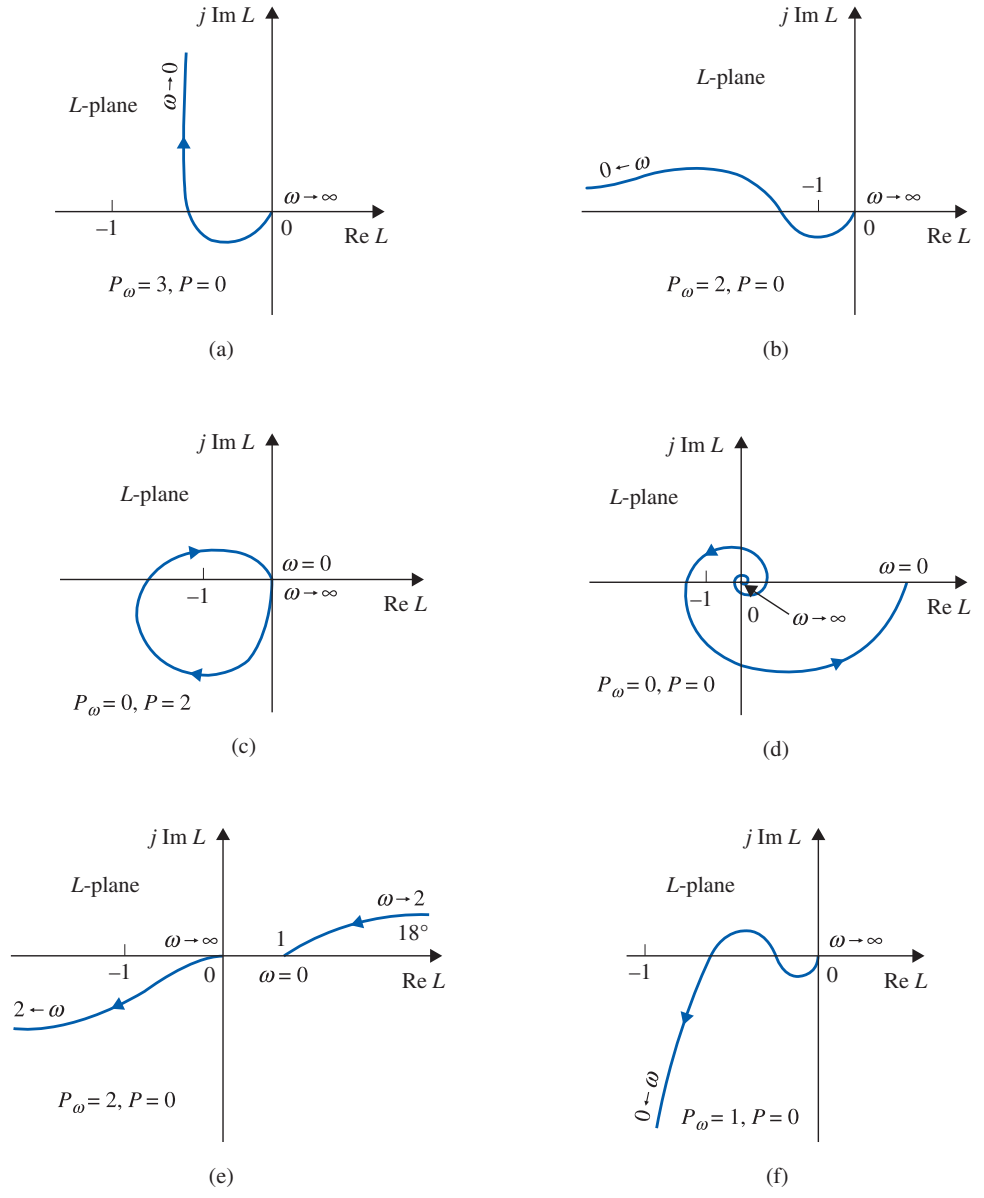

Figure FP-3

F-4. It was mentioned in the text that, when the function $L(j\omega)$ has more zeros than poles, it is necessary to plot the Nyquist plot of $1/L(j\omega)$ to apply the simplified Nyquist criterion. Determine the stability of the systems described by the function $1/L(j\omega)$ shown in Fig. FP-4. For each case, the values of P_ω and P for the function $1/L(j\omega)$ are given, where P_ω refers to the number of poles of $1/L(j\omega)$ that are on the $j\omega$-axis, and P refers to the number of poles of $1 + 1/L(j\omega)$ that are in the right-half s-plane.

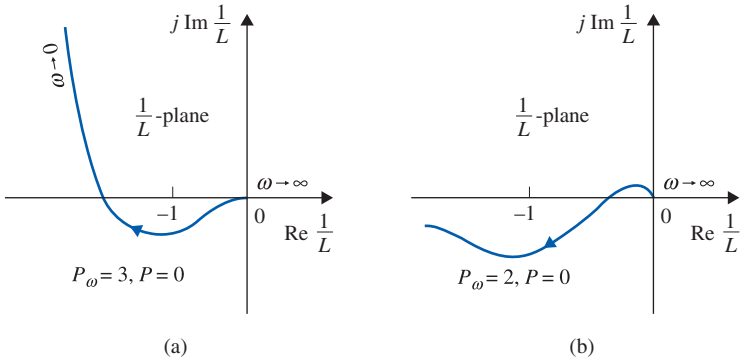

Figure FP-4

F-5. Fig. FP-5 shows the Nyquist plots of the loop transfer function $L(j\omega)$ for $\omega = 0$ to $\omega = \infty$ for single-feedback-loop control systems. The gain K appears as a multiplying factor in $L(s)$. The number of poles of $L(j\omega)$ that are on the $j\omega$-axis and in the right-half s-plane are indicated in each case. Determine the range(s) of K for closed-loop system stability.

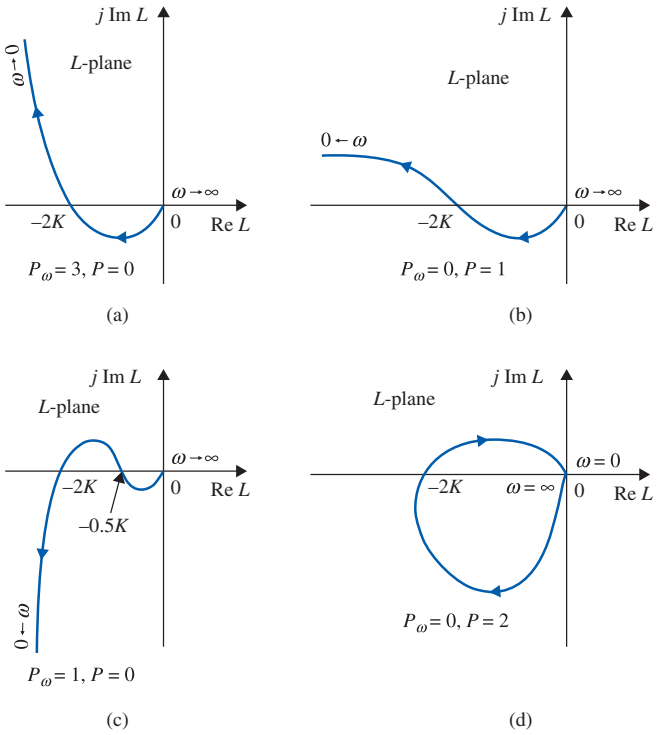

Figure FP-5

F-6. The characteristic equations of linear control systems are given below. Apply the Nyquist criterion to determine the values of K for system stability. Check the answers by means of the Routh-Hurwitz criterion.

(a) $s^3 + 4Ks^2 + (K + 5)s + 10 = 0$

(b) $s^3 + K(s^3 + 2s^2 + 1) = 0$

(c) $s(s + 1)(s^2 + 4) + K(s^2 + 1) = 0$

(d) $s^3 + 2s^2 + 20s + 10K = 0$

(e) $s(s^3 + 3s + 3) + K(s + 2) = 0$

▶ REFERENCES

1. K. S. Yeung, "A Reformulation of Nyquist's Criterion," *IEEE Trans. Educ.*, Vol. E-28, pp. 58–60, Feb. 1985.

APPENDIX G

ACSYS 2008: Description of the Software

This appendix introduces and describes the various components of the **A**utomatic **C**ontrol **Sys**tems (**ACSYS**) toolbox.

G-1 INSTALLATION OF ACSYS

The first-time user must follow steps (1) and (2) prior to using **ACSYS**.

1. Create a folder C:\ACSYS2008 (or any folder of your choice).
2. Select all MATLAB and image files from the ACSYS2008 folder on this book's companion Web site: www.wiley.com/college/Golnaraghi. Copy them into your C:\ACSYS2008 folder.

Please note that if you use another folder name, the preceding and the following directions should be modified accordingly.

To run ACSYS2008, simply start MATLAB.

(Note: screen resolution of 1024 by 768 pixels or higher is recommended for best graphics quality.) At the MATLAB prompt, type

cd C:\ACSYS2008

You may also use the MATLAB directory browser to move to the ACSYS2008 folder. See MATLAB documentation for instructions. Next, in the MATLAB command window, at the prompt (>>), type

Acsys

Fig. G-1 should appear. Upon clicking any button, the corresponding toolbox appears.

G-2 DESCRIPTION OF THE SOFTWARE

The Automatic Control Systems (**ACSYS**) tool consists of a number of m-files and GUIs (Graphical User Interface) for the analysis of simple control engineering transfer functions. In all, **ACSYS** is composed of six different toolboxes and a tool for converting engineering quantities between SI and other units. These components cover various topics encountered in the study of control systems. These include:

- The Transfer Function Symbolic or **tfsym**
- The State Space Tool or **Statetool**

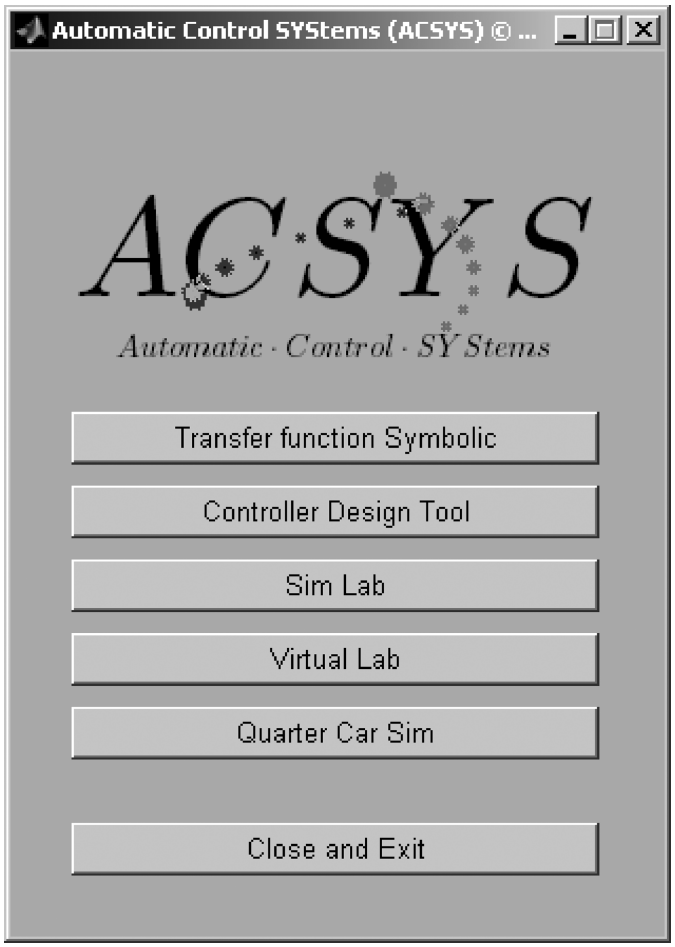

Figure G-1 The Automatic Control Systems (ACSYS) toolbox main window.

- The Controller Design Tool or **Controls**
- The SIMLab or **simlab**
- The Virtual Lab or **virtuallab**
- The Quarter Car Simulation or **quartercarsim**

G-2-1 tfsym

If you have access to the MATLAB Symbolics Tool, you may use the **ACSYS** Transfer Function Symbolic Tool by pressing the appropriate button in the **ACSYS** window. The Symbolic Tool window is shown in Fig. G-2. Click on the ''Help for 1st Time User'' button to see the instructions on how to use the toolbox. The instructions appear in a Help Dialog window, as shown in Fig. G-3. As instructed, you must run this program within the MATLAB command window after clicking the desired buttons. For a description of the ''Transfer Function and Inverse Laplace'' button, refer to Chapter 2, and, for the state-space applications, see Chapter 10.

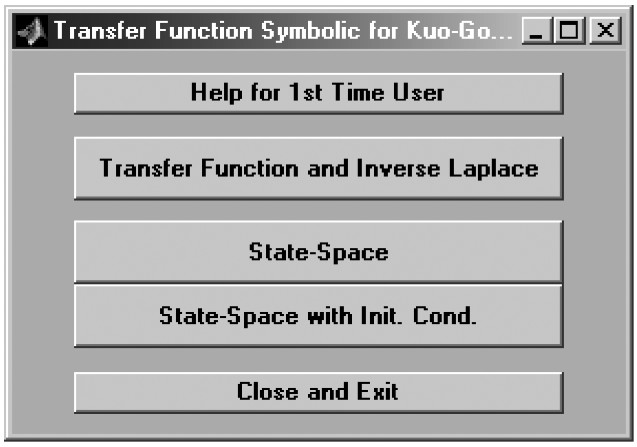

Figure G-2 The Transfer Function Symbolic window.

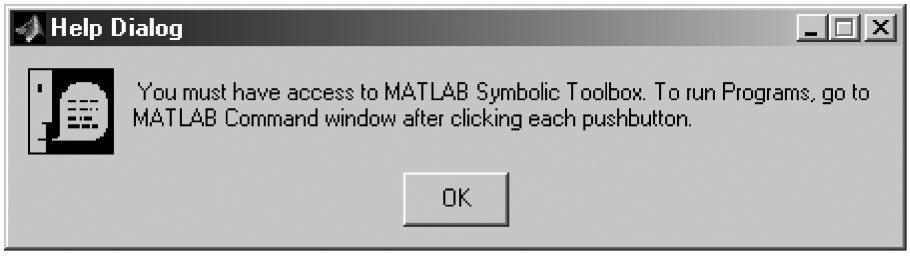

Figure G-3 The Symbolic Help Dialog window.

G-2-2 Statetool

The State Space Analysis Tool (Statetool) consists of a number of m-files and GUIs for the analysis of state-space systems. The Statetool can be invoked from the Automatic Control Systems (**ACSYS**) launch applet by clicking on the appropriate button. This program may be used for problems in Chapter 10 and allows the user to conduct the following tasks:

- Enter the state matrices.
- Find the system characteristic polynomial, eigenvalues, and eigenvectors.
- Find the similarity transformation matrices.
- Examine the system controllability and observability properties.
- Obtain the step, impulse, and natural (response to initial conditions) responses, as well as the time response to any function of time.

A detailed description of the program appears in Chapter 10.

G-2-3 Controls

The Controller Design Tool (Controls) consists of a number of m-files and GUIs for the time and frequency analysis and design of simple control systems using transfer functions. The controls tool can be invoked from the Automatic Control Systems (**ACSYS**) launch applet by clicking on the appropriate button. This comprehensive tool is a powerful tool to

solve most controller design problems (except for state-space examples) discussed in this book. This software allows the user to be able to conduct the following tasks:

- Enter the transfer function values in polynomial form (user must use the tftool to convert the transfer function from pole-zero-gain form into the polynomial form).
- Obtain the step, impulse, parabolic, ramp, or other type input time responses.
- Obtain the closed-loop frequency plots.
- Obtain the phase and gain margin Bode plots and the polar plot of the loop transfer functions (in a single feedback loop configuration).
- Understand the effect of adding zeros and poles to the closed-loop or open-loop transfer functions.
- Generate root contour and root locus plots to investigate the effect that varying certain controller parameters has on the poles of a system.
- Design and compare various controllers including PID, lead, and lag compensators.
- Convert quantities between different units using the Unit Conversion tool.

The reader can get a detailed discussion of this program in Chapter 9.

G-2-4 SIMLab and Virtual Lab

SIMLab (simlab) and Virtual Lab (virtuallab) are series of MATLAB and Simulink files that make up an educational tool for students learning about dc motors and control systems. The SIMLab or the Virtual Lab main menu windows can be called from the Automatic Control Systems (**ACSYS**) launch applet by clicking on the appropriate button.

SIMLab was created to allow students to understand the basic simulation model of a dc motor. The parameters of the motor can be adjusted to see how they affect the system. The Virtual Lab was designed to exhibit some of the key behaviors of real dc motor systems. Real motors have issues such as gear backlash and saturation that may cause the motor response to deviate from expected behavior. Users should be able to cope with these problems. The motor parameters cannot be modified in the Virtual Lab, as in a realistic scenario; a motor may not be modified unless it is replaced by a new one!

In both the SIMLab and the Virtual Lab, there are five experiments. In the first two experiments, feedback speed control and position control are explored. Open-loop step response of the motor appears in the third experiment. In the fourth experiment, the frequency response of the open-loop system can be examined by applying a sinusoidal input. A controller design project is the last experiment.

For a detailed discussion of these topics, the user is instructed to see Chapter 6.

▶ G-3 FINAL COMMENTS

This version of ACSYS2008 has been compiled using MATLAB R14 (version 7.0) or higher, and may possibly have problems with earlier MATLAB versions. The MATLAB Toolboxes used in this software include Control Systems, Simulink, Real-Time Workshop, and Windows Target. Students who want to use the Student Version of MATLAB should note that it contains MATLAB, Simulink, and the Symbolic Math Toolbox. Students can purchase and download additional products to use with their student version. The Control

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System Toolbox is one of these products (see the Mathworks Web page[1] for more information). For bug reports and enhancements, please send email to mfgolnar@sfu.ca.

Thank you for trying **ACSYS**.

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Simon Fraser University
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REFERENCES

1. B. C. Kuo and F. Golnaraghi, *Automatic Control Systems*, 8th Ed., John Wiley and Sons, NY, 2003.
2. M. F. Golnaraghi, D. Boulahbal, and R. L. Leask, *Solving Engineering Mechanics Problems with MATLAB*, Book and Software package. ISBN 0-13-021537-6, Prentice-Hall, NJ, 1999

[1] http://www.mathworks.com/products

APPENDIX H

Discrete-Data Control Systems

H-1 INTRODUCTION

In recent years discrete-data and digital control systems have become more important in industry, mainly because of advances made in microprocessors and microcomputers. In addition, there are distinct advantages working with digital versus analog signals.

The block diagram of a typical digital control system is shown in Fig. H-1. The system is characterized by digitally coded signals at various parts of the system. However, the output device of the system is usually an analog component, such as a dc motor, driven by analog signals. Therefore, a digital control system often requires the use of digital-to-analog (D/A) and analog-to-digital (A/D) converters.

H-2 THE *z*-TRANSFORM

Just as linear continuous-data systems are described by differential equations, linear digital control systems are described by difference equations (see Appendix B). We have seen that the Laplace transform is a powerful method of solving linear time-invariant differential equations. Similarly, *z*-transform is an operational method of solving linear time-invariant difference equations.

H-2-1 Definition of the *z*-Transform

Consider the sequence $y(k)$, $k = 0, 1, 2, \ldots$, where $y(k)$ could represent a sequence of numbers or events. The *z*-transform of $y(k)$ is defined as

$$Y(z) = z\text{-transform of } y(k) = \mathcal{Z}[y(k)] = \sum_{k=0}^{\infty} y(k)z^{-k} \tag{H-1}$$

where z is a complex variable with real and imaginary parts. The significance of this definition will become clear later. One important property of the *z*-transform is that it can convert a sequence of numbers in the real domain into an expression in the complex z-domain. The following examples illustrate the derivation of the *z*-transforms of two simple functions.

EXAMPLE H-2-1 Consider the sequence

$$y(zk) = e^{-\alpha k} \qquad k = 0, 1, 2, \ldots \tag{H-2}$$

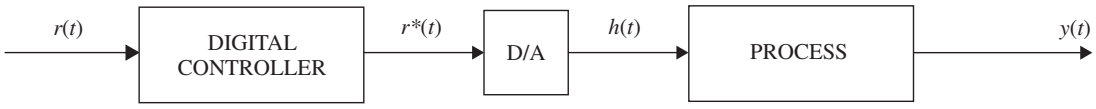

Figure H-1 Block diagram of a typical digital control system.

where α is a real constant. Applying Eq. (H-1), the z-transform of $y(k)$ is written

$$Y(z) = \sum_{k=0}^{\infty} e^{-\alpha k} z^{-k} = 1 + e^{-\alpha} z^{-1} + e^{-2\alpha} z^{-2} + \cdots \tag{H-3}$$

which converges for $\left|e^{-\alpha} z^{-1}\right| < 1$.

Multiplying both sides of the last equation by $e^{-\alpha} z^{-1}$, subtracting the resulting equation from Eq. (H-3), and solving for $Y(z)$, the latter is expressed in closed form as

$$Y(z) = \frac{1}{1 - e^{-\alpha} z^{-1}} = \frac{z}{z - e^{-\alpha}} \tag{H-4}$$

for $\left|e^{-\alpha} z^{-1}\right| < 1$. ◀

▶ **EXAMPLE H-2-2** In Example H-2-1, if $\alpha = 0$, we have

$$y(k) = 1 \quad k = 0, 1, 2, \ldots \tag{H-5}$$

which represents a sequence of ones. Then, the z-transform of $y(k)$ is

$$Y(z) = 1 + z^{-1} + z^{-2} + \cdots = \frac{z}{z - 1} \tag{H-6}$$

which converges for $|z| > 1$. ◀

H-2-2 Relationship between the Laplace Transform and the *z*-Transform

While the mathematicians like to talk about sequences, engineers feel more at home dealing with signals. It may be useful to represent the sequence $y(kT)$, $k = 0, 1, 2, \ldots$ as a train of impulses separated by the time interval T. The latter is defined as the **sampling period**. The impulse at the kth time instant, $\delta(t - kT)$, carries the value of $y(kT)$. This situation occurs quite often in digital and sampled-data control systems in which a signal $y(t)$ is digitized or sampled every T seconds to form a time sequence that represents the signal at the sampling instants. Thus, we can relate the sequence $y(kT)$ with a signal that can be expressed as

$$y^*(t) = \sum_{k=0}^{\infty} y(kT)\delta(t - kT) \tag{H-7}$$

Taking the Laplace transform on both sides of Eq. (H-7), we have

$$Y^*(s) = \mathcal{L}[y^*(t)] = \sum_{k=0}^{\infty} y(kT) e^{-kTs} \tag{H-8}$$

Comparing Eq. (H-8) with Eq. (H-1), we see that the z-transform may be related to the Laplace transform through

$$z = e^{Ts} \tag{H-9}$$

In fact, the z-transform as defined in Eq. (H-1) may be regarded as a special case with $T = 1$. The definition of the z-transform in Eq. (H-9) allows us to treat sampled systems and perform digital simulation of continuous-data systems. Thus, we can summarize the definition of the z-transform as

$$Y(z) = \mathcal{Z}\,[y(kT)] = \mathcal{Z}\,[y^*(t)] = \mathcal{Z}\,[Y^*(s)] \tag{H-10}$$

Or, we can write

$$Y(z) = \mathcal{Z}\,[y(t)] = \mathcal{Z}\,[Y(s)] \tag{H-11}$$

with the understanding that the function $y(t)$ is first sampled or discretized to get $y^*(t)$ before taking the z-transform.

▶ **EXAMPLE H-2-3** Consider the time function

$$y(t) = e^{-\alpha t} u_s(t) \tag{H-12}$$

The z-transform of $y(t)$ is obtained by performing the following steps:

1. Represent the values of $y(t)$ at the time instants $t = kT$, $k = 0, 1, 2, \ldots$, to form the function $y^*(t)$:

$$y^*(t) = \sum_{k=0}^{\infty} e^{-\alpha kT} \delta(t - kT) \tag{H-13}$$

2. Take the Laplace transform on both sides of Eq. (H-13):

$$Y^*(s) = \sum_{k=0}^{\infty} e^{-\alpha kT} e^{-kTs} = \sum_{k=0}^{\infty} e^{-(s+\alpha)kT} \tag{H-14}$$

3. Express $Y^*(s)$ in closed form and apply Eq. (H-9), giving the z-transform,

$$Y(z) = \frac{z}{z - e^{-\alpha T}} \tag{H-15}$$

In general, the z-transforms of more complex functions may be obtained with the help of some of the z-transform theorems that follow. For engineering purposes, a z-transform table such as that in Appendix D may be used to transform from $y(k)$ to $Y(z)$. ◀

H-2-3 Some Important Theorems of the z-Transform

Some of the commonly used theorems of the z-transform are stated in the following without proof. Just as in the case of the Laplace transform, these theorems are useful in many aspects of the z-transform analysis. For uniformity, the real sequence is expressed as $y(kT)$, and if a sampling period is not involved, T can be set to unity.

■ **Theorem 1.** ***Addition and Subtraction*** If $y_1(kT)$ and $y_2(kT)$ have z-transforms $Y_1(z)$ and $Y_2(z)$, respectively, then

$$\mathcal{Z}\,[y_1(kT) \pm y_2(kT)] = Y_1(z) \pm Y_2(z) \tag{H-16}$$

■ **Theorem 2.** ***Multiplication by a Constant***

$$\mathcal{Z}\,[\alpha y(kT)] = \alpha \mathcal{Z}\,[y(kT)] = \alpha Y(z) \tag{H-17}$$

where α is a constant.

Theorem 3. ***Real Translation (Time Delay and Time Advance)***

$$\mathcal{Z}\left[y(kT-nT)\right] = z^{-n}Y(z) \tag{H-18}$$

and

$$\mathcal{Z}\left[y(kT+nT)\right] = z^{n}\left[Y(z) - \sum_{k=0}^{n-1} y(kT)z^{-k}\right. \tag{H-19}$$

where n is a positive integer.

Eq. (H-18) represents the z-transform of a time sequence that is shifted to the right by nT, and Eq. (H-19) denotes that of a time sequence shifted to the left by nT. The reason that the right-hand side of Eq. (H-19) is not just $z^nY(z)$ is because the one-sided z-transform, similar to the Laplace transform, is defined only for $k \geq 0$. Thus, the second term on the right-hand side of Eq. (H-19) simply represents the sequence that is lost after it is shifted to the left of $k = 0$.

Theorem 4. ***Complex Translation***

$$\mathcal{Z}\left[e^{\mp \alpha kT}y(kT)\right] = Y\left(ze^{\pm \alpha T}\right) \tag{H-20}$$

where α is a constant. $Y(z)$ is the z-transform of $y(kT)$.

Theorem 5. ***Initial-Value Theorem***

$$\lim_{k \to 0} y(kT) = \lim_{z \to \infty} Y(z) \tag{H-21}$$

if the limit exists.

Theorem 6. ***Final-Value Theorem***

$$\lim_{k \to \infty} y(kT) = \lim_{z \to 1} \left(1 - z^{-1}\right)Y(z) \tag{H-22}$$

if the function $\left(1 - z^{-1}\right)Y(z)$ has no poles on or outside the unit circle $|z| = 1$ in the z-plane.

Theorem 7. ***Real Convolution***

$$\begin{aligned} Y_1(z)Y_2(z) &= \mathcal{Z}\left[\sum_{k=0}^{N} y_1(kT)y_2(NT-kT)\right] = \mathcal{Z}\left[\sum_{k=0}^{N} y_2(kT)y_1(NT-kT)\right] \\ &= \mathcal{Z}\left[y_1(kT)^* \, y_2(kT)\right] \end{aligned} \tag{H-23}$$

where "*" denotes real convolution in the discrete-time domain.

Thus, we see that, as in the Laplace transform, the z-transform of the product of two real functions $y_1(k)$ and $y_2(k)$ is **not** equal to the product of the z-transforms $Y_1(z)$ and $Y_2(z)$. One exception to this in the case of the z-transform is if one of the two functions is the delay e^{-NTs}, where N is a positive integer. Then

$$\mathcal{Z}\left[e^{-NTs}Y(s)\right] = \mathcal{Z}\left[e^{-NTs}\right]\mathcal{Z}\left[Y(s)\right] = z^{-N}Y(z) \tag{H-24}$$

Table H-1 summarizes the theorems on the z-transform just given. The following examples illustrate the usefulness of some of these theorems.

TABLE H-1 Theorems of z-Transforms

Addition and subtraction	$\mathcal{Z}[y_1(kT) \pm y_2(kT)] = Y_1(z) \pm Y_2(z)$
Multiplication by a constant	$\mathcal{Z}[\alpha y(kT)] = \alpha\mathcal{Z}[y(kT)] = \alpha Y(z)$
Real translation	$\mathcal{Z}[y(k-n)T] = z^{-n}Y(z)$ **(time delay)**
	$\mathcal{Z}[y(k+n)T] = z^n\left[Y(z) - \sum_{k=0}^{n-1} y(kT)z^{-k}\right]$ **(time advance)**
	where n = positive integer
Complex translation	$\mathcal{Z}\left[e^{\mp\alpha kT}y(kT)\right] = Y\left(ze^{\pm\alpha T}\right)$
Initial-value theorem	$\lim_{k\to 0} y(kT) = \lim_{z\to\infty} Y(z)$
Final-value theorem	$\lim_{k\to\infty} y(kT) = \lim_{z\to 1}\left(1 - z^{-1}\right)Y(z)$
	if $(1 - z^{-1})Y(z)$ has no poles on or inside $\lvert z\rvert = 1$.
Real convolution	$Y_1(z)Y_2(z) = \mathcal{Z}\left[\sum_{k=0}^{n} y_1(kT)y_2(NT - kT)\right]$
	$= \mathcal{Z}\left[\sum_{k=0}^{n} y_2(kT)y_1(NT - kT)\right]$
	$= \mathcal{Z}[y_1(kT)^* y_2(kT)]$

▶ **EXAMPLE H-2-4** (*Complex-translation theorem*) To find the z-transform of $y(t) = te^{-\alpha t}$, let $f(t) = t$, $t \geq 0$; then

$$F(z) = \mathcal{Z}[tu_s(t)] = \mathcal{Z}(kT) = \frac{Tz}{(z-1)^2} \tag{H-25}$$

Using the complex-translation theorem in Eq. (H-20), we obtain

$$Y(z) = \mathcal{Z}[te^{-\alpha t}u_s(t)] = F\left(ze^{\alpha T}\right) = \frac{Tze^{-\alpha T}}{(z - e^{-\alpha T})^2} \tag{H-26}$$

◀

▶ **EXAMPLE H-2-5** (*Final-value theorem*) Given the function

$$Y(z) = \frac{0.792z^2}{(z-1)(z^2 - 0.146z + 0.208)} \tag{H-27}$$

determine the value of $y(kT)$ as k approaches infinity.

Because the function $(1 - z^{-1})Y(z)$ does not have any pole on or outside the unit circle $\lvert z\rvert = 1$ in the z-plane, the final-value theorem in Eq. (H-22) can be applied. Thus,

$$\lim_{k\to\infty} y(kT) = \lim_{z\to 1} \frac{0.792z}{z^2 - 0.416z + 0.208} \tag{H-28}$$

◀

H-2-4 Inverse z-Transform

Just as in the Laplace transform, one of the major objectives of the z-transform is that algebraic manipulations can be made first in the z-domain and then the final time response determined by the inverse z-transform. *In general, the inverse z-transform of a function $Y(z)$ yields information on $y(kT)$ only, not on $y(t)$. In other words, the z-transform carries information only at the sampling instants.* With this in mind, the inverse z-transform can be carried out by one of the following three methods:

1. Partial-fraction expansion
2. Power-series method
3. The inverse formula

Partial-Fraction Expansion Method

The z-transform function $Y(z)$ is expanded by partial-fraction expansion into a sum of simple recognizable terms, and the z-transform table is used to determine the corresponding $y(kT)$. In carrying out the partial-fraction expansion, there is a slight difference between the z-transform and the Laplace transform procedures. With reference to the z-transform table, we note that practically all the z-transform functions have the term z in the numerator. Therefore, we should expand $Y(z)$ into the form of

$$Y(z) = \frac{K_1 z}{z - e^{-\alpha T}} + \frac{K_2 z}{z - e^{-\beta T}} + \cdots \tag{H-29}$$

To do this, first expand $Y(z)/z$ into fractions and then multiply by z to obtain the final expression. The following example will illustrate this procedure.

▶ **EXAMPLE H-2-6** Given the z-transform function

$$Y(z) = \frac{(1 - e^{-\alpha T})z}{(z - 1)(z - e^{-\alpha T})} \tag{H-30}$$

find the inverse z-transform. Expanding $Y(z)/z$ by partial-fraction expansion, we have

$$\frac{Y(z)}{z} = \frac{1}{z - 1} - \frac{1}{z - e^{-\alpha T}} \tag{H-31}$$

The final expanded expression for $Y(z)$ is

$$Y(z) = \frac{z}{z - 1} - \frac{z}{z - e^{-\alpha T}} \tag{H-32}$$

From the z-transform table in Appendix D, the corresponding inverse z-transform of $Y(z)$ is found to be

$$y(kT) = 1 - e^{-\alpha k T} \quad k = 0, 1, 2, \cdots \tag{H-33}$$

It should be pointed out that if $Y(z)$ does not contain any factors of z in the numerator, this usually means that the time sequence has a delay, and the partial-fraction expansion of $Y(z)$ should be carried out without first dividing the function by z. The following example illustrates this situation. ◀

▶ **EXAMPLE H-2-7** Consider the function

$$Y(z) = \frac{(1 - e^{-\alpha T})}{(z - 1)(z - e^{-\alpha T})} \tag{H-34}$$

which does not contain any powers of z as a factor in the numerator. In this case, the partial-fraction expansion of $Y(z)$ is carried out directly. We have

$$Y(z) = \frac{1}{z - 1} - \frac{1}{z - e^{-\alpha T}} \tag{H-35}$$

Although the z-transform table does not contain exact matches for the components in Eq. (H-35), we recognize that the inverse z-transform of the first term on the right-hand side can be written as

$$\mathcal{Z}^{-1}\left[\frac{1}{z - 1}\right] = \mathcal{Z}^{-1}\left[z^{-1}\frac{z}{z - 1}\right] = \sum_{k=1}^{\infty} z^{-1} \quad = u_k[(k - 1)T] \quad k = 1, 2, \cdots \tag{H-36}$$

Similarly, the second term on the right-hand side of Eq. (H-35) can be identified with a time delay of T seconds. Thus, the inverse z-transform of $Y(z)$ is written

$$y(kT) = \left(1 - e^{-\alpha(k-1)T}\right)u[(k-1)T] \qquad k = 1, 2, \cdots \tag{H-37}$$

◀

H-2-5 Computer Solution of the Partial-Fraction Expansion of $Y(z)/z$

Whether the function to be expanded by partial fraction is in the form of $Y(z)/z$ or $Y(z)$, the computer programs designed for performing the partial-fraction of Laplace transform functions can still be applied.

Power-Series Method

The definition of the z-transform in Eq. (H-1) gives a straightforward method of carrying out the inverse z-transform. Based on Eq. (H-1), we can clearly see that in the sampled case the coefficient of z^{-k} in $Y(z)$ is simply $y(kT)$. Thus, if we expand $Y(z)$ into a power series in powers of z^{-k}, we can find the values of $y(kT)$ for $k = 0, 1, 2, \ldots$.

▶ **EXAMPLE H-2-8** Consider the function $Y(z)$ given in Eq. (H-30), which can be expanded into a power series of z^{-1} by dividing the numerator polynomial by the denominator polynomial by long division. The result is

$$Y(z) = \left(1 - e^{-\alpha T}\right)z^{-1} + \left(1 - e^{-2\alpha T}\right)z^{-2} + \cdots + \left(1 - e^{-k\alpha T}\right)z^{-k} + \cdots \tag{H-38}$$

Thus, it is apparent that

$$y(kT) = 1 - e^{-\alpha kT} \qquad k = 0, 1, 2, \ldots \tag{H-39}$$

which is the same result as in Eq. (H-33). ◀

Inversion Formula

The time sequence $y(kT)$ can be determined from $Y(z)$ by use of the inversion formula:

$$y(kT) = \frac{1}{2\pi j}\oint_{\Gamma} Y(z)z^{k-1}dz \tag{H-40}$$

which is a contour integration along the path Γ, that is, a circle of radius $|z| = e^{cT}$ centered at the origin in the z-plane, and c is a value such that the poles of $Y(z)z^{k-1}$ are inside the circle. The inversion formula of the z-transform is similar to that of the inverse Laplace-transform integral given in Eq. (2-124). One way of evaluating the contour integration of Eq. (H-40) is to use the residue theorem of complex-variable theory (the details are not covered here).

H-2-6 Application of the z-Transform to the Solution of Linear Difference Equations

The z-transform can be used to solve linear difference equations. As a simple example, let us consider the first-order unforced difference equation

$$y(k+1) + y(k) = 0 \tag{H-41}$$

To solve this equation, we take the z-transform on both sides of the equation. By this, we mean that we multiply both sides of the equation by z^{-k} and take the sum from $k = 0$ to $k = \infty$. We have

$$\sum_{k=0}^{\infty} y(k+1)z^{-k} + \sum_{k=0}^{\infty} y(k)z^{-k} = 0 \tag{H-42}$$

By using the definition of $Y(z)$ and the real translation theorem of Eq. (H-19) for time advance, the last equation is written

$$z[Y(z) - y(0)] + Y(z) = 0 \tag{H-43}$$

Solving for $Y(z)$, we get

$$Y(z) = \frac{z}{z+1} y(0) \tag{H-44}$$

The inverse z-transform of the last equation can be obtained by expanding $Y(z)$ into a power series in z^{-1} by long division. We have

$$Y(z) = \left(1 - z^{-1} + z^{-2} - z^{-3} + \cdots\right) x(0) \tag{H-45}$$

Thus, $y(k)$ is written

$$y(k) = (-1)^k y(0) \quad k = 0, 1, 2, \ldots \tag{H-46}$$

Eq. (H-41) is recognized as a single-state equation. The z-transform solution of high-order discrete-data systems described by state equations is treated in Section H-3.

The following example shows the z-transform solution of a second-order difference equation.

▶ EXAMPLE H-2-9 Consider the second-order difference equation

$$y(k+2) + 0.5y(k+1) + 0.2y(k) = u(k) \tag{H-47}$$

where

$$u(k) = u(k) = 1 \quad \text{for } k = 0, 1, 2, \ldots \tag{H-48}$$

The initial conditions of $y(k)$ are $y(0) = 0$ and $y(1) = 0$.

Taking the z-transform on both sides of Eq. (H-47), we get

$$\left[z^2 Y(z) - z^2 y(0) - zy(1)\right] + 0.5[zY(z) - zy(0)] + 0.2Y(z) = U(z) \tag{H-49}$$

The z-transform of $u(k)$ is $U(z) = z/(z - 1)$. Substituting the initial conditions of $y(k)$ and the expression of $U(z)$ into Eq. (H-49) and solving for $Y(z)$, we have

$$Y(z) = \frac{z}{(z-1)(z^2 + 0.5z + 0.2)} \tag{H-50}$$

The partial-fraction expansion of $Y(z)/z$ is

$$\frac{Y(z)}{z} = \frac{0.588}{z-1} - \frac{1.036e^{j1.283}}{z + 0.25 + j0.37} - \frac{1.036e^{-j1.283}}{z + 0.25 - j0.37} \tag{H-51}$$

where the exponents in the numerator coefficients are in radians.

Taking the inverse z-transform of $Y(z)$, we get

$$\begin{aligned} y(k) &= 0.588 - 1.036(0.447)^k \left[e^{-j(2.165k - 1.283)} + e^{j(2.165k - 1.283)}\right] \\ &= 0.588 - 2.072(0.447)^k \cos(2.165k - 1.283) \qquad k \geq 0 \end{aligned} \tag{H-52}$$

◀

▶ H-3 TRANSFER FUNCTIONS OF DISCRETE-DATA SYSTEMS

Discrete-data control systems have the unique features that the signals in these systems either are in the form of pulse trains or are digitally coded, and the controlled processes often contain analog components. For instance, a dc motor, which is an analog device, can

be controlled either by a controller that outputs analog signals or by a digital controller that outputs digital data. In the latter case, an interface such as a digital-to-analog (D/A) converter is necessary to couple the digital component to the analog devices. The input and output of the discrete-data system in Fig. H-1 can be represented by number sequences with the numbers separated by the sampling period T.

For linear operation, the D/A converter can be represented by a **sample-and-hold (S/H)** device, which consists of a sampler and a data-hold device. The S/H that is most often used for the analysis of discrete-data systems consists of an **ideal sampler** and a **zero-order-hold (ZOH)** device. Thus, the system shown in Fig. H-1 can be functionally represented by the block diagram in Fig. H-2. Fig. H-3 shows the typical operation of an ideal sampler and a ZOH. The continuous data $r(t)$ is sampled with a sampling period T by the ideal sampler. The output of the ideal sampler $r^*(t)$ is a train of impulses with the magnitudes of $r(t)$ at T carried by the strengths of the impulses. Note that the ideal sampler is not a physical entity. It is devised simply to represent the discrete-time signal mathematically. In Fig. H-3, the arrows at the sampling instants represent impulses. Because, by definition, an impulse has zero pulse width and infinite height, the lengths of the arrows simply represent the areas under the impulses and are the magnitudes of the input signal $r(t)$ at the sampling instants. The ZOH simply holds the magnitude of the signal carried by the incoming impulse at a given time instant, say, kT, for the entire sampling period t until the next impulse arrives at $t = (k + 1)T$. The output of the ZOH is a

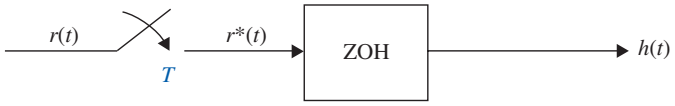

Figure H-2 Sample-and-hold (S/H) device.

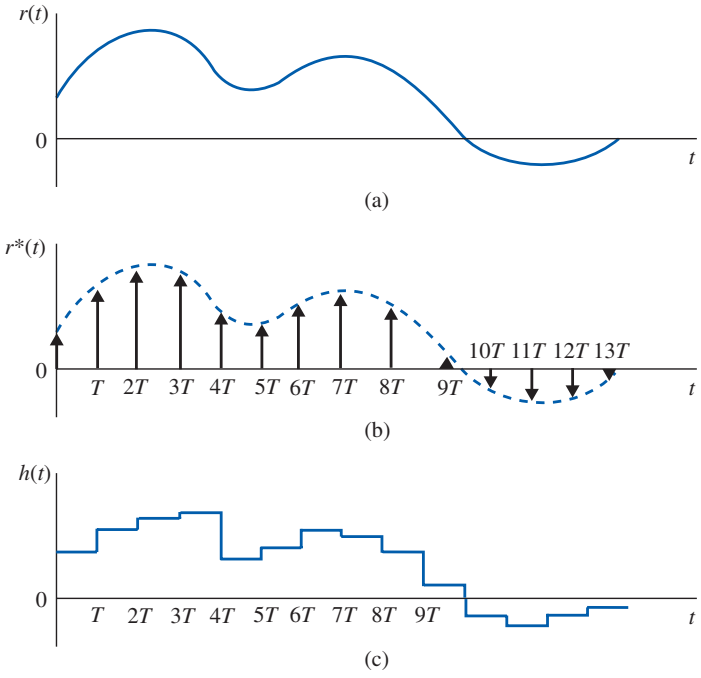

Figure H-3 (a) Input signal to an ideal sampler. (b) Output signal of an ideal sampler. (c) Output signal of a zero-order-hold (ZOH) device.

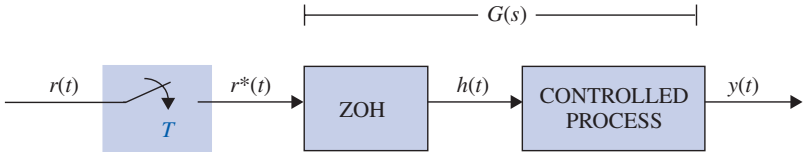

Figure H-4 Block diagram of a discrete-data system.

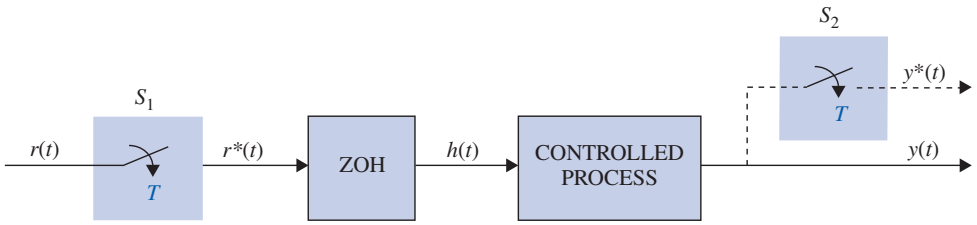

Figure H-5 Discrete-data system with a fictitious sampler.

staircase approximation of the input to the ideal sampler, $r(t)$. As the sampling period T approaches zero, the output of the ZOH, $h(t)$, approaches $r(t)$; that is,

$$\lim_{T \to 0} h(t) = r(t) \tag{H-53}$$

However, because the output of the sampler, $r^*(t)$, is an impulse train, its limit as T approaches zero *does not* have any physical meaning. Based on the preceding discussions, a typical open-loop discrete-data system is modeled as shown in Fig. H-4.

There are several ways of deriving the transfer-function representation of the system in Fig. H-5. The following derivation is based on the Fourier-series representation of the signal $r^*(t)$. We begin by writing

$$r^*(t) = r(t)\delta_T(t) \tag{H-54}$$

where $\delta_T(t)$ is the unit-impulse train,

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) \tag{H-55}$$

Because $\delta_T(t)$ is a periodic function with period T, it can be expressed as a Fourier series:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi nt/T} \tag{H-56}$$

where C_n is the Fourier coefficient and is given by

$$C_n = \frac{1}{T}\int_0^T \delta_T(t) e^{-jn\omega_s t} dt \tag{H-57}$$

where $\omega_s = 2\pi/T$ is the **sampling frequency** in rad/sec.

Because the unit impulse is defined as a pulse with a width of δ and a height of $1/\delta$, and $\delta \to 0$, C_n is written

$$C_n = \lim_{\delta \to 0} \frac{1}{T\delta}\int_0^{\delta} e^{-jn\omega_s t}\, dt = \lim_{\delta \to 0} \frac{1 - e^{-jn\omega_s \delta}}{jn\omega_s T\delta} = \frac{1}{T} \tag{H-58}$$

Substituting Eq. (H-58) in Eq. (H-56), and then the latter in Eq. (H-54), we get

$$r^*(t) = \frac{1}{T}\sum_{n=-\infty}^{\infty} r(t)e^{jn\omega_s t} \tag{H-59}$$

Taking the Laplace transform on both sides of Eq. (H-59), and using the complex shifting property of Eq. (2-137), we get

$$R^*(s) = \frac{1}{T}\sum_{n=-\infty}^{\infty} R(s - jn\omega_s) = \frac{1}{T}\sum_{n=-\infty}^{\infty} R(s + jn\omega_s) \tag{H-60}$$

Eq. (H-60) represents the Laplace transform of the sampled signal $r^*(t)$. It is an alternative expression to Eq. (H-8). From Eq. (H-8), $R^*(s)$ can be written as

$$R^*(s) = \sum_{k=0}^{\infty} r(kT)e^{-kTs} \tag{H-61}$$

Because the summing limits of $R^*(s)$ range from $-\infty$ to ∞, if s is replaced by $s + jm\omega_s$ in Eq. (H-60), where m is any integer, we have

$$R^*(s + jm\omega_s) = R^*(s) \tag{H-62}$$

Pulse-Transfer Function

Now we are ready to derive the transfer function of the discrete-data system shown in Fig. H-4. The Laplace transform of the system output $y(t)$ is written

$$Y(s) = G(s)R^*(s) \tag{H-63}$$

Although the output $y(t)$ is obtained from $Y(s)$ by taking the inverse Laplace transform on both sides of Eq. (H-63), this step is difficult to execute because $G(s)$ and $R^*(s)$ represent different types of signals. To overcome this problem, we apply a fictitious sampler at the output of the system, as shown in Fig. H-5. The fictitious sampler S_2 has the same sampling period T and is synchronized to the original sampler S_1. The sampled form of $y(t)$ is $y^*(t)$. Applying Eq. (H-60) to $y^*(t)$, and using Eq. (H-63), we have

$$Y^*(s) = \frac{1}{T}\sum_{n=-\infty}^{\infty} G(s + jn\omega_s)R^*(s + jn\omega_s) \tag{H-64}$$

In view of the relationship in Eq. (H-62), Eq. (H-64) is written

$$Y^*(s) = R^*(s)\frac{1}{T}\sum_{n=-\infty}^{\infty} G(s + jn\omega_s) = R^*(s)G^*(s) \tag{H-65}$$

where $G^*(s)$ is defined the same way as $R^*(s)$ in Eq. (H-60) and is called the **pulse-transfer function** of $G(s)$.

***z*-Transfer Function**

Now that all the functions in Eq. (H-65) are in sampled form, where $R^*(s)$, $G^*(s)$, and $Y^*(s)$ all have the form of Eq. (H-61), we can take the z-transform on both sides of the equation by substituting $z = e^{Ts}$. We have

$$Y(z) = G(z)R(z) \tag{H-66}$$

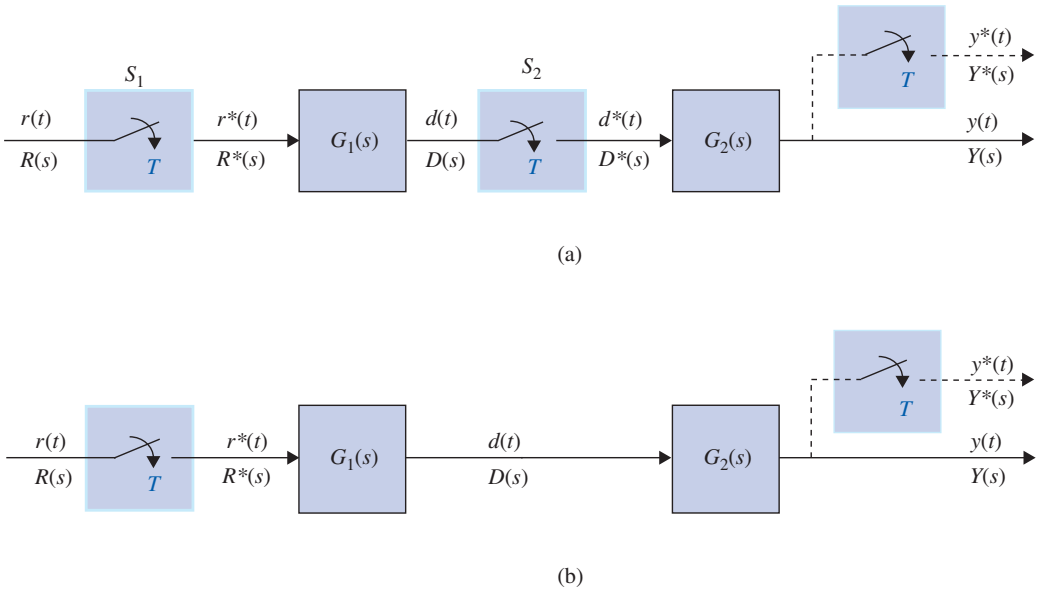

Figure H-6 (a) Discrete-data system with cascaded elements and a sampler separating the two elements. (b) Discrete-data system with cascaded elements and no sampler in between.

where $G(z)$ is defined as the ***z*-transfer function** of $G(s)$ and is given by

$$G(z) = \sum_{k=0}^{\infty} g(kT)z^{-k} \tag{H-67}$$

Thus, for the discrete-data system shown in Figs. H-5 and H-6, the z-transform of the output is equal to the z-transfer function of the process and the z-transform of the input.

H-3-1 Transfer Functions of Discrete-Data Systems with Cascade Elements

The transfer-function representation of discrete-data systems with elements connected in cascade is slightly more involved than that for continuous-data systems, because of the variation of having or not having samplers in between the elements. Fig. H-6 shows two different discrete-data systems that contain two elements connected in cascade. In the system of Fig. H-6(a), the two elements are separated by the sampler S_2, which is synchronized to, and has the same period as, the sampler S_1. The two elements in the system of Fig. H-6(b) are connected directly together. It is important to distinguish these two cases when deriving the pulse-transfer function and the z-transfer function. For the system in Fig. H-6(a), the output of $G_1(s)$ is written

$$D(s) = G_1(s)R^*(s) \tag{H-68}$$

and the system output is

$$Y(s) = G_2(s)D^*(s) \tag{H-69}$$

Taking the pulse transform on both sides of Eq. (H-68) and using Eq. (H-62), we have

$$D^*(s) = G_1^*(s)R^*(s) \tag{H-70}$$

Now substituting Eq. (H-70) in Eq. (H-69) and taking the pulse transform, we get

$$Y^*(s) = G_1^*(s)G_2^*(s)R^*(s) \tag{H-71}$$

The corresponding z-transform expression of Eq. (H-71) is

$$Y(z) = G_1(z)G_2(z)R(z) \tag{H-72}$$

We conclude that *the* z*-transform of two systems separated by a sampler is equal to the product of the* z*-transforms of the two systems.*

The Laplace transform of the output of the system in Fig. H-6(b) is

$$Y(s) = G_1(s)G_2(s)R^*(s) \tag{H-73}$$

Taking the pulse transform on both sides of the last equation, we get

$$Y^*(s) = [G_1(s)G_2(s)]^*R^*(s) \tag{H-74}$$

where

$$[G_1(s)G_2(s)]^* = \frac{1}{T}\sum_{n=-\infty}^{\infty} G_1(s + jn\omega_s)G_2(s + jn\omega_s) \tag{H-75}$$

Notice that because $G_1(s)$ and $G_2(s)$ are not separated by a sampler, they have to be treated as one system when taking the pulse transform.

Taking the z-transform on both sides of Eq. (H-74) gives

$$Y(z) = \mathcal{Z}\{[G_1(s)G_2(s)]^*\}R(z) \tag{H-76}$$

Let

$$\mathcal{Z}\{[G_1(s)G_2(s)]^*\} = G_1G_2(z) \tag{H-77}$$

Then, Eq. (H-76) is written

$$Y(z) = G_1G_2(z)R(z) \tag{H-78}$$

H-3-2 Transfer Function of the Zero-Order-Hold

Based on the description of the ZOH given earlier, its impulse response is shown in Fig. H-7. The transfer function of the ZOH is written

$$G_h(s) = \mathcal{L}[g_h(t)] = \frac{1 - e^{-Ts}}{s} \tag{H-79}$$

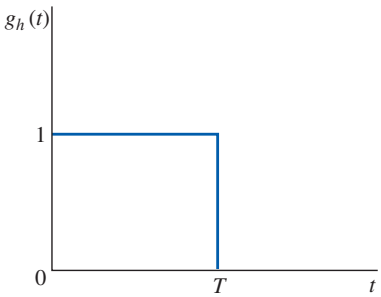

Figure H-7 Impulse response of the ZOH.

Thus, if the ZOH is connected in cascade with a linear process using transfer function $G_p(s)$, as shown in Fig. H-5, the z-transform of the combination is written

$$G(z) = \mathcal{Z}\left[G_h(s)G_p(s)\right] = \mathcal{Z}\left(\frac{1 - e^{-Ts}}{s}G_p(s)\right) \tag{H-80}$$

By using the time-delay property of z-transforms, Eq. (H-18), Eq. (H-80) is simplified to

$$G(z) = \left(1 - z^{-1}\right)\mathcal{Z}\left(\frac{G_p(s)}{s}\right) \tag{H-81}$$

▶ EXAMPLE H-3-1 Consider that, for the system shown in Fig. H-5,

$$G_p(s) = \frac{1}{s(s + 0.5)} \tag{H-82}$$

The sampling period is 1 second. The z-transfer function of the system between the input and the output is determined using Eq. (H-81).

$$\begin{aligned} G(z) &= \left(1 - z^{-1}\right)\mathcal{Z}\left(\frac{1}{s^2(s + 0.5)}\right) \\ &= \left(1 - z^{-1}\right)\mathcal{Z}\left(\frac{2}{s^2} - \frac{4}{s} + \frac{4}{s + 0.5}\right) \\ &= \frac{0.426z + 0.361}{z^2 - 1.606z + 0.606} \end{aligned} \tag{H-83}$$

◀

H-3-3 Transfer Functions of Closed-Loop Discrete-Data Systems

The transfer functions of closed-loop discrete-data systems are derived using the following procedures:

1. Regard the outputs of samplers as inputs to the system.
2. All other noninputs of the system are treated as outputs.
3. Write cause-and-effect equations between the inputs and the outputs of the system using the SFG gain formula.
4. Take the pulsed transform or the z-transform of the equations obtained in Step 3, and manipulate these equations to get the pulse-transfer function or the z-transfer function.

Reference [1] describes the sampled signal-flow graph that can be used to implement Step 4 using the SFG gain formula.

The following examples illustrate the algebraic procedure of finding the transfer functions of closed-loop discrete-data systems.

▶ EXAMPLE H-3-2 Consider the closed-loop discrete-data system shown in Fig. H-8. The output of the sampler is regarded as an input to the system. Thus, the system has inputs $R(s)$ and $E^*(s)$. The signals $E(s)$ and $Y(s)$ are regarded as the outputs of the system.

Writing the cause-and-effect equations for $E(s)$ and $Y(s)$ using the gain formula, we get

$$E(s) = R(s) - G(s)H(s)E^*(s) \tag{H-84}$$

$$Y(s) = G(s)E^*(s) \tag{H-85}$$

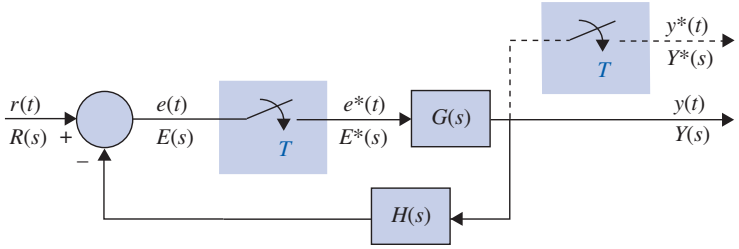

Figure H-8 Closed-loop discrete-data system.

Notice that the right-hand side of the last two equations contains only the inputs $R(s)$ and $E^*(s)$ and the transfer functions. Taking the pulse transform on both sides of Eq. (H-60) and solving for $E^*(s)$, we get

$$E^*(s) = \frac{R^*(s)}{1 + [G(s)H(s)]^*} \tag{H-86}$$

Substituting $E^*(s)$ from Eq. (H-86) into Eq. (H-85), we get

$$Y(s) = \frac{G(s)}{1 + [G(s)H(s)]^*} R^*(s) \tag{H-87}$$

Taking the pulse transform on both sides of Eq. (H-87) and using Eq. (H-62), we arrive at the pulse-transfer function of the closed-loop system,

$$\frac{Y^*(s)}{R^*(s)} = \frac{G^*(s)}{1 + [G(s)H(s)]^*} \tag{H-88}$$

Taking the z-transform on both sides of the last equation, we have

$$\frac{Y(z)}{R(z)} = \frac{G(z)}{1 + GH(z)} \tag{H-89}$$

◀

▶ **EXAMPLE H-3-3** We show in this example that, although it is possible to define an input–output transfer function for the system in Fig. H-8, this may not be possible for all discrete-data systems. Let us consider the system shown in Fig. H-9, which has a sampler in the feedback path. In this case, the outputs of the sampler $Y^*(s)$ and $R(s)$ are the inputs of the system; $Y(s)$ and $E(s)$ are regarded as the outputs. Writing $E(s)$ and $Y(s)$ in terms of the inputs using the gain formula, we get

$$Y(s) = G(s)E(s) \tag{H-90}$$

$$E(s) = R(s) - H(s)Y^*(s) \tag{H-91}$$

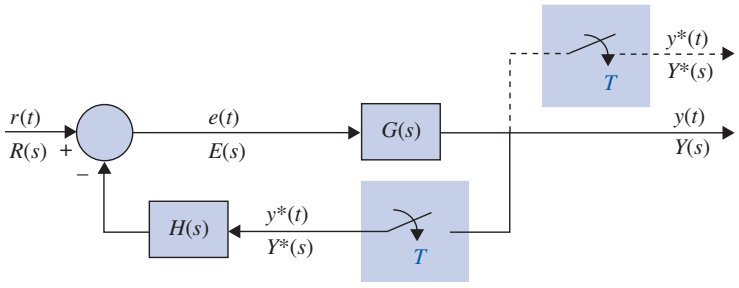

Figure H-9 Closed-loop discrete-data system.

Taking the pulse transform on both sides of the last two equations and after simple algebraic manipulations, the pulse transform of the output is written

$$Y^*(s) = \frac{[G(s)R(s)]^*}{1 + [G(s)H(s)]^*} \tag{H-92}$$

Note that the input $R(s)$ and the transfer function $G(s)$ are now combined as one function, $[G(s)R(s)]^*$, and we cannot define a transfer function in the form of $Y^*(s)/R^*(s)$. The z-transform of the output is written

$$Y(z) = \frac{GR(z)}{1 + GH(z)} \tag{H-93}$$

Although we have been able to arrive at the input–output transfer function and transfer relation of the systems in Figs. H-8 and H-9 by algebraic means without difficulty, for more complex system configurations, the algebraic method may become tedious. The signal-flow graph method may be extended to the analysis of discrete-data systems; refer to Reference [1] at the end of this appendix for details. ◀

▶ H-4 STATE EQUATIONS OF LINEAR DISCRETE-DATA SYSTEMS

Just as for continuous-data systems, the modern way of modeling a discrete-data system is by discrete state equations. As described earlier, when dealing with discrete-data systems, we often encounter two situations. The first one is that the system contains continuous-data components, but the signals at certain points of the system are discrete with respect to time because of sample-and-hold (S/H) operations. In this case, the components of the system are still described by differential equations, but because of the discrete-time data, the differential equations are discretized to yield a set of difference equations. The second situation involves systems that are completely discrete with respect to time, and the system dynamics should be difference equations from the outset.

H-4-1 Discrete State Equations

Let us consider the discrete-data control system with an S/H device, as shown in Fig. H-10. Typical signals that appear at various points in the system are shown in the figure. The output signal $y(t)$ ordinarily is a continuous-data signal. The output of the S/H, $h(t)$, is a sequence of steps. Therefore, we can write

$$h(t) = h(kT) = r(kT) \tag{H-94}$$

for $kT \leq t < (k + 1)T$, $k = 0, 1, 2, \ldots$.

Now we let the linear process G be described by the state equation and output equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}h(t) \tag{H-95}$$

$$\mathbf{y}(\mathbf{t}) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}h(t) \tag{H-96}$$

where $\mathbf{x}(t)$ is the $n \times n$ state vector, and $h(t)$ and $y(t)$ are the scalar input and output, respectively. The matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are coefficient matrices. By using Eq. (10-79), the state transition equation is

$$\mathbf{x}(t) = \boldsymbol{\phi}(t - t_0)\mathbf{x}(t_0) + \int_{t_0}^{t} \boldsymbol{\phi}(t - \tau)\mathbf{B}h(\tau)d\tau \tag{H-97}$$

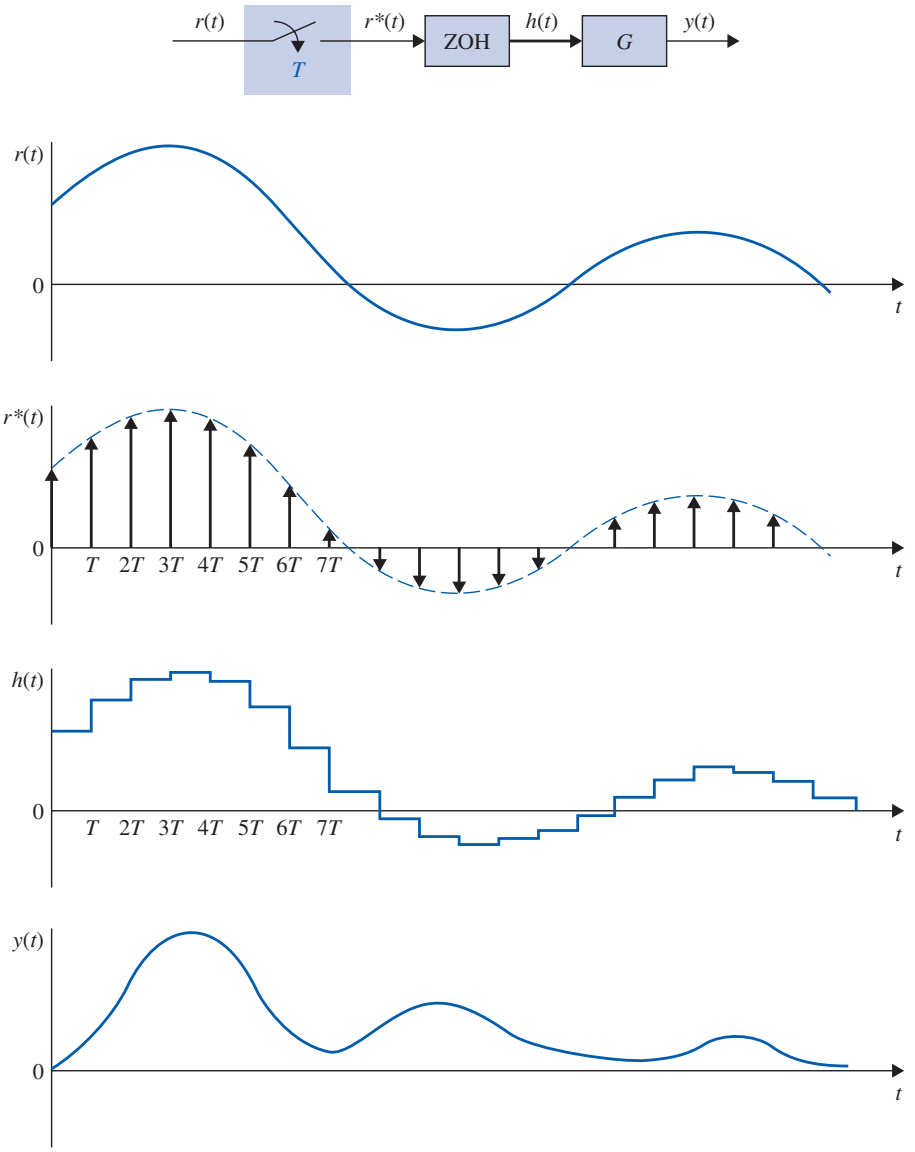

Figure H-10 Discrete-data system with sample-and-hold (S/H).

for $t \geq t_0$. If we are interested only in the responses at the sampling instants, we let $t = (k + 1)T$ and $t_0 = kT$. Then Eq. (H-97) becomes

$$\mathbf{x}[(k+1)T] = \boldsymbol{\phi}(T)\mathbf{x}(kT) + \int_{kT}^{(k+1)T} \boldsymbol{\phi}[(k+1)T - \tau]\mathbf{B}h(\tau)d\tau \qquad \text{(H-98)}$$

Because $h(t)$ is piecewise constant, as defined in Eq. (H-95), the input $h(\tau)$ in Eq. (H-98) can be taken outside of the integral sign. Eq. (H-98) is written

$$\mathbf{x}[(k+1)T] = \boldsymbol{\phi}(T)\mathbf{x}(kT) + \int_{kT}^{(k+1)T} \boldsymbol{\phi}[(k+1)T - \tau]\mathbf{B}d\tau h(kT) \qquad \text{(H-99)}$$

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or

$$\mathbf{x}[(k+1)T] = \boldsymbol{\phi}(T)\mathbf{x}(kT) + \boldsymbol{\theta}(T)h(kT) \tag{H-100}$$

where

$$\theta(T) = \int_{kT}^{(k+1)T} \boldsymbol{\phi}[(k+1)T - \tau]\mathbf{B}d\boldsymbol{\tau} = \int_0^T \boldsymbol{\phi}(T - \boldsymbol{\tau})\mathbf{B}d\tau \tag{H-101}$$

Eq. (H-100) is of the form of a set of linear first-order difference equations in vector-matrix form and is referred to as the **vector-matrix discrete state equation**.

H-4-2 Solutions of the Discrete State Equations: Discrete State-Transition Equations

The discrete state equations represented by Eq. (H-100) can be solved by using a simple recursion procedure. By setting $k = 0, 1, 2, \ldots$ successively in Eq. (H-100), the following equations result:

$$k = 0: \quad \mathbf{x}(T) = \boldsymbol{\phi}(T)\mathbf{x}(0) + \boldsymbol{\theta}(T)h(0) \tag{H-102}$$

$$k = 1: \quad \mathbf{x}(2T) = \boldsymbol{\phi}(2T) = \boldsymbol{\phi}(T)\mathbf{x}(T) + \boldsymbol{\theta}(T)h(T) \tag{H-103}$$

$$k = 2: \quad \mathbf{x}(3T) = \boldsymbol{\phi}(3T) = \boldsymbol{\phi}(T)\mathbf{x}(2T) + \boldsymbol{\theta}(T)h(2T) \tag{H-104}$$

$$\vdots \qquad \vdots$$

$$k = n-1: \quad \mathbf{x}(nT) = \boldsymbol{\phi}(T)\mathbf{x}[(n-1)T] + \boldsymbol{\theta}(T)h[(n-1)T] \tag{H-105}$$

Substituting Eq. (H-102) into Eq. (H-103), then Eq. (H-103) into Eq. (H-104), and so on, we obtain the following solution for Eq. (H-100):

$$\mathbf{x}(nT) = \phi^n(T)\mathbf{x}(0) + \sum_{i=0}^{n-1} \phi^{n-i-1}(T)\theta(T)h(iT) \tag{H-106}$$

where, from Eq. (5-36), $\boldsymbol{\phi}^n(T) = [\boldsymbol{\phi}(T)]^n = \boldsymbol{\phi}(nT)$.

Eq. (H-106) is defined as the **discrete state-transition equation** of the discrete-data system. It is interesting to note that Eq. (H-106) is analogous to its continuous-data counterpart in Eq. (5-54). The state-transition equation of Eq. (H-97) describes the state of the system of Fig. H-10 for all values of t, whereas the discrete state-transition equation in Eq. (H-106) describes the states only at the sampling instants $t = nT$, $n = 0, 1, 2, \ldots$.

With nT considered as the initial time, where n is any positive integer, the state-transition equation is

$$\mathbf{x}[(n+N)T] = \boldsymbol{\phi}^N(T)\mathbf{x}(nT) + \sum_{i=0}^{N-1} \boldsymbol{\phi}^{N-i-1}(T)\boldsymbol{\theta}(T)h[(n+i)T] \tag{H-107}$$

where N is a positive integer.

The output of the system at the sampling instants is obtained by substituting $t = nT$ and Eq. (H-106) into Eq. (H-96), yielding

$$\begin{aligned} y(nT) &= \mathbf{C}\mathbf{x}(nT) + \mathbf{D}h(nT) \\ &= \mathbf{C}\boldsymbol{\phi}(nT)\mathbf{x}(0) + \mathbf{C}\sum_{i=0}^{n-1} \boldsymbol{\phi}[(n-i-1)T]\boldsymbol{\theta}(T)h(iT) + \mathbf{D}\mathbf{h}(nT) \end{aligned} \tag{H-108}$$

An important advantage of the state-variable method over the z-transform method is that it can be modified easily to describe the states and the output between

sampling instants. In Eq. (H-97), if we let $t = (n + \Delta)T$, where $0 < \Delta \leq 1$ and $t_0 = nT$, we get

$$\begin{aligned}\mathbf{x}[(n+\Delta)T] &= \boldsymbol{\phi}(\Delta T)\mathbf{x}(nT) + \int_{nT}^{(n+\Delta)T} \boldsymbol{\phi}[(n+\Delta)T - \boldsymbol{\tau}]\mathbf{B}d\tau h(nT) \\ &= \boldsymbol{\phi}(\Delta T)x(nT) + \boldsymbol{\theta}(\Delta T)h(nT)\end{aligned} \tag{H-109}$$

By varying the value of Δ between 0 and 1, the information on the state variables between the sampling instants is completely described by Eq. (H-109).

When a linear system has only discrete data through the system, its dynamics can be described by a set of discrete state equations:

$$\mathbf{x}[(k+1)T] = \mathbf{A}\mathbf{x}(kT) + \mathbf{B}\mathbf{r}(kT) \tag{H-110}$$

and output equations:

$$\mathbf{y}(kT) = \mathbf{C}\mathbf{x}(kT) + \mathbf{D}\mathbf{r}(kT) \tag{H-111}$$

where **A**, **B**, **C**, and **D** are coefficient matrices of the appropriate dimensions. Notice that Eq. (H-110) is basically of the same form as Eq. (H-100). The only difference in the two situations is the starting point of system representation. In the case of Eq. (H-100), the starting point is the continuous-data state equations of Eq. (H-95); $\phi(T)$ and $\theta(T)$ are determined from the **A** and **B** matrices and must satisfy the conditions and properties of the state transition matrix. In the case of Eq. (H-110), the equation itself represents an outright description of the discrete-data system, and there are no restrictions on the matrices **A** and **B**.

The solution of Eq. (H-110) follows directly from that of Eq. (H-100) and is

$$\mathbf{x}(nT) = \mathbf{A}^n\mathbf{x}(0) + \sum_{i=0}^{n-1}\mathbf{A}^{n-i-1}\mathbf{B}\mathbf{r}(iT) \tag{H-112}$$

where

$$\begin{aligned}\mathbf{A}^n &= \mathbf{AAAA}\cdots\mathbf{A} \\ &\quad |\leftarrow\ n\ \rightarrow|\end{aligned} \tag{H-113}$$

H-4-3 *z*-Transform Solution of Discrete State Equations

In Section H-1-5, we illustrated the solution of a simple discrete state equation by the *z*-transform method. In this section, the discrete state equations in vector-matrix form of an *n*th-order system are solved by *z*-transformation. Consider the discrete state equations

$$\mathbf{x}[(k+1)T] = \mathbf{A}\mathbf{x}(kT) + \mathbf{B}\mathbf{r}(kT) \tag{H-114}$$

Taking the *z*-transform on both sides of the last equation, we get

$$z\mathbf{X}(z) - z\mathbf{x}(0) = \mathbf{A}\mathbf{X}(z) + \mathbf{B}\mathbf{R}(z) \tag{H-115}$$

Solving for $\mathbf{X}(z)$ from Eq. (H-115), we get

$$\mathbf{X}(z) = (z\mathbf{I} - \mathbf{A})^{-1}z\mathbf{x}(0) + (z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{R}(z) \tag{H-116}$$

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Taking the inverse z-transform on both sides of Eq. (H-116), we have

$$\mathbf{x}(nT) = \mathcal{Z}^{-1}\left[(z\mathbf{I} - \mathbf{A})^{-1}z\right]\mathbf{x}(0) + \mathcal{Z}^{-1}\left[(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{BR}(z)\right] \tag{H-117}$$

To carry out the inverse z-transform operation of the last equation, we write the z-transform of $\mathbf{A}^n$ as

$$\mathcal{Z}\,(\mathbf{A}^n) = \sum_{\mathbf{n}=0}^{\infty}\mathbf{A}^n\mathbf{z}^{-n} = \mathbf{I} + \mathbf{A}z^{-1} + \mathbf{A}^2z^{-2} + \cdots \tag{H-118}$$

Premultiplying both sides of Eq. (H-118) by $\mathbf{A}z^{-1}$ and subtracting the result from the last equation, we get

$$\left(\mathbf{I} - \mathbf{A}z^{-1}\right)\mathcal{Z}\,(\mathbf{A}^n) = \mathbf{I} \tag{H-119}$$

Therefore, solving for $\mathcal{Z}\,(\mathrm{A}^n)$ from the last equation yields

$$\mathcal{Z}\,(\mathbf{A}^n) = \left(\mathbf{I} - \mathbf{A}z^{-1}\right)^{-1} = (z\mathbf{I} - \mathbf{A})^{-1}z \tag{H-120}$$

or

$$\mathbf{A}^n = \mathcal{Z}^{-1}\left[(z\mathbf{I} - \mathbf{A})^{-1}z\right] \tag{H-121}$$

Eq. (H-121) represents a way of finding $\mathbf{A}^n$ by using the z-transform method. Similarly, we can prove that

$$\mathcal{Z}^{-1}\left[(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{BR}(z)\right] = \sum_{i=0}^{n-1}\mathbf{A}^{n-i-1}\mathbf{Br}(iT) \tag{H-122}$$

Now we substitute Eqs. (H-121) and (H-122) into Eq. (H-117), $\mathbf{x}(nT)$ becomes

$$\mathbf{x}(nT) = \mathbf{A}^n\mathbf{x}(0) + \sum_{i=0}^{n-1}\mathbf{A}^{n-i-1}\mathbf{Br}(iT) \tag{H-123}$$

which is identical to Eq. (H-112).

H-4-4 Transfer-Function Matrix and the Characteristic Equation

Once a discrete-data system is modeled by the dynamic equations of Eqs. (H-110) and (H-111), the transfer-function relation of the system can be expressed in terms of the coefficient matrices. By setting the initial state $\mathbf{x}(0)$ to zero, Eq. (H-116) becomes

$$\mathbf{X}(z) = (z\mathbf{I} - \mathbf{A})^{-1}\mathbf{BR}(z) \tag{H-124}$$

Substituting Eq. (H-124) into the z-transformed version of Eq. (H-111), we have

$$\mathbf{Y}(z) = \left[\mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}\right]\mathbf{R}(z) = \mathbf{G}(z)\mathbf{R}(z) \tag{H-125}$$

where the transfer-function matrix of the system is defined as

$$\mathbf{G}(z) = \mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \tag{H-126}$$

or

$$\mathbf{G}(z) = \frac{\mathbf{C}[\mathrm{adj}(z\mathbf{I} - \mathbf{A})\mathbf{B} + |z\mathbf{I} - \mathbf{A}|\mathbf{D}]}{|z\mathbf{I} - \mathbf{A}|} \tag{H-127}$$

The characteristic equation of the system is defined as

$$|z\mathbf{I} - \mathbf{A}| = 0 \tag{H-128}$$

In general, a linear time-invariant discrete-data system with one input and one output can be described by the following difference equation with constant coefficients:

$$\begin{aligned} &y[(k+n)T] + a_{n-1}y[(k+n-1)T] + a_{n-2}y[(k+n-2)T] \\ &\quad + \cdots + a_1 y[((k+1))T] + a_0 y(kT) \\ &= b_m r[(k+m)T] + b_{m-1}r[(k+m-1)T] \\ &\quad + \cdots + b_1 r[((k+1))T] + b_0 r(kT) \end{aligned} \tag{H-129}$$

Taking the z-transform on both sides of Eq. (H-129) and setting zero initial conditions, the transfer function of the system is written

$$\frac{Y(z)}{R(s)} = \frac{b_m z^m + b_{m-1}z^{m-1} + \cdots + b_1 z + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0} \qquad n \geq m \tag{H-130}$$

The characteristic equation is obtained by equating the denominator polynomial of the transfer function to zero.

$$z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0 = 0 \tag{H-131}$$

▶ EXAMPLE H-4-1 Consider that a discrete-data system is described by the difference equation

$$y(k+2) + 5y(k+1) + 3y(k) = r(k+1) + 2r(k) \tag{H-132}$$

The transfer function of the system is simply

$$\frac{Y(z)}{R(z)} = \frac{z+2}{z^2 + 5z + 3} \tag{H-133}$$

The characteristic equation is

$$z^2 + 5z + 3 = 0 \tag{H-134}$$

The state variables of the system may be defined as

$$x_1(k) = y(k) \tag{H-135}$$

$$x_2(k) = x_1(k+1) - r(k) \tag{H-136}$$

Substituting the last two equations into Eq. (H-132) gives the two state equations as

$$x_1(k+1) = x_2(k) + r(k) \tag{H-137}$$

$$x_2(k+1) = -3x_1(k) - 5x_2(k) - 3r(k) \tag{H-138}$$

from which we have the matrices $\mathbf{A}$ and $\mathbf{B}$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -3 & -5 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \tag{H-139}$$

The same characteristic equation as in Eq. (H-134) is obtained by using $|z\mathbf{I} - \mathbf{A}| = 0$. ◀

H-4-5 State Diagrams of Discrete-Data Systems

When a discrete-data system is described by difference or discrete state equations, a **discrete state diagram** may be constructed for the system. Similar to the relations between the analog-computer block diagram and the state diagram for continuous-data systems, the elements of a discrete state diagram resemble the computing elements of a digital computer. Some of the operations of a digital computer are **multiplication by a constant**, **addition of several variables**, and **time delay** or **shifting**. The discrete state diagram can be used to determine the transfer functions as well as for digital implementation of the system. The mathematical description of these basic digital computations and their corresponding z-transform expressions are as follows:

1. *Multiplication by a constant:*

$$x_2(kT) = ax_1(kT) \tag{H-140}$$

$$X_2(z) = aX_1(z) \tag{H-141}$$

2. *Summing:*

$$x_2(kT) = x_1(kT) + x_3(kT) \tag{H-142}$$

$$X_2(z) = X_1(z) + X_3(z) \tag{H-143}$$

3. *Shifting or time delay:*

$$x_2(kT) = x_1[(k+1)T] \tag{H-144}$$

$$X_2(z) = zX_1(z) - zx_1(0) \tag{H-145}$$

or

$$X_1(z) = z^{-1}X_2(z) + x_1(0) \tag{H-146}$$

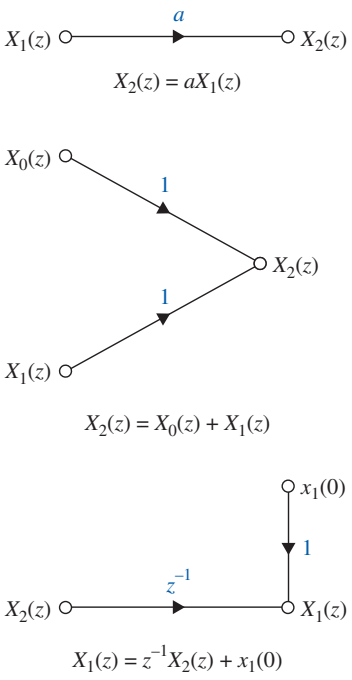


Figure H-11 Basic elements of a discrete state diagram.

The state diagram representation of these operations is illustrated in Fig. H-11. The initial time $t = 0$ in Eqs. (H-145) and (H-146) can be generalized to $t = t_0$. Then the equations represent the discrete-time state transition from $t \geq t_0$.

▶ **EXAMPLE H-4-2** Consider again the difference equation in Eq. (H-132), which is

$$y(k+2) + 5y(k+1) + 3y(k) = r(k+1) + 2r(k) \tag{H-147}$$

One way of constructing the discrete state diagram for the system is to use the state equations. In this case, the state equations are already defined in Eqs. (H-137) and (H-138). By using essentially the same principle as the state diagram for continuous-data systems, the state diagram for Eqs. (H-137) and (H-138) is shown in Fig. H-12. The time delay unit z^{-1} is used to relate $x_1(k+1)$ to $x_2(k)$. *The state variables are defined as the outputs of the delay units in the state diagram.*

The state-transition equations of the system can be obtained directly from the state diagram using the SFG gain formula. By referring to $X_1(z)$ and $X_2(z)$ as the output nodes and $x_1(0)$, $x_2(0)$, and $R(z)$ as input nodes in Fig. H-12, the state-transition equations are written as

$$\begin{bmatrix} X_1(z) \\ X_2(z) \end{bmatrix} = \frac{1}{\Delta}\begin{bmatrix} 1+5z^{-1} & z^{-1} \\ -3z^{-1} & 1 \end{bmatrix}\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \frac{1}{\Delta}\begin{bmatrix} z^{-1}(1+5z^{-1}) - 3z^{-2} \\ -3z^{-1} - 3z^{-2} \end{bmatrix} R(z) \tag{H-148}$$

where

$$\Delta = 1 + 5z^{-1} + 3z^{-2} \tag{H-149}$$

The same transfer function between $R(z)$ and $Y(z)$ as in Eq. (H-133) can be obtained directly from the state diagram in Fig. H-13 by applying the SFG gain formula between these two nodes.

As an alternative, the discrete state diagram can be drawn directly from the difference equation via the transfer function, using the decomposition schemes (Fig. H-13). The decomposition of a discrete-data transfer function follows basically the same procedure as that of an analog transfer function covered in Section 9-11, so the details are not repeated here.

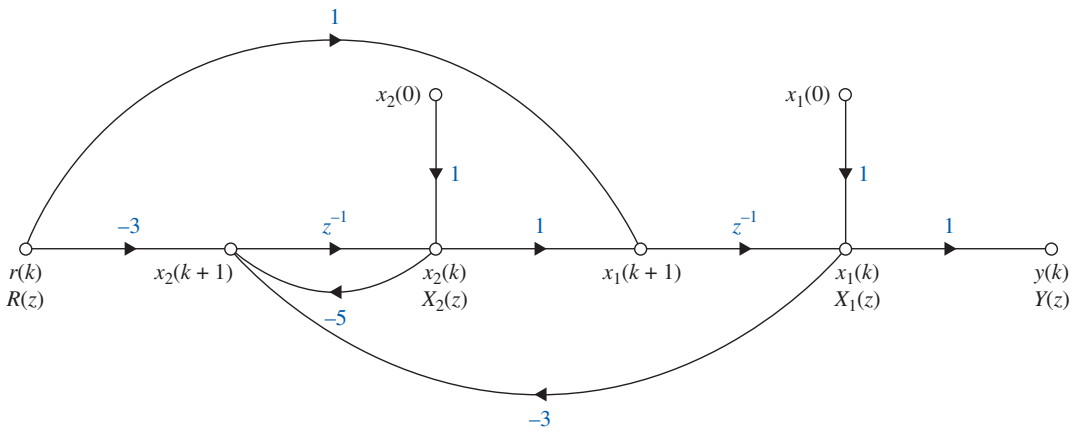
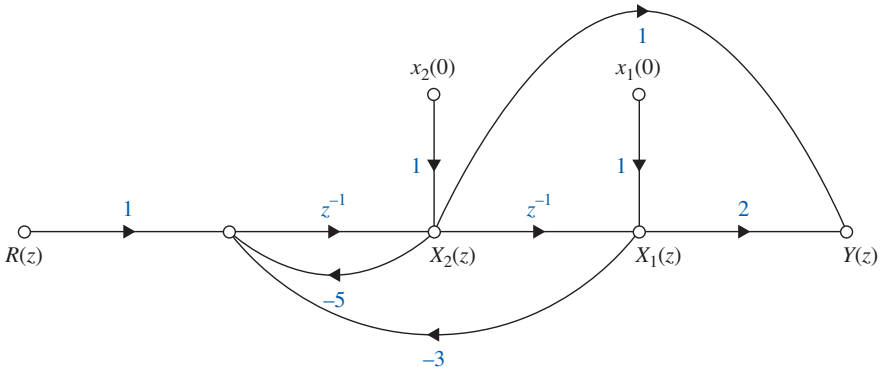


Figure H-12 Discrete state diagram of the system described by the difference equation of Eq. (H-132) or by the state equations of Eqs. (H-137) and (H-138). ◀

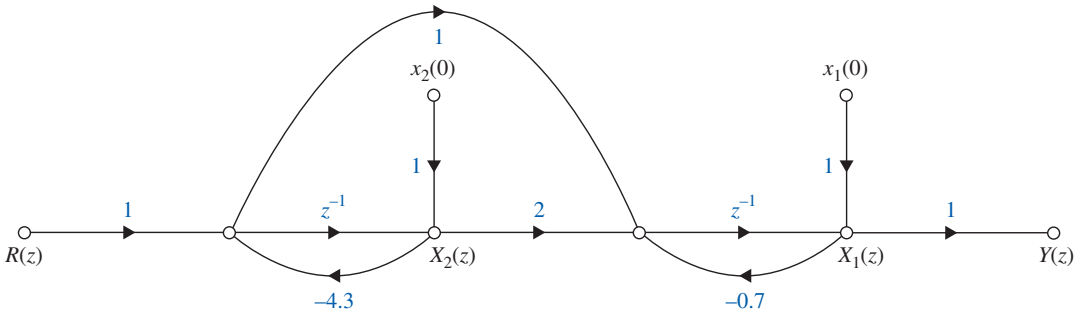
H-4-6 State Diagrams for Sampled-Data Systems

When a discrete-data system has continuous-data as well as discrete-data elements, with the two types of elements separated by sample-and-hold devices, a state diagram model for the sample-and-hold (zero-order-hold) must be established.

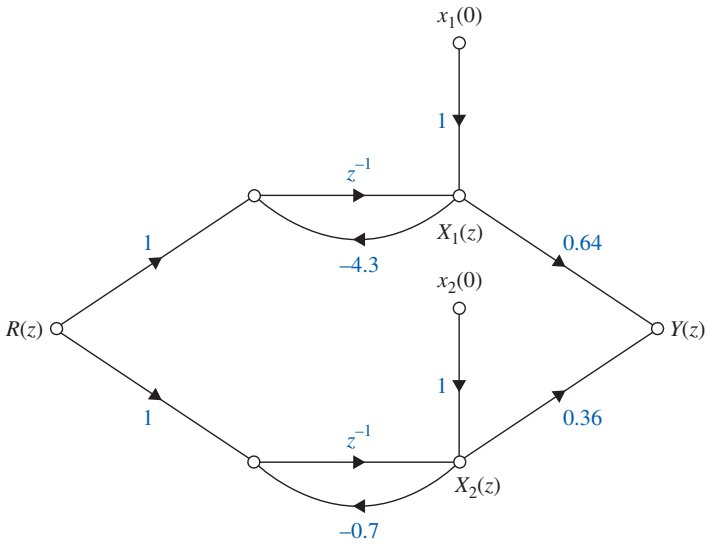
Consider that the input of the ZOH is denoted by $e^*(t)$, which is a train of impulses, and the output by $h(t)$. Because the ZOH simply holds the strength of the input impulse at the



(a) Direct decomposition



(b) Cascade decomposition



(c) Parallel decomposition

Figure H-13 State diagrams of the transfer function $\dfrac{Y(z)}{R(z)} = \dfrac{(z+2)}{(z^2+5z+3)}$.

sampling instant until the next input comes, the signal $h(t)$ is a sequence of steps. The input–output relation in the Laplace domain is

$$H(s) = \frac{1 - e^{-Ts}}{s} E^*(s) \tag{H-150}$$

In the time domain, the relation is simply

$$h(t) = e\left(kT^{+}\right) \tag{H-151}$$

for $kT \leq t < (k+1)T$.

In the state-diagram notation, we need the relation between $H(s)$ and $e(kT^{+})$. For this purpose, we take the Laplace transform on both sides of Eq. (H-151) to give

$$H(s) = \frac{e(kT^{+})}{s} \tag{H-152}$$

for $kT \leq t < (k+1)T$. The state-diagram representation of the zero-order-hold is shown in Fig. H-14.

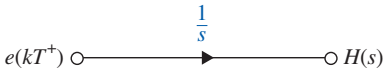


Figure H-14 State-diagram representation of the zero-order-hold (ZOH).

▶ EXAMPLE H-4-3 As an illustrative example on how the state diagram of a sampled-data system is constructed, let us consider the system in Fig. H-15. We shall demonstrate the various ways of modeling the input–output relations of the system. First, the Laplace transform of the output of the system is written in terms of the input to the ZOH.

$$Y(s) = \frac{1 - e^{-Ts}}{s} \frac{1}{s+1} E^{*}(s) \tag{H-153}$$

Taking the z-transform on both sides of Eq. (H-153), we get

$$Y(z) = \frac{1 - e^{-T}}{z - e^{-T}} E(z) \tag{H-154}$$

Fig. H-16 shows the state diagram for Eq. (H-154). The discrete dynamic equations of the system are written directly from the state diagram.

$$x_1[(k+1)T] = -e^{-T} x_1(kT) + \left(1 - e^{-T}\right) e\left(kT^{+}\right) \tag{H-155}$$

$$y(kT) = x_1(kT) \tag{H-156}$$

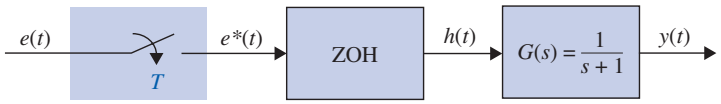


Figure H-15 Sampled-data system.

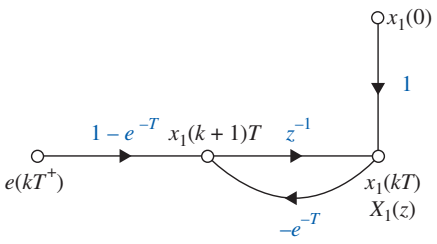


Figure H-16 Discrete state diagram of the system in Fig. H-15.

◀

▶ H-5 STABILITY OF DISCRETE-DATA SYSTEMS

The definitions of BIBO and zero-input stability can be readily extended to linear time-invariant SISO discrete-data control systems.

H-5-1 BIBO Stability

Let $u(kT)$, $y(kT)$, and $g(kT)$ be the input, output, and impulse sequence of a linear time-invariant SISO discrete-data system, respectively. With *zero initial conditions, the system is said to be BIBO stable, or simply stable, if its output sequence* $y(kT)$ *is bounded to a bounded input* $u(kT)$. As with the treatment in Section 2-8, we can show that, for the system to be BIBO stable, the following condition must be met:

$$\sum_{k=0}^{\infty} |g(kT)| < \infty \tag{H-157}$$

H-5-2 Zero-Input Stability

For zero-input stability, the output sequence of the system must satisfy the following conditions:

$$|y(kT)| \leq M < \infty \tag{H-158}$$

$$\lim_{k \to \infty} |y(kT)| = 0 \tag{H-159}$$

Thus, zero-input stability can also be referred to as **asymptotic stability**. We can show that *both the BIBO stability and the zero-input stability of discrete-data systems require that the roots of the characteristic equation lie inside the unit circle* $|z| = 1$ *in the z-plane.*

This is not surprising, because the $j\omega$-axis of the s-plane is mapped onto the unit circle in the z-plane. The regions of stability and instability for discrete-data systems in the z-plane are shown in Fig. H-17. Let the characteristic equation roots of a linear discrete-data time-invariant SISO system be z_i, $i = 1, 2, \ldots, n$. The possible stability conditions of the system are summarized in Table H-2 with respect to the roots of the characteristic equation.

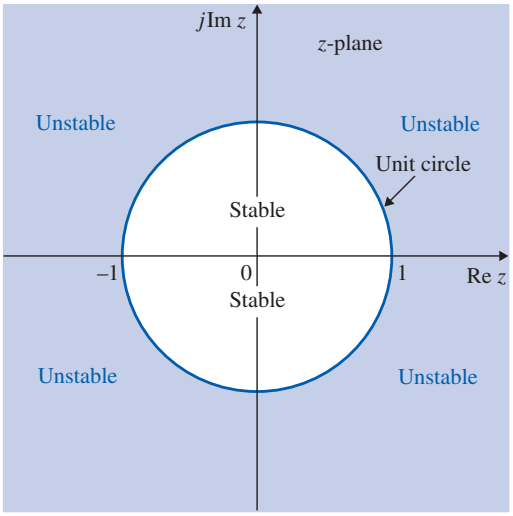


Figure H-17 Stable and unstable regions for discrete-data systems in the z-plane.

TABLE H-2 Stability Conditions of Linear Time-Invariant Discrete-Data SISO Systems

Stability condition	Root values
Asymptotically stable or simply stable	$\|z_i\| < 1$ for all i, $i = 1, 2, \ldots, n$ (all roots inside the unit circle)
Marginally stable or marginally unstable	$\|z_i\| = 1$ for any i for simple roots, and no $\|z_i\| > 1$ for $i = 1, 2, \ldots, n$ (at least one simple root, no multiple-order roots on the unit circle, and no roots outside the unit circle)
Unstable	$\|z_i\| > 1$ for any i, or $\|z_i\| = 1$ for any multiple-order root. $i = 1, 2, \ldots, n$ (at least one simple root outside the unit circle and at least one multiple-order root on the unit circle)

The following example illustrates the relationship between the closed-loop transfer-function poles, which are the characteristic equation roots, and the stability condition of the system.

▶ **EXAMPLE H-5-1**

$$M(z) = \frac{5z}{(z-0.2)(z-0.8)} \quad \text{Stable system}$$

$$M(z) = \frac{5z}{(z+1.2)(z-0.8)} \quad \text{Unstable system due to the pole at } z = -1.2$$

$$M(z) = \frac{5(z+1)}{z(z-1)(z-0.8)} \quad \text{Marginally stable due to } z = 1$$

$$M(z) = \frac{5(z+1.2)}{z^2(z+1)^2(z+0.1)} \quad \text{Unstable due to second-order pole at } z = -1$$

◀

H-5-3 Stability Tests of Discrete-Data Systems

We pointed out earlier in Section H-5 that the stability test of a linear discrete-data system is essentially a problem of investigating whether all the roots of the characteristic equation are inside the unit circle $|z| = 1$ in the z-plane. The Nyquist criterion, root-locus diagram, and Bode diagram, originally devised for continuous-data systems, can all be extended to the stability studies of discrete-data systems. One exception is the Routh-Hurwitz criterion, which in its original form is restricted to only the imaginary axis of the s-plane as the stability boundary and thus can be applied only to continuous-data systems.

Bilinear Transformation Method [1]

We can still apply the Routh-Hurwitz criterion to discrete-data systems if we can find a transformation that transforms the unit circle in the z-plane onto the imaginary axis of another complex plane. We cannot use the z-transform relation $z = \exp(Ts)$ or $s = (\ln z)/T$, since it would transform an algebraic equation in z into a nonalgebraic equation in s, and the Routh test still cannot be applied. However, there are many bilinear transformations of the form of

$$z = \frac{ar + b}{cr + d} \tag{H-160}$$

where a, b, c, and d are real constants, and r is a complex variable that will transform circles in the z-plane onto straight lines in the r-plane. One such transformation that transforms the interior of the unit circle of the z-plane onto the left half of the r-plane is

$$z = \frac{1 + r}{1 - r} \tag{H-161}$$

which is referred to as the ***r*-transformation**. Once the characteristic equation in z is transformed into the r domain using Eq. (H-161), the Routh-Hurwitz criterion can again be applied to the equation in r.

The r-transformation given in Eq. (H-161) is probably the simplest form that can be used for manual transformation of an equation $F(z)$ to an equation in r. Another transformation that is often used in discrete-data control-system design in the frequency domain is

$$z = \frac{(2/T) + w}{(2/T) - w} \tag{H-162}$$

or

$$w = \frac{2}{T}\frac{z - 1}{z + 1} \tag{H-163}$$

which is called the ***w*-transformation**. Note that the w-transformation becomes the r-transformation when $T = 2$. The advantage of the w-transformation over the r-transformation is that the imaginary axis of the w-plane resembles that of the s-plane. To show this, we substitute

$$z = e^{j\omega T} = \cos \omega T + j \sin \omega T \tag{H-164}$$

into Eq. (H-163), and we get

$$w = \frac{2}{T}\frac{\cos \omega T + j \sin \omega T - 1}{\cos \omega T + j \sin \omega T + 1} \tag{H-165}$$

Rationalizing the last equation and simplifying, we get

$$w = j\omega_w = j\frac{2}{T}\tan\frac{\omega T}{2} \tag{H-166}$$

Thus, the unit circle in the z-plane is mapped onto the imaginary axis $w = j\omega_w$ in the w-plane. The relationship between ω_w and ω, the real frequency, is

$$\omega_w = \frac{2}{\mathrm{T}}\tan\frac{\omega T}{2} = \frac{\omega_s}{\pi}\tan\frac{\pi\omega}{\omega_s} \tag{H-167}$$

where ω_s is the sampling frequency in rad/sec. The correlation between ω and ω_w is that they both go to 0 and ∞ at the same time. For the Routh-Hurwitz criterion, of course, the w-transformation is more difficult to use, especially since the sampling period T appears in Eq. (H-163). However, if computer programs are available for the transformations, the difference is insignificant.

The following examples illustrate the application of the r-transformation to a characteristic equation in z so that the equation can be tested by the Routh-Hurwitz criterion in the r-domain.

▶ **EXAMPLE H-5-2** Consider that the characteristic equation of a discrete-data control system is

$$z^3 + 5.94z^2 + 7.7z - 0.368 = 0 \tag{H-168}$$

Substituting Eq. (H-161) into the last equation and simplifying, we get

$$3.128r^3 - 11.74r^2 + 2.344r + 14.27 = 0 \tag{H-169}$$

Routh's tabulation of the last equation is

	r^3	3.128	2.344
Sign change			
	r^2	−11.74	14.27
Sign change			
	r^1	6.146	0
	r^0	14.27	

Because there are two sign changes in the first column of the tabulation, Eq. (H-169) has two roots in the right half of the r-plane. This corresponds to Eq. (H-168) having two roots outside the unit circle in the z-plane. This result can be checked by solving the two equations in z and r. For Eq. (H-168), the roots are $z = -2.0$, $z = -3.984$, and $z = 0.0461$. The three corresponding roots in the r-plane are $r = 3.0$, $r = 1.67$, and $r = -0.9117$, respectively. ◀

▶ **EXAMPLE H-5-3** Let us consider a design problem using the bilinear transformation and Routh-Hurwitz criterion. The characteristic equation of a linear discrete-data control system is given as

$$F(z) = z^3 + z^2 + z + K = 0 \tag{H-170}$$

where K is a real constant. The problem is to find the range of values of K so that the system is stable. We first transform $F(z)$ into an equation in r using the bilinear transformation of Eq. (H-161). The result is

$$(1 - K)r^3 + (1 + 3K)r^2 + 3(1 - K)r + 3 + K = 0 \tag{H-171}$$

Routh's tabulation of the last equation is

r^3	$1 - K$	$3(1 - K)$
r^2	$1 + 3K$	$3 + K$
r^1	$\dfrac{8K(1 - K)}{1 + 3K}$	0
r^0	$3 + K$	

For a stable system, the numbers in the first column of the tabulation must be of the same sign. We can show that these numbers cannot be all negative, since the conditions contradict each other. Next, for all the numbers to be positive, we have the following conditions:

$$1 - K > 0 \qquad 1 + 3K > 0 > 0 \qquad 3 + K > 0$$

which lead to the condition for stability:

$$0 < K < 1 \tag{H-172}$$

◀

Direct Stability Tests

There are stability tests that can be applied directly to the characteristic equation in z with reference to the unit circle in the z-plane. One of the first methods that gives the necessary and sufficient conditions for the characteristic equation roots to lie inside the unit circle is the Schur-Cohn criterion [2]. A simpler tabulation method was devised by Jury and Blanchard [3,4] and is called **Jury's stability criterion** [6]. R. H. Raible [5] devised an alternate tabular form of Jury's stability test. Unfortunately, these analytical tests all become very tedious for equations higher than the second order, especially when the equation has unknown parameter(s) in it. Then, there is no reason to use any of these tests if all the coefficients of the equation are known, since we can always use a root-finding program on a computer. Weighing all the pros and cons, this author believes that when the characteristic equation has at least one unknown parameter, the bilinear transformation method is still the best manual method for determining stability of linear discrete-data systems. However, it is useful to introduce the necessary condition of stability that can be checked by inspection.

Consider that the characteristic equation of a linear time-invariant discrete-data system is

$$F(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0 = 0 \tag{H-173}$$

where all the coefficients are real. Among all the conditions provided in Jury's test, the following *necessary* conditions must be satisfied for $F(z)$ to have no roots on or outside the unit circle.

$$\begin{aligned} &F(1) > 0 \\ &F(-1) > 0 \qquad \text{if } n = \text{even integer} \\ &F(-1) < 0 \qquad \text{if } n = \text{odd integer} \\ &|a_0| < a_n \end{aligned} \tag{H-174}$$

If an equation of the form of Eq. (H-173) violates any one of these conditions, then not all of the roots are inside the unit circle, and the system would not be stable. Apparently, these necessary conditions can be checked easily by inspection.

▶ EXAMPLE H-5-4 Consider the equation

$$F(z) = z^3 + z^2 + 0.5z + 0.25 = 0 \tag{H-175}$$

Applying the conditions in Eq. (H-174), we have

$$F(1) = 2.75 > 0 \qquad \text{and} \qquad F(-1) = -0.25 < 0 \qquad \text{for } n = 3, \text{ which is odd}$$
$$|a_0| = 0.25 < a_3 = 1$$

Thus, the conditions in Eq. (H-174) are all satisfied, but nothing can be said about the stability of the system. ◀

▶ EXAMPLE H-5-5 Consider the equation

$$F(z) = z^3 + z^2 + 0.5z + 1.25 = 0 \tag{H-176}$$

The conditions in Eq. (H-176) are

$$F(-1) = 0.75 > 0 \qquad \text{for } n = 3, \text{ which is odd}$$

$$|a_0| = 1.25, \text{ which is not less then } a_3, \text{ which equals 1.}$$

Because, for odd n, $F(-1)$ must be negative, the equation in Eq. (H-176) has at least one root outside the unit circle. The condition on the absolute value of a_0 is also not met. ◀

Second-Order Systems

The conditions in Eq. (H-174) become *necessary and sufficient* when the system is of the second order. That is, the necessary and sufficient conditions for the second-order equation

$$F(z) = a_2 z^2 + a_1 z + a_0 = 0 \tag{H-177}$$

to have no roots on or outside the unit circle are

$$\begin{aligned} &F(1) > 0 \\ &F(-1) > 0 \\ &|a_0| < a_2 \end{aligned} \tag{H-178}$$

▶ **EXAMPLE H-5-6** Consider the equation

$$F(z) = z^2 + z + 0.25 = 0 \tag{H-179}$$

Applying the conditions in Eq. (H-178), we have

$$F(1) = 2.25 > 0 \qquad F(-1) = 0.25 > 0 \qquad \text{for } n = 2, \text{ which is even}$$
$$|a_0| = 0.25 < a_2 = 1$$

Thus, the conditions in Eq. (H-178) are all satisfied. The two roots in Eq. (H-179) are all inside the unit circle, and the system is stable. ◀

▶ H-6 TIME-DOMAIN PROPERTIES OF DISCRETE-DATA SYSTEMS

H-6-1 Time Response of Discrete-Data Control Systems

To carry out the design of discrete-data control systems in the time domain or the z-domain, we must first study the time- and z-domain properties of these systems. We learned from the previous sections that the output responses of most discrete-data control systems are functions of the continuous-time variable t. Thus, the time-domain specifications such as the maximum overshoot, rise time, damping ratio, and so forth can still be applied to discrete-data systems. The only difference is that, to make use of the analytical tools such as z-transforms, the continuous data found in a discrete-data system are sampled so that the independent time variable is kT, where T is the sampling period in seconds. Also, instead of working in the s-plane, the transient performance of a discrete-data system is characterized by poles and zeros of the transfer function in the z-plane.

The objectives of the following sections are as follows:

1. To present methods of finding the discretized time responses of discrete-data control systems
2. To describe the important characteristics of the discretized time response $y(kT)$
3. To establish the significance of pole and zero locations in the z-plane
4. To provide comparison between time responses of continuous-data and discrete-data control systems

Let us refer to the block diagram of the discrete-data control system shown in Fig. H-18. The transfer function of the system is

$$\frac{Y(z)}{R(z)} = \frac{G(z)}{1 + GH(z)} \tag{H-180}$$

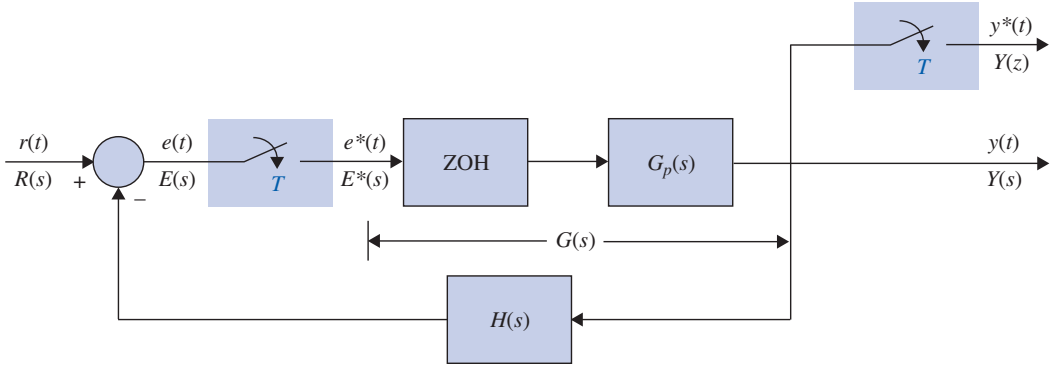


Figure H-18 Block diagram of a discrete-data control system.

where $GH(z)$ denotes the z-transform of $G(s)H(s)$. Once the input $R(z)$ is given, the output sequence $y(kT)$ can be determined using one of the following two methods:

1. Take the inverse z-transform of $Y(z)$ using the z-transform table.
2. Expand $Y(z)$ into a power series of z^{-k}.

The z-transform of the output is defined as

$$Y(z) = \sum_{k=0}^{\infty} y(kT)z^{-k} \tag{H-181}$$

The discrete-time response $y(kT)$ can be determined by referring to the coefficient of z^{-k} for $k = 0, 1, 2, \ldots$. Remember that $y(kT)$, $k = 0, 1, 2, \ldots$ contains only the sampled information on $y(t)$ at the sampling instants. If the sampling period is large relative to the most significant time constant of the system, $y(kT)$ may not be an accurate representation of $y(t)$.

▶ EXAMPLE H-6-1

Consider that the position-control system described in Section 5-8 has discrete data in the forward path, so that the system is now described by the block diagram of Fig. H-18. For $K = 14.5$, the transfer function of the controlled process shown in Eq. (5-132) is

$$G_p(s) = \frac{65,250}{s(s + 361.2)} \tag{H-182}$$

The forward-path transfer function of the discrete-data system is

$$G_{h0}G_p(z) = \mathcal{Z}\left[G_{h0}G_p(s)\right] = \left(1 - z^{-1}\right)\mathcal{Z}\left[\frac{G_p(s)}{s}\right] \tag{H-183}$$

For a sampling period of $T = 0.001$ second, the z-transfer function in Eq. (H-183) is evaluated as

$$G_{h0}G_p(z) = \frac{0.029z + 0.0257}{z^2 - 1.697z + 0.697} \tag{H-184}$$

The closed-loop transfer function of the system is

$$\frac{Y(z)}{R(z)} = \frac{G_{h0}G_p(z)}{1 + G_{h0}G_p(z)} = \frac{0.029z + 0.0257}{z^2 - 1.668z + 0.7226} \tag{H-185}$$

where $R(z)$ and $Y(z)$ represent the z-transforms of the input and the output, respectively. For a unit-step input, $R(z) = z/(z-1)$. The output transform $Y(z)$ becomes

$$Y(z) = \frac{z(0.029z + 0.0257)}{(z-1)(z^2 - 1.668z + 0.7226)} \tag{H-186}$$

The output sequence $y(kT)$ can be determined by dividing the numerator polynomial of $Y(z)$ by its denominator polynomial to yield a power series in z^{-1}. Fig. H-19 shows the plot of $y(kT)$ (dots) versus kT, when $T = 0.001$ second. For comparison, the unit-step response of the continuous-data system in Section 5-8 with $K = 14.5$ is shown in the same figure. As seen in Fig. H-19, when the sampling period is small, the output responses of the discrete-data and the continuous-data systems are very similar. The maximum value of $y(kT)$ is 1.0731, or a 7.31% maximum overshoot, as against the 4.3% maximum overshoot for the continuous-data system.

When the sampling period is increased to 0.01 second, the forward-path transfer function of the discrete-data system is

$$G_{h0}G_p(z) = \frac{1.3198z + 0.4379}{z^2 - 1.027z + 0.027} \tag{H-187}$$

and the closed-loop transfer function is

$$\frac{Y(z)}{R(z)} = \frac{1.3198z + 0.4379}{z^2 + 0.2929z + 0.4649} \tag{H-188}$$

The output sequence $y(kT)$ with $T = 0.01$ second is shown in Fig. H-19 with $k = 0, 1, 2, 3, 4$, and 5. The true continuous-time output of the discrete-data system is shown as the dotted curve. Notice that the maximum value of $y(kT)$ is 1.3712, but the true maximum overshoot is considerably higher than that. Thus, the larger sampling period only makes the system less stable, but the sampled output no longer gives an accurate measure of the true output.

When the sampling period is increased to 0.01658 second, the characteristic equation of the discrete-data system is

$$z^2 + 1.4938z + 0.4939 = 0 \tag{H-189}$$

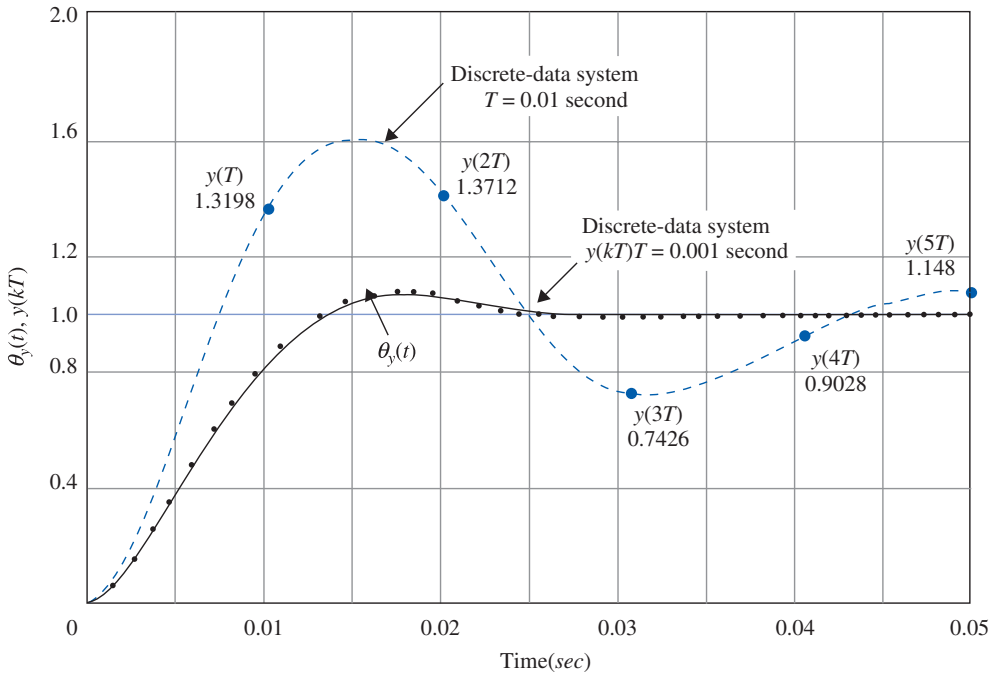


Figure H-19 Comparison of unit-step responses of discrete-data and continuous-data systems.

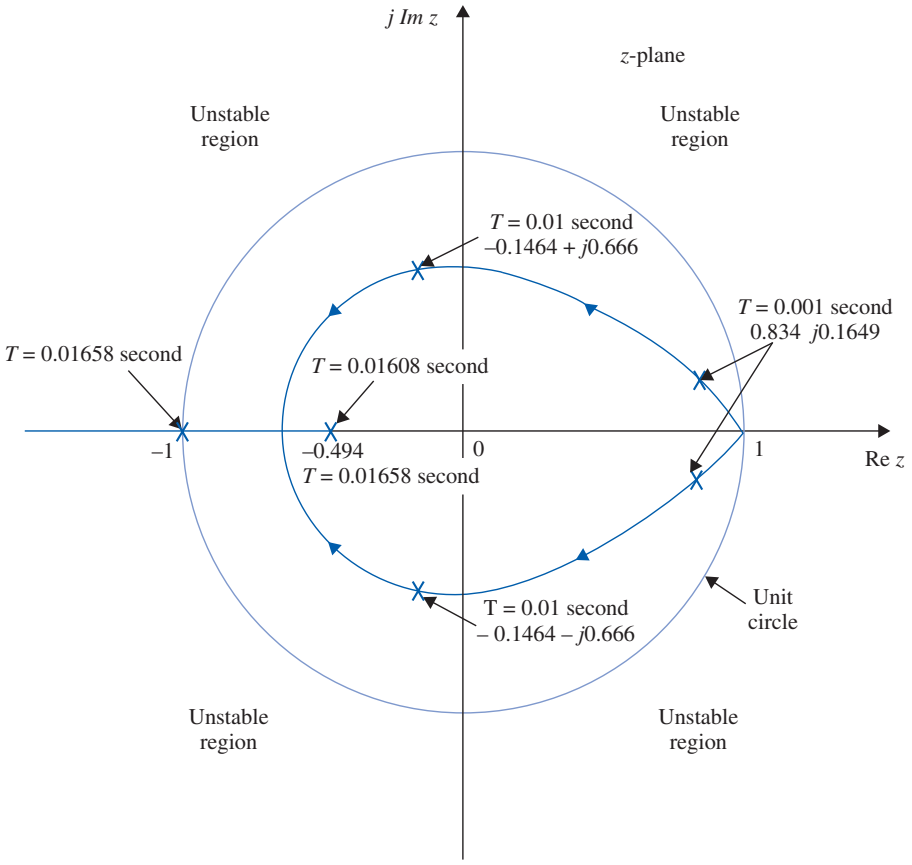


Figure H-20 Trajectories of roots of a second-order discrete-data control system as the sampling period T varies.

which has roots at $z = -0.494$ and $z = -1.000$. The root at -1.000 causes the step response of the system to oscillate with a constant amplitude, and the system is marginally stable. Thus, for all sampling periods greater than 0.01658 second, the discrete-data system will be unstable. From Section 5-8, we learned that the second-order continuous-data system is always stable for finite positive values of K. For the discrete-data system, the sample-and-hold has the effect of making the system less stable, and if the value of T is too large, the second-order system can become unstable. Fig. H-20 shows the trajectories of the two characteristic-equation roots of the discrete-data system as the sampling period T varies. Notice that, when the sampling period is very small, the two characteristic-equation roots are very close to the $z = 1$ point and are complex. When $T = 0.01608$ second, the two roots become equal and real and are negative. Unlike the continuous-data system, the case of two identical roots on the *negative* real axis in the z-plane *does not* correspond to critical damping. For discrete-data systems, when one or more characteristic-equation roots lie on the negative real axis of the z-plane, the system response will oscillate with positive and negative peaks. Fig. H-21 shows the oscillatory response of $y(kT)$ when $T = 0.01658$ second, which is the critical value for stability. Beyond this value of T, one root will move outside the unit circle, and the system will become unstable. ◀

H-6-2 Mapping between *s*-Plane and *z*-Plane Trajectories

For analysis and design purposes, it is important to study the relation between the location of the characteristic-equation roots in the z-plane and the time response of the discrete-data system. In Section H-2, the periodic property of the Laplace transform of the sampled

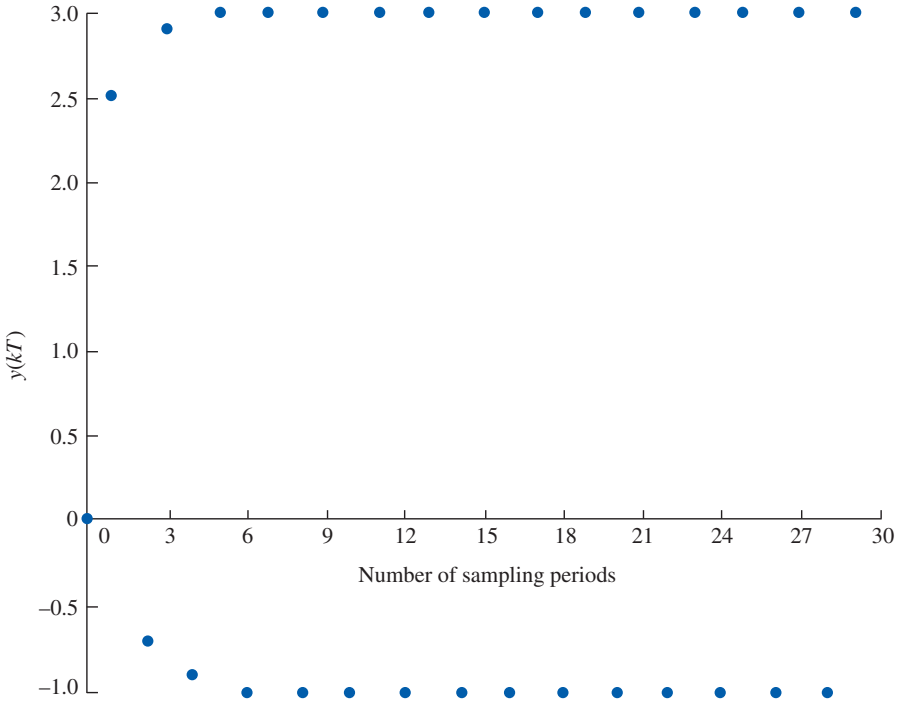


Figure H-21 Oscillatory response of a discrete-data system with a sampling period $T = 0.01658$ second.

signal $R^*(s)$ is established by Eq. (H-62); that is, $R^*(s + jm\omega_s) = R^*(s)$, where m is an integer. In other words, given any point s_1 in the s-plane, the function $R^*(s)$ has the same value at all periodic points $s = s_1 + jm\omega_s$. Thus, the s-plane is divided into an infinite number of periodic strips, as shown in Fig. H-22(a). The strip between $\omega = \omega_s/2$ is called the **primary strip**, and all others at higher frequencies are called the **complementary strips**. Fig. H-22(b) shows the mapping of the periodic strips from the s-plane to the z-plane, and the details are explained as follows.

1. The $j\omega$-axis in the s-plane is mapped onto the unit circle $|z| = 1$ in the z-plane.
2. The boundaries of the period strips, $s = jm\omega_s/2$, $m = \pm 1, \pm 3, \pm 5, \ldots$, are mapped onto the negative real axis of the z-plane. The portion inside the unit circle corresponds to $\sigma < 0$, and the portion outside the unit circle corresponds to $\sigma > 0$.
3. The center lines of the periodic strips, $s = jm\omega_s$, $m = 0, \pm 2, \pm 4, \ldots$, are mapped onto the positive real axis of the z-plane. The portion inside the unit circle corresponds to $\sigma < 0$, and the portion outside the unit circle corresponds to $\sigma > 0$.
4. Regions shown in the periodic strips in the left-half s-plane are mapped onto the interior of the unit circle in the z-plane.
5. The point $z = 1$ in the z-plane corresponds to the origin, $s = 0$, in the x-plane.
6. The origin, $z = 0$, in the z-plane corresponds to $s = -\infty$ in the s-plane.

In the time-domain analysis of continuous-data systems, we devise the damping factor α, the damping ratio ζ, and the natural undamped frequency ω_n to characterize the system dynamics. The same parameters can be defined for discrete-data systems with respect to the characteristic-equation roots in the z-plane. The loci of the constant-α, constant-ζ, constant-ω, and constant-ω_n in the z-plane are described in the following sections.

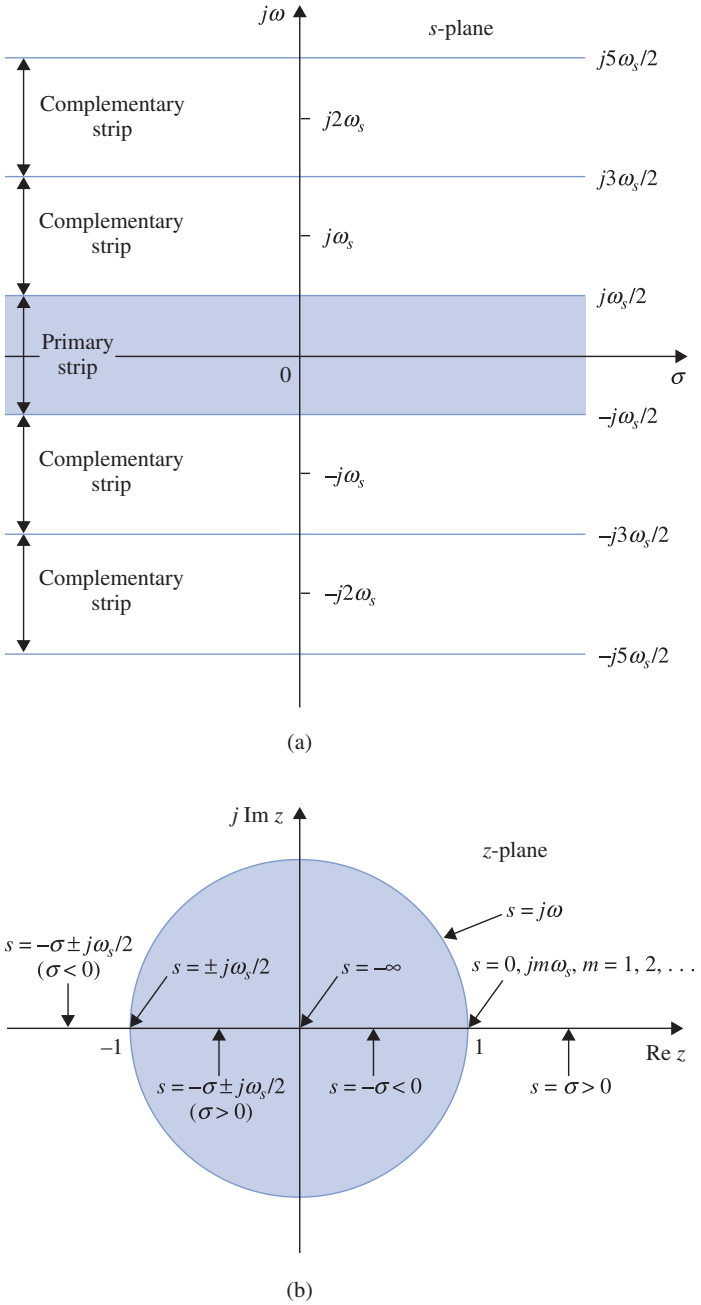


Figure H-22 Periodic strips in the s-plane and the corresponding points and lines between the s-plane and the z-plane.

Constant-Damping Loci: For a constant-damping factor $\sigma = \alpha$ in the s-plane, the corresponding trajectory in the z-plane is described by

$$z = e^{\alpha T} \tag{H-190}$$

which is a circle centered at the origin with a radius of $e^{\alpha T}$, as shown in Fig. H-23.

Constant-Frequency Loci: The constant-frequency $\omega = \omega_1$ locus in the s-plane is a horizontal line parallel to the σ-axis. The corresponding z-plane locus is a straight line emanating from the origin at an angle of $\theta = \omega_1 T$ radians, measured from the real axis, as shown in Fig. H-24.

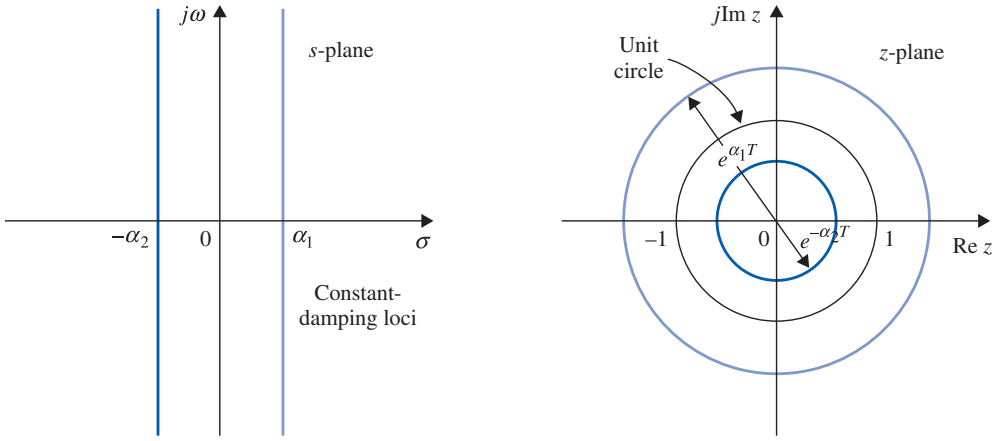


Figure H-23 Constant-damping loci in the s-plane and the z-plane.

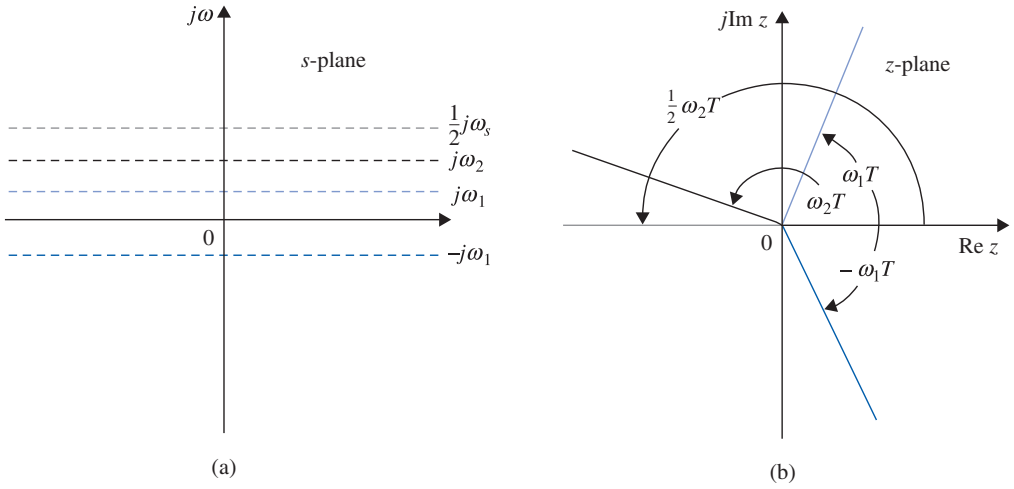


Figure H-24 Constant-frequency loci in the s-plane and the z-plane.

Constant Natural-Undamped Frequency Loci: The constant-ω_n loci in the s-plane are concentric circles with the center at the origin, and the radius is ω_n. The corresponding constant-ω_n loci in the z-plane are shown in Fig. H-25 for $\omega_n = \omega_s/16$ to $\omega_s/2$. Only the loci inside the unit circle are shown.

Constant-Damping Ratio Loci: For a constant-damping ratio ζ, the s-plane loci are described by

$$s = -\omega \tan\beta + j\omega \tag{H-191}$$

The constant-ζ loci in the z-plane are described by

$$z = e^{Ts} = e^{-2\pi(\tan\beta)/\omega_s \angle 2\pi\omega/\omega_s} \tag{H-192}$$

where

$$\beta = \sin^{-1}\zeta = \text{constant} \tag{H-193}$$

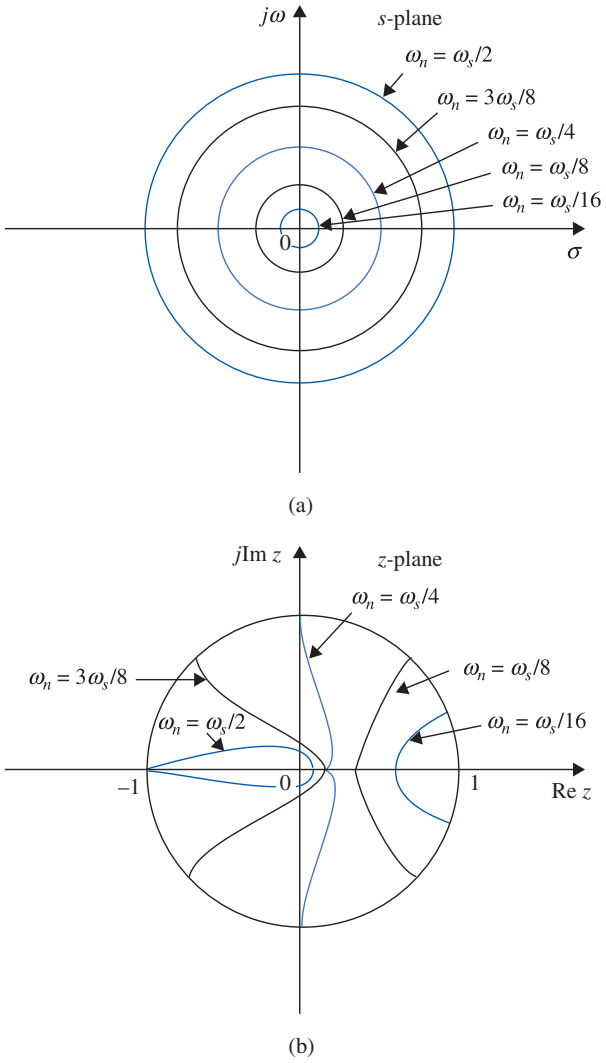


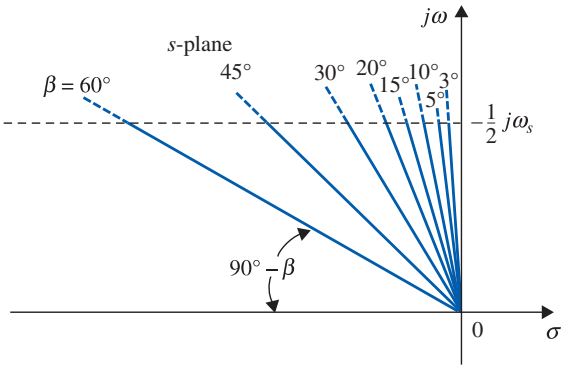
Figure H-25 Constant-natural-undamped frequency loci in the s-plane and the z-plane.

For a given value of β, the constant-ζ locus in the z-plane, described by Eq. (H-193), is a logarithmic spiral for $0° < \beta < 90°$. Fig. H-26 shows several typical constant-ζ loci in the top half of the z-plane.

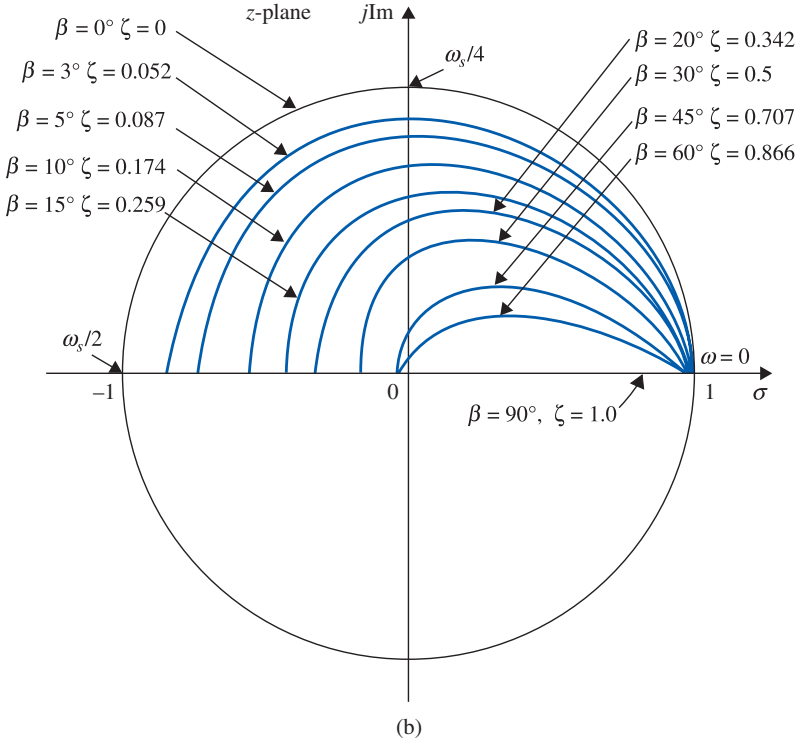
H-6-3 Relation between Characteristic-Equation Roots and Transient Response

Based on the discussions given in the last section, we can establish the basic relation between the characteristic-equation roots and the transient response of a discrete-data system, keeping in mind that, in general, the zeros of the closed-loop transfer function will also play an important role on the response, but not on the stability, of the system.

Roots on the Positive Real Axis in the z-Plane: Roots on the positive real axis inside the unit circle of the z-plane give rise to responses that decay exponentially with an increase of kT. Typical responses relative to the root locations are shown in Figs. H-27 and H-28. The roots closer to the unit circle will decay slower. When the root is at $z = 1$, the response has a



(a)



(b)

Figure H-26 Constant-damping-ratio loci in the s-plane and the z-plane.

constant amplitude. Roots outside the unit circle correspond to unstable systems, and the responses will increase with kT.

Roots on the Negative Real Axis in the z-Plane: The negative real axis of the z-plane corresponds to the boundaries of the periodic strips in the s-plane. For example, when $s = -\sigma_1 \pm j\omega_s/2$, the complex-conjugate points are on the boundaries of the primary strip in the s-plane. The corresponding z-plane points are

$$z = e^{-\sigma_1 T} e^{\pm j\omega_s T/2} = -e^{-\sigma_1 T} \tag{H-194}$$

which are on the negative real axis of the z-plane. For the frequency of $\omega_s/2$, the output sequence will have exactly one sample in each one-half period of the envelope.

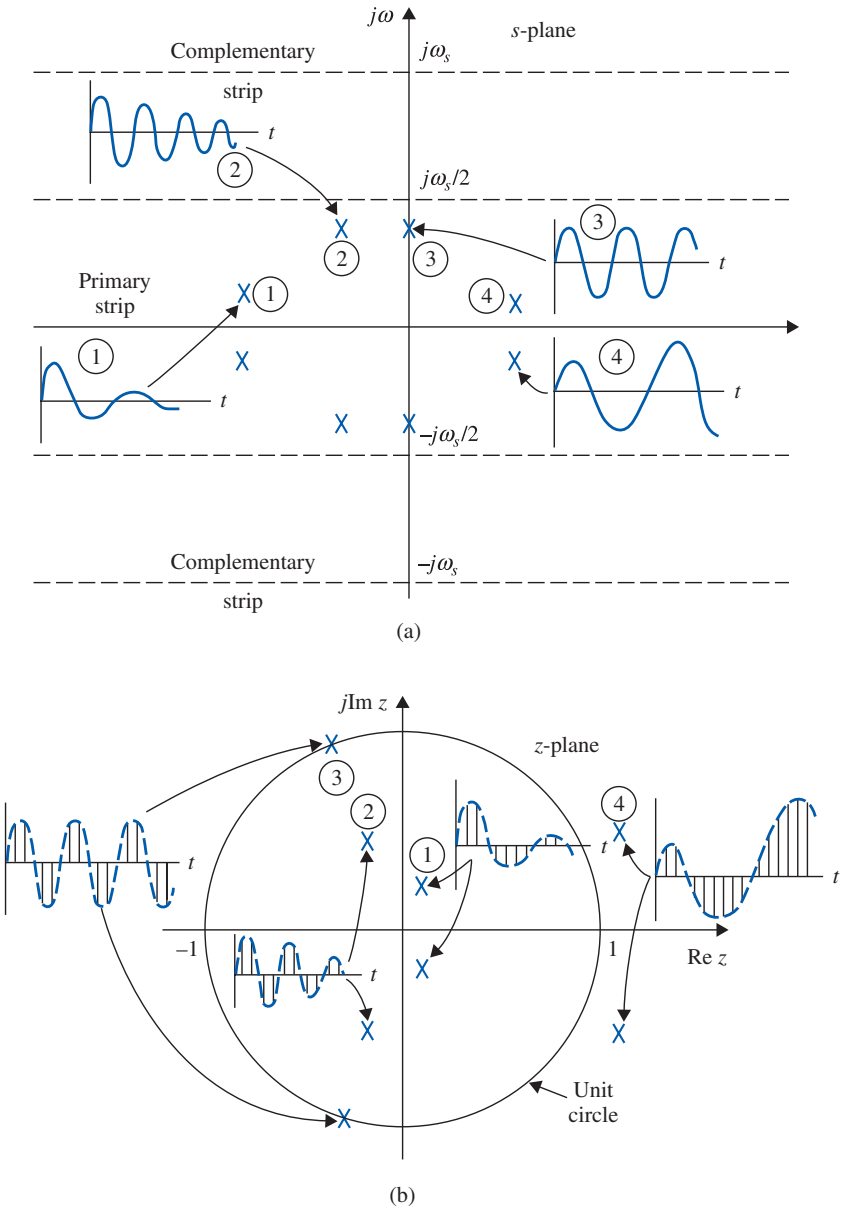
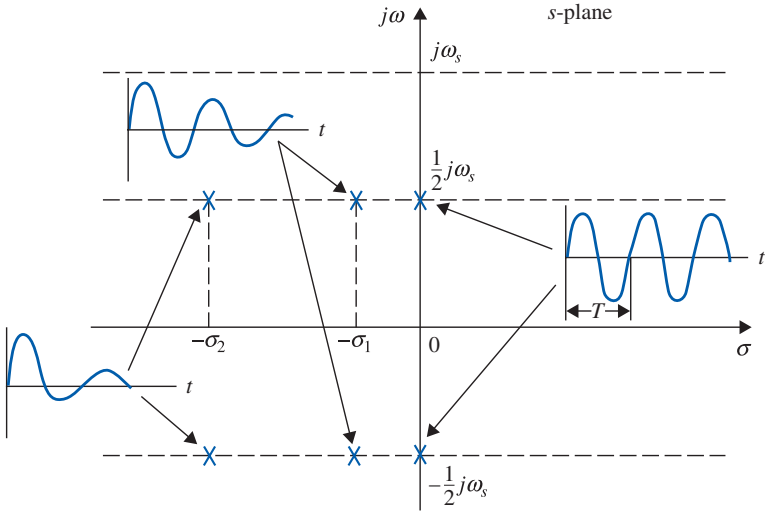


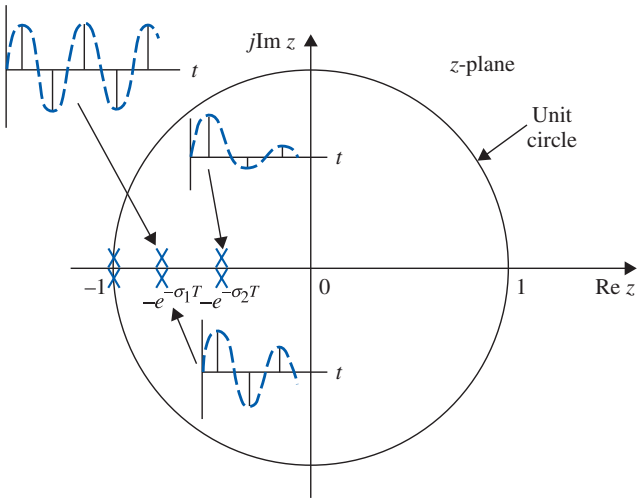
Figure H-27 (a) Transient responses corresponding to various pole locations of $Y^*(s)$ in the s-plane (complex-conjugate poles only). (b) Transient-response sequence corresponding to various pole locations of $Y(z)$ in the z-plane.

Thus, the output sequence will occur in alternating positive and negative pulses, as shown in Fig. H-28(b).

Complex-Conjugate Roots in the z-Plane: Complex-conjugate roots inside the unit circle in the z-plane correspond to oscillatory responses that decay with an increase in kT. Roots that are closer to the unit circle will decay slower. As the roots move toward the second and the third quadrants, the frequency of oscillation of the response increases. Refer to Figs. H-27 and H-28 for typical examples.



(a)



(b)

Figure H-28 (a) Transient responses corresponding to various pole locations of $Y^*(s)$ in the s-plane (complex-conjugate poles on the boundaries between periodic strips). (b) Transient-response sequence corresponding to various pole locations of $Y(z)$ in the z-plane.

▶ H-7 STEADY-STATE ERROR ANALYSIS OF DISCRETE-DATA CONTROL SYSTEMS

Because the input and output signals of a typical discrete-data control system are continuous-time functions, as shown in the block diagram of Fig. H-19, the error signal should still be defined as

$$e(t) = r(t) - y(t) \tag{H-195}$$

where $r(t)$ is the input and $y(t)$ is the output. The error analysis conducted here is only for unity-feedback systems with $H(s) = 1$. Due to the discrete data that appear inside the system, z-transform or difference equations are often used, so that the input and output are

represented in sampled form, $r(kT)$ and $y(kT)$, respectively. Thus, the error signal is more appropriately represented by $e^*(t)$ or $e(kT)$. That is,

$$e^*(t) = r^*(t) - y^*(t) \tag{H-196}$$

or

$$e(kT) = r(kT) - y(kT) \tag{H-197}$$

The steady-state error at the sampling instants is defined as

$$e_{ss}^* = \lim_{k \to \infty} e^*(t) = \lim_{k \to \infty} e(kT) \tag{H-198}$$

By using the final-value theorem of the z-transform, the steady-state error is

$$e_{ss}^* = \lim_{k \to \infty} e^*(kT) = \lim_{z \to 1} \left(1 - z^{-1}\right) E(z) \tag{H-199}$$

provided that the function $(1 - z^{-1})E(z)$ does not have any pole on or outside the unit circle in the z-plane. It should be pointed out that, because the true error of the system is $e(t)$, e_{ss}^* predicts only the steady-state error of the system at the sampling instants.

By expressing $E(z)$ in terms of $R(z)$ and $G_{ho}G_p(z)$, Eq. (H-199) is written

$$e_{ss}^* = \lim_{k \to \infty} e(kT) = \lim_{z \to 1} \left(1 - z^{-1}\right) \frac{R(z)}{1 + G_{ho}G_p(z)} \tag{H-200}$$

This expression shows that the steady-state error depends on the reference input $R(z)$ as well as the forward-path transfer function $G_{ho}G_p(z)$. Just as in the continuous-data systems, we shall consider only the three basic types of input signals and the associated error constants and relate e_{ss}^* to these and the type of the system.

Let the transfer function of the controlled process in the system of Fig. H-18 be of the form

$$G_p(s) = \frac{K(1 + T_a s)(1 + T_b s) \cdots (1 + T_m s)}{s^j (1 + T_1 s)(1 + T_2 s) \cdots (1 + T_n s)} \tag{H-201}$$

where $j = 0, 1, 2, \ldots$. The transfer function $G_{ho}G_p(z)$ is

$$G_{h0}G_p(z) = \left(1 - z^{-1}\right) \mathcal{Z} \left[\frac{K(1 + T_a s)(1 + T_b s) \cdots (1 + T_m s)}{s^{j+1} (1 + T_1 s)(1 + T_2 s) \cdots (1 + T_n s)} \right] \tag{H-202}$$

Steady-State Error Due to a Step-Function Input

When the input to the system, $r(t)$, in Fig. H-18 is a step function with magnitude R, the z-transform of $r(t)$ is

$$R(z) = \frac{Rz}{z - 1} \tag{H-203}$$

Substituting $R(z)$ into Eq. (H-200), we get

$$e_{ss}^* = \lim_{z \to 1} \frac{R}{1 + G_{ho}G_p(z)} = \frac{R}{1 + \lim_{z \to 1} G_{ho}G_p(z)} \tag{H-204}$$

Let the **step-error constant** be defined as

$$K_p^* = \lim_{z \to 1} G_{ho} G_p(z) \tag{H-205}$$

Eq. (H-204) becomes

$$e_{ss}^* = \frac{R}{1 + K_p^*} \tag{H-206}$$

Thus, we see that the steady-state error of the discrete-data control system in Fig. H-18 is related to the step-error constant K_p^* in the same way as in the continuous-data case, except that K_p^* is given by Eq. (H-205).

We can relate K_p^* to the system type as follows.

For a type-0 system, $j = 0$ in Eq. (H-202), and the equation becomes

$$G_{ho} G_p(z) = \left(1 - z^{-1}\right) \mathcal{Z} \left[\frac{K(1 + T_a s)(1 + T_b s) \cdots (1 + T_m s)}{s(1 + T_1 s)(1 + T_2 s) \cdots (1 + T_n s)}\right] \tag{H-207}$$

Performing partial-fraction expansion to the function inside the square brackets of the last equation, we get

$$\begin{aligned} G_{ho} G_p(z) &= \left(1 - z^{-1}\right) \mathcal{Z} \left[\frac{K}{s} + \text{terms due to the nonzero poles}\right] \\ &= \left(1 - z^{-1}\right) \left[\frac{Kz}{z - 1} + \text{terms due to the nonzero poles}\right] \end{aligned} \tag{H-208}$$

Because the terms due to the nonzero poles do not contain the term $(z - 1)$ in the denominator, the step-error constant is written

$$K_p^* = \lim_{z \to 1} G_{ho} G_p(z) = \lim_{z \to 1} \left(1 - z^{-1}\right) \frac{Kz}{z - 1} = K \tag{H-209}$$

Similarly, for a type-1 system, $G_{ho} G_p(z)$ will have an s^2 term in the denominator that corresponds to a term $(z - 1)^2$. This causes the step-error constant $K_p^*$to be infinite. The same is true for any system type greater than 1. The summary of the error constants and the steady-state error due to a step input is as follows:

System Type	K_p^*	e_{ss}^*
0	K	$R/(1 + K)$
1	∞	0
2	∞	0

Steady-State Error Due to a Ramp-Function Input

When the reference input to the system in Fig. H-18 is a ramp function of magnitude R, $r(t) = Rtu_s(t)$. The steady-state error in Eq. (H-200) becomes

$$e_{ss}^* = \lim_{z \to 1} \frac{RT}{(z - 1)\left[1 + G_{ho} G_p(z)\right]} = \frac{R}{\lim\limits_{z \to 1} \dfrac{z - 1}{T} G_{ho} G_p(z)} \tag{H-210}$$

Let the **ramp-error constant** be defined as

$$K_v^* = \frac{1}{T} \lim_{z \to 1} \left[(z - 1) G_{ho} G_p(z)\right] \tag{H-211}$$

Then, Eq. (H-210) becomes

$$e_{ss}^* = \frac{R}{K_v^*} \tag{H-212}$$

The ramp-error constant K_v^* is meaningful only when the input $r(t)$ is a ramp function and if the function $(z-1)G_{ho}G_p(z)$ in Eq. (H-211) does not have any poles on or outside the unit circle $|z| = 1$. The relations among the steady-state error e_{ss}^*, K_v^*, and the system type when the input is a ramp function with magnitude R are summarized as follows.

System Type	K_v^*	e_{ss}^*
0	0	∞
1	K	R/K
2	∞	0

Steady-State Error Due to a Parabolic-Function Input

When the input is a parabolic function, $r(t) = Rtu_s(t)/2$; the z-transform of $r(t)$ is

$$R(z) = \frac{RT^2 z(z+1)}{2(z-1)^3} \tag{H-213}$$

From Eq. (H-200), the steady-state error at the sampling instants is

$$\begin{aligned} e_{ss}^* &= \frac{T^2}{2} \lim_{z \to 1} \frac{R(z+1)}{(z-1)^2 \left[1 + G_{ho}G_p(z)\right]} \\ &= \frac{R}{\dfrac{1}{T^2} \lim\limits_{z \to 1} (z-1)^2 G_{ho}G_p(z)} \end{aligned} \tag{H-214}$$

By defining the **parabolic-error constant** as

$$K_a^* = \frac{1}{T^2} \lim_{z \to 1} \left[(z-1)^2 G_{ho}G_p(z)\right] \tag{H-215}$$

the steady-state error due to a parabolic-function input is

$$e_{ss}^* = \frac{R}{K_a^*} \tag{H-216}$$

The relations among the steady-state error e_{ss}^*, K_a^*, and the system type when the input is a parabolic function with its z-transform described by Eq. (H-213) are summarized as follows.

System Type	K_a^*	e_{ss}^*
0	0	∞
1	0	∞
2	K	R/K
3	∞	0

▶ H-8 ROOT LOCI OF DISCRETE-DATA SYSTEMS

The root-locus technique can be applied to discrete-data systems without any complications. With the z-transformed transfer function, the root loci for discrete-data systems are plotted in the z-plane, rather than in the s-plane. Let us consider the discrete-data control system shown in Fig. H-29. The characteristic equation roots of the system satisfy the following equation:

$$1 + GH^*(s) = 0 \tag{H-217}$$

in the s-plane, or

$$1 + GH(z) = 0 \tag{H-218}$$

in the z-plane. From Eq. (H-64), $GH^*(s)$ is written

$$GH^*(s) = \frac{1}{T}\sum_{n=-\infty}^{\infty} G(s + jn\omega_s)H(s + jn\omega_s) \tag{H-219}$$

which is an infinite series. Thus, the poles and zeros of $GH^*(s)$ in the s-plane will be infinite in number. This evidently makes the construction of the root loci of Eq. (H-217) in the s-plane quite complex. As an illustration, consider that, for the system of Fig. H-29,

$$G(s)H(s) = \frac{K}{s(s+1)} \tag{H-220}$$

Substituting Eq. (H-220) into Eq. (H-219), we get

$$GH^*(s) = \frac{1}{T}\sum_{n=-\infty}^{\infty} \frac{K}{(s + jn\omega_s)(s + jn\omega_s + 1)} \tag{H-221}$$

which has poles at $s = -jn\omega_s$ and $s = -1 - jn\omega_s$, where n takes on all integers between $-\infty$ and ∞. The pole configuration of $GH^*(s)$ is shown in Fig. H-30(a). By using the properties of the RL in the s-plane, RL of $1 + GH^*(s) = 0$ are drawn as shown in Fig. H-30(b) for the sampling period $T = 1$ s. The RL contain an infinite number of branches, and these clearly indicate that the closed-loop system is unstable for all values of K greater than 4.32. In contrast, it is well known that the same system without sampling is stable for all positive values of K.

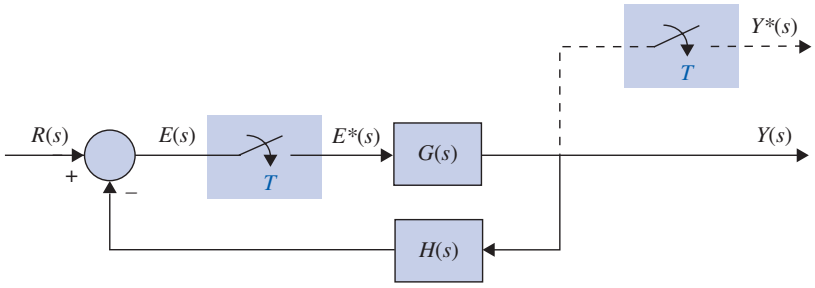


Figure H-29 Discrete-data control system.

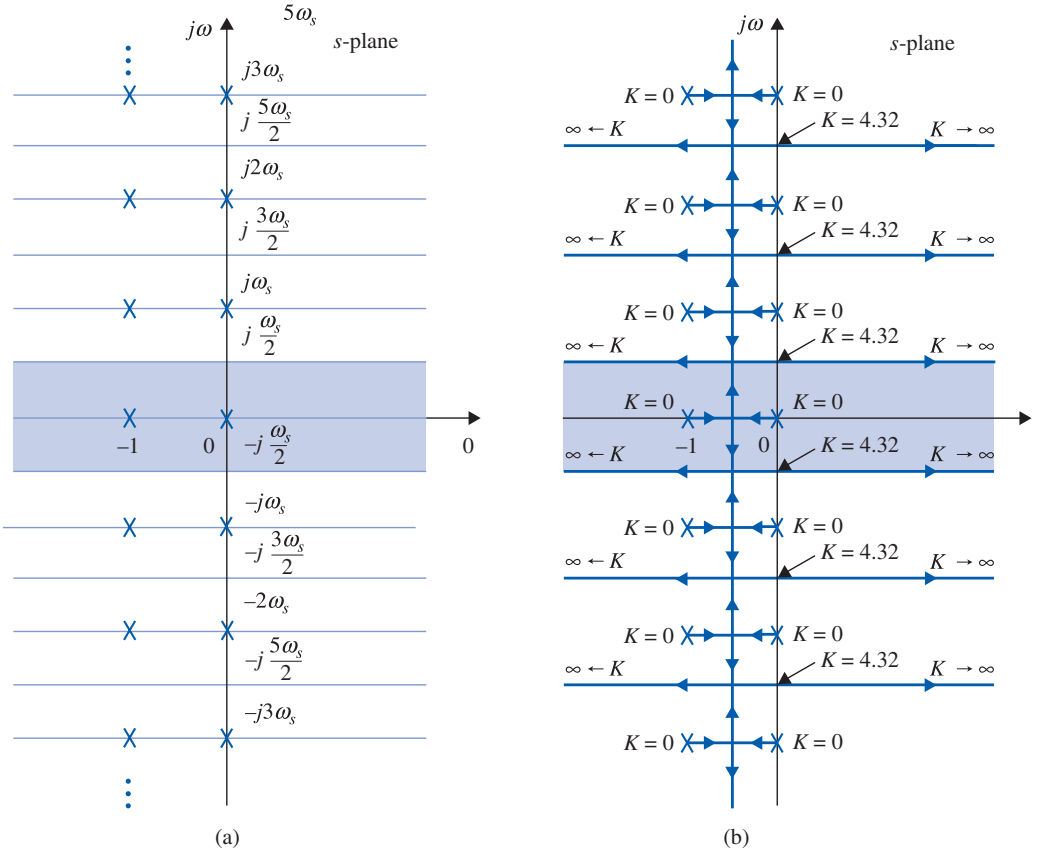


Figure H-30 Pole configuration of $GH*(s)$ and the root-locus diagram in the s-plane for the discrete-data system in Fig. H-29 with $G(s)H(s) = \dfrac{K}{s(s+1)}$, $T = 1$ sec.

The root-locus problem for discrete-data systems is simplified if the root loci are constructed in the z-plane using Eq. (H-218). Because Eq. (H-218) is, in general, a rational function in z with constant coefficients, its poles and zeros are finite in number, and the number of root loci is finite in the z-plane. The same procedures of construction for continuous-data systems are directly applicable in the z-plane for discrete-data systems. The following examples illustrate the constructions of root loci for discrete-data systems in the z-plane.

▶ EXAMPLE H-8-1 Consider that for the discrete-data system shown in Fig. H-29 the loop transfer function in the z domain is

$$GH(z) = \frac{0.632Kz}{(z-1)(z-0.368)} \tag{H-222}$$

The RL of the closed-loop characteristic equation are constructed based on the pole–zero configuration of $GH(z)$, as shown in Fig. H-31. Notice that, when the value of K exceeds 4.32, one of the two roots moves outside the unit circle, and the system becomes unstable. The constant-damping-ratio locus may be superimposed on the RL to determine the required value of K for a specified damping ratio. In Fig. H-31, the constant-damping-ratio locus for $\zeta = 0.5$ is drawn, and the intersection with the RL gives the desired value of $K = 1$. For the same system, if the sampling period T is increased to 2 seconds, the z-transform loop transfer function becomes

$$GH(z) = \frac{0.865Kz}{(z-1)(z-0.135)} \tag{H-223}$$

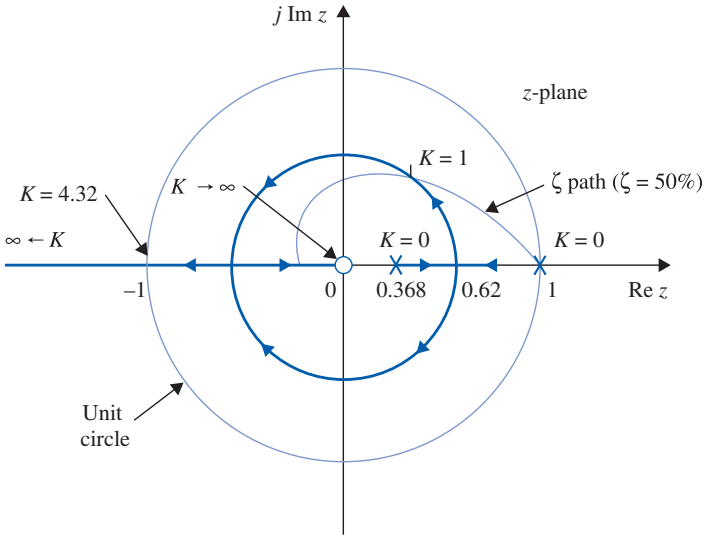


Figure H-31 Root-locus diagram of a discrete-data control system without zero-order-hold. $G(s)H(s) = \dfrac{K}{s(s+1)}$, $T = 1$ second.

The RL for this case are shown in Fig. H-32. Note that, although the complex part of the RL for $T = 2$ seconds takes the form of a smaller circle than that when $T = 1$ second, the system is actually less stable, because the marginal value of K for stability is 2.624, as compared with the marginal K of 4.32 for $T = 1$ second.

Next, let us consider that a zero-order-hold is inserted between the sampler and the controlled process $G(s)$ in the system of Fig. H-29. The loop transfer function of the system with the zero-order-hold is

$$G_{ho}GH(z) = \frac{K\left[(T - 1 + e^{-T})z - Te^{-T} + 1 - e^{-T}\right]}{(z-1)(z - e^{-T})} \tag{H-224}$$

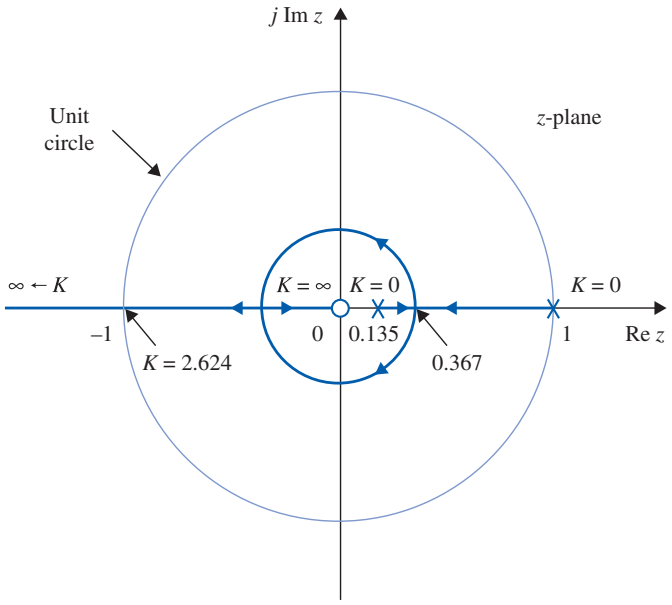
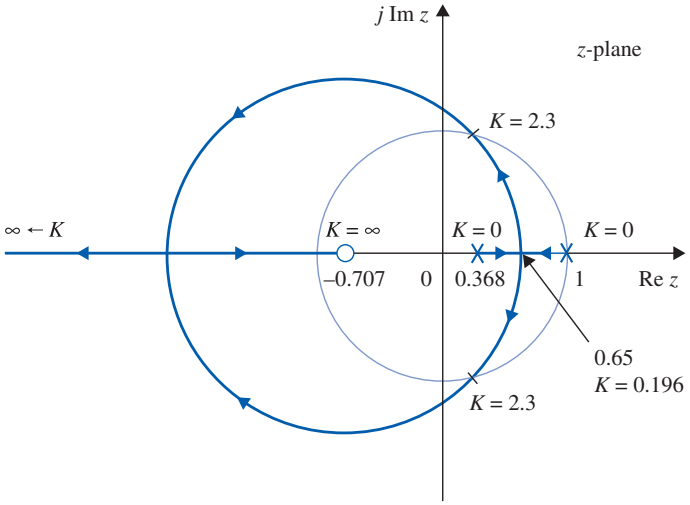
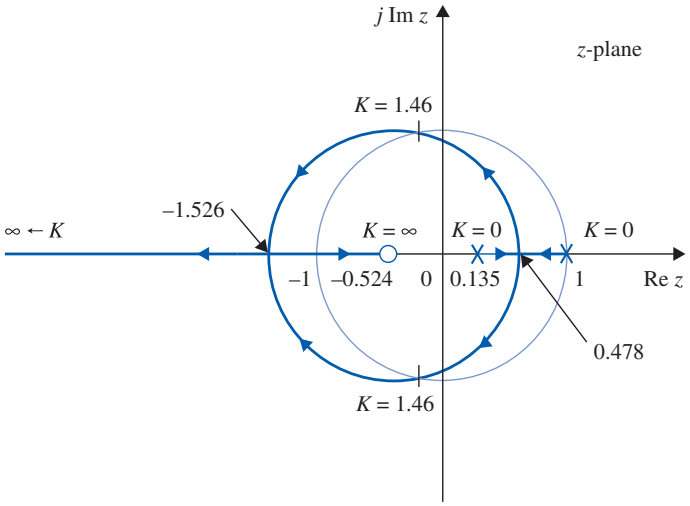


Figure H-32 Root-locus diagram of a discrete-data control system without zero-order-hold. $G(s)H(s) = \dfrac{K}{s(s+1)}$, $T = 2$ seconds.



(a) Root loci for $T = 1$ second



(b) Root loci for $T = 2$ seconds

Figure H-33 Root-locus diagram of a discrete-data control system with zero-order-hold. $G(s)H(s) = \dfrac{K}{s(s+1)}$.

The RL of the system with ZOH for $T = 1$ and 2 seconds are shown in Fig. H-33(a) and H-33(b), respectively. In this case, the marginal value of stability for K is 2.3 for $T = 1$ second and 1.46 for $T = 2$ seconds. Comparing the root loci of the system with and without the ZOH, we see that the ZOH reduces the stability margin of the discrete-data system.

In conclusion, the root loci of discrete-data systems can be constructed in the z-plane using essentially the same properties as those of the continuous-data systems in the s-plane. However, the absolute and relative stability conditions of the discrete-data system must be investigated with respect to the unit circle and the other interpretation of performance with respect to the regions in the z-plane. ◀

▶ H-9 FREQUENCY-DOMAIN ANALYSIS OF DISCRETE-DATA CONTROL SYSTEMS

All the frequency-domain methods discussed in the preceding sections can be extended to the analysis of discrete-data systems. Consider the discrete-data system shown in Fig. H-34. The closed-loop transfer function of the system is

$$\frac{Y(z)}{R(z)} = \frac{G_{ho}G(z)}{1 + G_{ho}G(z)} \tag{H-225}$$

where $G_{ho}G(z)$ is the z-transform of $G_{ho}(s)G(s)$. Just as in the case of continuous-data systems, the absolute and relative stability conditions of the closed-loop discrete-data system can be investigated by making the frequency-domain plots of $G_{ho}G(z)$. Because the positive $j\omega$-axis of the s-plane corresponds to real frequency, the frequency-domain plots of $G_{ho}G(z)$ are obtained by setting $z = e^{j\omega T}$ and then letting ω vary from 0 to ∞. This is also equivalent to mapping the points on the unit circle, $|z| = 1$, in the z-plane onto the $G_{ho}G(e^{j\omega T})$-plane. Because the unit circle repeats for every sampling frequency $\omega_s (= 2\pi/T)$, as shown in Fig. H-35, when ω is varied along the $j\omega$-axis, the frequency-domain plot of $G(e^{j\omega T})$ repeats for $\omega = n\omega_s$ to $(n + 1)\omega_s$, $n = 0, 1, 2, \ldots$. Thus, it is necessary to plot $G_{ho}G(e^{j\omega T})$ only for the range of $\omega = 0$ *to* $\omega = \omega_s$. In fact, because the unit circle in the z-plane is symmetrical about the real axis, the plot of $G_{ho}G(e^{j\omega T})$ in the polar coordinates for $\omega = 0$ to $\omega_s/2$ needs to be plotted.

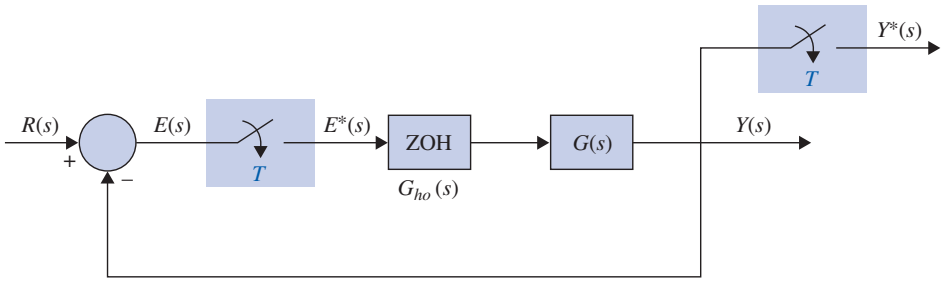


Figure H-34 Closed-loop discrete-data control system.

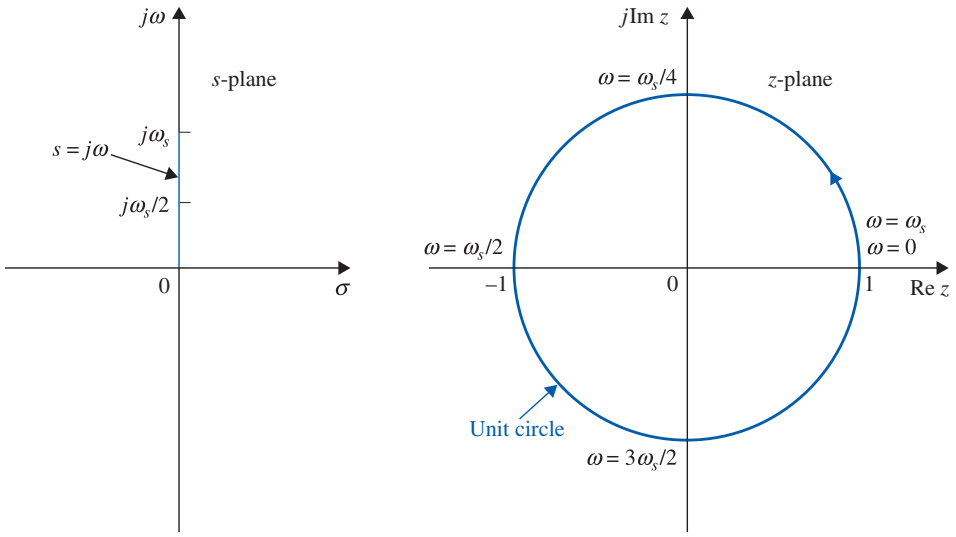


Figure H-35 Relation between the $j\omega$-axis in the s-plane and the unit circle in the z-plane.

H-9-1 Bode Plot with the *w*-Transformation

The w-transformation introduced in Eq. (H-162) can be used for frequency-domain analysis and design of discrete-data control systems. The transformation is

$$z = \frac{(2/T) + w}{(2/T) - w} \tag{H-226}$$

In the frequency domain, we set [Eq. (H-166)],

$$w = j\omega_w = j\,\frac{2}{T}\tan\frac{\omega T}{2} \tag{H-227}$$

For frequency-domain analysis of a discrete-data system, we substitute Eqs. (H-226) and (H-227) in $G(z)$ to get $G(j\omega_w)$; the latter can be used to form the Bode plot or the polar plot of the system.

▶ EXAMPLE H-9-1

As an illustrative example on frequency-domain plots of discrete-data control systems, let the transfer function of the process in the system in Fig. H-34 be

$$G(s) = \frac{1.57}{s(s+1)} \tag{H-228}$$

and the sampling frequency is 4 rad/sec. Let us first consider that the system does not have a zero-order-hold, so that

$$G_{ho}G(z) = G(z) = \frac{1.243z}{(z-1)(z-0.208)} \tag{H-229}$$

The frequency response of $G_{ho}G(z)$ is obtained by substituting $z = e^{j\omega T}$ in Eq. (H-228). The polar plot of $G_{ho}G(e^{j\omega T})$ for $\omega = 0$ *to* $\omega_s/2$ is shown in Fig. H-36. The mirror image of the locus shown, with the mirror placed on the real axis, represents the plot for $\omega = \omega_s/2$ to ω_s.

The Bode plot of $G_{ho}G(e^{j\omega T})$ consists of the graphs of $\left|G_{ho}G\left(e^{j\omega T}\right)\right|$ in dB versus ω, and $\angle G_{ho}G(e\,j\omega T)$ in degrees versus ω, as shown in Fig. H-37 for three decades of frequency with the plots ended at $\omega = \omega_s/2 = 2$ rad/sec.

For the sake of comparison, the forward-path transfer function of the system with a zero-order-hold is obtained:

$$G_{ho}G(z) = \frac{1.2215z + 0.7306}{(z-1)(z-0.208)} \tag{H-230}$$

The polar plot and the Bode plot of the last equation are shown in Figs. H-36 and H-37, respectively. Notice that the polar plot of the system with the ZOH intersects the negative real axis at a point that is closer to the $(-1, j0)$ point than that of the system without the ZOH. Thus, the system with the ZOH is less stable. Similarly, the phase of the Bode plot of the system with the ZOH is more negative than that of the system without the ZOH. The gain margin, phase margin, and peak resonance of the two systems are summarized as follows.

	Gain Margin (dB)	Phase Margin (deg)	M^r
Without ZOH	5.77	39.0	1.58
With ZOH	0.71	2.91	22.64

As an alternative, the Bode plot and polar plot of the forward-path transfer function can be done using the w-transformation of Eqs. (H-226). For the system with ZOH, the forward-path transfer function in the w-domain is

$$G_{ho}G(w) = \frac{1.57(1 + 0.504w)(1 - 1.0913w)}{w(1 + 1.197w)} \tag{H-231}$$

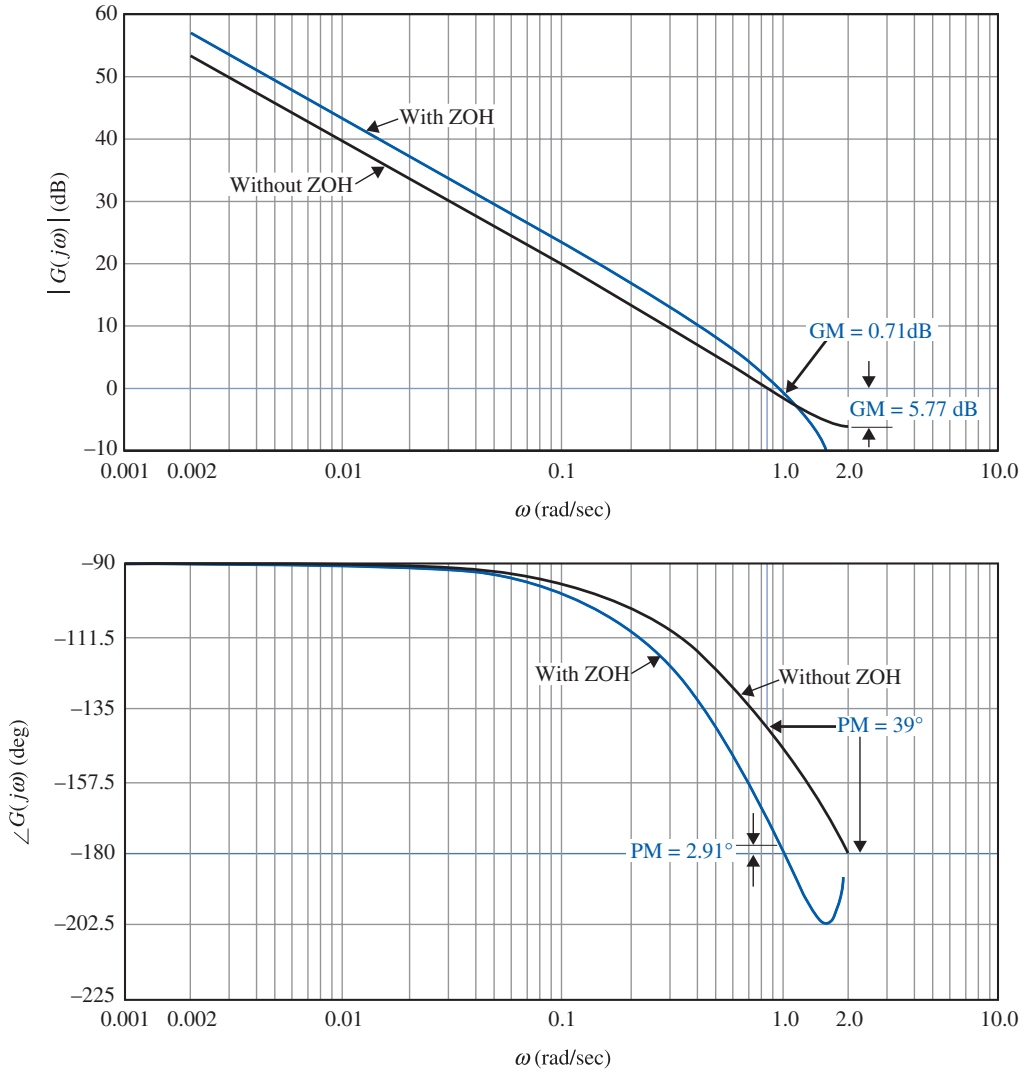


Figure H-36 Bode plot of $G_{ho}G(z)$ of the system in Fig. H-34, with $G(s) = 1.57/[s(s+1)]$. $T = 1.57$ sec, and with and without ZOH.

For the system without ZOH,

$$G_{ho}G(jw) = \frac{1 - 0.6163w^2}{w(1 + 1.978w)} \tag{H-232}$$

Substituting $w = j\omega_w$ into Eq. (H-232), the Bode plots are made as shown in Fig. H-38. Notice that the frequency coordinates in Fig. H-38 are ω_w, whereas those in Fig. H-36 are the real frequency ω. The two frequencies are related through Eq. (H-227).

The conclusion from this illustrative example is that, once z is replaced by $e^{j\omega T}$ in the z-domain transfer function, or if the w-transform is used, all the frequency-domain analysis techniques available for continuous-data systems can be applied to discrete-data systems. ◀

▶ H-10 DESIGN OF DISCRETE-DATA CONTROL SYSTEMS

H-10-1 Introduction

The design of discrete-data control systems is similar in principle to the design of continuous-data control systems. The design objective is basically that of determining the controller so that the system will perform in accordance with specifications. In fact, in

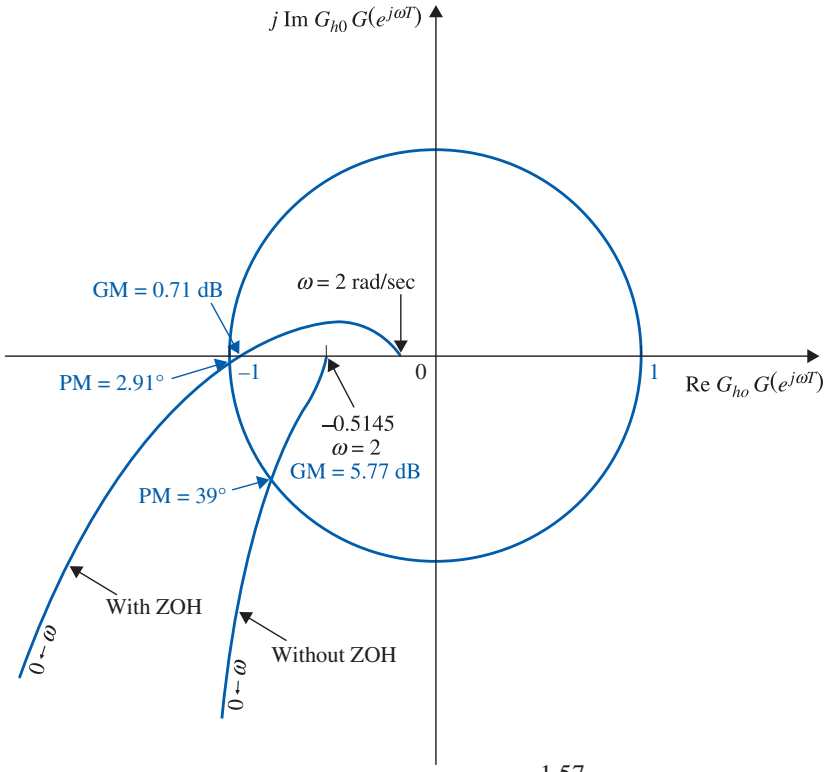


Figure H-37 Frequency-domain plot of $G(s) = \dfrac{1.57}{s(s+1)}$, $T = 1.57$ seconds, and with and without ZOH.

most situations, the controlled process is the same, except in discrete-data systems the controller is designed to process sampled or digital data.

The design of discrete-data control systems treated in this chapter is intended only for introductory purposes. An in-depth coverage of the subject may be found in books dedicated to digital control. In this chapter we deal only with the design of a control system with a cascade digital controller and a system with digital state feedback. Block diagrams of these systems are shown in Fig. H-39.

Just as with the design of continuous-data control systems, the design of discrete-data control systems can be carried out in either the frequency domain or the time domain. Using computer programs, digital control systems can be designed with a minimum amount of trial and error.

H-10-2 Digital Implementation of Analog Controllers

It seems that most people learn how to design continuous-data systems before they learn to design digital systems, if at all. Therefore, it is not surprising that most engineers prefer to design continuous-data systems. Ideally, if the designer intends to use digital control, the system should be designed so that the dynamics of the controller can be described by a z-transfer function or difference equations. However, there are situations in which the analog controller is already designed, but the availability and advantages of digital control suggest that the controller be implemented by digital elements. Thus, the

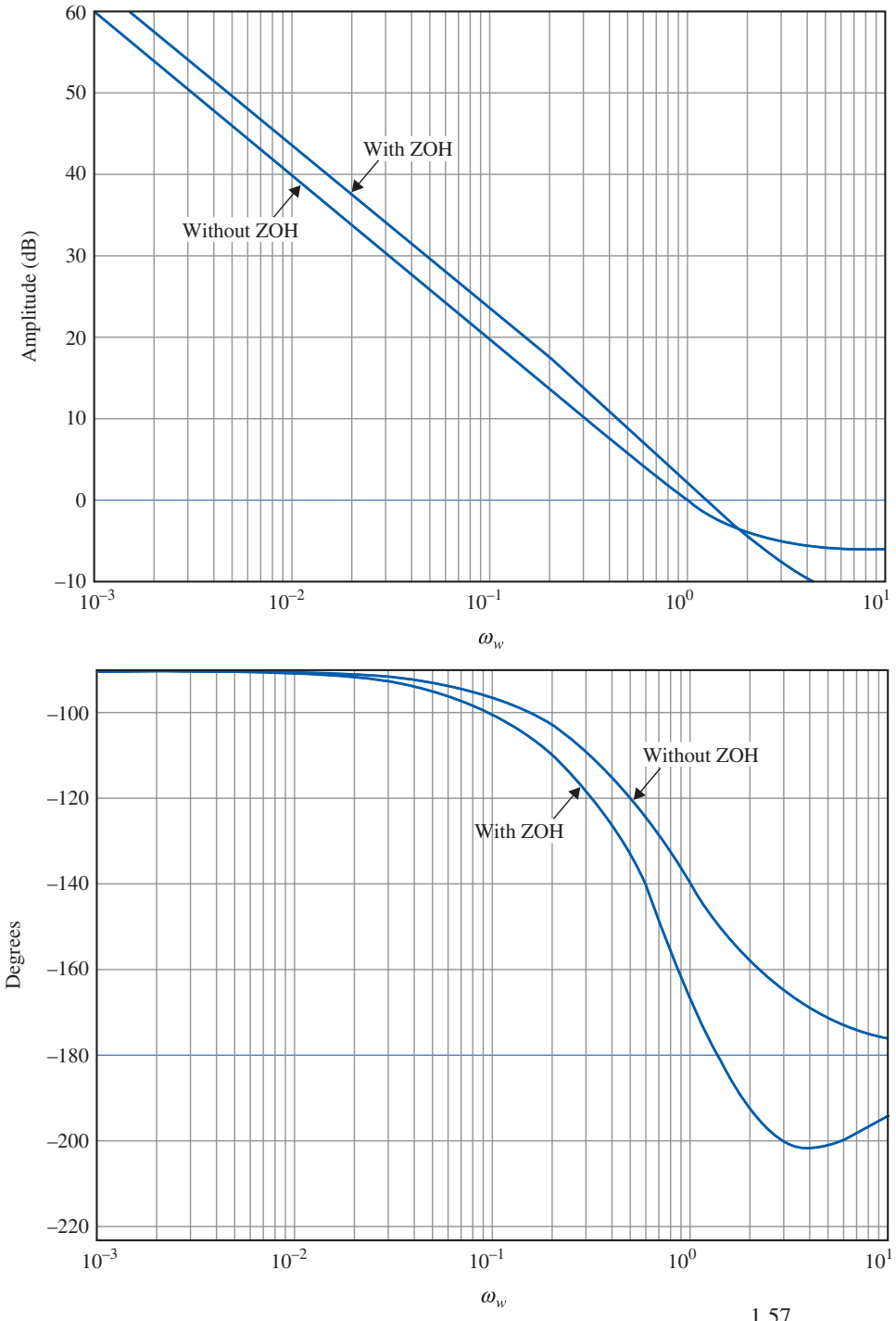


Figure H-38 Bode plot of $G_{ho}G(z)$ of the system in Fig. H-34 with $G(s) = \dfrac{1.57}{s(s+1)}, T = 1.57$ seconds, with and without ZOH. The plots are done with the w-transformation, $w = j\omega_w$.

problems discussed in this section are twofold: first how continuous-data controllers such as PID, phase-lead or phase-lag controllers, and others can be approximated by digital controllers and, second, the problem of implementing digital controllers by digital processors.

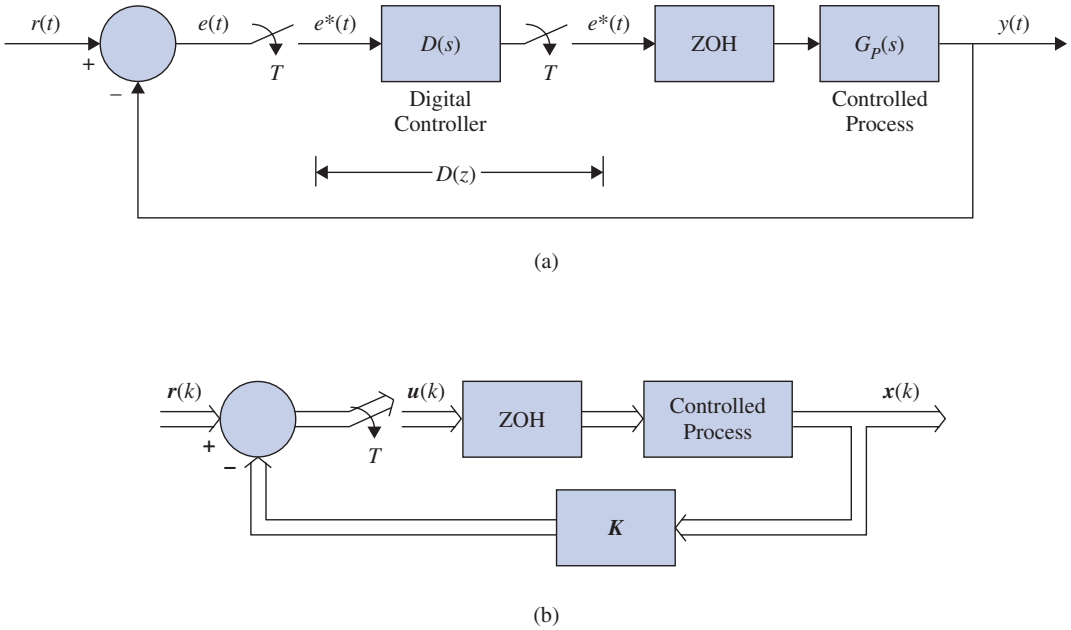


Figure H-39 (a) Digital control system with cascade digital controller. (b) Digital control system with state feedback.

H-10-3 Digital Implementation of the PID Controller

The PID controller in the continuous-data domain is described by

$$G_c(s) = K_P + K_D s + \frac{K_I}{s} \tag{H-233}$$

The proportional component K_P is implemented digitally by a constant gain K_P. Because a digital computer or processor has finite word length, the constant K_P cannot be realized with infinite resolution.

The time derivative of a function $f(t)$ at $t = kT$ can be approximated by the **backward-difference rule**, using the values of $f(t)$ measured at $t = kT$ and $(k - 1)\,T$; that is,

$$\left.\frac{df(t)}{dt}\right|_{t=kT} = \frac{1}{T}\,(\,f(kT) - f[(k-1)T]) \tag{H-234}$$

To find the z-transfer function of the derivative operation described before, we take the z-transform on both sides of Eq. (H-234). We have

$$\mathcal{Z}\left(\left.\frac{df(t)}{dt}\right|_{t=kT}\right) = \frac{1}{T}\left(1 - z^{-1}\right)F(z) = \frac{z-1}{Tz}F(z) \tag{H-235}$$

Thus, the z-transfer function of the digital differentiator is

$$G_D(z) = K_D\,\frac{z-1}{Tz} \tag{H-236}$$

where K_D is the proportional constant of the derivative controller. Replacing z by e^{Ts} in Eq. (H-236), we can show that, as the sampling period T approaches zero, $G_D(z)$ approaches $K_D s$, which is the transfer function of the analog derivative controller. In general, the choice of the sampling period is extremely important. The value of T should be sufficiently small so that the digital approximation is adequately accurate.

There are a number of numerical integration rules that can be used to digitally approximate the integral controller K_I/s. The three basic methods of approximating the area of a function numerically are **trapezoidal integration**, **forward-rectangular integration**, and **backward-rectangular integration**. These are described as follows.

Trapezoidal Integration

The trapezoidal-integration rule approximates the area under the function $f(t)$ by a series of trapezoids, as shown in Fig. H-40. Let the integral of $f(t)$ evaluated at $t = kT$ be designated as $u(kT)$. Then,

$$u(kT) = u[(k-1)T] + \frac{T}{2}\{f(kT) - f[(k-1)T]\} \tag{H-237}$$

where the area under $f(t)$ for $(k-1)T \leq t < kT$ is approximated by the area of the trapezoid in the interval. Taking the z-transform on both sides of Eq. (H-237), we have the transfer function of the digital integrator as

$$G_I(z) = K_I \frac{U(z)}{F(z)} = \frac{K_I T(z+1)}{2(z-1)} \tag{H-238}$$

where K_I is the proportional constant.

Forward-Rectangular Integration

For the forward-rectangular integration, we approximate the area under $f(t)$ by rectangles, as shown in Fig. H-41. The integral of $f(t)$ at $t = kT$ is approximated by

$$u(kT) = u[(k-1)T] + Tf(kT) \tag{H-239}$$

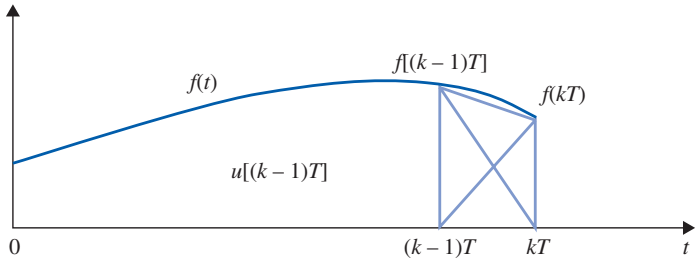


Figure H-40 Trapezoidal-integration rule.

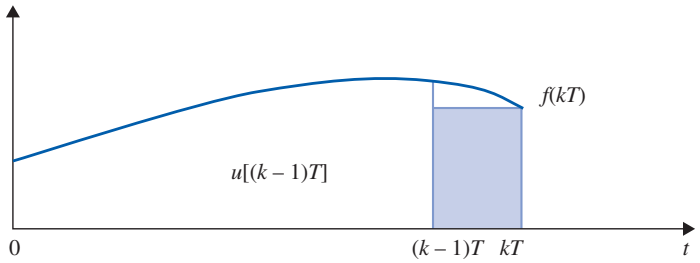


Figure H-41 Forward-rectangular integration rule.

By taking the z-transform on both sides of Eq. (H-239), the transfer function of the digital integrator using the forward-rectangular rule is

$$G_I(z) = K_I \frac{U(z)}{F(z)} = \frac{K_I T z}{z-1} \tag{H-240}$$

Backward-Rectangular Integration

For the backward-rectangular integration, the digital approximation rule is illustrated in Fig. H-42. The integral of $f(t)$ at $t = kT$ is approximated by

$$u(kT) = u[(k-1)T] + Tf[(k-1)T] \tag{H-241}$$

The z-transfer function of the digital integrator using the backward-rectangular integration rule is

$$G_I(z) = K_I \frac{U(z)}{F(z)} = \frac{K_I T}{z-1} \tag{H-242}$$

By combining the proportional, derivative, and integration operations described previously, the digital PID controller is modeled by the following transfer functions.

Trapezoidal Integration

$$G_c(z) = \frac{\left(K_P + \frac{TK_I}{2} + \frac{K_D}{T}\right)z^2 + \left(\frac{TK_I}{2} - K_P - \frac{2K_D}{T}\right)z + \frac{K_D}{T}}{z(z-1)} \tag{H-243}$$

Forward-Rectangular Integration

$$G_c(z) = \frac{\left(K_P + \frac{K_D}{T} + TK_I\right)z^2 - \left(K_P + \frac{2K_D}{T}\right)z + \frac{K_D}{T}}{z(z-1)} \tag{H-244}$$

Backward-Rectangular Integration

$$G_c(z) = \frac{\left(K_P + \frac{K_D}{T}\right)z^2 + \left(TK_I - K_P - \frac{2K_D}{T}\right)z + \frac{K_D}{T}}{z(z-1)} \tag{H-245}$$

When $K_I = 0$, the transfer function of the digital PD controller is

$$G_c(z) = \frac{\left(K_P + \frac{K_D}{T}\right)z - \frac{K_D}{T}}{z} \tag{H-246}$$

Once the transfer function of a digital controller is determined, the controller can be implemented by a digital processor or computer. The operator z^{-1} is interpreted as a time delay of T seconds. In practice, the time delay is implemented by storing a variable in some

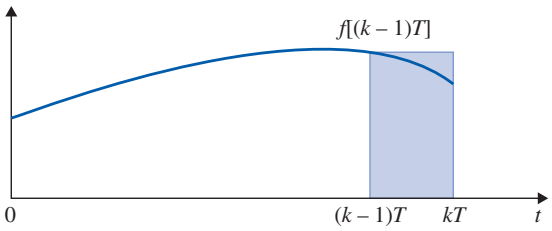


Figure H-42 Backward-rectangular integration rule.

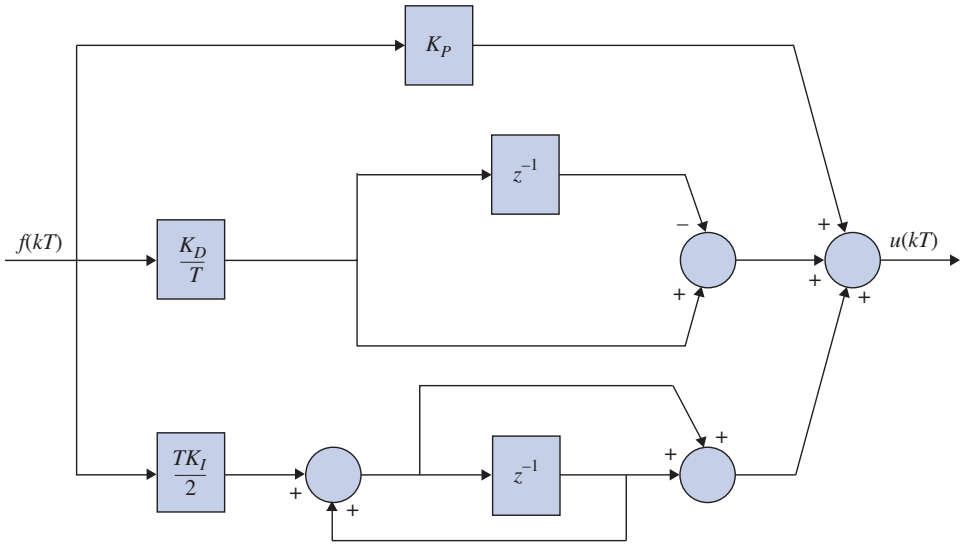


Figure H-43 Block diagram of a digital-program implementation of the PID controller.

storage location in the computer and then taking it out after T seconds have elapsed. Fig. H-43 illustrates a block diagram representation of the digital program of the PID controller using the trapezoidal-integration rule.

H-10-4 Digital Implementation of Lead and Lag Controllers

In principle, any continuous-data controller can be made into a digital controller simply by adding sample-and-hold units at the input and the output terminals of the controller and selecting a sampling frequency as small as is practical. Figure H-44 illustrates the basic scheme with $G_c(s)$, the transfer function of the continuous-data controller, and $G_c(z)$, the equivalent digital controller. The sampling period T should be sufficiently small so that the dynamic characteristics of the continuous-data controller are not lost through the digitization. The system configuration in Fig. H-44 actually suggests that, given the continuous-data controller $G_c(s)$, the equivalent digital controller $G_c(z)$ can be obtained by the arrangement shown. On the other hand, given the digital controller $G_c(z)$, we can realize it by using an analog controller $G_c(s)$ and sample-and-hold units, as shown in Fig. H-44.

▶ **EXAMPLE H-10-1** As an illustrative example, consider that the continuous-data controller in Fig. H-44 is represented by the transfer function

$$G_c(s) = \frac{s+1}{s+1.61} \tag{H-247}$$

From Fig. H-44, the transfer function of the digital controller is written

$$G_c(z) = \frac{U(z)}{F(z)} = (1-z^{-1})\mathcal{Z}\left[\frac{s+1}{s(s+1.61)}\right] = \frac{z-\left(0.62e^{-1.61T}+0.38\right)}{z-e^{-1.61T}} \tag{H-248}$$

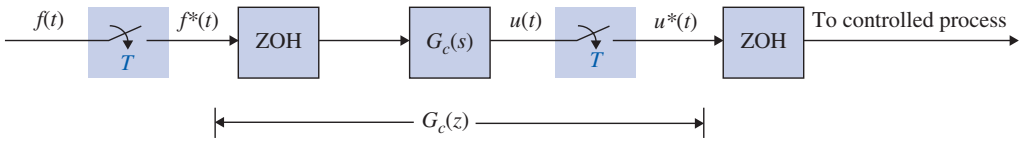


Figure H-44 Realization of a digital controller by an analog controller with sample-and-hold units.

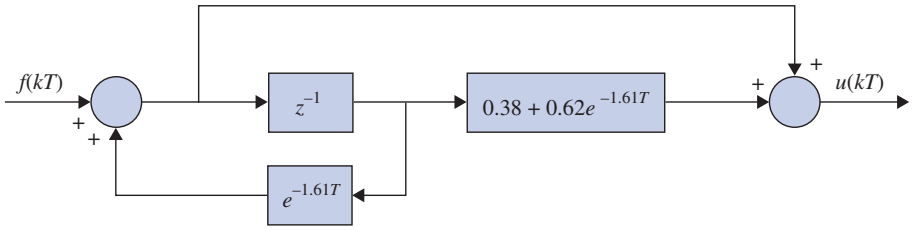


Figure H-45 Digital-program realization of Eq. (H-248).

The digital-program implementation of Eq. (H-248) is shown in Fig. H-45. ◀

▶ H-11 DIGITAL CONTROLLERS

Digital controllers can be realized by digital networks, digital computers, microprocessors, or digital signal processors (DSPs). A distinct advantage of digital controllers implemented by microprocessors or DSPs is that the control algorithm contained in the controller can be easily altered by changing the program. Changing the components of a continuous-data controller is rather difficult once the controller has been built.

H-11-1 Physical Realizability of Digital Controllers

The transfer function of a digital controller can be expressed as

$$G_c(z) = \frac{E_2(z)}{E_1(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_m z^{-m}}{a_0 + a_1 z^{-1} + \cdots + a_n z^{-n}} \tag{H-249}$$

where n and m are positive integers. The transfer function $G_c(z)$ is said to be physically realizable if its output does not precede any input. This means that the series expansion of $G_c(z)$ should not have any positive powers in z. In terms of the $G_c(z)$ given in Eq. (H-249), if $b_0 \neq 0$, then $a_0 \neq 0$. If $G_c(z)$ is expressed as

$$G_c(z) = \frac{b_m z^m + b_{m-1} z^{m-1} + \cdots + b_1 z + b_0}{a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0} \tag{H-250}$$

then the physical realizability requirement is $n \geq m$.

The decomposition techniques presented in Chapter 2 can be applied to realize the digital controller transfer function by a digital program. We consider that a digital program is capable of performing arithmetic operations of addition, subtraction, multiplication by a constant, and shifting. The three basic methods of decomposition for digital programming are discussed in the following sections.

Digital Program by Direct Decomposition

Applying direct decomposition to Eq. (H-249), we have the following equations:

$$E_2(z) = \frac{1}{a_0}\left(b_0 + b_1 z^{-1} + \cdots + b_m z^{-m}\right) X(z) \tag{H-251}$$

$$X(z) = \frac{1}{a_0} E_1(z) - \frac{1}{a_0}\left(a_1 z^{-1} + a_2 z^{-2} + \cdots + a_n z^{-n}\right) X(z) \tag{H-252}$$

Fig. H-46 shows the signal-flow graph of a direct digital program of Eq. (H-249) by direct decomposition for $m = 2$ and $n = 3$. The branches with gains of z^{-1} represent time delays or shifts of one sampling period.

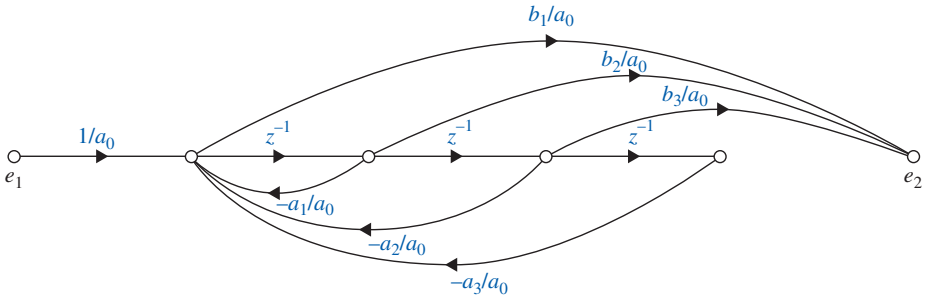


Figure H-46 Signal-flow graph of digital program by direct decomposition of Eq. (H-249) with $n = 3$ and $m = 2$.

Digital Program by Cascade Decomposition

The transfer function $G_c(z)$ can be written as a product of first- or second-order transfer functions, each realizable by a simple digital program. The digital program of the overall transfer function is then represented by these simple digital programs connected in cascade. Eq. (H-249) is written in factored form as

$$G_c(z) = G_{c1}(z)G_{c2}(z)\cdots G_{cn}(z) \tag{H-253}$$

where the individual factors can be expressed as

Real Pole and Zero

$$G_{ci}(z) = K_i \frac{1 + c_i z^{-1}}{1 + d_i z^{-1}} \tag{H-254}$$

Complex-Conjugate Poles (No Zeros)

$$G_{ci}(z) = \frac{K_i}{1 + d_{i1} z^{-1} + d_{i2} z^{-2}} \tag{H-255}$$

Complex-Conjugate Poles with One Zero

$$G_{ci}(z) = K_i \frac{1 + c_i z^{-1}}{1 + d_{i1} z^{-1} + d_{i2} z^{-2}} \tag{H-256}$$

and several other possible forms up to the second order.

Digital Program by Parallel Decomposition

The transfer function in Eq. (H-249) can be expanded into a sum of simple first- or second-order terms by partial-fraction expansion. These terms are then realized by digital programs connected in parallel.

▶ **EXAMPLE H-11-1** Consider the following transfer function of a digital controller:

$$G_c(z) = \frac{E_2(z)}{E_1(z)} = \frac{10\left(1 + 0.5z^{-1}\right)}{(1 - z^{-1})(1 - 0.2z^{-1})} \tag{H-257}$$

Because the leading coefficients of the numerator and denominator polynomials in z^{-1} are all constants, the transfer function is physically realizable. The transfer function $G_c(z)$ is realized by the three types of digital programs discussed earlier. ◀

Direct Digital Programming

Eq. (H-257) is written

$$G_c(z) = \frac{E_2(z)}{E_1(z)} = \frac{10(1 + 0.5z^{-1})X(z)}{(1 - z^{-1})(1 - 0.2z^{-1})X(z)} \tag{H-258}$$

Expanding the numerator and denominator of the last equation and equating, we have

$$E_2(z) = (10 + 5z^{-1})X(z) \tag{H-259}$$

$$X(z) = E_1(z) + 1.2z^{-1}X(z) - 0.2z^{-2}X(z) \tag{H-260}$$

The last two equations are realized by the digital program shown in Fig. H-47.

Cascade Digital Programming

The right-hand side of Eq. (H-257) is divided into two factors in one of several possible ways.

$$G_c(z) = \frac{E_2(z)}{E_1(z)} = \frac{1 + 0.5z^{-1}}{1 - z^{-1}} \frac{10}{1 - 0.2z^{-1}} \tag{H-261}$$

Fig. H-48 shows the signal-flow graph of the cascade digital program of the controller.

Parallel Digital Programming

The right-hand side of Eq. (H-257) is expanded by partial fraction into two separate terms.

$$G_c(z) = \frac{E_2(z)}{E_1(z)} = \frac{18.75}{1 - z^{-1}} - \frac{8.75}{1 - 0.2z^{-1}} \tag{H-262}$$

Fig. H-49 shows the signal-flow graph of the parallel digital program of the controller.

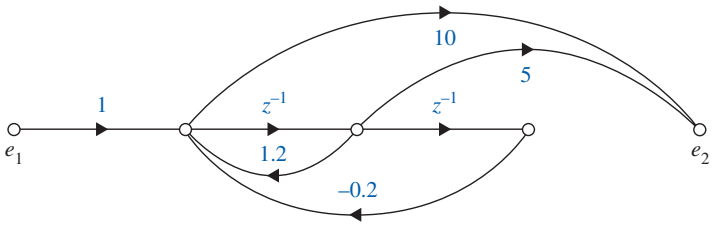


Figure H-47 Direct digital program of Eq. (H-257).

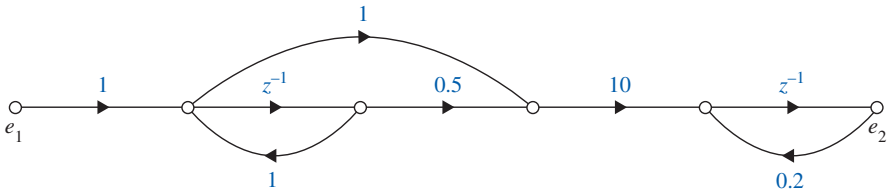


Figure H-48 Cascade digital program of Eq. (H-257).

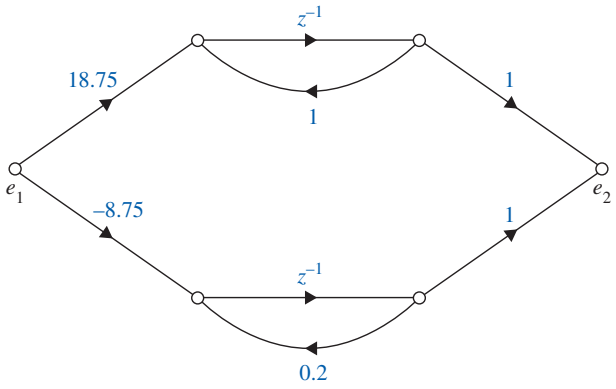


Figure H-49 Parallel digital program of Eq. (H-257).

▶ H-12 DESIGN OF DISCRETE-DATA CONTROL SYSTEMS IN THE FREQUENCY DOMAIN AND THE *z*-PLANE

The *w*-transformation introduced in Section H-5 can be used to carry out the design of discrete-data control systems in the frequency domain. Once the transfer function of the controlled process is transformed into the *w*-domain, all the design techniques for continuous-data control systems can be applied to the design of discrete-data systems.

H-12-1 Phase-Lead and Phase-Lag Controllers in the *w*-Domain

Just as in the *s*-domain, the single-stage phase-lead and phase-lag controllers in the *w*-domain can be expressed by the transfer function

$$G_c(w) = \frac{1 + a\tau w}{1 + \tau w} \tag{H-263}$$

where $a > 1$ corresponds to phase lead and $a < 1$ corresponds to phase lag. When w is replaced by $j\omega_w$, the Bode plots of Eq. (H-263) are identical to those of Figs. 9-28 and 9-45 for $a > 1$ and $a < 1$, respectively. Once the controller is designed in the *w*-domain, the *z*-domain controller is obtained by substituting the *w*-transformation relationship in Eq. (H-163); that is,

$$w = \frac{2}{T}\frac{z - 1}{z + 1} \tag{H-264}$$

The following example illustrates the design of a discrete-data control system using the *w*-transformation in the frequency domain and the *z*-plane.

▶ EXAMPLE H-12-1 Consider the sun-seeker control system described in Section 4-11 and shown in Fig. 9-29. Now let us assume that the system has discrete data so that there is a ZOH in the forward path. The sampling period is 0.01 second. The transfer function of the controlled process is

$$G_p(s) = \frac{2500}{s(s + 25)} \tag{H-265}$$

The *z*-transfer function of the forward path, including the sample-and-hold, is

$$G_{h0}G_p(z) = (1 - z^{-1})\mathcal{Z}\left(\frac{2500}{s^2(s + 25)}\right) \tag{H-266}$$

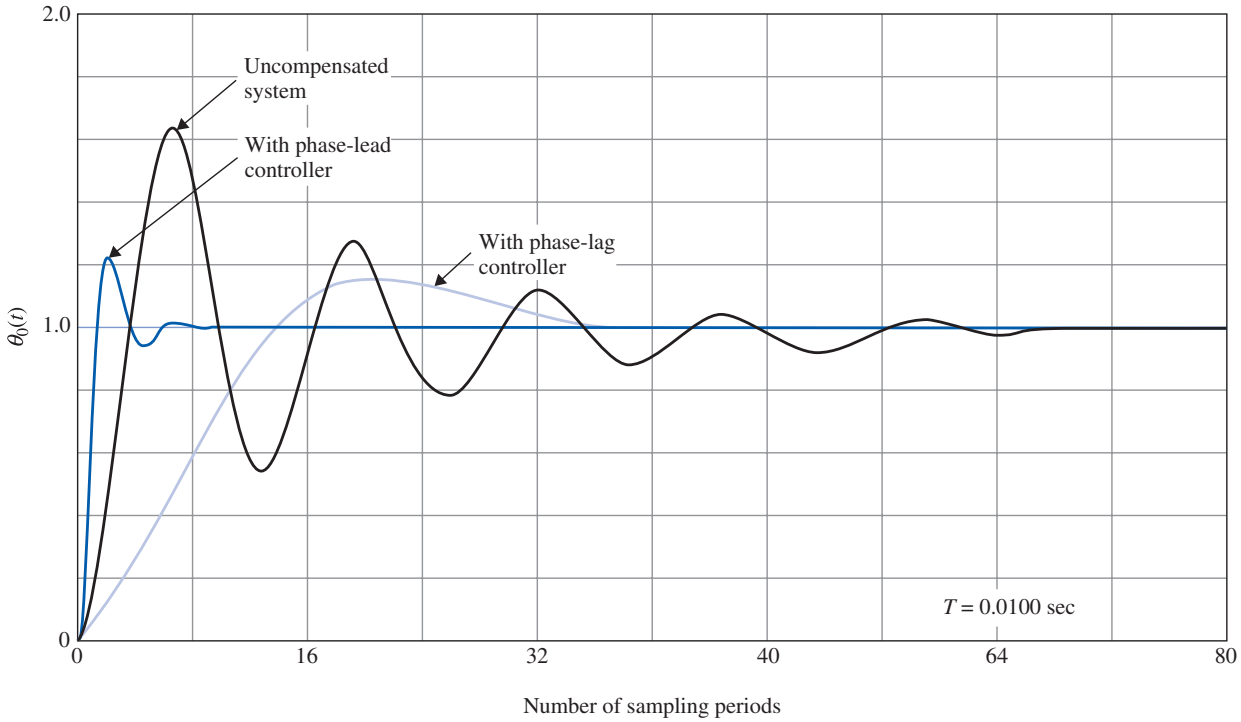


Figure H-50 Step responses of discrete-data sun-seeker system in Example H-12-1.

Carrying out the z-transform in the last equation with $T = 0.01$ second, we get

$$G_{h0}G_p(z) = \frac{0.1152z + 0.106}{(z-1)(z-0.7788)} \tag{H-267}$$

The closed-loop transfer function of the discrete-data system is

$$\frac{\Theta_o(z)}{\Theta_r(z)} = \frac{G_{h0}G_p(z)}{1 + G_{h0}G_p(z)} = \frac{0.1152z + 0.106}{z^2 - 1.6636z + 0.8848} \tag{H-268}$$

The unit-step response of the uncompensated system is shown in Fig. H-50. The maximum overshoot is 66%.

Let us carry out the design in the frequency domain using the w-transformation of Eq. (H-162),

$$z = \frac{(2/T) + w}{(2/T) - w} \tag{H-269}$$

Substituting Eq. (H-269) into Eq. (H-268), we have

$$G_{h0}G_p(w) = \frac{100(1 - 0.005w)(1 + 0.000208w)}{w(1 + 0.0402w)} \tag{H-270}$$

The Bode plot of the last equation is shown in Fig. H-51. The gain and phase margins of the uncompensated system are 6.39 dB and 14.77°, respectively.

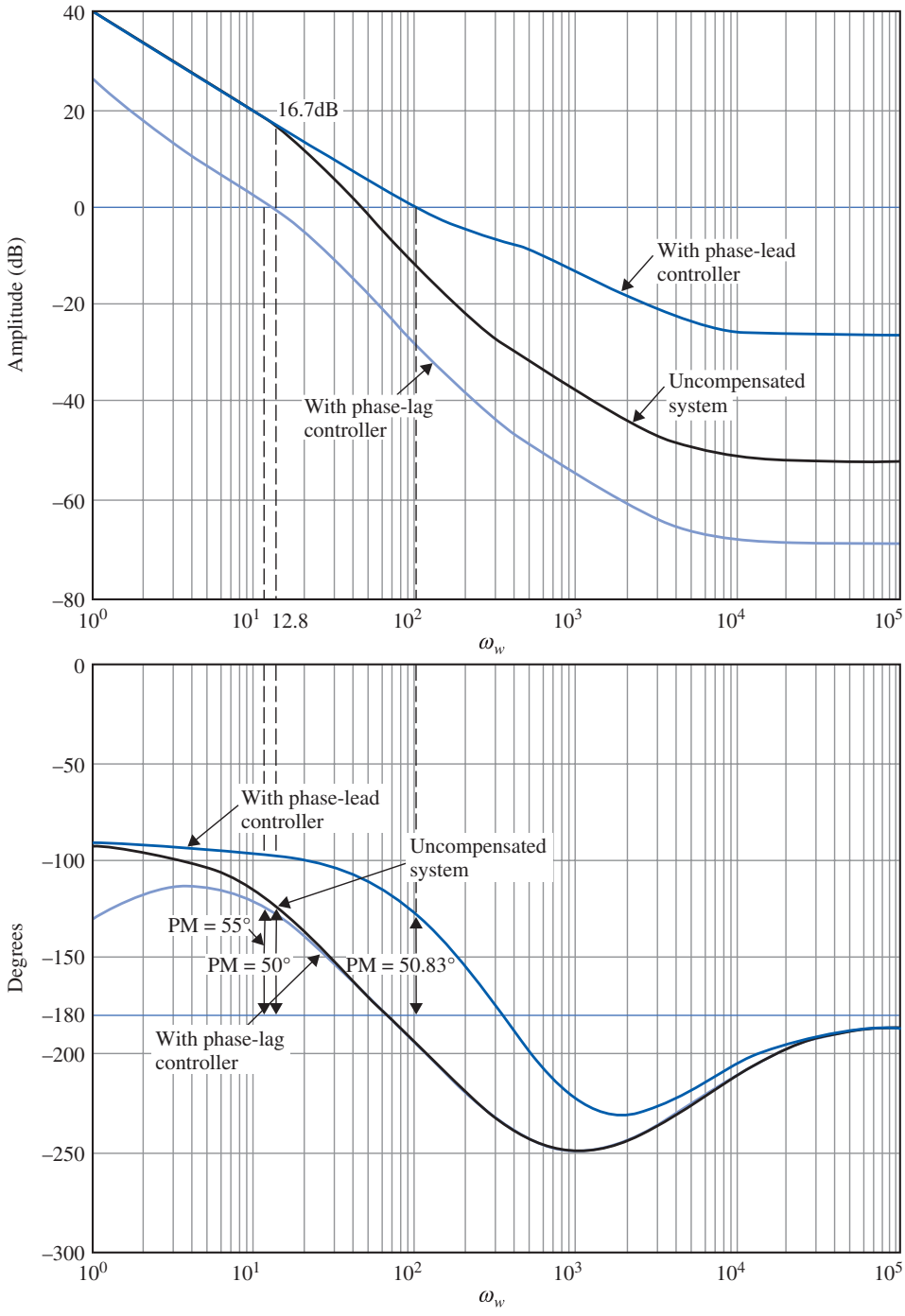


Figure H-51 Bode plots of discrete-data sun-seeker system in Example H-12-1.

Phase-Lag Controller Design in the Frequency Domain

Let us first design the system using a phase-lag controller with the transfer function given in Eq. (H-263) with $a < 1$. Let us require that the phase margin of the system be at least 50°.

From the Bode plot in Fig. H-51, a phase margin of 50° can be realized if the gain crossover point is at $\omega_w = 12.8$ and the gain of the magnitude curve of $G_{ho}G_p(j\omega_w)$ is 16.7 dB.

Thus, we need −16.7 dB of attenuation to bring the magnitude curve down so that it will cross the 0-dB-axis at $\omega_w = 12.8$. We set

$$20 \log_{10} a = -16.7 \text{ dB} \tag{H-271}$$

from which we get $a = 0.1462$. Next, we set $1/a\tau$ to be at least one decade below the gain-crossover point at $\omega_w = 12.8$. We set

$$\frac{1}{a\tau} = 1 \tag{H-272}$$

Thus,

$$\frac{1}{\tau} = a = 0.1462 \tag{H-273}$$

The phase-lag controller in the w-domain is

$$G_c(w) = \frac{1 + a\tau w}{1 + \tau w} = \frac{1 + w}{1 + 6.84w} \tag{H-274}$$

Substituting the z-w-transform relation, $w = (2/T)(z - 1)/(z + 1)$, in Eq. (H-274), the phase-lag controller in the z-domain is obtained:

$$G_c(z) = 0.1468 \frac{z - 0.99}{z - 0.9985} \tag{H-275}$$

The Bode plot of the forward-path transfer function with the phase-lag controller of Eq. (H-274) is shown in Fig. H-51. The phase margin of the compensated system is improved to 55°. The unit-step response of the phase-lag compensated system is shown in Fig. H-50. The maximum overshoot is reduced to 16%.

Phase-Lead Controller Design in the Frequency Domain

A phase-lead controller is obtained by using $a > 1$ in Eq. (H-263). The same principle for the design of phase-lead controllers of continuous-data systems described in Chapter 9 can be applied here. Because the slope of the phase curve near the gain crossover is rather steep, as shown in Fig. H-51, it is expected that some difficulty may be encountered in designing a phase-lead controller for the system. Nevertheless, we can assign a relatively large value for a, rather than using the amount of phase lead required as a guideline.

Let us set $a = 20$. The gain of the controller at high values of ω_w is

$$20 \log_{10} a = 20 \log_{10} 20 = 26 \text{ dB} \tag{H-276}$$

From the design technique outlined in Chapter 9, the new gain crossover should be located at the point where the magnitude curve is at $-26/2 = -13$ dB. Thus, the geometric mean of the two corner frequencies of the phase-lead controller should be at the point where the magnitude of $G_{ho}G_p(j\omega)$ is −13 dB. From Fig. H-51 this is found to be at $\omega_w = 115$. Thus,

$$\frac{1}{\tau} = 115\sqrt{a} = 514 \tag{H-277}$$

The w-domain transfer function of the phase-lead controller is

$$G_c(w) = \frac{1 + a\tau w}{1 + \tau w} = \frac{1 + 0.03888w}{1 + 0.001944w} \tag{H-278}$$

The transfer function of the phase-lead controller in the z-domain is

$$G_c(z) = \frac{8.7776z - 6.7776}{1.3888z + 0.6112} \tag{H-279}$$

The Bode plot of the phase-lead compensated system is shown in Fig. H-51. The phase margin of the compensated system is 50.83°. The unit-step response of the phase-lead compensated system is shown in Fig. H-50. The maximum overshoot is 27%, but the rise time is faster.

Digital PD-Controller Design in the *z*-Plane

The digital PD controller is described by the transfer function in Eq. (H-246) and is repeated as

$$G_c(z) = \frac{\left(K_P + \dfrac{K_D}{T}\right)z - \dfrac{K_D}{T}}{z} \tag{H-280}$$

To satisfy the condition that $G_c(1) = 1$ so that $G_c(z)$ does not affect the steady-state error of the system, we set $K_P = 1$. Applying the digital PD controller as a cascade controller to the sun-seeker system, the forward-path transfer function of the compensated system is

$$G_c(z)G_{ho}G_p(z) = \frac{(1 + 100K_D)z - 100K_D}{z} \frac{0.1152z + 0.106}{(z - 1)(z - 0.7788)} \tag{H-281}$$

We can use the root-contour method to investigate the effects of varying K_D. The characteristic equation of the closed-loop system is

$$z(z^2 - 1.6636z + 0.8848) + 11.52K_D(z - 1)(z + 0.9217) = 0 \tag{H-282}$$

Dividing both sides of the last equation by the terms that do not contain K_D, the equivalent forward-path transfer function with K_D appearing as a multiplying factor is

$$G_{eq}(z) = \frac{11.52K_D(z - 1)(z + 0.9217)}{z(z^2 - 1.6636z + 0.8848)} \tag{H-283}$$

The root contours of the system for $K_D > 0$ are shown in Fig. H-52. These root contours show that the effectiveness of the digital PD controller is limited for this system, since the contours do not dip low enough toward the real axis. In fact, we can show that the best value of K_D from the standpoint of overshoot is 0.022, and the maximum overshoot is 28%.

Digital PE-Controller Design in the *z*-Plane

The digital PI controller introduced in Section H-10-3 can be used to improve the steady-state performance by increasing the system type and, at the same time, improve the relative stability by using the *dipole* principle. Let us select the backward-rectangular integration implementation of the PID controller given by Eq. (H-245). With $K_D = 0$, the transfer function of the digital PI controller becomes

$$G_c(z) = \frac{K_P z - (K_P - K_I T)}{z - 1} = K_P \frac{z - \left(1 - \dfrac{K_I T}{K_P}\right)}{z - 1} \tag{H-284}$$

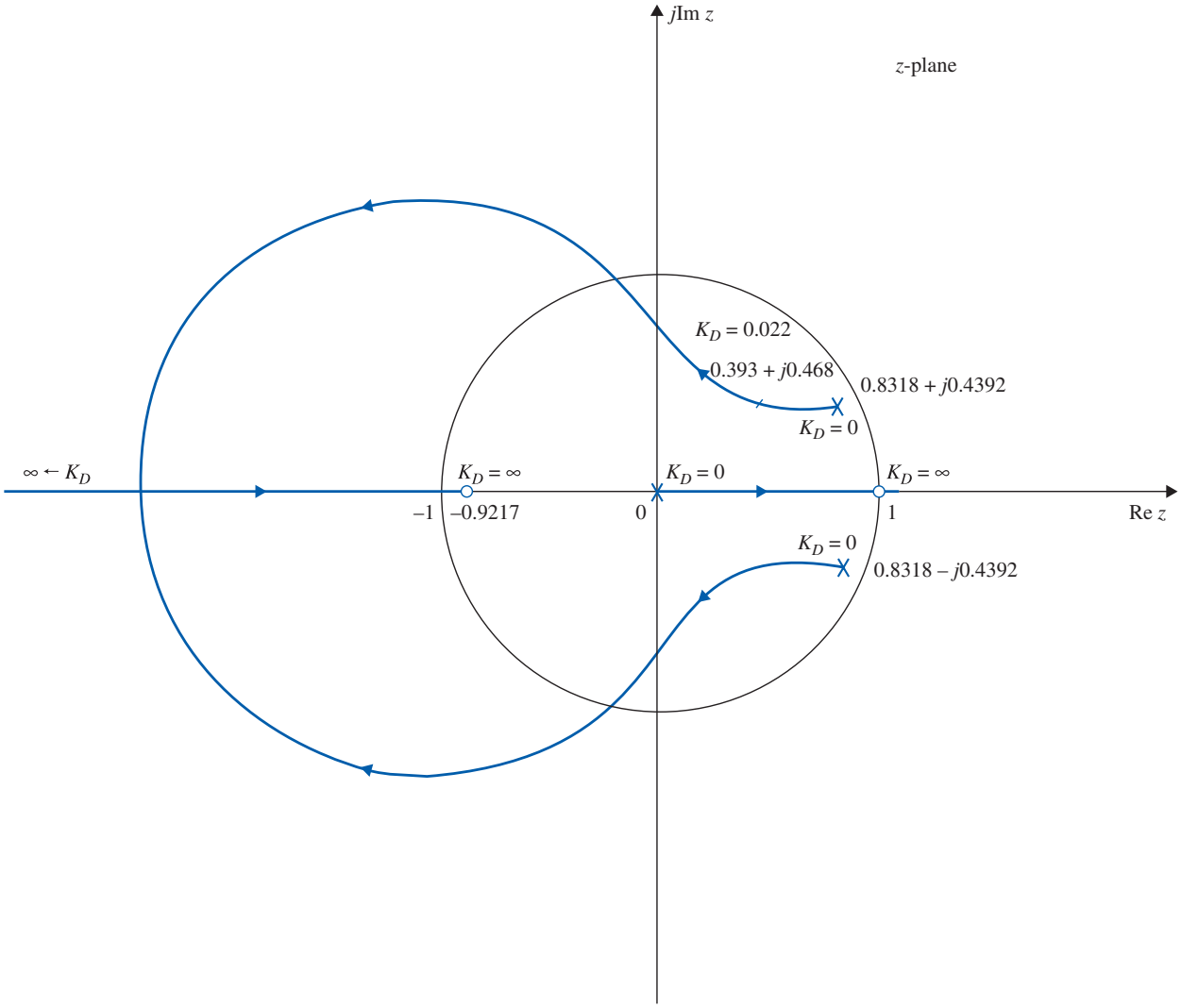


Figure H-52 Root contours of sun-seeker system in Example H-12-1 with digital PD controller. K_D varies.

The principle of the dipole design of the PI controller is to place the zero of $G_c(z)$ very close to the pole at $z = 1$. The effective gain provided by the controller is essentially equal to K_P.

To create a root-locus problem, the transfer function in Eq. (H-267) is written

$$G_{ho}G_p(z) = \frac{K(z + 0.9217)}{(z - 1)(z - 0.7788)} \tag{H-285}$$

where $K = 0.1152$. The root loci of the system are drawn as shown in Fig. H-53. The roots of the characteristic equation when $K = 0.1152$ are $0.8318 + j0.4392$ and $0.8318 - j0.4392$. As shown earlier, the maximum overshoot of the system is 66%. If K is reduced to 0.01152, the characteristic equation roots are at $0.8836 + j0.0926$ and $0.8836 - j0.0926$. The maximum overshoot is reduced to only 3%.

We can show that, if the gain in the numerator of Eq. (H-265) is reduced to 250, the maximum overshoot of the system would be reduced to 3%. This means to realize a

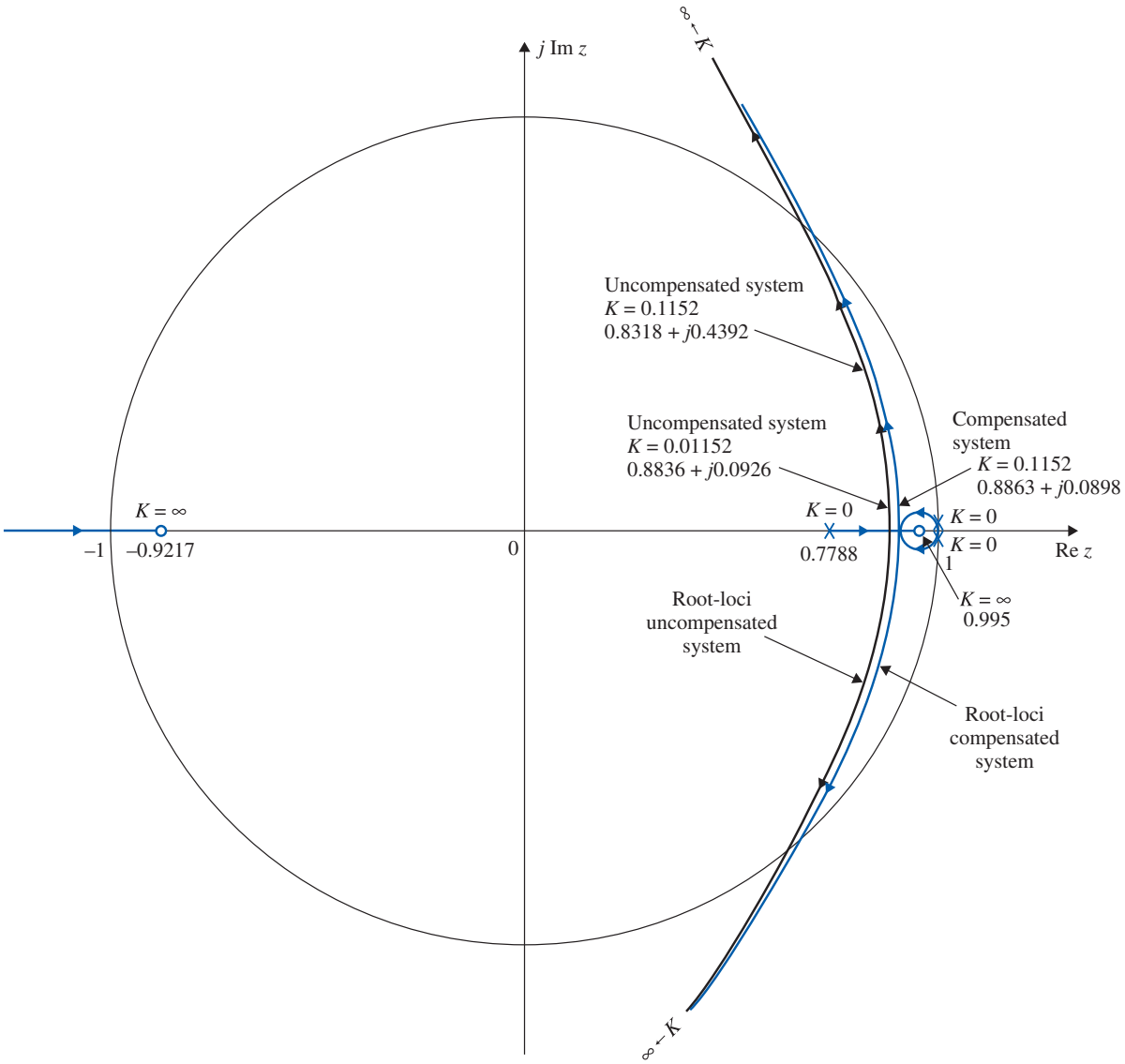


Figure H-53 Root loci of sun-seeker system in Example H-12-1 with and without digital PI controller.

similar improvement on the maximum overshoot, the value of K_P in Eq. (H-284) should be set to 0.1. At the same time, we let the zero of $G_c(z)$ be at 0.995. Thus,

$$G_c(z) = 0.1\,\frac{z - 0.995}{z - 1} \tag{H-286}$$

The corresponding value of K_I is 0.05. The forward-path transfer function of the system with the PI controller becomes

$$G_c(z)G_{ho}G_p(z) = \frac{0.1K(z + 0.995)(z + 0.9217)}{(z - 1)^2(z - 0.7788)} \tag{H-287}$$

where $K = 0.1152$. The root loci of the compensated system are shown in Fig. H-53. When $K = 0.1152$, the two dominant roots of the characteristic equation are at $0.8863 + j0.0898$

and $0.8863 - j0.0898$. The third root is at 0.9948, which is very close to the pole of $G_c(z)$ at $z = 1$, and thus the effect on the transient response is negligible. We can show that the actual maximum overshoot of the system with the forward-path transfer function in Eq. (H-287) is approximately 8%.

This design problem simply illustrates the mechanics of designing a phase-lead and phase-lag controller using the w-transformation method in the frequency domain and digital PD and PI controllers in the z-plane. No attempt was made in optimizing the system with respect to a set of performance specifications.

▶ H-13 DESIGN OF DISCRETE-DATA CONTROL SYSTEMS WITH DEADBEAT RESPONSE

One difference between a continuous-data control system and a discrete-data control system is that the latter is capable of exhibiting a **deadbeat response**. A deadbeat response is one that reaches the desired reference trajectory in a minimum amount of time without error. In contrast, a continuous-data system reaches the final steady-state trajectory or value theoretically only when time reaches infinity. The switching operation of sampling allows the discrete-data systems to have a finite transient period. Fig. H-54 shows a typical deadbeat response of a discrete-data system subject to a unit-step input. The output response reaches the desired steady state with zero error in a minimum number of sampling periods without intersampling oscillations.

H. R. Sirisena [10] showed that given a discrete-data control system with the controlled process described by

$$G_{h0}G_p(z^{-1}) = \frac{Q(z^{-1})}{P(z^{-1})} \tag{H-288}$$

for the system to have a deadbeat response to a step input, the transfer function of the digital controller is given by

$$G_c(z) = \frac{P(z^{-1})}{Q(1) - Q(z^{-1})} \tag{H-289}$$

where $Q(1)$ is the value of $Q(z^{-1})$ with $z^{-1} = 1$.

The following example illustrates the design of a discrete-data system with deadbeat response using Eq. (H-289).

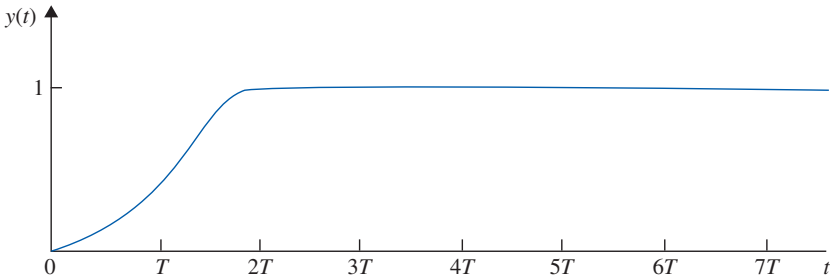


Figure H-54 A typical deadbeat response to a unit-step input.

▶ EXAMPLE H-13-1 Consider the discrete-data sun-seeker system discussed in Example H-12-1. The forward-path transfer function of the uncompensated system is given in Eq. (H-207) and is written as

$$G_{ho}G_p(z^{-1}) = \frac{Q(z^{-1})}{P(z^{-1})} = \frac{0.1152z^{-1}(1 + 0.9217z^{-1})}{(1 - z^{-1})(1 - 0.7788z^{-1})} \tag{H-290}$$

Thus,

$$Q(z^{-1}) = 0.1152z^{-1}(1 + 0.9217z^{-1}) \tag{H-291}$$

$$P(z^{-1}) = (1 - z^{-1})(1 - 0.7788z^{-1}) \tag{H-292}$$

and $Q(1) = 0.22138$.

The digital controller for a deadbeat response is obtained by using Eq. (H-289).

$$G_c(z^{-1}) = \frac{P(z^{-1})}{Q(1) - Q(z^{-1})} = \frac{(1 - z^{-1})(1 - 0.7788z^{-1})}{0.22138 - 0.1152z^{-1} - 0.106z^{-2}} \tag{H-293}$$

Thus,

$$G_c(z) = \frac{(z - 1)(z - 0.7788)}{0.22138z^2 - 0.1152z - 0.106} \tag{H-294}$$

The forward-path transfer function of the compensated system is

$$G_c(z)G_{ho}G_p(z) = \frac{0.1152(z + 0.9217)}{0.22138z^2 - 0.1152z - 0.106} \tag{H-295}$$

The closed-loop transfer function of the compensated system is

$$\begin{aligned} \frac{\Theta_o(z)}{\Theta_r(z)} &= \frac{G_c(z)G_{ho}G_p(z)}{1 + G_c(z)G_{ho}G_p(z)} \\ &= \frac{0.0.5204(z + 0.9217)}{z^2} \end{aligned} \tag{H-296}$$

For a unit-step function input, the output transform is

$$\begin{aligned} \Theta_o(z) &= \frac{0.5204(z + 0.9217)}{z(z - 1)} \\ &= 0.5204z^{-1} + z^{-2} + z^{-3} + \cdots \end{aligned} \tag{H-297}$$

Thus, the output response reaches the unit-step input in two sampling periods.

To show that the output response is without intersampling ripples, we evaluate the output velocity of the system; that is, $\omega_o(t) = d\theta_o(t)/dt$.

The z-transfer function of the output velocity is written

$$\begin{aligned} \frac{\Omega_o(z)}{\Theta_r(z)} &= \frac{G_c(z)(1 - z^{-1})\mathcal{Z}\left[\dfrac{2500}{s(s + 25)}\right]}{G_c(z)G_{ho}G_p(z)} \\ &= \frac{100(z - 1)}{z^2} \end{aligned} \tag{H-298}$$

The output velocity response due to a unit-step input is

$$\Omega_o(z) = \frac{100}{z} = 100z^{-1} \tag{H-299}$$

Thus, the output velocity becomes zero after the second sampling period, which proves that the position response is deadbeat without intersampling ripples. The responses of $\theta_o(t)$ and $\omega_o(t)$ are shown in Fig. H-55. The characteristic of a system with deadbeat response is that the poles

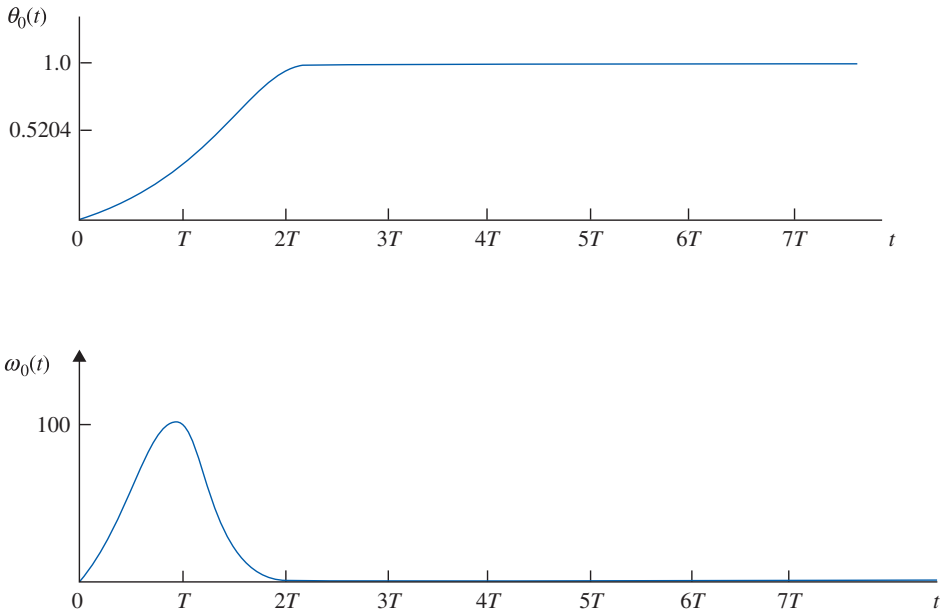


Figure H-55 Output position and velocity responses of discrete-data sun-seeker system in Example H-13-1.

of the closed-loop transfer function are all at $z = 0$. Because these are multiple-order poles, from the standpoint of root sensitivity discussed in Chapter 7, the root sensitivity of a system with deadbeat response is very high. ◀

▶ H-14 POLE-PLACEMENT DESIGN WITH STATE FEEDBACK

Just as for continuous-data systems, pole-placement design through state feedback can be applied to discrete-data systems. Let us consider the discrete-data system described by the following state equation:

$$\mathbf{x}[(k+1)T] = \mathrm{A}\mathbf{x}(kT) + \mathrm{B}u(kT) \tag{H-300}$$

where $\mathbf{x}(kT)$ is an $n \times 1$ state vector, and $u(kT)$ is the scalar control. The state-feedback control is

$$u(kT) = -\mathbf{K}\mathbf{x}(kT) + r(kT) \tag{H-301}$$

where $\mathbf{K}$ is the $1 \times n$ feedback matrix with constant-gain elements. By substituting Eq. (H-301) into Eq. (H-300), the closed-loop system is represented by the state equation

$$\mathrm{x}[(k+1)T] = (\mathbf{A} - \mathbf{BK})\mathbf{x}(kT) \tag{H-302}$$

Just as in the continuous-data case treated in Section 10-17, we can show that, if the pair $[\mathbf{A}, \mathbf{B}]$ is completely controllable, a matrix $\mathbf{K}$ exists that can give an arbitrary set of eigenvalues of $(\mathbf{A} - \mathbf{BK})$; that is, the n roots of the characteristic equation

$$|z\mathbf{I} - \mathbf{A} + \mathbf{BK}| = 0 \tag{H-303}$$

can be arbitrarily placed. The following example illustrates the design of a discrete-data control system with state feedback and pole placement.

▶ EXAMPLE H-14-1 Consider that the sun-seeker system described in Example 10-5-1 is subject to sampled data so that the state diagram of the system without feedback is shown in Fig. H-56(a). The sampling period is 0.01 second. The dynamics of the ZOH are represented by the branch with a transfer function of $1/s$. The closed-loop system with state feedback for the time interval $(kT) \le t \le (k+1)T$ is portrayed by the state diagram shown in Fig. H-56(b), where the feedback gains k_1 and k_2 form the feedback matrix

$$\mathbf{K} = [k_1 \; k_2] \tag{H-304}$$

Applying the SFG gain formula to Fig. H-56(b), with $X_1(s)$ and $X_2(s)$ as outputs and $x_1(kT)$ and $x_2(kT)$ as inputs, we have

$$X_1(s) = \left[\frac{1}{s} - \frac{2500k_1}{s^2(s+25)}\right]x_1(kT) + \left[\frac{1}{s(s+25)} - \frac{2500k_2}{s^2(s+25)}\right]x_2(kT) \tag{H-305}$$

$$X_2(s) = \frac{-2500k_1}{s(s+25)}x_1(kT) - \frac{2500k_2}{s(s+25)}x_2(kT) + \frac{1}{s+25}x_2(kT) \tag{H-306}$$

Taking the inverse Laplace transform on both sides of Eqs. (H-305) and (H-306) and letting $t = (k+1)T$, we have the discrete-data state equations as

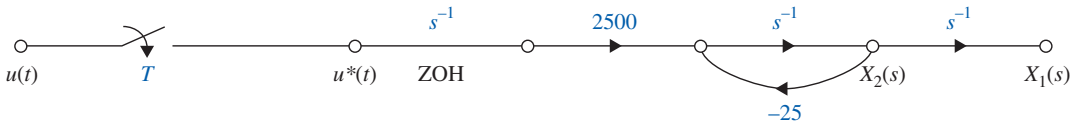
$$\begin{aligned} x_1[(k+1)T] &= (1 - 0.1152k_1)x_1(kT) + (0.2212 - 0.1152k_2)x_2(kT) \\ x_2[(k+1)T] &= -22.12k_1x_1(kT) + (0.7788 - 22.12k_2)x_2(kT) \end{aligned} \tag{H-307}$$

Thus, the coefficient matrix of the closed-loop system with state feedback is

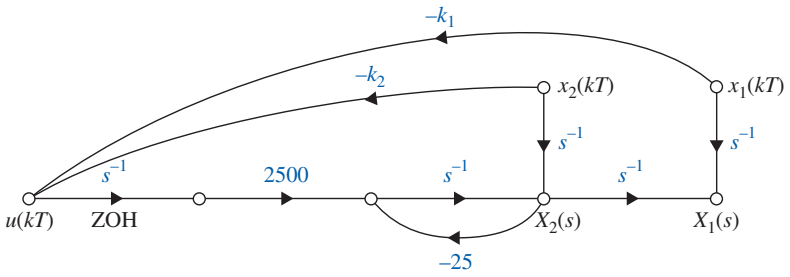
$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 1 - 0.1152k_1 & 0.2212 - 0.1152k_2 \\ -22.12k_1 & 0.7788 - 22.12k_2 \end{bmatrix} \tag{H-308}$$

The characteristic equation of the closed-loop system is

$$\begin{aligned} |z\mathbf{I} - \mathbf{A} + \mathbf{BK}| &= \begin{vmatrix} z - 1 + 0.1152k_1 & -0.2212 + 0.1152k_2 \\ 22.12k_1 & z - 0.7788 + 22.12k_2 \end{vmatrix} \\ &= z^2 + (-1.7788 + 0.1152k_1 + 22.12k_2)z + 0.7788 + 4.8032k_1 - 22.12k_2 \\ &= 0 \end{aligned} \tag{H-309}$$



(a)



(b)

Figure H-56 (a) Signal-flow graph of open-loop, discrete-data, sun-seeker system. (b) Signal-flow graph of discrete-data, sun-seeker system with state feedback.

Let the desired location of the characteristic equation roots be at $z = 0, 0$. The conditions on k_1 and k_2 are

$$-1.7788 + 0.1152k_1 + 22.12k_2 = 0 \tag{H-310}$$

$$0.7788 + 4.8032k_1 - 22.12k_2 = 0 \tag{H-311}$$

Solving for k_1 and k_2 from the last two equations, we get

$$k_1 = 0.2033 \qquad \text{and} \qquad k_2 = -0.07936 \tag{H-312}$$

◀

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▶ PROBLEMS

H-1. Find the z-transforms of the following functions.

(a) $f(k) = ke^{-3k}$ **(b)** $f(k) = k \sin 2k$

(c) $f(k) = e^{-2k} \sin 3k$ **(d)** $f(k) = k^2 e^{-2k}$

H-2. Determine the z-transforms of the following sequences.

(a) $f(kT) = kT \sin 2kT$

(b) $f(k) = \begin{cases} 1 & k = 0, 2, 4, 6, \ \ldots \text{, even integers} \\ -1 & k = 1, 3, 5, 7, \ \ldots \text{, odd integers} \end{cases}$

H-3. Perform the partial-fraction of the following functions, if applicable, and then find the z-transforms using the z-transform table.

(a) $F(s) = \dfrac{1}{(s+5)^3}$ **(b)** $F(s) = \dfrac{1}{s^3(s+1)}$

(c) $F(s) = \dfrac{10}{s(s+5)^2}$ **(d)** $F(s) = \dfrac{5}{s(s^2+2)}$

H-4. Find the inverse z-transforms $f(k)$ of the following functions. Apply partial-fraction expansion to $F(z)$ and then use the z-transform table.

(a) $F(z) = \dfrac{10z}{(z-1)(z-0.2)}$ **(b)** $F(z) = \dfrac{z}{(z-1)(z^2+z+1)}$

(c) $F(z) = \dfrac{z}{(z-1)(z+0.85)}$ **(d)** $F(z) = \dfrac{10}{(z-1)(z-0.5)}$

H-5. Given that $\mathcal{Z}[f(k)] = F(z)$, find the value of $f(k)$ as k approaches infinity without obtaining the inverse z-transform of $F(z)$. Use the final-value theorem of the z-transform if it is applicable.

(a) $F(z) = \dfrac{0.368z}{(z-1)(z^2 - 1.364z + 0.732)}$

(b) $F(z) = \dfrac{10z}{(z-1)(z+1)}$

(c) $F(z) = \dfrac{z^2}{(z-1)(z-0.5)}$

(d) $F(z) = \dfrac{z}{(z-1)(z-1.5)}$

Check the answers by carrying out the long division of $F(z)$ and express it in a power series of z^{-1}.

H-6. Solve the following difference equations by means of the z-transform.

(a) $x(k+2) - x(k+1) + 0.1x(k) = u_s(k) \qquad x(0) = x(1) = 0$

(b) $x(k+2) - x(k) = 0 \qquad x(0) = 1, \quad x(1) = 0$

H-7. This problem deals with the application of the difference equations and the z-transform to a loan-amortization problem. Consider that a new car is purchased with a load of P_0 dollars over a period of N months at a monthly interest rate of r percent. The principal and interest are to be paid back in N equal payments of u dollars each.

(a) Show that the difference equation that describes the loan process can be written as

$$P(k+1) = (1+r)P(k) - u$$

where

$P(k)$ = amount owed after the kth period, $k = 0, 1, 2, \ldots, N$.

$P(\mathrm{O}) = P_0$ = initial amount borrowed

$P(N) = 0$ (after N periods, owe nothing)

The last two conditions are also known as the boundary conditions.

(b) Solve the difference equation in part (a) by the recursive method, starting with $k = 0$, then $k = 1, 2, \ldots$, and substituting successively. Show that the solution of the equation is

$$u = \frac{(1+r)^N P_0 r}{(1+r)^N - 1}$$

(c) Solve the difference equation in part (a) by using the z-transform method.

(d) Consider that $P_0 = \$15{,}000$, $r = 0.01$ (1% per month), and $N = 48$ months. Find u, the monthly payment.

H-8. Perform the partial-fraction expansion to the following z-transfer functions.

(i) $G(z) = \dfrac{5z}{(z-1)(z-0.1)}$

(ii) $G(z) = \dfrac{10z(z-0.2)}{(z-1)(z-0.5)(z-0.8)}$

(iii) $G(z) = \dfrac{z}{(z-1)(z-0.5)^2}$

(iv) $G(z) = \dfrac{2z}{(z-1)(z^2 - z + 1)}$

Find $y(t)$ for $t \geq 0$ when the input is a unit-step function.

H-9. A linear time-invariant discrete-data system has an output that is described by the time sequence

$$y(kT) = 1 - e^{-2kT} \qquad k = 0, 1, 2, \ldots$$

when the system is subject to an input sequence described by $r(kT) = 1$ for all ≥ 0. Find the transfer function $G(z) = Y(z)/R(s)$.

H-10. Find the transfer functions $Y(z)/R(z)$ of the discrete-data systems shown in Fig. HP-10. The sampling period is 0.5 second.

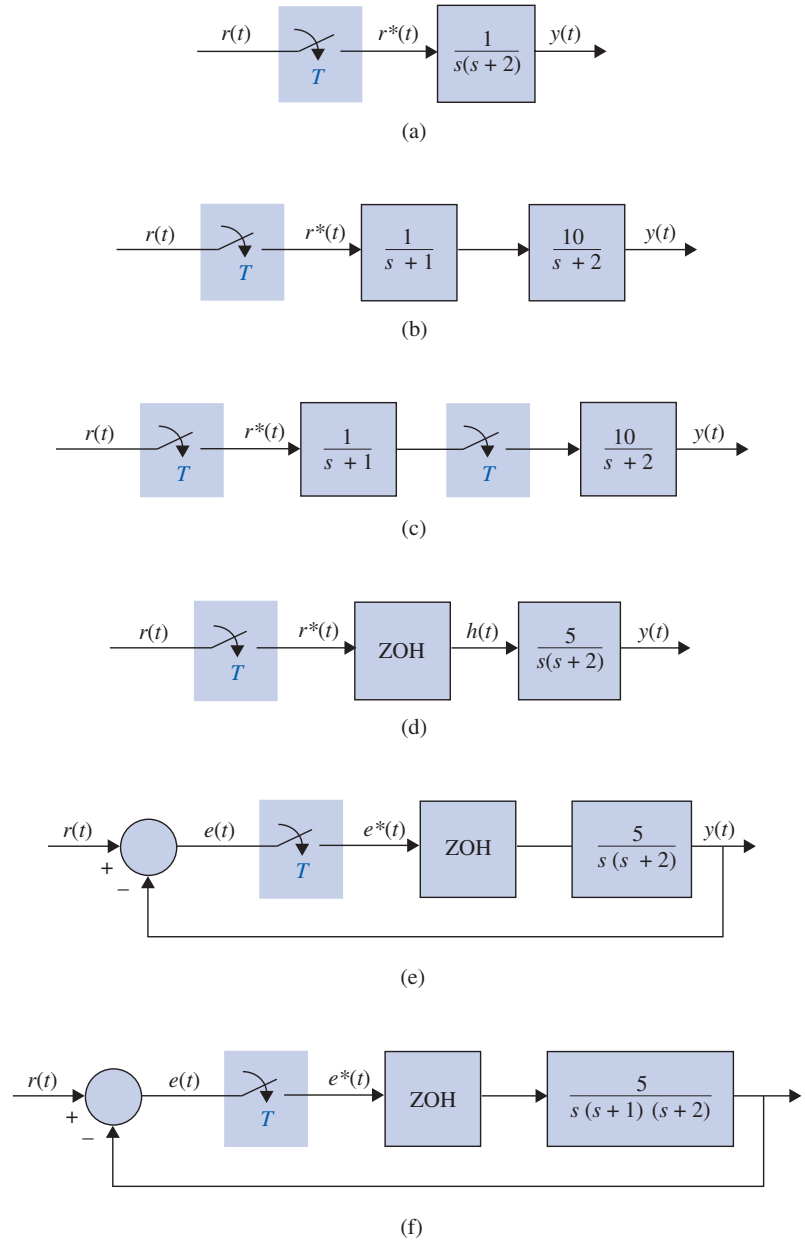


Figure HP-10

H-11. It is well known that the transfer function of an analog integrator is

$$G(s) = \frac{Y(s)}{X(s)} = \frac{1}{s}$$

where $X(s)$ and $Y(s)$ are the Laplace transforms of the input and the output of the integrator, respectively. There are many ways of implementing integration digitally. In a basic computer course, the rectangular integration is described by the schemes shown in Fig. HP-10. The continuous signal $x(t)$ is approximated by a staircase signal; T is the sampling period. The integral of $x(t)$, which is the area under $x(t)$, is approximated by the area under the rectangular approximation signal.

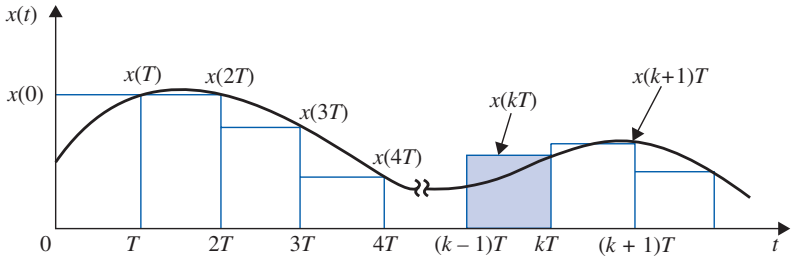
(a) Let $y(kT)$ denote the digital approximation of the integral of $x(t)$ from $t = 0$ to $t = kT$. Then $y(kT)$ can be written as

$$|z_i| > 1 \tag{1}$$

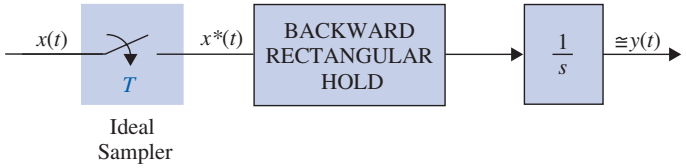
where $y[(k-1)T]$ denotes the area under $x(t)$ from $t = 0$ to $t = (k-1)T$. Take the z-transform on both sides of Eq. (H-1) and show that the transfer function of the digital integrator is

$$G(z) = \frac{Y(z)}{X(z)} = \frac{Tz}{z-1}$$

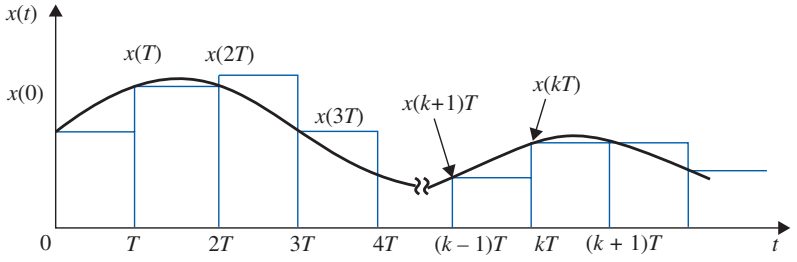
(b) The rectangular integration described in Fig. HP-11(a) can be interpreted as a sample-and-hold operation, as shown in Fig. HP-11(b). The signal $x(t)$ is first sent through an ideal sampler with sampling period T. The output of the sampler is the sequence $x(0)$, $x(T)$, . . . , $x(kT)$, ... These numbers are then sent through a "backward" hold device to give the rectangle of height $x(kT)$ during the time interval from $(k-1)T$ to kT. Verify the result obtained in part (a) for $G(z)$ using the "backward" sample-and-hold interpretation.



(a)



(b)



(c)

Figure HP-11

(c) As an alternative, we can use a "forward" rectangular hold, as shown in Fig. HP-11(c). Find the transfer function $G(z)$ for such a rectangular integrator.

H-12. The block diagram of a sampled-data system is shown in Fig. HP-12. The state equations of the controlled process are

$$\frac{dx_1(t)}{dt} = x_2(t) \qquad \frac{dx_2(t)}{dt} = -2x_1(t) - 3x_2(t) + h(t)$$

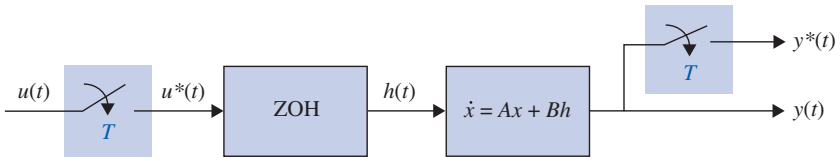


Figure HP-12

where $h(t)$ is the output of the sample-and-hold; that is, $u(t)$ is constant during the sampling period T.

(a) Find the vector-matrix discrete state equations in the form of

$$\mathbf{x}[(k+1)T] = \phi(T)\mathbf{x}(kT) + \theta(T)u(kT)$$

(b) Find $\mathbf{x}(NT)$ as functions of $\mathbf{x}(0)$ and $u(kT)$ for $k = 0, 1, 2, \ldots N$.

H-13. Repeat Problem H-12 for the linear sampled-data system with the following state equations.

$$\frac{dx_1(t)}{dt} = x_1(t) \qquad \frac{dx_2(t)}{dt} = u(t)$$

The sampling period is 0.001 second.

H-14.

(a) Find the transfer function $\mathbf{X}(z)/U(z)$ for the system described in Problem H-12.

(b) Find the characteristic equation of the system described in Problem H-12.

H-15.

(a) Find the transfer function $\mathbf{X}(z)/U(z)$ for the system described in Problem H-13.

(b) Find the characteristic equation and its roots of the system described in Problem H-13.

H-16. Draw a state diagram for the digital control system represented by the following dynamic equations:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k) \qquad y(k) = x_1(k)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & -1 \\ 0 & 1 & 2 \\ 5 & 3 & -1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find the characteristic equation of the system.

H-17. The state diagram of a digital control system is shown in Fig. HP-17. Write the dynamic equations.

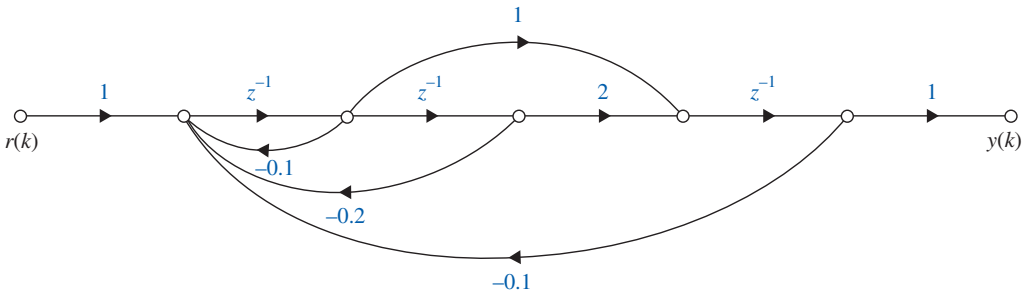


Figure HP-17

Find the transfer function $Y(z)/R(z)$.

H-18. The block diagram of a sampled-data system is shown in Fig. HP-18. Write the discrete state equations of the system. Draw a state diagram for the system.

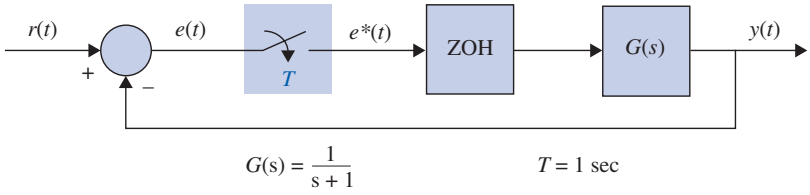


Figure HP-18

H-19. Apply the w-transform to the following characteristic equations of discrete-data control systems, and determine the conditions of stability (asymptotically stable, marginally stable, or unstable) using the Routh-Hurwitz criterion.

(a) $z^2 + 1.5z - 1 = 0$

(b) $z^3 + z^2 + 3z + 0.2 = 0$

(c) $z^3 - 1.2z^2 - 2z + 3 = 0$

(d) $z^3 - z^2 - 2z + 0.5 = 0$

Check the answers by solving for the roots of the equations using a root-finding computer program.

H-20. A digital control system is described by the state equation

$$x(k+1) = (0.368 - 0.632K)x(k) + Kr(k)$$

where $r(k)$ is the input, and $x(k)$ is the state variable. Determine the values of K for the system to be asymptotically stable.

H-21. The characteristic equation of a linear digital control system is

$$z^3 + z^2 + 1.5Kz - (K + 0.5) = 0$$

Determine the values of K for the system to be asymptotically stable.

H-22. The block diagram of a discrete-data control system is shown in Fig. HP-22.

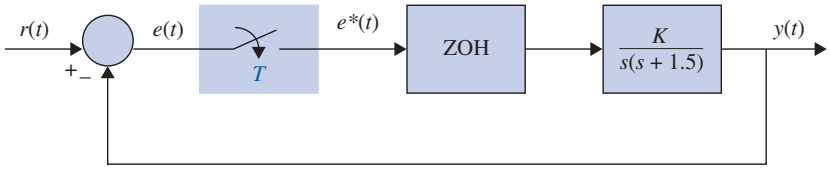


Figure HP-22

(a) For $T = 0.1$ second, find the values of K so that the system is asymptotically stable at the sampling instants.

(b) Repeat part (a) when the sampling period is 0.5 second.

(c) Repeat part (a) when the sampling period is 1.0 second.

H-23. Use a root-finding computer program to find the roots of the following characteristic equations of linear discrete-data control systems, and determine the stability condition of the systems.

(a) $z^3 + 2z^2 + 1.2z + 0.5 = 0$

(b) $z^3 + z^2 + z - 0.5 = 0$

(c) $0.5z^3 + z^2 + 1.5z + 0.5 = 0$

(d) $z^4 + 0.5z^3 + 0.25z^2 + 0.1z - 0.25 = 0$

H-24. The block diagram of a sampled-data control system is shown in Fig. HP-24.

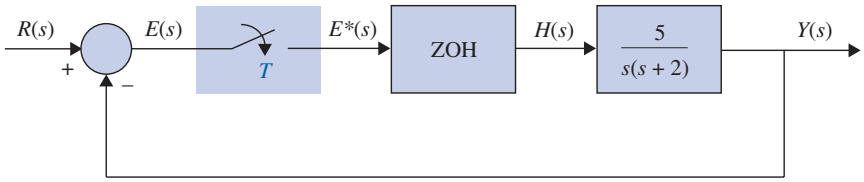


Figure HP-24

(a) Derive the forward-path and the closed-loop transfer functions of the system in z-transforms. The sampling period is 0.1 second.

(b) Compute the unit-step response $y(kT)$ for $k = 0$ to 100.

(c) Repeat parts (a) and (b) for $T = 0.05$ second.

H-25. The block diagram of a sampled-data control system is shown in Fig. HP-25.

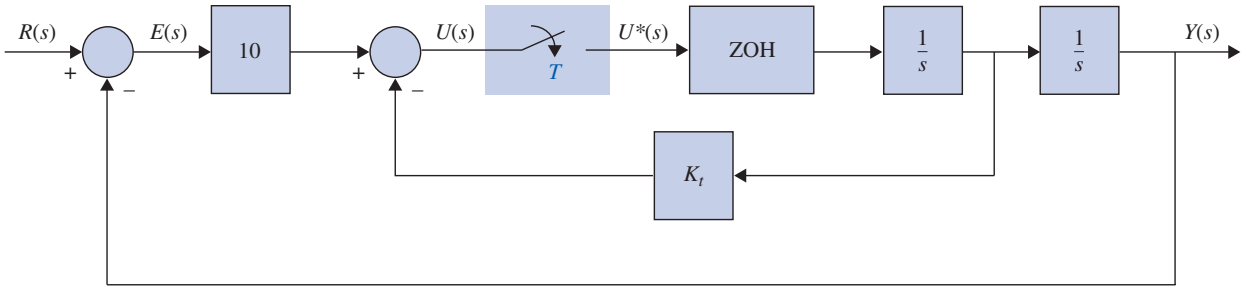


Figure HP-25

(a) Find the error constants K_p^*, K_v^*, and K_a^*.

(b) Derive the transfer functions $Y(z)/E(z)$ and $Y(z)/R(z)$.

(c) For $T = 0.1$ second, find the critical value of K for system stability.

(d) Compute the unit-step response $y(kT)$ for $k = 0$ to 50 for $T = 0.1$ second and $K_t = 5$.

(e) Repeat part (d) for $T = 0.1$ second and $K_t = 1$.

H-26. The forward-path dc-motor control system described in Problem 10-43 is now incorporated in a digital control system, as shown in Fig. HP-26(a). The microprocessor takes the information from the encoder and computes the velocity information. This generates the sequence of numbers, $\omega(kT)$ $k = 0, 1, 2, \ldots$. The microprocessor then generates the error signal $e(kT) = r(kT) - \omega(kT)$. The digital control is modeled by the block diagram shown in Fig. HP-26(b). Use the parameter values given in Problem 10-43.

(a) Find the transfer function $\Omega(z)/E(z)$ with the sampling period $T = 0.1$ second.

(b) Find the closed-loop transfer function $\Omega(z)/R(z)$. Find the characteristic equation and its roots. Locate these roots in the z-plane. Show that the closed-loop system is unstable when $T = 0.1$ second.

(c) Repeat parts (a) and (b) for $T = 0.01$ and 0.001 second. Use any computer simulation program.

(d) Find the error constants K_p^*, K_v^*, and K_a^*. Find the steady-state error $e(kT)$ as $k \to \infty$ when the input $r(t)$ is a unit-step function, a unit-ramp function, and a parabolic function $t^2 u_s(t)/2$.

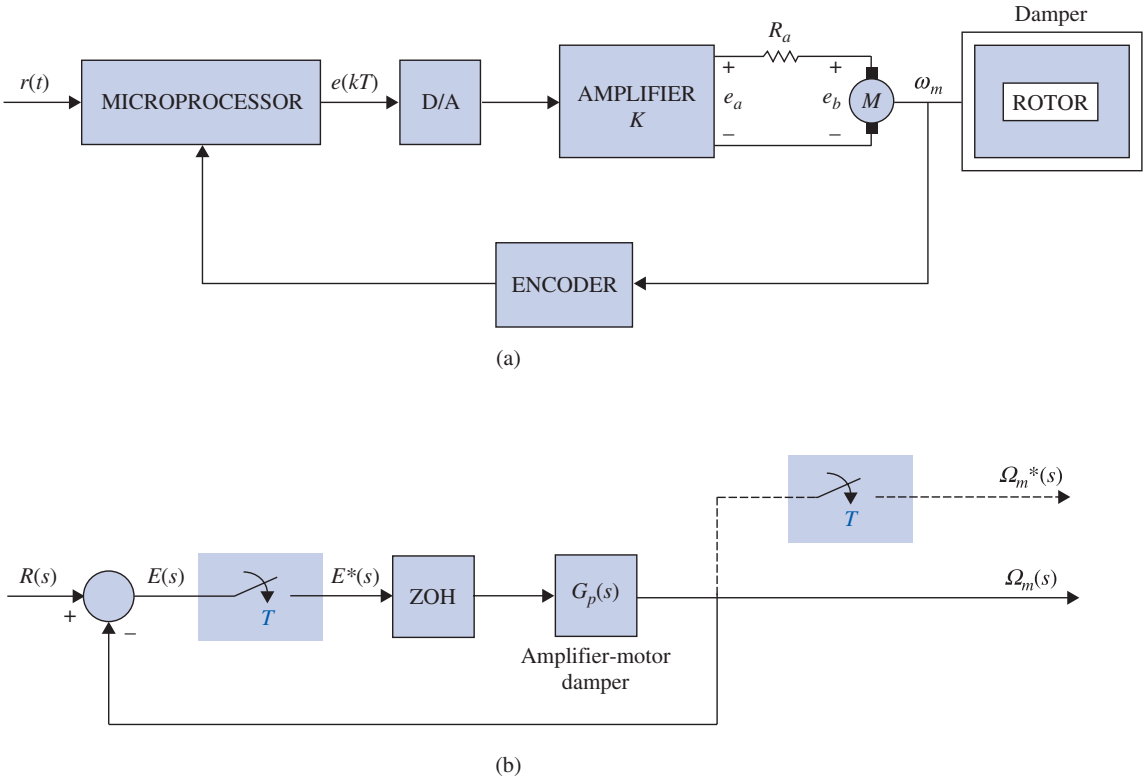


Figure HP-26

H-27. The block diagram of a sampled-data control system is shown in Fig. HP-27.

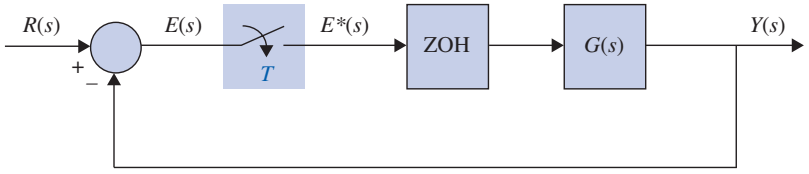


Figure HP-27

(a) Construct the root loci in the z-plane for the system for $K \geq 0$, without the zero-order hold, when $T = 0.5$ second, and then with $T = 0.1$ second. Find the marginal values of K for stability.

$$G(s) = \frac{K}{s(s+5)}$$

(b) Repeat part (a) when the system has a zero-order-hold, as shown in Fig. HP-27.

H-28. The system shown in Fig. HP-27 has the following transfer function:

$$G(s) = \frac{Ke^{-0.1s}}{s(s+1)(s+2)}$$

Construct the root loci in the z-plane for $K \geq 0$, with $T = 0.1$ second.

H-29. The characteristic equations of linear discrete-data control systems are given in the following equations. Construct the root loci for $K \geq 0$. Determine the marginal value of K for stability.

(a) $z^3 + Kz^2 + 1.5Kz - (K+1) = 0$

(b) $z^2 + (0.15K - 1.5)z + 1 = 0$

(c) $z^2 + (0.1K - 1)z + 0.5 = 0$

(d) $z^2 + (0.4 + 0.14K)z + (0.5 + 0.5K) = 0$

(e) $(z-1)(z^2 - z + 0.4) + 4 \times 10^{-5}K(z+1)(z+0.7) = 0$

H-30. The forward-path transfer function of a unity-feedback discrete-data control system with sample-and-hold is

$$G_{ho}G(z) = \frac{0.0952z}{(z-1)(z-0.905)}$$

The sampling period is $T = 0.1$ second.

(a) Plot the plot of $G_{ho}G(z)$ and determine the stability of the closed-loop system.

(b) Apply the w-transformation of Eq. (H-226) to $G_{ho}G(z)$ and plot the Bode plot of $G_{ho}G(w)$. Find the gain and phase margins of the system.

H-31. Consider that the liquid-level control system described in Problem 10-50 is now subject to sample-and-hold operation. The forward-path transfer function of the system is

$$G_{ho}G(z) = \frac{1 - e^{-Ts}}{s}\left(\frac{16.67N}{s(s+1)(s+12.5)}\right)$$

The sampling period is 0.05 second. The parameter N represents the number of inlet valves. Construct the Bode plot of $G_{ho}G(w)$ using the w-transformation of Eq. (H-226), and determine the limiting value of N (integer) for the closed-loop system to be stable.

H-32. Find the digital equivalents using the following integration rules for the controllers given. (a) Backward-rectangular integration rule, (b) forward-rectangular integration rule, and (c) trapezoidal-integration rule. Use the backward-difference rule for derivatives.

(i) $G_c(s) = 2 + \dfrac{200}{s}$ (ii) $G_c(s) = 10 + 0.1s$ (iii) $G_c(s) = 1 + 0.2s + \dfrac{5}{s}$

H-33. A continuous-data controller with sample-and-hold units is shown in Fig. HP-33. The sampling period is 0.1 second. Find the transfer function of the equivalent digital controller. Draw a digital-program implementation diagram for the digital controller. Carry out the analysis for the following continuous-data controllers.

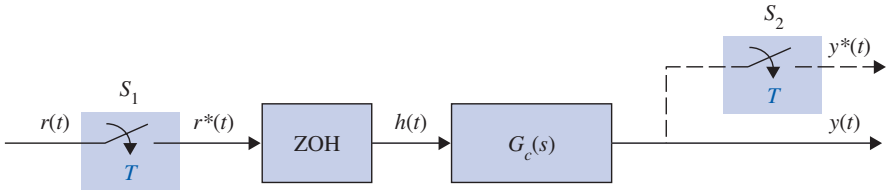


Figure HP-33

(a) $G_c(s) = \dfrac{10}{s+12}$ **(b)** $G_c(s) = \dfrac{10(s+1.5)}{(s+10)}$

(c) $G_c(s) = \dfrac{s}{s+1.55}$ **(d)** $G_c(s) = \dfrac{1+0.4s}{1+0.01s}$

H-34. Determine which of the following digital transfer functions are physically realizable.

(a) $G_c(z) = \dfrac{10(1 + 0.2z^{-1} + 0.5z^{-2})}{z^{-1} + z^{-2} + 1.5z^{-3}}$ **(b)** $G_c(z) = \dfrac{1.5z^{-1} - z^{-2}}{1 + z^{-1} + 2z^{-2}}$

(c) $G_c(z) = \dfrac{z + 1.5}{z^3 + z^2 + z + 1}$ **(d)** $G_c(z) = z^{-1} + 0.5z^{-3}$

(e) $G_c(z) = 0.1z + 1 + z^{-1}$ **(f)** $G_c(z) = \dfrac{z^{-1} + 2z^{-2} + 0.5z^{-3}}{z^{-1} + z^{-2}}$

H-35. The transfer function of the process of a control system is

$$G_p(s) = \frac{4}{s^2}$$

The block diagram of the system with a PD controller and sample-and-hold is shown in Fig. HP-35. Find the transfer function of the digital PD controller using the following equation.

$$G_c(z) = \frac{\left(K_P + \dfrac{K_D}{T}\right) z - \dfrac{K_D}{T}}{z}$$

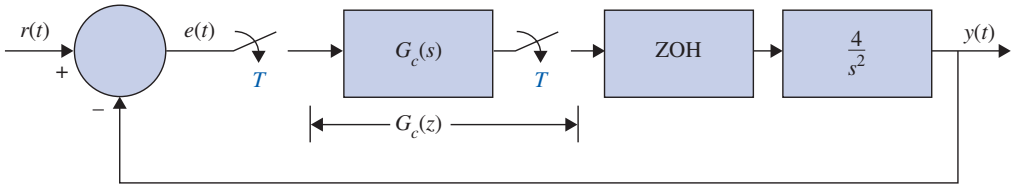


Figure HP-35

Select a sampling period T so that the maximum overshoot of $y(kT)$ will be less than 1%.

H-36. Fig. HP-35 shows the block diagram of the control system described in Problem H-35 with a digital PD controller. The sampling period is 0.01 second. Consider that the digital PD controller has the transfer function

$$G_c(z) = K_P + \frac{K_P(z-1)}{T_z}$$

(a) Find the values of K_P and K_D so that two of the three roots of the characteristic equation are at 0.5 and 0.5. Find the third root. Plot the output response $y(kT)$ for $k = 0, 1, 2, \ldots$.

(b) Set $K_P = 1$. Find the value of K_D so that the maximum overshoot of $y(kT)$ is a minimum.

H-37. For the inventory-control system described in Problem H-36, design a phase-lead controller using the w-transformation so that the phase-margin of the system is at least 60°. Can you design a phase-lag controller in the w-domain? If not, explain why not.

H-38. The process transfer function of the second-order aircraft attitude control system described in Chapter 9 is

$$G_p(s) = \frac{4500K}{s(s + 361.2)}$$

Consider that the system is to be compensated by a series digital controller $G_c(z)$ through a sample-and-hold.

(a) Find the value of K so that the discrete ramp-error constant K_v^* is 100.

(b) With the value of K found in part (a), plot the unit-step response of the output $y^*(t)$ and find maximum overshoot.

(c) Design the digital controller so that the output is a deadbeat response to a step input. Plot the unit-step response.

H-39. The sun-seeker system described in Example H-2-5 is considered to be controlled by a series digital controller with the transfer function $G_c(z)$. The sampling period is 0.01 second. Design the controller so that the output of the system is a deadbeat response to a unit-step input. Plot the unit-step response of the designed system.

H-40. Design the state-feedback control for the sun-seeker system in Example H-2-5 so that the characteristic equation roots are at $z = 0.5, 0.5$.

H-41. Consider the digital control system

$$\mathbf{x}[(k+1)T] = \mathbf{A}\mathbf{x}(kT) + \mathbf{B}u(kT)$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The state-feedback control is described by $u(kT) = -\mathbf{K}\mathbf{x}(kT)$, where $\mathbf{K} = [k_1 k_2]$. Find the values of k_1 and k_2 so that the roots of the characteristic equation of the closed-loop system are at 0.5 and 0.7.

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